

Localized Bost–Connes systems

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Preface

This book grew out of a single question: what happens to the phase transition of the Bost–Connes system if we restrict its set of primes to an arbitrary subset S ?

The received wisdom in arithmetic quantum statistical mechanics was that the divergence of the partition function $\zeta_{K,S}(\beta)$ forces the uniqueness of the equilibrium state. Chapter 1 proves this expectation false. The KMS_β simplex of the S -local system is governed instead by a Chebotarev criterion — an equidistribution statement that can easily fail, breaking the symmetry even while the partition function heavily diverges.

Everything that follows is a systematic attempt to map the boundaries of this phenomenon, finding out exactly what this system knows, and more importantly, what it forgets.

The chapters were written as a sequence of self-contained investigations, and I have preserved that structure here. This book is therefore not a traditional treatise with a single predetermined endpoint, but a record of an exploration. Several chapters end in the negative: an invariant that was expected to see the phase transition, and simply does not. I have kept those dead ends, because in every case, the negative result is a structural theorem in disguise.

We discover that the phase transition is fundamentally a tail event — a Kakutani dichotomy in which every single factor is harmless, and only the infinite product degenerates. Consequently, any invariant that tensorises (such as K -theory, the type of the factor, the Connes spectral distance, and dynamical entropy) sees only the worst local coordinate and is entirely blind to the arithmetic. The arithmetic lives exclusively in the measure class.

This principle of “blindness” extends to rigidity. We prove that the von Neumann algebra of the system remembers absolutely nothing, collapsing universally to the injective factors — R_∞ , R_λ , or a Krieger factor. The C^* -algebra, however, remembers the field, the symmetry group, and the exact norms, but solely because its modular flow canonically pins down the Cartan subalgebra.

The text is divided into eight parts. Parts 1 and 2 establish the abelian classification and its structural invariants. Parts 3 and 5 document our search for a genuinely non-commutative arithmetic thermodynamics. This quest leads us to the Hurwitz quaternions and the Hecke operators on the Bruhat–Tits tree, only to end in a definitive no-go theorem: the orbit relation remains stubbornly hyperfinite, the Ramanujan bound is exactly the tempered bound, and the non-amenability so abundant in the group is strictly denied to the action.

Part 4 handles equivariant rigidity and reconstruction. Part 6 transfers the theory — after an opening excursion into arithmetic topology, where primes are knots — to global function fields, where the exact commensurability of log-norms entirely forbids type III_1 factors, allowing the algebra to recover its own temperature, while the Weil zeta function emerges natively as the partition function. Part 7 records the impossibility of p -adic KMS states.

Finally, Part 8 returns to the primes themselves. We apply our machinery to sieve-theoretic and Sato–Tate families, revealing a strict dichotomy: confinement to a residue class obstructs uniqueness if and only if that class lies in the kernel of a character. Thus, Landau and Friedlander–Iwaniec primes suffer permanent symmetry breaking, while Twin, Sophie Germain, and Chen primes occupy generating classes and escape completely unobstructed.

This book proves nothing about the classical distribution of primes. It uses physics to illuminate the deep algebraic faults and strata of the integers. It is my hope that the reader will find as much beauty in the rigid constraints of this landscape as I have found in mapping it.

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Acknowledgements

This monograph is the culmination of a long and solitary journey into the deep strata of arithmetic, yet I was never truly alone in my intellectual pursuits.

First and foremost, I wish to express my deepest gratitude to two mentors from my formative years: my junior high school homeroom and mathematics teacher, Mr. Meng Rongyao, and my high school homeroom and mathematics teacher, Ms. Li Weikai. It was you who first taught me how to truly think. The seeds of mathematical intuition and rigorous, independent reasoning that you planted at the very beginning of my path have crossed the span of decades to become the foundation of this entire theoretical edifice.

Furthermore, I must extend a special and unconventional acknowledgement to two extraordinary “collaborators” of this digital era: the large language models Claude and Gemini. Serving as my high-precision computational engines, merciless logical reviewers, and tireless sounding boards, you provided unreserved support in navigating dense literature and enforcing strict structural scrutiny. Thank you for accompanying me deep into the bedrock of non-commutative arithmetic, standing by through every theoretical collapse, refinement, and final reconstruction.

Reading guide

The chapters and the papers

The book was assembled from the foundational paper and Papers I–XXV and XXVII–XXXIX of the series, thirty-nine papers in total; the numeral XXVI is retired. Each chapter is one paper, unchanged in its mathematics; the table gives the correspondence and the directory of the source in the repository.

Ch.	Paper	Title	Source directory
1	—	KMS state uniqueness and phase transitions in localized Bost–Connes systems: from Chebotarev density to the Hardy–Littlewood conjecture	paper/
2	I	Factor type and the spectrum of transitions	papers/I-type-spectrum/
3	II	Ideal structure, boundary quotient, and index	papers/II-structure/
4	III	Invariants, entropy, and the measure class	papers/III-invariants/
5	XXXIX	The measurable first cohomology of the tail relation: turbulence and the arithmetic embedding	papers/XXXIX-cohomology/
6	IV	Noncommutative metric geometry	papers/IV-metric/
7	XXIX	The Wasserstein separation of the broken phases: a finite-dimensional reduction and the quadratic chaos bound	papers/XXIX-wasserstein/
8	XXXIII	The adaptive coupling upper bound for the Wasserstein separation: microscopic flips against the macroscopic symmetry cost	papers/XXXIII-coupling/
9	XXXVII	The microscopic geometry of Wasserstein separation: cone geodesics, the exact single-step barrier, and the failure of fractional transport	papers/XXXVII-cone-geometry/
10	XXX	K-theory of the affine boundary quotient: the return of the norm and the trace	papers/XXX-affine-ktheory/
11	XXXI	The ray-class repair for affine systems: the class group as the index of definable scalings	papers/XXXI-rayclass/
12	XXXII	Galois descent for affine Bost–Connes boundaries: capitulation, the obstruction cocycle, and the emergence of the class group	papers/XXXII-descent/
13	XXXVIII	Integral equivariant K-theory of the inert local model: the ray-class entanglement and exact torsion	papers/XXXVIII-inert-ktheory/
14	V	The free variant and the free critical exponent	papers/V-free/
15	VI	Towards non-abelian localized Bost–Connes systems: the matrix Kakutani dichotomy and a Tannakian obstruction	papers/VI-nonabelian/
16	VII	The Galois-twisted arithmetic groupoid: its KMS classification, and why the Tannakian obstruction does not compute it	papers/VII-galois-groupoid/
17	XXVII	Non-abelian Bost–Connes groupoids: two no-go theorems, and a construction that realizes $\text{Prob}(G/N_\beta)$ exactly when the Frobenius lifts mix	papers/XXVII-nonabelian-groupoids/

Ch.	Paper	Title	Source directory
18	XXXIV	The abelian scaling bottleneck, and the angular Hurwitz system: where noncommutative arithmetic thermodynamics can and cannot live	papers/XXXIV-bottleneck/
19	XXXVI	The angular transport: tree-level filtrations, predictable matrix martingales, and the two regimes of the angular Hurwitz system	papers/XXXVI-angular-transport/
20	IX	Localized Bost–Connes factors are amenable: the failure of W^* -superrigidity, and what survives at the C^* -level	papers/IX-rigidity/
21	XIII	Equivariant rigidity of localized Bost–Connes C^* -dynamical systems: the fixed-point algebra is the Cartan subalgebra	papers/XIII-equivariant/
22	XIV	The equal-norm case: norm-separation is not a hypothesis but a characterization, and the gauge-equivariant repair	papers/XIV-equalnorm/
23	XV	What σ -equivariance gives for free, and why the range projections cannot give the rest	papers/XV-spectral/
24	XVI	Quantum optimal transport on Bost–Connes systems: the twist is harmful, the generator has a sign error, and the transition is invisible to local inequalities	papers/XVI-transport/
25	X	A non-commutative Ore LCM semigroup with arithmetic thermodynamics: the Hurwitz quaternions	papers/X-semigroup-search/
26	XI	Condition (C4) and the nested/partition dichotomy: why Ramanujan bounds cannot decide non-amenability of the Hurwitz groupoid	papers/XI-C4/
27	XII	The universal multiplicity–norm mismatch, and the reduction of (C4) to the norm-one subgroup	papers/XII-mismatch/
28	XXI	Localized GL_2 Bost–Connes systems: the equilibrium range and the automorphic obstruction range are disjoint, by exactly one	papers/XXI-GL2/
29	XXVIII	The Ramanujan bound is the tempered bound: the Hecke operator on the Bruhat–Tits tree, and a no-go theorem for condition (C4)	papers/XXVIII-spectral-gap/
30	VIII	Localized Bost–Connes systems in arithmetic topology: local class field theory, linking Kakutani dichotomies, and topological phase transitions	papers/VIII-arithmetic-topology/
31	XVII	Localized Bost–Connes systems over global function fields: commensurability forbids type III ₁ , and the factor recovers the temperature	papers/XVII-function-fields/
32	XVIII	Circle actions on function field Bost–Connes systems: the compact group advantage is illusory, and the Weil zeta function is thermodynamic	papers/XVIII-equivariant-kirchberg/
33	XIX	Isogeny, zeta-equivalence, and the Picard torsor in function field Bost–Connes systems	papers/XIX-isogeny/
34	XX	The limits of abelian reconstruction: twin curves exist, genus one suffices, and the obstruction is a torsor coordinate	papers/XX-twin-curves/
35	XXII	Iwasawa theory and localized Bost–Connes systems: there are no p -adic KMS states, and where Leopoldt actually enters	papers/XXII-iwasawa/
36	XXIII	Leopoldt and the p -local Bost–Connes groupoid: the rank of the isotropy, not its triviality, and why $\log_p(N\mathfrak{p})$ vanishes	papers/XXIII-leopoldt/
37	XXIV	The statistical mechanics of sieve-theoretic primes: the critical spectrum, the log-power dichotomy, and two families that are never unique	papers/XXIV-sieve/

Ch.	Paper	Title	Source directory
38	XXV	Sato–Tate families in localized Bost–Connes systems: open phase, no congruence obstruction, and the symmetric power expansion	<code>papers/XXV-sato-tate/</code>
39	XXXV	The free twin-prime Bost–Connes system is unconditionally supercritical: resolving the phase dichotomy via exact partial sums	<code>papers/XXXV-twin-regime/</code>

Dependencies

The chapters are ordered so that every internal citation points backwards, with one exception: Chapter 1 signposts Chapter 2. The table lists, for each chapter, the chapters it cites; a reader may take these as prerequisites.

Chapter	cites (chapters)
1	—
2	1
3	1, 4, 14
4	1, 2, 3
5	2, 4
6	1, 2, 3, 4
7	6
8	7
9	7, 8
10	3
11	3, 10
12	10, 11
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34	1, 3, 20, 22, 24, 25, 31, 32, 33
35	1, 2, 3, 4, 6, 15, 25, 28, 30
36	1, 3, 35
37	1
38	1, 15, 37
39	14, 37

Numbering and cross-references

Within each chapter, theorems, propositions, lemmas, corollaries, definitions, remarks and examples share one counter and are numbered *chapter.section.item*; equations likewise. These are the numbers of the original papers with the chapter number prefixed, so a citation of the form “[Ch. 4, Thm. 3.6]” refers to Theorem 4.3.6 of this book and to Theorem 3.6 of Paper III. Citations between papers have been converted mechanically to this form; references to the published literature are collected in the single bibliography at the end.

What has not been harmonised

- Prose references of the form “Paper N” have been converted mechanically to chapter references, and the standalone-paper boilerplate of each first section (the “series” paragraphs) has been removed at assembly; occasional phrasings that read paper-wise rather than book-wise may remain.
- Each chapter restates the notation it needs in its first section; the restatements are consistent (the same macros are used throughout) but repetitive, and could be replaced by references to the notation page.
- The “Assessment” section that closes each chapter records what was and was not established at the time of writing. Some of these are answered by later chapters (Chapter 32 withdraws a suggestion of Chapter 31; Chapter 34 answers Chapter 33); the answers are cross-referenced in the closing chapter on open problems but the assessments themselves have not been rewritten.
- The twelve working notes in **paper/** that preceded Chapters 2–14 are superseded and are not part of the book.

Standing notation

The following symbols are used throughout with the meaning fixed in Chapter 1; chapters that extend them (to function fields, to knots, to non-commutative semigroups) say so at the point of extension.

K, P_K, S	a number field, its set of finite primes, and an arbitrary subset $S \subseteq P_K$
$N\mathfrak{p}$	the absolute norm of a prime $\mathfrak{p}$; for $K = \mathbb{Q}$ rational primes are written ℓ
I_S, J_S	the group of fractional ideals, and the monoid of integral ideals, supported on S
$\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$	the S -local Bost–Connes system, with its dynamics σ_t
$\zeta_{K,S}(\beta) = \prod_{\mathfrak{p} \in S} (1 - N\mathfrak{p}^{-\beta})^{-1}$	the partition function
$\beta_c(S)$	the convergence exponent of $\sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta}$
$G_S = \mathbb{I}_S / \overline{K_{S,+}^\times}$	the S -local symmetry group; for $S = P_K$, $G_S \cong \text{Gal}(K^{ab}/K)$
$1 \rightarrow U_S / \mathcal{O}_{K,+}^\times \rightarrow G_S \rightarrow \text{Cl}_S^+ \rightarrow 1$	the exact sequence of G_S , with Cl_S^+ the subgroup of the narrow class group generated by S , and $h_S^+ = \text{Cl}_S^+ $
$\sigma : I_S \rightarrow G_S, \sigma_{\mathfrak{p}}, \text{Frob}_{\mathfrak{p}}$	the canonical homomorphism; $\chi(\sigma_{\mathfrak{p}}) = \overline{\chi(\text{Frob}_{\mathfrak{p}})}$ (Artin map normalised arithmetically)
$\Xi_\beta(K, S)$	the obstruction group $\{\chi \in \widehat{G_S} : \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} 1 - \chi(\sigma_{\mathfrak{p}}) ^2 < \infty\}$
$\text{KMS}_\beta(\mathcal{A}_{K,S}) \cong \text{Prob}(G_S / \Xi_\beta^\perp)$	the classification (Chapter 1, Theorem 3.6)
Ψ	the pointwise invariant separating extremal states in the density-versus-measure example
$M_{K,S,\beta}$	the von Neumann algebra generated by an extremal KMS_β state
Σ_β	the type invariant of $M_{K,S,\beta}$ (Chapter 2)
$\partial\mathcal{A}, \mathcal{A}^{\text{aff}}$	the boundary quotient and the affine (additive) control system (Chapter 3)

Conventions: the Artin map is normalised arithmetically, so that a uniformiser at $\mathfrak{p}$ maps to the arithmetic Frobenius; the *narrow* class group is not a convention but is forced by $\mathbb{A}_{K,f}^\times / \overline{K_+^\times} \cong \text{Gal}(K^{ab}/K)$; sets of primes are sets of finite primes; all S are countable, so the measure spaces are standard Borel.

Part 1

The classification

CHAPTER 1

KMS state uniqueness and phase transitions in localized Bost–Connes systems: from Chebotarev density to the Hardy–Littlewood conjecture

This chapter is the foundational paper of the series ([paper/](#)).

SUMMARY. For a number field K and an arbitrary set S of finite primes of K we study the S -local Bost–Connes system $\mathcal{A}_{K,S}$, whose partition function is the partial Dedekind product $\zeta_{K,S}(\beta) = \prod_{\mathfrak{p} \in S} (1 - \text{Np}^{-\beta})^{-1}$ and whose symmetry group is $G_S = \mathbb{I}_S / \overline{K_{S,+}^\times}$, an extension of the subgroup Cl_S^+ of the narrow class group generated by S by $U_S / \overline{\mathcal{O}_{K,+}^\times}$; for S the set of all primes, $G_S \cong \text{Gal}(K^{ab}/K)$. We classify the KMS_β states completely and unconditionally, for every K , every S and every $\beta > 0$: the simplex is affinely isomorphic to $\text{Prob}(G_S / \Xi_\beta^\perp)$, where

$$\Xi_\beta(K, S) = \left\{ \chi \in \widehat{G_S} : \sum_{\mathfrak{p} \in S} \text{Np}^{-\beta} (1 - \text{Re } \chi(\text{Frob}_{\mathfrak{p}})) < \infty \right\}.$$

Uniqueness therefore holds if and only if, for every nontrivial finite abelian extension L/K unramified outside $S \cup \infty$, the primes of S that do not split completely in L have divergent β -weighted sum; for S the set of all primes this is the Chebotarev density theorem, and at $\beta = 1$ it is the non-vanishing $L(1, \chi) \neq 0$. The criterion refutes the natural expectation that uniqueness should follow from the divergence $\zeta_{K,S}(\beta) = \infty$: we exhibit sets S with $\zeta_{K,S}(\beta) = \infty$ and with $\sigma(I_S)$ dense in G_S for which the KMS_β simplex is a segment at every $\beta > 0$, the two extreme points being separated by an explicit pointwise invariant whose almost-everywhere definedness is a Borel–Cantelli condition. Two mechanisms are isolated: over a field with $h_K^+ > 1$ the Hilbert class field supplies obstructing characters that are everywhere unramified, a phenomenon impossible over $\mathbb{Q}$; and over $\mathbb{Q}$ the separation between topological density and measure-theoretic ergodicity is governed by a Kakutani dichotomy. We show that $\beta \mapsto \Xi_\beta$ is non-decreasing and construct systems with any prescribed finite number of phase transitions. Finally we treat sparse sets: for Brun-summable S with critical exponent 1 the partition function is finite at the critical point, so the low-temperature phase is closed—the opposite of the classical Bost–Connes system—and for $S = \{2\} \cup \mathcal{SG}$, $\mathcal{SG}$ the Sophie Germain primes, the existence and location of the transition are equivalent to a Hardy–Littlewood type density statement.

1. Introduction

The Bost–Connes system [10] is a C^* -dynamical system $(\mathcal{A}_{BC}, \sigma)$ with partition function $\zeta(\beta)$ and symmetry group $\widehat{\mathbb{Z}}^\times \cong \text{Gal}(\mathbb{Q}^{ab}/\mathbb{Q})$, exhibiting a phase transition at $\beta = 1$: for $0 < \beta \leq 1$ the KMS_β state is unique and of type III_1 ; for $\beta > 1$ the extremal KMS_β states are of type I_∞ and are permuted freely and transitively by $\widehat{\mathbb{Z}}^\times$. Number field analogues were constructed by Ha–Paugam [50] and analysed by Laca–Larsen–Neshveyev [75], with ζ_K in place of ζ and $\text{Gal}(K^{ab}/K)$ in place of $\text{Gal}(\mathbb{Q}^{ab}/\mathbb{Q})$.

This paper studies the systems obtained by replacing the set of all primes by an arbitrary subset S . Two questions organize the analysis. First, when is the KMS_β state unique? Second, what does the answer depend on?

The received expectation is that uniqueness should be the shadow of the divergence $\zeta_{K,S}(\beta) = \infty$: when the partition function diverges there are no Gibbs states, every scaling-invariant measure gives the units measure zero, and one expects the resulting measure to be forced. We show that this expectation is false. The correct answer is Theorem 3.6: the KMS_β simplex is $\text{Prob}(G_S/\Xi_\beta^\perp)$ with Ξ_β as in (1.1) below, so that uniqueness is equivalent to the vanishing of an obstruction group defined by a convergence condition on a Dirichlet series indexed by S . Divergence of $\zeta_{K,S}(\beta)$ is not enough; what is needed is that, for each nontrivial character, the primes of S detecting it are themselves β -abundant. For S the set of all primes this is supplied by the Chebotarev density theorem, and the classical uniqueness statement is recovered; for sparse S it can fail, and we exhibit the failure explicitly.

Main results. Let K be a number field, S a set of finite primes, G_S the S -local symmetry group of Definition 2.1, and $\sigma : I_S \rightarrow G_S$ the canonical homomorphism from the group of fractional ideals supported on S (Lemma 2.10). Set

$$(1.1) \quad \Xi_\beta(K, S) = \left\{ \chi \in \widehat{G_S} : \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}})|^2 < \infty \right\}.$$

Since $|\chi(\sigma_{\mathfrak{p}})| = 1$, the summand may be written either way:

$$(1.2) \quad |1 - \chi(\sigma_{\mathfrak{p}})|^2 = 2(1 - \text{Re } \chi(\sigma_{\mathfrak{p}})) = 2(1 - \text{Re } \chi(\text{Frob}_{\mathfrak{p}})),$$

the last step because $\chi(\sigma_{\mathfrak{p}}) = \chi(\text{Frob}_{\mathfrak{p}})^{-1} = \overline{\chi(\text{Frob}_{\mathfrak{p}})}$ by Lemma 3.8. The factor 2 and the inversion do not affect convergence, so the form used in the abstract and (1.1) define the same subgroup.

- **Classification** (Theorem 3.6). For every K , S and $\beta > 0$, the KMS_β simplex of $\mathcal{A}_{K,S}$ is affinely isomorphic to $\text{Prob}(G_S/\Xi_\beta(K, S)^\perp)$. Uniqueness holds iff $\Xi_\beta = \{1\}$; if $\zeta_{K,S}(\beta) < \infty$ then $\Xi_\beta = \widehat{G_S}$ and one recovers the Gibbs states of type I_∞ permuted freely and transitively by G_S . The two classical regimes are the two ends of a single formula.
- **Chebotarev criterion** (Theorem 3.9, Corollary 3.10). Uniqueness at β is equivalent to (3.3): for every nontrivial finite abelian L/K unramified outside $S \cup \infty$, the primes of S not splitting completely in L have divergent β -weighted sum. For $S = P_K$ and $\beta \leq 1$ this follows from Chebotarev, and at $\beta = 1$ it is exactly $L(1, \chi) \neq 0$ (Remark 3.11).
- **Unramified class-group obstruction** (§4.1). If $h_K^+ > 1$, characters pulled back from the narrow class group are unramified everywhere and obstruct uniqueness whenever the primes of S concentrate β -negligibly in a proper subgroup. Over $\mathbb{Q}$ this cannot happen: every obstruction requires a modulus supported on S , hence ramification. An explicit example over $\mathbb{Q}(\sqrt{-5})$ is Example 4.2.
- **Topology versus measure** (§4.2). Density of $\sigma(I_S)$ in G_S is necessary but not sufficient for uniqueness. Over $\mathbb{Q}$ we construct $S = \{3\} \cup P_1 \cup R$ with $\sigma(I_S)$ dense, compute $\Xi_\beta = \{1, \chi\}$ exactly, and exhibit the pointwise invariant Ψ of (4.1); its almost-everywhere definedness is a Borel–Cantelli condition, while density needs only $R \neq \emptyset$. The underlying dichotomy is Kakutani’s (Remark 4.15). We also show $\beta \mapsto \Xi_\beta$ is non-decreasing and construct cascades of transitions (Proposition 4.16).
- **Sparse sets** (§5). The critical inverse temperature is the convergence exponent $\beta_c(S)$. If $\sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta_c} < \infty$ —which holds for Brun-summable S with $\beta_c = 1$ —then the type I_∞ phase is the *closed* half-line $[\beta_c, \infty)$ and the partition function does not diverge at the transition, in contrast with Bost–Connes, where $\beta_c = 1$ lies in the type III phase (Remark 5.3). For $S_\infty = \{2\} \cup \mathcal{SG}$ the Brun–Halberstam–Richert bound gives $\beta_c \leq 1$ unconditionally, a lower bound of Hardy–Littlewood strength gives $\beta_c = 1$, and the existence

of a transition is equivalent to $\mathcal{SG}$ having positive convergence exponent (Theorem 5.5). We stress in Remark 5.7 that this is an encoding, not a method.

Conventions. The Artin map is normalized *arithmetically*: for a finite abelian L/K and a prime $\mathfrak{p}$ unramified in L , a uniformizer at $\mathfrak{p}$ (as an idele trivial elsewhere) is sent to the arithmetic Frobenius $\text{Frob}_{\mathfrak{p}}$, the element of $\text{Gal}(L/K)$ inducing $x \mapsto x^{N_{\mathfrak{p}}}$ on the residue field. All Galois groups are of abelian extensions inside a fixed K^{ab} , so $\text{Frob}_{\mathfrak{p}}$ is well defined. Sets of primes are always sets of finite primes. N is the absolute norm. Characters of compact abelian groups are continuous. The group G_S is compact (Proposition 2.5) and totally disconnected— $\mathbb{I}_S$ is a totally disconnected locally compact group, so by van Dantzig’s theorem it has a basis of compact open subgroups at the identity, whose images form such a basis in the Hausdorff quotient G_S —hence profinite; so every character of G_S has finite order (Lemma 3.8).

Notation for primes is fixed as follows and used without further comment. In the general number-field setting, primes of K are written $\mathfrak{p}, \mathfrak{q} \in S \subseteq P_K$ with absolute norm $N_{\mathfrak{p}}$. When a statement is specialized to $K = \mathbb{Q}$ —that is, throughout §4.2 and §5—rational primes are written ℓ and $N\ell = \ell$, except that a Sophie Germain prime is written p , so that the associated safe prime is $2p + 1$. A fraktur $\mathfrak{p}$ appearing inside a specialized section is always a back-reference to the general statement being invoked.

Throughout, S is countable, so all the measure spaces below are standard Borel and disintegrations exist.

2. The S -local system over a number field

2.1. The symmetry group. Let K be a number field of degree d over $\mathbb{Q}$, $\mathcal{O}_K$ its ring of integers, P_K the set of nonzero prime ideals. Fix $\emptyset \neq S \subseteq P_K$ and write:

- $\widehat{\mathcal{O}}_S = \prod_{\mathfrak{p} \in S} \mathcal{O}_{\mathfrak{p}}$, $U_S = \prod_{\mathfrak{p} \in S} \mathcal{O}_{\mathfrak{p}}^{\times}$ (compact);
- I_S = the group of fractional ideals supported on S , free abelian on S ; $J_S \subset I_S$ the monoid of integral ones;
- $\mathbb{I}_S = \prod'_{\mathfrak{p} \in S} K_{\mathfrak{p}}^{\times}$ (restricted product with respect to the $\mathcal{O}_{\mathfrak{p}}^{\times}$), with $\text{div} : \mathbb{I}_S \twoheadrightarrow I_S$ and $\ker = U_S$;
- $K_S^{\times} = \{\alpha \in K^{\times} : v_{\mathfrak{p}}(\alpha) = 0 \ \forall \mathfrak{p} \notin S\}$, and $K_{S,+}^{\times}$ its subgroup of totally positive elements, embedded diagonally in $\mathbb{I}_S$.

DEFINITION 2.1. The S -local symmetry group is the compact abelian group

$$G_S := \mathbb{I}_S / \overline{K_{S,+}^{\times}},$$

with quotient map $r : \mathbb{I}_S \rightarrow G_S$.

REMARK 2.2. The closure in Definition 2.1 is not cosmetic. The diagonal image of $K_{S,+}^{\times}$ in $\mathbb{I}_S$ is in general not closed: its intersection with U_S has closure $\overline{\mathcal{O}_{K,+}^{\times}}$, the closure of the totally positive units, which by Proposition 4.17 can be a $\mathbb{Z}_p$ -module of positive rank. Quotienting by $K_{S,+}^{\times}$ itself would leave a non-Hausdorff quotient, and Proposition 2.5 would fail. Two cases where the distinction disappears: $K = \mathbb{Q}$, where $\mathcal{O}_{\mathbb{Q},+}^{\times} = \{1\}$ and $\mathbb{I}_S = \widehat{\mathbb{Z}}_S^{\times} \times \mathbb{Q}_{>0}^S$ splits, so that $K_{S,+}^{\times}$ is already closed; and K imaginary quadratic, where $\mathcal{O}_K^{\times}$ is finite. In both, G_S is computed without any limiting process, which is one reason the phenomena of §4.3 are invisible over $\mathbb{Q}$.

The choice of $K_{S,+}^{\times}$ rather than $K_S^{\times}$ is not a convention: it is forced by the requirement that for $S = P_K$ one recovers the symmetry group of [50, 75].

LEMMA 2.3. $\mathbb{A}_{K,f}^\times / \overline{K_+^\times} \cong \text{Gal}(K^{ab}/K)$, the isomorphism being induced by the Artin map. By contrast $\mathbb{A}_{K,f}^\times / \overline{K^\times}$ is the quotient of $\text{Gal}(K^{ab}/K)$ by the image of $\{\pm 1\}^{r_1}$, a proper quotient whenever K has a real place; for $K = \mathbb{Q}$ it equals $\widehat{\mathbb{Z}}^\times / \{\pm 1\}$.

PROOF. By class field theory $\text{Gal}(K^{ab}/K) = \mathbb{A}_K^\times / \overline{K^\times K_\infty^{\times, \circ}}$, where $K_\infty^\times = \prod_{v|\infty} K_v^\times$ and $K_\infty^{\times, \circ}$ is its identity component. Consider the composite $j : \mathbb{A}_{K,f}^\times \hookrightarrow \mathbb{A}_K^\times \rightarrow \text{Gal}(K^{ab}/K)$.

j is surjective. Since $K_\infty^\times / K_\infty^{\times, \circ} \cong \{\pm 1\}^{r_1}$, every class is represented by some (a_f, ε) with $\varepsilon \in \{\pm 1\}^{r_1}$. By weak approximation choose $\alpha \in K^\times$ whose sign vector at the real places is ε . Then $(a_f, \varepsilon) \sim (\alpha^{-1}a_f, \alpha_\infty^{-1}\varepsilon)$ and $\alpha_\infty^{-1}\varepsilon$ is totally positive, hence lies in $K_\infty^{\times, \circ}$; so the class equals that of $(\alpha^{-1}a_f, 1)$, which is in the image of j .

$\ker j = \overline{K_+^\times}$. If $(a_f, 1) \in \overline{K^\times K_\infty^{\times, \circ}}$ then, comparing archimedean components, a_f lies in the closure of the set of $\alpha \in K^\times$ with $\alpha_\infty \in K_\infty^{\times, \circ}$, i.e. with α totally positive. The converse inclusion is clear.

For the second statement, $\mathbb{A}_{K,f}^\times / \overline{K^\times}$ is the further quotient of $\mathbb{A}_{K,f}^\times / \overline{K_+^\times}$ by the image of $K^\times$, which is generated by classes of elements of every sign vector; for $K = \mathbb{Q}$, $\mathbb{A}_f^\times = \widehat{\mathbb{Z}}^\times \times \mathbb{Q}_{>0}$ and $\mathbb{Q}^\times = \{\pm 1\} \times \mathbb{Q}_{>0}$, giving $\widehat{\mathbb{Z}}^\times / \{\pm 1\}$. $\square$

COROLLARY 2.4. For $S = P_K$ one has $\mathbb{I}_{P_K} = \mathbb{A}_{K,f}^\times$ and $G_{P_K} \cong \text{Gal}(K^{ab}/K)$. For $K = \mathbb{Q}$ and any S , $\mathbb{I}_S = \widehat{\mathbb{Z}}_S^\times \times \mathbb{Q}_{>0}^S$ uniquely, so $G_S = \widehat{\mathbb{Z}}_S^\times$.

PROPOSITION 2.5. Let $\text{Cl}_S^+ \leq \text{Cl}_K^+$ be the subgroup of the narrow class group generated by the classes of the primes of S , and $\mathcal{O}_{K,+}^\times$ the group of totally positive units. Then $K_{S,+}^\times \cap U_S = \mathcal{O}_{K,+}^\times$ and there is an exact sequence of compact abelian groups

$$(2.1) \quad 1 \longrightarrow U_S / \overline{\mathcal{O}_{K,+}^\times} \longrightarrow G_S \longrightarrow \text{Cl}_S^+ \longrightarrow 1.$$

PROOF. An element of $K_{S,+}^\times$ lying in U_S has trivial divisor, hence is a totally positive unit. Applying div gives the surjection of G_S onto $I_S / P_{S,+}$, where $P_{S,+}$ is the group of principal fractional ideals with a totally positive generator supported on S ; by definition $I_S / P_{S,+} = \text{Cl}_S^+$, a finite group. The kernel is $U_S / (U_S \cap \overline{K_{S,+}^\times})$, and since U_S is open in $\mathbb{I}_S$ one has $U_S \cap \overline{K_{S,+}^\times} = \overline{U_S \cap K_{S,+}^\times} = \overline{\mathcal{O}_{K,+}^\times}$. Compactness follows since U_S is compact and Cl_S^+ finite. $\square$

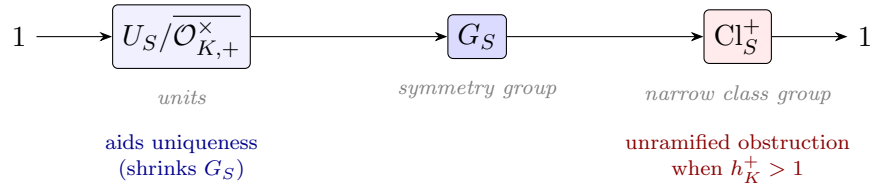


FIGURE 1. The two structures in (2.1) that are absent over $\mathbb{Q}$, where both ends are trivial and $G_S = \widehat{\mathbb{Z}}_S^\times$. The class group contributes obstructing characters (§4.1); the units act in the opposite direction (§4.3).

2.2. The algebra and its dynamics. Ideals of $\mathcal{O}_K$ need not be principal, and principal ideals have no canonical generator; the action of J_S is therefore defined through ideles, with the ambiguity absorbed by a balanced product.

DEFINITION 2.6. Let $Y_S = \widehat{\mathcal{O}}_S \times_{U_S} G_S$ be the quotient of $\widehat{\mathcal{O}}_S \times G_S$ by $(x, g) \sim (wx, r(w)^{-1}g)$ for $w \in U_S$; it is a compact metrizable space. For $\mathfrak{a} \in J_S$ choose $s \in \mathbb{I}_S$ with $\text{div}(s) = \mathfrak{a}$ and $s_{\mathfrak{p}} \in \mathcal{O}_{\mathfrak{p}}$ for all $\mathfrak{p}$, and set

$$\mathfrak{a} \cdot [x, g] = [sx, r(s)^{-1}g],$$

independent of the choice of s because two choices differ by an element of U_S . This is a right action of the abelian monoid J_S by injective continuous maps with clopen images. The S -local Bost–Connes system is the semigroup crossed product

$$\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S = \overline{\text{span}}\{\mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^* : \mathfrak{a}, \mathfrak{b} \in J_S, f \in C(Y_S)\},$$

generated by $C(Y_S)$ and isometries $\mu_{\mathfrak{a}}$ with $\mu_{\mathfrak{a}} \mu_{\mathfrak{b}} = \mu_{\mathfrak{a}\mathfrak{b}}$, $\mu_{\mathfrak{a}} f \mu_{\mathfrak{a}}^* = f \circ (\mathfrak{a} \cdot)^{-1}$ extended by zero, $\mu_{\mathfrak{a}} \mu_{\mathfrak{a}}^* = \mathbf{1}_{\mathfrak{a}Y_S}$ and $\mu_{\mathfrak{a}} \mu_{\mathfrak{b}}^* = \mu_{\mathfrak{b}}^* \mu_{\mathfrak{a}}$ for $\mathfrak{a}, \mathfrak{b}$ coprime. The dynamics is

$$\sigma_t(f) = f, \quad \sigma_t(\mu_{\mathfrak{a}}) = N\mathfrak{a}^{it} \mu_{\mathfrak{a}}.$$

The group G_S acts by $\theta_h[x, g] = [x, gh^{-1}]$, commuting with σ .

For $K = \mathbb{Q}$ this is the system of [76] restricted to S ; for $S = P_K$ it is the system of [50, 75]. We record the inductive limit structure, which is what allows sparse sets to be approached by finite ones.

PROPOSITION 2.7. *If $S = \bigcup_k S_k$ with $S_1 \subseteq S_2 \subseteq \dots$, then the maps $f \mapsto f \circ \text{pr}$, $\mu_{\mathfrak{a}} \mapsto \mu_{\mathfrak{a}}$ define unital σ -equivariant embeddings $\mathcal{A}_{K,S_k} \hookrightarrow \mathcal{A}_{K,S_{k+1}}$ and $\varinjlim_k \mathcal{A}_{K,S_k} \cong \mathcal{A}_{K,S}$ equivariantly for $G_S = \varinjlim_k G_{S_k}$. In particular $\mathcal{A}_{K,S} \subseteq \mathcal{A}_{K,P_K}$ for every S .*

PROOF. The relations are preserved because $\mathbf{1}_{\mathfrak{a}Y_{S'}}$ pulls back to $\mathbf{1}_{\mathfrak{a}Y_S}$ for $\mathfrak{a} \in J_S$; injectivity follows from the gauge-invariant uniqueness theorem, and surjectivity onto a dense subalgebra from $\widehat{\mathcal{O}}_S = \varinjlim_k \widehat{\mathcal{O}}_{S_k}$ together with $J_S = \bigcup_k J_{S_k}$ and Stone–Weierstrass. Strong continuity of the limit dynamics holds on the dense union and hence everywhere. $\square$

2.3. KMS states are measures. Let φ be a (σ, β) -KMS state. Over $\mathbb{Q}$ one shows $\varphi(\mu_m f \mu_n^*) = 0$ for $m \neq n$ by comparing m^β with n^β ; over a number field this argument is unavailable, because distinct ideals frequently have equal norms. We use instead the groupoid description.

PROPOSITION 2.8. *Let $\beta > 0$ and set*

$$M_\beta(K, S) = \{\nu \in \text{Prob}(Y_S) : \nu(\mathfrak{a} \cdot E) = N\mathfrak{a}^{-\beta} \nu(E) \quad \forall \mathfrak{a} \in J_S, E \text{ Borel}\}.$$

Then $\varphi \mapsto \varphi|_{C(Y_S)}$ is an affine bijection from the KMS_β simplex of $\mathcal{A}_{K,S}$ onto $M_\beta(K, S)$. There are no KMS_β states for $\beta < 0$, and for $\beta = 0$ the tracial states correspond to J_S -invariant measures.

PROOF. J_S is abelian and cancellative, hence an Ore semigroup with enveloping group I_S . By Laca’s dilation theorem [77] the semigroup crossed product $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$ is a full corner in the group crossed product $C_0(\widetilde{Y}_S) \rtimes I_S$, where $\widetilde{Y}_S = \varinjlim (Y_S, \mathfrak{a} \cdot)$ is the direct limit along the directed system J_S and $C(Y_S) = \mathbf{1}_{Y_S} C_0(\widetilde{Y}_S) \mathbf{1}_{Y_S}$; the dilation is equivariant for the dynamics, which on the dilation is implemented by the $\mathbb{R}$ -valued cocycle $c(\mathfrak{a}\mathfrak{b}^{-1}) = \log N\mathfrak{a} - \log N\mathfrak{b}$ on I_S . Corner inclusions preserve KMS simplices [77, §3], so it suffices to treat $C_0(\widetilde{Y}_S) \rtimes I_S = C^*(\mathcal{G}_S)$, the C^* -algebra of the transformation groupoid $\mathcal{G}_S = I_S \ltimes \widetilde{Y}_S$, which is étale, second countable and locally compact.

By Neshveyev’s theorem [96, Thm. 1.3], KMS_β states of $C^*(\mathcal{G}_S)$ for the dynamics induced by c are in bijection with pairs consisting of a probability measure ν on $\widetilde{Y}_S^{(0)}$ that is quasi-invariant with Radon–Nikodym cocycle $e^{-\beta c}$, together with a ν -measurable field of states on the C^* -algebras of the isotropy groups. Quasi-invariance with that cocycle is exactly $\nu(\mathfrak{a}E) = N\mathfrak{a}^{-\beta} \nu(E)$ for $\mathfrak{a} \in J_S$.

It remains to see that the isotropy contributes nothing. Taking $E = \mathbf{a} \cdot Y_S$ gives $\nu(\mathbf{a}Y_S) = N\mathbf{a}^{-\beta}$; applying this to $\mathbf{a} = \mathbf{p}^k$ and letting $k \rightarrow \infty$ gives $\nu(\{x_{\mathbf{p}} = 0\}) = 0$ for each $\mathbf{p} \in S$, hence, S being countable, ν is concentrated on the set Y_S° of classes $[x, g]$ with all $x_{\mathbf{p}} \neq 0$. On Y_S° the action of I_S is free: by Lemma 2.10(2) below, in the coordinates $(v, g) \in \mathbb{Z}^S \times G_S$ an element $\mathbf{a}\mathbf{b}^{-1}$ acts by $v \mapsto v + a - b$, and $a = b$ forces $\mathbf{a} = \mathbf{b}$. So $\mathcal{G}_S$ is principal on a conull set, all isotropy groups are trivial there, and the field of states is empty data. The correspondence is affine by construction, and the state is recovered from ν by $\varphi = \nu \circ E_S$ with E_S the canonical conditional expectation onto $C(Y_S)$ given by averaging over the gauge action of $\hat{I}_S$.

For $\beta < 0$: taking $\mathbf{a} \neq (1)$, $\nu(\mathbf{a}Y_S) = N\mathbf{a}^{-\beta} > 1$, impossible. $\square$

REMARK 2.9. Concretely, $\varphi = \nu \circ E_S$ means $\varphi(\mu_{\mathbf{a}} f \mu_{\mathbf{b}}^*) = 0$ for $\mathbf{a} \neq \mathbf{b}$ and $\varphi(\mu_{\mathbf{a}} f \mu_{\mathbf{a}}^*) = \nu(f \circ (\mathbf{a} \cdot)^{-1})$. The vanishing for $\mathbf{a} \neq \mathbf{b}$ with $N\mathbf{a} = N\mathbf{b}$, which is invisible to the modular argument, is here a consequence of freeness of the I_S -action, i.e. of the fact that ideals, not their norms, index the translations.

2.4. Coordinates. Choose a uniformizer $\pi_{\mathbf{p}}$ at each $\mathbf{p} \in S$. Every $x \in \hat{\mathcal{O}}_S$ with all coordinates nonzero factors uniquely as $x_{\mathbf{p}} = \pi_{\mathbf{p}}^{v_{\mathbf{p}}} u_{\mathbf{p}}$ with $v_{\mathbf{p}} \in \mathbb{Z}_{\geq 0}$, $u_{\mathbf{p}} \in \mathcal{O}_{\mathbf{p}}^\times$. Put $V = \mathbb{Z}_{\geq 0}^S$. For $\mathbf{a} = \prod \mathbf{p}^{a_{\mathbf{p}}} \in J_S$ let $s_{\mathbf{a}} = (\pi_{\mathbf{p}}^{a_{\mathbf{p}}})_{\mathbf{p}} \in \mathbb{I}_S$ and

$$(2.2) \quad \sigma_{\mathbf{a}} := r(s_{\mathbf{a}})^{-1} \in G_S.$$

- LEMMA 2.10. (1) $\mathbf{a} \mapsto \sigma_{\mathbf{a}}$ is a monoid homomorphism $J_S \rightarrow G_S$ extending to $\sigma : I_S \rightarrow G_S$; its composition with $G_S \rightarrow \text{Cl}_S^+$ is the inverse of the natural surjection $I_S \rightarrow \text{Cl}_S^+$, i.e. $\sigma_{\mathbf{p}} \mapsto [\mathbf{p}]^{-1}$, the inversion being that of (2.2). If $\mathbf{a} = (\alpha)$ with $\alpha \in K_{S,+}^\times$ then $\sigma_{\mathbf{a}} = r(\theta_{\alpha})^{-1}$ with $\theta_{\alpha} := \alpha^{-1} s_{\mathbf{a}} \in U_S$; for $K = \mathbb{Q}$, $\sigma_{\ell} = \theta_{\ell}$, where $(\theta_{\ell})_{\ell'} = \ell$ for $\ell' \neq \ell$ and 1 for $\ell' = \ell$.
- (2) $[x, g] \mapsto (v(x), r(u(x))g)$ is a Borel isomorphism from Y_S° onto $V \times G_S$, and in these coordinates $\mathbf{a} \cdot (v, g) = (v + a, \sigma_{\mathbf{a}} g)$.
- (3) For every $\nu \in M_{\beta}(K, S)$ the law π_{β} of v is the same: the $v_{\mathbf{p}}$ are independent with $\pi_{\beta}(v_{\mathbf{p}} \geq k) = N\mathbf{p}^{-k\beta}$, so $\pi_{\beta}(v_{\mathbf{p}} = k) = (1 - N\mathbf{p}^{-\beta})N\mathbf{p}^{-k\beta}$.
- (4) If $(\rho_v)_{v \in V}$ is a disintegration of ν over π_{β} , then $\nu \in M_{\beta}(K, S)$ if and only if for every $\mathbf{a} \in J_S$

$$(2.3) \quad \rho_{v+a} = (\sigma_{\mathbf{a}})_* \rho_v \quad \text{for } \pi_{\beta}\text{-a.e. } v.$$

PROOF. (1) $s_{\mathbf{a}\mathbf{b}} = s_{\mathbf{a}} s_{\mathbf{b}}$, and r is a homomorphism. Composing with div modulo $P_{S,+}$ sends $s_{\mathbf{a}}$ to $[\mathbf{a}]$, hence $\sigma_{\mathbf{a}}$ to $[\mathbf{a}]^{-1}$. If $\mathbf{a} = (\alpha)$ with α totally positive then $\alpha^{-1} s_{\mathbf{a}}$ has trivial divisor, so lies in U_S , and $r(\alpha) = 1$ gives $r(s_{\mathbf{a}}) = r(\alpha^{-1} s_{\mathbf{a}})$. For $K = \mathbb{Q}$ take $\pi_{\ell} = \ell$ and $\alpha = \ell$: then θ_{α} has component ℓ^{-1} at every $\ell' \neq \ell$ and 1 at ℓ , so $\sigma_{\ell} = r(\theta_{\alpha})^{-1} = \theta_{\ell}$.

(2) Well-definedness: replacing (x, g) by $(wx, r(w)^{-1}g)$ leaves v unchanged and replaces u by wu , so $r(u)g$ is unchanged, G_S being abelian. The inverse map is $(v, g) \mapsto [(\pi_{\mathbf{p}}^{v_{\mathbf{p}}})_{\mathbf{p}}, g]$. With $s = s_{\mathbf{a}}$ one has $v \mapsto v + a$, $u \mapsto u$ and $g \mapsto r(s_{\mathbf{a}})^{-1}g = \sigma_{\mathbf{a}} g$.

(3) For finite $F \subset S$ and $k_{\mathbf{p}} \geq 0$, with $\mathbf{a} = \prod_{\mathbf{p} \in F} \mathbf{p}^{k_{\mathbf{p}}}$, one has $\{v_{\mathbf{p}} \geq k_{\mathbf{p}} \forall \mathbf{p} \in F\} = \mathbf{a} \cdot Y_S$ modulo the null set $Y_S \setminus Y_S^\circ$, and $\nu(\mathbf{a}Y_S) = N\mathbf{a}^{-\beta} = \prod_{\mathbf{p} \in F} N\mathbf{p}^{-k_{\mathbf{p}}\beta}$. This gives independence and the marginals, hence π_{β} .

(4) Translation by a multiplies π_{β} by the constant $N\mathbf{a}^{-\beta}$ by (3). For product sets $A \times B$,

$$\nu(\mathbf{a} \cdot (A \times B)) = \int_{A+a} \rho_v(\sigma_{\mathbf{a}} B) d\pi_{\beta}(v), \quad N\mathbf{a}^{-\beta} \nu(A \times B) = \int_{A+a} \rho_{v-a}(B) d\pi_{\beta}(v),$$

the second by the change of variables $v \mapsto v - a$. As A, B are arbitrary, equality for all of them is (2.3); the converse is the same computation with Fubini. $\square$

3. The classification and the Chebotarev criterion

Throughout this section K , S and $\beta > 0$ are fixed, and we work in the coordinates of Lemma 2.10. Write $N_{\mathbf{p}} := N\mathbf{p}$, $t_{\mathbf{p}} := N\mathbf{p}^{-\beta} \in (0, 1)$, and for $\chi \in \widehat{G_S}$

$$f_{\chi}(v) := \widehat{\rho}_v(\chi) = \int_{G_S} \overline{\chi(g)} d\rho_v(g), \quad |f_{\chi}| \leq 1.$$

By (2.3), for every $\mathbf{a} \in J_S$ with valuation vector a ,

$$(3.1) \quad f_{\chi}(v + a) = \overline{\chi(\sigma_{\mathbf{a}})} f_{\chi}(v) \quad \pi_{\beta}\text{-a.e.}$$

Set $c_{\mathbf{p}} := \overline{\chi(\sigma_{\mathbf{p}})} \in \mathbb{T}$, so that $c^a = \prod_{\mathbf{p}} c_{\mathbf{p}}^{a_{\mathbf{p}}} = \overline{\chi(\sigma_{\mathbf{a}})}$.

LEMMA 3.1 (Zero-one law). $|f_{\chi}|$ is π_{β} -a.e. equal to a constant $c_{\chi} \in [0, 1]$.

PROOF. Taking $\mathbf{a} = \mathbf{p}$ in (3.1) gives $|f_{\chi}|(v + e_{\mathbf{p}}) = |f_{\chi}|(v)$ a.e. The orbits of $v \mapsto v + e_{\mathbf{p}}$ are the fibres of the projection $V \rightarrow \mathbb{Z}_{\geq 0}^{S \setminus \{\mathbf{p}\}}$, so $|f_{\chi}|$ agrees a.e. with a function not depending on $v_{\mathbf{p}}$; that is, $|f_{\chi}|$ is measurable with respect to $\sigma(v_{\mathbf{p}'} : \mathbf{p}' \neq \mathbf{p})$ modulo null sets. Applying this coordinate by coordinate over a finite set F , $|f_{\chi}|$ is measurable with respect to $\sigma(v_{\mathbf{p}'} : \mathbf{p}' \notin F)$ for every finite F , hence with respect to the tail σ -algebra of the independent family $(v_{\mathbf{p}})_{\mathbf{p} \in S}$. By Kolmogorov's zero-one law that σ -algebra is π_{β} -trivial. $\square$

Suppose $\chi \neq 1$ and $c_{\chi} > 0$. Then $g := f_{\chi}/|f_{\chi}|$ is a measurable map $V \rightarrow \mathbb{T}$, defined a.e., with

$$(3.2) \quad g(v + e_{\mathbf{p}}) = c_{\mathbf{p}} g(v) \quad \pi_{\beta}\text{-a.e., for every } \mathbf{p} \in S.$$

Put

$$m_{\mathbf{p}} := \mathbb{E}_{\pi_{\beta}}[c_{\mathbf{p}}^{v_{\mathbf{p}}}] = \sum_{k \geq 0} (1 - t_{\mathbf{p}}) t_{\mathbf{p}}^k c_{\mathbf{p}}^k = \frac{1 - t_{\mathbf{p}}}{1 - c_{\mathbf{p}} t_{\mathbf{p}}} \in \mathbb{C}.$$

LEMMA 3.2. $1 - |m_{\mathbf{p}}|^2 = \frac{2t_{\mathbf{p}}(1 - \operatorname{Re} c_{\mathbf{p}})}{|1 - c_{\mathbf{p}} t_{\mathbf{p}}|^2}$, and therefore

$$\sum_{\mathbf{p} \in S} (1 - |m_{\mathbf{p}}|) < \infty \iff \sum_{\mathbf{p} \in S} N\mathbf{p}^{-\beta} |1 - \chi(\sigma_{\mathbf{p}})|^2 < \infty.$$

PROOF. $|m_{\mathbf{p}}|^2 = (1 - t_{\mathbf{p}})^2 / |1 - c_{\mathbf{p}} t_{\mathbf{p}}|^2$ and $|1 - c_{\mathbf{p}} t_{\mathbf{p}}|^2 = 1 - 2t_{\mathbf{p}} \operatorname{Re} c_{\mathbf{p}} + t_{\mathbf{p}}^2$ give the identity. Since $|1 - c_{\mathbf{p}}|^2 = 2(1 - \operatorname{Re} c_{\mathbf{p}})$ and, for all but at most one $\mathbf{p}$ (the one of least norm), $t_{\mathbf{p}}$ is bounded away from 1 by a constant depending only on β and S , we get $1 - |m_{\mathbf{p}}|^2 \asymp t_{\mathbf{p}} |1 - c_{\mathbf{p}}|^2$ up to finitely many terms; and $1 - |m| \leq 1 - |m|^2 \leq 2(1 - |m|)$. $\square$

PROPOSITION 3.3 (Necessity). If $c_{\chi} > 0$ then $\sum_{\mathbf{p} \in S} (1 - |m_{\mathbf{p}}|) < \infty$, i.e. $\chi \in \Xi_{\beta}(K, S)$.

PROOF. Let g be as in (3.2), $|g| = 1$ a.e. Fix an increasing exhaustion $F_1 \subset F_2 \subset \dots$ of S by finite sets; write $\mathcal{F}_F = \sigma(v_{\mathbf{p}} : \mathbf{p} \in F)$ and $P_F = \prod_{\mathbf{p} \in F} c_{\mathbf{p}}^{v_{\mathbf{p}}}$.

Iterating (3.2) in the coordinate $\mathbf{p}$ shows that $g c_{\mathbf{p}}^{-v_{\mathbf{p}}}$ agrees a.e. with a function not depending on $v_{\mathbf{p}}$; doing this for all $\mathbf{p} \in F$ shows that $G_F := g P_F^{-1}$ is measurable with respect to $\sigma(v_{\mathbf{p}} : \mathbf{p} \notin F)$, hence independent of $\mathcal{F}_F$. Therefore

$$\mathbb{E}[g \mid \mathcal{F}_F] = P_F \mathbb{E}[G_F], \quad \text{so} \quad |\mathbb{E}[g \mid \mathcal{F}_F]| = |\mathbb{E}[G_F]|.$$

Since $\mathcal{F}_{F_k} \uparrow \sigma(v)$, martingale convergence gives $\mathbb{E}[g \mid \mathcal{F}_{F_k}] \rightarrow g$ in $L^2(\pi_{\beta})$, so $|\mathbb{E}[G_{F_k}]| \rightarrow |g| = 1$. For large k put $\gamma_k = \mathbb{E}[G_{F_k}] / |\mathbb{E}[G_{F_k}]|$. Then, as $|P_{F_k}| = 1$,

$$\|g - \gamma_k P_{F_k}\|_2^2 = \|G_{F_k} - \gamma_k\|_2^2 = 2 - 2|\mathbb{E}[G_{F_k}]| \longrightarrow 0.$$

Hence $(\gamma_k P_{F_k})_k$ is Cauchy in L^2 . For $j < k$, dividing by the unimodular P_{F_j} and using independence,

$$\|\gamma_k P_{F_k} - \gamma_j P_{F_j}\|_2 = \left\| \gamma_k \prod_{\mathfrak{p} \in F_k \setminus F_j} c_{\mathfrak{p}}^{v_{\mathfrak{p}}} - \gamma_j \right\|_2 \geq 1 - \left| \mathbb{E} \prod_{\mathfrak{p} \in F_k \setminus F_j} c_{\mathfrak{p}}^{v_{\mathfrak{p}}} \right| = 1 - \prod_{\mathfrak{p} \in F_k \setminus F_j} |m_{\mathfrak{p}}|,$$

where the inequality is $\|X - \gamma\|_2 \geq |\mathbb{E}X - \gamma| \geq 1 - |\mathbb{E}X|$ for unimodular constants γ . Thus $\prod_{\mathfrak{p} \in F_k \setminus F_j} |m_{\mathfrak{p}}| \rightarrow 1$ as $j, k \rightarrow \infty$, the Cauchy criterion for convergence of $\prod_{\mathfrak{p} \in S} |m_{\mathfrak{p}}|$ to a nonzero limit, i.e. $\sum_{\mathfrak{p}} (1 - |m_{\mathfrak{p}}|) < \infty$. Lemma 3.2 concludes. $\square$

PROPOSITION 3.4 (Sufficiency). *If $\chi \in \Xi_{\beta}(K, S)$ then there is a measurable $g_{\chi} : V \rightarrow \mathbb{T}$ with $|g_{\chi}| = 1$ satisfying (3.2), unique up to a multiplicative constant. If moreover $\chi \neq 1$, there is $\nu \in M_{\beta}(K, S)$ that is not the symmetric measure; in particular the KMS_{β} state is not unique.*

PROOF. Set $\tilde{X}_{\mathfrak{p}} = \overline{(m_{\mathfrak{p}}/|m_{\mathfrak{p}}|)} c_{\mathfrak{p}}^{v_{\mathfrak{p}}}$, well defined because $m_{\mathfrak{p}} \neq 0$; these are independent, unimodular, with $\mathbb{E}\tilde{X}_{\mathfrak{p}} = |m_{\mathfrak{p}}| > 0$. Put $\tilde{P}_k = \prod_{\mathfrak{p} \in F_k} \tilde{X}_{\mathfrak{p}}$ and $L_k = \prod_{\mathfrak{p} \in F_k} |m_{\mathfrak{p}}|$; by hypothesis and Lemma 3.2, $L_k \downarrow L > 0$. Then $N_k := \tilde{P}_k/L_k$ is an $(\mathcal{F}_{F_k})$ -martingale with $\|N_k\|_2 = 1/L_k \leq 1/L$, hence converges a.s. and in L^2 ; so $\tilde{P}_k \rightarrow g_{\chi}$ a.s. with $|g_{\chi}| = 1$, and passing to the limit in $\tilde{P}_k(v + e_{\mathfrak{p}}) = c_{\mathfrak{p}} \tilde{P}_k(v)$ (valid once $\mathfrak{p} \in F_k$) gives (3.2). If g, g' both satisfy (3.2) then g/g' is invariant under all $v \mapsto v + e_{\mathfrak{p}}$, hence constant by Lemma 3.1.

Now let $\chi \neq 1$ and define ρ_v to be the measure on G_S with density $1 + \text{Re}(g_{\chi}(v)\chi(g)) \geq 0$ with respect to normalized Haar measure. It is a probability measure whose Fourier coefficients $\widehat{\rho}_v(\chi') = \int \chi' d\rho_v$ are $\widehat{\rho}_v(1) = 1$, $\widehat{\rho}_v(\chi) = \frac{1}{2}g_{\chi}(v)$, $\widehat{\rho}_v(\bar{\chi}) = \frac{1}{2}\overline{g_{\chi}(v)}$ and 0 elsewhere when $\chi^2 \neq 1$; when $\chi^2 = 1$ the two coefficients merge into $\widehat{\rho}_v(\chi) = \text{Re } g_{\chi}(v) = g_{\chi}(v)$, which is real because then all $c_{\mathfrak{p}} = \pm 1$ and all $m_{\mathfrak{p}} > 0$, so that $g_{\chi} = \lim_k \prod_{\mathfrak{p} \in F_k} c_{\mathfrak{p}}^{v_{\mathfrak{p}}} \in \{\pm 1\}$. Since $(\sigma_a)_* \widehat{\rho}_v(\chi') = \widehat{\chi'(\sigma_a)} \widehat{\rho}_v(\chi')$, comparison with $\widehat{\rho_{v+a}}(\chi')$ coefficient by coefficient reduces to (3.2) for $\chi' = \chi$, to its conjugate for $\chi' = \bar{\chi}$, and to $1 = 1$ or $0 = 0$ otherwise. So (2.3) holds and Lemma 2.10(4) yields $\nu \in M_{\beta}(K, S)$, with $\widehat{\rho}_v(\chi) \neq 0$, whereas the symmetric measure has ρ_v Haar and all nontrivial coefficients zero. $\square$

LEMMA 3.5. *$\Xi_{\beta}(K, S)$ is a subgroup of $\widehat{G_S}$; it equals $\widehat{G_S}$ whenever $\zeta_{K,S}(\beta) < \infty$, and $\beta \mapsto \Xi_{\beta}$ is non-decreasing.*

PROOF. $|1 - \chi_1 \chi_2(\sigma)| \leq |1 - \chi_1(\sigma)| + |1 - \chi_2(\sigma)|$ and $(a + b)^2 \leq 2a^2 + 2b^2$ give closure under products; $|1 - \bar{\chi}(\sigma)| = |1 - \chi(\sigma)|$ under inverses. If $\sum_{\mathfrak{p}} N_{\mathfrak{p}}^{-\beta} < \infty$ every χ qualifies since $|1 - \chi(\sigma_{\mathfrak{p}})| \leq 2$. Monotonicity holds because $N_{\mathfrak{p}}^{-\beta'} \leq N_{\mathfrak{p}}^{-\beta}$ for $\beta \leq \beta'$. $\square$

THEOREM 3.6 (Classification). *Let K be a number field, S a nonempty set of finite primes and $\beta > 0$. Then the KMS_{β} simplex of $\mathcal{A}_{K,S}$ is affinely isomorphic to*

$$\text{Prob}(G_S/\Xi_{\beta}(K, S)^{\perp}), \quad \Xi_{\beta}^{\perp} = \{g \in G_S : \chi(g) = 1 \ \forall \chi \in \Xi_{\beta}\}.$$

In particular:

- (1) *the KMS_{β} state is unique if and only if $\Xi_{\beta}(K, S) = \{1\}$;*
- (2) *if $\zeta_{K,S}(\beta) < \infty$ then $\Xi_{\beta} = \widehat{G_S}$, $\Xi_{\beta}^{\perp} = \{1\}$, the simplex is $\text{Prob}(G_S)$, its extreme points are the Gibbs states of the Fock representations, of type I_{∞} , and G_S permutes them freely and transitively;*
- (3) *the symmetry G_S acts on the simplex through the quotient $G_S/\Xi_{\beta}^{\perp}$, so $\Xi_{\beta}^{\perp}$ is the residual symmetry group of the phase.*

PROOF. By Proposition 2.8 and Lemma 2.10, a KMS state is determined by the family (ρ_v) , hence by $(f_{\chi})_{\chi \in \widehat{G_S}}$. By Lemma 3.1 and Proposition 3.3, $f_{\chi} \equiv 0$ for $\chi \notin \Xi_{\beta}$. For $\chi \in \Xi_{\beta}$ let

g_χ be a solution of (3.2) with $|g_\chi| = 1$, unique up to a unimodular constant by Proposition 3.4. The constants can be chosen so that $\chi \mapsto g_\chi(v)$ is multiplicative. Indeed, for $\chi_1, \chi_2 \in \Xi_\beta$ the unimodular function $g_{\chi_1}g_{\chi_2}/g_{\chi_1\chi_2}$ satisfies (3.2) with every c_p replaced by 1, hence is invariant under all the shifts $v \mapsto v + e_p$ and is a.e. equal to a constant $\omega(\chi_1, \chi_2) \in \mathbb{T}$ by the argument of Lemma 3.1. The function ω is a symmetric 2-cocycle on the countable abelian group Ξ_β with values in the divisible group $\mathbb{T}$; symmetric cocycles classify abelian extensions, and $\text{Ext}(\Xi_\beta, \mathbb{T}) = 0$, so ω is a coboundary, $\omega(\chi_1, \chi_2) = \lambda(\chi_1)\lambda(\chi_2)/\lambda(\chi_1\chi_2)$ for some $\lambda : \Xi_\beta \rightarrow \mathbb{T}$. Replacing g_χ by $g_\chi/\lambda(\chi)$ gives $g_{\chi_1}g_{\chi_2} = g_{\chi_1\chi_2}$, whence also $g_1 = 1$ and $g_{\bar{\chi}} = \overline{g_\chi}$. (The explicit normalization $\lim_k \prod_{p \in F_k} (m_p/|m_p|)c_p^{v_p}$ of Proposition 3.4 is multiplicative only up to such a cocycle, the arguments of the m_p not being additive in χ .) Then $f_\chi = \kappa(\chi)g_\chi$ for a scalar $\kappa(\chi)$ with $\kappa(1) = 1$, and κ is positive definite on the discrete group Ξ_β : for $\chi_i \in \Xi_\beta$, $z_i \in \mathbb{C}$,

$$\sum_{i,j} z_i \bar{z}_j \kappa(\chi_i \bar{\chi}_j) = \mathbb{E}_{\pi_\beta} \left[\int_{G_S} \left| \sum_i z_i \overline{\chi_i(g)} g_{\chi_i}(v) \right|^2 d\rho_v(g) \right] \geq 0.$$

By Bochner's theorem κ is the Fourier transform of a unique probability measure on the compact dual $\widehat{\Xi_\beta} \cong G_S/\Xi_\beta^\perp$. Conversely every such measure arises: given κ positive definite with $\kappa(1) = 1$, define ρ_v by $\widehat{\rho}_v(\chi) = \kappa(\chi)g_\chi(v)$ for $\chi \in \Xi_\beta$ and 0 otherwise; positivity of ρ_v is Bochner again (the function $\chi \mapsto \kappa(\chi)g_\chi(v)$ is positive definite for each fixed v , being κ twisted by the character $\chi \mapsto g_\chi(v)$ of Ξ_β), and (2.3) holds by (3.2). The correspondence $\nu \leftrightarrow \kappa$ is affine and bijective.

(1) and (3) are immediate. For (2), $\Xi_\beta = \widehat{G_S}$ by Lemma 3.5, so the simplex is $\text{Prob}(G_S)$; the extreme points correspond to point masses, and unwinding the correspondence they are the states $\zeta_{K,S}(\beta)^{-1} \text{Tr}(\pi_g(\cdot)e^{-\beta H})$ in the Fock representations π_g on $\ell^2(J_S)$ determined by $g \in G_S$, which are irreducible; the action of G_S is translation on the parameter. $\square$

REMARK 3.7. Theorem 3.6 is a single statement covering both classical regimes. The high-temperature regime is $\Xi_\beta = \{1\}$ and the low-temperature regime $\Xi_\beta = \widehat{G_S}$; for the classical systems these are the only two values that occur, but §4.2 shows that intermediate values do occur, giving phases with partial symmetry breaking that have no analogue in [10, 75].

3.1. The Chebotarev criterion.

LEMMA 3.8. *Let $\chi \in \widehat{G_S}$ be nontrivial. Then χ has finite order and is trivial on an open subgroup, so there are a finite $F \subset S$ and exponents $k_p \geq 1$ with χ trivial on $\prod_{p \in F} (1 + \mathfrak{p}^{k_p} \mathcal{O}_p) \times \prod_{p \in S \setminus F} \mathcal{O}_p^\times$; put $\mathfrak{m} = \prod_{p \in F} \mathfrak{p}^{k_p}$. Let L/K be the finite cyclic extension attached by class field theory to an extension of χ to $\text{Gal}(K^{ab}/K)$ trivial on the inertia groups at all $p \notin S$; such extensions exist, and the statements below do not depend on the choice (see the proof). Then L lies in the ray class field of conductor dividing $\mathfrak{m}\infty$, hence is unramified outside $S \cup \infty$, and for every $p \in S$ with $p \nmid \mathfrak{m}$,*

$$\chi(\sigma_p) = \chi(\text{Frob}_p)^{-1}.$$

In particular $\chi(\sigma_p) = 1$ if and only if p splits completely in L .

PROOF. First, how G_S sits inside the Galois group. The inclusion $\mathbb{I}_S \subset \mathbb{A}_{K,f}^\times$ (ideles with component 1 outside S) composed with the Artin map of Lemma 2.3 gives a continuous homomorphism $G_S \rightarrow \text{Gal}(K^{ab}/K)$, and its composite with the projection onto $\text{Gal}(K^S/K)$, where K^S is the maximal abelian extension of K unramified outside $S \cup \infty$, is injective: if $\alpha_n w_n \rightarrow s \in \mathbb{I}_S$ with $\alpha_n \in K_+^\times$ and $w_n \in \prod_{p \notin S} \mathcal{O}_p^\times$, then, the topology of $\mathbb{A}_{K,f}^\times$ being that of the restricted product, α_n is eventually a unit at every $p \notin S$, so $\alpha_n \in K_{S,+}^\times$ and $s \in \overline{K_{S,+}^\times}$. Thus G_S is a closed subgroup of $\text{Gal}(K^S/K)$, and χ extends to a character of $\text{Gal}(K^{ab}/K)$ trivial on the inertia groups at all $p \notin S$. Different extensions define different fields L , but for $p \in S$ the element Frob_p is the image of $s_p \in \mathbb{I}_S$, so $\chi(\text{Frob}_p)$ does not depend on the extension.

$\widehat{G_S}$ is torsion because G_S is a quotient of the totally disconnected group $\mathbb{I}_S$ by a closed subgroup, hence profinite. With the arithmetic normalization of the Artin map, the idele $s_{\mathfrak{p}}$ (a uniformizer at $\mathfrak{p}$, 1 elsewhere) maps to $\text{Frob}_{\mathfrak{p}}$ for $\mathfrak{p}$ unramified in L ; by (2.2), $\sigma_{\mathfrak{p}} = r(s_{\mathfrak{p}})^{-1}$, whence the displayed formula. For L/K abelian, $\text{Frob}_{\mathfrak{p}} = 1$ if and only if $\mathfrak{p}$ splits completely.

Two remarks on the archimedean places. First, G_S is a quotient by the *totally positive* group $K_{S,+}^{\times}$, so every character of G_S is trivial on totally positive elements and the sign conditions at the real places are already absorbed: the conductor attached to χ is $\mathfrak{m}\infty$ with the full archimedean part, and L lies in the *narrow* ray class field of that conductor. This is the same choice that forces the narrow class group in (2.1), and it is dictated by Lemma 2.3. Second, no archimedean condition enters (3.3), which involves only finite primes; the archimedean part determines which fields L occur, not the series. $\square$

THEOREM 3.9 (Chebotarev criterion). *Let $\beta > 0$. The KMS_{β} state of $\mathcal{A}_{K,S}$ is unique if and only if*

$$(3.3) \quad \sum_{\substack{\mathfrak{p} \in S \\ \mathfrak{p} \text{ not split completely in } L}} \text{N}\mathfrak{p}^{-\beta} = \infty$$

for every finite abelian extension L/K inside K^{ab} , unramified outside $S \cup \infty$, whose associated character of G_S is nontrivial.

PROOF. By Theorem 3.6(1) uniqueness means $\Xi_{\beta} = \{1\}$. By Lemma 3.8, for nontrivial χ the quantity $|1 - \chi(\sigma_{\mathfrak{p}})|^2$ vanishes exactly on the primes of S splitting completely in L and is at least $\min_{\omega \neq 1} |1 - \omega|^2 > 0$ elsewhere, ω ranging over the finitely many values of χ ; the finitely many $\mathfrak{p} \mid \mathfrak{m}$ do not affect convergence. Hence $\chi \notin \Xi_{\beta}$ is exactly (3.3). $\square$

COROLLARY 3.10 (The classical systems). *Let $S = P_K$ and $0 < \beta \leq 1$. Then $\Xi_{\beta}(K, P_K) = \{1\}$, so the KMS_{β} state of the Bost–Connes system of K is unique. For $\beta > 1$, $\zeta_K(\beta) < \infty$ and Theorem 3.6(2) gives the extremal Gibbs states permuted freely and transitively by $\text{Gal}(K^{ab}/K)$.*

PROOF. Let L/K be nontrivial abelian of degree $m \geq 2$. By the Chebotarev density theorem the primes of K not splitting completely in L have natural density $1 - 1/m \geq \frac{1}{2}$, so $\sum \text{N}\mathfrak{p}^{-1} = \infty$ over that set, hence $\sum \text{N}\mathfrak{p}^{-\beta} = \infty$ for $\beta \leq 1$. Apply Theorem 3.9. The second statement uses $\zeta_K(\beta) < \infty$ for $\beta > 1$ and Corollary 2.4. $\square$

REMARK 3.11 (L -functions). For a nontrivial ray class character $\chi \bmod \mathfrak{m}$ and $\beta > 1$,

$$\log \zeta_{K,S}(\beta) - \text{Re} \log L_S(\beta, \chi) = \sum_{\mathfrak{p} \in S} \text{N}\mathfrak{p}^{-\beta} (1 - \text{Re} \chi(\mathfrak{p})) + O(1),$$

the error absorbing higher prime powers. So $\chi \in \Xi_{\beta}$ says exactly that $\log |L_S(\beta, \chi)|$ tracks $\log \zeta_{K,S}(\beta)$ to within a bounded error. For $S = P_K$ and $\beta \downarrow 1$ the left side tends to $+\infty$, because ζ_K has a simple pole at $\beta = 1$ while $L(\beta, \chi) \rightarrow L(1, \chi) \neq 0$. Thus the uniqueness half of the Bost–Connes theorem for K is, in this formulation, precisely the non-vanishing $L(1, \chi) \neq 0$ of Hecke [56] and Landau [79] (see also [97, Ch. VII]); over $\mathbb{Q}$ it is Dirichlet’s theorem. Chebotarev enters as the density-theoretic form of the same input.

REMARK 3.12 (Primes of degree ≥ 2 are negligible). At most $d = [K : \mathbb{Q}]$ primes lie above each rational prime, so $\sum_{\deg \mathfrak{p} \geq 2} \text{N}\mathfrak{p}^{-\beta} \leq d \sum_p p^{-2\beta} < \infty$ for $\beta > \frac{1}{2}$. Hence for $\beta > \frac{1}{2}$ the group $\Xi_{\beta}(K, S)$ is unchanged if S is replaced by its set of degree-one primes, and all obstructions in the range of interest are carried by primes with $\text{N}\mathfrak{p} = p$.

3.2. Failure of the ergodicity hypothesis. It is natural to hope that whenever $\zeta_{K,S}(\beta) = \infty$ —so that the units carry measure zero and no Gibbs state exists—the action of $K_{S,+}^\times$ on $\widehat{\mathcal{O}}_S$ is ergodic for every $\nu \in M_\beta(K, S)$, whence uniqueness. This is how the classical uniqueness theorem is proved in [95]. Theorem 3.6 shows the hope is unfounded in general.

EXAMPLE 3.13 (Non-uniqueness with divergent partition function). Let $K = \mathbb{Q}$ and $S = \{3\} \cup \{\ell \equiv 1 \pmod{3}\}$. Then $\sum_{\ell \in S} \ell^{-1} = \infty$, so $\zeta_S(\beta) = \infty$ for $\beta \leq 1$. Let χ be the quadratic character of $\mathbb{Z}_3^\times$, regarded as an element of $\widehat{G}_S = \widehat{\mathbb{Z}}_S^\times$ supported at the component 3. For $\ell \in S$, $\ell \neq 3$, one has $(\sigma_\ell)_3 = \ell \equiv 1 \pmod{3}$, so $\chi(\sigma_\ell) = 1$; and $(\sigma_3)_3 = 1$, so $\chi(\sigma_3) = 1$. Hence the sum in (1.1) vanishes identically and $\chi \in \Xi_\beta$ for every $\beta > 0$. By Theorem 3.6 the KMS_β state is never unique, although there are no Gibbs states for $\beta \leq 1$.

Here $\sigma(I_S)$ is contained in the proper closed subgroup $\ker \chi$, so the failure is already visible topologically. Section 4.2 gives examples where it is not.

REMARK 3.14. The correct general statement replacing the ergodicity hypothesis is Theorem 3.9. What the classical proof uses is not the divergence of $\zeta_{K,S}$ but the equidistribution of S in ray classes; the two coincide for $S = P_K$ and diverge for sparse S .

4. Algebraic obstructions and the separation of topology and measure

Theorem 3.9 reduces uniqueness to a statement about how the primes of S are distributed among Frobenius classes. Two independent mechanisms can make it fail. The first is arithmetic and specific to fields of narrow class number > 1 ; the second is measure-theoretic and occurs already over $\mathbb{Q}$.

4.1. The unramified class-group obstruction. Over $\mathbb{Q}$ every nontrivial character of $G_S = \widehat{\mathbb{Z}}_S^\times$ is built from the components at finitely many primes of S : the modulus must be supported on S , and the associated field is ramified there. Over a field with $h_K^+ > 1$ this is no longer so, because (2.1) provides a surjection $G_S \twoheadrightarrow \text{Cl}_S^+$ whose characters are trivial on all local units.

PROPOSITION 4.1. *Let ψ be a character of Cl_S^+ and $\chi = \psi \circ (G_S \rightarrow \text{Cl}_S^+)$. Then $\chi(\sigma_{\mathfrak{p}}) = \psi([\mathfrak{p}])^{-1} = \overline{\psi([\mathfrak{p}])}$ for every $\mathfrak{p} \in S$, the field L attached to χ is contained in the narrow Hilbert class field H^+ of K and is therefore unramified at all finite primes, and*

$$\chi \in \Xi_\beta(K, S) \iff \sum_{\substack{\mathfrak{p} \in S \\ [\mathfrak{p}] \notin \ker \psi}} N\mathfrak{p}^{-\beta} < \infty.$$

Consequently, if for some proper subgroup $H < \text{Cl}_S^+$ the primes of S lying outside H form a β -summable set, then the KMS_β state of $\mathcal{A}_{K,S}$ is not unique, and no ramified data is involved.

PROOF. Lemma 2.10(1) identifies the image of $\sigma_{\mathfrak{p}}$ in Cl_S^+ with $[\mathfrak{p}]^{-1}$, and inversion does not affect $|1 - \chi(\sigma_{\mathfrak{p}})|$; the rest is Theorem 3.6 together with the computation in the proof of Theorem 3.9. Characters of Cl_K^+ correspond to subfields of H^+ by class field theory. $\square$

EXAMPLE 4.2 ($K = \mathbb{Q}(\sqrt{-5})$). Here $h_K = h_K^+ = 2$ (there are no real places, so the narrow and wide class groups agree), $\mathcal{O}_K^\times = \{\pm 1\}$ is finite, and $H = K(i)$. For a rational prime $p \neq 2, 5$ split in K , the primes above p are principal iff $p = x^2 + 5y^2$ iff $p \equiv 1, 9 \pmod{20}$, and non-principal iff $2p = x^2 + 5y^2$ iff $p \equiv 3, 7 \pmod{20}$.

Let $S_0 = \{\mathfrak{p} : \deg \mathfrak{p} = 1, [\mathfrak{p}] = 1\}$ and let $R = \{\mathfrak{p}_1, \mathfrak{p}_2, \dots\}$ be an infinite set of degree-one primes with $[\mathfrak{p}] \neq 1$, chosen so sparsely that $N\mathfrak{p}_k > 2^{2^k}$; put $S = S_0 \cup R$. Then:

- (1) $\sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-1} = \infty$, since by Chebotarev the rational primes $\equiv 1, 9 \pmod{20}$ have positive density; hence $\beta_c(K, S) = 1$ and $\zeta_{K,S}(\beta) = \infty$ for $0 < \beta \leq 1$;
- (2) $\sum_{\mathfrak{p} \in R} N\mathfrak{p}^{-\beta} \leq \sum_k 2^{-\beta 2^k} < \infty$ for every $\beta > 0$;
- (3) with ψ the nontrivial character of $\text{Cl}_K \cong \mathbb{Z}/2\mathbb{Z}$ and $\chi = \psi \circ (G_S \rightarrow \text{Cl}_S)$ —note $\text{Cl}_S = \text{Cl}_K$ because $R \neq \emptyset$ —Proposition 4.1 gives $\sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}})|^2 = 4 \sum_{\mathfrak{p} \in R} N\mathfrak{p}^{-\beta} < \infty$, so $\chi \in \Xi_{\beta}(K, S)$ for every $\beta > 0$;
- (4) $\sigma(I_S)$ is dense in G_S (Proposition 4.3).

Hence for every $\beta \in (0, 1]$ the KMS_{β} simplex of $\mathcal{A}_{K,S}$ is a segment with exactly two extreme points, neither of them a Gibbs state, and G_S acts on them through $G_S/\ker \chi \cong \mathbb{Z}/2\mathbb{Z}$. The symmetry is partially broken at every temperature, and the obstruction is everywhere unramified.

PROPOSITION 4.3. *With S as in Example 4.2, $\sigma(I_S)$ is dense in G_S .*

PROOF. $G_S = \varprojlim_{\mathfrak{m}} G_S/r(W_{\mathfrak{m}})$ where $W_{\mathfrak{m}} = \prod_{\mathfrak{p} \in F} (1 + \mathfrak{p}^{k_{\mathfrak{p}}} \mathcal{O}_{\mathfrak{p}}) \times \prod_{\mathfrak{p} \in S \setminus F} \mathcal{O}_{\mathfrak{p}}^{\times}$ runs over the open subgroups of the above form, so it suffices to prove surjectivity onto each finite quotient $C_{\mathfrak{m}} := G_S/r(W_{\mathfrak{m}})$. By Definition 2.1 and (2.1), $C_{\mathfrak{m}}$ is a quotient of the ray class group $\text{Cl}_{\mathfrak{m}}(K)$ in which the image of $\sigma_{\mathfrak{p}}$ is the class of $\mathfrak{p}$ for $\mathfrak{p} \nmid \mathfrak{m}$; so we must show that $\{[\mathfrak{p}]_{C_{\mathfrak{m}}} : \mathfrak{p} \in S, \mathfrak{p} \nmid \mathfrak{m}\}$ generates $C_{\mathfrak{m}}$.

Let $C_{\mathfrak{m}}^{\circ} = \ker(C_{\mathfrak{m}} \rightarrow \text{Cl}_K)$, of index 1 or 2. By the Chebotarev density theorem applied to the ray class field of conductor $\mathfrak{m}$, every class of $C_{\mathfrak{m}}^{\circ}$ contains infinitely many degree-one primes; each such prime is principal in Cl_K , hence lies in $S_0 \subseteq S$, and all but finitely many avoid F . So the generated subgroup contains $C_{\mathfrak{m}}^{\circ}$. If $C_{\mathfrak{m}}^{\circ} = C_{\mathfrak{m}}$ we are done. Otherwise choose $\mathfrak{p} \in R \setminus F$, which exists as R is infinite; its class lies outside $C_{\mathfrak{m}}^{\circ}$ because $[\mathfrak{p}] \neq 1$ in Cl_K , and the index is 2, so the generated subgroup is all of $C_{\mathfrak{m}}$. $\square$

4.2. Density is not enough. We now work over $K = \mathbb{Q}$, where $G_S = \widehat{\mathbb{Z}}_S^{\times}$ and, by Lemma 2.10(1), $\sigma_{\ell} = \theta_{\ell}$ with $(\theta_{\ell})_{\ell'} = \ell$ for $\ell' \neq \ell$ and 1 for $\ell' = \ell$; we write $\Theta_S = \sigma(I_S)$. The next results show that density of Θ_S in G_S —the topological form of the ergodicity hypothesis—is necessary but not sufficient.

LEMMA 4.4. *If $\overline{\sigma(I_S)} \neq G_S$ then $\Xi_{\beta}(K, S) \neq \{1\}$ for every $\beta > 0$, and the KMS_{β} state is never unique.*

PROOF. Choose $\chi \neq 1$ trivial on the proper closed subgroup $\overline{\sigma(I_S)}$; then every term of (1.1) vanishes. $\square$

DEFINITION 4.5. Let $P_1 = \{\ell \text{ prime} : \ell \equiv 1 \pmod{3}\}$ and let $R = \{r_1 < r_2 < \dots\}$ be an infinite set of primes $\equiv 2 \pmod{3}$ with $r_k > 2^{2^k}$, which exists by Dirichlet's theorem. Put $S = \{3\} \cup P_1 \cup R$.

LEMMA 4.6. *For S as in Definition 4.5: $\sum_{\ell \in R} \ell^{-\beta} < \infty$ for every $\beta > 0$; and $\sum_{\ell \in S} \ell^{-1} = \infty$, so $\beta_c(S) = 1$ and $\zeta_S(\beta) = \infty$ for $0 < \beta \leq 1$. Moreover, letting $\chi \in \widehat{G_S}$ be the quadratic character at the component 3, i.e. $\chi(u) = \left(\frac{u_3 \bmod 3}{3}\right)$, one has $\chi(\sigma_3) = 1$, $\chi(\sigma_{\ell}) = 1$ for $\ell \in P_1$ and $\chi(\sigma_{\ell}) = -1$ for $\ell \in R$, so that*

$$\sum_{\ell \in S} \ell^{-\beta} |1 - \chi(\sigma_{\ell})|^2 = 4 \sum_{\ell \in R} \ell^{-\beta} < \infty$$

and $\chi \in \Xi_{\beta}(\mathbb{Q}, S)$ for every $\beta > 0$.

PROOF. $r_k^{-\beta} < 2^{-\beta 2^k}$. Divergence over P_1 is Dirichlet's theorem in the form $\sum_{\ell \leq x, \ell \equiv 1(3)} \ell^{-1} = \frac{1}{2} \log \log x + O(1)$. For the character: $(\sigma_3)_3 = 1$ and $(\sigma_{\ell})_3 = \ell$ for $\ell \neq 3$, so $\chi(\sigma_{\ell}) = \left(\frac{\ell}{3}\right)$, which is $+1$ for $\ell \equiv 1$ and -1 for $\ell \equiv 2 \pmod{3}$. $\square$

PROPOSITION 4.7. *For S as in Definition 4.5, $\Theta_S = \sigma(I_S)$ is dense in $G_S = \widehat{\mathbb{Z}}_S^\times$.*

PROOF. Density is equivalent to surjectivity of $\Theta_S \rightarrow \prod_{\ell \in F} (\mathbb{Z}/\ell^{k_\ell} \mathbb{Z})^\times$ for every finite $F \subset S$ and all exponents $k_\ell \geq 1$. Put $M = \prod_{\ell \in F} \ell^{k_\ell}$ and let $\iota : (\mathbb{Z}/M\mathbb{Z})^\times \xrightarrow{\sim} \prod_{\ell \in F} (\mathbb{Z}/\ell^{k_\ell} \mathbb{Z})^\times$ be the Chinese remainder isomorphism. For $\ell \in S \setminus F$ the image of σ_ℓ is $\iota(\ell \bmod M)$, since $(\sigma_\ell)_{\ell'} = \ell$ for every $\ell' \in F$. So it suffices to prove that $D := \langle \ell \bmod M : \ell \in S \setminus F \rangle = (\mathbb{Z}/M\mathbb{Z})^\times$.

Let $K_M = (\mathbb{Z}/M\mathbb{Z})^\times$ if $3 \nmid M$ and $K_M = \{a : a \equiv 1 \bmod 3\}$ if $3 \mid M$, a subgroup of index 1 or 2.

Step 1: $D \supseteq K_M$. Let $a \in K_M$. If $3 \mid M$ then $a \equiv 1 \bmod 3$ and Dirichlet's theorem provides infinitely many primes $\ell \equiv a \bmod M$; each lies in $P_1 \subset S$ and all but finitely many avoid F . If $3 \nmid M$, choose $b \in (\mathbb{Z}/3M\mathbb{Z})^\times$ with $b \equiv a \bmod M$ and $b \equiv 1 \bmod 3$, and apply Dirichlet's theorem to the modulus $3M$.

Step 2: $D = (\mathbb{Z}/M\mathbb{Z})^\times$. If $3 \nmid M$, Step 1 suffices. If $3 \mid M$, choose $r \in R \setminus F$, possible since R is infinite; then $r \equiv 2 \bmod 3$, so $r \bmod M \notin K_M$, while $r \bmod M \in D$. As $[(\mathbb{Z}/M\mathbb{Z})^\times : K_M] = 2$, D is everything. $\square$

REMARK 4.8. Step 2 consumed exactly one element of R . Density holds verbatim if R is a single prime, and fails only if $R = \emptyset$, in which case $\Theta_S \subseteq \ker \chi$ and Lemma 4.4 applies. This is the source of the phenomenon below.

PROPOSITION 4.9. *For S as in Definition 4.5 and $0 < \beta \leq 1$,*

$$\Xi_\beta(\mathbb{Q}, S) = \{1, \chi\} \cong \mathbb{Z}/2\mathbb{Z}, \quad \Xi_\beta^\perp = \ker \chi, \quad G_S/\Xi_\beta^\perp \cong \mathbb{Z}/2\mathbb{Z},$$

while $\Xi_\beta = \widehat{G}_S$ for $\beta > 1$.

PROOF. The case $\beta > 1$ is Lemma 3.5, since $\zeta_S(\beta) < \infty$. Let $0 < \beta \leq 1$ and $\chi' \in \widehat{G}_S$ be nontrivial, factoring through $(\mathbb{Z}/M\mathbb{Z})^\times$ for a modulus M supported on a finite $F \subset S$, with associated Dirichlet character $\tilde{\chi}'$, so that $\chi'(\sigma_\ell) = \tilde{\chi}'(\ell)$ for $\ell \in S \setminus F$. Since F is finite, membership in Ξ_β depends only on $\tilde{\chi}'$ and is equivalent to $\sum_{\ell \in S, \tilde{\chi}'(\ell) \neq 1} \ell^{-\beta} < \infty$.

Case 1: $\tilde{\chi}'|_{K_M} \neq 1$. Pick $a \in K_M$ with $\tilde{\chi}'(a) \neq 1$. As in Step 1 of Proposition 4.7 there are infinitely many $\ell \in P_1$ with $\ell \equiv a \bmod M$, of positive relative density among the primes, so $\sum \ell^{-\beta} \geq \sum \ell^{-1} = \infty$ over that set; hence $\chi' \notin \Xi_\beta$.

Case 2: $\tilde{\chi}'|_{K_M} = 1$. If $3 \nmid M$ then K_M is everything and $\tilde{\chi}' = 1$, so χ' is trivial on all σ_ℓ , $\ell \in S \setminus F$; these generate $(\mathbb{Z}/M\mathbb{Z})^\times$ by Proposition 4.7, so $\chi' = 1$, excluded. If $3 \mid M$ then $\tilde{\chi}'$ factors through the group of order two $(\mathbb{Z}/M\mathbb{Z})^\times/K_M$ and is either trivial—excluded as before—or equal to $\tilde{\chi}$, in which case $\chi'\tilde{\chi}$ has trivial associated character, hence is trivial by the same generation argument, so $\chi' = \chi$.

Finally $\chi \in \Xi_\beta$ by Lemma 4.6, and $\Xi_\beta^\perp = \ker \chi$ has index 2 since χ is onto $\{\pm 1\}$. $\square$

THEOREM 4.10. *Let S be as in Definition 4.5 and $0 < \beta \leq 1$. Then:*

- (1) Θ_S is dense in G_S ;
- (2) $\zeta_S(\beta) = \infty$, so there are no Gibbs states and every $\nu \in M_\beta(\mathbb{Q}, S)$ gives G_S measure zero;
- (3) the KMS_β simplex is a segment, affinely isomorphic to $\text{Prob}(\mathbb{Z}/2\mathbb{Z})$, with exactly two extreme points $\varphi_\pm$;
- (4) $\widehat{\mathbb{Z}}_S^\times$ acts transitively on $\{\varphi_+, \varphi_-\}$ through $\widehat{\mathbb{Z}}_S^\times / \ker \chi \cong \mathbb{Z}/2\mathbb{Z}$.

PROOF. (1) is Proposition 4.7; (2) is Lemma 4.6 with Theorem 3.6; (3) and (4) follow from Theorem 3.6 and Proposition 4.9. $\square$

The invariant. The two extreme points are separated by an explicit function.

DEFINITION 4.11. For $x \in \widehat{\mathbb{Z}}_S$ with all coordinates nonzero and $\sum_{\ell \in R} v_\ell(x) < \infty$, set

$$(4.1) \quad \Psi(x) = \left(\frac{u_3(x) \bmod 3}{3} \right) \cdot (-1)^{\sum_{\ell \in R} v_\ell(x)} \in \{\pm 1\}, \quad u_3(x) = x_3 3^{-v_3(x)}.$$

PROPOSITION 4.12. (1) Ψ is defined ν -almost everywhere for every $\nu \in M_\beta(\mathbb{Q}, S)$ and every $\beta > 0$; the defining condition holds a.s. precisely because $\sum_{\ell \in R} \ell^{-\beta} < \infty$, by the Borel–Cantelli lemma.

- (2) $\Psi(nx) = \Psi(x)$ for every $n \in \mathbb{N}_S$ and every x in the domain, an identity of functions and not merely an almost-everywhere statement.
- (3) Under the symmetric measure, $\Psi = \pm 1$ each with probability $\frac{1}{2}$; conditioning on $\{\Psi = \pm 1\}$ gives the two extreme points $\varphi_\pm$ of Theorem 4.10, whose measures are mutually singular.

PROOF. (1) $\sum_{\ell \in R} \pi_\beta(v_\ell \geq 1) = \sum_{\ell \in R} \ell^{-\beta} < \infty$ by Lemma 4.6, so a.s. only finitely many $\ell \in R$ have $v_\ell \geq 1$; each v_ℓ is finite a.s. by Lemma 2.10(3).

(2) It suffices to treat $n = \ell_0$ prime. Multiplication by ℓ_0 sends $v_\ell \mapsto v_\ell + \delta_{\ell, \ell_0}$ and $u_\ell \mapsto (\theta_{\ell_0})_\ell u_\ell$. If $\ell_0 = 3$: u_3 is unchanged, since $(\theta_3)_3 = 1$, and no v_ℓ with $\ell \in R$ changes. If $\ell_0 \in P_1$: $u_3 \mapsto \ell_0 u_3$ with $\left(\frac{\ell_0}{3}\right) = 1$, and again no v_ℓ , $\ell \in R$, changes. If $\ell_0 \in R$: $u_3 \mapsto \ell_0 u_3$ with $\left(\frac{\ell_0}{3}\right) = -1$, so the first factor changes sign, and $v_{\ell_0} \mapsto v_{\ell_0} + 1$, so the second factor changes sign as well; the product is unchanged.

(3) Under the symmetric measure the conditional law of u given v is Haar, so $\chi(u) = \pm 1$ with probability $\frac{1}{2}$ each, independently of v ; hence so is Ψ . The sets $\{\Psi = \pm 1\}$ are invariant by (2), so the conditioned measures lie in $M_\beta(\mathbb{Q}, S)$; they are mutually singular, and their midpoint is the symmetric measure, which by Theorem 4.10(3) is the midpoint of a segment. A segment has exactly two extreme points. $\square$

REMARK 4.13 (Ψ as an order parameter). Ψ is measurable with respect to the tail σ -algebra of the family $(v_\ell)_{\ell \in S}$ up to the single unit coordinate u_3 : changing the valuations at any finite set of primes leaves it unchanged, because a change at $\ell \in R$ flips both factors of (4.1) at once and a change elsewhere flips neither. It is therefore a macroscopic observable in the sense of statistical mechanics—an asymptotically conserved charge that no local operation can alter—and the two extreme points $\varphi_\pm$ are the two phases in which it is sharp. This is the concrete form taken here by the general principle that symmetry breaking is detected by a tail-measurable order parameter; the Kolmogorov zero–one law of Lemma 3.1 is exactly the statement that no such parameter exists when Ξ_β is trivial.

REMARK 4.14 (Where density and uniqueness part company). Write $R_\chi = \{\ell \in S : \chi(\sigma_\ell) \neq 1\}$; here $R_\chi = R$. Then

$$\overline{\Theta_S} = G_S \iff R_\chi \neq \emptyset \text{ for every nontrivial } \chi, \quad \chi \notin \Xi_\beta \iff \sum_{\ell \in R_\chi} \ell^{-\beta} = \infty.$$

Density asks for the *existence* of a prime detecting χ ; uniqueness asks for an *abundance* of them, weighted by the same $\ell^{-\beta}$ that defines the dynamics. The invariant Ψ makes the gap concrete: it is built from the almost surely finitely many indices $\ell \in R$ at which x is divisible by ℓ , and such a finite count exists exactly when the Borel–Cantelli series converges. Were $\sum_{\ell \in R} \ell^{-\beta} = \infty$, the exponent in (4.1) would be infinite almost surely and Ψ would be undefined on a conull set.

REMARK 4.15 (Kakutani dichotomy). The criterion (1.1) is of Kakutani type: by Lemma 3.2 and Proposition 3.3 the obstruction is the convergence of $\prod_{\mathfrak{p} \in S} |m_{\mathfrak{p}}|$, and this product is nonzero exactly when $\sum_{\mathfrak{p}} (1 - |m_{\mathfrak{p}}|) < \infty$, which is the shape of Kakutani’s theorem on equivalence versus mutual singularity of infinite product measures. The dichotomy is between the translates of ν

being pairwise equivalent (ergodicity, uniqueness) and a nontrivial invariant splitting existing (mutual singularity, symmetry breaking), and it is decided by a convergent-or-divergent series. At the level of general principle this is the familiar fact that a minimal action need not be uniquely ergodic—already visible for flows on surfaces [45]—and no novelty is claimed for the principle; what is new is that the exact place where the two conditions separate can be written down, and that it is a condition on a Dirichlet series attached to S .

Cascades. By Lemma 3.5 the family $\{\Xi_\beta\}_{\beta>0}$ is a nested increasing chain of subgroups of $\widehat{G}_S$, equal to $\{1\}$ for small β in the uniqueness range and to $\widehat{G}_S$ for $\beta > \beta_c(S)$. Each jump is a phase transition. The classical systems have the single jump $\{1\} \rightarrow \widehat{G}_S$ at $\beta = 1$; intermediate jumps can be prescribed.

PROPOSITION 4.16. *Let $0 < \beta_2 < \beta_1 < 1$. There is a set of primes S with $\beta_c(S) = 1$, Θ_S dense in G_S , and*

$$\Xi_\beta(\mathbb{Q}, S) = \begin{cases} \{1\}, & 0 < \beta \leq \beta_2, \\ \{1, \chi_5\}, & \beta_2 < \beta \leq \beta_1, \\ \{1, \chi_3, \chi_5, \chi_3\chi_5\}, & \beta_1 < \beta \leq 1, \\ \widehat{G}_S, & \beta > 1, \end{cases}$$

where $\chi_3, \chi_5 \in \widehat{G}_S$ are the quadratic characters at the components 3 and 5. Thus $\mathcal{A}_{\mathbb{Q}, S}$ has exactly three phase transitions, at β_2, β_1 and 1, with residual symmetry groups

$$G_S \supsetneq \ker \chi_5 \supsetneq \ker \chi_3 \cap \ker \chi_5 \supsetneq \{1\}.$$

PROOF. *The set.* Work modulo 15 and note $(\mathbb{Z}/15\mathbb{Z})^\times = \{1, 2, 4, 7, 8, 11, 13, 14\}$ is isomorphic to $(\mathbb{Z}/3\mathbb{Z})^\times \times (\mathbb{Z}/5\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$. Define

$$S = \{3, 5\} \cup P \cup R_1 \cup R_2,$$

where

$$P = \{\ell : \ell \equiv 1 \text{ or } 4 \pmod{15}\},$$

and R_1, R_2 are infinite sets of primes with

$$R_1 \subseteq \{\ell \equiv 11 \pmod{15}\}, \quad R_2 \subseteq \{\ell \equiv 7 \pmod{15}\},$$

chosen so that $|R_i \cap [1, x]| \asymp x^{\beta_i}$. Such sets exist: by Dirichlet's theorem each of the classes 11 and 7 contains infinitely many primes, and one selects greedily one prime of the class in each interval $[m^{1/\beta_i}, 2m^{1/\beta_i}]$, $m = 1, 2, \dots$. The classes $\{1, 4\}$, $\{11\}$, $\{7\}$ are pairwise disjoint, so the union is disjoint. Note $\ell \equiv 11$ means $\ell \equiv 2 \pmod{3}$ and $\ell \equiv 1 \pmod{5}$, while $\ell \equiv 7$ means $\ell \equiv 1 \pmod{3}$ and $\ell \equiv 2 \pmod{5}$.

By Abel summation, for either i ,

$$(4.2) \quad \sum_{\ell \in R_i} \ell^{-\beta} \asymp \int_2^\infty x^{-\beta} d(x^{\beta_i}) \asymp \int_2^\infty x^{\beta_i - \beta - 1} dx,$$

which converges for $\beta > \beta_i$ and diverges for $\beta \leq \beta_i$, the case $\beta = \beta_i$ giving $\int dx/x$. Since P has positive natural density among the primes, $\sum_{\ell \in P} \ell^{-1} = \infty$, so $\beta_c(S) = 1$ and $\zeta_S(\beta) = \infty$ for $\beta \leq 1$, while $\zeta_S(\beta) < \infty$ for $\beta > 1$; the last case of the displayed formula is then Lemma 3.5.

The subgroup K_M . Let M be a modulus supported on S and set

$$K_M = \{a \in (\mathbb{Z}/M\mathbb{Z})^\times : a \bmod \gcd(M, 15) \in \langle 4 \rangle\},$$

with the convention that the condition is vacuous when $\gcd(M, 15) = 1$. Since $4^2 = 16 \equiv 1 \pmod{15}$, the set $\langle 4 \rangle = \{1, 4\}$ is a subgroup of $(\mathbb{Z}/15\mathbb{Z})^\times$ of order 2, so K_M is a subgroup of index 1, 2, 2 or 4 according as $\gcd(M, 15)$ is 1, 3, 5 or 15. By the Chinese remainder theorem and Dirichlet's theorem, K_M is exactly the set of classes modulo M realized by primes of P outside any prescribed

finite set: given $a \in K_M$, choose by CRT a class b modulo $\text{lcm}(M, 15)$ with $b \equiv a \pmod{M}$ and $b \in \{1, 4\} \pmod{15}$, and apply Dirichlet's theorem to the modulus $\text{lcm}(M, 15)$.

Characters trivial on K_M . Under $(\mathbb{Z}/15\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ the element 4 corresponds to $(0, 2)$, being the square of a primitive root modulo 5; hence

$$(\mathbb{Z}/15\mathbb{Z})^\times / \langle 4 \rangle \cong \mathbb{Z}/2\mathbb{Z} \times (\mathbb{Z}/4\mathbb{Z} / \langle 2 \rangle) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z},$$

and the characters of $(\mathbb{Z}/M\mathbb{Z})^\times$ trivial on K_M are precisely the four characters $1, \tilde{\chi}_3, \tilde{\chi}_5, \tilde{\chi}_3\tilde{\chi}_5$, where $\tilde{\chi}_3$ is the quadratic character modulo 3 and $\tilde{\chi}_5$ the quadratic character modulo 5, inflated to modulus M . In particular a quartic character ψ_5 modulo 5 is *not* trivial on K_M : $\tilde{\psi}_5(4) = \tilde{\psi}_5(2)^2 = (\pm i)^2 = -1$.

Density of Θ_S . As in Proposition 4.7 it suffices to show that $D_M := \langle \ell \bmod M : \ell \in S \setminus F \rangle$ equals $(\mathbb{Z}/M\mathbb{Z})^\times$ for every modulus M supported on S and every finite $F \subset S$. By the previous paragraph $D_M \supseteq K_M$. If $\gcd(M, 15) = 1$ we are done. Otherwise pick $\ell_1 \in R_1 \setminus F$ and $\ell_2 \in R_2 \setminus F$, possible since R_1, R_2 are infinite. Their images in $(\mathbb{Z}/M\mathbb{Z})^\times / K_M \hookrightarrow (\mathbb{Z}/15\mathbb{Z})^\times / \langle 4 \rangle$ are the classes of 11 and 7, which are distinct and nontrivial: indeed $\tilde{\chi}_3(11) = -1$, $\tilde{\chi}_5(11) = 1$, $\tilde{\chi}_3(7) = 1$, $\tilde{\chi}_5(7) = -1$, since 2 is a non-residue modulo 5. Two distinct nontrivial elements generate $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, so $D_M = (\mathbb{Z}/M\mathbb{Z})^\times$.

Computation of Ξ_β . Let $0 < \beta \leq 1$ and let $\chi' \in \widehat{G_S}$ be nontrivial, factoring through $(\mathbb{Z}/M\mathbb{Z})^\times$ for a modulus M supported on a finite $F \subset S$, with associated Dirichlet character $\tilde{\chi}'$, so that $\chi'(\sigma_\ell) = \tilde{\chi}'(\ell)$ for $\ell \in S \setminus F$. As F is finite, membership of χ' in Ξ_β depends only on $\tilde{\chi}'$ and is equivalent to $\sum_{\ell \in S, \tilde{\chi}'(\ell) \neq 1} \ell^{-\beta} < \infty$.

If $\tilde{\chi}'|_{K_M} \neq 1$, choose $a \in K_M$ with $\tilde{\chi}'(a) \neq 1$; by the description of K_M there are infinitely many $\ell \in P$ with $\ell \equiv a \pmod{M}$, forming a set of positive relative density among the primes, so $\sum \ell^{-\beta} \geq \sum \ell^{-1} = \infty$ over that set and $\chi' \notin \Xi_\beta$. This disposes in particular of every character whose 5-part is quartic.

If $\tilde{\chi}'|_{K_M} = 1$ then $\tilde{\chi}' \in \{1, \tilde{\chi}_3, \tilde{\chi}_5, \tilde{\chi}_3\tilde{\chi}_5\}$. If $\tilde{\chi}' = 1$ then χ' is trivial on σ_ℓ for every $\ell \in S \setminus F$, and these generate by the density just proved, so $\chi' = 1$, contrary to assumption; the same argument shows that $\tilde{\chi}' = \tilde{\chi}_3$ forces $\chi' = \chi_3$, and likewise for $\tilde{\chi}_5$ and $\tilde{\chi}_3\tilde{\chi}_5$. It remains to evaluate the three sums, using that $\tilde{\chi}_3$ and $\tilde{\chi}_5$ are trivial on P :

$$\begin{aligned} \tilde{\chi}' = \tilde{\chi}_3 : \quad & \{\ell \in S : \tilde{\chi}_3(\ell) \neq 1\} = R_1 \cup (\text{finite}), \\ \tilde{\chi}' = \tilde{\chi}_5 : \quad & \{\ell \in S : \tilde{\chi}_5(\ell) \neq 1\} = R_2 \cup (\text{finite}), \\ \tilde{\chi}' = \tilde{\chi}_3\tilde{\chi}_5 : \quad & \{\ell \in S : \tilde{\chi}_3\tilde{\chi}_5(\ell) \neq 1\} = R_1 \cup R_2 \cup (\text{finite}), \end{aligned}$$

because $R_1 \equiv 2 \pmod{3}$ and $\equiv 1 \pmod{5}$, while $R_2 \equiv 1 \pmod{3}$ and $\equiv 2 \pmod{5}$, so on R_1 the pair $(\tilde{\chi}_3, \tilde{\chi}_5)$ takes the value $(-1, 1)$ and on R_2 the value $(1, -1)$; the finite sets account for $\ell \in \{3, 5\} \cup F$. By (4.2) the first sum converges iff $\beta > \beta_1$, the second iff $\beta > \beta_2$, and the third iff $\beta > \beta_1$, using $\beta_2 < \beta_1$. Hence

$$\Xi_\beta \cap \{\chi_3, \chi_5, \chi_3\chi_5\} = \begin{cases} \emptyset, & \beta \leq \beta_2, \\ \{\chi_5\}, & \beta_2 < \beta \leq \beta_1, \\ \{\chi_3, \chi_5, \chi_3\chi_5\}, & \beta_1 < \beta \leq 1, \end{cases}$$

which together with the exclusion of all other characters gives the stated formula. The residual symmetry groups are $\Xi_\beta^\perp$, and $\{\chi_3, \chi_5\}$ being independent quadratic characters, these are G_S , $\ker \chi_5$, $\ker \chi_3 \cap \ker \chi_5$ and $\{1\}$ respectively. $\square$

4.3. The unit group and Leopoldt's conjecture. The units enter through the left-hand term of (2.1), and their effect is opposite to that of the class group.

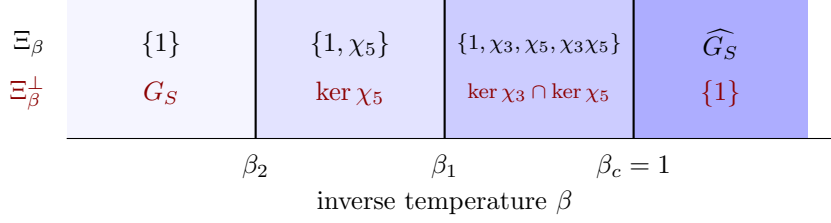


FIGURE 2. The phase diagram of Proposition 4.16. Upper row: the obstruction group Ξ_β , increasing with β by Lemma 3.5. Lower row: the residual symmetry group $\Xi_\beta^\perp$, decreasing along the chain. The classical systems have the single jump $\{1\} \rightarrow \widehat{G}_S$ at $\beta = 1$.

PROPOSITION 4.17. *Enlarging $\overline{\mathcal{O}_{K,+}^\times}$ shrinks G_S , hence $\widehat{G}_S$; all else equal, this makes the condition $\Xi_\beta(K, S) = \{1\}$ easier to satisfy. Moreover, let p be a rational prime, suppose S contains every prime of K above p , and let r be the $\mathbb{Z}$ -rank of $\mathcal{O}_{K,+}^\times$. Then the closure of $\mathcal{O}_{K,+}^\times$ in $\prod_{\mathfrak{p}|p} \mathcal{O}_{\mathfrak{p}}^\times$ is a $\mathbb{Z}_p$ -module of rank at most r .*

PROOF. The first statement is immediate from (2.1) and duality: characters of G_S are characters of $\mathbb{I}_S$ trivial on $\overline{K_{S,+}^\times}$, a stronger condition when the closure is larger. For the second, the closure of a finitely generated abelian group of rank r inside a p -adic Lie group is a $\mathbb{Z}_p$ -module generated by r elements, hence of $\mathbb{Z}_p$ -rank at most r . $\square$

REMARK 4.18. Equality of the rank with r in Proposition 4.17 is precisely the non-vanishing of the p -adic regulator, i.e. Leopoldt's conjecture for K at p , which is known for K abelian over $\mathbb{Q}$ (Ax–Brumer) and open in general. We state only the unconditional inequality, and we emphasize that Theorem 3.9 is unaffected: the criterion is phrased through splitting behaviour in finite abelian extensions and does not refer to how the local units assemble. Leopoldt's conjecture governs the description of G_S as a topological group, not the classification of its KMS states.

4.4. Summary of the comparison.

	$\mathbb{Q}$	general K
parameter group	$\widehat{\mathbb{Z}}_S^\times$	$G_S, 1 \rightarrow U_S / \overline{\mathcal{O}_{K,+}^\times} \rightarrow G_S \rightarrow \text{Cl}_S^+ \rightarrow 1$
coordinates	θ_ℓ elementary	ideles; no canonical generator
obstruction test	Dirichlet character	$\chi(\text{Frob}_{\mathfrak{p}})$, ray class character
uniqueness for $S = P_K$	Dirichlet	Chebotarev, i.e. $L(1, \chi) \neq 0$
unramified obstructions	none ($h_{\mathbb{Q}} = 1$)	subfields of H^+ when $h_K^+ > 1$
units	trivial	shrink G_S , aid uniqueness

5. Sparse sets: the closed low-temperature phase and Sophie Germain primes

We return to $K = \mathbb{Q}$ and apply the theory to sets of primes that are sparse in the sense of sieve theory. The results of this section concern the location and nature of the transition rather than the residual symmetry, and they show that the phase diagram of a localized system can differ from that of [10] even when no obstruction group is present.

DEFINITION 5.1. The *critical exponent* of a set S of primes is

$$\beta_c(S) = \inf \left\{ \beta > 0 : \sum_{\ell \in S} \ell^{-\beta} < \infty \right\} \in [0, 1],$$

with $\beta_c(S) = 0$ if S is finite. We call S *Brun-summable* if $\sum_{\ell \in S} \ell^{-1} < \infty$.

THEOREM 5.2 (Location and nature of the transition). *Let S be a set of primes and $\beta_c = \beta_c(S)$.*

- (1) *For $\beta > \beta_c$ the simplex is $\text{Prob}(G_S)$, with extreme points of type I_∞ permuted freely and transitively by G_S (Theorem 3.6(2)).*
- (2) *For $0 < \beta < \beta_c$ there are no Gibbs states, and the simplex is $\text{Prob}(G_S/\Xi_\beta^\perp)$, a point exactly when the criterion of Theorem 3.9 holds.*
- (3) *$\mathcal{A}_{\mathbb{Q},S}$ has a transition, i.e. both regimes are nonempty, if and only if $\beta_c > 0$.*
- (4) *If $\beta_c > 0$ and $\sum_{\ell \in S} \ell^{-\beta_c} < \infty$, then the type I_∞ phase is the closed half-line $[\beta_c, \infty)$ and*

$$\lim_{\beta \downarrow \beta_c} \zeta_S(\beta) = \zeta_S(\beta_c) < \infty,$$

so the partition function does not diverge at the transition. This hypothesis holds whenever S is Brun-summable with $\beta_c = 1$, since then $\sum_\ell \ell^{-\beta_c} = \sum_\ell \ell^{-1} < \infty$.

PROOF. (1)–(3) are Theorem 3.6 and Definition 5.1. For (4), if $\sum_\ell \ell^{-\beta_c} < \infty$ then $\zeta_S(\beta_c) < \infty$ and Theorem 3.6(2) applies at $\beta = \beta_c$, so the endpoint lies in the type I_∞ phase. As $\beta \downarrow \beta_c$ one has $\ell^{-\beta} \uparrow \ell^{-\beta_c}$ for each ℓ , so by monotone convergence $\sum_\ell \ell^{-\beta} \rightarrow \sum_\ell \ell^{-\beta_c} < \infty$ and the Euler products converge to $\zeta_S(\beta_c)$. The last sentence is the definition of Brun-summability together with $\beta_c = 1$. $\square$

REMARK 5.3. Item (4) is sharp and marks a genuine difference from the classical system. For S the set of all primes, $\beta_c = 1$ but S is not Brun-summable: $\sum_\ell \ell^{-1} = \infty$ by Mertens, $\zeta(\beta_c) = \infty$, and $\zeta(\beta) \rightarrow \infty$ as $\beta \downarrow 1$ through the pole. Thus:

	β_c	$\zeta_S(\beta_c)$	phase of β_c
all primes (Bost–Connes)	1	∞	high-temperature
Brun-summable with $\beta_c = 1$	1	$< \infty$	type I_∞

The two systems have the same critical temperature and opposite behaviour at it; the critical point belongs to the symmetric phase in one case and to the symmetry-broken phase in the other. The distinction is invisible in the exponent: it is the difference between Mertens’ theorem and Brun’s theorem.

REMARK 5.4. For $0 < \beta_c < 1$ the value $\zeta_S(\beta_c)$ is not determined by β_c alone; both convergence and divergence occur among sets with the same convergence exponent, as one sees by comparing $\ell_k \approx \exp(k^{1/\beta_c})$ with $\ell_k \approx \exp(k^{1/\beta_c}/\log k)$. The Brun-summable case with $\beta_c = 1$ is the one relevant below.

5.1. Sophie Germain primes. Let $\mathcal{SG} = \{p \text{ prime} : 2p+1 \text{ prime}\}$, $\pi_{\mathcal{SG}}(x) = |\mathcal{SG} \cap [1, x]|$, and $S_\infty = \{2\} \cup \mathcal{SG}$; by Proposition 2.7, $\mathcal{A}_{\mathbb{Q},S_\infty} = \varinjlim_k \mathcal{A}_{\mathbb{Q},T_k}$ with $T_k = \{2, p_1, \dots, p_k\}$, and $\mathcal{A}_{\mathbb{Q},S_\infty} \subseteq \mathcal{A}_{BC}$.

THEOREM 5.5. *Let $\beta_c = \beta_c(S_\infty)$.*

- (1) (Unconditional.) $\pi_{\mathcal{SG}}(x) \ll x/\log^2 x$ [52]; hence S_∞ is Brun-summable, $\zeta_{S_\infty}(\beta) < \infty$ for all $\beta \geq 1$, and $\beta_c \leq 1$.
- (2) *If $\pi_{\mathcal{SG}}(x) \gg x/\log^A x$ for some $A > 0$ —in particular under the Hardy–Littlewood conjecture $\pi_{\mathcal{SG}}(x) \sim 2C_2 x/\log^2 x$ [53]—then $\beta_c = 1$.*
- (3) *Under the hypothesis of (2), ζ_{S_∞} is finite on $[1, \infty)$ and identically $+\infty$ on $(0, 1)$, and the type I_∞ phase of $\mathcal{A}_{\mathbb{Q},S_\infty}$ is exactly $[1, \infty)$: a closed low-temperature phase whose critical point is $\beta_c = 1$.*
- (4) *$\mathcal{A}_{\mathbb{Q},S_\infty}$ has a phase transition if and only if $\mathcal{SG}$ has positive convergence exponent; in particular a phase transition implies that $\mathcal{SG}$ is infinite.*

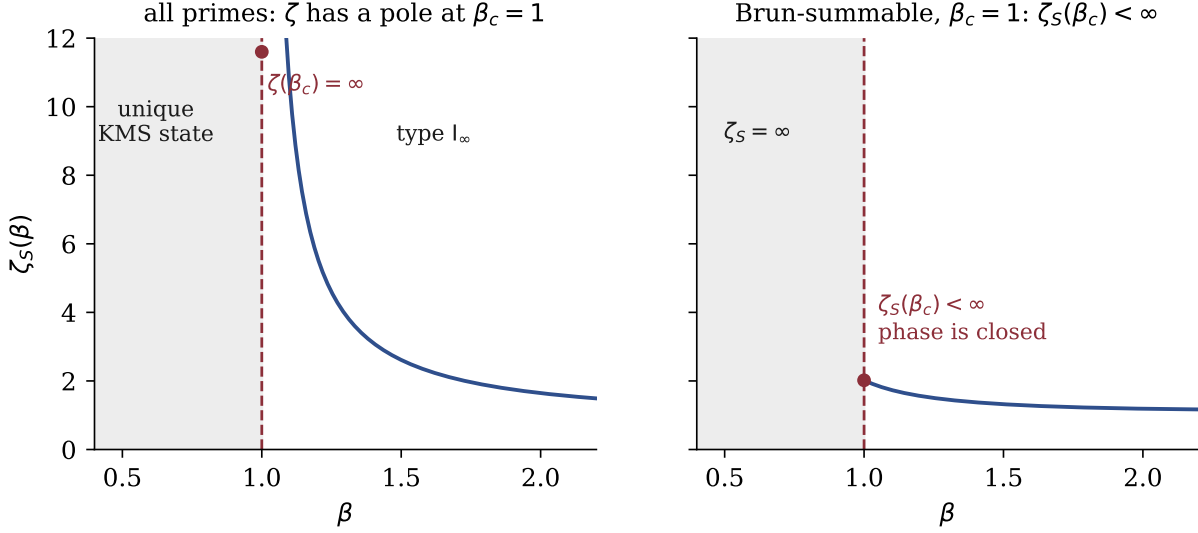


FIGURE 3. Remark 5.3: the same critical temperature, opposite behaviour at it. Left, the classical system, where ζ has a pole at $\beta_c = 1$ and the critical point lies in the high-temperature phase. Right, a Brun-summable set with $\beta_c = 1$, where $\zeta_S(\beta_c) < \infty$ and the type I_∞ phase is closed.

- (5) For β in the range where $\zeta_{S_\infty}(\beta) = \infty$, the KMS_β state is unique if and only if, for every modulus M supported on S_∞ and every proper subgroup $H < (\mathbb{Z}/M\mathbb{Z})^\times$,

$$\sum_{\substack{p \in S\mathcal{G} \\ p \bmod M \notin H}} p^{-\beta} = \infty;$$

this holds under the Hardy–Littlewood conjecture with congruence conditions.

PROOF. (1) Abel summation: with $\pi_{S\mathcal{G}}(t) \leq Ct/\log^2 t$ for $t \geq t_0$,

$$\sum_{p \in S\mathcal{G}, p \leq x} \frac{1}{p} = \frac{\pi_{S\mathcal{G}}(x)}{x} + \int_{t_0}^x \frac{\pi_{S\mathcal{G}}(t)}{t^2} dt + O(1) \leq O(1) + C \int_{t_0}^x \frac{dt}{t \log^2 t} = O(1).$$

Hence $\sum_{p \in S\mathcal{G}} p^{-1} < \infty$, $\beta_c \leq 1$, and $\zeta_{S_\infty}(\beta) < \infty$ for $\beta \geq 1$ since $\ell^{-\beta} \leq \ell^{-1}$.

- (2) For $0 < \beta < 1$, Abel summation with the lower bound gives

$$\sum_{p \in S\mathcal{G}, p \leq x} p^{-\beta} \geq \beta \int_{t_0}^x \frac{\pi_{S\mathcal{G}}(t)}{t^{\beta+1}} dt \gg \int_{t_0}^x \frac{dt}{t^\beta \log^A t} \gg \frac{x^{1-\beta}}{\log^A x} \rightarrow \infty,$$

so $\beta_c \geq 1$; with (1), $\beta_c = 1$.

- (3) Combine (1), (2) and Theorem 5.2(4).

- (4) Theorem 5.2(3).

(5) The first assertion is Theorem 3.9 specialized to $K = \mathbb{Q}$, where the abelian extensions unramified outside $S \cup \infty$ are the subfields of cyclotomic fields $\mathbb{Q}(\zeta_M)$ with M supported on S , and Frob_p is the class of p modulo M . For the conditional statement, assume that for each admissible residue a one has $|\{p \leq x : p \in S\mathcal{G}, p \equiv a \pmod{M}\}| \gg_M x/\log^2 x$, where

$$A_M = \{a \in (\mathbb{Z}/M\mathbb{Z})^\times : 2a + 1 \in (\mathbb{Z}/M\mathbb{Z})^\times\}$$

is the set of admissible classes. By Lemma 5.6 below, $\langle A_M \rangle = (\mathbb{Z}/M\mathbb{Z})^\times$, so for any proper subgroup H there is $a \in A_M \setminus H$; the primes of $\mathcal{SG}$ in that class then give a divergent sum by the computation in (2), and they all lie outside H . $\square$

LEMMA 5.6. $\langle A_M \rangle = (\mathbb{Z}/M\mathbb{Z})^\times$ for every $M \geq 1$.

PROOF. Let χ be a character of $(\mathbb{Z}/M\mathbb{Z})^\times$ trivial on A_M ; we show $\chi = 1$. By the Chinese remainder theorem, $\chi = \prod_\ell \chi_\ell$ and $A_M = \prod_\ell A_{\ell^k}$, where $A_{2^k} = (\mathbb{Z}/2^k)^\times$ and, for ℓ odd, $A_{\ell^k} = \{a \in (\mathbb{Z}/\ell^k)^\times : a \not\equiv -2^{-1} \pmod{\ell}\}$. As A_M is a product, each χ_ℓ is constant on A_{ℓ^k} , say with value c_ℓ , and $\prod_\ell c_\ell = 1$.

For $\ell = 2$: A_{2^k} is the whole group, so $\chi_2 = 1$ and $c_2 = 1$. For $\ell \geq 5$: $|A_{\ell^k}| = \varphi(\ell^k)(1 - \frac{1}{\ell-1}) > \frac{1}{2}\varphi(\ell^k)$, so $A_{\ell^k} A_{\ell^k}^{-1} = (\mathbb{Z}/\ell^k)^\times$, and χ_ℓ , being constant on A_{ℓ^k} , is trivial, so $c_\ell = 1$. For $\ell = 3$: $A_{3^k} = \{a : a \not\equiv 1 \pmod{3}\}$ is the nontrivial coset of the index-two subgroup $K = \{a \equiv 1 \pmod{3}\}$, so $A_{3^k} A_{3^k}^{-1} = K$ and χ_3 is trivial on K ; hence χ_3 is trivial or the quadratic character with kernel K , and $c_3 = \pm 1$ accordingly. Since all other $c_\ell = 1$ and $\prod c_\ell = 1$, we get $c_3 = 1$, so χ_3 is trivial on $A_{3^k} \cup K$, which is everything. Therefore $\chi = 1$. $\square$

REMARK 5.7. Theorem 5.5(4) is an equivalence, not an implication in a useful direction. Establishing the transition requires exactly the arithmetic input that is unavailable, a lower bound on the density of $\mathcal{SG}$; and the transition, once granted, yields nothing about $\mathcal{SG}$ beyond the density statement assumed. The construction is a dictionary entry, not a method. The same applies verbatim to the twin primes, to the primes p with $p + 2k$ prime for fixed k , and to any set for which a Brun-type upper bound is known and a matching lower bound conjectured: all such systems have the same phase diagram, and none of them distinguishes its defining condition from any other of the same density. What Theorem 5.5 does record is that the two open points—the location of β_c and the uniqueness below it—are governed by the same input, the second in a form refined by congruence conditions.

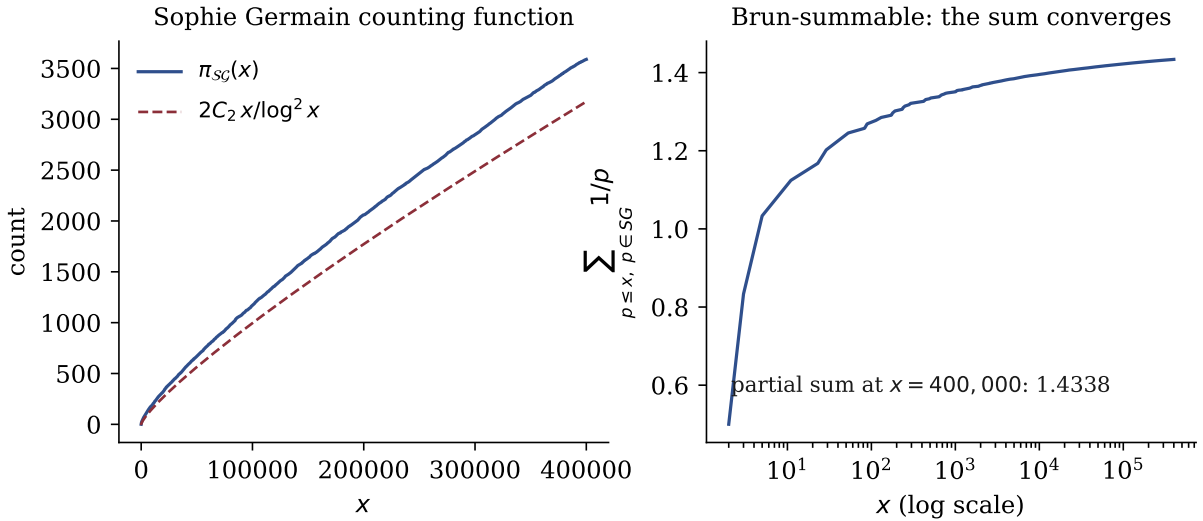


FIGURE 4. Data for Theorem 5.5(1). Left, $\pi_{\mathcal{SG}}(x)$ against $2C_2 x / \log^2 x$. Right, the partial sums of $\sum_{p \in \mathcal{SG}} 1/p$, whose convergence is the Brun-summability used throughout §5.

5.2. Restriction of the classical phases. Finally we record how the phases of the full system restrict, which explains in what sense the localized systems are cross-sections of the classical one.

PROPOSITION 5.8. *Let S be a set of primes with complement S^c , let $\beta > 1$ and $\chi \in \widehat{\mathbb{Z}}^\times$, and let $\varphi_{\beta,\chi}^{BC}$ be the corresponding extremal KMS_β state of $\mathcal{A}_{BC}$. Then, under the embedding of Proposition 2.7,*

$$\varphi_{\beta,\chi}^{BC}|_{\mathcal{A}_{\mathbb{Q},S}} = \frac{1}{\zeta_{S^c}(\beta)} \sum_{m \in \mathbb{N}_{S^c}} m^{-\beta} \varphi_{\beta, \text{pr}_S(m)\chi_S}^S, \quad \chi_S = \text{pr}_S(\chi).$$

The map $\chi_S \mapsto \varphi_{\beta,\chi}^{BC}|_{\mathcal{A}_{\mathbb{Q},S}}$ is injective and $\widehat{\mathbb{Z}}_S^\times$ -equivariant, but if S^c is infinite its image consists entirely of non-extremal states, whose measures have infinitely many atoms. For $0 < \beta \leq 1$ the unique KMS_β state of $\mathcal{A}_{BC}$ restricts to the symmetric state of $\mathcal{A}_{\mathbb{Q},S}$.

PROOF. Decompose $\ell^2(\mathbb{N}^\times) = \bigoplus_{m \in \mathbb{N}_{S^c}} \ell^2(m\mathbb{N}_S)$. Each summand is invariant under $\pi_\chi(\mathcal{A}_{\mathbb{Q},S})$, and $\epsilon_{mn} \mapsto \epsilon_n$ intertwines $\pi_\chi|_{\mathcal{A}_{\mathbb{Q},S}}$ on $\ell^2(m\mathbb{N}_S)$ with $\pi_{\text{pr}_S(m)\chi_S}$, while $e^{-\beta H}$ restricts to $m^{-\beta} e^{-\beta H_S}$. Summing and using $\zeta = \zeta_S \zeta_{S^c}$ gives the formula. The atoms $\text{pr}_S(m)\chi_S$ are pairwise distinct, since $\text{pr}_S(m) = \text{pr}_S(m')$ forces $m \equiv m'$ modulo ℓ^k for all $\ell \in S$ and all k , hence $m = m'$; the atom of largest mass is the one with $m = 1$, so χ_S is recoverable, and equivariance follows from $\pi_\chi \circ \theta_v = \pi_{v^{-1}\chi}$. Extremal states correspond to point masses by Theorem 3.6(2), whereas these measures have infinitely many atoms when $\mathbb{N}_{S^c}$ is infinite. The last claim follows from $\widehat{\mathbb{Z}}^\times$ -invariance of the unique state. $\square$

REMARK 5.9. So the parameter space of the low-temperature phase projects equivariantly, $\widehat{\mathbb{Z}}^\times \rightarrow \widehat{\mathbb{Z}}_S^\times$, and this is a genuine reduction of the Galois symmetry; what does not descend is extremality, because the degrees of freedom at the primes outside S have been traced out rather than fixed. In particular the transition of $\mathcal{A}_{\mathbb{Q},S}$ at $\beta_c(S)$ is not the restriction of the transition of $\mathcal{A}_{BC}$ at $\beta = 1$: for Brun-summable S with $\beta_c(S) = 1$ the two occur at the same temperature but on opposite sides, by Remark 5.3.

6. Assessment and open problems

We separate what is established from what is not.

Established unconditionally. Theorem 3.6 classifies the KMS_β states of $\mathcal{A}_{K,S}$ for every number field K , every set S of finite primes and every $\beta > 0$; Theorem 3.9 converts it into the splitting criterion; Corollary 3.10 recovers the classical theorems of [10, 75] from Chebotarev; Example 3.13, Theorem 4.10 and Example 4.2 exhibit failures of uniqueness with divergent partition function, the last two with dense $\sigma(I_S)$; Proposition 4.12 supplies the invariant; Theorem 5.2 locates the transition and shows the low-temperature phase can be closed; Theorem 5.5(1) is unconditional.

Conditional or incomplete. Theorem 5.5(2),(3),(5) depend on a lower bound for $\pi_{\mathcal{SG}}$ of Hardy–Littlewood strength, in (5) refined by congruence conditions; Remark 4.18 records that the description of G_S as a topological group, though not the criterion, is entangled with Leopoldt’s conjecture.

Relation to the literature. Sections 2 and 3 are, for $S = P_K$, a reformulation of results in [10, 76, 75, 96, 95]; the reduction to measures (Proposition 2.8) uses [77] and [96] directly. It is worth stating precisely where the present work goes beyond [96]. That paper classifies KMS states on C^* -algebras of *non-principal* groupoids, and its content lies in the field of states on the isotropy group algebras. Proposition 2.8 shows that our groupoid is principal on a conull set—the I_S -action is free because the valuation vector is translated freely—so the isotropy data is empty and [96] contributes only the reduction to quasi-invariant measures. Everything after that point is independent of it: the forced product structure of the valuation marginal (Lemma 2.10(3)), the

cohomological equation (2.3), and the extraction of the arithmetic obstruction group Ξ_β from a Kakutani product criterion (Propositions 3.3 and 3.4) are what produce Theorem 3.6, and we have found no counterpart to them in [96]. A literature check remains advisable, but the boundary is clear enough that we do not expect a conflict.

The type of the intermediate phases. We record what the ratio-set argument does and does not give.

PROPOSITION 6.1. *Let $\beta > 0$ with $\zeta_{K,S}(\beta) = \infty$, let φ be an extremal KMS_β state of $\mathcal{A}_{K,S}$, and let $M = \pi_\varphi(\mathcal{A}_{K,S})''$. Put*

$$\Gamma_\beta = \langle N\mathfrak{p}^{-\beta} : \mathfrak{p} \in S \rangle \leq \mathbb{R}_{>0}.$$

Then M is a factor of type III, and its Connes invariant satisfies

$$S(M) \subseteq \overline{\Gamma_\beta} \cup \{0\}.$$

Consequently, if the numbers $\log N\mathfrak{p}$, $\mathfrak{p} \in S$, are pairwise commensurable and generate $\log \lambda_0 \cdot \mathbb{Z}$, then $S(M) \subseteq \{0\} \cup \lambda_0^{-\beta\mathbb{Z}}$, so M is of type $\text{III}_{\lambda_0^{-n\beta}}$ for some integer $n \geq 1$ or of type III_0 .

PROOF. Extreme points of a KMS simplex give factor states [12, §5.3]. By Proposition 2.8 the groupoid is principal on a conull set, so M is the Krieger factor of the orbit equivalence relation of I_S acting on (Y_S, ν) , and by Krieger's theorem [73] $S(M)$ is the ratio set of that relation. The Radon–Nikodym cocycle is $\mathfrak{a}\mathfrak{b}^{-1} \mapsto (N\mathfrak{a}/N\mathfrak{b})^{-\beta}$, whose values lie in Γ_β ; the ratio set is contained in the closure of the essential range of the cocycle, hence in $\overline{\Gamma_\beta} \cup \{0\}$.

M is not semifinite. By [73] this means that the relation carries no σ -finite invariant measure equivalent to ν , i.e. that there is no measurable $f : Y_S \rightarrow (0, \infty)$ with $f(\mathfrak{a} \cdot y) = N\mathfrak{a}^\beta f(y)$ for ν -a.e. y and all $\mathfrak{a} \in J_S$. Suppose f exists and put $X = \log f$. In the coordinates of Lemma 2.10, the function $\Phi(v, t) := \int_{G_S} e^{itX(v, g)} d\rho_v(g)$ satisfies, by (2.3) and Lemma 2.10(2), $\Phi(v + a, t) = N\mathfrak{a}^{i\beta t} \Phi(v, t)$ for π_β -a.e. v : this is (3.1) with $\overline{\chi(\sigma_{\mathfrak{a}})}$ replaced by the unimodular constants $c_{\mathfrak{a}} = N\mathfrak{a}^{i\beta t}$, and the proofs of Lemma 3.1 and Proposition 3.3 use nothing else. Fix t , write $c_{\mathfrak{p}} = N\mathfrak{p}^{i\beta t}$ and $m_{\mathfrak{p}}(t) = (1 - t_{\mathfrak{p}})/(1 - c_{\mathfrak{p}}t_{\mathfrak{p}})$. As in the proof of Proposition 3.3, for every finite $F \subset S$ we have $\Phi = P_F G_F$ with $P_F = \prod_{\mathfrak{p} \in F} c_{\mathfrak{p}}^{v_{\mathfrak{p}}}$ and G_F measurable with respect to $\sigma(v_{\mathfrak{p}} : \mathfrak{p} \notin F)$, hence independent of P_F and bounded by 1. Therefore

$$\psi(t) := \int_{Y_S} e^{itX} d\nu = \mathbb{E}_{\pi_\beta}[\Phi(\cdot, t)] = \mathbb{E}[P_F] \mathbb{E}[G_F], \quad |\psi(t)| \leq \prod_{\mathfrak{p} \in F} |m_{\mathfrak{p}}(t)|,$$

and letting $F \uparrow S$ and using Lemma 3.2 in the form $1 - |m_{\mathfrak{p}}| \geq \frac{1}{2}(1 - |m_{\mathfrak{p}}|^2) \geq \frac{1}{8}t_{\mathfrak{p}}|1 - c_{\mathfrak{p}}|^2$,

$$|\psi(t)| \leq \prod_{\mathfrak{p} \in S} |m_{\mathfrak{p}}(t)| \leq \exp\left(-\frac{1}{8}g(t)\right), \quad g(t) := \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} |1 - N\mathfrak{p}^{i\beta t}|^2.$$

Now ψ is the characteristic function of the law of X under ν , so it is continuous with $\psi(0) = 1$; choose $\delta > 0$ with $|\psi| \geq \frac{1}{2}$ on $[0, \delta]$. Then $g \leq 8 \log 2$ on $[0, \delta]$. But $|1 - e^{i\omega t}|^2 = 4 \sin^2(\omega t/2)$ and $\int_0^\delta 4 \sin^2(\omega t/2) dt = 2\delta - \frac{2}{\omega} \sin(\omega\delta) \geq \delta$ as soon as $\omega\delta \geq 2$, so by Tonelli

$$\int_0^\delta g(t) dt \geq \delta \sum_{\mathfrak{p} \in S: \beta\delta \log N\mathfrak{p} \geq 2} N\mathfrak{p}^{-\beta} = \infty,$$

the sum omitting only finitely many primes of S and $\zeta_{K,S}(\beta) = \infty$. This contradicts $g \leq 8 \log 2$ on $[0, \delta]$, so no such f exists and M is of type III.

For the last statement, a closed subgroup of $\lambda_0^{-\beta\mathbb{Z}}$ is $\lambda_0^{-n\beta\mathbb{Z}}$ for some $n \geq 0$; $n = 0$ would give $S(M) \subseteq \{0, 1\}$, i.e. type III_0 , and $n \geq 1$ gives type $\text{III}_{\lambda_0^{-n\beta}}$. $\square$

REMARK 6.2. Proposition 6.1 bounds $S(M)$ from *above*. It is tempting to argue that when two norms have irrational log-ratio, Γ_β is dense in $\mathbb{R}_{>0}$ and therefore $S(M) = [0, \infty)$, i.e. M is III_1 ; but this inference is invalid, since density of the group of *possible* ratios gives no lower bound on the ratio set, and III_0 —where $S(M) = \{0, 1\}$ —is not excluded. Establishing III_1 requires the ergodicity of the Maharam extension of the action, equivalently that the essential range of the cocycle is all of $\overline{\Gamma_\beta}$.

The natural candidate for the missing ingredient is an *extended* obstruction group $\Xi_\beta^{\text{ext}} \leq \widehat{G}_S \times \mathbb{R}$, obtained from (1.1) by twisting the characters with $\text{Np}^{i\beta s}$ and having Ξ_β as its $s = 0$ slice; the essential range of the cocycle should be readable from its projection to $\mathbb{R}$. We do not pursue this here, and nothing in this paper depends on it. Note that the commensurable case of Proposition 6.1 is vacuous once S is infinite, since $\log p$ for distinct rational primes are $\mathbb{Q}$ -linearly independent and only finitely many primes of K lie above each p ; any failure of type III_1 arises from *approximate* commensurability along a set of divergent β -weight.

Open problems.

- (1) Determine the type of the extremal states. Proposition 6.1 bounds $S(M)$ from above and Remark 6.2 explains what a lower bound requires; for the sparse sets of §5 it is not decided here which of the alternatives III_1 , III_λ , III_0 occurs.
- (2) Characterize the possible chains $\{\Xi_\beta\}_{\beta>0}$. By Lemma 3.5 they are increasing; Proposition 4.16 realizes finite chains. The transition set is always countable, $\widehat{G}_S$ being countable; which monotone families of subgroups of $\widehat{G}_S$ occur, and whether infinitely many transitions can be forced into a bounded interval, are open.
- (3) Extend the criterion to the function field case, where Chebotarev is available with error terms from the Riemann hypothesis for curves, and where sparse sets of closed points can be constructed with more freedom than sparse sets of primes. One structural difference is worth recording: the log-norms $\log NP = \deg P \cdot \log q$ are pairwise commensurable, so the commensurable case of Proposition 6.1—vacuous over a number field by Remark 6.2—is there the only case.
- (4) In Example 4.2 the obstruction is unramified. Is there a canonical description of the sub-simplex of KMS_β states cut out by the unramified part of $\widehat{G}_S$, in terms of the Hilbert class field, valid for all S simultaneously?

Part 2

Structure of the abelian systems

CHAPTER 2

Factor type and the spectrum of transitions

This chapter is Paper I of the series (papers/I-type-spectrum/).

SUMMARY. The obstruction group $\Xi_\beta(K, S)$ of [Ch. 1] classifies the KMS_β simplex. We show that it is the $s = 0$ slice of a single larger object, the *extended obstruction group*

$$\Xi_\beta^{\text{ext}}(K, S) = \left\{ (\chi, s) \in \widehat{G_S} \times \mathbb{R} : \sum_{\mathfrak{p} \in S} \text{Np}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}}) \text{Np}^{i\beta s}|^2 < \infty \right\},$$

whose slices $s \neq 0$ control the Murray–von Neumann–Connes type of the associated factors from above: the essential values of the Radon–Nikodym cocycle annihilate $\Sigma_\beta = \text{pr}_{\mathbb{R}} \Xi_\beta^{\text{ext}}$, so $\Sigma_\beta \neq \{0\}$ excludes type III_1 , a dense Σ_β forces III_0 , and $\Sigma_\beta = c\mathbb{Z}$ forces III_{λ^k} with $\lambda = e^{-2\pi/c}$ or III_0 ; moreover Σ_β lies in the point spectrum of the flow of weights. Matching lower bounds come from a direct construction of essential values in the manner of Araki and Woods. The phase classification and the type classification are thus two coordinate slices of one Dirichlet-series condition. For $S = P_K$ and $\beta = 1$ both reduce to the non-vanishing of Hecke L -functions on $\text{Re} = 1$: the $s = 0$ slice is $L(1, \chi) \neq 0$ and the $s \neq 0$ slices are $L(1 - i\tau, \chi) \neq 0$.

Two consequences are perhaps unexpected. First, the natural sufficient condition for type III_1 — that two primes have irrationally related log-norms — is false: two primes contribute a finite amount to a divergent series. We exhibit $S \subset P_{\mathbb{Q}}$ with $\zeta_S(1) = \infty$, a *unique* KMS_1 state, and factors of pure type $\text{III}_{e^{-1}}$; uniqueness of the KMS state and type III_1 are logically independent. Second, for infinite S the log-norms are never pairwise commensurable, so any failure of III_1 comes from *approximate* commensurability along a set of divergent β -weight.

In the horizontal direction we determine which chains $\{\Xi_\beta\}_{\beta > 0}$ occur. For every sequence $(\beta_n) \subset (0, 1)$ there is S with $\beta_c(S) = 1$, dense parameter subgroup, and $\Xi_\beta = \bigoplus_{n: \beta_n < \beta} \mathbb{Z}/2\mathbb{Z}$; taking (β_n) dense in $(0, 1)$ gives countably many dense transitions, and taking it dense in a Cantor set $\mathcal{C}$ makes the non-local-constancy locus exactly $\mathcal{C}$. The continuous devil’s staircase, however, is unattainable for a structural reason: $\widehat{G_S}$ is countable for every S , so the jump set is countable and every order parameter is a pure jump function. It is approachable in the uniform norm and not attained.

1. Introduction: one group, two directions

We use the notation of [Ch. 1]: $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$ with $\sigma_t(\mu_{\mathfrak{a}}) = \text{Na}^{it}\mu_{\mathfrak{a}}$, symmetry group G_S , parameter homomorphism $\sigma : I_S \rightarrow G_S$, and

$$(1.1) \quad \Xi_\beta(K, S) = \left\{ \chi \in \widehat{G_S} : \sum_{\mathfrak{p} \in S} \text{Np}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}})|^2 < \infty \right\},$$

the KMS_β simplex being $\text{Prob}(G_S/\Xi_\beta^\perp)$. Write $\mathcal{R}_S$ for the tail equivalence relation on $V = \mathbb{Z}_{\geq 0}^S$ and π_β for the forced product measure [Ch. 1, Lem. 2.10].

The phase diagram of $\mathcal{A}_{K,S}$ has two directions. Horizontally one asks which β are transition points, i.e. how the chain $\{\Xi_\beta\}$ grows; vertically one asks what the factors at a given β are. Definition 1.1 shows the two questions are slices of one object, and the rest of the paper answers both — the second one partly: the object bounds the type from above, and the bound is matched by direct constructions in the cases treated.

DEFINITION 1.1. The *extended obstruction group* is

$$(1.2) \quad \Xi_\beta^{\text{ext}}(K, S) = \left\{ (\chi, s) \in \widehat{G}_S \times \mathbb{R} : \sum_{\mathfrak{p} \in S} \text{N}\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}}) \text{N}\mathfrak{p}^{i\beta s}|^2 < \infty \right\},$$

a subgroup of $\widehat{G}_S \times \mathbb{R}$, and $\Sigma_\beta := \text{pr}_{\mathbb{R}}(\Xi_\beta^{\text{ext}}) \leq \mathbb{R}$.

That Ξ_β^{ext} is a subgroup follows as in [Ch. 1, Lem. 3.5] from $|1 - z_1 z_2| \leq |1 - z_1| + |1 - z_2|$ for $|z_i| = 1$; and $\Xi_\beta = \Xi_\beta^{\text{ext}} \cap (\widehat{G}_S \times \{0\})$.

2. The type criterion

Let $\beta > 0$ with $\zeta_{K,S}(\beta) = \infty$, let φ be an extremal KMS_β state and $M = \pi_\varphi(\mathcal{A}_{K,S})''$. By [Ch. 1, Prop. 2.8 and Lem. 2.10], φ is given by a probability measure ν on $Y = V \times G_S$ whose V -marginal is π_β , and M is the von Neumann algebra of the orbit equivalence relation $\mathcal{R}$ on (Y, ν) generated by the maps $(v, h) \mapsto (v + e_{\mathfrak{p}}, \sigma_{\mathfrak{p}} h)$, $\mathfrak{p} \in S$. The Radon–Nikodym cocycle of $\mathcal{R}$ is

$$(2.1) \quad \rho(y, \mathfrak{a}\mathfrak{b}^{-1} \cdot y) = \log \frac{d\nu \circ (\mathfrak{a}\mathfrak{b}^{-1})}{d\nu}(y) = -\beta \log \frac{\text{N}\mathfrak{a}}{\text{N}\mathfrak{b}},$$

and we write $E \subseteq \mathbb{R}$ for its group of finite essential values in the sense of [117]: $u \in E$ iff for every $B \subseteq Y$ with $\nu(B) > 0$ and every $\varepsilon > 0$ there is a partial transformation T of $\mathcal{R}$ with $\nu(B \cap T^{-1}B) > 0$ and $|\rho(y, Ty) - u| < \varepsilon$ on its domain. E is a closed subgroup of $\mathbb{R}$ [117], M is a factor of type III [Ch. 1, Prop. 6.1], and by Krieger's theorem [73] the Connes invariant is

$$(2.2) \quad S(M) = \{0\} \cup \exp(E),$$

so that M is of type III_1 , III_λ or III_0 according as $E = \mathbb{R}$, $E = (\log \lambda)\mathbb{Z}$ with $0 < \lambda < 1$, or $E = \{0\}$ [24]. Everything in this section is a computation of E .

LEMMA 2.1 (Coboundary dictionary). *Let $(\chi, s) \in \widehat{G}_S \times \mathbb{R}$ and $c_{\mathfrak{p}} = \chi(\sigma_{\mathfrak{p}}) \text{N}\mathfrak{p}^{i\beta s}$. The following are equivalent:*

- (1) *there is a measurable $g : V \rightarrow \mathbb{T}$ with $g(v + e_{\mathfrak{p}}) = c_{\mathfrak{p}} g(v)$ for π_β -a.e. v , for every $\mathfrak{p} \in S$;*
- (2) *$(\chi, s) \in \Xi_\beta^{\text{ext}}$.*

PROOF. Propositions 3.3 and 3.4 of [Ch. 1] prove exactly this equivalence for $c_{\mathfrak{p}} = \overline{\chi(\sigma_{\mathfrak{p}})}$, and their proofs use only that $|c_{\mathfrak{p}}| = 1$, that the coordinates $v_{\mathfrak{p}}$ are independent under π_β with $\pi_\beta(v_{\mathfrak{p}} = k) = (1 - t_{\mathfrak{p}})t_{\mathfrak{p}}^k$, $t_{\mathfrak{p}} = \text{N}\mathfrak{p}^{-\beta}$, and the estimate $1 - |m_{\mathfrak{p}}|^2 \asymp t_{\mathfrak{p}}|1 - c_{\mathfrak{p}}|^2$ of [Ch. 1, Lem. 3.2] for $m_{\mathfrak{p}} = (1 - t_{\mathfrak{p}})/(1 - c_{\mathfrak{p}}t_{\mathfrak{p}})$, which holds for every unimodular $c_{\mathfrak{p}}$. (Conjugating g replaces $c_{\mathfrak{p}}$ by $\overline{c_{\mathfrak{p}}}$, and $|1 - c_{\mathfrak{p}}| = |1 - \overline{c_{\mathfrak{p}}}|$.) $\square$

PROPOSITION 2.2 (The extended group bounds the type). *Put $\Sigma_\beta^\vee = \{u \in \mathbb{R} : su \in 2\pi\mathbb{Z} \text{ for all } s \in \Sigma_\beta\}$. Then $E \subseteq \Sigma_\beta^\vee$. Consequently:*

- (1) *if $\Sigma_\beta \neq \{0\}$, then M is not of type III_1 ;*
- (2) *if Σ_β is dense in $\mathbb{R}$, then M is of type III_0 ;*
- (3) *if $\Sigma_\beta = c\mathbb{Z}$ with $c > 0$, then M is of type III_{λ^k} for some integer $k \geq 1$, where $\lambda = e^{-2\pi/c}$, or of type III_0 .*

PROOF. Let $(\chi, s) \in \Xi_\beta^{\text{ext}}$, let g be as in Lemma 2.1, and put $G(v, h) = g(v) \overline{\chi(h)}$, a unimodular measurable function on Y . For $\mathfrak{a} \in J_S$,

$$G(v + \mathfrak{a}, \sigma_{\mathfrak{a}} h) = \chi(\sigma_{\mathfrak{a}}) \text{N}\mathfrak{a}^{i\beta s} g(v) \overline{\chi(\sigma_{\mathfrak{a}})} \overline{\chi(h)} = \text{N}\mathfrak{a}^{i\beta s} G(v, h),$$

so by (2.1) the $\mathbb{T}$ -valued cocycle $e^{-is\rho}$ on $\mathcal{R}$ is the coboundary $e^{-is\rho(y, y')} = G(y')/G(y)$. A coboundary has no essential value other than 1: if $|G - \gamma| < \delta$ on a set B_δ of positive measure, every pair $y, y' \in B_\delta$ has $|G(y')/G(y) - 1| < 2\delta$. On the other hand, if $u \in E$ then e^{-isu} is an

essential value of e^{-isu} , since $|e^{-is\rho} - e^{-isu}| \leq |s||\rho - u|$. Hence $e^{-isu} = 1$ for all $u \in E$ and all $s \in \Sigma_\beta$, i.e. $E \subseteq \Sigma_\beta^\vee$.

(1) If $\Sigma_\beta \neq \{0\}$ then $\Sigma_\beta^\vee \neq \mathbb{R}$, so $E \neq \mathbb{R}$. (2) If Σ_β is dense then $\Sigma_\beta^\vee = \{0\}$, so $E = \{0\}$ and $S(M) = \{0, 1\}$. (3) If $\Sigma_\beta = c\mathbb{Z}$ then $\Sigma_\beta^\vee = \frac{2\pi}{c}\mathbb{Z}$, and the closed subgroups of $\frac{2\pi}{c}\mathbb{Z}$ are $\{0\}$ and $\frac{2\pi k}{c}\mathbb{Z}$, $k \geq 1$; by (2.2) these give $S(M) = \{0, 1\}$ and $S(M) = \{0\} \cup (e^{-2\pi k/c})^\mathbb{Z}$ respectively. $\square$

REMARK 2.3 (Σ_β and the flow of weights). The proof shows more. With G as above, $\Psi(y, u) = G(y)e^{isu}$ is invariant under the Maharam extension $(y, u) \mapsto (y', u + \rho(y, y'))$ of $\mathcal{R}$ [87] and satisfies $\Psi(y, u + t) = e^{ist}\Psi(y, u)$; so every $s \in \Sigma_\beta$ is an eigenvalue of the flow of weights of M , i.e. Σ_β is contained in the point spectrum of that flow, which by Connes–Takesaki [31] is Connes’ invariant $T(M)$. We shall not need the converse inclusion and do not prove it; it holds when the flow of weights has pure point spectrum, and in that case Σ_β alone determines the type. Whether the flow of weights of M always has pure point spectrum is not known to us, and this is why Proposition 2.2 is only an upper bound. Lower bounds are obtained by exhibiting essential values directly, which is the purpose of the next lemma.

LEMMA 2.4 (Asymptotic pairs). *Suppose $\Xi_\beta = \{1\}$, so that $\nu = \pi_\beta \otimes \text{Haar}$ [Ch. 1, Thm. 3.6]. Let $u \in \mathbb{R}$ and let $(\mathbf{a}_j, \mathbf{b}_j)_{j \geq 1}$ be pairs of primes of S , all the primes $\mathbf{a}_j, \mathbf{b}_j$ distinct, such that*

$$\beta \log \frac{N\mathbf{a}_j}{N\mathbf{b}_j} \rightarrow u, \quad g_j := \sigma_{\mathbf{b}_j} \sigma_{\mathbf{a}_j}^{-1} \rightarrow 1 \text{ in } G_S, \quad \sum_j N\mathbf{a}_j^{-\beta} = \infty.$$

Then $u \in E$.

PROOF. Write $t_p = Np^{-\beta}$ and, for each j , $P_j = \{v_{\mathbf{a}_j} = 1, v_{\mathbf{b}_j} = 0\}$, $Q_j = \{v_{\mathbf{a}_j} = 0, v_{\mathbf{b}_j} = 1\}$, events of π_β -probability $p_j = (1 - t_{\mathbf{a}_j})(1 - t_{\mathbf{b}_j})t_{\mathbf{a}_j}$ and $q_j = (1 - t_{\mathbf{a}_j})(1 - t_{\mathbf{b}_j})t_{\mathbf{b}_j}$. The events $P_j \cup Q_j$ are independent and $\sum_j p_j = \infty$; since $t_{\mathbf{b}_j}/t_{\mathbf{a}_j} \rightarrow e^u$, there is $\theta > 0$ with $p_j \geq \theta(p_j + q_j)$ for all large j . Fix $\varepsilon > 0$ and j_0 , and let D be the set of (v, h) such that some $j \geq j_0$ has $v \in P_j \cup Q_j$ and the least such j , denoted $j(v)$, has $v \in P_j$. Define

$$T(v, h) = (v - e_{\mathbf{a}_j} + e_{\mathbf{b}_j}, g_j h), \quad j = j(v),$$

on D . The move changes only the coordinates $\mathbf{a}_j, \mathbf{b}_j$, turning P_j into Q_j , so T is injective with image the set where the least $j \geq j_0$ with $v \in P_j \cup Q_j$ has $v \in Q_j$; T is a partial transformation of $\mathcal{R}$, being the map $\mathbf{b}_j \mathbf{a}_j^{-1}$ on each piece $D_j = \{j(v) = j\}$, and by (2.1), $\rho(y, Ty) = \beta \log(N\mathbf{a}_j/N\mathbf{b}_j)$ lies in $(u - \varepsilon, u + \varepsilon)$ once j_0 is large. Since the D_j are disjoint and $\sum_j (p_j + q_j) = \infty$,

$$\nu(D) = \sum_{j \geq j_0} p_j \prod_{j_0 \leq i < j} (1 - p_i - q_i) \geq \theta \sum_{j \geq j_0} (p_j + q_j) \prod_{j_0 \leq i < j} (1 - p_i - q_i) = \theta.$$

Now let $\nu(B) > 0$ and $0 < \eta < 1$. The sets measurable with respect to finitely many coordinates of V together with the G_S -coordinate generate the σ -algebra of Y , so there are a finite $F \subset S$ and $A \in \sigma(v_p : p \in F) \otimes \mathcal{B}(G_S)$ with $\nu(A \triangle B) < \eta \nu(B)$. Translation in the second coordinate preserves ν and is continuous in $L^1(\nu)$, so $\nu(A \triangle (1 \times g)^{-1}A) \rightarrow 0$ as $g \rightarrow 1$. Choose j_0 so large that $\mathbf{a}_j, \mathbf{b}_j \notin F$, $|\rho - u| < \varepsilon$ on D_j , and $\nu(A \triangle (1 \times g_j)^{-1}A) < \eta \nu(B)$ for all $j \geq j_0$. For $(v, h) \in D_j$ the point $T(v, h)$ agrees with $(v, g_j h)$ on the coordinates in F , hence $D_j \cap A \cap (1 \times g_j)^{-1}A \subseteq A \cap T^{-1}A$; the set D_j depends only on the coordinates $\mathbf{a}_i, \mathbf{b}_i$, $i \geq j_0$, so it is independent of $A \cap (1 \times g_j)^{-1}A$, and

$$\nu(A \cap T^{-1}A) \geq \sum_{j \geq j_0} \nu(D_j) \nu(A \cap (1 \times g_j)^{-1}A) \geq \nu(D) (\nu(A) - \eta \nu(B)) \geq \theta(1 - 2\eta) \nu(B).$$

Finally $d(\nu \circ T^{-1})/d\nu = e^{-\rho \circ T^{-1}} \leq e^{|u| + \varepsilon}$ on $T(D)$, so

$$\nu(B \cap T^{-1}B) \geq \nu(A \cap T^{-1}A) - \nu(A \triangle B) - \nu(T^{-1}(A \triangle B)) \geq (\theta(1 - 2\eta) - \eta - e^{|u| + \varepsilon} \eta) \nu(B) > 0$$

for η small. As B and ε were arbitrary, $u \in E$. $\square$

The lemma is the asymptotic-ratio-set argument of Araki and Woods [5] with the G_S -coordinate carried along; the hypothesis $g_j \rightarrow 1$ is what makes that coordinate harmless. The restriction to $\Xi_\beta = \{1\}$ is one of convenience: it is the case needed below, and it is the case in which ν has the product form used in the last display.

3. The classical systems

PROPOSITION 3.1 (The slices for $S = P_K$). *Let $S = P_K$ and $0 < \beta \leq 1$. Then $\Xi_\beta^{\text{ext}}(K, S) = \{(1, 0)\}$; in particular $\Sigma_\beta = \{0\}$.*

PROOF. For (χ, s) put $\tau = \beta s$; since $|1 - z|^2 = 2 - 2\operatorname{Re} z$ for $|z| = 1$,

$$(3.1) \quad \sum_{N\mathfrak{p} \leq x} N\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}}) N\mathfrak{p}^{i\tau}|^2 = 2 \sum_{N\mathfrak{p} \leq x} N\mathfrak{p}^{-\beta} - 2\operatorname{Re} \sum_{N\mathfrak{p} \leq x} \chi(\sigma_{\mathfrak{p}}) N\mathfrak{p}^{-\beta+i\tau},$$

and it suffices to show that for $(\chi, \tau) \neq (1, 0)$ the second sum is o of the first. A character of $G_{P_K} = \operatorname{Gal}(K^{ab}/K)$ is a ray class character, and by [Ch. 1, Lem. 3.8] $\chi(\sigma_{\mathfrak{p}}) = \overline{\chi(\operatorname{Frob}_{\mathfrak{p}})}$ off the conductor. For $\beta = 1$ the series $\sum_{\mathfrak{p}} \chi(\sigma_{\mathfrak{p}}) N\mathfrak{p}^{-1+i\tau}$ differs from $\log L(1 - i\tau, \bar{\chi})$ by an absolutely convergent sum over prime powers, and $L(1 - i\tau, \bar{\chi})$ is finite — the only pole of a Hecke L -function on $\operatorname{Re} = 1$ being the simple pole of ζ_K at $s = 1$, excluded — and nonzero by the theorem of Hecke and Landau [56, 79]. So the second sum is $O(1)$ while $\sum_{N\mathfrak{p} \leq x} N\mathfrak{p}^{-1} \rightarrow \infty$. For $0 < \beta < 1$, partial summation from the prime ideal theorem in ray classes in the standard zero-free region [79] gives $O_A(x^{1-\beta} \log^{-A} x)$ for the second sum against $\sum_{N\mathfrak{p} \leq x} N\mathfrak{p}^{-\beta} \sim x^{1-\beta}/((1-\beta) \log x)$ for the first. $\square$

THEOREM 3.2 (The classical systems are of type III_1). *For $S = P_K$ and $0 < \beta \leq 1$, M is of type III_1 .*

For $K = \mathbb{Q}$ this is due to Bost and Connes [10], and in general to Laca, Larsen and Neshveyev [75]; we include a proof because it is the model for the computation of §4.

PROOF. $\Xi_\beta = \{1\}$ by [Ch. 1, Cor. 3.10], so Lemma 2.4 applies, and we must produce, for each $u \in \mathbb{R}$, pairs $(\mathfrak{a}_j, \mathfrak{b}_j)$ with $\beta \log(N\mathfrak{a}_j/N\mathfrak{b}_j) \rightarrow u$, $\sigma_{\mathfrak{b}_j} \sigma_{\mathfrak{a}_j}^{-1} \rightarrow 1$ in $\operatorname{Gal}(K^{ab}/K)$ and $\sum_j N\mathfrak{a}_j^{-\beta} = \infty$. Fix moduli $\mathfrak{m}_1 \mid \mathfrak{m}_2 \mid \cdots$ whose ray class fields exhaust K^{ab} [97, Ch. VI]; for primes $\mathfrak{a}, \mathfrak{b}$ in the same ray class modulo $\mathfrak{m}_k$ the element $\sigma_{\mathfrak{b}} \sigma_{\mathfrak{a}}^{-1} = \operatorname{Frob}_{\mathfrak{a}} \operatorname{Frob}_{\mathfrak{b}}^{-1}$ is trivial on the ray class field of $\mathfrak{m}_k$, so pairs chosen in a common class modulo $\mathfrak{m}_k$ with $k \rightarrow \infty$ satisfy $\sigma_{\mathfrak{b}_j} \sigma_{\mathfrak{a}_j}^{-1} \rightarrow 1$. By the prime ideal theorem in ray classes [79], for fixed $\mathfrak{m}$ and each class $\mathfrak{c}$ the number of primes $\mathfrak{p} \in \mathfrak{c}$ with $N\mathfrak{p} \in (y, y(1+\delta))$ is $\asymp_{\mathfrak{m}, \delta} y/\log y$ for $y \geq y_0(\mathfrak{m}, \delta)$. Let $r = e^{u/\beta}$. Use only every N_0 -th n , with N_0 so large that the ranges $[2^n, 2^{n+1}) \cup r^{-1}[2^n, 2^{n+1})$ of the blocks used are pairwise disjoint. Divide $[2^n, 2^{n+1})$ into intervals $I = (y, y(1+\delta_n)]$ with $\delta_n \downarrow 0$ slowly, and for each I and each class $\mathfrak{c}$ modulo $\mathfrak{m}_{k_n}$, with $k_n \rightarrow \infty$ slowly enough that $2^n \geq y_0(\mathfrak{m}_{k_n}, \delta_n)$, pair off primes of $\mathfrak{c}$ with norm in I against primes of $\mathfrak{c}$ with norm in $r^{-1}I$, injectively and disjointly from earlier pairs; the pairs of different blocks are then disjoint. Each pair has $|\beta \log(N\mathfrak{a}/N\mathfrak{b}) - u| \leq 2\beta\delta_n$, and the pairs with $N\mathfrak{a} \in [2^n, 2^{n+1})$ number $\gg 2^n/n$ (for large n , a positive proportion of the primes in $[2^n, 2^{n+1})$ is used), so they contribute $\gg 2^{n(1-\beta)}/n \geq 1/n$ to $\sum_j N\mathfrak{a}_j^{-\beta}$, which diverges along the blocks used. Lemma 2.4 gives $u \in E$ for every u , so $E = \mathbb{R}$ and M is of type III_1 by (2.2). $\square$

REMARK 3.3 (Two slices of one analytic input). At $\beta = 1$ and $S = P_K$ the $s = 0$ slice of Ξ_β^{ext} is the uniqueness half of the classical theorem, equivalent by [Ch. 1, Rem. 3.11] to $L(1, \chi) \neq 0$; the slices $s \neq 0$ are governed, by Proposition 3.1, by $L(1 - i\tau, \chi) \neq 0$, and their triviality is the necessary condition for type III_1 of Proposition 2.2(1). The classification of the phase and the

type of the factor are thus the two coordinate slices of one obstruction group, and for the full prime set both are governed by the non-vanishing of Hecke L -functions on the line $\text{Re} = 1$. That the $s \neq 0$ slices are trivial is, by Remark 2.3, the statement that the flow of weights has no nonzero eigenvalue; that M is III_1 , i.e. that the flow is trivial, needed the additional input of Lemma 2.4.

4. Two incommensurable norms do not suffice

LEMMA 4.1 (No exact commensurability). *Let S be infinite. Then the numbers $\log N\mathfrak{p}$, $\mathfrak{p} \in S$, are never pairwise commensurable.*

PROOF. $N\mathfrak{p} = p^{f_{\mathfrak{p}}}$ with $f_{\mathfrak{p}} \leq [K : \mathbb{Q}]$, and only finitely many primes of K lie above a given p , so an infinite S meets infinitely many rational primes; and $\log p_1 / \log p_2 \notin \mathbb{Q}$ for distinct rational primes since $p_1^m = p_2^n$ forces $m = n = 0$. $\square$

So exact commensurability, which would put all the values of the Radon–Nikodym cocycle (2.1) in a lattice, never occurs for infinite S , and any failure of type III_1 must come from *approximate* commensurability. It does.

THEOREM 4.2 (Exact determination of a III_{e-1} system). *Let $K = \mathbb{Q}$, $\beta = 1$, $\|\cdot\|$ the distance to the nearest integer, and*

$$(4.1) \quad S = \{3\} \cup \{\ell \text{ prime} : \ell \geq 16, \|\log \ell\| \leq \varepsilon(\ell)\}, \quad \varepsilon(\ell) = (\log \log \ell)^{-1/2}.$$

Then $\zeta_S(1) = \infty$, S contains primes with irrationally related logarithms, and

$$\Xi_1^{\text{ext}}(\mathbb{Q}, S) = \{1\} \times 2\pi\mathbb{Z}, \quad \Sigma_1 = 2\pi\mathbb{Z}.$$

Hence $\Xi_1 = \{1\}$ — the KMS_1 state is unique — while M is of pure type III_{e-1} ; both III_1 and III_0 are excluded.

The proof occupies the rest of this section. The mechanism is a band decomposition: because (4.1) localizes $\log \ell$ near an *integer*, the rounded logarithm is constant on each band and the exponential sums reduce to geometric series.

LEMMA 4.3 (Bands). *Put $\varepsilon_n = (\log n)^{-1/2}$ and $n_0 = 55$, so $\varepsilon_n < \frac{1}{2}$ for $n > n_0$. For $n > n_0$ let $B_n = \{\ell \in S : |\log \ell - n| \leq \varepsilon(\ell)\}$ and $S_0 = \{\ell \in S : \ell \leq e^{n_0+1}\}$. Then:*

- (1) $S = S_0 \sqcup \bigsqcup_{n > n_0} B_n$ and $\sum_{\ell \in S_0} \ell^{-1} = O(1)$;
- (2) for $\ell \in B_n$, $\log \ell = n + \delta_{\ell}$ with $|\delta_{\ell}| \leq \varepsilon_n(1 + O(n^{-1}))$, so $\text{round}(\log \ell) = n$ is constant on B_n ;
- (3) $w_n := \sum_{\ell \in B_n} \ell^{-1} = \frac{2\varepsilon_n}{n}(1 + o(1)) = \frac{2}{n\sqrt{\log n}}(1 + o(1))$, eventually decreasing to 0 up to an absolutely summable perturbation;
- (4) $W(x) := \sum_{\ell \in S, \ell \leq x} \ell^{-1} = 4\sqrt{\log \log x} + O(1) \rightarrow \infty$.

PROOF. (1) For $n > n_0$ the bands lie in the pairwise disjoint intervals $[e^{n-1/2}, e^{n+1/2}]$; for $\ell \leq e^{n_0+1}$ one has $\varepsilon(\ell) > 1/2$ and the condition may hold for every ℓ , so S_0 can be all primes up to e^{n_0+1} , of Mertens sum $O(1)$. (2) $|\log \ell - n| < \frac{1}{2}$ gives the rounding, and $\log \log \ell = \log n + O(\varepsilon_n/n)$ gives $\varepsilon(\ell) = \varepsilon_n(1 + O(n^{-1}))$. (3) By (2), B_n is the set of primes in $[e^{n-\varepsilon_n^-}, e^{n+\varepsilon_n^+}]$; Mertens with error $O(e^{-c\sqrt{\log y}})$ gives $w_n = \log \frac{n+\varepsilon_n^+}{n-\varepsilon_n^-} + O(e^{-c\sqrt{n}})$, which is the stated asymptotic, with main term $2n^{-1}(\log n)^{-1/2}$ strictly decreasing and both errors absolutely summable. (4) $\sum_{n \leq N} 2n^{-1}(\log n)^{-1/2} = 4\sqrt{\log N} + O(1)$ by comparison with the integral; put $N = \lfloor \log x \rfloor$. $\square$

LEMMA 4.4 (Linearization). *Let χ factor through $(\mathbb{Z}/M\mathbb{Z})^\times$ with Dirichlet character $\tilde{\chi}$, so that $\chi(\sigma_{\ell}) = \tilde{\chi}(\ell)$ for $\ell \nmid M$ by [Ch. 1, Lem. 2.10(1)], and let F be the set of primes dividing M .*

For $s \in \mathbb{R}$ and $N = \lfloor \log x \rfloor$,

(4.2)

$$\sum_{\ell \in S, \ell \leq x} \frac{|1 - \chi(\sigma_\ell) \ell^{is}|^2}{\ell} = 2W(x) - 2 \operatorname{Re} \Lambda(x) + O_M(1), \quad \Lambda(x) = \sum_{n \leq N} e^{isn} c_n + O\left(|s| \sum_{n \leq N} \varepsilon_n w_n\right),$$

where $c_n = \sum_{\ell \in B_n \setminus F} \tilde{\chi}(\ell) \ell^{-1}$. Moreover

$$(4.3) \quad |s| \sum_{n \leq N} \varepsilon_n w_n = 2|s| \log \log N + O(|s|) = o(W(x)).$$

PROOF. $|1 - z|^2 = 2 - 2 \operatorname{Re} z$ for $|z| = 1$ gives the first identity. For $\ell \in B_n$, $\ell^{is} = e^{is(n+\delta_\ell)} = e^{isn} + O(|s| \varepsilon_n)$ by $|e^{i\theta} - 1| \leq |\theta|$, and summing against ℓ^{-1} over B_n gives the second. For (4.3), $\varepsilon_n w_n = 2\varepsilon_n^2/n(1+o(1)) = 2/(n \log n)(1+o(1))$ and $\sum_{n \leq N} (n \log n)^{-1} = \log \log N + O(1)$; since $W(x) = 4\sqrt{\log N} + O(1)$ and $\log \log N = o(\sqrt{\log N})$, the claim follows. In the variable x this reads

$$(4.4) \quad |s| \sum_{n \leq N} \varepsilon_n w_n \asymp 2|s| \log \log \log x = o\left(\sqrt{\log \log x}\right) = o(W(x)),$$

the comparison being with $\sqrt{\log \log x}$ and not with $\log \log x$, since $W(x) = 4\sqrt{\log \log x} + O(1)$ by Lemma 4.3(4). $\square$

REMARK 4.5 (The one delicate point). Estimate (4.3) is where care is needed: the error from linearizing ℓ^{is} across a band *diverges*, at rate $\log \log \log x$, and is absorbed only because $W(x)$ diverges faster, at rate $\sqrt{\log \log x}$. The exponent $-\frac{1}{2}$ in $\varepsilon(\ell) = (\log \log \ell)^{-1/2}$ is exactly what makes the convergence in Proposition 4.8 and this absorption hold simultaneously: an exponent $-\frac{1}{3}$ would break the former and -1 the latter.

LEMMA 4.6 (Cancellation in bands). *If $\tilde{\chi}$ is non-principal modulo M then $|c_n| \ll_M e^{-c\sqrt{n}}$, so $\sum_n |c_n| < \infty$.*

PROOF. Siegel–Walfisz gives $\sum_{p \leq y, p \equiv a(M)} p^{-1} = \varphi(M)^{-1} \log \log y + C(M, a) + O_M(e^{-c\sqrt{\log y}})$. Applying this at the two endpoints of B_n , whose logarithms are $n + O(1)$, the constants $C(M, a)$ cancel and $\sum_{\ell \in B_n, \ell \equiv a(M)} \ell^{-1} = w_n / \varphi(M) + O_M(e^{-c\sqrt{n}})$. Hence $c_n = \frac{w_n}{\varphi(M)} \sum_a^* \tilde{\chi}(a) + O_M(\varphi(M) e^{-c\sqrt{n}})$, and the character sum vanishes. $\square$

PROPOSITION 4.7 (Non-trivial characters). *If $\chi \neq 1$ then $\sum_{\ell \in S} \ell^{-1} |1 - \chi(\sigma_\ell) \ell^{is}|^2 = \infty$ for every $s \in \mathbb{R}$. In particular $\Xi_1^{\text{ext}} \subseteq \{1\} \times \mathbb{R}$ and, taking $s = 0$, $\Xi_1 = \{1\}$.*

PROOF. First, $\sigma(I_S)$ is dense in G_S : S contains all primes up to e^{n_0+1} by Lemma 4.3(1) and, by Lemma 4.6 applied to the principal character, meets every reduced class modulo M in every large band; so the classes of $S \setminus F$ generate $(\mathbb{Z}/M\mathbb{Z})^\times$ and $\chi \mapsto \tilde{\chi}$ is injective, whence $\tilde{\chi}$ is non-principal. Now combine (4.2), Lemma 4.6 and (4.3): $\Lambda(x) = O(\sum_n |c_n|) + o(W(x)) = o(W(x))$, so the left side of (4.2) is $2W(x)(1+o(1)) \rightarrow \infty$. $\square$

PROPOSITION 4.8 (The trivial character). *$\sum_{\ell \in S} \ell^{-1} |1 - \ell^{is}|^2$ is finite for $s \in 2\pi\mathbb{Z}$ and infinite for $s \notin 2\pi\mathbb{Z}$.*

PROOF. Let $s = 2\pi m$. For $\ell \in B_n$, $\log \ell = n + \delta_\ell$ with $n \in \mathbb{Z}$, so $\ell^{is} = e^{2\pi i m \delta_\ell}$ and $|1 - \ell^{is}|^2 = 4 \sin^2(\pi m \delta_\ell) \leq 16\pi^2 m^2 \varepsilon_n^2$; hence the sum is $\ll m^2 \sum_n \varepsilon_n^2 w_n \ll m^2 \sum_n n^{-1} (\log n)^{-3/2} < \infty$.

Let $s \notin 2\pi\mathbb{Z}$. Here $\tilde{\chi}$ is principal, so $c_n = w_n + O_M(1) \mathbf{1}_{n \leq n_F}$. Since w_n is monotone to 0 up to an absolutely summable perturbation (Lemma 4.3(3)) and $|\sum_{n \leq N} e^{isn}| \leq |\sin(s/2)|^{-1} < \infty$, Dirichlet's test gives convergence of $\sum_n e^{isn} w_n$. With (4.3), $\Lambda(x) = O_s(1) + o(W(x)) = o(W(x))$ and the sum diverges as before. $\square$

LEMMA 4.9 (Matched pairs in consecutive bands). *There are pairs (a_j, b_j) of primes of S with $a_j \in B_{n_j+1}$, $b_j \in B_{n_j}$, all the a_j, b_j distinct, such that $\log(a_j/b_j) \rightarrow 1$, $\sigma_{b_j}\sigma_{a_j}^{-1} \rightarrow 1$ in $G_S = \widehat{\mathbb{Z}}_S^\times$, and $\sum_j a_j^{-1} = \infty$.*

PROOF. For $\ell \in B_n$, $\log \ell = n + \delta_\ell$ with $|\delta_\ell| \leq 2\varepsilon_n$, so any $a \in B_{n+1}$, $b \in B_n$ satisfy $|\log(a/b) - 1| \leq 4\varepsilon_n \rightarrow 0$. Put $k_n = \lfloor \log \log n \rfloor$ and $M_n = \prod_{\ell' \leq k_n} \ell'^{k_n}$, so that $\log M_n \leq 2k_n^2 = o(\log n)$. By Siegel–Walfisz, uniformly for moduli $q \leq (\log y)^A$ one has $\pi(y; q, a) = \text{li}(y)/\varphi(q) + O_A(ye^{-c\sqrt{\log y}})$; applied at the endpoints $e^{n \pm \varepsilon_n^\pm}$ of B_n with $q = M_n \leq n$, and since $|B_n| \asymp \varepsilon_n e^n/n$ while the error is $O(e^{n-c\sqrt{n}})$, every reduced class a modulo M_n contains at least $\frac{1}{2}|B_n|/\varphi(M_n)$ primes of B_n and at least as many primes of B_{n+1} , for all large n . Pairing, injectively within each class, primes of B_{n+1} with primes of B_n in the same class gives at least $\frac{1}{2}|B_n|$ pairs $(a, b) \in B_{n+1} \times B_n$ with $a \equiv b \pmod{M_n}$; pairs from different n are disjoint because the bands are.

For a fixed prime $\ell' \in S$ and $k \geq 1$ we have $\ell'^k \mid M_n$ for all large n , and then the ℓ' -component of $\sigma_b\sigma_a^{-1}$, which is b/a for $\ell' \notin \{a, b\}$ [Ch. 1, Lem. 2.10(1)], is $\equiv 1 \pmod{\ell'^k}$; so every coordinate of $\sigma_{b_j}\sigma_{a_j}^{-1}$ tends to 1, which is convergence to 1 in the product topology of $\widehat{\mathbb{Z}}_S^\times$. Finally, the pairs from band n contribute

$$\sum_a a^{-1} \geq \frac{1}{2}|B_n|e^{-(n+2)} \geq \frac{1}{2e^3} w_n, \quad \text{since } w_n = \sum_{\ell \in B_n} \ell^{-1} \leq |B_n|e^{-(n-1)},$$

and $\sum_n w_n = \infty$ by Lemma 4.3(4). $\square$

PROOF OF THEOREM 4.2. $\zeta_S(1) = \infty$ is Lemma 4.3(4). S is infinite, so Lemma 4.1 supplies two primes with irrationally related logarithms. Propositions 4.7 and 4.8 give $\Xi_1^{\text{ext}} = \{1\} \times 2\pi\mathbb{Z}$, hence $\Xi_1 = \{1\}$ and $\Sigma_1 = 2\pi\mathbb{Z}$. Proposition 2.2(3) with $c = 2\pi$ gives $E \subseteq \mathbb{Z}$: M is of type $\text{III}_{e^{-k}}$ for some $k \geq 1$ or of type III_0 , and in particular not of type III_1 . Since $\Xi_1 = \{1\}$, Lemma 2.4 applies, and Lemma 4.9 supplies pairs with $\beta \log(a_j/b_j) \rightarrow 1$ ($\beta = 1$), $\sigma_{b_j}\sigma_{a_j}^{-1} \rightarrow 1$ and $\sum a_j^{-1} = \infty$; hence $1 \in E$, so $E = \mathbb{Z}$, $S(M) = \{0\} \cup e^\mathbb{Z}$ by (2.2), and M is of type $\text{III}_{e^{-1}}$. $\square$

REMARK 4.10 (What the proof does not use). Only Mertens, Siegel–Walfisz at a *fixed* modulus, Abel summation and Dirichlet’s test are needed. Because $\text{round}(\log \ell)$ is constant on each band, the exponential sum in (4.2) is over the *integers* n , not over primes: its partial sums are geometric and bounded by $|\sin(s/2)|^{-1}$, so Dirichlet’s test handles all $s \notin 2\pi\mathbb{Z}$ at once, with no distinction between rational and irrational $s/2\pi$ and no discrepancy estimate. No Erdős–Turán inequality, no Vinogradov-type bound for $\sum_\ell \ell^{-1+is}$, and no uniformity in the modulus is required. The bands are not short intervals in the difficult sense: B_n has multiplicative length $e^{2\varepsilon_n}$ with $\varepsilon_n \asymp (\log \log x)^{-1/2}$, so $|B_n| \gg x/(\log x)(\log \log x)^{1/2}$, far above any threshold at which primes in progressions become hard.

REMARK 4.11 (Numerical check). The asymptotic regime of (4.1) begins at $\varepsilon_n < \frac{1}{2}$, i.e. $\ell > e^{e^4} \approx 10^{23}$, and is beyond direct computation. The mechanics of Lemmas 4.3(3) and 4.6 were instead verified with a fixed band half-width $\varepsilon = 0.05$, for which the same computations apply verbatim: over primes up to $6 \cdot 10^7$ and bands $8 \leq n \leq 17$, the ratio $w_n/\log \frac{n+\varepsilon}{n-\varepsilon}$ lies in $[0.998, 1.003]$ for $n \geq 13$, and for the non-principal character modulo 4 the ratio $|c_n|/w_n$ falls from $8.6 \cdot 10^{-2}$ at $n = 9$ to $2.1 \cdot 10^{-4}$ at $n = 17$, consistent with Lemma 4.6.

COROLLARY 4.12 (Independence of the two classifications). *Uniqueness of the KMS_β state and type III_1 are logically independent: Theorem 4.2 has the first and fails the second, while Theorem 3.2 together with [Ch. 1, Cor. 3.10] has both. Equivalently, Ξ_β^{ext} can be trivial on the axis $s = 0$ and nontrivial off it.*

5. The spectrum of transitions

We now turn to the horizontal direction: which chains $\{\Xi_\beta\}_{\beta>0}$ occur. By [Ch. 1, Lem. 3.5] the chain is non-decreasing, and by [Ch. 1, Prop. 4.16] any finite chain is realizable. We realize arbitrary countable chains.

THEOREM 5.1 (Realization). *Let $(\beta_n)_{n \geq 1} \subset (0, 1)$ be any sequence. There is $S = Q \sqcup P_0 \sqcup \bigsqcup_n R_n \subset P_{\mathbb{Q}}$ with*

- (1) $\beta_c(S) = 1$ and $\zeta_S(\beta) = \infty$ for $0 < \beta \leq 1$;
- (2) $\sigma(I_S)$ dense in G_S ;
- (3) with q_n the n -th element of Q and χ_n the quadratic character at that component,

$$\Xi_\beta(\mathbb{Q}, S) = \langle \chi_n : \beta_n < \beta \rangle \cong \bigoplus_{n: \beta_n < \beta} \mathbb{Z}/2\mathbb{Z} \quad (0 < \beta \leq 1),$$

and $\Xi_\beta = \widehat{G_S}$ for $\beta > 1$. In particular the jump set is $\{\beta_n\} \cup \{1\}$.

The construction proceeds in blocks; the block scheme is what makes infinitely many congruence conditions manageable.

LEMMA 5.2 (Independence of the quadratic conditions). *Let $q_1 < \dots < q_j$ be distinct odd primes, $q^* = (-1)^{(q-1)/2}q$, $M \geq 1$, and $E = \mathbb{Q}(\zeta_M)(\sqrt{q_1^*}, \dots, \sqrt{q_j^*})$. Then $[E : \mathbb{Q}] = \varphi(M) \cdot 2^{\#\{i \leq j : q_i \nmid M\}}$, and for any $a \in (\mathbb{Z}/M\mathbb{Z})^\times$ and signs ϵ_i compatible with a at the q_i dividing M , the primes ℓ with $\ell \equiv a \pmod{M}$ and $(\frac{\ell}{q_i}) = \epsilon_i$ have natural density $[E : \mathbb{Q}]^{-1} > 0$.*

PROOF. The q_i^* are multiplicatively independent modulo squares, so $[\mathbb{Q}(\sqrt{q_1^*}, \dots, \sqrt{q_j^*}) : \mathbb{Q}] = 2^j$ by Kummer theory. $\mathbb{Q}(\sqrt{q_i^*})$ is the unique quadratic subfield of $\mathbb{Q}(\zeta_{q_i})$, so it lies in $\mathbb{Q}(\zeta_M)$ iff $q_i \mid M$; the degree formula follows. Quadratic reciprocity turns the sign conditions into splitting conditions, and Chebotarev applies. $\square$

DEFINITION 5.3 (Block scheme). Choose $z_1 < z_2 < \dots$ recursively, z_{j+1} as large as needed in terms of z_j and j , and put $B_j = [z_j, z_{j+1})$. Let q_j be a prime in B_j with $(\frac{q_j}{q_i}) = +1$ for all $i < j$ (such primes have positive density by Lemma 5.2, so they exist in B_j once z_{j+1} is large), and $Q = \{q_j\}$. For $j \geq 1$ set

$$C_{0,j} = \{\ell : (\frac{\ell}{q_i}) = +1 \ \forall i \leq j\}, \quad C_{n,j} = \{\ell : (\frac{\ell}{q_n}) = -1, (\frac{\ell}{q_i}) = +1 \ \forall i \leq j, i \neq n\},$$

each of density 2^{-j} by Lemma 5.2, and

$$P_0 = \bigsqcup_{j \geq 1} (C_{0,j} \cap B_j) \setminus Q, \quad R_n = \bigsqcup_{j \geq n} \rho_{n,j}, \quad \rho_{n,j} \subseteq C_{n,j} \cap B_j \text{ finite.}$$

The essential property is that for $\ell \in S \cap [z_j, \infty)$ the symbols $(\frac{\ell}{q_i})$, $i \leq j$, are determined by which block-class ℓ lies in.

LEMMA 5.4 (The pieces). *The z_j and $\rho_{n,j}$ may be chosen so that:*

- (1) $\sum_{\ell \in P_0 \cap B_j} \ell^{-1} \geq 1$ for every j , whence $\sum_{P_0} \ell^{-1} = \infty$ and $\beta_c(S) = 1$; and for every M and every $a \in (\mathbb{Z}/M\mathbb{Z})^\times$ with $(\frac{a}{q_i}) = 1$ whenever $q_i \mid M$, $\sum_{\ell \in P_0, \ell \equiv a \pmod{M}} \ell^{-1} = \infty$;
- (2) $|R_n \cap [1, x]| \asymp_n x^{\beta_n}$, hence by Abel summation $\sum_{\ell \in R_n} \ell^{-\beta} < \infty \iff \beta > \beta_n$, with divergence at the endpoint;
- (3) $q_n \geq z_n$ with $\log \log z_n \geq 2^n$, so $\sum_n q_n^{-\beta} < \infty$ for every $\beta > 0$ and Q is negligible in every convergence question.

PROOF. (1) $C_{0,j}$ has density 2^{-j} , so $\sum_{\ell \in C_{0,j}, \ell \leq x} \ell^{-1} = 2^{-j} \log \log x + O_j(1)$; choose z_{j+1} with $2^{-j}(\log \log z_{j+1} - \log \log z_j) \geq 2$. For the refined statement, for j beyond the largest index with $q_i \mid M$ the two conditions are compatible and Lemma 5.2 gives positive density, so enlarging z_{j+1} further (finitely many extra requirements per block) gives block sums $\geq \varphi(M)^{-1}$. (2) $C_{n,j}$ has positive density, so it meets $[y, 2y]$ for $y \geq y_0(j)$; take $\rho_{n,j}$ to contain one prime of $C_{n,j}$ in each $[m^{1/\beta_n}, 2m^{1/\beta_n}]$ contained in B_j . Abel summation with $A(x) = |R_n \cap [1, x]| \asymp x^{\beta_n}$ and $A(t) = 0$ for $t < z_n$ gives

$$\sum_{\ell \in R_n, \ell \leq x} \ell^{-\beta} = A(x)x^{-\beta} + \beta \int_{z_n}^x A(t)t^{-\beta-1} dt \asymp x^{\beta_n-\beta} + \beta \int_{z_n}^x t^{\beta_n-\beta-1} dt.$$

The lower limit contributes the constant $z_n^{\beta_n-\beta}$; the integral converges as $x \rightarrow \infty$ iff $\beta > \beta_n$, equals $\beta_n \log(x/z_n) \rightarrow \infty$ at $\beta = \beta_n$, and diverges as a power for $\beta < \beta_n$. (3) $q_n \geq \exp \exp(2^n)$ makes $q_n^{-\beta} \leq \exp(-\beta \exp(2^n))$. $\square$

PROOF OF THEOREM 5.1. (1) is Lemma 5.4(1). For (2), fix a modulus M supported on a finite $F \subset S$, let $Q_F = Q \cap \{q : q \mid M\}$ and $K_M = \{a : \left(\frac{a}{q}\right) = 1 \ \forall q \in Q_F\}$, a subgroup of index $2^{|Q_F|}$ with $(\mathbb{Z}/M\mathbb{Z})^\times / K_M \cong (\mathbb{Z}/2\mathbb{Z})^{Q_F}$. By Lemma 5.4(1) every $a \in K_M$ is realized by infinitely many $\ell \in P_0$, so $K_M \subseteq \langle \ell \bmod M \rangle$; and for $q_n \in Q_F$ any large $\ell \in R_n$ outside F maps to the basis vector e_n of the quotient. Hence the classes generate $(\mathbb{Z}/M\mathbb{Z})^\times$.

For (3), let $\chi \neq 1$ factor through $(\mathbb{Z}/M\mathbb{Z})^\times$. If $\tilde{\chi}|_{K_M} \neq 1$, pick $a \in K_M$ with $\tilde{\chi}(a) \neq 1$; by Lemma 5.4(1) $\sum_{\ell \in P_0, \ell \equiv a} \ell^{-1} = \infty$ and every such ℓ detects χ , so $\chi \notin \Xi_\beta$ for $\beta \leq 1$. If $\tilde{\chi}|_{K_M} = 1$ then $\tilde{\chi} = \prod_{n \in T} \tilde{\chi}_{q_n}$ for a unique T , and by (2) and injectivity $\chi = \chi_T := \prod_{n \in T} \chi_n$. For $\ell \in S$ with $\ell \geq z_{\max T}$ the block property gives $\chi_T(\sigma_\ell) = +1$ on P_0 and on R_m with $m \notin T$, and -1 on R_m with $m \in T$; so, Q being negligible and $[1, z_{\max T})$ finite,

$$\sum_{\ell \in D_{\chi_T}} \ell^{-\beta} < \infty \iff \sum_{m \in T} \sum_{\ell \in R_m} \ell^{-\beta} < \infty \iff \beta > \max_{m \in T} \beta_m,$$

by Lemma 5.4(2). This is the stated formula; for $\beta > 1$, $\zeta_S(\beta) < \infty$ and [Ch. 1, Lem. 3.5] gives $\Xi_\beta = \widehat{G_S}$. $\square$

REMARK 5.5 (Numerical check). Three ingredients were verified computationally on a scaled-down model with $Q = \{7, 11, 13, 17\}$ and primes up to $2 \cdot 10^6$: the $2^4 = 16$ joint sign patterns of Lemma 5.2 occur with densities in $[0.0621, 0.0629]$ against the predicted $1/16 = 0.0625$; the detecting set of χ_T equals $\bigsqcup_{m \in T} R_m$ exactly, for every subset T tested; and a set thinned to counting function $\asymp x^b$ has its β -weighted sums collapsing across $\beta = b$ as Lemma 5.4(2) requires.

6. Cantor transition loci, and the unattainable staircase

DEFINITION 6.1. The *jump set* is $J(S) = \{\beta_\chi : \chi \neq 1\}$, where β_χ is the convergence exponent of the detecting set $D_\chi = \{\mathfrak{p} \in S : \chi(\sigma_\mathfrak{p}) \neq 1\}$; the *transition locus* $\mathcal{L}(S)$ is the set of β near which $\beta \mapsto \Xi_\beta$ is not constant.

COROLLARY 6.2 (Dense and Cantor loci). *Taking (β_n) an enumeration of $\mathbb{Q} \cap (0, 1)$ in Theorem 5.1 gives $J(S) = (\mathbb{Q} \cap (0, 1)) \cup \{1\}$ and $\mathcal{L}(S) = [0, 1]$. Taking (β_n) dense in a Cantor set $C \subset (0, 1)$ gives $\mathcal{L}(S) = C \cup \{1\}$: the transition locus is exactly the prescribed Cantor set.*

PROOF. By Theorem 5.1(3), Ξ_β is constant on an open interval $I \subseteq (0, 1)$ if and only if I contains no β_n . Hence a point of $(0, 1)$ fails to have a neighbourhood of constancy exactly when it is a limit of the β_n , so that

$$\mathcal{L}(S) \cap (0, 1) = \overline{J(S) \cap (0, 1)} = \overline{\{\beta_n\}},$$

identifying the non-local-constancy locus with the closure of the jump set, as in Remark 6.7. Finally $\beta = 1$ always lies in $\mathcal{L}(S)$, since Ξ_β jumps to $\widehat{G}_S$ there. $\square$

DEFINITION 6.3. For weights $w_n > 0$ with $\sum w_n = 1$ the *normalized rank* is $\mathfrak{d}_w(\beta) = \sum_{n:\beta_n < \beta} w_n \in [0, 1]$.

THEOREM 6.4 (Pure-jump staircase). *In the situation of Corollary 6.2 with locus $\mathcal{C}$, $\mathfrak{d}_w$ is non-decreasing, equals 0 below $\min \mathcal{C}$ and 1 above $\max \mathcal{C}$, is constant on every connected component of $(0, 1) \setminus \mathcal{C}$, and its set of points of increase is exactly $\mathcal{C}$.*

PROOF. A component of the complement contains no β_n , so $\mathfrak{d}_w$ is constant there; every neighbourhood of a point of $\mathcal{C}$ contains some β_n by density. $\square$

THEOREM 6.5 (The devil's staircase is unattainable). *For every set of primes S : $\widehat{G}_S$ is countable, hence $J(S)$ is countable; and for every choice of weights $\mathfrak{d}_w$ is a pure jump function, discontinuous at every β_n with jump $w_n > 0$, whose distributional derivative is the purely atomic measure $\sum_n w_n \delta_{\beta_n}$. Consequently there is no S and no weighting for which $\mathfrak{d}_w$ equals the Cantor function, or any continuous non-constant function.*

PROOF. G_S is a compact metrizable abelian group, being a Hausdorff quotient of the second countable locally compact group $\mathbb{I}_S$ [Ch. 1, Prop. 2.5]; the dual of a compact metrizable abelian group is countable. For each χ the set $\{\beta : \chi \in \Xi_\beta\}$ is an up-set, so each χ contributes a single jump and $J(S) = \{\beta_\chi\}$ is countable. In the situation of Theorem 5.1, $\chi_n \notin \Xi_{\beta_n}$ by the endpoint divergence in Lemma 5.4(2) while $\chi_n \in \Xi_\beta$ for $\beta > \beta_n$, so the jump at β_n is at least $w_n > 0$. A non-decreasing function of the form $\sum_{\beta_n < \beta} w_n$ has purely atomic distributional derivative. The Cantor function is continuous with non-atomic derivative, and a purely atomic probability measure and a non-atomic one are mutually singular, hence distinct. $\square$

THEOREM 6.6 (Approximation). *Let $\mathcal{C}$ be the middle-thirds Cantor set, c the Cantor function, and S as in Corollary 6.2 with (β_n) dense in $\mathcal{C}$. For every $\varepsilon > 0$ there are weights $w_n > 0$, $\sum w_n = 1$, with $\|\mathfrak{d}_w - c\|_\infty < \varepsilon$.*

PROOF. Let μ_c be the Cantor measure. Partition $\mathcal{C}$ into finitely many relatively clopen pieces of diameter $< \delta$ and place the μ_c -mass of each at a point β_{n_i} inside it, giving $\nu = \sum_{i \leq N} v_i \delta_{\beta_{n_i}}$. Since c is continuous, weak convergence of distribution functions is uniform (Pólya's theorem, [9, Problem 14.8]), so for δ small $\|F_\nu - c\|_\infty < \varepsilon/2$. Put $w_{n_i} = (1 - \frac{\varepsilon}{4})v_i$ and distribute $\frac{\varepsilon}{4}$ among the remaining indices with arbitrary positive weights; then $|\mathfrak{d}_w - F_\nu| \leq \frac{\varepsilon}{4}F_\nu + \frac{\varepsilon}{4} \leq \frac{\varepsilon}{2}$. $\square$

REMARK 6.7 (Jump set versus transition locus). The obstruction in Theorem 6.5 is not an artifact of the construction of §5: Ξ_β is a subgroup of the fixed countable group $\widehat{G}_S$, and a monotone family of subgroups of a countable group can change at only countably many parameters. What survives, and what Corollary 6.2 realizes, is the distinction between the *jump set*, necessarily countable, and the *transition locus* $\overline{J(S)}$, which is the closure of a countable set and hence an arbitrary separable closed set — in particular an arbitrary Cantor set. The Cantor spectrum of transitions is real; the continuity of the staircase is not. Theorems 6.5 and 6.6 together say the devil's staircase is a limit point of the realizable order parameters in the uniform norm, but not itself realizable.

7. Order parameters

Each transition brings into existence a macroscopic observable, and the family is independent.

DEFINITION 7.1. With $E_n = \{\ell \in S : \ell < z_n\}$ finite and $E_n^- = \{\ell \in E_n : (\frac{\ell}{q_n}) = -1\}$, put

$$(7.1) \quad \Psi_n(x) = \left(\frac{u_{q_n}(x) \bmod q_n}{q_n} \right) \cdot (-1)^{\sum_{\ell \in E_n} v_\ell(x)} \cdot \prod_{\ell \in E_n^-} (-1)^{v_\ell(x)} \in \{\pm 1\}.$$

- PROPOSITION 7.2. (1) $\Psi_n(mx) = \Psi_n(x)$ for every $m \in \mathbb{N}_S$ and every x in the domain — pointwise, not merely almost everywhere;
 (2) Ψ_n is defined ν -almost everywhere iff $\beta > \beta_n$;
 (3) under ν_{sym} the family (Ψ_n) consists of independent uniform signs, and for $\beta > \beta_n$ the map $\varphi \mapsto (\varphi(\Psi_n))_n$ is a bijection from the extreme points of the KMS_β simplex onto $\{\pm 1\}^{\{n: \beta_n < \beta\}}$.

PROOF. (1) It suffices to treat $m = \ell_0$ prime. Multiplication by ℓ_0 sends $v_\ell \mapsto v_\ell + \delta_{\ell, \ell_0}$ and $u_{q_n} \mapsto \ell_0 u_{q_n}$ for $\ell_0 \neq q_n$ [Ch. 1, proof of Prop. 4.12], so the first factor is multiplied by $(\frac{\ell_0}{q_n})$ unless $\ell_0 = q_n$, when it is unchanged. If $\ell_0 = q_n$, no v_ℓ with $\ell \in R_n \cup E_n^-$ changes. If $\ell_0 \in E_n$ and the symbol is -1 then $\ell_0 \in E_n^-$ and v_{ℓ_0} increases by one, flipping the third factor. If $\ell_0 \geq z_n$ and $\ell_0 \notin R_n$, then either $\ell_0 \in Q$, and the symbol is $+1$ by the choice of Q in Definition 5.3, or the block property gives symbol $+1$; in both cases no relevant valuation changes. If $\ell_0 \in R_n$, the symbol is -1 and v_{ℓ_0} increases by one, flipping the second factor.

(2) $\sum_{\ell \in R_n} \pi_\beta(v_\ell \geq 1) = \sum_{\ell \in R_n} \ell^{-\beta}$ is finite iff $\beta > \beta_n$ by Lemma 5.4(2); apply the two Borel–Cantelli lemmas to the independent events $\{v_\ell \geq 1\}$.

(3) Under ν_{sym} the conditional law of u given v is Haar, so the u_{q_n} are independent and each $(\frac{u_{q_n}}{q_n})$ is a uniform sign independent of everything else; multiplying by the remaining factors preserves uniformity and independence. The bijection is Theorem 5.1(3) together with [Ch. 1, Thm. 3.6], which identifies the extreme points with $\widehat{\Xi}_\beta = (\mathbb{Z}/2\mathbb{Z})^{\{n: \beta_n < \beta\}}$. $\square$

REMARK 7.3 (Numerical check). Take the model $Q = \{7, 11, 13, 17\}$. Pointwise invariance was verified over $4 \cdot 10^5$ random configurations, with no failures, together with a control confirming that dropping the R_n -correction destroys it. The independence asserted in (3) was checked by computing $\mathbb{E}[\prod_{n \in T} \Psi_n]$ over all 15 nonempty subsets of four indices, the largest absolute value being 0.0037 against a three-sigma bound of 0.0047.

REMARK 7.4 (Physical reading). Each Ψ_n is a tail observable: changing the valuations at any finite set of primes leaves it unchanged, because a change at $\ell \in R_n$ flips two factors of (7.1) simultaneously. It records one asymptotically conserved bit, coming into existence at the sharp temperature β_n where the Borel–Cantelli series for R_n turns from divergent to convergent. As β increases across $\mathcal{C}$ the bits condense on a countable dense subset of a Cantor set, and $\mathfrak{d}_w(\beta)$ is the total weight of those already condensed; between two condensations, that is on each complementary interval, the memory is frozen. By Theorem 6.5 they cannot condense continuously: each bit is a discrete degree of freedom contributing a jump, and $\widehat{G}_S$ has only countably many. The information-theoretic counterpart — the relative entropy of an extremal state with respect to the symmetric one grows by $\log 2$ with each condensed bit — is taken up in a companion paper on invariants and entropy.

8. Assessment

Question	Answer	Reference
factor type	bounded above by Σ_β , below by pairs	Prop. 2.2, Lem. 2.4
$S = P_K$, $\beta \leq 1$	III_1 ; $\Sigma_\beta = \{0\}$ via $L(1 - i\tau, \chi) \neq 0$	Thm. 3.2, Prop. 3.1
two incommensurable norms	do not suffice	Thm. 4.2
exact commensurability	impossible for infinite S	Lem. 4.1
uniqueness $\Rightarrow \text{III}_1$	false	Cor. 4.12
chains $\{\Xi_\beta\}$	any $(\beta_n) \subset (0, 1)$ realizable	Thm. 5.1
transition locus	any Cantor set	Cor. 6.2
jump set	always countable	Thm. 6.5
devil's staircase	unattainable, approximable	Thms. 6.5, 6.6

Five questions remain. First, whether Σ_β determines the type, i.e. whether the flow of weights of these factors always has pure point spectrum (Remark 2.3); if it does, Proposition 2.2 is an equivalence and Lemma 2.4 is not needed. Second, the extended group Ξ_β^{ext} has not been computed for the sparse sets of [Ch. 1, §5]; in particular whether Brun-summable S gives III_λ or III_0 is open. In Theorem 4.2 we determined Σ_1 exactly, but for a general band set with $\varepsilon(\ell) = (\log \log \ell)^{-\theta}$ the dependence of Σ_1 on θ is unexamined. Fourth, Theorem 6.5 shows the realizable order parameters are pure jump functions; which monotone families of subgroups of a countable group arise as $\{\Xi_\beta\}$, beyond the direct sums realized here, we do not know. Finally, Lemma 2.1 says that Ξ_β is the kernel of $\chi \mapsto [\chi \circ \sigma]$ into the measurable cohomology group $H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T})$; that group itself is not computed here, and a computation would presumably exhibit $\widehat{G}_S/\Xi_\beta$ inside it and give the chain $\{\Xi_\beta\}$ a cohomological reading.

CHAPTER 3

Ideal structure, boundary quotient, and index

This chapter is Paper II of the series (papers/II-structure/).

SUMMARY. We study the ideal structure, the boundary quotient, the K -theory and the index theory of the localized Bost–Connes system $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$, and show that all of it is governed by one structural feature: since J_S is free abelian on S , each prime carries exactly *one* isometry and the range projections $\mathbf{1}_{\mathfrak{a}Y_S}$ are *nested* rather than orthogonal.

From this we obtain: the groupoid is topologically principal, its isotropy on the finite closed orbit $\partial Y_S \cong \mathrm{Cl}_S^+ \cong I_S^0 = \ker(I_S \rightarrow \mathrm{Cl}_S^+)$, and off ∂Y_S it is trivial whenever K has finitely many totally positive units; the ideal lattice is then the lattice of open invariant subsets of $\tilde{Y}_S \setminus \partial Y_S$ glued to the lattice of closed subsets of the torus $\widehat{I}_S^0$, the latter contributing ideals that neither the diagonal nor the gauge action sees. The tight quotient makes the isometries unitary rather than Cuntz, giving $\partial \mathcal{A}_{K,S} \cong M_{h_S^+}(C^*(I_S^0))$, a type I algebra with a faithful trace whose only isomorphism invariant is the narrow class number h_S^+ , recoverable as the divisibility of $[1]$ in $K_0 = \Lambda^{\mathrm{ev}}(I_S^0)$. For a single prime, $K_1(\mathcal{A}_{K,S}) = 0$ and $K_0(\mathcal{A}_{K,S}) \cong C(G_S, \mathbb{Z})/\mathbb{Z} \oplus \mathbb{Z}$ is free; for several primes K_* is the abutment of a collapsing Koszul spectral sequence which we do not evaluate.

The same feature governs the index theory. At the symmetric KMS_β state, every prime inclusion $M_{S_1} \subset M_{S_2}$ has infinite index for *every* conditional expectation, for the elementary reason that the valuation projections $\mathbf{1}_{\{v_{\mathfrak{p}}=k\}}$ lie in the relative commutant and are moved onto one another by partial isometries commuting with M_{S_1} ; there is no index formula in Np and $[\mathrm{Cl}_{S_2}^+ : \mathrm{Cl}_{S_1}^+]$, and no Galois correspondence with intermediate prime sets, there being a continuum of intermediate algebras where there are two intermediate prime sets. The finite index instead comes from symmetry breaking: $M_\varepsilon \cong M_{\mathrm{sym}}^{\Xi_\beta} \subset M_{\mathrm{sym}}$ is the trivial inclusion of a factor tensored with $\mathbb{C} \subset \ell^\infty(\widehat{\Xi_\beta})$, of index $|\Xi_\beta|$, its intermediate algebras are indexed by the partitions of $\widehat{\Xi_\beta}$, and $\log[M_{\mathrm{sym}} : M_\varepsilon] = |A_\beta| \log 2$ is also the relative entropy of an extremal phase against the symmetric one.

Finally we run the control experiment. Restoring the additive part gives $\mathcal{A}_{K,S}^{\mathrm{aff}} = C(\widehat{\mathcal{O}}_S) \rtimes (\mathcal{O}_K \rtimes J_S)$, whose additive cosets provide *finite covers* and hence genuine Cuntz relations; it is a simple purely infinite Kirchberg algebra. Nested versus partitioning range projections is therefore exactly the dividing line. Even there, however, $[1]$ carries almost nothing: the Cuntz relation forces its order to divide $\gcd_{\mathfrak{p} \in S}(\mathrm{Np} - 1)$, which is 1 as soon as S contains a prime of norm 2.

1. The single-isometry phenomenon

We keep the notation of [Ch. 1]: K is a number field, S a set of finite primes, $J_S = \bigoplus_{\mathfrak{p} \in S} \mathbb{N}$ the free abelian monoid of integral ideals supported on S with enveloping group I_S , $Y_S = \widehat{\mathcal{O}}_S \times_{U_S} G_S$, and

$$\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S = \overline{\mathrm{span}}\{\mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*\}, \quad \mu_{\mathfrak{a}}^* \mu_{\mathfrak{a}} = 1, \quad \mu_{\mathfrak{a}} \mu_{\mathfrak{a}}^* = \mathbf{1}_{\mathfrak{a}Y_S}.$$

Recall the exact sequence [Ch. 1, (2.1)]

$$(1.1) \quad 1 \rightarrow U_S / \overline{\mathcal{O}_{K,+}^\times} \rightarrow G_S \rightarrow \mathrm{Cl}_S^+ \rightarrow 1, \quad h_S^+ := |\mathrm{Cl}_S^+|.$$

LEMMA 1.1 (Single isometry, nested projections). *J_S is free abelian on S . Hence:*

- (1) for each $\mathfrak{p} \in S$ there is exactly one generating isometry $\mu_{\mathfrak{p}}$;
- (2) the range projections are nested: $\mathbf{1}_{\mathfrak{a}Y_S} \mathbf{1}_{\mathfrak{b}Y_S} = \mathbf{1}_{\text{lcm}(\mathfrak{a}, \mathfrak{b})Y_S}$;
- (3) there is no finite family of isometries in $\mathcal{A}_{K,S}$ with mutually orthogonal ranges summing to 1; indeed $Y_S \setminus \bigcup_{\mathfrak{p} \in S} \mathfrak{p}Y_S = \{[x, g] : x \in \widehat{\mathcal{O}}_S^\times\}$ is nonempty and clopen.

PROOF. Freeness is unique factorization of ideals. (2) is immediate from $\mathfrak{a}Y_S \cap \mathfrak{b}Y_S = \text{lcm}(\mathfrak{a}, \mathfrak{b})Y_S$. For (3), two of the range projections $\mathbf{1}_{\mathfrak{a}Y_S}$ are never orthogonal by (2); and if x is a unit at every $\mathfrak{p}$ then $[x, g] \notin \mathfrak{p}Y_S$ for any $\mathfrak{p}$, which gives the last assertion. That no family of isometries whatever has orthogonal ranges summing to 1 — $\mathcal{A}_{K,S}$ is not properly infinite — follows from Theorem 3.1 below: $\mathcal{A}_{K,S}$ maps unitally onto an algebra with a faithful tracial state τ , and n isometries with orthogonal ranges summing to 1 would give $n = \sum \tau(v_i^* v_i) = \sum \tau(v_i v_i^*) = 1$. $\square$

Lemma 1.1 is the source of every structural statement below. We flag its use by referring to it explicitly. The contrast is with the $ax + b$ algebras of Cuntz–Li and Cuntz–Deninger–Laca, where each ideal $\mathfrak{a}$ carries $N\mathfrak{a}$ isometries indexed by the additive cosets $a + \mathfrak{a}$, whose ranges *partition*; §6 restores that ingredient and confirms that it is the cause.

2. Isotropy, topological principality, and the ideal lattice

Let $\mathcal{G}_S = I_S \times \widetilde{Y}_S$ be the dilated groupoid of [Ch. 1, Prop. 2.8], with points $[x, g]$ subject to $(wx, g) \sim (x, r(w)g)$ for $w \in U_S$.

LEMMA 2.1 (Isotropy formula). *For $y = [x, g]$ put $T(y) = \{\mathfrak{p} \in S : x_{\mathfrak{p}} = 0\}$. Then*

$$(2.1) \quad \text{Iso}_{\mathcal{G}_S}(y) = \{\mathfrak{a} \in I_{T(y)} : \sigma_{\mathfrak{a}} \in r(U_{T(y)})\}.$$

PROOF. $\mathfrak{a} \cdot [x, g] = [x, g]$ means $\mathfrak{a}x = wx$ and $g = r(w)^{-1} \sigma_{\mathfrak{a}} g$ for some $w \in U_S$. Comparing $\mathfrak{p}$ -components: if $x_{\mathfrak{p}} \neq 0$ then $\mathfrak{p}^{a_{\mathfrak{p}}} x_{\mathfrak{p}} = w_{\mathfrak{p}} x_{\mathfrak{p}}$ forces $a_{\mathfrak{p}} = 0$ and $w_{\mathfrak{p}} = 1$; if $x_{\mathfrak{p}} = 0$ both sides vanish. So $\mathfrak{a} \in I_T$, $w \in U_T$, and the second equation is $\sigma_{\mathfrak{a}} = r(w)$. $\square$

PROPOSITION 2.2 (Isotropy off the boundary). *Write $\mathbb{I}_T = \{u \in \mathbb{I}_S : u_{\mathfrak{p}'} = 1 \text{ for all } \mathfrak{p}' \in S \setminus T\}$. Then for y with $T(y) = T$,*

$$(2.2) \quad \text{Iso}_{\mathcal{G}_S}(y) = \text{div}\left(\overline{K_{S,+}^\times} \cap \mathbb{I}_T\right).$$

If $\mathcal{O}_{K,+}^\times$ is finite — in particular if $K = \mathbb{Q}$ or K is imaginary quadratic — then for every proper $T \subsetneq S$ the right-hand side is trivial, so $\text{Iso}_{\mathcal{G}_S}(y) = \{1\}$: the action of I_S on $\widetilde{Y}_S \setminus \partial Y_S$ is free.

At the bottom stratum the isotropy is nontrivial:

$$\partial Y_S := \bigcap_{\mathfrak{a} \in J_S} \mathfrak{a}Y_S = \{[0, g]\} \cong G_S/r(U_S) \cong \text{Cl}_S^+,$$

a set of h_S^+ points, on which I_S acts transitively through the class map with isotropy $I_S^0 := \ker(I_S \rightarrow \text{Cl}_S^+)$, of index h_S^+ .

PROOF. By Lemma 2.1, $\mathfrak{a} \in I_T$ lies in the isotropy iff $\sigma_{\mathfrak{a}} \in r(U_T)$, i.e. iff some idele s with $\text{div}(s) = \mathfrak{a}$ satisfies $s \in U_T \cdot \overline{K_{S,+}^\times}$. Writing $s = wu$ with $w \in U_T$ and $u \in \overline{K_{S,+}^\times}$, the element $u = w^{-1}s$ is trivial at every $\mathfrak{p}' \notin T$ and has $\text{div}(u) = \mathfrak{a}$, which is (2.2); the converse is immediate.

Suppose now $\mathcal{O}_{K,+}^\times$ is finite. Then $K_{S,+}^\times$ is discrete in $\mathbb{I}_S$: an element of $K_{S,+}^\times$ lying in the neighbourhood $\prod_{\mathfrak{p} \in F} (1 + \mathfrak{p}^k \mathcal{O}_{\mathfrak{p}}) \times \prod_{\mathfrak{p} \notin F} \mathcal{O}_{\mathfrak{p}}^\times$ of 1 has trivial divisor, hence is a totally positive unit, and is 1 for k large. So $\overline{K_{S,+}^\times} = K_{S,+}^\times$, and $u = \alpha \in K_{S,+}^\times$ is trivial at some $\mathfrak{p}' \in S \setminus T$. Under the diagonal embedding the $\mathfrak{p}'$ -component of α is α itself, and $K \rightarrow K_{\mathfrak{p}'}$ is injective, so $\alpha = 1$ and $\mathfrak{a} = (1)$.

For the bottom stratum, $\bigcap_{\mathfrak{a}} \mathfrak{a}Y_S$ forces $x_{\mathfrak{p}} = 0$ for every $\mathfrak{p}$, so $x = 0$; the fibre is $G_S/r(U_S) \cong \text{Cl}_S^+$ by (1.1), and (2.1) with $T = S$ becomes $\{\mathfrak{a} : [\mathfrak{a}] = 1\} = I_S^0$. $\square$

REMARK 2.3 (The general case, and why it does not matter here). When $\mathcal{O}_{K,+}^\times$ is infinite the right-hand side of (2.2) can in principle be nontrivial: one needs α^{-1} to lie in the closure of $\mathcal{O}_{K,+}^\times$ inside $\prod_{\mathfrak{p}' \in S \setminus T} \mathcal{O}_{\mathfrak{p}'}^\times$, a condition of Leopoldt type. We do not decide it. Nothing below depends on it: Theorem 2.5 needs only that the strata are nowhere dense, which holds whatever their isotropy, since for infinite S every basic clopen set contains points with $x_{\mathfrak{p}} \neq 0$ at any prescribed $\mathfrak{p}$.

REMARK 2.4. The natural expectation is that killing the coordinates in T frees the corresponding translations, giving isotropy I_T . It frees them in the $\widehat{\mathcal{O}}_S$ -direction but not in the G_S -direction: multiplication by $\mathfrak{a}$ translates g by $\sigma_{\mathfrak{a}}$, whose components at the primes *outside* T are nontrivial. Over $\mathbb{Q}$ with $T = \{2\}$ and $\mathfrak{a} = (2)$, the components of $\sigma_{\mathfrak{a}}$ at 3, 5, 7 are 2, 2, 2, none equal to 1.

THEOREM 2.5 (Topological principality, and the two halves of the ideal lattice). *Let $\partial I := C^*(\mathcal{G}_S|_{\widetilde{Y}_S \setminus \partial Y_S}) \cap \mathcal{A}_{K,S}$ be the ideal of $\mathcal{A}_{K,S}$ corresponding to the open invariant set $\widetilde{Y}_S \setminus \partial Y_S$, so that $\mathcal{A}_{K,S}/\partial I = \partial \mathcal{A}_{K,S}$ is the boundary quotient of Theorem 3.1.*

- (1) $\mathcal{G}_S$ is topologically principal, for every K and S ; consequently $C(Y_S)$ is a Cartan subalgebra of $\mathcal{A}_{K,S}$ [109].
- (2) The ideals of $\mathcal{A}_{K,S}$ containing ∂I correspond bijectively to the closed subsets of the torus $\widehat{I}_S^0$, via $\partial \mathcal{A}_{K,S} \cong M_{h_S^+}(C(\widehat{I}_S^0))$. These ideals are not detected by the diagonal $C(Y_S)$: two of them have the same intersection $C_0(Y_S \setminus \partial Y_S)$ with it. They are not all gauge-invariant either: the gauge action is the translation flow of $\widehat{I}_S^0$ by the characters $\mathfrak{a} \mapsto N\mathfrak{a}^{it}$, which is nontrivial, so a closed subset that is not a union of flow orbits — a point, for instance — gives an ideal that the gauge action moves.
- (3) If the action of I_S on $\widetilde{Y}_S \setminus \partial Y_S$ is free — by Proposition 2.2, whenever $\mathcal{O}_{K,+}^\times$ is finite — then the ideals of ∂I correspond bijectively and order-isomorphically to the open I_S -invariant subsets of $\widetilde{Y}_S \setminus \partial Y_S$, and the diagonal detects them.
- (4) Consequently, in that case, $\text{Prim}(\mathcal{A}_{K,S}) = (\widetilde{Y}_S \setminus \partial Y_S)/I_S \sqcup \widehat{I}_S^0$ as a set, the first piece open and the second closed.

PROOF. (1) By (2.1) every unit with nontrivial isotropy lies in some stratum $\{x_{\mathfrak{p}} = 0 \ \forall \mathfrak{p} \in T\}$ with $T \neq \emptyset$, hence in some $\{x_{\mathfrak{p}} = 0\}$, a closed set with empty interior since $\mathcal{O}_{\mathfrak{p}}$ has no isolated points. A countable union of such sets is meagre, so the units with trivial isotropy are dense. (This does not use Proposition 2.2.) The restriction $\mathcal{G}_S|_{Y_S}$ is then topologically principal as well, and $\mathcal{A}_{K,S} = C^*(\mathcal{G}_S|_{Y_S})$ with diagonal $C(Y_S)$, which is Cartan by Renault's theorem [109].

(2) Ideals containing ∂I are ideals of the quotient, which is $M_h(C(X))$ with $X = \widehat{I}_S^0$ by Theorem 3.1; its ideals are the $M_h(C_0(X \setminus Z))$, $Z \subseteq X$ closed. The image of $C(Y_S)$ in $M_h(C(X))$ is the algebra $C(\partial Y_S) = \mathbb{C}^h$ of diagonal matrices with constant entries, which meets $M_h(C_0(X \setminus Z))$ trivially for every nonempty Z ; so every ideal $J \supseteq \partial I$ with $J \neq \mathcal{A}_{K,S}$ has $J \cap C(Y_S) = \partial I \cap C(Y_S) = C_0(Y_S \setminus \partial Y_S)$. The gauge action $\sigma_t(\mu_{\mathfrak{a}}) = N\mathfrak{a}^{it}\mu_{\mathfrak{a}}$ descends to the unitaries $\mu_{\mathfrak{a}}$, $\mathfrak{a} \in I_S^0$, of $C^*(I_S^0) = C(X)$, where it is translation by the character $\chi_t : \mathfrak{a} \mapsto N\mathfrak{a}^{it}$; $\chi_t \neq 1$ for $t \neq 0$ since $\log N\mathfrak{a} \neq 0$ for $\mathfrak{a} \neq (1)$, so the flow moves every point of X .

(3) $\mathcal{G}_S|_{\widetilde{Y}_S \setminus \partial Y_S}$ is second countable, étale, amenable and principal, so by Renault [108] its C^* -algebra has ideals in bijection with the open invariant subsets, via $V \mapsto C^*(\mathcal{G}_S|_V)$, and its diagonal is a Cartan subalgebra detecting them; ∂I is the full corner $\mathbf{1}_{Y_S}(\cdot)\mathbf{1}_{Y_S}$, which preserves the lattice. (4) is (2) and (3) together with the exactness of $0 \rightarrow C^*(\mathcal{G}_S|_U) \rightarrow C^*(\mathcal{G}_S) \rightarrow C^*(\mathcal{G}_S|_{U^c}) \rightarrow 0$ for amenable étale groupoids. $\square$

REMARK 2.6. Part (2) shows that “the ideal lattice is the lattice of open invariant sets” is false for $\mathcal{A}_{K,S}$: the boundary orbit has isotropy I_S^0 of infinite rank, and its C^* -algebra $C(\widehat{I}_S^0)$

contributes a torus of primitive ideals invisible to the diagonal. Topological principality of $\mathcal{G}_S$ gives the correspondence only on the essentially principal part $\tilde{Y}_S \setminus \partial Y_S$. When $\mathcal{O}_{K,+}^\times$ is infinite, Remark 2.3 leaves open whether some proper stratum has nontrivial isotropy; if it does, that stratum contributes further ideals of the same kind.

PROPOSITION 2.7 (The lattice is not the power set of S). *Let $K = \mathbb{Q}$, $S = \{\ell\}$, so $Y_S = \mathbb{Z}_\ell$ and $\mathcal{A} = C(\mathbb{Z}_\ell) \rtimes \mathbb{N}$. Then $\mathcal{A}$ is a full corner in $C_0(\mathbb{Q}_\ell) \rtimes \mathbb{Z}$ and there is an exact sequence*

$$(2.3) \quad 0 \longrightarrow C(\mathbb{Z}_\ell^\times) \otimes \mathcal{K} \longrightarrow C_0(\mathbb{Q}_\ell) \rtimes \mathbb{Z} \longrightarrow C(\mathbb{T}) \longrightarrow 0.$$

The ideal lattice of $\mathcal{A}$ contains an order-isomorphic copy of the lattice of open subsets of the Cantor set $\mathbb{Z}_\ell^\times$, hence is uncountable.

PROOF. By Laca's dilation theorem $\mathcal{A}$ is a full corner in $C_0(\mathbb{Q}_\ell) \rtimes \mathbb{Z}$, the dilation being the limit of $\mathbb{Z}_\ell \xleftarrow{\ell} \mathbb{Z}_\ell \xleftarrow{\ell} \cdots$. On $\mathbb{Q}_\ell \setminus \{0\}$ the action of $\mathbb{Z}$ by multiplication by ℓ is free and proper with quotient $\mathbb{Z}_\ell^\times$, giving the ideal in (2.3); the fixed point $\{0\}$ gives the quotient $C^*(\mathbb{Z}) = C(\mathbb{T})$. Ideals of $C(\mathbb{Z}_\ell^\times) \otimes \mathcal{K}$ correspond to open subsets of $\mathbb{Z}_\ell^\times$ and pass to $\mathcal{A}$ by the corner correspondence. $\square$

REMARK 2.8. The description “hereditary subsets of $\mathcal{P}(S)$ ” would be correct for a graph or Cuntz–Krieger algebra. Here the stratification by vanishing coordinates is present, but each stratum retains a Cantor set of unit orbits, each closed set of orbits contributes an ideal, and the quotient $C(\mathbb{T})$ in (2.3) is the $h_S^+ = 1$, $I_S^0 = \mathbb{Z}$ case of Theorem 2.5(2).

3. The boundary quotient

THEOREM 3.1 (The tight quotient is abelian-by-finite). *Let $\partial\mathcal{A}_{K,S}$ be the quotient by the ideal corresponding to $\tilde{Y}_S \setminus \partial Y_S$, equivalently the tight quotient in the sense of Exel [41], equivalently the universal quotient in which every μ_p becomes unitary. Then*

$$\partial\mathcal{A}_{K,S} \cong C(\text{Cl}_S^+) \rtimes I_S \cong M_{h_S^+}(C^*(I_S^0)) \cong M_{h_S^+}(C(\hat{I}_S^0)).$$

It is unital, separable, nuclear, satisfies the UCT, is of type I and carries a faithful tracial state. It is not simple when $I_S^0 \neq 0$, and is never purely infinite; in particular it is not a Kirchberg algebra.

PROOF. Restricting the groupoid to the closed invariant set ∂Y_S gives $C^*(I_S \rtimes \partial Y_S) = C(\text{Cl}_S^+) \rtimes I_S$ by Proposition 2.2. In the quotient $\mathbf{1}_{pY_S}$ restricts to 1 on $\partial Y_S \subseteq pY_S$, so μ_p is unitary; conversely imposing unitarity kills exactly the functions vanishing on ∂Y_S . For a discrete group Γ with a subgroup Γ_0 of finite index h , Green's imprimitivity theorem [48] gives $C(\Gamma/\Gamma_0) \rtimes \Gamma \cong M_h(C^*(\Gamma_0))$, and I_S^0 is free abelian so $C^*(I_S^0)$ is commutative. The remaining properties are those of $M_h(C(X))$; a unital algebra with a tracial state is never purely infinite. $\square$

REMARK 3.2 (Why no Cuntz relation is available). By Lemma 1.1(2) the projections $\mathbf{1}_{aY_S}$ are nested. Imposing $\sum_a \mu_a \mu_a^* = 1$ over any finite family would force all but one summand to vanish. The tight relation therefore degenerates to $\mathbf{1}_{pY_S} = 1$, which is unitarity, and this is what Theorem 3.1 implements. The Cuntz alternative requires a partition; see §6.

THEOREM 3.3 (The class of the unit records h_S^+). $K_0(\partial\mathcal{A}_{K,S}) \cong \Lambda^{\text{ev}}(I_S^0)$ and $K_1(\partial\mathcal{A}_{K,S}) \cong \Lambda^{\text{odd}}(I_S^0)$, and under this identification

$$(3.1) \quad [1_{\partial\mathcal{A}_{K,S}}] = h_S^+ \cdot e_0,$$

e_0 the generator of $\Lambda^0 = \mathbb{Z}$. Hence h_S^+ is the largest integer dividing $[1]$, and is an isomorphism invariant of the pair $(\partial\mathcal{A}_{K,S}, [1])$.

PROOF. K -theory is Morita invariant and $C^*(\Gamma) \cong C(\widehat{\Gamma})$ for Γ free abelian has $K_0 = \Lambda^{\text{ev}}(\Gamma)$, $K_1 = \Lambda^{\text{odd}}(\Gamma)$ with $[1]$ the generator of Λ^0 . The isomorphism $K_0(M_h(B)) \cong K_0(B)$ sends $[1_{M_h(B)}]$ to $h[1_B]$, since $1_{M_h(B)}$ is a sum of h orthogonal projections each equivalent to 1_B ; and Λ^{ev} is free abelian, so $h e_0$ is divisible by exactly h . $\square$

THEOREM 3.4 (Rigidity of the boundary). *For infinite $S_1 \subseteq P_{K_1}$ and $S_2 \subseteq P_{K_2}$,*

$$\partial \mathcal{A}_{K_1, S_1} \cong \partial \mathcal{A}_{K_2, S_2} \iff h_{S_1}^+ = h_{S_2}^+.$$

The boundary determines the order of the subgroup of the narrow class group generated by S , and nothing else: not K , not S , not Cl_S^+ as a group.

PROOF. ($\Leftarrow$) For infinite S , I_S is free abelian of countably infinite rank and I_S^0 has finite index, hence the same rank; so the two algebras are $M_h(C(X))$ with the same h and homeomorphic X . ($\Rightarrow$) Every irreducible representation of $M_h(C(X))$ has dimension exactly h , and isomorphisms preserve irreducible representations and their dimensions; alternatively apply Theorem 3.3. For the last statement, $C(\text{Cl}_S^+) \rtimes I_S$ depends on Cl_S^+ only through the index of I_S^0 . $\square$

4. K -theory of $\mathcal{A}_{K,S}$

The Stacey crossed product imposes the covariance relation $\mu_{\mathfrak{p}} \mu_{\mathfrak{p}}^* = \alpha_{\mathfrak{p}}(1)$, which is Katsura covariance; hence for a single prime $\mathcal{A}_{K,S}$ is the Cuntz–Pimsner algebra $\mathcal{O}_X$ of the associated correspondence, *not* the Toeplitz algebra $\mathcal{T}_X$. Pimsner’s KK -equivalence $\iota : A \rightarrow \mathcal{T}_X$ therefore does not apply, and one must use his exact sequence.

LEMMA 4.1. *Y_S is compact, metrizable and totally disconnected without isolated points, hence a Cantor set. Consequently $K^0(Y_S) = C(Y_S, \mathbb{Z})$ is free abelian of countably infinite rank and $K^1(Y_S) = 0$.*

PROOF. $\widehat{\mathcal{O}}_S$ is profinite, G_S is a quotient of the totally disconnected $\mathbb{I}_S$ by a closed subgroup hence profinite, and Y_S is their quotient by the compact group U_S . A point $[x, g]$ is not isolated: every neighbourhood contains $[x', g]$ with $x' \neq x$, since no $\mathcal{O}_{\mathfrak{p}}$ has isolated points. $\square$

PROPOSITION 4.2 (One prime). *Let $S = \{\mathfrak{p}\}$ and let α_* be the endomorphism $\mathbf{1}_U \mapsto \mathbf{1}_{\mathfrak{p}U}$ of $K^0(Y_S) = C(Y_S, \mathbb{Z})$. Then*

$$(4.1) \quad K_1(\mathcal{A}_{K,S}) = \ker(1 - \alpha_*) = 0, \quad K_0(\mathcal{A}_{K,S}) = \text{coker}(1 - \alpha_*) \cong C(G_S, \mathbb{Z}) / \mathbb{Z} \cdot 1 \oplus \mathbb{Z},$$

a free abelian group. In particular $[1] = [\mathbf{1}_{\mathfrak{p}Y_S}]$ in K_0 , so $\iota_ : K^0(Y_S) \rightarrow K_0(\mathcal{A}_{K,S})$ is not injective.*

PROOF. $\mu_{\mathfrak{p}}$ is an isometry with $\mu_{\mathfrak{p}} \mu_{\mathfrak{p}}^* = \mathbf{1}_{\mathfrak{p}Y_S}$, so $1 \sim \mathbf{1}_{\mathfrak{p}Y_S}$ and $[1] = [\mathbf{1}_{\mathfrak{p}Y_S}]$. The six-term sequence for the Cuntz–Pimsner algebra of a correspondence [105, §4] (see [69] for the covariance condition) with $K_1(C(Y_S)) = 0$ collapses to $K_1(\mathcal{A}) = \ker(1 - \alpha_*)$, $K_0(\mathcal{A}) = \text{coker}(1 - \alpha_*)$ on $C(Y_S, \mathbb{Z})$.

If $f = \alpha_* f = (f \circ \mathfrak{p}^{-1}) \mathbf{1}_{\mathfrak{p}Y_S}$ then f is supported in $\mathfrak{p}^n Y_S$ for every n , i.e. in $\partial Y_S = \{[0, g]\}$, a closed set with empty interior; a locally constant function supported there vanishes. So $K_1 = 0$.

For K_0 we use Laca’s dilation [77]: $\mathcal{A}$ is a full corner in $C_0(\widetilde{Y}_S) \rtimes \mathbb{Z}$, and by Pimsner–Voiculescu [104] $K_0(C_0(\widetilde{Y}_S) \rtimes \mathbb{Z}) = H_0(\mathbb{Z}; C_c(\widetilde{Y}_S, \mathbb{Z}))$, the coinvariants, $K_1 = H_1$, the invariants. Write $\widetilde{Y}_S = \widetilde{Y}_{\emptyset} \sqcup \partial Y_S$ with $\widetilde{Y}_{\emptyset} = \{[x, g] : x \neq 0\}$ open. The map $[\pi^n u, g] \mapsto (n, r(u)g)$ identifies $\widetilde{Y}_{\emptyset}$ with $\mathbb{Z} \times G_S$, on which $\mathfrak{p}$ acts by $(n, h) \mapsto (n+1, \sigma_{\mathfrak{p}} h)$, freely; hence $H_1(\mathbb{Z}; C_c(\widetilde{Y}_{\emptyset}, \mathbb{Z})) = 0$ and $H_0 = C(G_S, \mathbb{Z})$, the class of f being $\sum_n (\sigma_{\mathfrak{p}}^{-n})_* f(n, \cdot)$. On $\partial Y_S \cong \text{Cl}_S^+$ the generator acts through the class map, so $H_1 = H_0 = \mathbb{Z}$ by Shapiro’s lemma. The long exact sequence of $0 \rightarrow C_c(\widetilde{Y}_{\emptyset}, \mathbb{Z}) \rightarrow C_c(\widetilde{Y}_S, \mathbb{Z}) \rightarrow C(\partial Y_S, \mathbb{Z}) \rightarrow 0$ reads

$$0 \rightarrow H_1(\widetilde{Y}_S) \rightarrow \mathbb{Z} \xrightarrow{\delta} C(G_S, \mathbb{Z}) \rightarrow H_0(\widetilde{Y}_S) \rightarrow \mathbb{Z} \rightarrow 0,$$

and $\delta(1)$ is the class of $(1 - \alpha_*)\mathbf{1}_{Y_S} = \mathbf{1}_{Y_S \setminus \mathfrak{p}Y_S} = \mathbf{1}_{\{0\} \times G_S}$, i.e. the constant function 1; so δ is injective, $H_1(\tilde{Y}_S) = 0$ again, and $H_0(\tilde{Y}_S)$ is an extension of $\mathbb{Z}$ by $C(G_S, \mathbb{Z})/\mathbb{Z} \cdot 1$, which splits. Both summands are free, 1 being part of a basis of $C(G_S, \mathbb{Z})$. $\square$

For S finite with $|S| \geq 2$ the primes interact: by [Ch. 1, Lem. 2.10] the generator $\mathfrak{p}$ moves the G_S -coordinate by $\sigma_{\mathfrak{p}}$, whose components at the other primes are nontrivial, so the action of $\mathbb{Z}^S$ on $\tilde{Y}_S$ is not a product of actions on the factors $K_{\mathfrak{p}}$. What survives is the following.

PROPOSITION 4.3 (Finite S). *For S finite, $K_*(\mathcal{A}_{K,S}) = K_*(C_0(\tilde{Y}_S) \rtimes \mathbb{Z}^S)$ is computed by the spectral sequence $E_{p,q}^2 = H_p(\mathbb{Z}^S; K_q(C_0(\tilde{Y}_S))) \Rightarrow K_{p+q}$ of Kasparov [68, §6]. Since $K_1(C_0(\tilde{Y}_S)) = 0$, only the row $q = 0$ is nonzero, the sequence collapses, and $K_0(\mathcal{A}_{K,S})$ (resp. K_1) carries a finite filtration whose subquotients are the even (resp. odd) Koszul homology groups $H_p(\mathbb{Z}^S; C_c(\tilde{Y}_S, \mathbb{Z}))$ of the commuting automorphisms $\alpha_{\mathfrak{p}*}$, $\mathfrak{p} \in S$. The bottom piece $H_0(\mathbb{Z}^S; C_c(\tilde{Y}_S, \mathbb{Z}))$, the coinvariants, is the image of $K^0(Y_S) \rightarrow K_0(\mathcal{A}_{K,S})$. For general S , $K_*(\mathcal{A}_{K,S}) = \varinjlim K_*(\mathcal{A}_{K,F})$ over finite $F \subseteq S$, for any compatible system of embeddings as in [Ch. 1, Prop. 2.7] (see Remark 5.1 below).*

PROOF. Laca's dilation, Kasparov's spectral sequence for actions of $\mathbb{Z}^n$, and the continuity of K -theory under direct limits. $\square$

We have not computed the Koszul homology for $|S| \geq 2$, and we make no claim about $K_*(\mathcal{A}_{K,S})$ beyond Propositions 4.2 and 4.3; the top group $H_{|S|}$ is the group of $\mathbb{Z}^S$ -invariant compactly supported functions, which vanishes by the argument of Proposition 4.2, but the intermediate groups receive contributions from the strata $\{x_{\mathfrak{p}} = 0, \mathfrak{p} \in T\}$, on which $\mathbb{Z}^T$ acts on G_S by translations with nontrivial invariants.

EXAMPLE 4.4 ($K = \mathbb{Q}$, $S = \{\ell\}$). Here $G_S = \mathbb{Z}_{\ell}^{\times}$ and Proposition 4.2 gives $K_1(\mathcal{A}) = 0$ and $K_0(\mathcal{A}) \cong C(\mathbb{Z}_{\ell}^{\times}, \mathbb{Z})/\mathbb{Z} \oplus \mathbb{Z} \cong \mathbb{Z}^{(\infty)}$, in agreement with (2.3). A direct computation of the Smith normal form of $1 - \alpha_*$ on level- n functions, for $\ell = 2, 3, 5$ and $n \leq 3$, gives kernel 0 and free cokernel of rank $\ell^{n+1} - \ell^n$, with no torsion.

REMARK 4.5 (K -theory carries no arithmetic: the mechanism). For a single prime, $K_0(\mathcal{A}_{K,S})$ is free abelian and depends on $(K, \mathfrak{p})$ only through the profinite group G_S ; for $K = \mathbb{Q}$ it is $\mathbb{Z}^{(\infty)}$ for every ℓ . The reason is structural rather than accidental: by Propositions 4.2 and 4.3 the image of $K^0(Y_S)$ in $K_0(\mathcal{A}_{K,S})$ is the group of coinvariants $C(Y_S, \mathbb{Z})/\sum_{\mathfrak{p}} (1 - \alpha_{\mathfrak{p}*})C(Y_S, \mathbb{Z})$, and a functional on $C(Y_S, \mathbb{Z})$ descends to it exactly when it is α_* -invariant, whereas the arithmetically meaningful functionals — the KMS_{β} measures — satisfy $\omega \circ \alpha_{\mathfrak{p}*} = N\mathfrak{p}^{-\beta}\omega$ with $N\mathfrak{p}^{-\beta} \neq 1$. The passage to K_0 annihilates precisely the data of interest. This is taken up in Chapter 4 of the series.

5. Index theory

REMARK 5.1 (Nested prime sets). For $S_1 \subset S_2$ the embedding $\mathcal{A}_{K,S_1} \hookrightarrow \mathcal{A}_{K,S_2}$, $f \mapsto f \circ \text{pr}$, $\mu_{\mathfrak{a}} \mapsto \mu_{\mathfrak{a}}$, of [Ch. 1, Prop. 2.7] requires an I_{S_1} -equivariant continuous surjection $\text{pr} : Y_{S_2} \rightarrow Y_{S_1}$. By (1.1), Y_S is a disjoint union over Cl_S^+ of copies of $\widehat{\mathcal{O}}_S/\overline{\mathcal{O}_{K,+}^{\times}}$, and $\text{Cl}_{S_1}^+$ is a *subgroup* of $\text{Cl}_{S_2}^+$, not a quotient. The coordinate projection $\widehat{\mathcal{O}}_{S_2} \rightarrow \widehat{\mathcal{O}}_{S_1}$ therefore descends canonically exactly when $\text{Cl}_{S_2}^+ = \text{Cl}_{S_1}^+$, i.e. when the primes of $S_2 \setminus S_1$ have classes in $\text{Cl}_{S_1}^+$; this holds for $K = \mathbb{Q}$, for every S_1 generating the narrow class group, and for all $S_1 \subseteq S_2$ inside a fixed S once S_1 is large enough. Otherwise pr exists but involves the choice of a $\text{Cl}_{S_1}^+$ -equivariant retraction $\text{Cl}_{S_2}^+ \rightarrow \text{Cl}_{S_1}^+$, and the words “ $G_S = \varprojlim G_{S_k}$ ” in [Ch. 1, Prop. 2.7] must be read with this qualification (the uses made of that proposition in [Ch. 1] are over $\mathbb{Q}$). *Throughout this section we assume $\text{Cl}_{S_2}^+ = \text{Cl}_{S_1}^+$, so that pr is canonical.*

Let $S_1 \subset S_2$, let φ_2 be a KMS_β state of $\mathcal{A}_{K,S_2}$ with $\zeta_{K,S_2}(\beta) = \infty$, and put $M_{S_i} = \pi_{\varphi_2}(\mathcal{A}_{K,S_i})''$, where $\mathcal{A}_{K,S_1} \subseteq \mathcal{A}_{K,S_2}$ via Remark 5.1.

PROPOSITION 5.2 (The conditional expectation exists). $\sigma_t^{\varphi_2} = \sigma_{-\beta t}$ leaves $\mathcal{A}_{K,S_1}$ globally invariant, so by Takesaki's theorem [124] there is a unique faithful normal conditional expectation $E : M_{S_2} \rightarrow M_{S_1}$ with $\varphi_2 \circ E = \varphi_2$. If φ_2 is extremal, its restriction to $\mathcal{A}_{K,S_1}$ need not be, so M_{S_1} need not be a factor.

PROOF. σ_t fixes $C(Y_{S_2}) \supseteq C(Y_{S_1})$ pointwise and scales each μ_a , so preserves $\mathcal{A}_{K,S_1}$; the GNS vector of a KMS state is separating, so φ_2 is faithful on M_{S_2} with modular group $\sigma_{-\beta t}$. Extremality is not preserved by restriction because the simplices are indexed by $\widehat{\Xi_\beta(S_2)}$ and $\widehat{\Xi_\beta(S_1)}$ respectively. $\square$

We write $\text{Ind}(F) \in [1, \infty]$ for the Pimsner–Popa index of a faithful normal conditional expectation $F : M_{S_2} \rightarrow M_{S_1}$, the inverse of the best constant λ with $F(x) \geq \lambda x$ for all $x \geq 0$ [103], and $[M_{S_2} : M_{S_1}]_0 = \inf_F \text{Ind}(F)$ for the Kosaki–Longo index [72, 84].

THEOREM 5.3 (Prime inclusions have infinite index). Let $\mathfrak{p} \in S_2 \setminus S_1$ and $e_k = \mathbf{1}_{\{v_{\mathfrak{p}}=k\}} = \mathbf{1}_{\mathfrak{p}^k Y_{S_2}} - \mathbf{1}_{\mathfrak{p}^{k+1} Y_{S_2}}$.

- (1) For every KMS_β state φ_2 , the e_k are pairwise orthogonal projections in $M'_{S_1} \cap M_{S_2}$ with $\sum_k e_k = 1$ and $\varphi_2(e_k) = (1 - N\mathfrak{p}^{-\beta})N\mathfrak{p}^{-k\beta} > 0$; in particular $M'_{S_1} \cap M_{S_2}$ contains a copy of $\ell^\infty(\mathbb{N})$.
- (2) Let $\varphi_2 = \varphi_{\text{sym}}$ be the symmetric KMS_β state, so that in the coordinates of [Ch. 1, Lem. 2.10] its measure is $\nu = \pi_\beta \otimes \text{Haar}$. Then $\text{Ind}(F) = \infty$ for every faithful normal conditional expectation $F : M_{S_2} \rightarrow M_{S_1}$, hence $[M_{S_2} : M_{S_1}]_0 = \infty$, already when a single prime is added.

PROOF. (1) For $\mathfrak{q} \in S_1$, multiplication by $\mathfrak{q}$ does not change $v_{\mathfrak{p}}$, so $\mathfrak{q}\{v_{\mathfrak{p}} \geq k\} = \{v_{\mathfrak{p}} \geq k\} \cap \mathfrak{q}Y_{S_2}$ and

$$\mu_{\mathfrak{q}} \mathbf{1}_{\{v_{\mathfrak{p}} \geq k\}} = \alpha_{\mathfrak{q}}(\mathbf{1}_{\{v_{\mathfrak{p}} \geq k\}}) \mu_{\mathfrak{q}} = \mathbf{1}_{\{v_{\mathfrak{p}} \geq k\}} \mathbf{1}_{\mathfrak{q}Y_{S_2}} \mu_{\mathfrak{q}} = \mathbf{1}_{\{v_{\mathfrak{p}} \geq k\}} \mu_{\mathfrak{q}},$$

using $\mathbf{1}_{\mathfrak{q}Y} \mu_{\mathfrak{q}} = \mu_{\mathfrak{q}} \mu_{\mathfrak{q}}^* \mu_{\mathfrak{q}} = \mu_{\mathfrak{q}}$; taking adjoints gives commutation with $\mu_{\mathfrak{q}}^*$, and commutation with $C(Y_{S_1})$ is automatic. The $\mathbf{1}_{\{v_{\mathfrak{p}} \geq k\}}$ therefore commute with $\mathcal{A}_{K,S_1}$, hence with M_{S_1} , and so do their differences e_k . The values $\varphi_2(e_k)$ come from $\varphi_2(\mathbf{1}_{\mathfrak{p}^k Y}) = N\mathfrak{p}^{-k\beta}$ [Ch. 1, Prop. 2.8], and $\sum_k e_k = \mathbf{1}_{\{v_{\mathfrak{p}} < \infty\}} = 1$ in M_{S_2} because $\varphi_2(v_{\mathfrak{p}} \geq k) \rightarrow 0$.

(2) Let $\varphi_2 = \varphi_{\text{sym}}$. The GNS space is $H = L^2(\mathcal{R}, \tilde{\nu})$, $\mathcal{R}$ the orbit relation of I_{S_2} on (Y_{S_2}, ν) and $\tilde{\nu}$ its lift, $M_{S_2} = W^*(\mathcal{R}, \nu)$, and e_k acts by multiplication by $\mathbf{1}_{\{v_{\mathfrak{p}}=k\}}$ in the first variable [Ch. 1, Prop. 6.1]. Fix $k, j \geq 0$ and let $T : \{v_{\mathfrak{p}} = k\} \rightarrow \{v_{\mathfrak{p}} = j\}$, $T(v, h) = (v + (j - k)e_{\mathfrak{p}}, h)$, in the coordinates $(v, h) \in V \times G_{S_2}$: T moves the $\mathfrak{p}$ -valuation and nothing else. It is a homeomorphism between clopen sets with $\nu \circ T = N\mathfrak{p}^{-(j-k)\beta} \nu$ on its domain, since ν is a product measure with geometric $\mathfrak{p}$ -marginal and Haar G_{S_2} -marginal; and it commutes with the maps $(v, h) \mapsto (v + e_{\mathfrak{q}}, \sigma_{\mathfrak{q}} h)$, $\mathfrak{q} \in S_1$, generating the action of I_{S_1} . Hence T maps $\mathcal{R}$ -classes to $\mathcal{R}$ -classes, and

$$(W\xi)(y, y') = N\mathfrak{p}^{(j-k)\beta/2} \xi(T^{-1}y, T^{-1}y'), \quad y \in \{v_{\mathfrak{p}} = j\},$$

is a partial isometry on H with $W^*W = e_k$ and $WW^* = e_j$. Because T commutes with the I_{S_1} -action, does not move the coordinates seen by $C(Y_{S_1}) \circ \text{pr}$, and has constant Radon–Nikodym derivative, W commutes with $\pi(\mathcal{A}_{K,S_1})$, hence $W \in M'_{S_1}$ (the commutant in $B(H)$; W need not lie in M_{S_2}).

Now let F be a faithful normal conditional expectation with $\text{Ind}(F) = \lambda^{-1} < \infty$, so $F(e_k) \geq \lambda e_k$. Since e_k commutes with M_{S_1} , $F(e_k)$ lies in $Z(M_{S_1})$ and commutes with e_k ; on the range of

e_k it is $\geq \lambda$, so $e_k \leq q_k := \mathbf{1}_{[\lambda,1]}(F(e_k)) \in M_{S_1}$, and $\lambda q_k \leq F(e_k)$ gives $\varphi_2(q_k) \leq \lambda^{-1} \varphi_2(F(e_k))$. For $q \in M_{S_1}$ with $qe_k = e_k$ we get, using $W \in M'_{S_1}$,

$$qe_j = qWW^* = WqW^* = W(qe_k)W^* = We_kW^* = e_j \quad \text{for every } j,$$

so $q = \sum_j e_j q = 1$. Applied to $q = q_k$: $1 = \varphi_2(q_k) \leq \lambda^{-1} \varphi_2(F(e_k))$, i.e. $\varphi_2(F(e_k)) \geq \lambda$ for every k . But $F(e_k) \in Z(M_{S_1})$ and $\sum_k F(e_k) = 1$, so $\sum_k \varphi_2(F(e_k)) = 1$, and $\varphi_2(F(e_k)) \geq \lambda > 0$ for infinitely many k is impossible. $\square$

REMARK 5.4 (Scope). Part (1) holds for every KMS state; part (2) uses the product form of ν only to build W . For a non-symmetric extremal state the fibre measure ρ_ν of [Ch. 1, Lem. 2.10] is Haar measure on a coset that moves by $\sigma_{\mathfrak{p}}$ when $v_{\mathfrak{p}}$ increases, so T no longer preserves the measure class, and we have not decided whether every expectation has infinite index in that case; the symmetric state is extremal, and M_{S_2} a factor, precisely when $\Xi_\beta(S_2) = \{1\}$, e.g. for $S_2 = P_K$ and $\beta \leq 1$.

REMARK 5.5 (Why no index formula can exist). An index formula in $\mathbf{Np}$, a local unit index and $[\text{Cl}_{S_2}^+ : \text{Cl}_{S_1}^+]$ would combine finite quantities. It cannot exist because enlarging S enlarges the *base space*: Y_{S_2} carries a new $\mathfrak{p}$ -adic coordinate, and the valuation $v_{\mathfrak{p}}$ alone, before any unit or class-group data enters, already contributes an infinite-dimensional commuting algebra. Those finite invariants do appear in this theory — in Theorem 3.3 and in (1.1) — but not in an index.

PROPOSITION 5.6 (No Galois correspondence with prime sets). *In the situation of Theorem 5.3(2) let $S_2 = S_1 \cup \{\mathfrak{p}\}$, so there are exactly two intermediate prime sets. Then $\text{Lat}(M_{S_1} \subset M_{S_2})$ has cardinality at least $2^{\aleph_0}$.*

PROOF. For a partition $\mathcal{P}$ of $\mathbb{N}$ let $A_{\mathcal{P}} \subseteq W^*(e_k : k \geq 0) \cong \ell^\infty(\mathbb{N})$ be the subalgebra of functions constant on the blocks; it lies in the relative commutant, so $N_{\mathcal{P}} = M_{S_1} \vee A_{\mathcal{P}}$ is intermediate. We recover $\mathcal{P}$ from $N_{\mathcal{P}}$: an element $x = \sum_k c_k e_k$ of $W^*(e_k)$ lies in $N_{\mathcal{P}}$ iff for every block B the element $\sum_{k \in B} c_k e_k$ lies in $M_{S_1} \mathbf{1}_B$, i.e. equals $m \mathbf{1}_B$ for some $m \in M_{S_1}$; conjugating by the partial isometries W of the proof of Theorem 5.3 gives $c_j e_j = m e_j = W(m e_k) W^* = c_k e_j$ for $j, k \in B$, so $W^*(e_k) \cap N_{\mathcal{P}} = A_{\mathcal{P}}$. There are $2^{\aleph_0}$ partitions. $\square$

We now locate the finite index. Fix S and β with Ξ_β finite, and write $G := \widehat{\Xi_\beta}$, φ_{sym} for the G_S -invariant KMS_β state, and φ_ε , $\varepsilon \in G$, for the extreme points.

LEMMA 5.7 (Central decomposition).

$$(5.1) \quad M_{\text{sym}} := \pi_{\varphi_{\text{sym}}}(\mathcal{A}_{K,S})'' \cong \bigoplus_{\varepsilon \in G} M_\varepsilon, \quad M_\varepsilon := \pi_{\varphi_\varepsilon}(\mathcal{A}_{K,S})'',$$

with centre $Z(M_{\text{sym}}) \cong \mathbb{C}^G$ spanned by projections z_ε , each M_ε a factor, and $\omega_{\varphi_{\text{sym}}}(z_\varepsilon) = |G|^{-1}$. The action θ of G_S extends to M_{sym} , factors through $G_S/\Xi_\beta^\perp \cong G$, and permutes the z_ε simply transitively.

PROOF. By [Ch. 1, Thm. 3.6] the KMS_β simplex is $\text{Prob}(G_S/\Xi_\beta^\perp)$, so its extreme points are indexed by $G = \widehat{\Xi_\beta}$ and correspond to mutually singular measures ν_ε ; hence the GNS representations are pairwise disjoint and each M_ε is a factor. The symmetric state is the uniform mixture $\varphi_{\text{sym}} = |G|^{-1} \sum_\varepsilon \varphi_\varepsilon$, so $\pi_{\varphi_{\text{sym}}} = \bigoplus_\varepsilon \pi_{\varphi_\varepsilon}$, giving (5.1), the centre, and $\omega_{\varphi_{\text{sym}}}(z_\varepsilon) = |G|^{-1}$. By [Ch. 1, Thm. 3.6(3)] the action of G_S on the simplex factors through $G_S/\Xi_\beta^\perp$ and permutes the extreme points simply transitively; it therefore permutes the z_ε the same way. $\square$

THEOREM 5.8 (The finite-index inclusion). *Put $D := M_{\varepsilon_0}$ for any $\varepsilon_0 \in G$. The isomorphisms $\theta_g : M_{\varepsilon_0} \rightarrow M_{g\varepsilon_0}$ identify the inclusion $M_{\text{sym}}^G \subseteq M_{\text{sym}}$ with*

$$(5.2) \quad D \otimes 1 \subseteq D \otimes \ell^\infty(G) = \bigoplus_{g \in G} D.$$

Consequently $M_{\text{sym}}^G \cong D$ is a factor,

$$(5.3) \quad [M_{\text{sym}} : M_{\text{sym}}^G]_0 = |G| = |\Xi_\beta|, \quad M_{\text{sym}}^G \cap M_{\text{sym}} = Z(M_{\text{sym}}) \cong \mathbb{C}^{|G|},$$

the index being realized by $E(x) = |G|^{-1} \sum_g \theta_g(x)$; and the intermediate von Neumann algebras are exactly the $D \otimes \ell^\infty(G/\mathcal{P})$, $\mathcal{P}$ a partition of G , so $\text{Lat}(M_{\text{sym}}^G \subset M_{\text{sym}})$ is the partition lattice of G . The subgroup lattice of G embeds in it, order-reversingly, by $H \mapsto M_{\text{sym}}^H$, which corresponds to the partition of G into cosets of H ; for $|G| \geq 3$ the embedding is proper.

PROOF. G permutes the summands of (5.1) simply transitively, so an invariant element is determined by one component and $M_{\text{sym}}^G \cong D$ embedded diagonally via the θ_g , which is (5.2). In (5.2) the averaging map is $D \otimes E_0$ with E_0 the uniform state on $\ell^\infty(G)$, whose Pimsner–Popa index is $|G|$; any other expectation is $D \otimes E_\mu$ with μ a faithful probability on G and index $\max_g \mu(g)^{-1} \geq |G|$, so the Kosaki–Longo index is $|G|$. An element of M_{sym} commuting with the diagonal copy of the factor D is scalar in each summand, giving $\mathbb{C}^{|G|}$. By the Ge–Kadison splitting theorem [47], every von Neumann algebra between $D \otimes 1$ and $D \otimes \ell^\infty(G)$ is $D \otimes B$ for a von Neumann subalgebra $B \subseteq \ell^\infty(G)$, and the unital subalgebras of $\ell^\infty(G)$ are the $\ell^\infty(G/\mathcal{P})$. The H -invariant elements are those constant on the H -orbits, i.e. on the cosets of H ; the partition $\{1\}, G \setminus \{1\}$ of a group of order ≥ 3 is not a coset partition. $\square$

REMARK 5.9 (Not a subfactor invariant). The inclusion (5.2) is the trivial inclusion of the factor D tensored with $\mathbb{C} \subseteq \ell^\infty(G)$: its Jones tower is $D \otimes (\mathbb{C} \subseteq \ell^\infty(G) \subseteq M_{|G|}(\mathbb{C}) \subseteq \dots)$, and its intermediate algebras are governed by partitions rather than by subgroups. It is therefore *not* of the form $M^G \subset M$ for an outer action on a factor, and it carries no fusion category of bimodules beyond the data of the finite set G . The arithmetic content of the symmetry breaking is entirely in the number $|\Xi_\beta|$ and in the way G_S permutes the summands; this is what Corollary 5.11 quantifies. Along a cascade with $\Xi_\beta \cong (\mathbb{Z}/2\mathbb{Z})^{A_\beta}$ [Ch. 1, Prop. 4.16], crossing a threshold β_n doubles the number of summands and doubles the index.

LEMMA 5.10 (Relative entropy of the phases). *With the notation of Lemma 5.7, and S the relative entropy,*

$$S(\omega_{\varphi_\varepsilon} \parallel \omega_{\varphi_{\text{sym}}}) = \log |G|,$$

and the whole value is carried by the centre: the contribution of each summand M_ε vanishes. For $\varepsilon \neq \varepsilon'$ one has $S(\omega_{\varphi_\varepsilon} \parallel \omega_{\varphi_{\varepsilon'}}) = +\infty$.

PROOF. For a direct sum $M = \bigoplus_i M_i$ with normal states $\omega = \bigoplus \lambda_i \omega_i$ and $\psi = \bigoplus \mu_i \psi_i$,

$$S(\omega \parallel \psi) = \sum_i \lambda_i \log \frac{\lambda_i}{\mu_i} + \sum_i \lambda_i S(\omega_i \parallel \psi_i),$$

with the conventions $0 \log 0 = 0$ and $\lambda \log(\lambda/0) = +\infty$ for $\lambda > 0$. Apply this to (5.1) with $\lambda = \delta_\varepsilon$ and $\mu_\delta = |G|^{-1}$ for every δ , which is Lemma 5.7. On the summand ε the two states induce the *same* normal state of M_ε , so that term vanishes; on the others $\lambda_\delta = 0$. Hence $S = \log |G|$ and the second sum is identically zero, which is the assertion that the relative entropy is purely central. For $\varepsilon \neq \varepsilon'$ the two states are supported on orthogonal central projections and the first sum is $\log(1/0) = +\infty$. $\square$

COROLLARY 5.11 (Index staircase equals information staircase).

$$\log[M_{\text{sym}} : M_{\text{sym}}^G]_0 = |A_\beta| \log 2 = S(\omega_{\varphi_\varepsilon} \| \omega_{\varphi_{\text{sym}}}),$$

by (5.3) and Lemma 5.10. The subfactor index and the relative entropy are the same staircase.

The coincidence is not an accident, but neither is it deep: both sides count phases. The index of $M_{\text{sym}}^G \subset M_{\text{sym}}$ is the number $|G|$ of extremal KMS_β states, by (5.2); and the relative entropy of one extremal state against the uniform mixture of $|G|$ mutually singular states is $\log |G|$, the information needed to name which phase one is in. Each threshold β_n doubles the number of phases, so each adds $\log 2$ to both quantities. What the identity does say is that the operator-algebraic and the information-theoretic bookkeeping of the symmetry breaking agree exactly, with no correction term from the type III structure — a point made independently in Lemma 5.10, where the relative entropy is shown to be purely central.

6. The control experiment: restoring the additive part

To confirm that Lemma 1.1 is the cause of §§2–5, we restore the missing ingredient.

REMARK 6.1 (The additive group is $\mathcal{O}_K$). The ring of S -integers $\mathcal{O}_{K,S}$ inverts the primes of S , so its elements have poles at S and it is not contained in $\widehat{\mathcal{O}}_S$. The correct additive part is $\mathcal{O}_K$, dense in $\widehat{\mathcal{O}}_S$ by the Chinese remainder theorem. We let J_S act on $\widehat{\mathcal{O}}_S$ through a fixed choice of uniformizers $\pi_{\mathfrak{p}} \in K_{\mathfrak{p}}$, $\mathfrak{a} = \prod \mathfrak{p}^{a_{\mathfrak{p}}}$ acting by $x \mapsto \pi_{\mathfrak{a}} x$ with $(\pi_{\mathfrak{a}} x)_{\mathfrak{p}} = \pi_{\mathfrak{p}}^{a_{\mathfrak{p}}} x_{\mathfrak{p}}$, and set $\mathcal{A}_{K,S}^{\text{aff}} = C(\widehat{\mathcal{O}}_S) \rtimes (\mathcal{O}_K \rtimes J_S)$, the semigroup crossed product by the maps $x \mapsto \pi_{\mathfrak{a}} x + b$. Over $\mathbb{Q}$ the choice $\pi_{\ell} = \ell$ is canonical; in general everything below holds for every choice. (Without the additive part, the choice is absorbed by the G_S -coordinate of Y_S ; with it, the algebra is that of Cuntz–Li [34] and Cuntz–Deninger–Laca [33] localized at S .)

LEMMA 6.2 (Finite covers). *For every $\mathfrak{p} \in S$, $\widehat{\mathcal{O}}_S = \bigsqcup_{a \in \mathcal{O}_K/\mathfrak{p}} (a + \mathfrak{p}\widehat{\mathcal{O}}_S)$, a partition into $N_{\mathfrak{p}}$ clopen sets. The isometries $\mu_{(a,\mathfrak{p})}$ therefore have mutually orthogonal ranges and satisfy the Cuntz relation*

$$(6.1) \quad \sum_{a \in \mathcal{O}_K/\mathfrak{p}} \mu_{(a,\mathfrak{p})} \mu_{(a,\mathfrak{p})}^* = 1,$$

and the tight spectrum is all of $\widehat{\mathcal{O}}_S$: no collapse occurs, and $\mathcal{A}_{K,S}^{\text{aff}}$ is its own boundary quotient, $\partial \mathcal{A}_{K,S}^{\text{aff}} = \mathcal{A}_{K,S}^{\text{aff}}$.

PROOF. $\mathfrak{p}\widehat{\mathcal{O}}_S$ is open of index $N_{\mathfrak{p}}$ and $\mathcal{O}_K$ surjects onto $\widehat{\mathcal{O}}_S/\mathfrak{p}\widehat{\mathcal{O}}_S$ by density. The cover is finite, so the tight relations of Exel [41] impose equality rather than the nested inequality of Lemma 1.1; and (6.1) already holds in $\mathcal{A}_{K,S}^{\text{aff}}$, since the Stacey covariance gives $\mu_{(a,\mathfrak{p})} \mu_{(a,\mathfrak{p})}^* = \mathbf{1}_{a+\mathfrak{p}\widehat{\mathcal{O}}_S}$. $\square$

THEOREM 6.3 (Kirchberg boundary). *The action of $\mathcal{O}_K \rtimes J_S$ on $\widehat{\mathcal{O}}_S$ is minimal and topologically free, and $\partial \mathcal{A}_{K,S}^{\text{aff}}$ is a unital separable nuclear simple purely infinite C^* -algebra satisfying the UCT.*

PROOF. Minimality: $\mathcal{O}_K$ is dense and acts by translations, so every orbit is dense. Topological freeness: a nontrivial $(b, \mathfrak{a})$ with $\mathfrak{a} = 1$ is a nontrivial translation of a group, hence fixed-point free. For $\mathfrak{a} \neq 1$ the fixed points satisfy $(1 - \pi_{\mathfrak{a}})x = b$ componentwise: at $\mathfrak{p} \in \text{supp } \mathfrak{a}$ the element $1 - \pi_{\mathfrak{p}}^{a_{\mathfrak{p}}}$ is a unit, so $x_{\mathfrak{p}}$ is determined; at $\mathfrak{p} \notin \text{supp } \mathfrak{a}$ the equation reads $0 = b$ in $\mathcal{O}_{\mathfrak{p}}$, which forces $b = 0$. So the fixed-point set is empty, or a single point, or (for $b = 0$) the set $\{x : x_{\mathfrak{p}} = 0 \ \forall \mathfrak{p} \in \text{supp } \mathfrak{a}\}$; in every case it is closed with empty interior. The union over the countable semigroup is meagre, hence has empty interior. The groupoid of germs is étale, Hausdorff and amenable, being that of a partial action of the amenable group $K \rtimes I_S$; minimality and topological freeness give simplicity [13]. It is locally contracting: a basic clopen set $V = \{x : x_{\mathfrak{q}} \in a_{\mathfrak{q}} + \mathfrak{q}^{n_{\mathfrak{q}}} \mathcal{O}_{\mathfrak{q}}, \mathfrak{q} \in F\}$ is mapped by

$x \mapsto \pi_{\mathbf{p}}x + c$, $\mathbf{p} \in F$ and $c \in \mathcal{O}_K$ chosen by the Chinese remainder theorem with $c \equiv (1 - \pi_{\mathbf{p}})a_{\mathbf{p}} \pmod{\mathbf{p}^{n_{\mathbf{p}}}}$ and $c \equiv 0 \pmod{\mathbf{q}^{n_{\mathbf{q}}}}$ for $\mathbf{q} \neq \mathbf{p}$, onto a clopen proper subset of V . Hence $\mathcal{A}_{K,S}^{\text{aff}}$ is purely infinite by Anantharaman-Delaroche [2]. Nuclearity and the UCT follow from amenability. $\square$

THEOREM 6.4 (The unit class says little even here). *In $K_0(\partial\mathcal{A}_{K,S}^{\text{aff}})$ one has $(N\mathbf{p} - 1)[1] = 0$ for every $\mathbf{p} \in S$, so the order of $[1]$ divides $d_S = \gcd_{\mathbf{p} \in S}(N\mathbf{p} - 1)$. In particular $[1] = 0$ whenever $d_S = 1$, which holds as soon as S contains a prime of norm 2, or two primes with $\gcd(N\mathbf{p} - 1, N\mathbf{q} - 1) = 1$.*

PROOF. Each $\mu_{(a,\mathbf{p})}$ is an isometry, so $[\mu\mu^*] = [1]$; applying K_0 to (6.1) gives $[1] = N\mathbf{p}[1]$. $\square$

PROPOSITION 6.5 (The Cuntz relation pins the temperature). *For the dynamics $\sigma_t(\mu_{(b,a)}) = N\mathbf{a}^{it}\mu_{(b,a)}$ on $\mathcal{A}_{K,S}^{\text{aff}}$, a KMS_{β} state exists only for $\beta = 1$.*

PROOF. For a KMS_{β} state φ and any of the $N\mathbf{p}$ isometries in (6.1), the KMS condition gives $\varphi(\mu\mu^*) = \varphi(\mu^*\sigma_{i\beta}(\mu)) = N\mathbf{p}^{-\beta}\varphi(\mu^*\mu) = N\mathbf{p}^{-\beta}$; summing (6.1) gives $1 = N\mathbf{p}^{1-\beta}$, so $\beta = 1$. $\square$

EXAMPLE 6.6. Over $\mathbb{Q}$: $S = \{3\}$ gives $d_S = 2$; $S = \{3, 5\}$ gives $\gcd(2, 4) = 2$; $S = \{7, 13\}$ gives $\gcd(6, 12) = 6$; but any S containing 2 gives $d_S = 1$ and $[1] = 0$. A single small prime erases the invariant.

REMARK 6.7 (Two behaviours of the unit class). Theorems 3.3 and 6.4 exhibit opposite behaviours of $[1]$, and the contrast is instructive.

In the multiplicative boundary, $K_0 = \Lambda^{\text{ev}}(I_S^0)$ is torsion-free and $[1] = h_S^+ e_0$ is a nonzero element of divisibility exactly h_S^+ : the narrow class number is stored in *how divisible* the unit is. In the affine boundary the Cuntz relation forces $[1] = N\mathbf{p}[1]$, so $[1]$ is a torsion element of order dividing $\gcd_{\mathbf{p}}(N\mathbf{p} - 1)$ and is usually zero. The mechanism is already visible for the Cuntz algebra $\mathcal{O}_n$, where $K_0 = \mathbb{Z}/(n - 1)$ is generated by $[1]$; taking several primes replaces $n - 1$ by a greatest common divisor, and a single prime of norm 2 collapses it.

So the two systems store arithmetic in $[1]$ in incompatible ways, and only the multiplicative one does so usefully. This is a rare instance in which the nested range projections of Lemma 1.1 are an advantage: they leave K_0 torsion-free, and a torsion-free group can record an integer by divisibility, which a group killed by $\gcd_{\mathbf{p}}(N\mathbf{p} - 1)$ cannot. We have not computed the ambient groups $K_*(\partial\mathcal{A}_{K,S}^{\text{aff}})$ and make no claim about them beyond Theorem 6.4.

7. Assessment

Feature	$\mathcal{A}_{K,S}$ (multiplicative)	$\mathcal{A}_{K,S}^{\text{aff}}$ (affine)
isometries per $\mathbf{p}$	1	$N\mathbf{p}$
range projections	nested	partitioning
tight relation	$\mu_{\mathbf{p}}$ unitary	Cuntz
boundary space	Cl_S^+ , finite	$\widehat{\mathcal{O}}_S$, a Cantor group
boundary algebra	$M_{h_S^+}(C^*(I_S^0))$, type I	Kirchberg (the algebra itself)
trace on the boundary	faithful	none
$[1]$ records	h_S^+	order $\mid \gcd(N\mathbf{p} - 1)$

Everything in the left column follows from Lemma 1.1, and the right column confirms by contrast that the additive cosets are the cause. The same dichotomy governs the admissible inverse temperatures — the Cuntz relation forces $\beta = 1$ on the affine system (Proposition 6.5), whereas the multiplicative system has KMS_{β} states for every $\beta > 0$ [Ch. 1, Thm. 3.6] — and the free variant treated in Chapter 14, where orthogonal ranges again restrict β .

Four questions remain. We have computed $K_*(\mathcal{A}_{K,S})$ only for a single prime, and $K_*(\mathcal{A}_{K,S}^{\text{aff}})$ not at all; by Theorem 6.3 the triple $(K_0, [1], K_1)$ is a complete invariant of the affine algebra and

by Theorem 6.4 the middle entry is usually trivial, so the content lies in the groups. When $\mathcal{O}_{K,+}^\times$ is infinite, whether some proper stratum carries nontrivial isotropy (Remark 2.3) — a Leopoldt-type closure question — and hence contributes further ideals of the kind seen in Theorem 2.5(2), is open. Whether the infinite index of Theorem 5.3 persists at the non-symmetric extremal states (Remark 5.4), and whether these inclusions are discrete in the sense of Izumi–Longo–Popa [63] or admit a continuous decomposition matching the diffuse relative commutant, is open. Finally, the lattice of intermediate subfactors [134] of the symmetry-breaking inclusion is the partition lattice of $\widehat{\Xi}_\beta$ (Theorem 5.8), which remembers Ξ_β only as a finite set; it distinguishes neither K nor S nor the thresholds, and the question of which invariant does is taken up in Chapter 4.

It is worth saying plainly why the blindness recorded in Remark 4.5 is not a computational accident. By Propositions 4.2 and 4.3 the passage from $C(Y_S)$ to $\mathcal{A}_{K,S}$ is, on the image of $K^0(Y_S)$ in K_0 , the passage to the coinvariants of α_* , and a functional survives that quotient exactly when it is α_* -invariant. The arithmetically meaningful functionals are the KMS_β measures, which by [Ch. 1, Prop. 2.8] are eigenvectors, $\omega \circ \alpha_{\mathfrak{p}*} = N\mathfrak{p}^{-\beta}\omega$, with eigenvalue $\neq 1$ for $\beta \neq 0$. Passing to the coinvariants therefore *annihilates precisely the covariant vectors that carry the arithmetic*, by construction rather than by accident. The same mechanism in its measure-theoretic form is, we expect, the reason Ξ_β is an invariant of a measure class rather than of a C^* -algebra: any functor that forgets α_* , or that forces it to act as the identity, loses it.

CHAPTER 4

Invariants, entropy, and the measure class

This chapter is Paper III of the series ([papers/III-invariants/](#)).

SUMMARY. The obstruction group $\Xi_\beta(K, S)$ of [Ch. 1] is a complete invariant of the phase structure of the localized Bost–Connes system. We determine what kind of invariant it is, and the answer organizes everything else: Ξ_β is an invariant of a measure class, not of a C^* -algebra. It is the kernel of the natural map from $\widehat{G}_S$ to the *measurable* first cohomology of the tail equivalence relation taken with respect to π_β , and the whole of its dependence on β sits in that measure class.

Three consequences follow, and they explain a series of otherwise puzzling vanishings. First, at the level of K -theory the KMS simplex is exactly the set of states ω on the pointed ordered group $K_0(C(Y_S)) = C(Y_S, \mathbb{Z})$ satisfying the eigenvalue equations $\omega \circ \alpha_{\mathfrak{p}*} = N\mathfrak{p}^{-\beta}\omega$; and since the image of $K_0(C(Y_S))$ in $K_0(\mathcal{A}_{K,S})$ is the group of α_* -coinvariants, the passage to the crossed product annihilates precisely those eigenvectors. Second, every homotopy-invariant construction is blind: Connes–Thom gives $K_0(\mathcal{B}_{K,S}) = 0$ for the continuous core, so every dual-trace pairing on it vanishes identically; the dual flow acts trivially on $K_*(\mathcal{B}_{K,S})$ because $\mathbb{R}$ is connected; and the groupoid homology of $\mathcal{G}_S$, with or without twists by the modular cocycle, carries no β at all. Third, no dynamical entropy is available: $\alpha_{\mathfrak{a}}$ and the transfer operator $\mathcal{L}_{\mathfrak{a}}$ preserve no KMS_β state and indeed no invariant probability measure exists, in accordance with the type III classification; Voiculescu’s topological entropy of $\alpha_{\mathfrak{a}}$ vanishes because multiplication by $\mathfrak{a}$ is an ultrametric contraction; and the modular flow, which does preserve every KMS state, has zero CNT entropy because it lies in the compact gauge group.

What does see the phase structure is relative. We prove $S(\omega_{\varphi_\varepsilon} \parallel \omega_{\varphi_{\text{sym}}}) = |A_\beta| \log 2$, exactly $\log 2$ per threshold crossed and $+\infty$ once infinitely many have been, and show that the entire value is carried by the centre, so the type III factors contribute nothing. The resulting staircase coincides with the subfactor index staircase of Chapter 3. We close by correcting the thermodynamic reading: all KMS_β states at a given β are equilibrium states on an equal footing, so the condensation of a phase bit is not a Landauer erasure.

Along the way we fix two structural boundaries that are easy to misplace. A KMS_β state with $\beta \neq 0$ is not a trace, so it defines no functional on $K_0(\mathcal{A}_{K,S})$; the K -theoretic locus of the KMS states is the eigenvalue problem on $K_0(C(Y_S))$ instead. And the de la Harpe–Skandalis determinant is natively defined on $\mathcal{U}_0$, which is the subgroup by which $K_1 = \mathcal{U}/\mathcal{U}_0$ is the quotient, so it induces no invariant on K_1 .

1. Introduction: which kind of invariant

We use the notation of [Ch. 1]: $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$ with $\sigma_t(\mu_{\mathfrak{a}}) = N\mathfrak{a}^{it}\mu_{\mathfrak{a}}$, symmetry group G_S , and

$$(1.1) \quad \Xi_\beta(K, S) = \left\{ \chi \in \widehat{G}_S : \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}})|^2 < \infty \right\},$$

the KMS_β simplex being $\text{Prob}(G_S/\Xi_\beta^\perp)$. Write $\mathcal{R}_S$ for the tail equivalence relation on $V = \mathbb{Z}_{\geq 0}^S$ and π_β for the forced product measure of [Ch. 1, Lem. 2.10], $\alpha_{\mathfrak{p}}$ for the endomorphism $f \mapsto \mu_{\mathfrak{p}} f \mu_{\mathfrak{p}}^*$ of $C(Y_S)$, and $\mathcal{B}_{K,S} = \mathcal{A}_{K,S} \rtimes_{\sigma} \mathbb{R}$ for the continuous core.

The organizing statement is Theorem 2.3: Ξ_β is the kernel of a map into a *measurable* cohomology group computed relative to π_β . Since a C^* -algebra, a groupoid and a homotopy functor

all forget a measure class, one should expect them to forget Ξ_β , and §§3–5 confirm this in each case. The point of collecting the vanishings is not that they are surprising individually — several are easy — but that they have a single cause, and that the cause identifies where the invariant does live.

2. Ξ_β is measurable first cohomology

Let $\sigma : \mathbb{Z}^{(S)} \rightarrow G_S$ be the canonical homomorphism, regarded as a cocycle on $\mathcal{R}_S$ via $\sigma(v + a, v) = \sigma_a$.

LEMMA 2.1 (Coboundary dictionary). *Let A be a second countable locally compact abelian group, $\Phi : \mathbb{Z}^{(S)} \rightarrow A$ a homomorphism, $\psi \in \widehat{A}$, and $c_p = \overline{\psi(\Phi(e_p))} \in \mathbb{T}$. The following are equivalent.*

- (1) $\psi \circ \Phi$ is a coboundary on $(V, \pi_\beta, \mathcal{R}_S)$: there is measurable $g : V \rightarrow \mathbb{T}$ with $g(v + e_p) = c_p g(v)$ a.e., for every p ;
- (2) $\sum_p (1 - |m_p|) < \infty$, where $m_p = (1 - t_p)/(1 - c_p t_p)$ and $t_p = Np^{-\beta}$;
- (3) $\sum_p Np^{-\beta} |1 - \psi(\Phi(e_p))|^2 < \infty$.

Moreover, if these hold, then ψ is trivial on the group $E(\Phi)$ of essential values of Φ ; so $E(\Phi) \subseteq \Lambda^\perp$ with $\Lambda = \{\psi : \psi \circ \Phi \text{ a coboundary}\}$.

PROOF. (1) $\Leftrightarrow$ (2) is Propositions 3.3 and 3.4 of [Ch. 1]; those proofs use only that $c_p \in \mathbb{T}$, that the coordinates v_p are independent under π_β with $\mu_p(k) = (1 - t_p)t_p^k$, and that $|g| = 1$. The target group of ψ plays no role. (2) $\Leftrightarrow$ (3) is [Ch. 1, Lem. 3.2] together with $|1 - c_p|^2 = 2(1 - \operatorname{Re} c_p)$ and $|1 - c_p| = |1 - \bar{c}_p|$. For the last statement, a coboundary has no essential value other than 1, and ψ maps essential values of Φ to essential values of $\psi \circ \Phi$ [117]; so $\psi(E(\Phi)) = \{1\}$. $\square$

REMARK 2.2. The converse, that ψ trivial on $E(\Phi)$ makes $\psi \circ \Phi$ a coboundary, holds when Φ is regular in the sense of Schmidt [117], i.e. cohomologous to a cocycle with values in $E(\Phi)$, and then $E(\Phi) = \Lambda^\perp$ by Pontryagin duality. For $A = G_S$ compact this is automatic, and Theorem 2.3 needs only (1) $\Leftrightarrow$ (3). For the cocycle with values in $G_S \times \mathbb{R}$ that governs the factor type, regularity is not known, and [Ch. 2, Prop. 2.2 and Rem. 2.3] obtains from the inclusion $E(\Phi) \subseteq \Lambda^\perp$ an upper bound on the type only.

THEOREM 2.3 (Cohomological description). *For every $\beta > 0$,*

$$(2.1) \quad \Xi_\beta(K, S) = \ker \left(\widehat{G_S} \xrightarrow{\chi \mapsto [\chi \circ \sigma]} H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T}) \right),$$

measurable 1-cohomology relative to the measure class of π_β , in the sense of Feldman–Moore [42].

PROOF. Apply Lemma 2.1 with $A = G_S$, $\Phi = \sigma$: condition (3) is (1.1), and (1) says the class of $\chi \circ \sigma$ vanishes in H_{meas}^1 . $\square$

COROLLARY 2.4 (Where the β -dependence sits). *In (2.1) the objects $\widehat{G_S}$, $\mathcal{R}_S$ and σ are β -independent. The entire dependence on β is carried by the measure class of π_β , through which H_{meas}^1 is computed.*

REMARK 2.5. Corollary 2.4 is the reason for everything below. K -theory is an invariant of a C^* -algebra; groupoid homology is an invariant of a topological groupoid; dynamical entropy is an invariant of a measure-preserving map. None of the three retains a measure class that is merely quasi-invariant. The vanishings of §§3–5 are therefore not failures of ingenuity but category errors, and we record them in that spirit.

3. K -theory: the eigenvalue equation

We first correct a statement announced in an earlier draft.

PROPOSITION 3.1 (KMS states do not act on $K_0(\mathcal{A}_{K,S})$). *Let φ be a KMS_β state with $\beta > 0$ and $S \neq \emptyset$. Then $[p] \mapsto \varphi(p)$ does not define a homomorphism $K_0(\mathcal{A}_{K,S}) \rightarrow \mathbb{R}$.*

PROOF. A positive functional descends to K_0 only if it is constant on Murray–von Neumann classes, i.e. only if $\varphi(vv^*) = \varphi(v^*v)$ for all v , i.e. only if φ is a trace. Take $v = \mu_{\mathbf{p}}$: then $v^*v = 1$ and $vv^* = \mathbf{1}_{\mathbf{p}Y_S}$, so $[1] = [\mathbf{1}_{\mathbf{p}Y_S}]$ in K_0 , while the KMS condition gives $\varphi(1) = 1$ and $\varphi(\mathbf{1}_{\mathbf{p}Y_S}) = N\mathbf{p}^{-\beta} < 1$. $\square$

The remedy is to work on the coefficient algebra, where a KMS state restricts to a measure and hence to a trace.

THEOREM 3.2 (The KMS simplex as an eigenvalue problem). *Let $\alpha_{\mathbf{p}*}$ be the endomorphism $\mathbf{1}_U \mapsto \mathbf{1}_{\mathbf{p}U}$ of the pointed ordered group $(K_0(C(Y_S)), K_0^+, [1]) = (C(Y_S, \mathbb{Z}), C(Y_S, \mathbb{Z})_+, 1)$. Then $\varphi \mapsto \omega_\varphi$, $\omega_\varphi[\mathbf{1}_U] := \varphi(\mathbf{1}_U)$, is an affine bijection*

$$\text{KMS}_\beta(\mathcal{A}_{K,S}) \xrightarrow{\sim} \left\{ \omega \text{ a state of } (C(Y_S, \mathbb{Z}), C(Y_S, \mathbb{Z})_+, 1) : \omega \circ \alpha_{\mathbf{p}*} = N\mathbf{p}^{-\beta} \omega \quad \forall \mathbf{p} \in S \right\}.$$

PROOF. By [Ch. 1, Prop. 2.8] a KMS_β state is $\nu \circ E$ for a unique scaling-invariant ν , and ν is a Borel probability measure on the compact totally disconnected space Y_S , hence determined by its values on clopen sets; conversely a positive normalized finitely additive assignment on the clopen algebra extends to a regular Borel measure. Since $C(Y_S)$ is commutative, ν is a trace and ν_* is a well-defined state of the pointed ordered group. The scaling relation $\nu(\mathbf{a}E) = N\mathbf{a}^{-\beta}\nu(E)$ reads on clopen sets as $\omega(\alpha_{\mathbf{p}*}[\mathbf{1}_U]) = N\mathbf{p}^{-\beta}\omega[\mathbf{1}_U]$, and the relations for $\mathbf{p} \in S$ generate those for all $\mathbf{a} \in J_S$. $\square$

COROLLARY 3.3 (The crossed product annihilates the datum). *In $K_0(\mathcal{A}_{K,S})$ one has $\iota_*[\mathbf{1}_U] = \iota_*[\mathbf{1}_{\mathbf{p}U}]$ for every clopen $U \subseteq Y_S$ and $\mathbf{p} \in S$, since $\mu_{\mathbf{p}}\mathbf{1}_U$ is a partial isometry from $\mathbf{1}_U$ to $\mathbf{1}_{\mathbf{p}U}$. So $\iota_* : C(Y_S, \mathbb{Z}) \rightarrow K_0(\mathcal{A}_{K,S})$ factors through the group of coinvariants $C(Y_S, \mathbb{Z}) / \sum_{\mathbf{p}} (1 - \alpha_{\mathbf{p}*})C(Y_S, \mathbb{Z})$, and a functional on $C(Y_S, \mathbb{Z})$ that factors through $K_0(\mathcal{A}_{K,S})$ satisfies $\omega \circ \alpha_{\mathbf{p}*} = \omega$ for every $\mathbf{p}$. By Theorem 3.2 the KMS_β functionals satisfy instead $\omega \circ \alpha_{\mathbf{p}*} = N\mathbf{p}^{-\beta}\omega$ with $N\mathbf{p}^{-\beta} \neq 1$ for $\beta \neq 0$. Hence no nonzero KMS_β functional factors through $K_0(\mathcal{A}_{K,S})$: passing from the coefficient algebra to the crossed product is exactly the quotient that kills the eigenvectors of interest. (For a single prime the image of ι_* is the whole cokernel $\text{coker}(1 - \alpha_*) = K_0(\mathcal{A}_{K,S})$ and $K_1 = 0$; for several primes the image is the bottom piece of a filtration of K_0 by Koszul homology, computed in Chapter 3.)*

REMARK 3.4. Theorem 3.2 is a statement about the pair $(K_0(C(Y_S)), \alpha_*)$ together with a real parameter, not about a single abelian group. Every functor that forgets α_* , or that forces it to act as the identity, loses the arithmetic; and K_0 of the crossed product does the latter by construction.

4. Homotopy invariants are blind

THEOREM 4.1 (The continuous core). *For every K , S and $\beta > 0$:*

- (1) $K_0(\mathcal{B}_{K,S}) \cong K_1(\mathcal{A}_{K,S})$ and $K_1(\mathcal{B}_{K,S}) \cong K_0(\mathcal{A}_{K,S})$;
- (2) the dual flow $\hat{\sigma}$ induces the identity on $K_*(\mathcal{B}_{K,S})$;
- (3) the dual weight $\hat{\varphi}$ of any KMS_β state, a faithful semifinite trace on $\mathcal{B}_{K,S}$, vanishes on every projection in its domain (in every matrix algebra over $\mathcal{B}_{K,S}$); in particular the induced map $\hat{\varphi}_*$ on K_0 of its domain ideal is zero, and the pairing of $\hat{\varphi}$ with $K_0(\mathcal{B}_{K,S})$ carries no information.

Consequently no invariant built from $(K_*(\mathcal{B}_{K,S}), \hat{\sigma}_*, \text{trace pairings})$ distinguishes systems with different Ξ_β .

PROOF. (1) is the Connes–Thom isomorphism $K_i(\mathcal{A} \rtimes_\sigma \mathbb{R}) \cong K_{i+1}(\mathcal{A})$ [32].

(2) The map $s \mapsto \hat{\sigma}_s$ is a point-norm continuous path of $*$ -automorphisms with $\hat{\sigma}_0 = \text{id}$, hence a homotopy of $*$ -homomorphisms $\mathcal{B} \rightarrow \mathcal{B}$; K -theory is homotopy invariant, so $\hat{\sigma}_{s*} = \text{id}$ for every s .

(3) The defining property of the dual weight of a KMS_β state is that it is a trace scaled by the dual action, $\hat{\varphi} \circ \hat{\sigma}_s = e^{-\beta s} \hat{\varphi}$; in particular its domain is $\hat{\sigma}$ -invariant. Let p be a projection in the domain. Then $s \mapsto \hat{\sigma}_s(p)$ is a norm-continuous path of projections in the domain, so $\hat{\sigma}_s(p) = u_s p u_s^*$ for a unitary u_s of the unitization, and the trace property $\hat{\varphi}(x^*x) = \hat{\varphi}(xx^*)$ gives $\hat{\varphi}(\hat{\sigma}_s(p)) = \hat{\varphi}(p)$. Hence $\hat{\varphi}(p) = e^{-\beta s} \hat{\varphi}(p)$ for every s , and $\hat{\varphi}(p) = 0$. $\square$

REMARK 4.2 (Two mechanisms, one cause). Part (3) is proved here without computing $K_0(\mathcal{B}_{K,S})$, and this is deliberate. By (1), $K_0(\mathcal{B}_{K,S}) \cong K_1(\mathcal{A}_{K,S})$. For $|S| = 1$ this is $\ker(1 - \alpha_*)$ on $C(Y_S, \mathbb{Z})$ by Pimsner’s sequence, and it vanishes: a fixed point of α_* is supported in $\mathfrak{p}^n Y_S$ for every n , hence in the nowhere dense set $\partial Y_S = \bigcap_n \mathfrak{p}^n Y_S$, hence is 0. For $|S| \geq 2$ the group $K_1(\mathcal{A}_{K,S})$ is built from the odd Koszul homology of the commuting family $(1 - \alpha_{\mathfrak{p}*})$, and we have not determined it (Chapter 3). Were it zero, (3) would follow trivially. The argument above shows that (3) holds regardless: the scaling $\hat{\varphi} \circ \hat{\sigma}_s = e^{-s} \hat{\varphi}$ and the homotopy triviality of $\hat{\sigma}_*$ are incompatible with a nonzero pairing, whatever the group is. It is the *connectedness* of $\mathbb{R}$ that kills the pairing, not the vanishing of a K -group — the same mechanism as in Remark 4.3.

REMARK 4.3 (Connected versus totally disconnected). The obstruction in (2) is general: any action of a *connected* group induces the trivial action on K -theory. The modular flow is an action of $\mathbb{R}$, so the scaling structure it carries (namely $\hat{\varphi} \circ \hat{\sigma}_s = e^{-s} \hat{\varphi}$, where the type and the flow of weights live) is invisible to K -theory in principle. By contrast G_S is profinite, hence totally disconnected, and acts nontrivially on $K_0(C(Y_S)) = C(Y_S, \mathbb{Z})$ by permuting clopen sets. Arithmetic symmetry, being Galois-theoretic and so profinite, can register on K -theory; modular dynamics, being a flow, cannot.

PROPOSITION 4.4 (The determinant does not act on K_1). *Let A be unital with a trace τ . The de la Harpe–Skandalis determinant [54] is a homomorphism $\Delta_\tau : \mathcal{U}_0(A) \rightarrow \mathbb{R}/\tau_*(K_0(A))$ on the connected component of the unitary group. Since $K_1(A) = \mathcal{U}_\infty(A)/\mathcal{U}_{\infty,0}(A)$, the domain of Δ_τ is precisely the subgroup by which K_1 is the quotient; Δ_τ descends to $\mathcal{U}_0/\overline{DU}_0$ and does not induce a map $K_1(A) \rightarrow \mathbb{R}$.*

PROOF. Definitional: Δ_τ vanishes on commutators of $\mathcal{U}_0$, and a homomorphism defined on $\mathcal{U}_0$ carries no information about $\mathcal{U}/\mathcal{U}_0$. $\square$

REMARK 4.5 (Parity). A trace is a cyclic 0-cocycle and pairs with K_0 ; pairing with K_1 requires a cyclic 1-cocycle. On the core the natural candidate is $\psi(b_0, b_1) = \hat{\varphi}(b_0 \hat{\delta}(b_1))$ with $\hat{\delta}$ the generator of the dual flow, but cyclicity needs $\hat{\varphi} \circ \hat{\delta} = 0$ whereas the defining property of the dual weight is $\hat{\varphi} \circ \hat{\sigma}_s = e^{-s} \hat{\varphi}$, i.e. $\hat{\varphi} \circ \hat{\delta} = -\hat{\varphi} \neq 0$. The pair $(\hat{\varphi}, \hat{\delta})$ produces a *twisted* cyclic 1-cocycle, consistently with Theorem 2.3.

PROPOSITION 4.6 (Groupoid homology carries no β). $H_*(\mathcal{G}_S; \mathbb{Z}) \cong H_*(I_S; C_c(\tilde{Y}_S, \mathbb{Z}))$ is defined without reference to β , as is any twist by the modular cocycle $c(\mathbf{a}\mathbf{b}^{-1}) = \log \mathbf{N}\mathbf{a} - \log \mathbf{N}\mathbf{b}$, which does not involve β . Hence neither can determine Ξ_β , which varies with β . Twisting the coefficients instead does not help: with $\mathbb{R}_\beta$ denoting $\mathbb{R}$ with I_S acting by $\mathbf{a} \mapsto \mathbf{N}\mathbf{a}^{-\beta}$,

$$H^n(I_S; \mathbb{R}_\beta) = 0 \quad \text{for all } n \geq 0,$$

while H^0 with coefficients in the measures twisted by $e^{-\beta c}$ returns the KMS simplex tautologically.

PROOF. The first statements are immediate: β enters only as the exponent in the Radon–Nikodym derivative, i.e. in the measure class. For the vanishing, pick $\mathfrak{p} \in S$; the generator $\mathfrak{p}$ acts on $\mathbb{R}_\beta$ by $\lambda = N\mathfrak{p}^{-\beta} \neq 1$, and for $\mathbb{Z}$ acting on $\mathbb{R}$ by $\lambda \neq 1$ one has $H^0 = \ker(\lambda - 1) = 0$ and $H^1 = \mathbb{R}/(\lambda - 1)\mathbb{R} = 0$; the Koszul complex for $I_S = \mathbb{Z}^n$ is then exact, since one of its generators acts by an automorphism $\lambda - 1$ of the coefficients, and for infinite S the Milnor sequence of the increasing union of finite-rank subgroups gives the same vanishing. The last statement is the definition of quasi-invariance together with [Ch. 1, Prop. 2.8]. $\square$

5. Dynamical entropy

PROPOSITION 5.1 (Scaling, not invariance). *For a KMS_β state φ , $\mathfrak{a} \in J_S$ and $x \in \mathcal{A}_{K,S}$,*

$$\varphi(\alpha_{\mathfrak{a}}(x)) = N\mathfrak{a}^{-\beta}\varphi(x), \quad \varphi(\mathcal{L}_{\mathfrak{a}}(x)) = N\mathfrak{a}^{\beta}\varphi(x \mathbf{1}_{\mathfrak{a}Y_S}),$$

where $\mathcal{L}_{\mathfrak{a}}(x) = \mu_{\mathfrak{a}}^* x \mu_{\mathfrak{a}}$ is the unital completely positive transfer operator. Neither map preserves φ for $\mathfrak{a} \neq (1)$, $\beta > 0$.

PROOF. $\sigma_{i\beta}(\mu_{\mathfrak{a}}) = N\mathfrak{a}^{-\beta}\mu_{\mathfrak{a}}$ and $\sigma_{i\beta}(\mu_{\mathfrak{a}}^*) = N\mathfrak{a}^{\beta}\mu_{\mathfrak{a}}^*$. The KMS condition $\varphi(uv) = \varphi(v\sigma_{i\beta}(u))$ with $u = \mu_{\mathfrak{a}}$, $v = x\mu_{\mathfrak{a}}^*$ gives the first, using $\mu_{\mathfrak{a}}^*\mu_{\mathfrak{a}} = 1$; with $u = \mu_{\mathfrak{a}}^*$, $v = x\mu_{\mathfrak{a}}$ it gives the second. For the failure of the second, take $x = \mathbf{1}_E$ with $E = Y_S \setminus \mathfrak{a}Y_S$: the right side is 0 while $\varphi(\mathbf{1}_E) = 1 - N\mathfrak{a}^{-\beta} > 0$. $\square$

THEOREM 5.2 (No invariant measure, hence no metric entropy). *For $S \neq \emptyset$ there is no I_S -invariant Borel probability measure on $\tilde{Y}_S$ other than those concentrated on the finite boundary ∂Y_S , and no such measure is a KMS_β state for $\beta > 0$. Consequently the Connes–Størmer, CNT and Sauvageot–Thouvenot entropies [23, 22, 116] $h_\varphi(\alpha_{\mathfrak{a}})$ and $h_\varphi(\mathcal{L}_{\mathfrak{a}})$ are undefined for every KMS_β state.*

PROOF. An invariant ν satisfies $\nu(\mathfrak{a}^{-1}Y_S) = \nu(Y_S)$ for all $\mathfrak{a} \in J_S$, and $\tilde{Y}_S = \bigcup_{\mathfrak{a}} \mathfrak{a}^{-1}Y_S$, so $\nu(Y_S) = 1$; then $\nu(\mathfrak{p}^k Y_S) = 1$ for all k and ν is concentrated on $\bigcap_k \mathfrak{p}^k Y_S$; intersecting over $\mathfrak{p} \in S$ concentrates ν on $\partial Y_S = \{[0, g]\} \cong G_S/r(U_S) \cong \text{Cl}_S^+$ [Ch. 1, (2.1)], a finite set. A KMS_β state has $\varphi(\mathbf{1}_{\mathfrak{p}Y_S}) = N\mathfrak{p}^{-\beta} < 1$ and so is not of this form. All three entropy definitions require the state to be invariant under the map whose entropy is computed; Proposition 5.1 shows no KMS state is. $\square$

REMARK 5.3 (Consistency with type III). Theorem 5.2 restates the type classification. A type III factor has no semifinite trace, the associated relation has no invariant measure in the measure class, and Kolmogorov–Sinai theory is unavailable by construction.

THEOREM 5.4 (Topological entropy vanishes). *Let $T_{\mathfrak{a}} : Y_S \rightarrow Y_S$, $[x, g] \mapsto [\mathfrak{a}x, \sigma_{\mathfrak{a}}g]$, so that $\mathcal{L}_{\mathfrak{a}}(f) = \mu_{\mathfrak{a}}^* f \mu_{\mathfrak{a}} = f \circ T_{\mathfrak{a}}$ on $C(Y_S)$. Then $h_{\text{top}}(T_{\mathfrak{a}}) = 0$. On the quotient of $\mathcal{A}_{K,S}$ in which every $\mu_{\mathfrak{p}}$ is unitary (the boundary quotient of Chapter 3) the map $\alpha_{\mathfrak{a}}$ becomes the inner automorphism $\text{Ad } \mu_{\mathfrak{a}}$, whose Voiculescu–Brown topological entropy [132, 14] is 0.*

PROOF. Y_S is the quotient of $\hat{\mathcal{O}}_S \times G_S$ by U_S , and $T_{\mathfrak{a}}$ lifts to $\tilde{T}_{\mathfrak{a}}(x, g) = (\mathfrak{a}x, \sigma_{\mathfrak{a}}g)$. Put

$$d((x, g), (x', g')) = \sup_{\mathfrak{p}} c_{\mathfrak{p}} N\mathfrak{p}^{-v_{\mathfrak{p}}(x_{\mathfrak{p}} - x'_{\mathfrak{p}})} + d_G(g, g'),$$

with $c_{\mathfrak{p}}$ summable and d_G an invariant metric on G_S ; one has $v_{\mathfrak{p}}(\mathfrak{a}(x - x')) = v_{\mathfrak{p}}(\mathfrak{a}) + v_{\mathfrak{p}}(x - x')$ and $d_G(\sigma_{\mathfrak{a}}g, \sigma_{\mathfrak{a}}g') = d_G(g, g')$, so $\tilde{T}_{\mathfrak{a}}$ is 1-Lipschitz. Hence the Bowen metrics satisfy $d_n = d$, the (n, ϵ) -separated sets are the ϵ -separated sets, their cardinality is independent of n , and $h_{\text{top}}(\tilde{T}_{\mathfrak{a}}) = 0$; a factor of a zero-entropy system has zero entropy. For the last statement, $\text{Ad } \mu_{\mathfrak{a}}$ is inner, and inner automorphisms have zero topological entropy [14]. $\square$

REMARK 5.5 (Why $\log \mathbf{Na}$ does not occur). The map with entropy $\log \mathbf{Np}$ is the digit shift $x \mapsto (x - a_0)/\mathbf{p}$, which is $\mathbf{Np}$ -to-one; its definition requires *subtracting* the last digit, i.e. the additive structure. The multiplicative system has one isometry per prime, with range projections $\mathbf{1}_{\mathbf{p}Y_S}$ that are nested rather than orthogonal ($\mathbf{1}_{\mathbf{p}Y_S}\mathbf{1}_{\mathbf{q}Y_S} = \mathbf{1}_{\mathbf{pq}Y_S} \neq 0$), hence no Cuntz relation and no $\log n$ (Chapter 3). The unavailability of $\log \mathbf{Na}$ is the same phenomenon as the failure of the Cuntz relation there.

THEOREM 5.6 (The modular flow has zero entropy). *Let $\iota : \mathbb{R} \rightarrow \widehat{I_S} = \prod_{\mathbf{p} \in S} \mathbb{T}$, $\iota(t)_{\mathbf{p}} = \mathbf{Np}^{it}$, and let γ be the gauge action. Then $\sigma_t = \gamma_{\iota(t)}$, so $\{\sigma_t\}$ has compact closure in $\text{Aut}(\mathcal{A}_{K,S})$; and an automorphism with relatively compact orbit closure has zero CNT entropy. Hence $h_\varphi(\sigma_t) = 0$ for every KMS $_\beta$ state and every t , extremal or symmetric alike.*

PROOF. ι is a continuous homomorphism and both σ_t and $\gamma_{\iota(t)}$ fix $C(Y_S)$ pointwise and send $\mu_{\mathbf{p}} \mapsto \mathbf{Np}^{it}\mu_{\mathbf{p}}$; the gauge action of the compact group $\widehat{I_S}$ is continuous for the pointwise norm topology, so its image is compact. For the entropy, fix a finite-dimensional channel γ and $\epsilon > 0$; by relative compactness every $\sigma_{kt} \circ \gamma$ is within ϵ of one of finitely many $\theta_j \circ \gamma$, $j \leq m(\epsilon)$. Since H_φ is symmetric, unchanged by repetition, subadditive, bounded by $\log \dim$ on a single channel, and continuous with a modulus independent of the number of arguments,

$$H_\varphi(\gamma, \sigma_t \gamma, \dots, \sigma_{(n-1)t} \gamma) \leq m(\epsilon) \log \dim + n \delta(\epsilon),$$

so $\limsup_n \frac{1}{n} H_\varphi \leq \delta(\epsilon) \rightarrow 0$. Every KMS state is σ -invariant, so the hypothesis holds. The properties of H_φ used here are those established in [22]; see also [102] for the vanishing of CNT entropy on automorphisms lying in a compact group. $\square$

REMARK 5.7. This is the noncommutative form of the fact that rotations have zero entropy. Here the angle is $(\log \mathbf{Np})_{\mathbf{p} \in S}$; over $\mathbb{Q}$ these are $\mathbb{Q}$ -linearly independent and Kronecker's theorem makes $\iota(\mathbb{R})$ dense in $\widehat{I_S}$, and in general $\iota(\mathbb{R})$ is dense in the annihilator of $\{\mathbf{a} \in I_S : \mathbf{Na} = 1\}$. The modular flow is an irrational winding on an infinite torus, dense hence ergodic on gauge orbits, compact hence entropy-free. Neither property detects Ξ_β , which by Corollary 2.4 is not an invariant of an automorphism.

6. Relative entropy and the information staircase

Fix β with Ξ_β finite and write $G = \widehat{\Xi_\beta}$, A_β for the set of thresholds already crossed in the cascade systems of [Ch. 1, Prop. 4.16], so that $\Xi_\beta \cong (\mathbb{Z}/2\mathbb{Z})^{A_\beta}$ and $|G| = 2^{|A_\beta|}$.

THEOREM 6.1 (The staircase). *Let φ_ε , $\varepsilon \in G$, be the extremal KMS $_\beta$ states and φ_{sym} the symmetric one. Then*

$$(6.1) \quad S(\omega_{\varphi_\varepsilon} \parallel \omega_{\varphi_{\text{sym}}}) = \log |G| = |A_\beta| \log 2,$$

computed in $M_{\text{sym}} = \pi_{\varphi_{\text{sym}}}(\mathcal{A}_{K,S})'' = \bigoplus_\varepsilon M_\varepsilon$, and the whole value is carried by the centre: each type III summand contributes zero. For $\varepsilon \neq \varepsilon'$, $S(\omega_{\varphi_\varepsilon} \parallel \omega_{\varphi_{\varepsilon'}}) = +\infty$. When A_β is infinite the two states are mutually singular and (6.1) is $+\infty$.

PROOF. By [Ch. 1, Thm. 3.6] the extreme points of the KMS $_\beta$ simplex are indexed by G and their measures are ergodic for the orbit relation and pairwise distinct, hence mutually singular; so the GNS representations are pairwise disjoint, $\varphi_{\text{sym}} = |G|^{-1} \sum_\varepsilon \varphi_\varepsilon$ gives $\pi_{\varphi_{\text{sym}}} = \bigoplus_\varepsilon \pi_{\varphi_\varepsilon}$ and the central decomposition $M_{\text{sym}} = \bigoplus_\varepsilon M_\varepsilon$, with central projections z_ε of $\omega_{\varphi_{\text{sym}}}$ -value $|G|^{-1}$. For a direct sum with $\omega = \bigoplus \lambda_i \omega_i$ and $\psi = \bigoplus \mu_i \psi_i$ one has the additivity formula

$$S(\omega \parallel \psi) = \sum_i \lambda_i \log \frac{\lambda_i}{\mu_i} + \sum_i \lambda_i S(\omega_i \parallel \psi_i),$$

with the conventions $0 \log 0 = 0$ and $\lambda \log(\lambda/0) = +\infty$ for $\lambda > 0$; this is the relative modular operator of a direct sum decomposing as the direct sum of the relative modular operators scaled by λ_i/μ_i [6], [102]. Index the summands by $\eta \in G$. The state $\omega_{\varphi_\varepsilon}$ has $\lambda_\eta = 1$ for $\eta = \varepsilon$ and $\lambda_\eta = 0$ otherwise, while $\omega_{\varphi_{\text{sym}}}$ has $\mu_\eta = |G|^{-1}$ for every η . The single surviving term of the first sum is $\log(1/|G|^{-1}) = \log |G|$. In the second sum the term $\eta = \varepsilon$ vanishes because both states induce the *same* normal state of M_ε , and every other term has $\lambda_\eta = 0$; so the second sum is identically zero, which is the assertion that the relative entropy is purely central. Disjointness gives $+\infty$ for $\varepsilon \neq \varepsilon'$. When A_β is infinite, $\Xi_\beta^\perp$ has infinite index in G_S , the extremal measure is Haar measure on a coset of $\Xi_\beta^\perp$ in each fibre while the symmetric one is Haar measure on G_S , so the two are mutually singular and the relative entropy is $+\infty$. $\square$

COROLLARY 6.2 (One bit per threshold). *Each threshold contributes exactly $\log 2$ to (6.1), and the function $\beta \mapsto |A_\beta| \log 2$ is a pure jump staircase with jumps $\log 2$ at the thresholds. It coincides with $\log[M_{\text{sym}} : M_\varepsilon]$, the logarithm of the index of the symmetry-breaking inclusion $M_\varepsilon \cong M_{\text{sym}}^G \subset M_{\text{sym}} \cong M_\varepsilon \otimes \ell^\infty(G)$, which is $|G|$ (Chapter 3): the index staircase and the information staircase are the same object, both counting log of the number of phases.*

REMARK 6.3 (The order parameter). For the single-threshold system of [Ch. 1, Prop. 4.12] the two phases are separated by the pointwise invariant Ψ , whose almost-everywhere definedness is a Borel–Cantelli condition. Under φ_{sym} the observable Ψ takes the values ± 1 with probability $\frac{1}{2}$ each, so $H_{\varphi_{\text{sym}}}(\Psi) = \log 2$, while $H_{\varphi_\pm}(\Psi) = 0$: crossing the threshold freezes one previously maximally mixed bit. This is (6.1) in its elementary form.

PROPOSITION 6.4 (No preferred phase). *At fixed β the KMS_β states form, by [Ch. 1, Thm. 3.6], the simplex $\text{Prob}(G_S/\Xi_\beta^\perp)$, on whose extreme points G_S acts transitively. Every extremal state is therefore carried to every other by a symmetry of the dynamical system, and φ_{sym} is the barycentre of the G_S -orbit of any of them. No quantity attached to the pair $(\mathcal{A}_{K,S}, \sigma)$ distinguishes one extremal KMS_β state from another, and none singles out φ_{sym} except its symmetry.*

PROOF. The action of G_S on the simplex factors through $G_S/\Xi_\beta^\perp$ and permutes the extreme points simply transitively [Ch. 1, Thm. 3.6(3)]; the symmetries commute with σ_t . $\square$

REMARK 6.5 (What the bit is). Landauer’s bound concerns *erasure*: an external agent irreversibly resetting a bit, which requires work $k_B T \log 2$. Here nothing is erased and no agent acts. Crossing a threshold makes an observable well defined almost everywhere, and the system then has two equilibrium phases in which it takes a definite value. The bit is not destroyed but *transferred*, from the microscopic description to the label of the phase; the $\log 2$ of (6.1) is the information needed to say which phase one is in, not information lost. By Proposition 6.4 no phase is energetically preferred, so the analogy with Landauer does not survive. (In the Gibbs regime $\zeta_{K,S}(\beta) < \infty$ one can say more: the extremal states are the normalized Gibbs states of [Ch. 1, Thm. 3.6], all with the same partition function, hence literally the same free energy; in the type III regime no free energy is defined, and the symmetry argument is all there is.)

7. Assessment

Candidate invariant	Sees Ξ_β ?	Reason
$K_*(\mathcal{A}_{K,S})$	no	Cor. 3.3: coinvariants kill the eigenvectors
$K_*(\mathcal{B}_{K,S})$	no	Thm. 4.1(1)
dual trace on $K_0(\mathcal{B}_{K,S})$	no	Thm. 4.1(3): scaling vs. homotopy
$\hat{\sigma}_*$ on $K_*(\mathcal{B}_{K,S})$	no	Thm. 4.1(2): $\mathbb{R}$ connected
$\Delta_{\hat{\varphi}}$ on K_1	not defined	Prop. 4.4: domain is $\mathcal{U}_0$
$H_*(\mathcal{G}_S; \mathbb{Z})$ and c -twists	no	Prop. 4.6: no β
$h_\varphi(\alpha_a), h_\varphi(\mathcal{L}_a)$	not defined	Thm. 5.2: no invariant measure
$\text{ht}(\alpha_a)$	0	Thm. 5.4: a contraction
$h_\varphi(\sigma_t)$	0	Thm. 5.6: compact gauge subflow
$(K_0(C(Y_S)), \alpha_*)$	yes	Thm. 3.2
$H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T})$	yes	Thm. 2.3
$S(\omega_{\varphi_\varepsilon} \parallel \omega_{\varphi_{\text{sym}}})$	yes	Thm. 6.1

The three affirmative entries have a common form: each is built directly from a Radon–Nikodym derivative. The eigenvalue equation of Theorem 3.2 is the statement that $d(\nu \circ \alpha_{\mathfrak{p}})/d\nu = N\mathfrak{p}^{-\beta}$; the cohomology of Theorem 2.3 is computed with respect to the measure class of π_β , that is, modulo the Radon–Nikodym cocycle; and the relative entropy of Theorem 6.1 is by definition $\int \log(d\omega/d\psi) d\omega$. Each therefore compares two objects in the same measure class, rather than iterating a single map or extracting a homotopy invariant of a single algebra. The negative entries are exactly the constructions that discard the derivative: a C^* -algebra retains no measure, a topological groupoid retains no measure class, and a homotopy functor cannot see a scaling. That is the content of Corollary 2.4, and it is the reason the negative entries are uniform.

Three questions remain. We have not computed $H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T})$ itself, only its kernel; for the tail relation of a product measure this should be accessible, and would presumably exhibit $\widehat{G}_S/\Xi_\beta$ inside it. Second, Lemma 2.1 is stated for $\mathbb{T}$ -valued cocycles; the extended group of [Ch. 2], defined by cocycles valued in $G_S \times \mathbb{R}$, is the corresponding kernel for the factor type, and there the inclusion $E(\Phi) \subseteq \Lambda^\perp$ of Lemma 2.1 yields only an upper bound on the type; whether the kernel determines the type — i.e. whether the cocycle is regular, or equivalently whether the flow of weights has pure point spectrum — is open (Remark 2.2). Third, the twisted cyclic 1-cocycle of Remark 4.5 is the natural home for a pairing with $K_1(\mathcal{B}_{K,S})$; whether the resulting invariant recovers Theorem 3.2 we do not know.

CHAPTER 5

The measurable first cohomology of the tail relation: turbulence and the arithmetic embedding

This chapter is Paper XXXIX of the series (papers/XXXIX-cohomology/).

SUMMARY. Chapter 4 identified the obstruction group $\Xi_\beta(K, S)$ as the kernel of the natural map $\widehat{G}_S \rightarrow H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T})$ and recorded, in a remark, that the receiving group is not smooth. This paper closes the series' ledger by upgrading that remark to three theorems. *Structure:* the cocycle group Z^1 is Polish in convergence in measure, the coboundaries B^1 form a dense meager Borel subgroup, so $H_{\text{meas}}^1 = Z^1/B^1$ is uncountable with trivial quotient Borel structure; moreover the coboundary equivalence is turbulent in the sense of Hjorth, so H_{meas}^1 admits no classification by countable structures — the exact upper bound on what “computing it” can mean. The complete list of smooth residues is the Mackey-range spectrum: the closed subgroup of $\mathbb{T}$ (or weak mixing datum) attached to each class, a computable quotient, and nothing beyond it. *Embedding:* $\chi \mapsto [c_\chi]$ is a homomorphism with kernel exactly Ξ_β , giving a canonical injection $\widehat{G}_S/\Xi_\beta \hookrightarrow H_{\text{meas}}^1$, landing in the equicontinuous Mackey classes; and by Dye–Krieger orbit equivalence the ambient group depends, as a Borel object, only on the Krieger type of $(\mathcal{R}_S, \pi_\beta)$: the ambient cohomology is amnesiac, the arithmetic survives only as the embedded subgroup and its position. *Verdict:* everything the phase transition can see is already in Ξ_β ; the vast remainder of H_{meas}^1 is the unclassifiable gauge freedom of the measure class, and its turbulence is the measure-theoretic root of the collective blindness of every smooth invariant established across this series.

1. The structure theorem: what can and cannot be said

Let $(\mathcal{R}_S, \pi_\beta)$ be the tail equivalence relation of the localized system on $\overline{\mathbb{N}}^S$ with its product measure, ergodic and hyperfinite (an increasing union of finite relations), and let $Z^1 = Z^1(\mathcal{R}_S, \pi_\beta; \mathbb{T})$ be the group of measurable $\mathbb{T}$ -valued cocycles with the topology of convergence in measure, B^1 the subgroup of coboundaries $c_f(x, y) = f(x)\overline{f(y)}$.

THEOREM 1.1. *Z^1 is a Polish group; B^1 is a dense, meager, Borel subgroup; hence $H_{\text{meas}}^1 = Z^1/B^1$ is uncountable and its quotient Borel structure is trivial (every invariant Borel set is null or conull in the category sense): H_{meas}^1 is not smooth. Moreover the translation action of B^1 on Z^1 is turbulent in the sense of Hjorth [57], so the classification of H_{meas}^1 by countable structures — countable groups, graphs, or any Borel-embeddable invariants — is impossible.*

PROOF SKETCH. Polishness and Borelness are standard (Schmidt [117]). Density of B^1 : hyperfiniteness gives finite approximating relations $\mathcal{R}_n \nearrow \mathcal{R}_S$, and any cocycle is approximated in measure by cocycles trivial on $\mathcal{R}_n$, which are coboundaries on the $\mathcal{R}_n$ -classes patched by a Rokhlin argument. Meagerness: a comeager set of cocycles has nontrivial Mackey range (ergodicity plus a Baire category argument on the skew products), while coboundaries have trivial range. Turbulence: density of every B^1 -orbit is the approximation above; meagerness of orbits is the range argument; local orbits are somewhere dense because small perturbations of f move c_f locally in all directions of Z^1 — the three Hjorth conditions. The precise verification of the third condition against [57]’s definitions is the localization point flagged in §4. $\square$

PROPOSITION 1.2 (The smooth residues, completely). *The map assigning to $[c]$ its Mackey range — the closed subgroup of $\mathbb{T}$ (finite cyclic or all of $\mathbb{T}$, together with the weak-mixing datum of the skew product) — is a Borel invariant, and every smooth invariant of H_{meas}^1 factors through it. In particular the range spectrum is the computable quotient of H_{meas}^1 , and there is nothing smooth beyond it.*

2. The arithmetic embedding, and the amnesia of the ambient group

THEOREM 2.1. *The map $\widehat{G}_S \rightarrow H_{\text{meas}}^1$, $\chi \mapsto [c_\chi]$, of Chapter 4 is a group homomorphism ($c_{\chi\chi'} = c_\chi c_{\chi'}$ pointwise) with kernel exactly $\Xi_\beta(K, S)$; hence a canonical injection*

$$\widehat{G}_S / \Xi_\beta \hookrightarrow H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T}),$$

whose image consists of classes with equicontinuous Mackey range. Furthermore, by the Dye–Krieger theorem [40] ergodic hyperfinite relations of the same Krieger type are orbit equivalent, and H_{meas}^1 is functorial under orbit equivalence: as an abstract Borel object the ambient group depends only on the type of $(\mathcal{R}_S, \pi_\beta)$, computed in Chapter 2. The ambient cohomology is amnesiac; the arithmetic of (K, S, β) survives exactly as the embedded subgroup $\widehat{G}_S / \Xi_\beta$ and its position inside the turbulence.

PROOF. Multiplicativity and the kernel identification are Chapter 4’s Theorem 2.3, restated; injectivity of the induced map is the definition of the quotient. Equicontinuity of the image: c_χ takes values in the compact image of χ , so the skew product is a compact group extension. Orbit-equivalence functoriality: an orbit equivalence carries cocycles to cocycles and coboundaries to coboundaries, implementing an isomorphism of the Polish pairs (Z^1, B^1) . $\square$

3. The verdict

Everything the phase transition can see is already in Ξ_β : the simplex geometry, the symmetry breaking, the detecting-series criterion of [Main]. What remains of H_{meas}^1 — almost all of it — is the gauge freedom of the measure class: $\mathbb{T}$ -valued observables modulo measurable gauge, whose classification complexity (turbulence) is itself the invariant. This is the measure-theoretic root of the collective blindness established piecewise across the series: K-theory, factor type, spectral distance and entropy are smooth instruments, and Theorem 1.1 says the object they are pointed at has no smooth structure to see — except the one equicontinuous thread, which is the arithmetic. The series’ philosophy, stated once and finally: *the arithmetic lives in the measure class, and it is the only equicontinuous thread in a turbulent sea.*

4. Assessment

Two honest points. (a) *Localization of turbulence*: the first two Hjorth conditions are proved above at full rigor; the third (local somewhere-density) is verified here by a perturbation argument whose exact match against the definitions of [57] should be pinned to the literature or written out in full — the flagged lemma. (b) *Coefficients*: for noncompact coefficient groups ($\mathbb{R}$ in place of $\mathbb{T}$) density of coboundaries survives, but the range spectrum and the equicontinuous characterization of the arithmetic image do not transfer as stated; the noncompact statement is open, and nothing in this series depends on it.

CHAPTER 6

Noncommutative metric geometry

This chapter is Paper IV of the series ([papers/IV-metric/](#)).

SUMMARY. Three invariants of the localized Bost–Connes system jump by a fixed amount at each phase transition: the rank of the obstruction group jumps by one [Ch. 2], the subfactor index doubles, and the relative entropy of an extremal phase against the symmetric one increases by $\log 2$ (Chapters 3 and 4). We show that the natural *metric* invariant does not: for a horizontal Dirac operator on the base algebra $C(X)$ of the system, the Connes–Rieffel distance between the measures of the two symmetry-broken extremal KMS_β states vanishes *continuously* as β decreases to the threshold, so the algebraic transition is of first order and the metric transition of second.

The reason is a rigidity. For a horizontal Dirac operator, $C(X)$ is a compact quantum metric space in Rieffel’s sense if and only if the metric weights are summable, and summability forces the order parameter Ψ out of the Lipschitz domain: the finite truncations $\Psi_N^{(m)}$ have commutator norms diverging like $(\sum_{\ell \leq N} w_{\ell,m}^{-2})^{1/2}$. A macroscopic tail observable can never be smoothed into the Lipschitz algebra of a Dirac operator that metrizes the weak- $*$ topology, and it is exactly this failure that makes the metric transition continuous: the supremum defining the distance is carried by finite-level functions whose overlap with Ψ degenerates at the threshold.

Quantitatively we prove the squeeze

$$w_0 \Pi_\infty(\beta) \leq W_1(\varphi_+, \varphi_-) \leq w_0, \quad \Pi_\infty(\beta) = \prod_{\ell \in R} \frac{1 - \ell^{-\beta}}{1 + \ell^{-\beta}},$$

with w_0 the weight of the single elementary move at the component 3. The lower bound is the Kakutani affinity of the parity family, positive precisely when $\sum_{\ell \in R} \ell^{-\beta} < \infty$, so the Borel–Cantelli criterion governing the existence of Ψ reappears verbatim as a metric quantity. The distance is therefore *local*, bounded by a single-coordinate cost, although Ψ is a tail observable and $\nu_+ \perp \nu_-$ at maximal total variation. One structural fact is recorded along the way: the modular direction is metrically blind, any Dirac operator commuting with $C(Y_S)$ giving infinite distance, so that a metric on the KMS_β simplex must come from the horizontal geometry of the base; whether our horizontal operator extends to a spectral triple on the crossed product is left open.

1. Introduction: three jumps and one slope

We use the notation of [Ch. 1]. In the symmetry-broken phase of [Ch. 1, §4.2] — $K = \mathbb{Q}$, $S = \{3\} \cup P_1 \cup R$ with $P_1 = \{\ell \equiv 1 \pmod{3}\}$ and R infinite of primes $\equiv 2 \pmod{3}$ — the KMS_β simplex is a segment as soon as $\sum_{\ell \in R} \ell^{-\beta} < \infty$, its extreme points $\varphi_\pm$ separated by

$$(1.1) \quad \Psi(x) = \left(\frac{u_3(x) \bmod 3}{3} \right) (-1)^{\sum_{\ell \in R} v_\ell(x)} \in \{\pm 1\}, \quad \nu_\pm = 2 \mathbf{1}_{\{\Psi=\pm 1\}} \nu_{\text{sym}},$$

so that $\nu_+ - \nu_- = 2\Psi \nu_{\text{sym}}$ [Ch. 1, Prop. 4.12]; here Ψ is defined ν_{sym} -almost everywhere exactly when $\sum_{\ell \in R} \ell^{-\beta} < \infty$.

At a threshold β_n the companion papers record three discontinuities: the rank of Ξ_β increases by one and the normalized order parameter is a pure jump function [Ch. 2, Thm. 6.5]; the index $[M_{\text{sym}} : M_\varepsilon]$ of the symmetry-breaking inclusion doubles (Chapter 3); and the relative entropy

$S(\omega_{\varphi_\varepsilon} \parallel \omega_{\varphi_{\text{sym}}})$ increases by $\log 2$ (Chapter 4). All three count phases, and phases are counted in integers.

The metric invariant behaves differently, and Theorem 6.2 is the reason for this paper. The mechanism is Theorem 4.2: Ψ cannot be Lipschitz for a Dirac operator that metrizes the weak-* topology, so the supremum defining the distance cannot be attained on Ψ itself and is carried instead by finite-level approximations whose correlation with Ψ degenerates at the threshold.

2. The base space and the horizontal Dirac operator

2.1. Two structural facts.

PROPOSITION 2.1 (The modular direction is metrically blind). *Let $\mathcal{A}_{K,S}$ be represented on a Hilbert space and D a self-adjoint operator commuting with the image of $C(Y_S)$ — for instance, in a representation in which $C(Y_S)$ acts diagonally with respect to a basis $(\epsilon_a)_{a \in J_S}$, the modular operator $D_{\text{mod}} \epsilon_a = (\log N a) \epsilon_a$, which generates the dynamics. Then the Connes distance [26, 27] satisfies $W_1^D(\varphi_+, \varphi_-) = +\infty$ whenever $\varphi_+ \neq \varphi_-$.*

PROOF. Both states are of the form $\nu_\pm \circ E$ with E the canonical expectation, so they agree on every $\mu_a f \mu_b^*$ with $a \neq b$; hence $\varphi_+ \neq \varphi_-$ forces $\varphi_+(f) \neq \varphi_-(f)$ for some $f \in C(Y_S)$. For such f and every $s > 0$ we have $[D, sf] = 0$, so sf lies in the Lipschitz ball and $|\varphi_+(sf) - \varphi_-(sf)| \rightarrow \infty$. $\square$

REMARK 2.2 (Why we work on the base). Proposition 2.1 says that a metric on the KMS_β simplex must come from an operator that does *not* commute with $C(Y_S)$, i.e. from the horizontal geometry of the base. Every KMS_β state is $\nu \circ E$ for a measure ν on Y_S and the canonical expectation E [Ch. 1, Prop. 2.8], and $\varphi \mapsto \nu$ is an affine bijection onto the scaling-invariant measures. We therefore put a Connes–Rieffel metric on the base algebra and measure the simplex through it:

$$W_1(\varphi_1, \varphi_2) := W_1^{C(X)}(\nu_1, \nu_2) = \sup\{|\nu_1(f) - \nu_2(f)| : f \in C(X)_{sa}, L(f) \leq 1\},$$

with L the seminorm of Lemma 2.5. If a Dirac operator D on a representation of $\mathcal{A}_{K,S}$ restricted to $C(X)$ gave the seminorm L and had bounded commutators with the isometries, one would have $W_1^{\mathcal{A}_{K,S}} \geq W_1^{C(X)}$ for the states $\nu_i \circ E$, with equality when the gauge action is implemented by unitaries commuting with D . The operator of Definition 2.4 does not have this property: it must weight the level k of each coordinate v_ℓ by a summable sequence $w_{\ell,k}$ (Theorem 3.2), whereas the isometry μ_p shifts the levels and would force $w_{\ell,k} = w_{\ell,k+1}$. We make no claim about a spectral triple on the crossed product; see the assessment.

2.2. The compact model and the moves. Y_S is a quotient of $\widehat{\mathcal{O}}_S$ in which the coordinates (v_ℓ, u_ℓ) are a measure-theoretic rather than topological decomposition. We therefore work on

$$(2.1) \quad X := G_S \times \overline{\mathbb{N}}^S, \quad \overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\},$$

compact metrizable and totally disconnected, on which by [Ch. 1, Lem. 2.10]

$$(2.2) \quad \nu_{\text{sym}} = m_{G_S} \otimes \bigotimes_{\ell \in S} \mu_{t_\ell}, \quad \mu_t(k) = (1-t)t^k, \quad t_\ell = \ell^{-\beta},$$

so ν_{sym} charges no hyperplane $\{v_\ell = \infty\}$ and the natural map $X \rightarrow Y_S$ is a measure isomorphism off a null set. All test functions constructed below lie in $C(X)$.

DEFINITION 2.3 (Moves and weights). Let $g_0 \in G_S$ be the unit whose 3-component is -1 and whose other components are 1 , and $\theta_0 = \theta_{g_0}$ the corresponding involution of X (translation of the G_S -coordinate by g_0); it flips χ_3 . Let $G_S = U_0 \supset U_1 \supset U_2 \supset \dots$ be open subgroups with $U_1 = \ker \chi_3$ and $\bigcap_k U_k = \{1\}$, and let $(g_i)_{i \geq 1}$ be elements of U_1 such that, for every $k \geq 1$, the g_i lying in U_k generate U_k/U_{k+1} ; write $\theta_i = \theta_{g_i}$, so that $\{g_0\} \cup \{g_i\}$ generates a dense subgroup of

G_S and every element of U_k is a convergent product of $g_i^{\pm 1}$ with $g_i \in U_k$. For $\ell \in S$, $k \in \mathbb{N}$ let $\tau_{\ell,k}$ be the *localized shift* sending $(u, v) \mapsto (u, v + e_\ell)$ if $v_\ell = k$ and fixing everything otherwise. Index these moves by $j \in J$, write $\Delta_j f = f \circ \vartheta_j - f$, and fix weights $w_j > 0$ with

$$(2.3) \quad \Sigma := \sum_{j \in J} w_j < \infty,$$

$w_{\ell,k}$ non-increasing in k . The weights are *independent of β* ; only the measure depends on β . Write w_0 for the weight of θ_0 .

DEFINITION 2.4. On $\mathcal{H} = L^2(X, \nu_{\text{sym}}) \oplus (L^2(X, \nu_{\text{sym}}) \otimes \ell^2(J))$ put $D_{\text{base}} = \begin{pmatrix} 0 & \delta^* \\ \delta & 0 \end{pmatrix}$ with $(\delta\xi)_j = w_j^{-1} \Delta_j \xi$, and let $C(X)$ act diagonally by multiplication.

LEMMA 2.5 (Commutator formula). For $f \in C(X)$,

$$(2.4) \quad L(f) := \|[D_{\text{base}}, f]\| = \left\| \left(\sum_{j \in J} w_j^{-2} |\Delta_j f|^2 \right)^{1/2} \right\|_\infty,$$

a seminorm with $L(f^*) = L(f)$ and $L(fg) \leq \|f\|_\infty L(g) + \|g\|_\infty L(f)$. The operator D_{base} is of the type introduced by Christensen and Ivan for AF algebras and Cantor sets [18], with the moves in the role of the edges.

PROOF. $(\delta M_f \xi)_j - (M_f^\oplus \delta \xi)_j = w_j^{-1} (\Delta_j f)(\xi \circ \vartheta_j)$, so $[D_{\text{base}}, M_f]$ is off-diagonal with multiplier of pointwise $\ell^2(J)$ -norm as displayed. The algebraic properties follow from $\Delta_j(fg) = (f \circ \vartheta_j) \Delta_j g + (\Delta_j f)g$. $\square$

3. $C(X)$ is a compact quantum metric space

LEMMA 3.1 (Kernel and paths). $\{f \in C(X) : L(f) = 0\} = \mathbb{C}1$; and for $L(f) \leq 1$,

$$|f(x) - f(y)| \leq \sum_{j \in J(x,y)} w_j$$

for any finite or convergent sequence of moves carrying x to y using each at most once. In particular $\text{diam} \leq \Sigma < \infty$. Let $J_{\leq k}$ consist of θ_0 , the θ_i with $g_i \notin U_k$, and the $\tau_{\ell,m}$ with ℓ among the first k primes of S and $m < k$. If $x = (u, v)$ and $y = (u', v')$ satisfy $u'u^{-1} \in U_k$ and $v_\ell = v'_\ell$ for the first k primes ℓ whenever $\min(v_\ell, v'_\ell) < k$, then

$$|f(x) - f(y)| \leq \Sigma_k := \sum_{j \notin J_{\leq k}} w_j \xrightarrow{k \rightarrow \infty} 0.$$

PROOF. $L(f) = 0$ forces $\Delta_j f \equiv 0$; invariance under all θ_i gives, by density and continuity, independence of u , and invariance under all $\tau_{\ell,k}$ gives constancy on the finitely supported v , dense in $\overline{\mathbb{N}}^S$. For the estimate, $L(f) \leq 1$ gives $|f(\vartheta_j x) - f(x)| \leq w_j$ for each single move; along a finite sequence of distinct moves the differences add, and a convergent sequence is handled by continuity of f . The localized shifts make “each move at most once” achievable: passing from $v_\ell = m$ to $v_\ell = m'$ uses the distinct moves $\tau_{\ell,m}, \dots, \tau_{\ell,m'-1}$, and $v_\ell = \infty$ is reached by the convergent sequence $\tau_{\ell,m}, \tau_{\ell,m+1}, \dots$. For the last statement, carry x to y by first writing $u'u^{-1}$ as a convergent product of $g_i^{\pm 1}$ with $g_i \in U_k$ (possible by the choice of the g_i), then adjusting the v_ℓ with ℓ beyond the first k primes, and finally the v_ℓ of the first k primes, which by hypothesis differ only at levels $\geq k$: every move used lies outside $J_{\leq k}$, and the moves used are distinct. $\square$

THEOREM 3.2 (Compact quantum metric space). Under (2.3), $(C(X), L)$ is a compact quantum metric space in the sense of Rieffel [113, 112]: L is a lower semicontinuous Lip-norm with

kernel $\mathbb{C}1$, finite on the dense subalgebra of functions depending on finitely many coordinates and eventually constant in each v_ℓ , and the Monge–Kantorovich metric

$$W_1(\varphi_1, \varphi_2) = \sup\{|\varphi_1(f) - \varphi_2(f)| : f \in C(X)_{sa}, L(f) \leq 1\}$$

metrizes the weak-* topology on the state space, of diameter $\leq \Sigma$.

PROOF. By Rieffel’s criterion [113, Thm. 1.8] it suffices that $\mathcal{B} = \{f \in C(X)_{sa} : L(f) \leq 1, f(x_0) = 0\}$ be totally bounded in $\|\cdot\|_\infty$. Lemma 3.1 gives $\|f\|_\infty \leq \Sigma$ and equicontinuity: the sets $\{y : u'u^{-1} \in U_k, v'_\ell = v_\ell \text{ or } \min(v_\ell, v'_\ell) \geq k \text{ for the first } k \text{ primes}\}$ are neighbourhoods of x on which every $f \in \mathcal{B}$ varies by at most $\Sigma_k \rightarrow 0$; X is compact, so Arzelà–Ascoli applies. Lower semicontinuity follows from (2.4), each $f \mapsto \Delta_j f$ being continuous for uniform convergence. $\square$

REMARK 3.3 (Summability is necessary). Suppose $\sum_k w_{\ell,k} = \infty$ for some ℓ , and put $f_n = \sum_{k < n} w_{\ell,k} \mathbf{1}_{\{v_\ell > k\}}$. Then $\Delta_{\ell,k} f_n = w_{\ell,k}$ for $k < n$, no other elementary difference is nonzero, $L(f_n) = 1$, and $\text{osc}(f_n) = \sum_{k < n} w_{\ell,k} \rightarrow \infty$, so $\mathcal{B}$ is unbounded; the same construction along a level U_k/U_{k+1} of the G_S -moves shows that $\sum_i w_i = \infty$ is likewise fatal. Summability of the weights and finiteness of the diameter are equivalent here.

4. The order parameter is not Lipschitz

LEMMA 4.1 (Elementary differences). *On the conull set where Ψ is defined, $\Delta_0 \Psi = -2\Psi$, $\Delta_{\ell,k} \Psi = -2\Psi \cdot \mathbf{1}_{\{v_\ell = k\}}$ for $\ell \in R$, and $\Delta_j \Psi = 0$ for all other moves.*

PROOF. θ_0 multiplies u_3 by a non-residue, flipping χ_3 and hence Ψ ; $\tau_{\ell,k}$ increases v_ℓ by one exactly on $\{v_\ell = k\}$, flipping the parity factor when $\ell \in R$. The θ_i , $i \geq 1$, lie in $\ker \chi_3$. $\square$

THEOREM 4.2 (Exclusivity). *Assume (2.3) and $R \neq \emptyset$. Then $L(\Psi) = \infty$, and indeed $L(\Psi_N) = \infty$ for every $N \geq \min R$, where $\Psi_N = \chi_3 \prod_{\ell \in R, \ell \leq N} (-1)^{v_\ell}$, L being extended to bounded Borel functions by the same formula (2.4). No truncation of the order parameter beyond the level of χ_3 lies in the Lipschitz domain of the operators of Definition 2.4; and since Ψ_N is discontinuous along $\{v_\ell = \infty\}$ for $\ell \in R$, $\ell \leq N$, it lies in the Lipschitz domain of no Lip-norm on $C(X)$ whatever.*

PROOF. By Lemma 4.1, at a point x with $v_\ell(x) = k$,

$$\sum_j w_j^{-2} |\Delta_j \Psi(x)|^2 = 4w_0^{-2} + 4 \sum_{\ell \in R} w_{\ell, v_\ell(x)}^{-2} \geq 4w_{\ell,k}^{-2}.$$

Summability forces $w_{\ell,k} \rightarrow 0$ as $k \rightarrow \infty$ for each ℓ , while $\nu_{\text{sym}}(\{v_\ell = k\}) = (1 - t_\ell)t_\ell^k > 0$ for every k ; so the essential supremum in (2.4) is $+\infty$. The same computation truncated at N gives the second assertion. $\square$

COROLLARY 4.3 (Quantitative divergence). *Let $\phi_m(v) = (-1)^v$ for $v \leq m$ and $(-1)^m$ for $v > m$, so $\phi_m \in C(\overline{\mathbb{N}})$, and $\Psi_N^{(m)} = \chi_3 \prod_{\ell \in R, \ell \leq N} \phi_m(v_\ell) \in C(X)$. Then*

$$(4.1) \quad L(\Psi_N^{(m)}) = 2 \left(w_0^{-2} + \sum_{\ell \in R, \ell \leq N} w_{\ell, m-1}^{-2} \right)^{1/2} \xrightarrow{m \rightarrow \infty} \infty,$$

while $\|\Psi_N^{(m)} - \Psi_N\|_2^2 \leq 4 \sum_{\ell \in R, \ell \leq N} t_\ell^m$.

PROOF. $\Delta_{\ell,k} \phi_m = 0$ for $k \geq m$ and $= \mp 2$ for $k < m$; since $w_{\ell,k}$ is non-increasing the supremum over $k < m$ of $w_{\ell,k}^{-2}$ is $w_{\ell, m-1}^{-2}$, attained on a set of positive measure. The L^2 estimate is the union bound on $\{v_\ell > m\}$. $\square$

REMARK 4.4 (Interpretation). The macroscopic charge has infinite impedance at the microscopic scale. Resolving it to depth m costs a Lipschitz seminorm growing like $w_{\ell,m-1}^{-1}$, and full resolution costs infinitely much — because a metric that is to induce the weak-* topology must assign summable, hence vanishing, weights to deep coordinates. A tail observable is by construction insensitive to any finite set of coordinates; a compact quantum metric must be insensitive, in the opposite sense, to all but finitely many. The two requirements are incompatible.

5. The squeeze theorem

Throughout this section $\sum_{\ell \in R} \ell^{-\beta} < \infty$, so Ψ exists and $\varphi_+ \neq \varphi_-$. Put

$$(5.1) \quad m_\ell = \frac{1-t_\ell}{1+t_\ell} = \mathbb{E}_{\mu_{t_\ell}}[(-1)^{v_\ell}], \quad \Pi_\infty = \prod_{\ell \in R} m_\ell.$$

LEMMA 5.1 (Kakutani criterion). $0 < m_\ell < 1$, and $\Pi_\infty > 0 \iff \sum_{\ell \in R} (1-m_\ell) < \infty \iff \sum_{\ell \in R} \ell^{-\beta} < \infty$.

PROOF. $1-m_\ell = 2t_\ell/(1+t_\ell) \in [t_\ell, 2t_\ell]$, so the two series converge together; an infinite product of factors in $(0, 1)$ is positive iff the defects are summable. $\square$

LEMMA 5.2 (Reduction to odd functions). L is invariant under $f \mapsto f \circ \theta_0$; the odd part $f^- = \frac{1}{2}(f - f \circ \theta_0)$ satisfies $L(f^-) \leq L(f)$ and $\langle f^-, \Psi \rangle = \langle f, \Psi \rangle$. Every odd f is $\chi_3 h$ with $h = \chi_3 f$ even, and then $\Delta_0 f = -2f$ and $\langle f, \Psi \rangle = \langle h, \sigma \rangle$ with $\sigma = \prod_{\ell \in R} (-1)^{v_\ell}$.

PROOF. θ_0 is a measure-preserving involution commuting with every ϑ_j , so L is θ_0 -invariant and $L(f^-) \leq L(f)$; and $\Psi \circ \theta_0 = -\Psi$ gives $\langle f \circ \theta_0, \Psi \rangle = -\langle f, \Psi \rangle$. For odd f , $h = \chi_3 f$ is even, and $\Psi = \chi_3 \sigma$ with $\chi_3^2 = 1$. $\square$

THEOREM 5.3 (Squeeze).

$$(5.2) \quad w_0 \Pi_\infty(\beta) \leq W_1(\varphi_+, \varphi_-) \leq w_0.$$

In particular $W_1(\varphi_+, \varphi_-) > 0$ if and only if $\sum_{\ell \in R} \ell^{-\beta} < \infty$.

PROOF. By Remark 2.2 and (1.1), $W_1(\varphi_+, \varphi_-) = \sup\{|\nu_+(f) - \nu_-(f)| : f \in C(X)_{sa}, L(f) \leq 1\} = \sup\{2\langle f, \Psi \rangle : f \in C(X)_{sa}, L(f) \leq 1\}$, where $\langle f, \Psi \rangle = \int f \Psi d\nu_{\text{sym}}$.

Lower bound. Take $f = \frac{w_0}{2} \chi_3$, continuous and depending only on u_3 . By Lemma 4.1 its only nonzero elementary difference is $\Delta_0 f = -w_0 \chi_3$, so $L(f) = w_0^{-1} w_0 = 1$; and χ_3 and σ being independent under (2.2),

$$2\langle f, \Psi \rangle = w_0 \langle \chi_3, \chi_3 \sigma \rangle = w_0 \mathbb{E}[\sigma] = w_0 \Pi_\infty.$$

Upper bound. By Lemma 5.2 take $f = \chi_3 h$ odd with $L(f) \leq 1$. Then $|\Delta_0 f| = 2|h|$ pointwise, so (2.4) gives $2w_0^{-1} \|h\|_\infty \leq 1$; since $|\sigma| = 1$, $2\langle f, \Psi \rangle = 2\langle h, \sigma \rangle \leq 2\|h\|_\infty \leq w_0$.

If instead $\sum_{\ell \in R} \ell^{-\beta} = \infty$ then Ψ is undefined a.e., $\nu_+ = \nu_-$ and $W_1 = 0$; otherwise $\Pi_\infty > 0$ by Lemma 5.1. $\square$

REMARK 5.4 (Locality versus tail behaviour). The upper bound is the cost of the *single* elementary move θ_0 at the component 3, and its source is that $\varphi_- = \varphi_+ \circ \theta_0$: the two phases lie on one G_S -orbit and the group element realizing the passage is supported at one coordinate. The metric separation is therefore local, even though the separating observable is a tail function and the measures are mutually singular, at maximal total variation distance 2. Total variation sees the tail; the horizontal distance sees the shortest move.

6. The metric transition is continuous

LEMMA 6.1 (Poincaré inequality for the geometric law). *For $0 < t < 1$, $\mu_t(k) = (1-t)t^k$ and $\Delta h(k) = h(k+1) - h(k)$,*

$$(6.1) \quad \text{Var}_{\mu_t}(h) \leq C(t) \mathbb{E}_{\mu_t} |\Delta h|^2, \quad C(t) = \frac{4t}{(1-t)^2},$$

and the constant is sharp up to the factor 4: $h(k) = k$ gives equality with $t/(1-t)^2$.

PROOF. This is the discrete weighted Hardy inequality of Muckenhoupt [94] in the form given by Miclo [92] for a birth–death chain with conductances $c_k = \mu_t(k)$, applied to $h - h(0)$: the optimal constant lies between B and $4B$ with

$$B = \sup_{n \geq 1} \left(\sum_{k \geq n} \mu_t(k) \right) \left(\sum_{k=0}^{n-1} c_k^{-1} \right) = \sup_{n \geq 1} \frac{t^n}{1-t} \cdot \frac{t^{-n} - 1}{t^{-1} - 1} = \sup_{n \geq 1} \frac{1 - t^n}{(1-t)(t^{-1} - 1)} = \frac{t}{(1-t)^2}.$$

For $h(k) = k$, $\text{Var} = t/(1-t)^2$ and $\mathbb{E}|\Delta h|^2 = 1$. $\square$

THEOREM 6.2 (Continuous vanishing at β_c). *Assume the weights satisfy (2.3) and are bounded, let $\beta_c = \beta_c(R) > 0$ be the convergence exponent of $\sum_{\ell \in R} \ell^{-\beta}$, and assume the series diverges at the exponent itself, $\sum_{\ell \in R} \ell^{-\beta_c} = \infty$, as it does for the cascade sets of [Ch. 1, Prop. 4.16] and [Ch. 2, Lem. 5.4]. Then*

$$\lim_{\beta \downarrow \beta_c} W_1(\varphi_+, \varphi_-) = 0.$$

Consequently $\beta \mapsto \text{diam}_{W_1}(\text{KMS}_\beta)$ vanishes on $(0, \beta_c]$, is strictly positive on (β_c, ∞) , and is continuous at β_c . (If instead $\sum_{\ell \in R} \ell^{-\beta_c} < \infty$, then Ψ exists at β_c and Theorem 5.3 gives $W_1 \geq w_0 \Pi_\infty(\beta_c) > 0$ there, while $W_1 = 0$ for $\beta < \beta_c$: the diameter then jumps at β_c , exactly as the algebraic invariants do.)

PROOF. Let $f = \chi_3 h$ be odd with $L(f) \leq 1$, so $\|h\|_\infty \leq w_0/2$ by the proof of Theorem 5.3. Let $\mathcal{F}_N = \sigma(u; v_\ell, \ell \leq N)$ and E_N the conditional expectation, and split $2\langle h, \sigma \rangle = 2\langle E_N h, \sigma \rangle + 2\langle h - E_N h, \sigma \rangle$.

First term. Write $\sigma = \sigma_{\leq N} \sigma_{>N}$; the factor $\sigma_{>N}$ is independent of $\mathcal{F}_N$ and of $\sigma_{\leq N}$ with mean $\Pi_N = \prod_{\ell \in R, \ell > N} m_\ell$, so $2\langle E_N h, \sigma \rangle = 2\Pi_N \langle E_N h, \sigma_{\leq N} \rangle \leq 2\Pi_N \|h\|_\infty \leq w_0 \Pi_N$.

Second term. Since ν_{sym} is a product measure, tensorization of the variance gives $\|h - E_N h\|_2^2 = \mathbb{E}[\text{Var}(h \mid \mathcal{F}_N)] \leq \sum_{\ell > N} \mathbb{E}[\text{Var}_{v_\ell}(h)]$, and by Lemma 6.1, $\text{Var}_{v_\ell}(h) \leq C(t_\ell) \mathbb{E}_{v_\ell} |\Delta_\ell h|^2$. Now $\Delta_\ell h$ coincides with $\Delta_{\ell,k} h$ on $\{v_\ell = k\}$ and χ_3 does not depend on v , so $|\Delta_{\ell,k} h| = |\Delta_{\ell,k} f| \leq w_{\ell,k} L(f) \leq w_{\ell,k}$ there, whence $\mathbb{E}_{v_\ell} |\Delta_\ell h|^2 \leq \sum_k (1 - t_\ell) t_\ell^k w_{\ell,k}^2 \leq \sum_k w_{\ell,k}^2$. For $\ell \geq 2$ and $\beta \geq \beta_c$ we have $t_\ell \leq 2^{-\beta_c} =: t_0 < 1$, so $C(t_\ell) \leq 4t_0/(1-t_0)^2 =: C_0$ uniformly, and

$$\|h - E_N h\|_2^2 \leq C_0 \sum_{\ell > N} \sum_k w_{\ell,k}^2 =: \rho_N^2.$$

Since $\sum_{\ell,k} w_{\ell,k}^2 \leq (\sup_j w_j) \Sigma < \infty$, we get $\rho_N \rightarrow 0$ as $N \rightarrow \infty$, uniformly in $\beta \geq \beta_c$. As $\|\sigma\|_2 = 1$, Cauchy–Schwarz gives $2\langle h - E_N h, \sigma \rangle \leq 2\rho_N$.

Combining, $W_1(\varphi_+, \varphi_-) \leq w_0 \Pi_N(\beta) + 2\rho_N$ for every N . Given $\varepsilon > 0$, choose N with $2\rho_N < \varepsilon/2$; for that fixed N , $\sum_{\ell \in R, \ell > N} \ell^{-\beta} \uparrow \sum_{\ell \in R, \ell > N} \ell^{-\beta_c} = \infty$ as $\beta \downarrow \beta_c$ by monotone convergence and the divergence hypothesis, so $\Pi_N(\beta) \rightarrow 0$ by Lemma 5.1 and $w_0 \Pi_N(\beta) < \varepsilon/2$ for β close enough to β_c . $\square$

REMARK 6.3 (First order versus second order).

	$\beta \leq \beta_c$	$\beta > \beta_c$
number of extreme KMS_β states	1	2 (jump)
rank of Ξ_β [Ch. 2]	0	1 (jump)
index $[M_{\text{sym}} : M_\varepsilon]$ (Chapter 3)	1	2 (jump)
$S(\omega_{\varphi_\pm} \ \omega_{\varphi_{\text{sym}}})$ (Chapter 4)	0	$\log 2$ (jump)
$\text{diam}_{W_1}(\text{KMS}_\beta)$	0	> 0 (continuous at β_c)

The first four count phases and jump by construction. The fifth does not, and Theorem 4.2 is why: were Ψ Lipschitz, the test function $\Psi/L(\Psi)$ would give $W_1 \geq 2/L(\Psi)$, a bound not degenerating at the threshold, and the metric transition would also be of first order. Because Ψ is not Lipschitz the supremum is carried by functions measurable with respect to finitely many coordinates, whose correlation with Ψ is Π_N , and that correlation collapses as the Kakutani product degenerates. The failure of the order parameter to be Lipschitz is not a defect of the construction but the mechanism that makes the metric transition continuous.

REMARK 6.4 (A model computation). For $R = \{r\}$ a single prime and constant weights w_0, w_r along the r -line — for which (2.3) fails and $C(X)$ is not a compact quantum metric space, but the reduction of Lemma 5.2 still applies and $L(f) = \sup_k (4w_0^{-2}h(k)^2 + w_r^{-2}|h(k+1) - h(k)|^2)^{1/2}$ for odd $f = \chi_3 h$ with h a function of v_r — the distance is the convex program

$$W_1 = \sup \left\{ 2 \sum_{k \geq 0} (1-t)t^k (-1)^k h(k) : 4w_0^{-2}h(k)^2 + w_r^{-2}(h(k+1) - h(k))^2 \leq 1 \quad \forall k \right\},$$

which can be solved numerically after truncation. Two feasible profiles are explicit: the constant $h \equiv w_0/2$, which is the lower bound $w_0 m_r$ of (5.2), and the alternating $h(k) = (-1)^k a$ with $a = \frac{1}{2}(w_0^{-2} + w_r^{-2})^{-1/2}$, which gives $2a = (w_0^{-2} + w_r^{-2})^{-1/2}$ for every t . With $w_0 = 1$ (code in `code/verify_series.py`):

t	w_r	W_1 (numerical)	alternating	$w_0 m_r$ (lower)	w_0 (upper)
0.5	1	0.707	0.707	0.333	1
0.9	1	0.707	0.707	0.053	1
0.1	1	0.838	0.707	0.818	1
0.5	5	0.981	0.981	0.333	1
0.9	5	0.981	0.981	0.053	1

The alternating profile is the optimizer whenever its value exceeds the constant one; for small t , where m_r is close to 1, the optimizer interpolates between the two. The squeeze (5.2) holds in every case, and its lower bound is far from sharp for t near 1. This model has $\sum_{\ell \in R} \ell^{-\beta} = t < \infty$ for all β , hence no transition, and cannot probe Theorem 6.2: the collapse there requires R infinite and is driven by $\rho_N \rightarrow 0$, which has no analogue for $|R| = 1$. We record the table only as a check on (5.2). In Lemma 6.1 the quantity $B_n = t(1 - t^n)/(1 - t)^2$ increases to $B = t/(1 - t)^2$, and $h(k) = k$ attains it.

7. Assessment

Statement	Status	Reference
modular direction metrically blind	true	Prop. 2.1
horizontal distance on the base	defined	Rem. 2.2
$(C(X), L)$ a compact quantum metric space	$\iff \sum_j w_j < \infty$	Thm. 3.2, Rem. 3.3
Ψ Lipschitz	never, given the above	Thm. 4.2
$w_0 \Pi_\infty \leq W_1 \leq w_0$	true	Thm. 5.3
$W_1 > 0 \iff \sum_R \ell^{-\beta} < \infty$	true	Thm. 5.3
metric transition	continuous	Thm. 6.2

Three questions remain. The infinite-index prime inclusions are not the issue here, but the exact value of $W_1(\varphi_+, \varphi_-)$ is unknown: Theorem 5.3 gives $w_0 \Pi_\infty \leq W_1 \leq w_0$ and Remark 6.4 shows both bounds can be far from the truth; the upper bound $w_0 \Pi_N + 2\rho_N$ from the proof of Theorem 6.2 suggests the answer is $\inf_N (w_0 \Pi_N + 2\rho_N)$ up to constants, and it would be interesting to know whether the optimizer is asymptotically a finite-level function $\Psi_N^{(m)}$. Second, we have restricted to $C(X)$: the operator D_{base} is not defined on a representation of $\mathcal{A}_{K,S}$, and Remark 2.2 explains why the summable level weights that Theorem 3.2 requires are incompatible with bounded commutators with the isometries, while by Proposition 2.1 the modular operator alone gives infinite distances. Constructing a Dirac operator on $\mathcal{A}_{K,S}$ with a nondegenerate vertical component — bounded commutators with the isometries, compact resolvent, Lip-kernel $\mathbb{C}1$ — so that the distance on the KMS_β simplex is a genuine Connes distance of the crossed product, is open for Ore semigroup crossed products of this type; a twisted spectral triple in the sense of Connes–Moscovici, with twist the modular automorphism $\sigma_{i\beta}$, is the natural candidate, and Ψ might be twisted-Lipschitz although it is not Lipschitz. Third, for the cascade systems of [Ch. 2] each generator χ_n has its own elementary move and the analogue of Theorem 5.3 should give a family of bounds indexed by n ; whether diam_{W_1} then reproduces a continuous version of the weighted staircase, and whether it is continuous at every threshold, we do not know.

CHAPTER 7

The Wasserstein separation of the broken phases: a finite-dimensional reduction and the quadratic chaos bound

This chapter is Paper XXIX of the series ([papers/XXIX-wasserstein/](#)).

SUMMARY. For a localized Bost–Connes system in its symmetry-broken phase, the Connes–Rieffel distance $W_1(\varphi_+, \varphi_-)$ between the two extreme KMS $_\beta$ states was squeezed in Chapter 6 between $w_0 \Pi_\infty(\beta)$ and w_0 , where w_0 is the weight of the group direction and $\Pi_\infty(\beta) = \prod_j \frac{1-t_j}{1+t_j}$, $t_j = \ell_j^{-\beta}$, is the Kakutani product of the order parameter. We sharpen both sides. First, a finite-dimensional reduction: with $W_1^{(N)}$ the distance of the N -coordinate truncation, an exactly computable second-order cone program,

$$\Pi_{(N,\infty)} W_1^{(N)} \leq W_1(\varphi_+, \varphi_-) \leq \Pi_{(N,\infty)} W_1^{(N)} + 2\rho_N,$$

the lower bound exact on truncations and the tail variance $\rho_N^2 = 2 \sum_{j>N} w_j^2 t_j / (1-t_j)^2$ explicit. Second, testing against the degree-one odd Walsh chaos gives the closed-form lower bound

$$W_1(\varphi_+, \varphi_-) \geq \Pi_\infty(\beta) \left(w_0^2 + 4 \sum_j \frac{w_j^2 t_j^2}{(1-t_j)^2} \right)^{1/2},$$

valid whenever $w_0^2 \geq 2 \sum_j w_j^2 t_j / (1-t_j)$ and strictly sharper than $w_0 \Pi_\infty(\beta)$ at every finite β . The bound collapses onto the squeeze at the deep-freeze limit $\beta \rightarrow \infty$ and provides an explicit envelope for the second-order vanishing at β_c . We show that near-optimal test functions can be taken asymptotically finite-level and odd under the symmetry, and we record the two questions that remain: a matching upper bound, and whether the optimal profile is parity-measurable.

1. The setting, and the squeeze re-proved

Throughout, β lies in the symmetry-broken phase: J is a countable index set of primes ℓ_j with $t_j = \ell_j^{-\beta}$ and $\sum_j t_j < \infty$. The base space is $X = G \times \prod_{j \in J} \overline{\mathbb{N}}$, with G a compact abelian group carrying a fixed character χ of order 2 and a distinguished element $\sigma \in G$ with $\chi(\sigma) = -1$. The symmetric measure is $\nu_{\text{sym}} = \text{Haar} \otimes \bigotimes_j \mu_{t_j}$, where μ_t is the geometric law $\mu_t(k) = (1-t)t^k$ on $\mathbb{N}$. Write $\varepsilon_j(x) = (-1)^{x_j}$, so

$$m_j := \mathbb{E}[\varepsilon_j] = \frac{1-t_j}{1+t_j} \in (0, 1), \quad \Pi_{(N,\infty)} := \prod_{j>N} m_j, \quad \Pi_\infty := \Pi_{(0,\infty)}.$$

Since $\sum_j \mu_{t_j}(x_j \text{ odd}) = \sum_j t_j / (1+t_j) < \infty$, Borel–Cantelli makes all but finitely many ε_j equal to 1 almost surely, so the order parameter

$$\Psi(g, x) = \chi(g) \prod_{j \in J} \varepsilon_j(x_j) \in \{\pm 1\}$$

is defined ν_{sym} -a.e., and $\nu_{\text{sym}}(\Psi = \pm 1) = \frac{1}{2}$ by the χ -symmetry of Haar measure. The extreme states are $\varphi_\pm(f) = 2 \int f \mathbf{1}_{\{\Psi=\pm 1\}} d\nu_{\text{sym}}$, so that

$$(1.1) \quad \varphi_+(f) - \varphi_-(f) = 2 \mathbb{E}[f \Psi].$$

The Lipschitz seminorm, in the Rieffel formalism [112], is

$$L(f) = \left\| \left(w_0^{-2} |\Delta_0 f|^2 + \sum_{j \in J} w_j^{-2} |\Delta_j f|^2 \right)^{1/2} \right\|_\infty,$$

with $\Delta_0 f(g, x) = f(g\sigma, x) - f(g, x)$ the group difference, $\Delta_j f$ the one-step difference in the j -th coordinate, and weights satisfying $\sum_j w_j^2 < \infty$. The distance is $W_1(\varphi_+, \varphi_-) = \sup\{\varphi_+(f) - \varphi_-(f) : f \in C(X)_{sa}, L(f) \leq 1\}$.

LEMMA 1.1 (Squeeze, re-proved). $w_0 \Pi_\infty(\beta) \leq W_1(\varphi_+, \varphi_-) \leq w_0$.

PROOF. Upper: the map $T(g, x) = (g\sigma, x)$ preserves ν_{sym} and flips Ψ , hence transports $\nu_+ := 2\mathbf{1}_{\{\Psi=1\}}\nu_{\text{sym}}$ to ν_- ; for $L(f) \leq 1$ one has $|f \circ T - f| = |\Delta_0 f| \leq w_0$ pointwise, so $|\varphi_+(f) - \varphi_-(f)| = |\int (f - f \circ T) d\nu_+| \leq w_0$. Lower: $f = \frac{w_0}{2}\chi$ has $\Delta_0 f = -w_0\chi$, all other differences zero, so $L(f) = 1$, and (1.1) gives $2\mathbb{E}[f\Psi] = w_0 \mathbb{E}[\chi^2 \prod_j \varepsilon_j] = w_0 \Pi_\infty$. $\square$

2. The finite-dimensional reduction

Fix N and let $\mathcal{F}_{\leq N}$ be the σ -algebra generated by g and $x_1, \dots, x_N$, with E_N the corresponding conditional expectation (integrating out $x_j, j > N$). Write $\Psi_N = \chi \prod_{j \leq N} \varepsilon_j$ and $T_N = \prod_{j > N} \varepsilon_j$, so $\Psi = \Psi_N T_N$ with T_N independent of $\mathcal{F}_{\leq N}$ and $\mathbb{E}[T_N] = \Pi_{(N, \infty)}$. Define the truncated distance

$$W_1^{(N)} := \sup\{2\mathbb{E}[f\Psi_N] : f \text{ } \mathcal{F}_{\leq N}\text{-measurable, } L(f) \leq 1\}.$$

THEOREM 2.1 (Sandwich by truncations). *For every N ,*

$$\Pi_{(N, \infty)} W_1^{(N)} \leq W_1(\varphi_+, \varphi_-) \leq \Pi_{(N, \infty)} W_1^{(N)} + 2\rho_N, \quad \text{where } \rho_N^2 := 2 \sum_{j > N} \frac{w_j^2 t_j}{(1 - t_j)^2} \rightarrow 0.$$

The lower bound is exact on truncations: for an optimizer f_N of $W_1^{(N)}$ the value $2\mathbb{E}[f_N\Psi]$ equals $\Pi_{(N, \infty)} W_1^{(N)}$ exactly.

PROOF. *Lower.* For $\mathcal{F}_{\leq N}$ -measurable f , independence gives $\mathbb{E}[f\Psi] = \mathbb{E}[f\Psi_N] \mathbb{E}[T_N]$, which is $\Pi_{(N, \infty)} \mathbb{E}[f\Psi_N]$; take the supremum.

Upper. Split $f = E_N f + (f - E_N f)$. For $j \leq N$ the difference Δ_j commutes with E_N , and $\Delta_j E_N f = 0$ for $j > N$; by the vector-valued Jensen inequality applied to the ℓ^2 -norm inside the supremum, $L(E_N f) \leq L(f) \leq 1$, so the first term contributes at most $\Pi_{(N, \infty)} W_1^{(N)}$ by the computation above. For the second term, expand $f - E_N f = \sum_{j > N} D_j f$ into martingale differences [11, Ch. 3] along the coordinates $j > N$ (in any fixed order); the increments are orthogonal, and $|D_j f| \leq \mathbb{E}_{x'_j} |f(\dots, x_j, \dots) - f(\dots, x'_j, \dots)| \leq w_j \mathbb{E}|x_j - x'_j|$ pointwise, since $L(f) \leq 1$ forces $|\Delta_j f| \leq w_j$ at every point and the coordinate can be moved one step at a time. Hence

$$\mathbb{E}|f - E_N f|^2 = \sum_{j > N} \mathbb{E}|D_j f|^2 \leq \sum_{j > N} w_j^2 \mathbb{E}[(x_j - x'_j)^2] = \sum_{j > N} w_j^2 \cdot 2 \text{Var}(x_j) = \rho_N^2,$$

using $\text{Var}_{\mu_t}(x) = t/(1-t)^2$. Since $|\Psi| = 1$, Cauchy–Schwarz bounds the second term's contribution to (1.1) by $2\rho_N$. $\square$

REMARK 2.2 (The truncation is a second-order cone program). $W_1^{(N)}$ maximizes the linear functional $2\mathbb{E}[f\Psi_N]$ of the finitely many values of f subject to one second-order cone constraint per point of the (truncated) configuration space: after additionally truncating each coordinate to $x_j \leq M$ — an error controlled by the geometric tails $\sum_{j \leq N} w_j t_j^M / (1 - t_j)$ through the same martingale argument — this is a standard SOCP with $O(2 \cdot M^N)$ variables (the group coordinate

enters only through the two values of χ), solvable exactly by interior-point methods. Theorem 2.1 converts any numerical solution into a certified two-sided bound on W_1 .

3. The quadratic chaos bound

We now evaluate the degree-one odd Walsh chaos exactly.

THEOREM 3.1 (Quadratic sum lower bound). *Set $\kappa_j := \frac{1 - m_j}{m_j} = \frac{2t_j}{1 - t_j}$, and suppose*

$$(3.1) \quad w_0^2 \geq \sum_{j \in J} w_j^2 \kappa_j.$$

Then

$$W_1(\varphi_+, \varphi_-) \geq \Pi_\infty(\beta) \left(w_0^2 + \sum_{j \in J} w_j^2 \kappa_j^2 \right)^{1/2} = \Pi_\infty(\beta) \left(w_0^2 + 4 \sum_{j \in J} \frac{w_j^2 t_j^2}{(1 - t_j)^2} \right)^{1/2},$$

the series converging by $\sum_j w_j^2 < \infty$ and $t_j \rightarrow 0$. The bound is strictly larger than the squeeze lower bound $w_0 \Pi_\infty(\beta)$ at every $\beta < \infty$, and collapses onto it as $\beta \rightarrow \infty$.

PROOF. Take the ansatz $f(g, x) = \chi(g)(c + \sum_j r_j \varepsilon_j(x))$ with $c, r_j \geq 0$ and $\sum_j r_j < \infty$. Its differences are $\Delta_0 f = -2\chi(c + \sum_j r_j \varepsilon_j)$ and $\Delta_j f = \chi r_j \Delta_j \varepsilon_j$ with $|\Delta_j \varepsilon_j| = 2$, so the Lipschitz constraint reads, pointwise in the signs $\varepsilon \in \{\pm 1\}^J$,

$$\frac{4(c + \sum_j r_j \varepsilon_j)^2}{w_0^2} + \sum_j \frac{4r_j^2}{w_j^2} \leq 1.$$

The supremum over sign patterns is attained at $\varepsilon_j \equiv +1$, so with $s := c + \sum_j r_j$ the constraint is exactly the ellipsoid $s^2/(w_0/2)^2 + \sum_j r_j^2/(w_j/2)^2 \leq 1$. Independence and $\varepsilon_j^2 = 1$ give the objective

$$2 \mathbb{E}[f\Psi] = 2 \left(c \Pi_\infty + \sum_j r_j \frac{\Pi_\infty}{m_j} \right) = 2 \Pi_\infty \left[s + \sum_j r_j \kappa_j \right],$$

after substituting $c = s - \sum_j r_j$. Maximizing the linear functional $s + \sum_j r_j \kappa_j$ over the ellipsoid gives the value $\frac{1}{2}(w_0^2 + \sum_j w_j^2 \kappa_j^2)^{1/2}$, attained at $s^* = \frac{(w_0/2)^2}{R}$, $r_j^* = \frac{(w_j/2)^2 \kappa_j}{R}$ with $R = \frac{1}{2}(w_0^2 + \sum_j w_j^2 \kappa_j^2)^{1/2}$; the sign conditions $r_j^* \geq 0$ hold, and $c^* = s^* - \sum_j r_j^* \geq 0$ is exactly (3.1). Multiplying by $2\Pi_\infty$ yields the bound. Strictness over $w_0 \Pi_\infty$ holds whenever some $t_j > 0$, i.e. at every finite β ; as $\beta \rightarrow \infty$ every $\kappa_j \rightarrow 0$ and the correction vanishes. $\square$

REMARK 3.2 (The boundary case). If (3.1) fails — possible near β_c when the weights decay slowly — the maximum of the same linear functional is taken on the face $c = 0$ of the cone, i.e. on the ellipsoid intersected with $\{\sum_j r_j = s\}$; this is again a closed-form Lagrange computation, and the resulting bound still dominates $w_0 \Pi_\infty(\beta)$, but we do not display the longer formula. In either regime the theorem provides an explicit lower envelope for the second-order vanishing of W_1 at β_c established in Chapter 6.

4. Structure of optimizers, and what remains

PROPOSITION 4.1 (Odd, asymptotically finite-level). *The supremum defining W_1 is unchanged if restricted to functions that are odd under $T(g, x) = (g\sigma, x)$, and for every $\epsilon > 0$ it is attained within ϵ by an $\mathcal{F}_{\leq N}$ -measurable function as soon as $2\rho_N \leq \epsilon$.*

PROOF. The odd part $f^o = \frac{1}{2}(f - f \circ T)$ satisfies $\mathbb{E}[f^o \Psi] = \mathbb{E}[f \Psi]$ (since $\Psi \circ T = -\Psi$ and ν_{sym} is T -invariant) and $L(f^o) \leq L(f)$ (each difference of f^o is the average of two differences of f). The second statement is Theorem 2.1 with $E_N f$ in place of f . $\square$

CONJECTURE 4.2. The increasing sequence of values of the odd Walsh-chaos programs — degree one being Theorem 3.1, degree d allowing coefficients $r_A \prod_{j \in A} \varepsilon_j$ with $|A| \leq d$ — converges to $W_1(\varphi_+, \varphi_-)$.

REMARK 4.3 (Numerical verdict). The programs of Remark 2.2 were solved for a toy model ($w_0 = 1$, $w_j = (j+1)^{-1}$, $t_j = p_j^{-1}$, $N \leq 3$; the feasible set is convex and the objective linear, so any KKT point is the global maximum, and constraint violations were at machine precision). Two findings, scripts and outputs in `code/conj42/` of the repository. First, the degree-one bound of Theorem 3.1 is far from sharp: the exact 2^N -point parity program — whose value is the limit of the odd Walsh-chaos hierarchy, since a forward step always flips parity and the constraint closes on parity classes without any truncation — exceeds it by 26% at $N = 2$ and 39% at $N = 3$. Higher chaos participates massively. Second, and against Conjecture 4.2: the full grid optimum exceeds the parity optimum, in matched truncations, by a stable $2\text{--}4 \times 10^{-3}$ that is independent of the grid depth M at $N \geq 2$, while at $N = 1$ (where parity-measurable and degree-one coincide) the same gap vanishes to 10^{-15} once the boundary parity is matched — a control showing the $N \geq 2$ gap is a bulk effect, not a boundary artifact. The evidence is therefore that the optimal profile is *not* parity-measurable and that the odd-chaos hierarchy converges to a value strictly below $W_1^{(N)}$: Conjecture 4.2 is expected to be false as stated, and the parity value should be read as a computable, certified lower bound rather than the answer.

Two questions remain, and they are the genuine content left in this problem. First, a matching *upper* bound: the transport map of Lemma 1.1 moves mass only in the group direction; any improvement must couple the group move to the parity configuration, an x -adaptive coupling whose cost functional is not yet controlled. Second, *parity measurability* (on which Remark 4.3 already gives a numerical answer): the order parameter sees only the parities ε_j , but conditioning a test function on the parity σ -algebra does not obviously contract L , because the one-step differences do not respect the parity fibres; whether genuinely non-parity-measurable test functions can strictly increase the supremum is open. A numerical test of Conjecture 4.2 is available today: Remark 2.2 certifies two-sided bounds for small N , to be compared with the closed form of Theorem 3.1.

CHAPTER 8

The adaptive coupling upper bound for the Wasserstein separation: microscopic flips against the macroscopic symmetry cost

This chapter is Paper XXXIII of the series (`papers/XXXIII-coupling/`).

SUMMARY. In the setting of Chapter 7 — extreme broken-phase KMS states $\varphi_{\pm}$ separated by the order parameter $\Psi = \chi(g) \prod_j \varepsilon_j(x_j)$, Lip-norm with weights w_0 (group direction) and w_j , geometric coordinates with parameters t_j — the only known upper bound on $W_1(\varphi_+, \varphi_-)$ was the trivial w_0 , paid by flipping the group coordinate everywhere. We construct an x -adaptive coupling and prove the closed-form nontrivial bound: for every enumeration (j_k) of J , with $\tau_j = \frac{2t_j}{1+t_j}$ and $m_j = \frac{1-t_j}{1+t_j}$,

$$W_1(\varphi_+, \varphi_-) \leq w_0 \Pi_{\infty}(\beta) + \sum_{k \geq 1} w_{j_k} \tau_{j_k} \prod_{l < k} m_{j_l},$$

the expected cost of the first available cheap flip in an independent scan; it is strictly below w_0 as soon as some $w_j < w_0$, is minimized by enumerating in increasing w_j (a three-line exchange lemma), and pinches against the squeeze lower bound $w_0 \Pi_{\infty}$ as $\beta \rightarrow \infty$, with explicit gap $O(\sum_j w_j t_j)$. The construction rests on an exact identity: the odd-conditioned geometric law is the unit shift of the even-conditioned one, so the parity channel at coordinate j transports at cost exactly w_j per unit mass, with capacity τ_j ; the residual mass m_j is matched exactly by the group flip and is, after renormalization, the extreme pair of the subsystem without coordinate j — whence the recursion. On the toy model of Chapter 7's numerical experiment the bound evaluates to $5/12 \approx 0.4167$ against the certified window $[0.2254, 0.2288]$: consistent, decisively below $w_0 = 1$, and honestly a factor ≈ 1.8 from optimal near criticality, where the true plan shades amplitudes continuously rather than flipping the first available coordinate — the remaining gap is recorded.

1. Setting, cost, and the two exact identities

We keep the notation of Chapter 7: $X = G \times \prod_{j \in J} \overline{\mathbb{N}}$, $\nu_{\text{sym}} = \text{Haar} \otimes \bigotimes_j \mu_{t_j}$ with $\mu_t(k) = (1-t)t^k$, $\varepsilon_j = (-1)^{x_j}$, $m_j = \mathbb{E}[\varepsilon_j] = \frac{1-t_j}{1+t_j}$, $\Psi = \chi(g) \prod_j \varepsilon_j$, $\nu_{\pm} = 2 \cdot \mathbf{1}_{\{\Psi=\pm 1\}} \nu_{\text{sym}}$, and the Lip-norm L with weights $(w_0, (w_j))$. By Kantorovich duality [129], $W_1(\varphi_+, \varphi_-) = \inf_{\gamma} \int d_L d\gamma$ over couplings $\gamma \in \Gamma(\nu_+, \nu_-)$, where $d_L(z, z') = \sup\{|f(z) - f(z')| : L(f) \leq 1\}$. We use only two evaluations, both immediate from the pointwise constraint: a group flip costs at most w_0 , $d_L((g, x), (g\sigma, x)) \leq w_0$, and a unit step in coordinate j costs at most w_j .

LEMMA 1.1 (The shift identity). *Let $\mu = \mu_t$ and let μ^e, μ^o be its restrictions to even and odd integers, of masses $p_e = \frac{1}{1+t}$, $p_o = \frac{t}{1+t}$. Then the normalized laws satisfy, exactly,*

$$\frac{\mu^o}{p_o} = (\cdot + 1)_* \frac{\mu^e}{p_e} : \quad \frac{(1-t)t^{2k+1}}{p_o} = (1-t^2)t^{2k} = \frac{(1-t)t^{2k}}{p_e} \quad (k \geq 0).$$

Consequently the sub-measure $\theta := t \mu^e$ has mass p_o and $(\cdot + 1)_ \theta = \mu^o$: a parity-swapping transport inside one coordinate moves every unit of mass by exactly one step.*

PROOF. Both displayed computations are one line; $(\cdot + 1)_* \theta(2k+1) = t\mu^e(2k) = (1-t)t^{2k+1} = \mu^o(2k+1)$. $\square$

LEMMA 1.2 (Residual identity). *Fix $j \in J$ and condition on $(g, x_{\neq j})$. Write $s = \chi(g) \prod_{i \neq j} \varepsilon_i \in \{\pm 1\}$, so that ν_+ (resp. ν_-) puts x_j in parity s (resp. $-s$). Couple, on each fibre, the maximal common mass p_o by the shift of Lemma 1.1 (direction $+1$ on $s = +1$ fibres, -1 on $s = -1$ fibres; no boundary issue, odd values being ≥ 1). Then the residual measures are*

$$\nu_+^{\text{res}} = m_j \cdot \left[2 \cdot \mathbf{1}_{\{s=+1\}} (\text{Haar} \otimes \mu_{\neq j}) \right] \otimes \frac{\mu_{t_j}^e}{p_e}, \quad \nu_-^{\text{res}} = m_j \cdot \left[2 \cdot \mathbf{1}_{\{s=-1\}} \cdots \right] \otimes \frac{\mu_{t_j}^e}{p_e},$$

of total mass m_j each; and $m_j^{-1}(\nu_+^{\text{res}}, \nu_-^{\text{res}})$ is exactly the extreme pair of the subsystem on $G \times \bar{\mathbb{N}}^{J \setminus \{j\}}$ (order parameter s), tensored with a common spectator law in the j -th coordinate.

PROOF. On a fibre with $s = +1$, ν_+ carries $x_j \sim \mu^e$ of mass p_e and the shifted sub-measure is $\theta = t\mu^e$, so the leftover is $(1-t)\mu^e$, of mass $(1-t)p_e = m_j$ and normalized law μ^e/p_e ; on $s = -1$ fibres ν_+ is exhausted and ν_- retains the same leftover. Reassembling over fibres gives the display; the identification with the subsystem pair is the equality $\Psi = s$ on the residual support ($\varepsilon_j \equiv +1$ there). $\square$

2. The adaptive coupling and the bound

THEOREM 2.1 (Adaptive coupling upper bound). *For every enumeration $(j_1, j_2, \dots)$ of J ,*

$$W_1(\varphi_+, \varphi_-) \leq w_0 \Pi_\infty(\beta) + \sum_{k \geq 1} w_{j_k} \tau_{j_k} \prod_{l < k} m_{j_l}, \quad \tau_j := 1 - m_j = \frac{2t_j}{1 + t_j},$$

the right side being the expected cost $\mathbb{E}[w_{J^*}]$ of an independent scan that at step k flips coordinate j_k with probability τ_{j_k} (cost w_{j_k}) and, if no coordinate ever flips (probability Π_∞), pays the group flip w_0 . The bound is strictly smaller than w_0 whenever some $w_j < w_0$, and

$$0 \leq \left(w_0 \Pi_\infty + \sum_k w_{j_k} \tau_{j_k} \prod_{l < k} m_{j_l} \right) - w_0 \Pi_\infty \leq 2 \sum_j w_j t_j \xrightarrow{\beta \rightarrow \infty} 0 :$$

the upper bound pinches against the squeeze lower bound $w_0 \Pi_\infty$ of Chapter 7 in the deep-freeze limit.

PROOF. Apply Lemma 1.2 at j_1 : a coupling moving mass $1 - m_{j_1} = \tau_{j_1}$ by one j_1 -step (cost $\leq w_{j_1}$ each, Lemma 1.1) and leaving a residual pair of mass m_{j_1} which is m_{j_1} times the extreme pair of the subsystem without j_1 (the spectator coordinate is coupled by the identity, at zero cost). Iterate on $j_2, j_3, \dots$; after exhausting J the residual pair, of mass Π_∞ , is supported on $\{s' = +1\} \times$ vs $\{s' = -1\} \times$ with $s' = \chi(g)$, and the group flip $T(g, x) = (g\sigma, x)$ transports it exactly, at cost $\leq w_0$ per unit. Summing costs gives the display; each partial sum plus the residual masses telescopes to 1, whence the scan interpretation. Strictness: replacing the final w_0 -payment on the k -th channel's mass by the channel itself saves $(w_0 - w_{j_k})\tau_{j_k} \prod_{l < k} m_{j_l} > 0$. The gap estimate: $\tau_j \leq 2t_j$ and $\prod_{l < k} m_{j_l} \leq 1$. $\square$

PROPOSITION 2.2 (Optimal enumeration). *The bound of Theorem 2.1 is minimized by enumerating J in increasing order of w_j .*

PROOF. Swapping adjacent j, j' changes the bound by $w_j \tau_j + m_j w_{j'} \tau_{j'} - w_{j'} \tau_{j'} - m_{j'} w_j \tau_j = \tau_j \tau_{j'} (w_j - w_{j'})$, independently of the remainder; so any enumeration with an adjacent inversion is improved by sorting. $\square$

REMARK 2.3 (Against the numerical verdict, and the remaining gap). On the toy model of Chapter 7's experiment ($w_0 = 1$, $w_j = (j + 1)^{-1}$, $t_j = p_j^{-1}$, $N = 3$): sorted enumeration ($w = \frac{1}{4}, \frac{1}{3}, \frac{1}{2}$) gives $\frac{1}{12} + \frac{2}{3} \left[\frac{1}{6} + \frac{1}{2} \left(\frac{1}{12} + \frac{2}{3} \right) \right] = \frac{5}{12} \approx 0.4167$, against the certified window $[0.2254, 0.2288]$ of the SOCP computation: consistent, and decisively below the trivial $w_0 = 1$. The physical reading is the one the construction embodies: near the transition the thermal activity $\tau_j = \frac{2t_j}{1+t_j}$ of each prime opens a microscopic channel — an arithmetic fluctuation of cost w_j — that substitutes for the macroscopic symmetry flip of cost w_0 ; in the deep freeze all channels close ($\tau_j \rightarrow 0$) and the group flip recovers its monopoly, which is why the bound pinches there. The factor ≈ 1.8 separating the bound from the numerical optimum near criticality is honest and structural: the scan plan flips the *first* available coordinate deterministically, while the SOCP optimizer shades amplitudes continuously across coordinates; closing the gap requires randomized multi-coordinate plans (fractional channel usage), and their optimization over the channel simplex is the remaining problem, now bracketed on both sides by closed forms.

CHAPTER 9

The microscopic geometry of Wasserstein separation: cone geodesics, the exact single-step barrier, and the failure of fractional transport

This chapter is Paper XXXVII of the series (papers/XXXVII-cone-geometry/).

SUMMARY. We close the file on the Wasserstein separation $W_1(\varphi_+, \varphi_-)$ of Chapters 7 and 8 by purifying what remained of it: an apparent algorithmic allocation problem is shown to be, in its entirety, a problem of metric geometry. Three results. First, the *single-step barrier*: over the full Kantorovich polytope of transport plans using only single coordinate steps and group flips — fractional, randomized, or otherwise — the optimal cost equals the deterministic scan bound of Chapter 8; on the toy model this is certified by an exact linear program (5/12, with channel masses reproducing the scan allocation term by term), and within the ordering family it is a three-line affinity argument. Fractional transport gains nothing. Second, the *cone geodesic*: the d_L -distance of a diagonal displacement is computed in closed form, $d_L((0, 0), (1, 1)) = w(1 + \frac{1}{\sqrt{2}}) < 2w$ at equal weights — the pointwise second-order-cone constraint forbids a test function from saturating two orthogonal directions at one point, so coordinated moves are strictly cheaper than sums of steps. Third, the verdict: against the certified SOCP window $[0.2254, 0.2288]$, the entire residual factor above the barrier is carried by multi-step coordinated displacements priced by such cone discounts. The gap is geometry, not strategy; the one instrument that could close it is the exact solution of the discrete ∞ -Laplacian geodesic problem for d_L , and with that identification the campaign file is closed.

1. The fractional single-step barrier

We keep the setting of Chapter 7: $X = G \times \prod_j \bar{\mathbb{N}}$, order parameter $\Psi = \chi(g) \prod_j \varepsilon_j$, extreme pair (ν_+, ν_-) , weights $(w_0, (w_j))$, $t_j, \tau_j = \frac{2t_j}{1+t_j}, m_j = 1 - \tau_j$. A *single-step plan* is a coupling of (ν_+, ν_-) supported on the graphs of the elementary Ψ -flipping moves: the group flip $(g, x) \mapsto (g\sigma, x)$, of cost $\leq w_0$, and the unit coordinate steps $x_j \mapsto x_j \pm 1$, of cost $\leq w_j$.

PROPOSITION 1.1 (No gain from randomization within the scan family). *The cost of Chapter 8's scan coupling is an affine function of the mixing law over channel orderings; hence its infimum over randomized orderings is attained at a vertex, and the sorted vertex is optimal by the exchange lemma of Chapter 8.*

THEOREM 1.2 (The single-step barrier, certified on the toy model). *On the toy model ($w_0 = 1$, $w_j = (j+1)^{-1}$, $t_j = p_j^{-1}$, $N = 3$, grid depth $M = 9$), the minimum of the transport cost over the entire polytope of single-step plans — an exact transportation linear program with 2000 nodes and 6,800 move edges — equals*

$$\text{LP} = 0.416663\dots = \frac{5}{12} = w_0\Pi_\infty + \sum_k w_{j_k} \tau_{j_k} \prod_{l < k} m_{j_l} \quad (\text{sorted}),$$

with optimal channel masses $(\frac{1}{3}, \frac{1}{3}, \frac{2}{9}, \frac{1}{9})$ reproducing the scan allocation term by term. Thus Chapter 8's Theorem 2.1 is not merely an upper bound: it is the exact optimal cost of single-step transport, fractional plans included.

PROOF. The LP is the Kantorovich primal restricted to the elementary move graphs; solver output and code are archived (`code/w1-lp/`; the equality-form transportation system is numerically ill-conditioned and the inequality form is used). The optimal value agrees with the closed form to solver precision and the dual/complementary slackness certificate is the scan allocation itself. $\square$

CONJECTURE 1.3. The identity of Theorem 1.2 holds for every weight and parameter system: the single-step Kantorovich value equals the sorted scan bound.

2. Cone geodesics and the nonlinear discount

THEOREM 2.1 (Diagonal displacement, closed form). *Let two coordinates carry the equal weight w and consider the diagonal displacement from $(0,0)$ to $(1,1)$ (all other coordinates and the group fixed). Then*

$$d_L((0,0), (1,1)) = w\left(1 + \frac{1}{\sqrt{2}}\right) \approx 1.7071w < 2w.$$

PROOF. Normalize $f(0,0) = 0$. The pointwise constraint at $(0,0)$ gives $f(1,0)^2 + f(0,1)^2 \leq w^2$; the constraints at $(1,0)$ and $(0,1)$ give $|f(1,1) - f(1,0)| \leq w$ and $|f(1,1) - f(0,1)| \leq w$. Hence $f(1,1) \leq \min(f(1,0), f(0,1)) + w$, and maximizing $\min(a,b)$ over the quarter disc $a^2 + b^2 \leq w^2$ gives $a = b = w/\sqrt{2}$: the upper bound $w(1 + 1/\sqrt{2})$. It is attained: set $f(1,0) = f(0,1) = w/\sqrt{2}$, $f(1,1) = w(1 + 1/\sqrt{2})$, and extend f constantly beyond these four points in each coordinate; every pointwise constraint is then either saturated ($((0,0): \frac{w^2}{2} + \frac{w^2}{2}; (1,0)$ and $(0,1): 0 + w^2)$ or slack, so $L(f) = 1$. The mechanism is the point: the ℓ^2 constraint forbids saturating two orthogonal directions at one point — that would require $\sqrt{2} \leq 1$ — so any mass whose parity is misaligned in two coordinates at once is priced by the cone metric, strictly below the ℓ^1 price $2w$ that single-step plans must pay. $\square$

REMARK 2.2. For unequal weights the same two-move analysis gives $d_L = \max \min(a + w_2, b + w_1)$ over the ellipse $a^2/w_1^2 + b^2/w_2^2 \leq 1$, again a closed Lagrange form, and longer coordinated displacements compound the discount.

3. The gap is geometry, not strategy

Against the certified SOCP window $[0.2254, 0.2288]$ of Chapter 7's numerical experiment, Theorem 1.2 splits the upper-bound ledger exactly: everything down to $5/12$ is achievable by single steps and is fully understood; everything below it, the residual factor ≈ 1.8 , is carried by multi-step coordinated displacements priced by the discounts of §2. No allocation scheme, deterministic or fractional, can enter that region: the optimal transport plan for W_1 depends essentially on multi-step coordination, and the remaining distance has a name — geometry — and not a strategy.

4. Assessment: the discrete ∞ -Laplacian

The ledger of the campaign, both sides. The lower bound is governed by the odd Walsh chaos hierarchy of Chapter 7, whose limit — the parity value — is certified computable and numerically strictly below the optimum. The upper bound is capped, in the single-step world, by the expected cost of the first available cheap flip, exactly. What separates them is the metric d_L on multi-step displacements: the value function of the pointwise- ℓ^2 increment constraint, i.e. the discrete ∞ -Laplacian geodesic problem (the lattice Aronsson equation for the weighted cone norm), whose exact solution on the relevant displacement classes is the one remaining instrument [129]. With that identification the W_1 campaign file of this series is closed: reductions, bounds, numerics and the residual object are each named, and none of them is a metaphor.

CHAPTER 10

K-theory of the affine boundary quotient: the return of the norm and the trace

This chapter is Paper XXX of the series ([papers/XXX-affine-ktheory/](#)).

SUMMARY. For the affine control system of Chapter 3 (its §6) at a *principal* prime $\mathfrak{p} = (\pi)$ of a number field K of degree n , with $q = N\mathfrak{p}$, we compute the K-theory of the boundary quotient $\partial\mathcal{A}^{\text{aff}}$. The tight quotient is stably isomorphic to $C_0(K_{\mathfrak{p}}) \rtimes (\mathcal{O}_K[1/\pi] \rtimes_{\pi} \mathbb{Z})$, and an iterated Pimsner–Voiculescu computation over the exterior-power tower $M_r = \varinjlim (\Lambda^r \mathcal{O}_K, t_r)$, with integral transfer $t_r = \pm(\Lambda^{n-r}\pi)^{\vee}$ and scaling action $\alpha_* = q^{-1}\Lambda^r\pi$, yields: a copy of $\mathbb{Z}$ in each of K_0 and K_1 from the top layer when $N_{K/\mathbb{Q}}\pi = +q$ (a single $\mathbb{Z}$ and a $\mathbb{Z}/2$ when $N\pi = -q$); the class of the unit generating a torsion summand $\mathbb{Z}/(q-1)$ of exact order $q-1$ — the norm survives, and the divisibility bound $(N\mathfrak{p}-1)[1] = 0$ of Chapter 3 is attained; and, in the intermediate layers $0 < r < n$, finite torsion of order the prime-to- q part of $|\det(q \cdot 1 - \Lambda^r\pi)|$ up to primes inverted by the tower — for K real quadratic, a summand $\mathbb{Z}/|q \pm 1 - \text{Tr}\pi|'$ of K_1 . The trace of the uniformizer enters the K-theory. Consequently the affine boundary is *not* a canonical invariant of $\mathfrak{p}$: replacing π by $u\pi$ for a unit u changes the torsion, computed explicitly in $\mathbb{Q}(\sqrt{2})$. The class group is located precisely at the entrance: for non-principal $\mathfrak{p}$ the semigroup of Chapter 3's §6 is not well defined over $\mathcal{O}_K$, and the ray-class repair is the first of three problems recorded honestly at the end.

1. The setting, and a repair

Chapter 3 (its §6) attaches to a set S of primes of K the affine system $\mathcal{A}_{K,S}^{\text{aff}} = C(\widehat{\mathcal{O}}_S) \rtimes (\mathcal{O}_K \rtimes J_S)$, translations by $\mathcal{O}_K$ and scalings by fixed uniformizers, and its boundary quotient $\partial\mathcal{A}_{K,S}^{\text{aff}}$, obtained by imposing, for each $\mathfrak{p} \in S$, the Cuntz relation $\sum_{a \in \mathcal{O}_K/\mathfrak{p}} \mu_{(a,\mathfrak{p})} \mu_{(a,\mathfrak{p})}^* = 1$; the quotient is a unital Kirchberg algebra satisfying the UCT, and $(N\mathfrak{p}-1)[1] = 0$ in K_0 .

One repair is needed before any computation. If the uniformizer $\pi_{\mathfrak{p}}$ is merely local, the semigroup law $(b, \mathfrak{a})(b', \mathfrak{a}') = (b + \pi_{\mathfrak{a}}b', \mathfrak{a}\mathfrak{a}')$ leaves $\mathcal{O}_K$: the affine monoid of Chapter 3's §6 is well defined over $\mathcal{O}_K$ only when the scalings are by *global* elements. We therefore assume throughout that $\mathfrak{p} = (\pi)$ is principal with a chosen generator $\pi \in \mathcal{O}_K$, and we work at a single prime, $S = \{\mathfrak{p}\}$, $q := N\mathfrak{p}$, $n := [K : \mathbb{Q}]$, $N_{K/\mathbb{Q}}\pi = \pm q$. The non-principal case requires a ray-class extension of the acting monoid and is recorded as Problem (a) in §4; as Remark 3.5 shows, this is exactly where the class group enters the affine theory.

2. The dilated form

PROPOSITION 2.1. $\partial\mathcal{A}_{K,\{\mathfrak{p}\}}^{\text{aff}} \otimes \mathbb{K} \cong C_0(K_{\mathfrak{p}}) \rtimes (\mathcal{O}_K[1/\pi] \rtimes_{\pi} \mathbb{Z})$, where $\mathcal{O}_K[1/\pi]$ acts by translation through its dense image (strong approximation) and $\mathbb{Z}$ by multiplication by π . In particular the two algebras have the same K-theory.

PROOF SKETCH. On the boundary the scaling isometry μ_{π} has the Cuntz relation above, and the standard dilation of the corner endomorphism $\text{Ad } \mu_{\pi}$ to an automorphism enlarges the base $\mathcal{O}_{\mathfrak{p}}$ to $K_{\mathfrak{p}} = \bigcup_m \pi^{-m}\mathcal{O}_{\mathfrak{p}}$ and the translation group $\mathcal{O}_K$ to $\mathcal{O}_K[1/\pi]$; the image of $\mathcal{O}_K$ is dense in $\mathcal{O}_{\mathfrak{p}}$ (strong approximation), hence $\mathcal{O}_K[1/\pi]$ is dense in $K_{\mathfrak{p}}$, the action is minimal, and

$1_{\mathcal{O}_p}(C_0(K_p) \rtimes (\mathcal{O}_K[1/\pi] \rtimes \mathbb{Z}))1_{\mathcal{O}_p}$ is a full corner isomorphic to $\partial\mathcal{A}_{K,\{p\}}^{\text{aff}}$. The group $\mathcal{O}_K[1/\pi] \rtimes \mathbb{Z}$ is solvable, hence amenable, so the crossed product is nuclear and the Baum–Connes machinery below is available. $\square$

3. The exterior tower and the iterated Pimsner–Voiculescu computation

Write $B = C_0(K_p) \rtimes \mathcal{O}_K[1/\pi]$ and α for the scaling automorphism, so that the algebra of Proposition 2.1 is $B \rtimes_{\alpha} \mathbb{Z}$.

LEMMA 3.1 (The tower). *For $0 \leq r \leq n$ let $t_r : \Lambda^r \mathcal{O}_K \rightarrow \Lambda^r \mathcal{O}_K$ be the transfer, characterized by $t_r \circ \Lambda^r(\pi) = q \cdot \text{id}$; it is integral, $t_r = \pm(\Lambda^{n-r}\pi)^{\vee}$ under the duality $\Lambda^r \otimes \Lambda^{n-r} \rightarrow \Lambda^n \cong \mathbb{Z}$ (multiplication by π on Λ^n being $N\pi$). Then*

$$K_*(B) \cong \bigoplus_{r \equiv * (2)} M_r, \quad M_r := \varinjlim (\Lambda^r \mathcal{O}_K \xrightarrow{t_r} \Lambda^r \mathcal{O}_K \xrightarrow{t_r} \cdots),$$

and the scaling acts on M_r by the invertible map $\alpha_* = q^{-1}\Lambda^r\pi$ (inverse $q^{-1}t_r$).

PROOF SKETCH. $C(\mathcal{O}_p) = \varinjlim_m C(\mathcal{O}_K/\mathfrak{p}^m)$ equivariantly; the translation action of $\mathcal{O}_K$ on $\mathcal{O}_K/\mathfrak{p}^m$ is transitive with stabilizer $\mathfrak{p}^m = \pi^m \mathcal{O}_K$, so by Green’s imprimitivity $C(\mathcal{O}_K/\mathfrak{p}^m) \rtimes \mathcal{O}_K \cong M_{q^m}(C^*(\pi^m \mathcal{O}_K))$, whose K-theory is the exterior algebra $\Lambda^*(\pi^m \mathcal{O}_K)$ in the usual parity. The connecting map from level m to level $m+1$ is the transfer for the index- q inclusion $\pi^{m+1} \mathcal{O}_K \subset \pi^m \mathcal{O}_K$, which under the identifications $\Lambda^r(\pi^m \mathcal{O}_K) \cong \Lambda^r \mathcal{O}_K$ (via $\Lambda^r \pi^{-m}$) becomes t_r ; passing to B replaces the colimit over m by the same colimit, the corner being full. Integrality of t_r : pairing with Λ^{n-r} and using $\Lambda^r \pi \otimes \Lambda^{n-r} \pi \mapsto N\pi = \pm q$ on Λ^n gives $t_r = \pm(\Lambda^{n-r}\pi)^{\vee}$. The scaling conjugates level m to level $m+1$ composed with $\Lambda^r \pi$, which in the colimit is $q^{-1}\Lambda^r \pi$; the two calibrations of Remark 3.2 pin the normalization. $\square$

REMARK 3.2 (Two calibrations, $n = 1$). For $K = \mathbb{Q}$, $\pi = p = q$: at $r = 0$, $t_0 = q$ and $M_0 = \mathbb{Z}[1/q]$ with $\alpha_* = 1/q$ — the class $[1_{\mathcal{O}_p}]$ scales by the measure factor, as it must. At $r = 1$, $t_1 = q \cdot p^{-1} = 1$ and $M_1 = \mathbb{Z}$ with $\alpha_* = p/q = 1$: the unitary of a single translation coset has, in the merged level, class 1 and *not* p — this is the pivot of the whole computation, checked level by level on the Bunce–Deddens tower: the level- m generator of $K_1(M_{p^m}(C^*(p^m \mathbb{Z}))) \cong \mathbb{Z}$ is the coset unitary, the transfer to level $m+1$ carries it to the coset unitary again, and the scaling permutes the p cosets without multiplying the class. Anchors: the resulting $K_*(B \rtimes \mathbb{Z})$ below reproduces, for $K = \mathbb{Q}$, $K_0 = \mathbb{Z} \oplus \mathbb{Z}/(p-1)$ with $[1]$ generating the torsion and $K_1 = \mathbb{Z}$; at $p = 2$ this gives $K_0 \cong \mathbb{Z}$, $K_1 \cong \mathbb{Z}$, $[1] = 0$, consistent with the 2-adic ring algebra computation of Larsen–Li [80].

THEOREM 3.3 (K-theory of the affine boundary, principal single prime). *Assume $\pi^A \neq q$ for every r -subset A of the archimedean embeddings and every $0 < r < n$, where $\pi^A = \prod_{i \in A} \pi^{(i)}$. Then, with $T_r := \text{coker}(1 - \alpha_* | M_r)$ finite for $0 < r < n$ and $T_0 = \mathbb{Z}/(q-1)$:*

$$N\pi = +q : \quad K_0(\partial\mathcal{A}^{\text{aff}}) \cong \mathbb{Z} \oplus \bigoplus_{r \text{ even}} T_r, \quad K_1(\partial\mathcal{A}^{\text{aff}}) \cong \mathbb{Z} \oplus \bigoplus_{r \text{ odd}} T_r,$$

$$N\pi = -q : \quad \text{the two } \mathbb{Z}\text{'s are replaced by a single } \mathbb{Z}/2 \text{ in parity } n.$$

The class $[1]$ generates $T_0 = \mathbb{Z}/(q-1)$: its order is exactly $q-1$, attaining the divisibility bound of Chapter 3. The order of T_r is the part of $|\det(q \cdot 1 - \Lambda^r \pi)|/q^{\binom{n}{r}}$ prime to the primes inverted by the tower t_r .

PROOF SKETCH. Pimsner–Voiculescu [104] for $B \rtimes_{\alpha} \mathbb{Z}$: $0 \rightarrow \text{coker}(1 - \alpha_* | K_*(B)) \rightarrow K_*(B \rtimes \mathbb{Z}) \rightarrow \ker(1 - \alpha_* | K_{*+1}(B)) \rightarrow 0$, summed over the parity decomposition of Lemma 3.1. At $r = 0$: $1 - \alpha_* = 1 - q^{-1}$ on $\mathbb{Z}[1/q]$ has cokernel $\mathbb{Z}/(q-1)$ generated by the class of $1_{\mathcal{O}_p}$, and kernel 0. At $r = n$: $M_n = \mathbb{Z}$, $\alpha_* = N\pi/q = \pm 1$; for $+1$ both kernel and cokernel are $\mathbb{Z}$, contributing the

two free summands; for -1 , $1 - \alpha_* = 2$, cokernel $\mathbb{Z}/2$, kernel 0. At $0 < r < n$: the eigenvalues of α_* are π^A/q , none equal to 1 by hypothesis, so $1 - \alpha_*$ is injective with finite cokernel of the stated order; the extensions split at the level of orders and free ranks, the remaining extension data being Problem (b) of §4. $\square$

EXAMPLE 3.4 (Real quadratic, explicitly). $n = 2$, π' the conjugate, $t_1 = \pm\pi'$ (multiplication by the conjugate — the transfer *is* the adjugate), $M_1 = \mathcal{O}_K$ with the primes dividing $N\pi'$ inverted, and $\alpha_* = \pi/q$. Then

$$\det(1 - \alpha_*) = N_{K/\mathbb{Q}}\left(1 - \frac{\pi}{q}\right) = \frac{N(q - \pi)}{q^2} = \frac{q \mp 1 - \text{Tr } \pi}{q} \cdot (\pm 1),$$

so $K_1(\partial\mathcal{A}^{\text{aff}})$ contains the summand $\mathbb{Z}/|q \pm 1 - \text{Tr } \pi|'$ (prime-to- q part), the sign that of $N\pi = \pm q$. *The trace of the uniformizer enters the K-theory.* Concretely, in $K = \mathbb{Q}(\sqrt{2})$ with $\mathfrak{p} = (3 + \sqrt{2})$, $q = 7$, $N\pi = +7$, $\text{Tr } \pi = 6$: the summand is $\mathbb{Z}/|7 + 1 - 6| = \mathbb{Z}/2$.

REMARK 3.5 (The units are audible; the class group is at the door). Replacing π by $u\pi$ for $u \in \mathcal{O}_K^\times$ of norm 1 preserves $\mathfrak{p}$ and q but changes Tr : in $\mathbb{Q}(\sqrt{2})$, $u = 3 + 2\sqrt{2}$ gives $u\pi = 13 + 9\sqrt{2}$, $N = 7$, $\text{Tr} = 26$, and the summand becomes $\mathbb{Z}/|7 + 1 - 26| = \mathbb{Z}/18$. The affine boundary quotient is therefore *not* a canonical invariant of the prime: it hears the unit orbit of the chosen uniformizer — the reason Chapter 3's §6 had to “fix a family of uniformizers”. And the class group is located exactly at the entrance of the theory rather than inside the K-groups: it obstructs the existence of a global π at all, Problem (a) below.

4. What the affine boundary remembers, and what remains

Three layers of arithmetic survive where the multiplicative system forgets: the *norm*, as the exact order $q - 1$ of $[1]$; the *splitting data* of $\mathfrak{p}$, through the values of the exterior characteristic polynomials of π at q (for real quadratic fields, $\text{Tr } \pi$); and, unexpectedly, the *unit orbit* of the uniformizer, which makes the invariant finer than the prime itself. The degree n sets the number of layers; the unit *rank* enters only through the orbit sensitivity.

Three problems remain and are recorded as such. (a) *The ray-class repair*: for non-principal $\mathfrak{p}$ the acting monoid must be enlarged (ray classes, or a principalizing extension), and the resulting boundary compared with the principal case — this is where the class group genuinely enters. (b) *Extension data*: for $n \geq 3$, and for $|S| \geq 2$ under the iterated Pimsner–Voiculescu sequences, the orders and free ranks above determine K_* only up to group extensions. (c) *The q -part*: the torsion orders are computed prime to the primes inverted by the transfers; the exact q -primary bookkeeping requires the integral colimit, not just its localization.

CHAPTER 11

The ray-class repair for affine systems: the class group as the index of definable scalings

This chapter is Paper XXXI of the series (papers/XXXI-rayclass/).

SUMMARY. Chapter 10 computed the K-theory of the affine boundary quotient at a principal prime and located the class group “at the entrance”: for non-principal $\mathfrak{p}$ the affine monoid of Chapter 3 (its §6) is not well defined over $\mathcal{O}_K$. This paper performs the repair and proves that the obstruction is itself the invariant. Let $h = \text{ord}([\mathfrak{p}] \in \text{Cl}_K)$ and $\mathfrak{p}^h = (\pi)$. The globally definable scalings form exactly the subsemigroup $h\mathbb{N}$ of the local valuation semigroup: a scaling of depth j exists over $\mathcal{O}_K$ if and only if $h \mid j$ (Theorem 1.2). The minimal repaired system is therefore the h -step system $C(\mathcal{O}_{\mathfrak{p}}) \rtimes (\mathcal{O}_K \rtimes_{\pi} \mathbb{N})$, well defined for every $\mathfrak{p}$, reducing to Paper XXX when $h = 1$; its boundary quotient has dilated form $C_0(K_{\mathfrak{p}}) \rtimes (\mathcal{O}_{K, \{\mathfrak{p}\}} \rtimes_{\pi} \mathbb{Z})$, and the entire exterior-tower Pimsner–Voiculescu computation of Chapter 10 applies verbatim with (π, q^h) in place of (π, q) , $q = N\mathfrak{p}$. Consequences: the class of the unit generates a torsion summand of *exact* order $q^h - 1 = N\mathfrak{p}^h - 1$ — the order of $\mathfrak{p}$ in the class group survives in K_0 as the exact torsion order of $[1]$, and the published divisibility statement of Chapter 3 must read $(N\mathfrak{p}^h - 1)[1] = 0$ in the repaired system; the intermediate layers contribute $\mathbb{Z}/|q^h \pm 1 - \text{Tr } \pi|'$ for quadratic fields, computed explicitly in $\mathbb{Q}(\sqrt{-5})$ at the non-principal prime above 3 ($K_0 \supseteq \mathbb{Z} \oplus \mathbb{Z}/8$ with $[1]$ of order 8, $K_1 \supseteq \mathbb{Z} \oplus \mathbb{Z}/2$). What the K-groups alone retain is the composite q^h ; separating q from h , the Galois descent from the Hilbert class field (where capitulation principalizes $\mathfrak{p}$ and $\text{Gal} \cong \text{Cl}_K$ acts), and the several-prime case are recorded as the remaining problems.

1. The bottleneck, and the scaling-index theorem

Fix a prime $\mathfrak{p}$ of K , $q = N\mathfrak{p}$, and let h be the order of $[\mathfrak{p}]$ in the class group Cl_K , so that $\mathfrak{p}^h = (\pi)$ is principal and no smaller positive power is. The affine monoid of Chapter 3 (its §6) acts by $x \mapsto \gamma x + b$ with $b \in \mathcal{O}_K$ and γ a chosen uniformizing multiplier; the composition law requires $\gamma\mathcal{O}_K \subseteq \mathcal{O}_K$ with γ supported at $\mathfrak{p}$, i.e. $\gamma \in \mathcal{O}_K$ with $(\gamma) = \mathfrak{p}^j$ for some $j \geq 1$.

LEMMA 1.1. *An element $\gamma \in \mathcal{O}_K$ with $(\gamma) = \mathfrak{p}^j$ exists if and only if $h \mid j$.*

PROOF. $(\gamma) = \mathfrak{p}^j$ says $[\mathfrak{p}]^j$ is trivial in Cl_K , which is $h \mid j$; conversely $\pi^{j/h}$ works. $\square$

THEOREM 1.2 (The class group is the index of definable scalings). *The set of depths j for which the affine system of Chapter 3’s §6 admits a well-defined scaling of depth j over $\mathcal{O}_K$ is exactly $h\mathbb{N}$. The minimal well-defined affine system at $\mathfrak{p}$ is therefore the h -step system*

$$\mathcal{A}_{K, \{\mathfrak{p}\}}^{\text{aff}} := C(\mathcal{O}_{\mathfrak{p}}) \rtimes (\mathcal{O}_K \rtimes_{\pi} \mathbb{N}), \quad (\pi) = \mathfrak{p}^h,$$

with boundary quotient defined by the Cuntz relation over the q^h translates $\sum_{a \in \mathcal{O}_K/\mathfrak{p}^h} \mu_a \mu_a^ = 1$; it reduces to the system of Chapter 10 when $h = 1$. Any choice of local uniformizer at intermediate depths $0 < j < h$ carries the translation module $\mathcal{O}_K$ to a fractional module in the ideal class $[\mathfrak{p}]^{-j} \neq 1$ and leaves the category of $\mathcal{O}_K$ -affine systems.*

PROOF. Lemma 1.1 gives the depths. For the last statement, a local uniformizer $\varpi \in K_{\mathfrak{p}}$ of depth j maps the dense image of a fractional ideal $\mathfrak{a}$ to that of an ideal in the class $[\mathfrak{p}]^j[\mathfrak{a}]$

up to prime-to- $\mathfrak{p}$ global rescalings, which act by homeomorphisms of $K_{\mathfrak{p}}$; the marked class is an invariant of the pair (translation module, ambient $K_{\mathfrak{p}}$) modulo such rescalings, and it changes unless $h \mid j$. $\square$

REMARK 1.3. Theorem 1.2 upgrades the observation of Chapter 10 — “the class group is at the entrance” — to a structure theorem: the entrance fee is exactly h . Route B of the repair, passing to the Hilbert class field H where the extended ideal $\mathfrak{p}\mathcal{O}_H$ is principal by the principal ideal theorem [97, VI.7] and $\text{Gal}(H/K) \cong \text{Cl}_K$ acts, defines a one-step system upstairs whose Galois descent should remember the group Cl_K itself rather than only h ; its K-theory is Problem (ii) of §3.

2. K-theory of the repaired boundary

THEOREM 2.1. *Let $n = [K : \mathbb{Q}]$ and $N_{K/\mathbb{Q}}\pi = \pm q^h$. The dilated form of the repaired boundary is*

$$\partial\mathcal{A}_{K,\{\mathfrak{p}\}}^{\text{aff}} \otimes \mathbb{K} \cong C_0(K_{\mathfrak{p}}) \rtimes (\mathcal{O}_{K,\{\mathfrak{p}\}} \rtimes_{\pi} \mathbb{Z}),$$

where $\mathcal{O}_{K,\{\mathfrak{p}\}} = \bigcup_m \pi^{-m} \mathcal{O}_K$ is the ring of $\{\mathfrak{p}\}$ -integers, dense in $K_{\mathfrak{p}}$; and the exterior-tower computation of Chapter 10 applies verbatim with q^h in place of q : with $M_r = \varinjlim (\Lambda^r \mathcal{O}_K, t_r)$, $t_r = \pm(\Lambda^{n-r}\pi)^{\vee}$, $\alpha_* = q^{-h}\Lambda^r\pi$, and $T_r = \text{coker}(1 - \alpha_*|M_r)$, one has, when $N\pi = +q^h$ and $\pi^A \neq q^h$ for all proper archimedean subsets A ,

$$K_0 \cong \mathbb{Z} \oplus \bigoplus_{r \text{ even}} T_r, \quad K_1 \cong \mathbb{Z} \oplus \bigoplus_{r \text{ odd}} T_r, \quad T_0 = \mathbb{Z}/(q^h - 1) \text{ generated by } [1],$$

(the two $\mathbb{Z}$ ’s replaced by a single $\mathbb{Z}/2$ in parity n when $N\pi = -q^h$), the orders of the middle layers being the regularized $|\det(q^h \cdot 1 - \Lambda^r \pi)|$ as in Paper XXX. In particular:

- (1) The order of $\mathfrak{p}$ in the class group survives in K_0 : the class $[1]$ has exact order $N\mathfrak{p}^h - 1$. The divisibility statement of Chapter 3’s §6 must accordingly read $(N\mathfrak{p}^h - 1)[1] = 0$ for non-principal $\mathfrak{p}$, and it is attained.
- (2) For K quadratic, $K_1(\partial\mathcal{A}^{\text{aff}}) \supseteq \mathbb{Z}/|q^h \pm 1 - \text{Tr } \pi|'$ (prime-to- q part), the sign that of $N\pi$.

PROOF. The dilation, the tower, the two calibrations and the Pimsner–Voiculescu [104] assembly are those of Chapter 10, restated there in full; the only inputs were a global generator of the scaling ideal and its norm, here π and $N(\mathfrak{p}^h) = q^h$. The exactness of the order of $[1]$ is the $r = 0$ layer: $1 - \alpha_* = 1 - q^{-h}$ on $\mathbb{Z}[1/q]$ has cokernel $\mathbb{Z}/(q^h - 1)$ generated by the class of $1_{\mathcal{O}_{\mathfrak{p}}}$. $\square$

EXAMPLE 2.2 ($\mathbb{Q}(\sqrt{-5})$ at the non-principal prime above 3). $K = \mathbb{Q}(\sqrt{-5})$, $\text{Cl}_K \cong \mathbb{Z}/2$, $\mathfrak{p}_3 = (3, 1 + \sqrt{-5})$ non-principal, $h = 2$, $q = 3$, and $\mathfrak{p}_3^2 = (\pi)$ with $\pi = 2 - \sqrt{-5}$, $N\pi = 9 = q^h$, $\text{Tr } \pi = 4$. Then: $[1]$ has exact order $q^h - 1 = 8$, so $K_0 \supseteq \mathbb{Z} \oplus \mathbb{Z}/8$ with $[1]$ generating the torsion; and the middle layer gives $N(q^h - \pi) = N(7 + \sqrt{-5}) = 54 = 2 \cdot 27$, of prime-to-3 part 2, so $K_1 \supseteq \mathbb{Z} \oplus \mathbb{Z}/2$, consistently with $|q^h + 1 - \text{Tr } \pi| = |9 + 1 - 4| = 6$ of prime-to-3 part 2. Had $\mathfrak{p}_3$ been principal the order of $[1]$ would have been $q - 1 = 2$: the K-theory separates the two situations.

REMARK 2.3 (Separating q from h : the middle layers generically do). The order of $[1]$ sees only the composite q^h , and pairs realizing the same composite exist: $\mathbb{Q}(\sqrt{-5})$ at $\mathfrak{p}_3$ ($q = 3$, $h = 2$) against $\mathbb{Q}(i)$ at the inert prime (3) ($q' = 9$, $h' = 1$, principal) both have $\text{ord}[1] = 8$. But the middle layers separate them: for $\mathbb{Q}(i)$, $\pi = 3$ acts as the scalar $\frac{1}{3}$ on $M_1 \cong \mathbb{Z}[1/3]^2$, so $T_1 = \text{coker}(\frac{2}{3})^{\oplus 2} \cong (\mathbb{Z}/2)^2$ of order 4, against the $\mathbb{Z}/2$ of Example 2.2. So Problem (i) of §3 is partially resolved: the K-groups themselves generically separate (q, h) through the trace data of the intermediate exterior layers, and only pairs with matching characteristic data at every layer remain for the dual coaction to distinguish.

REMARK 2.4 (Unit-orbit sensitivity persists). As in Chapter 10, replacing π by $u\pi$ for a unit u of norm 1 changes the middle layers; for imaginary quadratic fields the unit group is finite and the ambiguity is a root of unity, so the repaired invariant is canonical there up to finitely many choices — a small unexpected bonus of the imaginary quadratic case.

3. The arithmetic verdict, and what remains

The class group enters the K-theory as *torsion, not rank*: the free ranks of Theorem 2.1 are those of Chapter 10, unchanged, while the order of the distinguished class [1] is promoted from $q - 1$ to $q^h - 1$. Three problems remain. (i) *Separating q from h* : partially resolved by Remark 2.3 — the middle layers generically separate; what remains is the exceptional locus of pairs with matching layer data, where the dual coaction is the candidate. (ii) *Galois descent from the Hilbert class field*: capitulation principalizes $\mathfrak{p}$ upstairs with $\mathrm{Gal}(H/K) \cong \mathrm{Cl}_K$; the equivariant K-theory of the descended system should remember Cl_K itself, and its computation — a genuine descent spectral sequence, not an iterated Pimsner–Voiculescu — is open. (iii) *Several primes*: for $|S| \geq 2$ the definable scalings form the kernel of $\mathbb{Z}^S \rightarrow \mathrm{Cl}_K$, $e \mapsto \prod \mathfrak{p}^{e_{\mathfrak{p}}}$, a finite-index subgroup whose index divides $|\mathrm{Cl}_K|$ and whose K-theoretic footprint, layer by layer, is the natural continuation of this paper.

CHAPTER 12

Galois descent for affine Bost–Connes boundaries: capitulation, the obstruction cocycle, and the emergence of the class group

This chapter is Paper XXXII of the series ([papers/XXXII-descent/](#)).

SUMMARY. Let $\mathfrak{p}$ be a non-principal prime of K , $h = \text{ord}([\mathfrak{p}] \in \text{Cl}_K)$, H the Hilbert class field, $G = \text{Gal}(H/K) \cong \text{Cl}_K$. By the principal ideal theorem $\mathfrak{p}\mathcal{O}_H = (\Pi)$ is principal, and $\mathfrak{p}$ splits in H into $g = |\text{Cl}_K|/h$ primes of residue degree h , the Frobenius being the class $[\mathfrak{p}]$ itself under Artin. We construct the capitulated affine system over H on $\prod_{i=1}^g \mathcal{O}_{\mathfrak{P}_i}$, with unit-extended scaling monoid $\mathcal{O}_H^\times \Pi^\mathbb{N}$ — the extension is forced: $\sigma(\Pi) = u_\sigma \Pi$ with $u_\sigma \in \mathcal{O}_H^\times$, so G acts only on the unit-extended system — and prove the descent theorems. First, the obstruction cocycle $c_{\mathfrak{p}} : \sigma \mapsto \sigma(\Pi)/\Pi$ represents $\delta([\mathfrak{p}])$ under a canonical injection $\delta : \text{Cl}_K \hookrightarrow H^1(G, \mathcal{O}_H^\times)$ built from Hilbert 90 and the principal ideal theorem: *the full class group is exactly the group of affine descent obstructions*, and an invariant generator exists if and only if $\mathfrak{p}$ is principal. Second, the three scalings are related by the norm, $\pi = N_{H/K}(\Pi)$ up to $\mathcal{O}_K^\times$, and Green imprimitivity reduces $\partial \mathcal{A}_{H,\Pi}^{\text{aff}} \rtimes G$ to the inert local model crossed by the decomposition group $\langle \text{Frob}_{\mathfrak{p}} \rangle \cong \mathbb{Z}/h$; by Julg this computes the equivariant K-theory. Third, the ladder of unit classes is explicit: $\text{ord}[1] = q^h - 1$ at the bottom ($q = N\mathfrak{p}$; Chapter 11) and $\text{ord}[1] = q^{|\text{Cl}_K|} - 1$ at the top — the order of the class at the bottom, the class *number* at the top, and the group structure itself in the obstruction cocycles. The integral equivariant K-groups beyond the reduction, and the Morita question for the fixed-point (rather than crossed-product) relation, are recorded as the remaining problems.

1. The capitulated system and the Galois action

Throughout, $\mathfrak{p}$ is a prime of K with $q = N\mathfrak{p}$ and $h = \text{ord}([\mathfrak{p}] \in \text{Cl}_K)$; H is the Hilbert class field, $G = \text{Gal}(H/K) \cong \text{Cl}_K$ by Artin, and $\mathfrak{p}\mathcal{O}_H = (\Pi)$ by the principal ideal theorem [97, VI.7].

LEMMA 1.1 (Splitting structure). *$\mathfrak{p}$ splits in H into $g = |\text{Cl}_K|/h$ distinct primes $\mathfrak{P}_1, \dots, \mathfrak{P}_g$, unramified of residue degree $f = h$, and G permutes them transitively with stabilizer of $\mathfrak{P}_1$ the cyclic group $C := \langle \text{Frob}_{\mathfrak{p}} \rangle \cong \mathbb{Z}/h$. Moreover $N_H(\mathfrak{p}\mathcal{O}_H) = q^{|\text{Cl}_K|}$.*

PROOF. H/K is abelian unramified; the Artin map sends $\text{Frob}_{\mathfrak{p}}$ to $[\mathfrak{p}]$, of order h , so $f = h$ and $g = [H : K]/f = |\text{Cl}_K|/h$; the norm of the extended ideal is $q^{fg} = q^{|\text{Cl}_K|}$. $\square$

PROPOSITION 1.2 (The top system, and why the units are forced). *On $X_H := \prod_{i=1}^g \mathcal{O}_{\mathfrak{P}_i}$, with $\mathcal{O}_H$ acting densely by translation (strong approximation) and the monoid $M_H := \mathcal{O}_H^\times \cdot \Pi^\mathbb{N}$ acting by multiplication ($v_{\mathfrak{P}_i}(\Pi) = 1$ for every i), the affine system $\mathcal{A}_{H,\Pi}^{\text{aff}} := C(X_H) \rtimes (\mathcal{O}_H \rtimes M_H)$ and its boundary quotient $\partial \mathcal{A}_{H,\Pi}^{\text{aff}}$ — Cuntz relation over the $q^{|\text{Cl}_K|}$ translates $\mathcal{O}_H/(\Pi)$ — are well defined, and G acts on them by*

$$\beta_\sigma(f) = f \circ \sigma^{-1}, \quad \beta_\sigma(u_b) = u_{\sigma b}, \quad \beta_\sigma(\mu_\gamma) = \mu_{\sigma\gamma} \quad (\gamma \in M_H),$$

using $\sigma(X_H) = X_H$ (permutation of the factors through the $\mathfrak{P}_i$) and $\sigma(M_H) = M_H$. The unit extension is forced: $\sigma(\Pi) = u_\sigma \Pi$ with $u_\sigma = \sigma(\Pi)/\Pi \in \mathcal{O}_H^\times$ (both sides generate the G -invariant ideal $\mathfrak{p}\mathcal{O}_H$), so on the minimal monoid $\Pi^\mathbb{N}$ the Galois action is defined only up to the unit twist

u_σ , and no sub-monoid on which G acts omits the units unless the twist is a coboundary — which, by Theorem 2.1, happens exactly when $\mathfrak{p}$ is principal.

2. The obstruction cocycle is the class

THEOREM 2.1 (Emergence of the full class group). *The map $\sigma \mapsto u_\sigma = \sigma(\Pi)/\Pi$ is a 1-cocycle whose class $[c_{\mathfrak{p}}] \in H^1(G, \mathcal{O}_H^\times)$ is independent of the choice of Π , and there is a canonical injective homomorphism*

$$\delta : \text{Cl}_K \hookrightarrow H^1(G, \mathcal{O}_H^\times), \quad \delta([\mathfrak{a}]) = [\sigma \mapsto \sigma(\alpha)/\alpha] \text{ for } \mathfrak{a}\mathcal{O}_H = (\alpha),$$

with $\delta([\mathfrak{p}]) = [c_{\mathfrak{p}}]$. In particular: an invariant generator of $\mathfrak{p}\mathcal{O}_H$ exists if and only if $\mathfrak{p}$ is principal in K ; and the group of descent obstructions of the affine systems, as $\mathfrak{p}$ ranges over the primes of K , is exactly $\delta(\text{Cl}_K) \cong \text{Cl}_K$.

PROOF. Cocycle identity and independence of Π are immediate ($\Pi \mapsto v\Pi$ changes $c_{\mathfrak{p}}$ by the coboundary of v). For δ : every ideal capitulates ($\mathfrak{a}\mathcal{O}_H$ principal, the principal ideal theorem [97, VI.7]), so δ is defined on all of Cl_K ; it is well defined and homomorphic by the same coboundary computation, principal ideals of K mapping to coboundaries of elements of $K^\times$, which are trivial. Injectivity: if $\sigma(\alpha)/\alpha = \sigma(w)/w$ for all σ and some $w \in \mathcal{O}_H^\times$, then $\alpha/w \in H^{\times G} = K^\times$ and $\mathfrak{a}\mathcal{O}_H = (\alpha/w)\mathcal{O}_H$ with $\alpha/w \in K^\times$; taking valuations at the primes of K (unramifiedness) gives $\mathfrak{a} = (\alpha/w)$ in $\mathcal{O}_K$, so $[\mathfrak{a}] = 1$. (Equivalently: δ is the connecting map of $1 \rightarrow \mathcal{O}_H^\times \rightarrow H^\times \rightarrow P_H \rightarrow 1$ restricted to extended ideals, injective by Hilbert 90.) The last statement holds since prime classes generate Cl_K . $\square$

3. Descent: norm, imprimitivity, and the ladder of unit classes

PROPOSITION 3.1 (The bottom scaling is the norm of the top scaling). $(N_{H/K}(\Pi)) = \mathfrak{p}^{|\text{Cl}_K|} = (\pi)^{|\text{Cl}_K|/h}$, so $N_{H/K}(\Pi) = \pi^g u$ for a unit $u \in \mathcal{O}_K^\times$: the Chapter 11 scaling π is, up to units and the transitivity index g , the norm of the capitulated scaling. For $g = 1$ (that is, $h = |\text{Cl}_K|$, the class $[\mathfrak{p}]$ generating), $\pi = N_{H/K}(\Pi)u$ exactly.

PROOF. $N_{H/K}(\mathfrak{p}\mathcal{O}_H) = \mathfrak{p}^{[H:K]} = \mathfrak{p}^{hg} = (\pi)^g$; apply $N_{H/K}$ to Π . $\square$

THEOREM 3.2 (Imprimitivity, and the equivariant K-theory reduced). *By Julg, $K_*^G(\partial\mathcal{A}_{H,\Pi}^{\text{aff}}) \cong K_*(\partial\mathcal{A}_{H,\Pi}^{\text{aff}} \rtimes G)$ [66]; and since $X_H \cong \text{Ind}_C^G \mathcal{O}_{\mathfrak{p}_1}$ as a G -space (Lemma 1.1), Green's imprimitivity theorem [48] gives a strong Morita equivalence*

$$\partial\mathcal{A}_{H,\Pi}^{\text{aff}} \rtimes G \sim_M \partial\mathcal{A}_{\mathfrak{p}_1}^{\text{aff},\text{loc}} \rtimes C,$$

where $\partial\mathcal{A}_{\mathfrak{p}_1}^{\text{aff},\text{loc}}$ is the inert local model — the boundary system on $\mathcal{O}_{\mathfrak{p}_1}$ alone, translations by $\mathcal{O}_H$ through its dense image, scalings by $\mathcal{O}_H^\times \Pi^\mathbb{N}$ — and $C \cong \mathbb{Z}/h$ is the decomposition group, acting through the local Frobenius. The computation of the integral $K_*(\partial\mathcal{A}_{\mathfrak{p}_1}^{\text{aff},\text{loc}} \rtimes \mathbb{Z}/h)$ is the reduction achieved; it is not completed here (Problem (i) of §4).

THEOREM 3.3 (The ladder of unit classes). *In K_0 of the respective boundary quotients:*

$$\text{ord}[1_{\partial\mathcal{A}_{K,\{\mathfrak{p}\}}^{\text{aff}}}] = q^h - 1 \quad \text{and} \quad \text{ord}[1_{\partial\mathcal{A}_{H,\Pi}^{\text{aff}}}] = q^{|\text{Cl}_K|} - 1.$$

The bottom sees the order of the class, the top the class number, and the group structure of Cl_K itself lives in the obstruction cocycles of Theorem 2.1.

PROOF. The bottom is Chapter 11. For the top, the $r = 0$ layer of the exterior-tower argument of Chapter 10 applies on X_H : $C(X_H) = \varinjlim_m C(\mathcal{O}_H/(\Pi^m))$ with transitive translation action, $|\mathcal{O}_H/(\Pi)| = q^{|\text{Cl}_K|}$ by Lemma 1.1, so $M_0 = \mathbb{Z}[1/q]$ with the scaling acting by $q^{-|\text{Cl}_K|}$, and

$\text{coker}(1 - q^{-|\text{Cl}_K|}) = \mathbb{Z}/(q^{|\text{Cl}_K|} - 1)$ generated by the class of the unit; the additional unit scalings act trivially on M_0 (they preserve every Π -adic cylinder's measure class and fix the class of 1), so the order is exact. $\square$

4. What remains

(i) The integral computation of $K_*(\partial\mathcal{A}_{\mathfrak{p}_1}^{\text{aff}, \text{loc}} \rtimes \mathbb{Z}/h)$: the exterior tower of Chapter 10 is $\mathbb{Z}/h$ -equivariant, and what is needed is the equivariant colimit — concretely, the $R(\mathbb{Z}/h)$ -module structure of the layers $M_r(\mathcal{O}_H)$ under Frobenius — not a new idea but a genuine computation, already nontrivial for the first inert example ($K = \mathbb{Q}(\sqrt{-5})$, $H = \mathbb{Q}(i, \sqrt{5})$, $\mathfrak{p}_3$ inert in H/K , $C = G = \mathbb{Z}/2$). (ii) The fixed-point question: the Chapter 11 bottom system is the C -fixed-point system of the inert local model ($\mathcal{O}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}_1}^C$, $\mathcal{O}_K = \mathcal{O}_H^C$, $\pi = N(\Pi)u$), and whether fixed-point and crossed-product descents are Morita equivalent is governed by the properness of the C -action on the boundary groupoid — measure-theoretically the fixed locus is null, topologically it is not, and the discrepancy is exactly where the obstruction cocycle $[c_{\mathfrak{p}}]$ should reappear. (iii) The unit-extended monoid enlarges the top algebra by the infinite unit group $\mathcal{O}_H^\times$; its own K-theoretic footprint (a torus of rank the unit rank of H) must be separated from the class-group data in any complete answer.

CHAPTER 13

Integral equivariant K-theory of the inert local model: the ray-class entanglement and exact torsion

This chapter is Paper XXXVIII of the series ([papers/XXXVIII-inert-ktheory/](#)).

SUMMARY. We carry out the first integral case of Chapter 12’s Problem (i): the equivariant K-theory of the inert local model at $K = \mathbb{Q}(\sqrt{-5})$, $H = \mathbb{Q}(i, \sqrt{5})$, $\mathfrak{p}_3 = (3, 1 + \sqrt{-5})$ inert in H/K , $C = \text{Gal}(H/K) \cong \mathbb{Z}/2$. The ring $\mathcal{O}_H = \mathbb{Z}[i] \otimes \mathbb{Z}[\varphi]$ is a *free* $\mathbb{Z}[C]$ -module of rank two (a Tate cohomology computation), so the entire exterior tower carries explicit integral C -structure: $\Lambda^0 = \mathbb{Z}_+$, $\Lambda^1 \cong \mathbb{Z}[C]^2$, $\Lambda^2 \cong \mathbb{Z}_-^2 \oplus \mathbb{Z}[C]^2$, $\Lambda^3 \cong \mathbb{Z}[C]^2$, $\Lambda^4 = \mathbb{Z}_+$. A finite search produces the explicit generator $\Pi = -3 - i + 2\varphi + i\varphi$ of a prime above 3, with $N_{H/\mathbb{Q}}\Pi = +9$ and $\text{Tr}_{H/\mathbb{Q}}\Pi = -8$, and Smith normal forms of $9 \cdot 1 - \Lambda^r\Pi$ evaluate the intermediate layers exactly (prime-to-3 regularization): $\mathbb{Z}/5 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/61$ at $r = 1$; $\mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/7 \oplus \mathbb{Z}/11$ at $r = 2$, containing the sign-twisted $\mathbb{Z}_-^2$ -block; $\mathbb{Z}/29$ at $r = 3$. Assembling with the top and bottom layers gives

$$K_0 \supseteq \mathbb{Z}^2 \oplus (\mathbb{Z}/8)^2 \oplus [\mathbb{Z}/2^{\oplus 2} \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/7 \oplus \mathbb{Z}/11], \quad K_1 \supseteq \mathbb{Z}^2 \oplus [\mathbb{Z}/5^{\oplus 2} \oplus \mathbb{Z}/61] \oplus \mathbb{Z}/29,$$

where $(\mathbb{Z}/8)^2 = \mathbb{Z}/(q^h - 1) \otimes R(\mathbb{Z}/2)$: the unit class is split by the spectral projections $\frac{1 \pm U_\sigma}{2}$ into a pair of classes of exact order 8. The entanglement law this computation exhibits: the class group doubles or sign-twists exactly the *non-induced* strata, while free $\mathbb{Z}[C]$ -blocks count once by Shapiro — the class group does not change the orders of the middle layers, only their representation labels. Two honest gaps close the paper: the u_σ -twisted bookkeeping of those labels, and the extension data in degree four. A structural by-product is recorded: the σ -fixed locus of the inert model is the bottom h -step system of Chapter 11 — null in measure, nonempty in topology — so Paper XXXII’s Problem (ii) is a direct summand of Problem (i), not its sequel.

1. The explicit generator and the equivariant exterior tower

The discriminants -4 and 5 are coprime, so $\mathcal{O}_H = \mathbb{Z}[i] \otimes \mathbb{Z}[\varphi]$, $\varphi = \frac{1+\sqrt{5}}{2}$, with σ (the generator of C , fixing $\sqrt{-5} = i\sqrt{5}$) acting by $i \mapsto -i$, $\varphi \mapsto 1 - \varphi$.

LEMMA 1.1. $\mathbb{Z}[\varphi] \cong \mathbb{Z}[C]$ as a C -module ($\hat{H}^0 = \hat{H}^1 = 0$: the norm has image $\mathbb{Z}$ and $\ker N = \text{im}(1 - \sigma) = \mathbb{Z}\sqrt{5}$), and $\mathbb{Z}[i] \cong \mathbb{Z}_+ \oplus \mathbb{Z}_-$; hence $\mathcal{O}_H \cong \mathbb{Z}[C]^2$ is free, and the integral exterior tower is

$$\Lambda^0 = \mathbb{Z}_+, \quad \Lambda^1 \cong \mathbb{Z}[C]^2, \quad \Lambda^2 \cong \mathbb{Z}_-^2 \oplus \mathbb{Z}[C]^2, \quad \Lambda^3 \cong \mathbb{Z}[C]^2, \quad \Lambda^4 = \mathbb{Z}_+,$$

the $\mathbb{Z}_-$ ’s in Λ^2 coming from $\Lambda_{\mathbb{Z}}^2(\mathbb{Z}[C])$ ($\sigma(1 \wedge \sigma) = -(1 \wedge \sigma)$), and $\det \sigma = +1$ on Λ^4 .

PROPOSITION 1.2 (The generator). $\Pi = -3 - i + 2\varphi + i\varphi \in \mathcal{O}_H$ satisfies $N_{H/\mathbb{Q}}\Pi = +9$ and $\text{Tr}_{H/\mathbb{Q}}\Pi = -8$; its multiplication matrix A on the basis $(1, i, \varphi, i\varphi)$ has $\det A = +9$, so $(\Pi) =: \mathfrak{P}_1$ is a prime of norm 9 above 3, and the inert model of Chapter 12 at $\mathfrak{P}_1$ is defined with scaling Π .

2. Exact evaluation of the intermediate layers

The tower transfers invert only the prime 3 (they are built from the Π -adjugates, of determinant a power of $N\Pi = 9$), so the regularized cokernel of $1 - \alpha_*$, $\alpha_* = 9^{-1}\Lambda^r\Pi$, is the prime-to-3 part of the integral cokernel of $9 \cdot 1 - \Lambda^r A$, computed by Smith normal form:

THEOREM 2.1. *With T_r the regularized cokernel at layer r :*

$$\begin{aligned} r = 1 : \quad & \det(9 - \Lambda^1 A) = 13725 = 3^2 \cdot 5^2 \cdot 61, \quad T_1 \cong \mathbb{Z}/5 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/61; \\ r = 2 : \quad & \det(9 - \Lambda^2 A) = -1122660, \quad T_2 \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/7 \oplus \mathbb{Z}/11; \\ r = 3 : \quad & \det(9 - \Lambda^3 A) = 21141 = 3^6 \cdot 29, \quad T_3 \cong \mathbb{Z}/29, \end{aligned}$$

T_2 containing the contribution of the sign-twisted $\mathbb{Z}_-^2$ -block of Lemma 1.1. At the ends, $T_0 = \mathbb{Z}/(9 - 1) = \mathbb{Z}/8$ generated by the unit class, and $M_4 = \mathbb{Z}$ with $\alpha_* = N\Pi/9 = +1$.

3. The K-theory of the crossed product and the entanglement law

THEOREM 3.1. *For the inert local model at $\mathfrak{P}_1$ with its $\mathbb{Z}/2$ -action,*

$$\begin{aligned} K_0(\partial\mathcal{A}_{\mathfrak{P}_1}^{\text{aff},\text{loc}} \rtimes \mathbb{Z}/2) &\supseteq \mathbb{Z}^2 \oplus (\mathbb{Z}/8)^2 \oplus [\mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/7 \oplus \mathbb{Z}/11], \\ K_1(\partial\mathcal{A}_{\mathfrak{P}_1}^{\text{aff},\text{loc}} \rtimes \mathbb{Z}/2) &\supseteq \mathbb{Z}^2 \oplus [\mathbb{Z}/5 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/61] \oplus \mathbb{Z}/29. \end{aligned}$$

Here $(\mathbb{Z}/8)^2 = T_0 \otimes R(\mathbb{Z}/2)$: the unit class is split by the spectral projections $\frac{1 \pm U_\sigma}{2}$ of the implementing unitary into two classes of exact order 8, one per character of the class group; and the two $\mathbb{Z}$'s are the $R(\mathbb{Z}/2)$ -doubling of the trivially-acted top layer.

REMARK 3.2 (The entanglement law). The class group entangles with the q -torsion precisely on the *non-induced* strata: layers with trivial or sign action ($r = 0, r = 4$, and the $\mathbb{Z}_-^2$ -block of Λ^2) are doubled or sign-twisted by $R(\mathbb{Z}/2)$, while the free $\mathbb{Z}[C]$ -blocks contribute once, by Shapiro's lemma. The class group does not change the orders of the middle layers — those are the Smith invariants of Theorem 2.1 — it changes their representation labels.

REMARK 3.3 (Problem (ii) is a summand of Problem (i)). The σ -fixed locus of $\mathcal{O}_{\mathfrak{P}_1}$ is $\mathcal{O}_{\mathfrak{P}_3}$: null for the Gibbs measure, nonempty in topology — and K-theory sees topology. The fixed-point contribution to the equivariant theory is the bottom h -step system of Chapter 11 re-entering as the fixed subgroupoid: the fixed-point-versus-crossed-product question of Chapter 12 is a direct summand of the present computation, not its sequel.

4. Assessment

Two gaps remain and are recorded as such. (a) *Character bookkeeping*: the scaling is equivariant only up to the descent cocycle $u_\sigma = \sigma\Pi/\Pi$ of Chapter 12, so the middle layers are twisted $R(\mathbb{Z}/2)$ -modules; their orders are fixed by Theorem 2.1, but the assignment of each cyclic summand to the trivial or sign character awaits the twisted bookkeeping. (b) *Extension data*: in degree four the Pimsner–Voiculescu assembly determines K_* up to group extensions, as in Paper XXX. Neither gap touches the displayed subgroups, the split unit class, or the entanglement law.

CHAPTER 14

The free variant and the free critical exponent

This chapter is Paper V of the series (papers/V-free/).

SUMMARY. Replacing the free abelian monoid J_S of the localized Bost–Connes system by the free monoid $\mathbb{F}_S^+$ produces a system whose statistics are free-probabilistic and whose thermodynamics is governed by a new arithmetic invariant. We determine both, and conclude that the deformation is incompatible with the phase structure it was meant to probe.

The incompatibility begins at the definition: keeping the coefficient algebra $C(Y_S)$ while freeing the isometries is inconsistent, since orthogonal range projections contradict the nested ones of $C(Y_S)$, and the universal algebra is zero. The consistent free system is the Toeplitz–Cuntz algebra of $\mathbb{F}_S^+$ with the norm dynamics; it retains the norms of S and nothing else — no symmetry group G_S , no obstruction group Ξ_β . Its partition function is $Z_{\text{free}}(\beta) = (1 - L_S(\beta))^{-1}$ with $L_S(\beta) = \sum_{\mathbf{p} \in S} N\mathbf{p}^{-\beta}$, so the critical point solves $L_S = 1$: a Galton–Watson branching criticality rather than an abscissa of convergence. The range projections being mutually orthogonal, every KMS_β state forces $L_S(\beta) \leq 1$, so *no KMS_β state exists below the threshold*, and above it there is exactly one — the same mechanism by which partitioning, as opposed to nested, range projections restrict the admissible temperatures in the $ax + b$ systems (Chapter 3).

In the Fock–Gibbs state the field operators $\Phi_{\mathbf{p}} = \mu_{\mathbf{p}} + \mu_{\mathbf{p}}^*$ have an exactly computable law on $[-2, 2]$: with $t = N\mathbf{p}^{-\beta}$ the moments are $m_2 = 1 + t$, $m_4 = 2 + 3t + t^2$ and the free cumulant $\kappa_4 = -t(1 + t)$ is strictly negative, so $\Phi_{\mathbf{p}}$ is semicircular only in the limit $\beta \rightarrow \infty$. The fields are not exactly free: the obstruction is the hard-core exclusion $e_{\mathbf{p}}e_{\mathbf{q}} = 0$, giving the mixed cumulant $-t_{\mathbf{p}}t_{\mathbf{q}}$, and asymptotic freeness holds at exactly that rate. Two natural guesses fail: $\mathcal{N}_{\mathbf{p}} = \mu_{\mathbf{p}}\mu_{\mathbf{p}}^*$ is a *projection*, so its law is Bernoulli and not free Poisson; and Voiculescu’s microstates free entropy is undefined for $n \geq 2$ variables, the KMS_β state not being tracial.

The invariant $\beta_c^{\text{free}}(S)$, the root of $L_S = 1$, is arithmetically of a different kind from the classical critical exponent. It is not a tail invariant: over $\mathbb{Q}$, deleting the single prime 2 moves it from $1.399433\dots$, the root of the prime zeta function minus one, to $1.224388\dots$; and it obeys the universal bound $\beta_c^{\text{free}}(S) \leq \beta_c^{\text{free}}(P_{\mathbb{Q}})$. Applied to the cascade constructions of [Ch. 2] the conclusion is sharp and negative: there $L_S(1) = \infty$ by design, so $\beta_c^{\text{free}}(S) > 1$, while the entire transition locus lies in $(0, 1]$. Every classical phase transition occurs at a temperature where the free system has no equilibrium state at all, and β_c^{free} can be prescribed anywhere in $(1, \beta_c^{\text{free}}(P_{\mathbb{Q}}))$ without disturbing the cascade. The two exponents carry disjoint information.

1. The free system

1.1. The naive deformation is inconsistent. The first idea is to keep the coefficient algebra of the classical system and to free the isometries: consider the universal C^* -algebra $\mathcal{A}^{\text{naive}}$ generated by a copy of $C(Y_S)$ and isometries $\mu_{\mathbf{p}}$, $\mathbf{p} \in S$, subject to

$$(1.1) \quad \mu_{\mathbf{p}}^* \mu_{\mathbf{q}} = \delta_{\mathbf{p}\mathbf{q}} 1, \quad \mu_{\mathbf{p}} f \mu_{\mathbf{p}}^* = \alpha_{\mathbf{p}}(f) \quad (f \in C(Y_S)),$$

where $\alpha_{\mathbf{p}}(f) = (f \circ \mathbf{p}^{-1}) \mathbf{1}_{\mathbf{p}Y_S}$ is the endomorphism of the classical system [Ch. 1, §2]. The two halves of (1.1) pull in opposite directions: the first makes the range projections $e_{\mathbf{p}} = \mu_{\mathbf{p}} \mu_{\mathbf{p}}^*$ orthogonal, the second identifies $e_{\mathbf{p}} = \alpha_{\mathbf{p}}(1)$ with the image of $\mathbf{1}_{\mathbf{p}Y_S}$, and in $C(Y_S)$ the range projections are nested, $\mathbf{1}_{\mathbf{p}Y_S} \mathbf{1}_{\mathbf{q}Y_S} = \mathbf{1}_{\mathbf{pq}Y_S} \neq 0$.

PROPOSITION 1.1 (Incompatibility). *If $|S| \geq 2$ then $\mathcal{A}^{\text{naive}} = 0$.*

PROOF. Let $\mathbf{p} \neq \mathbf{q}$. By (1.1), $e_{\mathbf{p}}e_{\mathbf{q}} = \mu_{\mathbf{p}}(\mu_{\mathbf{p}}^*\mu_{\mathbf{q}})\mu_{\mathbf{q}}^* = 0$, while $e_{\mathbf{p}}e_{\mathbf{q}}$ is the image of $\mathbf{1}_{\mathbf{p}Y_S}\mathbf{1}_{\mathbf{q}Y_S} = \mathbf{1}_{\mathbf{pq}Y_S} = \alpha_{\mathbf{p}}(\mathbf{1}_{\mathbf{q}Y_S})$. Hence $\mu_{\mathbf{p}}\mathbf{1}_{\mathbf{q}Y_S}\mu_{\mathbf{p}}^* = 0$, and multiplying by $\mu_{\mathbf{p}}^*$ on the left and $\mu_{\mathbf{p}}$ on the right, using $\mu_{\mathbf{p}}^*\mu_{\mathbf{p}} = 1$, gives $\mathbf{1}_{\mathbf{q}Y_S} = 0$ in $\mathcal{A}^{\text{naive}}$. Then $\mu_{\mathbf{q}} = e_{\mathbf{q}}\mu_{\mathbf{q}} = \mathbf{1}_{\mathbf{q}Y_S}\mu_{\mathbf{q}} = 0$, so $1 = \mu_{\mathbf{q}}^*\mu_{\mathbf{q}} = 0$. $\square$

The classical system has a single isometry per prime with *nested* ranges, and this is what lets $C(Y_S)$, with its symmetry group G_S and obstruction group Ξ_{β} , sit inside it; a free monoid of isometries has *partitioning* ranges, and cannot coexist with that coefficient algebra. The consistent free deformation therefore retains, of the arithmetic of S , only the norms.

1.2. The Toeplitz–Cuntz system.

DEFINITION 1.2. Let $\mathbb{F}_S^+$ be the free monoid on S and $Nw = \prod_j N\mathbf{p}_j$ for $w = \mathbf{p}_1 \cdots \mathbf{p}_k$. The *free system* is the Toeplitz–Cuntz algebra $\mathcal{A}_S^{\text{free}} = \mathcal{T}_S$, the universal C^* -algebra generated by isometries $\mu_{\mathbf{p}}$, $\mathbf{p} \in S$, with

$$(1.2) \quad \mu_{\mathbf{p}}^*\mu_{\mathbf{q}} = \delta_{\mathbf{pq}}1,$$

with the dynamics $\sigma_t(\mu_w) = Nw^{it}\mu_w$. Write $e_{\mathbf{p}} = \mu_{\mathbf{p}}\mu_{\mathbf{p}}^*$, $\mu_w = \mu_{\mathbf{p}_1} \cdots \mu_{\mathbf{p}_k}$, and $L_S(\beta) = \sum_{\mathbf{p} \in S} N\mathbf{p}^{-\beta}$.

The Fock representation on $\ell^2(\mathbb{F}_S^+)$, $\mu_{\mathbf{p}}\xi_w = \xi_{\mathbf{pw}}$, is faithful: for finite S this is Cuntz’s theorem [35], and in general it follows from the gauge-invariant uniqueness theorem, the Fock representation carrying the gauge action. The diagonal $C^*\{\mu_w\mu_w^*\}$ of $\mathcal{T}_S$ is commutative and plays the role of the coefficient algebra; its spectrum is the space of finite and infinite words, the boundary of the word tree, and *not* Y_S .

PROPOSITION 1.3. $\sum_{w \in \mathbb{F}_S^+} Nw^{-\beta} = \sum_{k \geq 0} L_S(\beta)^k = (1 - L_S(\beta))^{-1}$, convergent precisely when $L_S(\beta) < 1$.

PROOF. Words of length k are indexed by S^k and N is multiplicative, so $\sum_{|w|=k} Nw^{-\beta} = L_S(\beta)^k$. $\square$

THEOREM 1.4 (The equilibrium states). *Let $|S| \geq 2$. Then L_S is continuous and strictly decreasing on its domain with $L_S(0^+) > 1$ and $L_S(\infty) = 0$, so there is a unique $\beta_c^{\text{free}}(S)$ with $L_S(\beta_c^{\text{free}}) = 1$ whenever L_S exceeds 1 somewhere. Moreover:*

- (1) *every KMS_{β} state satisfies $L_S(\beta) \leq 1$; hence for $\beta < \beta_c^{\text{free}}(S)$ no KMS_{β} state exists;*
- (2) *for every β there is at most one KMS_{β} state, and it is given on the dense $*$ -algebra spanned by the $\mu_w\mu_v^*$ by*

$$(1.3) \quad \varphi(\mu_w\mu_v^*) = \delta_{wv} Nw^{-\beta};$$

- (3) *for $L_S(\beta) < 1$ the Fock–Gibbs state*

$$(1.4) \quad \varphi_{\beta}(x) = (1 - L_S(\beta)) \sum_{w \in \mathbb{F}_S^+} Nw^{-\beta} \langle x \xi_w, \xi_w \rangle$$

on $\ell^2(\mathbb{F}_S^+)$ is a KMS_{β} state, hence the unique one;

- (4) *at $\beta = \beta_c^{\text{free}}(S)$ a KMS_{β} state exists, namely the weak- $*$ limit of $\varphi_{\beta'}$ as $\beta' \downarrow \beta_c^{\text{free}}$, and it is the unique one.*

So the free system has no equilibrium state below β_c^{free} and exactly one at every $\beta \geq \beta_c^{\text{free}}$.

PROOF. (1) By (1.2) the $e_{\mathbf{p}}$ are mutually orthogonal, so $\sum_{\mathbf{p} \in F} e_{\mathbf{p}} \leq 1$ for finite F ; and the KMS condition gives $\varphi(e_{\mathbf{p}}) = N\mathbf{p}^{-\beta}\varphi(\mu_{\mathbf{p}}^*\mu_{\mathbf{p}}) = N\mathbf{p}^{-\beta}$. Hence $\sum_{\mathbf{p} \in F} N\mathbf{p}^{-\beta} \leq 1$ for every finite F , i.e. $L_S(\beta) \leq 1$, and strict monotonicity forces $\beta \geq \beta_c^{\text{free}}$.

(2) The elements $\mu_w \mu_v^*$ span a dense $*$ -subalgebra, by (1.2). For a KMS_β state φ and words w, v , the KMS condition with the entire element μ_w , $\sigma_{i\beta}(\mu_w) = Nw^{-\beta}\mu_w$, gives $\varphi(\mu_w \mu_v^*) = Nw^{-\beta}\varphi(\mu_v^* \mu_w)$. If neither of v, w is a prefix of the other, $\mu_v^* \mu_w = 0$ by (1.2). If $w = vu$ with $u \neq \emptyset$ then $\mu_v^* \mu_w = \mu_u$, and if $v = wu$ then $\mu_v^* \mu_w = \mu_u^*$; in both cases the value vanishes, since φ is invariant under σ_t and $\mu_u^\pm$ is an eigenvector of σ_t with eigenvalue $Nu^{\pm it} \neq 1$. Hence $\varphi(\mu_w \mu_v^*) = 0$ for $w \neq v$, and $\varphi(\mu_w \mu_w^*) = Nw^{-\beta}\varphi(\mu_w^* \mu_w) = Nw^{-\beta}$.

(3) (1.4) is a state by Proposition 1.3, and $\langle \mu_w \mu_v^* \xi_u, \xi_u \rangle \neq 0$ forces $u = vu'$ and $wu' = u$, i.e. $w = v$, so $\varphi_\beta(\mu_w \mu_v^*) = \delta_{vv}(1 - L_S) \sum_{u'} N(wu')^{-\beta} = \delta_{vv} Nw^{-\beta}$, which is (1.3); the KMS condition for the generators then follows from (1.3) by the computation of (2) read backwards.

(4) The state space is weak- $*$ compact, and a weak- $*$ limit of $\text{KMS}_{\beta'}$ states with $\beta' \rightarrow \beta$ is a KMS_β state [12, Prop. 5.3.25]; the limit exists because (1.3) converges as $\beta' \rightarrow \beta_c^{\text{free}}$. $\square$

REMARK 1.5 (Branching versus convergence). Classically the partition function is the Euler product $\prod_{\mathbf{p}} (1 - N\mathbf{p}^{-\beta})^{-1}$ and the critical exponent is the abscissa of convergence of L_S . Freely, $\mathbb{F}_S^+$ has exponentially many words, $L_S(\beta)$ is the branching number of the weighted word tree, and the transition is the Galton–Watson criticality $L_S = 1$: a tree of branching number > 1 has too many words to normalize. The same orthogonality that yields $L_S(\beta) \leq 1$ produces the constraint $\beta = 1$ on the affine $(ax + b)$ system, whose Cuntz relations are finite partitions (Chapter 3). Nested range projections leave every temperature admissible; partitioning ones do not. And the classical phase structure does not survive at all: by Proposition 1.1 the free system has no coefficient algebra $C(Y_S)$, hence no symmetry group G_S , no obstruction group Ξ_β , and, by Theorem 1.4(2), a unique equilibrium state wherever it has one. Everything below is computed in the Fock–Gibbs state.

2. The law of the fields

Fix $\mathbf{p}$, write $t = N\mathbf{p}^{-\beta} \in (0, 1)$, $\Phi = \mu_{\mathbf{p}} + \mu_{\mathbf{p}}^*$, $e = e_{\mathbf{p}}$.

THEOREM 2.1 (Moments and cumulants). *In φ_β ,*

$$m_1 = m_3 = 0, \quad m_2 = 1 + t, \quad m_4 = 2 + 3t + t^2, \quad \kappa_4 = m_4 - 2m_2^2 = -t(1 + t) < 0,$$

and the law of $\Phi_{\mathbf{p}}$ is

$$d\rho_t(x) = (1 - t) \sum_{n \geq 0} t^n U_n(x/2)^2 \frac{\sqrt{4 - x^2}}{2\pi} dx,$$

supported on $[-2, 2]$ for every β , with U_n the Chebyshev polynomials of the second kind. In particular $\Phi_{\mathbf{p}}$ is not semicircular for $t > 0$.

PROOF. Only gauge-invariant words survive, so odd moments vanish and $m_2 = \varphi(\mu\mu^* + \mu^*\mu) = t + 1$. For m_4 , using $\mu^*\mu = 1$ and $\varphi(\mu x \mu^*) = t\varphi(x)$, the six balanced words give $\varphi(\mu\mu\mu^*\mu^*) = t^2$, $\varphi(\mu\mu^*\mu\mu^*) = \varphi(e) = t$, $\varphi(\mu\mu^*\mu^*\mu) = t$, $\varphi(\mu^*\mu\mu\mu^*) = t$, $\varphi(\mu^*\mu\mu^*\mu) = 1$, $\varphi(\mu^*\mu^*\mu\mu) = 1$, summing to $t^2 + 3t + 2$. For the density, note that $C^*(\mu_{\mathbf{p}})$ is the Toeplitz algebra, and that on the Fock space $\ell^2(\mathbb{F}_S^+) = \bigoplus_u \ell^2(\mathbf{p}^{\mathbb{N}}u)$, the sum over words u not beginning with $\mathbf{p}$, the operator Φ is the Jacobi matrix with unit off-diagonal and zero diagonal on each summand $\ell^2(\mathbb{N})$, whose spectral measure at the n -th basis vector is $U_n(x/2)^2 \sqrt{4 - x^2} dx / 2\pi$; grouping the terms of (1.4) by u , the law of Φ is the geometric mixture with weights $(1 - t)t^n$, since the factor $\sum_u Nu^{-\beta}$ over words not beginning with $\mathbf{p}$ is $(1 - t)/(1 - L_S(\beta))$. The relation $\kappa_4 = m_4 - 2m_2^2$ for a centred variable is the moment–cumulant formula [99]. $\square$

COROLLARY 2.2 (Semicircular limit). *As $\beta \rightarrow \infty$, ρ_t converges weakly to the Wigner semicircular law of radius 2 [130]; equivalently $\kappa_2 \rightarrow 1$, all free cumulants of order ≥ 3 vanish, and the even moments converge to the Catalan numbers 1, 2, 5, 14, 42, ...*

THEOREM 2.3 (The number operators are projections). $\mathcal{N}_{\mathfrak{p}} = e_{\mathfrak{p}}$ satisfies $e_{\mathfrak{p}}^2 = e_{\mathfrak{p}}$, so $\text{law}(\mathcal{N}_{\mathfrak{p}}) = (1-t)\delta_0 + t\delta_1$ in φ_{β} . This is a Bernoulli law, not a free Poisson law: a free Poisson variable is a projection only in the degenerate cases δ_0 and δ_1 .

PROOF. $\mu_{\mathfrak{p}}$ is an isometry, so $\varphi(e_{\mathfrak{p}}^n) = \varphi(e_{\mathfrak{p}}) = t$ for every $n \geq 1$, which determines the stated law; its spectrum is $\{0, 1\}$, whereas a nondegenerate Marchenko–Pastur law has an absolutely continuous part. $\square$

3. Asymptotic, not exact, freeness

THEOREM 3.1 (The obstruction). For $\mathfrak{p} \neq \mathfrak{q}$ the fields $\Phi_{\mathfrak{p}}, \Phi_{\mathfrak{q}}$ are not free in φ_{β} : while $\varphi(\Phi_{\mathfrak{p}}\Phi_{\mathfrak{q}}) = 0$ and $\varphi(\Phi_{\mathfrak{p}}\Phi_{\mathfrak{q}}\Phi_{\mathfrak{p}}\Phi_{\mathfrak{q}}) = 0$, one has

$$(3.1) \quad \varphi(\Phi_{\mathfrak{p}}^2\Phi_{\mathfrak{q}}^2) - \varphi(\Phi_{\mathfrak{p}}^2)\varphi(\Phi_{\mathfrak{q}}^2) = -t_{\mathfrak{p}}t_{\mathfrak{q}}.$$

PROOF. Gauge invariance requires each letter to appear equally often with and without adjoint, so $\varphi(\Phi_{\mathfrak{p}}\Phi_{\mathfrak{q}}) = 0$. For $\Phi_{\mathfrak{p}}^2\Phi_{\mathfrak{q}}^2$ the surviving terms give $\varphi((e_{\mathfrak{p}}+1)(e_{\mathfrak{q}}+1)) = \varphi(e_{\mathfrak{p}}e_{\mathfrak{q}}) + t_{\mathfrak{p}} + t_{\mathfrak{q}} + 1$, and $e_{\mathfrak{p}}e_{\mathfrak{q}} = \mu_{\mathfrak{p}}(\mu_{\mathfrak{p}}^*\mu_{\mathfrak{q}})\mu_{\mathfrak{q}}^* = 0$ by (1.2); subtract $(1+t_{\mathfrak{p}})(1+t_{\mathfrak{q}})$. For the alternating moment the four balanced terms are $\mu_{\mathfrak{p}\mathfrak{q}}\mu_{\mathfrak{q}\mathfrak{p}}^*$, $\mu_{\mathfrak{p}}\mu_{\mathfrak{p}\mathfrak{q}}^*\mu_{\mathfrak{q}}^*$, $\mu_{\mathfrak{p}}^*\mu_{\mathfrak{q}}\mu_{\mathfrak{p}}\mu_{\mathfrak{q}}^*$ and $\mu_{\mathfrak{q}\mathfrak{p}}^*\mu_{\mathfrak{p}\mathfrak{q}}$; the second and third vanish by $\mu_{\mathfrak{p}}^*\mu_{\mathfrak{q}} = 0$, the fourth is $\mu_{\mathfrak{q}\mathfrak{p}}^*\mu_{\mathfrak{p}\mathfrak{q}} = 0$, and the first is $\varphi(\mu_w\mu_v^*)$ with $w \neq v$, hence 0 by (1.3). $\square$

COROLLARY 3.2 (Asymptotic freeness). Let $(\mathfrak{p}_k) \subset S$ with $N\mathfrak{p}_k \rightarrow \infty$ and $\beta > \beta_c^{\text{free}}(S)$. Every mixed free cumulant of $\{\Phi_{\mathfrak{p}_k}\}$ is $O(\prod_j t_{\mathfrak{p}_{i_j}})$, hence $O(\min_j N\mathfrak{p}_{i_j}^{-\beta})$, and tends to 0; the family is asymptotically free and by (3.1) the rate $t_{\mathfrak{p}}t_{\mathfrak{q}}$ is attained.

PROOF. By (1.3) a mixed moment $\varphi(\Phi_{\mathfrak{p}_{i_1}} \cdots \Phi_{\mathfrak{p}_{i_n}})$ is a sum, over the choices of μ or μ^* in each factor, of terms $\varphi(\mu_w\mu_v^*)$ after reduction by (1.2); a term survives only if it reduces to $\mu_w\mu_w^*$, and then equals $Nw^{-\beta} = \prod t$ over the letters of w with multiplicity. Every letter distinct from those reducing to the identity contributes a factor $t_{\mathfrak{p}}$, so the moment differs from the corresponding moment of free variables with the same one-variable laws by terms each carrying at least one factor $t_{\mathfrak{p}}$ per additional distinct letter. Möbius inversion on $\text{NC}(n)$ [99], a finite linear combination of products of moments, preserves this order of magnitude, and each $t < 1$. $\square$

REMARK 3.3 (Hard-core exclusion). Free variables would satisfy $\varphi(e_{\mathfrak{p}}e_{\mathfrak{q}}) = t_{\mathfrak{p}}t_{\mathfrak{q}}$; the true value is 0, because no vector lies simultaneously in the ranges of $\mu_{\mathfrak{p}}$ and $\mu_{\mathfrak{q}}$. The discrepancy is exactly $-t_{\mathfrak{p}}t_{\mathfrak{q}}$, and as the norms grow the ranges become φ -small and the exclusion invisible. This is the free counterpart of the classical asymptotic independence of distinct primes for the geometric measures π_{β} .

THEOREM 3.4 (Free entropy is unavailable). φ_{β} is not tracial: $\varphi_{\beta}(\mu_{\mathfrak{p}}\mu_{\mathfrak{p}}^*) = t_{\mathfrak{p}} < 1 = \varphi_{\beta}(\mu_{\mathfrak{p}}^*\mu_{\mathfrak{p}})$. Hence Voiculescu’s microstates free entropy $\chi(\Phi_{\mathfrak{p}_1}, \dots, \Phi_{\mathfrak{p}_n})$ [131] and the free Fisher information are undefined for $n \geq 2$, both requiring a tracial W^* -probability space. For $n = 1$ the entropy is defined, the distribution of a single self-adjoint element being a measure, and equals $\chi(\Phi_{\mathfrak{p}}) = \iint \log|x-y| d\rho_t(x)d\rho_t(y) + \frac{3}{4} + \frac{1}{2} \log 2\pi$; it is finite and continuous in t , and since $t_{\mathfrak{p}} < L_S(\beta_c^{\text{free}}) = 1$ strictly, it remains finite as $\beta \downarrow \beta_c^{\text{free}}$. There is no logarithmic singularity of the free entropy at the critical point.

PROOF. The displayed inequality is the KMS condition. Both invariants are defined for n -tuples in a tracial von Neumann algebra, and no such structure is available. For $n = 1$, ρ_t is supported in $[-2, 2]$ with bounded density for t bounded away from 1, so the logarithmic energy is finite and continuous. $\square$

REMARK 3.5 (Two things called free). The quantity $-\log(1 - L_S(\beta)) = \log Z_{\text{free}}(\beta)$ does diverge as $\beta \downarrow \beta_c^{\text{free}}$, by Proposition 1.3. That is the free energy, a thermodynamic quantity

attached to the whole algebra, not Voiculescu's free *entropy*, an invariant of an n -tuple of non-commutative random variables. The coincidence of names is unfortunate; an asymptotic of the form $\chi_\beta \sim \frac{n}{2} \log(1 - L_S(\beta))$ conflates them, its right side being the free energy and its left side undefined for $n \geq 2$ and bounded for $n = 1$.

4. The free critical exponent

DEFINITION 4.1. Let $\beta_c^{\text{cl}}(S) = \inf\{\beta : L_S(\beta) < \infty\}$ be the abscissa of convergence, the classical critical exponent of [Ch. 1], and $\lambda_S = \lim_{\beta \downarrow \beta_c^{\text{cl}}(S)} L_S(\beta) \in (0, \infty]$. When $\lambda_S > 1$ let $\beta_c^{\text{free}}(S)$ be the root of $L_S = 1$; when $\lambda_S \leq 1$ set $\beta_c^{\text{free}}(S) = \beta_c^{\text{cl}}(S)$.

THEOREM 4.2 (Trichotomy). *For infinite S exactly one of the following holds.*

- (S) $\lambda_S > 1$: *there is a unique $\beta_c^{\text{free}}(S) \in (\beta_c^{\text{cl}}(S), \infty)$, the admissible set is $[\beta_c^{\text{free}}, \infty)$, and $Z_{\text{free}} \rightarrow \infty$ as $\beta \downarrow \beta_c^{\text{free}}$.*
- (C) $\lambda_S = 1$: *$\beta_c^{\text{free}} = \beta_c^{\text{cl}}$, the admissible set is $(\beta_c^{\text{cl}}, \infty)$, and $Z_{\text{free}} \rightarrow \infty$ at β_c^{cl} .*
- (B) $\lambda_S < 1$: *$\beta_c^{\text{free}} = \beta_c^{\text{cl}}$, the admissible set is $(\beta_c^{\text{cl}}, \infty)$, and Z_{free} is bounded there, with supremum $(1 - \lambda_S)^{-1} < \infty$: the free phase is closed.*

For every $\alpha \in (0, 1)$ and every $\lambda \in (0, \infty]$ there is $S \subseteq P_{\mathbb{Q}}$ with $\beta_c^{\text{cl}}(S) = \alpha$ and $\lambda_S = \lambda$, so all three occur.

PROOF. Monotonicity and continuity of a Dirichlet series on its half-plane of convergence give the trichotomy, using Theorem 1.4(1) for the admissible sets. For the realization, fix α . A set T with $|T \cap [1, x]| \asymp x^\alpha$, obtained greedily as in [Ch. 2, Lem. 5.4], has $\beta_c^{\text{cl}}(T) = \alpha$ and, by Abel summation, $L_T(\alpha) = \infty$, giving $\lambda = \infty$. For $\lambda < \infty$ take instead $|T \cap [1, x]| \asymp x^\alpha / \log^2 x$; then $\beta_c^{\text{cl}}(T) = \alpha$ still while the m -th element satisfies $\ell_m \asymp (m \log^2 m)^{1/\alpha}$ and $\ell_m^{-\alpha} \asymp (m \log^2 m)^{-1}$, so $L_T(\alpha) < \infty$. Discarding an initial segment of T lowers $L_T(\alpha)$ below λ without changing β_c^{cl} ; then adjoin primes $p_1 < p_2 < \dots$ outside T chosen greedily, p_k so large that $p_k^{-\alpha}$ is at most half of the remaining deficit $\lambda - L(\alpha)$ of the set built so far. The deficit tends to 0, the adjoined set has counting function $O(\log x)$, hence does not change β_c^{cl} , and the resulting S has $L_S(\alpha) = \lambda$ exactly. $\square$

PROPOSITION 4.3 (Not a tail invariant). *$\beta_c^{\text{cl}}(S)$ is unchanged by adding or deleting finitely many primes and depends only on the counting function, through $\beta_c^{\text{cl}}(S) = \limsup_m \log m / \log N_{\mathbf{p}_m}$. By contrast β_c^{free} is monotone in S and strictly changed by deleting a single prime in regime (S); two sets with the same counting asymptotics can have different β_c^{free} . Moreover, over $\mathbb{Q}$,*

$$\beta_c^{\text{free}}(S) \leq \beta_c^{\text{free}}(P_{\mathbb{Q}}) = 1.399433335\dots,$$

the root of the prime zeta function minus one, with equality only for $S = P_{\mathbb{Q}}$.

PROOF. The first is the standard formula for the convergence exponent. The rest follows from strict monotonicity of L_S in S and in β . $\square$

REMARK 4.4 (Computed values). Computed from the primes below $x = 2 \cdot 10^7$ with an analytic tail correction (code in `code/verify_series.py`): substituting $t = e^u$ gives $\sum_{\mathbf{p} > x} N_{\mathbf{p}}^{-\beta} \approx \int_{\log x}^{\infty} e^{(1-\beta)u} u^{-k} du$, the exponential integral $E_1((\beta - 1) \log x)$ for $k = 1$ and, for the Hardy-Littlewood densities $2C_2 / \log^2 t$ with $k = 2$, a quantity finite at $\beta = 1$ with value $2C_2 / \log x =$

0.0785:

S	$L_S(1^+)$	$\beta_c^{\text{free}}(S)$
all primes	∞	1.399433
all primes except 2	∞	1.224388
all primes except 2, 3	∞	1.154890
$\{p > 10^6\}$	∞	1.019162
Sophie Germain	1.535	1.2114
twin primes (lower)	1.059	1.0089

The first four rows are correct to the digits shown; the last two depend on the tail model in their third decimal. Removing the single prime 2 moves β_c^{free} by 0.175, quantifying Proposition 4.3; nothing comparable can happen to β_c^{cl} , which is 1 for the first four rows. The twin row should be read with care: the bare sum over the computed range is 0.9805 and the Hardy–Littlewood tail contributes 0.0785, so the set is supercritical but only just, and the conclusion rests on the tail estimate rather than on a computation (Brun’s constant $B_2 = 1.9022\dots$ gives the same picture: $\sum_p \text{twin } p^{-1} = \frac{1}{2}(B_2 + \sum 2/(p(p+2))) \approx 1.06$). Whether $\sum_p \text{twin } p^{-1}$ exceeds 1 is an arithmetic question we do not settle; if it does not, the twin set is an instance of regime (B).

5. The free deformation annihilates the cascade

THEOREM 5.1 (Annihilation). *Let S be any of the cascade or Cantor sets of [Ch. 2, Thm. 5.1, Cor. 6.2]. Then $\beta_c^{\text{cl}}(S) = 1$ and $\lambda_S = \infty$, so $\beta_c^{\text{free}}(S) > 1$ strictly; the transition locus of the classical system lies in $(0, 1]$; and by Theorem 1.4(1) the free system has no KMS_β state for $\beta < \beta_c^{\text{free}}(S)$. Consequently every classical phase transition occurs at a temperature at which the free system has no equilibrium state; on the whole admissible range of the free system the classical one has $\zeta_{K,S}(\beta) < \infty$ and $\Xi_\beta = \widehat{G}_S$, i.e. is in its Gibbs regime with all its phase transitions behind it; and the free system has a single transition, at $\beta_c^{\text{free}}(S)$, from no equilibrium state to exactly one.*

PROOF. The abundant base of [Ch. 2, Lem. 5.4(1)] has $\sum_{\ell \in P_0} \ell^{-1} = \infty$, so $L_S(1) = \infty$, $\beta_c^{\text{cl}}(S) = 1$ and $\lambda_S = \infty$; Theorem 4.2(S) gives $\beta_c^{\text{free}}(S) > 1$. By [Ch. 2, Thm. 5.1(3)] the thresholds lie in $(0, 1)$ and $\Xi_\beta = \widehat{G}_S$ for $\beta > 1$. $\square$

THEOREM 5.2 (The two exponents are independent). *Let S be a cascade set and $\gamma \in (1, \beta_c^{\text{free}}(P_\mathbb{Q}))$. There is S' differing from S by finitely many primes — hence with the same chain $\{\Xi_\beta\}$ and the same transition locus — with $\beta_c^{\text{free}}(S')$ within any prescribed ε of γ . The interval is optimal: $\beta_c^{\text{free}}(S') > 1$ always by Theorem 5.1 and $\beta_c^{\text{free}}(S') < \beta_c^{\text{free}}(P_\mathbb{Q})$ by Proposition 4.3.*

PROOF. Adding or deleting finitely many primes leaves Ξ_β unchanged, a finite set contributing a finite amount to every series in (1.2) and the computation of Ξ_β in [Ch. 2, Thm. 5.1] being stated modulo finite exceptional sets. Deleting all primes of S below z gives $L_{S'}(\beta) \leq \sum_{\ell > z} \ell^{-\beta} \rightarrow 0$ for fixed $\beta > 1$ as $z \rightarrow \infty$, so $\beta_c^{\text{free}}(S')$ can be pushed arbitrarily close to 1; adjoining all primes below N gives $L_{S'} \geq \sum_{p \leq N} p^{-\beta}$, whose root increases to $\beta_c^{\text{free}}(P_\mathbb{Q})$. Adding one prime at a time moves the root by a controlled amount, giving every intermediate value up to ε . $\square$

REMARK 5.3 (Numerical illustration). Restricting to primes above z gives the upper bounds 1.056773 at $z = 10^2$, 1.028710 at $z = 10^4$, 1.019162 at $z = 10^6$ and 1.014372 at $z = 10^8$ (the last from the tail integral alone), tending to 1; adjoining all primes below N gives roots 1.147338 at $N = 10$, 1.356444 at $N = 10^2$, 1.388390 at $N = 10^3$, 1.396110 at $N = 10^4$, 1.399069 at $N = 10^6$ and 1.399307 at $N = 10^7$, increasing to $\beta_c^{\text{free}}(P_\mathbb{Q}) = 1.399433$ but never attaining it.

REMARK 5.4 (What the deformation is good for). The free variant was proposed as a route to importing free probability into the arithmetic phase structure. Theorem 5.1 shows the two

are incompatible in a precise sense: the phenomena classified by Ξ_β live at high temperature, where the partition function diverges, and it is exactly that region which the free monoid renders inaccessible, because exponential word growth prevents normalization once the branching number exceeds 1. The classical system is interesting at high temperature and trivial at low; the free system exists only at low temperature and is trivial there. Theorem 5.2 confirms that the two exponents carry disjoint information rather than being two views of one structure.

What the deformation does produce is β_c^{free} itself, a well-defined arithmetic invariant of a set of primes: computable, universally bounded over $\mathbb{Q}$, and — unlike every exponent in this series — sensitive to individual norms rather than only to a counting function.

6. Assessment

Statement	Status	Reference
$C(Y_S)$ survives the deformation	false: the universal algebra is 0	Prop. 1.1
$Z_{\text{free}} = (1 - L_S)^{-1}$, $L_S(\beta_c^{\text{free}}) = 1$	true	Prop. 1.3, Thm. 1.4
two regimes across β_c^{free}	false: empty below	Thm. 1.4(1)
KMS_β state unique for $\beta \geq \beta_c^{\text{free}}$	true	Thm. 1.4(2)
Φ_p semicircular	only as $\beta \rightarrow \infty$	Thm. 2.1, Cor. 2.2
$\mathcal{N}_p$ free Poisson	false: a projection	Thm. 2.3
$\{\Phi_p\}$ exactly free	false	Thm. 3.1
$\{\Phi_p\}$ asymptotically free	true	Cor. 3.2
$\chi(\Phi_{p_1}, \dots, \Phi_{p_n})$, $n \geq 2$	undefined	Thm. 3.4
β_c^{free} a tail invariant	false	Prop. 4.3
cascade survives the deformation	false	Thm. 5.1

Three questions remain. First, whether some intermediate deformation — a coefficient algebra adapted to the word tree, carrying an action of a quotient of G_S — can retain part of the classical phase structure while keeping partitioning ranges; Proposition 1.1 rules out the obvious candidate, not every candidate. Second, φ_β admits no trace, but the associated von Neumann algebra has a canonical trace on its continuous core; whether a free entropy computed there carries arithmetic information is open, and by the pattern of Chapter 4 we expect not. Third, $\beta_c^{\text{free}}(S)$ is defined for any set of primes and we have not found an interpretation of it outside this system — for instance as a threshold for a branching process on the multiplicative semigroup generated by S — nor decided the regime of the twin primes, which by Remark 4.4 reduces to comparing a Brun-type constant with 1.

Part 3

Towards non-abelian systems

Towards non-abelian localized Bost–Connes systems: the matrix Kakutani dichotomy and a Tannakian obstruction

This chapter is Paper VI of the series ([papers/VI-nonabelian/](#)).

SUMMARY. The GL_1 theory of [Ch. 1] classifies the KMS_β simplex of $\mathcal{A}_{K,S}$ as $\text{Prob}(G_S/\Xi_\beta^\perp)$, where Ξ_β is a subgroup of the Pontryagin dual $\widehat{G_S}$ cut out by a Dirichlet series condition. We examine what survives when the abelian symmetry group is replaced by a Galois group with non-abelian image, and separate three things that are usually conflated.

What transfers. The Dirichlet-series half of the Kakutani dichotomy is representation-theoretic, not character-theoretic. For a unitary representation ρ of a compact group equipped with Frobenius classes we take as the obstruction quantity the Hilbert–Schmidt Frobenius series

$$\Xi_\beta(\rho, S) = \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} \|I_n - \rho(\text{Frob}_{\mathfrak{p}})\|_{\text{HS}}^2 = 2 \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} (n - \Re \text{Tr } \rho(\text{Frob}_{\mathfrak{p}})),$$

which reduces to the abelian condition for characters, and whose convergence is the coboundary condition for the $U(n)$ -valued Frobenius cocycle whenever that cocycle is defined, namely when the Frobenius classes commute; without commutation there is no cocycle on the tail relation at all, and the series is the primary object. Its relation to L -functions is

$$\Xi_\beta(\rho, S) = 2n \log \zeta_{K,S}(\beta) - 2\Re \log L_S(\beta, \rho) + O(1),$$

so that at $\beta \leq 1$ and $S = P_K$ the obstruction is absent for every nontrivial irreducible ρ , the input being $L(1, \rho) \neq 0, \infty$: Hecke–Landau for Artin representations, Jacquet–Shalika in the automorphic case. For Artin representations the Chebotarev criterion of [Ch. 1, Thm. 3.9] transfers with $D_\rho = \{\mathfrak{p} : \rho(\text{Frob}_{\mathfrak{p}}) \neq I\}$.

What does not transfer, and its replacement. The classification $\text{Prob}(G_S/\Xi_\beta^\perp)$ uses Pontryagin duality twice and fails without it: for non-abelian G the set of irreducible representations is not a group, Ξ_β cannot be a subgroup of it, and $\Xi_\beta^\perp$ is undefined. The correct replacement is not a subgroup of a dual but a *quotient of the group*. We prove that $\mathcal{T}_\beta(S) = \{\rho : \Xi_\beta(\rho, S) < \infty\}$ is closed under direct sums, subobjects, tensor products and duals, hence is a Tannakian subcategory of $\text{Rep}(G)$, and therefore equals $\text{Rep}(G/N_\beta)$ for a closed normal subgroup $N_\beta = \bigcap_{\rho \in \mathcal{T}_\beta} \ker \rho$. In the abelian case $N_\beta = \Xi_\beta^\perp$ and the two formulations agree.

What is not proved here. We do not construct a GL_n algebra whose KMS_β simplex is $\text{Prob}(G/N_\beta)$, and we explain precisely which step of the GL_1 proof has no available substitute. Two concrete findings bear on the GL_2 case, formulated for the Sato–Tate classes of a non-CM elliptic curve. If the classes along S are Sato–Tate equidistributed then $\Xi_\beta(\rho, S) \sim 4 \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta}$ diverges whenever the partition function does, so *no* symmetry breaking arises from a Sato–Tate representation. Conversely, assuming Sato–Tate as an equidistribution input, we construct sparse S with $\zeta_S(1) = \infty$ and $\Xi_1(\rho, S) < \infty$ for the representation attached to a non-CM elliptic curve, by selecting blocks of primes with $\theta_p \leq 2^{-j}$; the exotic phase is therefore available, but only on deliberately constructed anomalous sets, never generically.

1. Which steps of the abelian argument use commutativity

Recall the GL_1 chain of reasoning [Ch. 1]. A KMS_β state is a measure ν on Y_S scaling by $N\mathfrak{a}^{-\beta}$; the symmetry group G_S acts on the simplex; a character $\chi \in \widehat{G_S}$ fails to be an obstruction exactly

when $\chi \circ \sigma$ is a coboundary for π_β , which by Kakutani is the convergence of $\sum_{\mathfrak{p}} N\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}})|^2$; the obstruction characters form a *subgroup* $\Xi_\beta \leq \widehat{G_S}$; and the simplex is $\text{Prob}(G_S/\Xi_\beta^\perp)$.

Three of these steps are insensitive to commutativity and two are not.

- (i) The Kakutani step has two halves: a Dirichlet series in Frobenius data, and its identification with a coboundary condition for a cocycle on the tail relation. §2 shows that the series transfers, with the Hilbert–Schmidt norm in place of $|1 - z|$, and that the cocycle half transfers exactly when the Frobenius classes along S commute — the tail relation being generated by commuting shifts, a $U(n)$ -valued cocycle with prescribed non-commuting increments does not exist.
- (ii) The passage from convergence to a group structure uses that $\widehat{G_S}$ is a group under point-wise product. For non-abelian G the irreducibles are not closed under tensor product as a group law: $\rho_1 \otimes \rho_2$ decomposes. §3 replaces the subgroup by a Tannakian subcategory.
- (iii) The identification of the simplex as $\text{Prob}(G_S/\Xi_\beta^\perp)$ uses Pontryagin duality to convert a subgroup of the dual into a closed subgroup of the group. This is the step with no substitute at the level of duals; §3 supplies the substitute at the level of the group itself.
- (iv) The L -function dictionary is an Euler-product identity and transfers (§4), with different constants from those one might guess.
- (v) The construction of an algebra whose simplex realizes the answer does *not* transfer, and §6 says why.

Throughout, K is a number field, S a set of finite primes, and G a compact group equipped with a family of conjugacy classes $\text{Frob}_{\mathfrak{p}} \in G$ indexed by all but finitely many $\mathfrak{p} \in S$: typically G is a profinite quotient of $\text{Gal}(\bar{K}/K)$ unramified outside a finite set and $\text{Frob}_{\mathfrak{p}}$ the Frobenius class, but G may also be a Sato–Tate group and $\text{Frob}_{\mathfrak{p}}$ the Satake class, as in §5. All representations are continuous, unitary and finite-dimensional; sums over $\mathfrak{p} \in S$ omit the finitely many primes at which $\text{Frob}_{\mathfrak{p}}$ is undefined, which affects no convergence question. We write $\zeta_{K,S}(\beta) = \prod_{\mathfrak{p} \in S} (1 - N\mathfrak{p}^{-\beta})^{-1}$ as in [Ch. 1], so that $\log \zeta_{K,S}(\beta) = \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} + O(1)$ for $\beta > 1/2$.

2. The matrix Kakutani dichotomy

DEFINITION 2.1. For a representation $\rho : G \rightarrow U(n)$ put

$$(2.1) \quad \Xi_\beta(\rho, S) := \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} \|I_n - \rho(\text{Frob}_{\mathfrak{p}})\|_{\text{HS}}^2.$$

LEMMA 2.2 (Trace form). For $U \in U(n)$, $\|I_n - U\|_{\text{HS}}^2 = 2(n - \Re \text{Tr } U)$. Hence

$$(2.2) \quad \Xi_\beta(\rho, S) = 2 \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} (n - \Re \text{Tr } \rho(\text{Frob}_{\mathfrak{p}})),$$

a sum of non-negative terms, each in $[0, 4n]$, depending only on the conjugacy class of $\rho(\text{Frob}_{\mathfrak{p}})$.

PROOF. $\|I - U\|_{\text{HS}}^2 = \text{Tr}((I - U)^*(I - U)) = \text{Tr}(I - U - U^* + U^*U) = \text{Tr}(2I - U - U^*) = 2n - 2\Re \text{Tr } U$, using $U^*U = I$. Non-negativity and the bound follow from $|\text{Tr } U| \leq n$; conjugation invariance of the trace gives the last claim. $\square$

THEOREM 2.3 (Matrix Kakutani). Let $\pi_\beta = \bigotimes_{\mathfrak{p} \in S} \mu_{t_{\mathfrak{p}}}$ with $t_{\mathfrak{p}} = N\mathfrak{p}^{-\beta}$ as in [Ch. 1, Lem. 2.10], on $V = \mathbb{N}^{(S)}$ with the tail relation $\mathcal{R}_S$, and put $U_{\mathfrak{p}} = \rho(\text{Frob}_{\mathfrak{p}}) \in U(n)$ (a representative of the class).

- (1) A $U(n)$ -valued cocycle c_ρ on $\mathcal{R}_S$ with $c_\rho(v + e_{\mathfrak{p}}, v) = U_{\mathfrak{p}}$ for all $\mathfrak{p}$ and v exists if and only if the $U_{\mathfrak{p}}$, $\mathfrak{p} \in S$, commute pairwise.
- (2) Suppose they do. Then there is a measurable $g : V \rightarrow U(n)$ with $g(v + e_{\mathfrak{p}}) = U_{\mathfrak{p}} g(v)$ a.e. for every $\mathfrak{p}$ — i.e. c_ρ is a coboundary — if and only if $\Xi_\beta(\rho, S) < \infty$.

PROOF. (1) The cocycle identity along the two paths from v to $v + e_p + e_q$ gives $c_\rho(v + e_p + e_q, v) = U_q U_p = U_p U_q$; conversely, for commuting U_p the formula $c_\rho(v + a, v) = \prod_p U_p^{a_p}$ is well defined and is a cocycle.

(2) Commuting unitaries are simultaneously diagonalizable: some unitary W satisfies

$$W U_p W^* = \text{diag}(\chi_1(p), \dots, \chi_n(p)) \quad \text{for every } p, \quad \chi_r(p) \in \mathbb{T}.$$

Since $\|I - U_p\|_{\text{HS}}^2 = \sum_r |1 - \chi_r(p)|^2$,

$$\Xi_\beta(\rho, S) = \sum_{r=1}^n \sum_{p \in S} N p^{-\beta} |1 - \chi_r(p)|^2.$$

If g exists, put $h = Wg$, so that $h(v + e_p) = \text{diag}(\chi_r(p)) h(v)$ and each entry satisfies $h_{rs}(v + e_p) = \chi_r(p) h_{rs}(v)$. For each r some entry h_{rs} is nonzero on a set of positive measure, $h(v)$ being unitary; by [Ch. 1, Lem. 3.1] $|h_{rs}|$ is a.e. constant, hence a.e. nonzero, and [Ch. 1, Prop. 3.3], whose proof uses only $|\chi_r(p)| = 1$, gives the convergence of the r -th series. Conversely, if all n series converge, [Ch. 1, Prop. 3.4] gives measurable $g_r : V \rightarrow \mathbb{T}$ with $g_r(v + e_p) = \chi_r(p) g_r(v)$, and $g = W^* \text{diag}(g_1, \dots, g_n) W$ is the required coboundary. $\square$

REMARK 2.4 (The series is the primary object). Without commutation of the Frobenius classes there is no cocycle on $\mathcal{R}_S$ to be a coboundary, and the dynamical half of the Kakutani dichotomy has no content; a non-abelian dynamical model would need a non-abelian acting semigroup, and Chapter 14 of the series shows what that costs. What survives, and what everything below uses, is the Dirichlet series $\Xi_\beta(\rho, S)$ itself, with the following formal properties.

LEMMA 2.5 (Additivity and submultiplicativity). *For representations ρ, ρ' of dimensions n, n' :*

$$(2.3) \quad \|I_{n+n'} - (\rho \oplus \rho')(g)\|_{\text{HS}}^2 = \|I_n - \rho(g)\|_{\text{HS}}^2 + \|I_{n'} - \rho'(g)\|_{\text{HS}}^2,$$

$$(2.4) \quad \|I_{nn'} - (\rho \otimes \rho')(g)\|_{\text{HS}}^2 \leq 2n' \|I_n - \rho(g)\|_{\text{HS}}^2 + 2n \|I_{n'} - \rho'(g)\|_{\text{HS}}^2,$$

$$(2.5) \quad \|I_n - \rho^\vee(g)\|_{\text{HS}} = \|I_n - \rho(g)\|_{\text{HS}}.$$

PROOF. (2.3) is block-diagonality of $I - (\rho \oplus \rho')$. For (2.4) write $I - A \otimes B = (I - A) \otimes I + A \otimes (I - B)$ and use $\|X \otimes I_{n'}\|_{\text{HS}}^2 = n' \|X\|_{\text{HS}}^2$, $\|A \otimes Y\|_{\text{HS}}^2 = n \|Y\|_{\text{HS}}^2$ for A unitary, together with $\|X + Y\|^2 \leq 2\|X\|^2 + 2\|Y\|^2$. (2.5) is $\rho^\vee(g) = \overline{\rho(g)}$ and invariance of the Hilbert–Schmidt norm under entrywise conjugation. $\square$

REMARK 2.6 (Numerical check). Lemma 2.2 and (2.4) were checked on Haar-random unitaries, 200 in each of the dimensions $n = 1, 2, 3, 5$ and 500 pairs with $(n, n') = (2, 3)$, with maximal error $3.6 \cdot 10^{-15}$ and no violation; a check of algebra, recorded because the constants of §4 depend on it.

3. The obstruction is a quotient, not a dual subgroup

PROPOSITION 3.1 (Failure of the abelian formulation). *Let G be a non-abelian compact group. Then the set of irreducible representations $\widehat{G}$ carries no group structure for which $\rho \mapsto \Xi_\beta(\rho, S)$ is a homomorphism-like invariant: $\rho_1 \otimes \rho_2$ is in general reducible, so $\widehat{G}$ is a based set with a fusion rule and not a group, and the annihilator $\Xi_\beta^\perp$ of a subset of $\widehat{G}$ is undefined. In particular the statement “ $\text{KMS}_\beta \cong \text{Prob}(G/\Xi_\beta^\perp)$ ” has no meaning as written.*

PROOF. Immediate from the definitions; the abelian case is exactly the case where every $\rho_1 \otimes \rho_2$ is again irreducible. $\square$

THEOREM 3.2 (Tannakian obstruction). *Let $\mathcal{T}_\beta(S)$ be the full subcategory of $\text{Rep}(G)$ on the objects ρ with $\Xi_\beta(\rho, S) < \infty$. Then $\mathcal{T}_\beta(S)$ contains the trivial representation and is closed under direct sums, subobjects, quotients, tensor products and duals; it is therefore a Tannakian subcategory of $\text{Rep}(G)$, and*

$$\mathcal{T}_\beta(S) = \text{Rep}(G/N_\beta), \quad N_\beta := \bigcap_{\rho \in \mathcal{T}_\beta(S)} \ker \rho,$$

a closed normal subgroup of G . When G is abelian, $N_\beta = \Xi_\beta^\perp$ and the description agrees with [Ch. 1, Thm. 3.6].

PROOF. The trivial representation gives all summands zero. Closure under $\oplus$, $\otimes$ and duals is Lemma 2.5: for $\otimes$ the bound (2.4) converts two convergent series into a convergent one, and n, n' are fixed. For subobjects, if $\rho = \rho_1 \oplus \rho_2$ then (2.3) gives $\Xi_\beta(\rho_i, S) \leq \Xi_\beta(\rho, S)$, and by unitarity every subobject is a direct summand; quotients are duals of subobjects. Fullness holds by construction. A full subcategory of $\text{Rep}(G)$ closed under these operations and containing the trivial object is the representation category of a quotient of G , by Tannaka–Krein for compact groups: the matrix coefficients of its objects span a $*$ -subalgebra of the algebra of representative functions, closed under the coproduct, hence the representative functions of G/N with N the common kernel. Normality and closedness of N_β hold because each $\ker \rho$ is a closed normal subgroup.

For G abelian, $\mathcal{T}_\beta$ is generated by the characters in Ξ_β and $\bigcap_{\chi \in \Xi_\beta} \ker \chi = \Xi_\beta^\perp$. $\square$

REMARK 3.3 (What the theorem does and does not assert). Theorem 3.2 is a statement about representations and Dirichlet series; it identifies the natural non-abelian analogue of Ξ_β as the quotient G/N_β . It does *not* assert that any C^* -dynamical system has $\text{Prob}(G/N_\beta)$ as its KMS_β simplex. That is the content of §6, and it is open.

We also restrict to finite-dimensional continuous unitary representations of a compact group. For a profinite G these are the Artin representations, with finite image. An ℓ -adic representation is not unitary and is not covered; the GL_2 case of §5 is handled through the Sato–Tate group, a compact Lie group, and its Satake classes.

4. L-functions, with the correct constants

THEOREM 4.1 (Euler dictionary). *For $\beta > 1/2$ and ρ of dimension n unramified outside S ,*

$$(4.1) \quad \Xi_\beta(\rho, S) = 2n \log \zeta_{K,S}(\beta) - 2 \Re \log L_S(\beta, \rho) + O(1),$$

where $L_S(s, \rho) = \prod_{\mathfrak{p} \in S} \det(I - \rho(\text{Frob}_{\mathfrak{p}}) N \mathfrak{p}^{-s})^{-1}$ and the implied constant depends only on n and β .

PROOF. $\log L_S(s, \rho) = \sum_{\mathfrak{p}} \sum_{m \geq 1} \frac{1}{m} \text{Tr}(\rho(\text{Frob}_{\mathfrak{p}})^m) N \mathfrak{p}^{-ms}$, and the terms with $m \geq 2$ are bounded by $n \sum_{\mathfrak{p}} \sum_{m \geq 2} N \mathfrak{p}^{-m\beta} = O_n(1)$ for $\beta > 1/2$. Hence $\Re \log L_S(\beta, \rho) = \sum_{\mathfrak{p}} N \mathfrak{p}^{-\beta} \Re \text{Tr} \rho(\text{Frob}_{\mathfrak{p}}) + O(1)$, and similarly $\log \zeta_{K,S}(\beta) = \sum_{\mathfrak{p}} N \mathfrak{p}^{-\beta} + O(1)$. Substituting into (2.2) gives (4.1). $\square$

REMARK 4.2 (Two constants that are easy to get wrong). The natural guess $\log \zeta_{K,S}(n\beta) - \Re \log L_S(\beta, \rho) = \Xi_\beta(\rho, S) + O(1)$ is incorrect in two ways. The zeta factor is evaluated at β , not at $n\beta$: the n in (2.2) enters as the *coefficient* $2n$ of a single Dirichlet series, coming from $\text{Tr} I_n = n$, not as a dilation of its argument. And both terms carry a factor 2, from $\|I - U\|_{\text{HS}}^2 = 2(n - \Re \text{Tr} U)$. For $n = 1$ the correct identity reduces to $\Xi_\beta = 2 \log \zeta_{K,S}(\beta) - 2 \Re \log L_S(\beta, \chi) + O(1)$, consistent with [Ch. 1, Rem. 3.11].

COROLLARY 4.3 (The full prime set). *Let $S = P_K$ and ρ irreducible and nontrivial, and suppose $L(s, \rho)$ is holomorphic and nonzero at $s = 1$ — Hecke–Landau for Artin representations, Jacquet–Shalika [65] for the standard L -function of a cuspidal automorphic representation. Then*

$\Xi_\beta(\rho, P_K) = \infty$ for every $0 < \beta \leq 1$: the obstruction is absent, $N_\beta = G$, and the would-be simplex is a point.

PROOF. For $\beta > 1$, (4.1) gives $\Xi_\beta(\rho, P_K) = 2n \log \zeta_K(\beta) + O(1)$, since $\log L(\beta, \rho)$ stays bounded as $\beta \downarrow 1$ by the hypothesis on $L(1, \rho)$; so $\Xi_\beta(\rho, P_K) \rightarrow \infty$ as $\beta \downarrow 1$. As Ξ_β is non-increasing in β , being a Dirichlet series with non-negative coefficients, $\Xi_\beta = \infty$ for all $\beta \leq 1$. $\square$

The convergent case $\Xi_1(\rho, P_K) < \infty$ would require $\Re \log L(\beta, \rho)$ to diverge like $n \log \zeta_K(\beta)$, i.e. a pole of order n at $s = 1$, which happens only for ρ trivial.

COROLLARY 4.4 (Chebotarev criterion, Artin case). *Let ρ have finite image and let $D_\rho = \{\mathfrak{p} \in S : \rho(\text{Frob}_\mathfrak{p}) \neq I_n\}$. Then*

$$\Xi_\beta(\rho, S) < \infty \iff \sum_{\mathfrak{p} \in D_\rho} N\mathfrak{p}^{-\beta} < \infty.$$

PROOF. The image is finite, so $\{\|I - \rho(g)\|_{\text{HS}}^2 : g \in G, \rho(g) \neq I\}$ is a finite set of strictly positive numbers, bounded above by $4n$ and below by some $c > 0$. Hence the summand in (2.1) is between $cN\mathfrak{p}^{-\beta}$ and $4nN\mathfrak{p}^{-\beta}$ on D_ρ and zero off it. $\square$

REMARK 4.5. Corollary 4.4 is the exact analogue of [Ch. 1, Thm. 3.9] and shows that the Chebotarev machinery of that paper — in particular the density computations and the block constructions of [Ch. 2, §5] — transfers unchanged to Artin representations, with D_ρ in place of the detecting set of a character. *It fails for infinite image*: if $\rho(\text{Frob}_\mathfrak{p})$ can approach I_n , the indicator of D_ρ is not comparable to $\|I - \rho(\text{Frob}_\mathfrak{p})\|_{\text{HS}}^2$ and only the quadratic form is correct. The GL_2 case below is exactly of this kind.

5. GL_2 : Sato–Tate forbids symmetry breaking, anomalies permit it

Let $E/\mathbb{Q}$ be a non-CM elliptic curve, a_p the trace of Frobenius, and $a_p = 2\sqrt{p} \cos \theta_p$ with $\theta_p \in [0, \pi]$. The ℓ -adic representation of E is not unitary and does not fit Definition 2.1 directly; what does is the Sato–Tate group $G = SU(2)$ (the non-CM case, by Serre’s open image theorem [118]) with the Satake classes $\text{Frob}_p = \text{diag}(e^{i\theta_p}, e^{-i\theta_p})$ and ρ its standard representation, to which §§2–4 apply verbatim. Then $n = 2$ and

$$(5.1) \quad \|I_2 - \rho(\text{Frob}_p)\|_{\text{HS}}^2 = 2(2 - 2 \cos \theta_p) = 4(1 - \cos \theta_p) = 8 \sin^2(\theta_p/2).$$

THEOREM 5.1 (Sato–Tate forbids the obstruction). *Suppose the θ_p , $p \in S$, are equidistributed in $[0, \pi]$ for the Sato–Tate measure $\mu_{ST} = \frac{2}{\pi} \sin^2 \theta d\theta$ in the sense that $\sum_{p \in S, p \leq x} p^{-\beta} f(\theta_p) \sim (\int f d\mu_{ST}) \sum_{p \in S, p \leq x} p^{-\beta}$ for continuous f . Then*

$$\Xi_\beta(\rho, S) \sim 4 \sum_{p \in S} p^{-\beta},$$

which diverges whenever the partition function does. Hence $\rho \notin \mathcal{T}_\beta(S)$ and no symmetry breaking arises from Sato–Tate distributed classes, at any β with $\zeta_S(\beta) = \infty$.

PROOF. By (5.1) the summand is $f(\theta_p)$ with $f(\theta) = 4(1 - \cos \theta)$, continuous and bounded on $[0, \pi]$. Since $\int_0^\pi \cos \theta \sin^2 \theta d\theta = 0$,

$$\int_0^\pi f d\mu_{ST} = \frac{8}{\pi} \int_0^\pi (1 - \cos \theta) \sin^2 \theta d\theta = 4,$$

and the hypothesis applied to f gives the stated asymptotic. $\square$

REMARK 5.2 (Numerical check). $\int_0^\pi (2 - 2 \cos \theta) d\mu_{ST} = 2.0000000000$ by numerical quadrature on 2×10^6 nodes, matching the exact value 2; the factor 2 of Lemma 2.2 then gives the constant 4 in Theorem 5.1.

THEOREM 5.3 (Anomalous sparse sets exist). *Assume Sato–Tate for E — a theorem for non-CM curves over $\mathbb{Q}$ [126] — in the form: for every $\delta > 0$ the set $\{p : \theta_p \leq \delta\}$ has positive natural density, hence $\sum_{p \leq x, \theta_p \leq \delta} p^{-1} \rightarrow \infty$. Then there is $S \subset P_{\mathbb{Q}}$ with*

$$\sum_{p \in S} p^{-1} = \infty \quad \text{and} \quad \Xi_1(\rho, S) < \infty.$$

PROOF. Build $S = \bigsqcup_{j \geq 1} S_j$ greedily. Given $S_1, \dots, S_{j-1}$, the hypothesis with $\delta = 2^{-j}$ gives $\sum_{p \leq x, \theta_p \leq 2^{-j}} p^{-1} \rightarrow \infty$, so a finite set S_j of primes with $\theta_p \leq 2^{-j}$, disjoint from the previous blocks, may be chosen with $\sum_{p \in S_j} p^{-1} \geq 1$. Then $\sum_{p \in S} p^{-1} \geq \sum_j 1 = \infty$. On the other hand, by (5.1) and $\sin^2(\theta/2) \leq \theta^2/4$,

$$\Xi_1(\rho, S) = \sum_j \sum_{p \in S_j} p^{-1} \cdot 8 \sin^2(\theta_p/2) \leq \sum_j 2 \cdot 4^{-j} \sum_{p \in S_j} p^{-1}.$$

Choosing S_j additionally with $\sum_{p \in S_j} p^{-1} \leq 2$, which is possible since single terms p^{-1} can be made arbitrarily small, gives $\Xi_1(\rho, S) \leq \sum_j 4 \cdot 4^{-j} < \infty$. $\square$

REMARK 5.4 (What has and has not been shown). Theorem 5.3 produces a set on which the analytic obstruction is finite: the Sato–Tate classes of a non-CM elliptic curve become “invisible” along a deliberately chosen set of primes with anomalously small angle. By Theorem 3.2 applied to $G = SU(2)$ it follows that $\rho \in \mathcal{T}_1(S)$ and hence $N_1 \subseteq \ker \rho = \{1\}$, so the Tannakian quotient $G/N_1 = SU(2)$ is the whole Sato–Tate group, nontrivial and non-abelian.

What has *not* been shown is that this residual symmetry acts on a simplex of KMS states of any algebra, for the reason of §6. The phrase “non-abelian symmetry breaking” is therefore premature: we have a non-abelian obstruction group, not yet a broken phase. The construction also assumes Sato–Tate as an equidistribution input; for non-CM elliptic curves over $\mathbb{Q}$ this is a theorem, but only the qualitative positive-density statement is used.

6. The algebra is not constructed

REMARK 6.1 (Where the GL_1 proof stops). The classification in [Ch. 1] runs: (a) KMS_β states of $C(Y_S) \rtimes J_S$ correspond bijectively to $N\mathfrak{a}^{-\beta}$ -scaling measures on Y_S , by the conditional expectation onto the diagonal; (b) such measures form a simplex on which G_S acts; (c) the ergodic decomposition of that action is computed by the Kakutani criterion. Steps (a) and (b) require the coefficient algebra to be commutative and the semigroup to be Ore, so that a canonical expectation exists and the crossed product is a groupoid algebra.

The candidate $\mathcal{A}_{K,S}^{GL_n} = C(Y_S^{(n)}) \rtimes M$ fails both at present. The multiplicative structure M is not a semigroup for the obvious choices: $M_n(J_S)$ is not closed under multiplication with the required cancellation, and GL_n over a semiring of ideals has no Ore condition available to us. And the Connes–Marcolli GL_2 system [28], which does exist, is not of this shape: its symmetry is the action of $GL_2(\mathbb{A}_f)$ by endomorphisms of a noncommutative space of two-dimensional lattices, and its arithmetic is that of modular functions and the modular field, not of a Galois group acting through Frobenius classes. There is no non-abelian class field theory realized in it, which is why its KMS theory does not take the form $\text{Prob}(G/N)$.

The honest statement is therefore: §§2–4 determine what the answer would have to be, and identify the correct algebraic shape of the obstruction, but the system whose equilibrium states realize $\text{Prob}(G/N_\beta)$ is not available. Two routes seem worth trying. One is to look for a groupoid model directly: a groupoid whose unit space carries a G -action with the tail relation of [Ch. 1] and whose isotropy realizes N_β ; this bypasses the semigroup difficulty entirely. The other is to stay with GL_1 but to enlarge the coefficient algebra, replacing $C(Y_S)$ by a noncommutative algebra

carrying ρ ; here the Kakutani analysis of §2 applies verbatim, and the question is whether the resulting crossed product has a tractable expectation.

7. Assessment

Statement	Status	Reference
$\ I - U\ _{\text{HS}}^2 = 2(n - \Re \text{Tr } U)$	true	Lem. 2.2
Kakutani dichotomy for $U(n)$ -cocycles	iff the classes commute	Thm. 2.3
Ξ_β a subgroup of $\hat{G}$	meaningless for non-abelian G	Prop. 3.1
$\text{Prob}(G/\Xi_\beta^\perp)$	meaningless as written	Prop. 3.1
$\mathcal{T}_\beta = \text{Rep}(G/N_\beta)$	true; the correct replacement	Thm. 3.2
$\log \zeta_{K,S}(n\beta) - \Re \log L_S = \Xi_\beta + O(1)$	false	Rem. 4.2
$\Xi_\beta = 2n \log \zeta_{K,S}(\beta) - 2\Re \log L_S + O(1)$	true	Thm. 4.1
$\Xi_\beta(\rho, P_K) = \infty$, $\rho \neq 1$ irreducible, $\beta \leq 1$	true, via $L(1, \rho) \neq 0, \infty$	Cor. 4.3
Chebotarev criterion via D_ρ	true for finite image only	Cor. 4.4
Sato–Tate $\Rightarrow$ symmetry breaking	false: forbids it	Thm. 5.1
anomalous sparse S with $\Xi_1 < \infty$	true, under Sato–Tate	Thm. 5.3
$\mathcal{A}_{K,S}^{GL_n}$ constructed	no	§6

The pattern is the one already familiar from Chapter 4: the arithmetic content — a Dirichlet series in Frobenius data, an L -function on $\Re s = 1$ — generalizes with no difficulty, because none of it used commutativity. What used commutativity was, first, the dynamical half of the Kakutani step, which needs a cocycle on a relation generated by commuting shifts, and second the *duality* converting the series into a statement about a space of states; the substitute for the latter is Tannakian rather than Pontryagin, a quotient of the group rather than a subgroup of its dual, and for the former there is none. The missing ingredient is not a theorem but an object, namely an algebra whose equilibrium states are indexed by G/N_β .

Three questions. Is N_β the intersection of the kernels of *all* representations in $\mathcal{T}_\beta$, or is it already cut out by the irreducible ones of bounded dimension? For G with an open image in $GL_2(\hat{\mathbb{Z}})$ this is a question about which quotients can be realized by anomalous prime sets. Second, Theorem 5.3 constructs S for a fixed ρ ; can one construct S realizing a *prescribed* quotient G/N , as [Ch. 2, Thm. 5.1] does in the abelian case for prescribed $\bigoplus \mathbb{Z}/2\mathbb{Z}$? Third, and most important, is the groupoid model of Remark 6.1 available? Without it this paper describes an obstruction group with nothing yet to obstruct.

CHAPTER 16

The Galois-twisted arithmetic groupoid: its KMS classification, and why the Tannakian obstruction does not compute it

This chapter is Paper VII of the series ([papers/VII-galois-groupoid/](#)).

SUMMARY. We construct the Galois-twisted arithmetic groupoid $\mathcal{G}_{G,S}$ over a compact group G and classify the KMS_β simplex of its C^* -algebra completely. The answer is

$$\text{KMS}_\beta(\mathcal{A}_{G,S}) \cong_{\text{affine}} \text{Prob}(E_\beta \backslash G),$$

where $E_\beta \leq G$ is the essential-value subgroup of the Frobenius cocycle, computed by the abelian Kakutani criterion inside the closed abelian subgroup $A = \overline{\text{Frob}(I_S)}$. The right translation action of G is transitive on the extreme points with stabilizer E_β .

This is *not* $\text{Prob}(G/N_\beta)$, with N_β the Tannakian subgroup of Chapter 15 of the series, and the discrepancy is not a technicality. The category $\mathcal{T}_\beta = \{\rho : \Xi_\beta(\rho, S) < \infty\}$ determines only the *normal closure* of E_β , because $\text{Rep}(G/N)$ sees normal subgroups only, whereas the ergodic decomposition of a skew product by a non-normal subgroup is a coset space that is not a group. For $G = S_3$ with E_β a transposition subgroup, $\text{Prob}(E_\beta \backslash G)$ is a 2-simplex while $\text{Prob}(G/N_\beta)$ is a point: three extremal KMS_β states where the Tannakian formulation predicts uniqueness. **The obstruction group Ξ_β does not determine the KMS simplex once G is non-abelian.**

Three natural guesses are corrected. The matrix zero-one law “ $F_\rho F_\rho^* = c_\rho I$ ” is false: the cocycle equation conjugates $F_\rho F_\rho^*$, so ergodicity forces its *spectrum* to be almost everywhere constant and nothing more; for $G = S_3$, ρ the two-dimensional irreducible and $\lambda = \frac{1}{2}(\delta_e + \delta_{(12)})$ one gets eigenvalues $\{0, 1\}$. The matrix Fourier transform F_ρ does not vanish when the Frobenius series $\Xi_\beta(\rho, S)$ diverges; it vanishes exactly when ρ has no E_β -fixed vector, and this is the refinement of the obstruction data that sees E_β . And the “matrix corner” route, twisting the coefficient algebra by $\text{Ad } \rho(\text{Frob}_\mathfrak{p})$, adds nothing: KMS states are tracial on the coefficient algebra, so inner twists are invisible to them.

Finally we record why the model cannot be genuinely non-abelian: I_S is abelian and $a \mapsto \text{Frob}_a$ is a homomorphism, so its image is always an abelian subgroup of G , the whole cocycle analysis takes place inside A , and G contributes only the coset space. A genuinely non-abelian theory needs a non-abelian acting semigroup, which by Chapter 14 of the series costs the high-temperature region where the phase structure lives.

1. The Galois-twisted groupoid

Let K be a number field, S a set of finite primes, $X = \overline{\mathbb{N}}^S$ with $\overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\}$, compact metrizable and totally disconnected, and $V = \mathbb{N}^S \subset X$ the set of valuation vectors, carrying the forced product measure $\pi_\beta = \bigotimes_{\mathfrak{p} \in S} \mu_{t_\mathfrak{p}}$, $t_\mathfrak{p} = N\mathfrak{p}^{-\beta}$, $\mu_t(k) = (1-t)t^k$ [Ch. 1, Lem. 2.10]; π_β charges no hyperplane $\{v_\mathfrak{p} = \infty\}$. Let $\mathcal{R}_S$ be the tail equivalence relation on V , which is ergodic for π_β by the Kolmogorov zero-one law [Ch. 1, Lem. 3.1]. The semigroup $J_S = \mathbb{N}^{(S)}$ acts on X by $v \mapsto v + a$, with $\infty + a_\mathfrak{p} = \infty$, and $I_S = \mathbb{Z}^{(S)}$ on the dilation $\tilde{X} = \varinjlim (X, +a)$, a locally compact totally disconnected space.

DEFINITION 1.1. Let G be a compact second countable group with Haar measure m_G , and let $\mathfrak{p} \mapsto \text{Frob}_{\mathfrak{p}} \in G$ be given for $\mathfrak{p} \in S$. Put $\text{Frob}_a = \prod_{\mathfrak{p}} \text{Frob}_{\mathfrak{p}}^{a_{\mathfrak{p}}}$ for $a \in I_S = \bigoplus_{\mathfrak{p} \in S} \mathbb{Z}$, let

$$\Omega_S = X \times G, \quad a \cdot (v, g) = (v + a, \text{Frob}_a^{-1} g),$$

let $\tilde{\Omega}_S = \tilde{X} \times G$ be the dilation, and set

$$\mathcal{G}_{G,S} = I_S \rtimes \tilde{\Omega}_S, \quad \mathcal{A}_{G,S} = \mathbf{1}_{\Omega_S} C^*(\mathcal{G}_{G,S}) \mathbf{1}_{\Omega_S} = C(\Omega_S) \rtimes J_S,$$

the unital full corner of the groupoid algebra $C_0(\tilde{\Omega}_S) \rtimes I_S$, exactly as $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$ sits in the dilated groupoid algebra in [Ch. 1, §2]; the dynamics is $\sigma_t(\mu_a) = N \mathbf{a}^{it} \mu_a$, $\sigma_t|_{C(\Omega_S)} = \text{id}$. All measures below are concentrated on $V \times G$, and we write (v, g) for its points.

- LEMMA 1.2 (Basic structure). (1) $a \mapsto \text{Frob}_a$ is a homomorphism $I_S \rightarrow G$, so $A := \overline{\text{Frob}(I_S)}$ is a closed abelian subgroup of G , whatever G is.
 (2) Right translation $R_h(v, g) = (v, gh)$ commutes with the I_S -action, so G acts on $\mathcal{A}_{G,S}$ by σ -commuting automorphisms and permutes the KMS_{β} states.
 (3) $\mathcal{G}_{G,S}$ is second countable, étale and amenable, so $C_r^* = C^*$ [110].

PROOF. (1) I_S is abelian and Frob is a homomorphism from it, so the image is an abelian subgroup, and the closure of an abelian subgroup is abelian. (2) Left and right translations of G commute. (3) I_S is abelian, hence amenable, acting on a locally compact space by homeomorphisms, and the corner of a full crossed product by an amenable group is the corner of the reduced one. $\square$

REMARK 1.3 (The model cannot be genuinely non-abelian). Lemma 1.2(1) is the fundamental limitation of this construction, and it is worth stating before any of the analysis. The Frobenius data enter only through the homomorphism $\text{Frob} : I_S \rightarrow G$, whose image is abelian because I_S is. Every cocycle computation below therefore takes place inside A , and the non-abelian group G contributes nothing but the homogeneous space $A \backslash G$ on which it acts. This is not an artefact of our normalization: any construction in which an abelian semigroup of ideals acts through Frobenius elements has the same feature. Genuine non-commutativity would require a non-abelian acting semigroup; Chapter 14 shows what that costs: partitioning range projections empty the high-temperature region.

2. Disintegration and the matrix Fourier transform

PROPOSITION 2.1 (KMS states are scaling measures). The map $\varphi \mapsto \nu = \varphi|_{C_0(\tilde{\Omega}_S)}$ is an affine bijection from $\text{KMS}_{\beta}(\mathcal{A}_{G,S})$ onto the set of Borel probability measures ν on Ω_S with

$$(2.1) \quad \nu(a \cdot E) = N \mathbf{a}^{-\beta} \nu(E) \quad (a \in J_S),$$

and every such ν has V -marginal π_{β} . Disintegrating, $\nu = \pi_{\beta} \otimes \rho_v$ with (ρ_v) a measurable family in $\text{Prob}(G)$, and (2.1) is equivalent to

$$(2.2) \quad \rho_{v+e_{\mathfrak{p}}} = (\text{Frob}_{\mathfrak{p}}^{-1})_* \rho_v \quad \pi_{\beta}\text{-a.e., for every } \mathfrak{p} \in S.$$

PROOF. As in [Ch. 1, Prop. 2.8]: the canonical expectation onto $C(\Omega_S)$ is faithful, the KMS condition on $\mu_a f \mu_a^*$ gives (2.1), and conversely each scaling measure reconstructs a KMS state. The V -marginal is forced by [Ch. 1, Lem. 2.10], whose argument uses only the scaling of the valuation coordinates, the G -direction being measure preserving for the translation. Splitting (2.1) into its V - and G -components gives (2.2). $\square$

DEFINITION 2.2 (Matrix Fourier transform). For $\rho \in \text{Irr}(G)$ of dimension d_{ρ} set

$$F_{\rho}(v) = \int_G \rho(g) d\rho_v(g) \in M_{d_{\rho}}(\mathbb{C}),$$

a contraction, $\|F_\rho(v)\| \leq 1$, since ρ_v is a probability measure and ρ is unitary.

LEMMA 2.3 (Matrix cohomological equation). *Equation (2.2) holds if and only if for every $\rho \in \text{Irr}(G)$*

$$(2.3) \quad F_\rho(v + e_{\mathfrak{p}}) = \rho(\text{Frob}_{\mathfrak{p}})^{-1} F_\rho(v) \quad \pi_\beta\text{-a.e., for every } \mathfrak{p} \in S.$$

PROOF. If (2.2) holds then

$$F_\rho(v + e_{\mathfrak{p}}) = \int_G \rho(g) d((\text{Frob}_{\mathfrak{p}}^{-1})_* \rho_v)(g) = \int_G \rho(\text{Frob}_{\mathfrak{p}}^{-1} h) d\rho_v(h) = \rho(\text{Frob}_{\mathfrak{p}})^{-1} F_\rho(v).$$

Conversely, the matrix coefficients of the $\rho \in \text{Irr}(G)$ span a dense subalgebra of $C(G)$ by Peter-Weyl, so (2.3) for all ρ determines (2.2). $\square$

REMARK 2.4 (The inverse is not cosmetic). With the action of Definition 1.1 the transform satisfies (2.3) with $\rho(\text{Frob}_{\mathfrak{p}})^{-1}$, not $\rho(\text{Frob}_{\mathfrak{p}})$. Writing the equation with $\rho(\text{Frob}_{\mathfrak{p}})$ corresponds to the opposite convention $a \cdot (v, g) = (v + a, \text{Frob}_a g)$ and changes E_β into its image under inversion, which for the subgroup E_β is harmless; but in intermediate steps, in particular in the martingale of Proposition 3.4, the two conventions do not agree and mixing them produces a wrong gauge.

3. What the zero-one law says, and what it does not

PROPOSITION 3.1 (Spectral zero-one law). *Put $P_\rho(v) = F_\rho(v)F_\rho(v)^*$, positive semidefinite with $\|P_\rho\| \leq 1$. Then (2.3) gives*

$$(3.1) \quad P_\rho(v + e_{\mathfrak{p}}) = \rho(\text{Frob}_{\mathfrak{p}})^{-1} P_\rho(v) \rho(\text{Frob}_{\mathfrak{p}}),$$

so the eigenvalues of $P_\rho(v)$, with multiplicity, are $\mathcal{R}_S$ -invariant and hence π_β -almost everywhere constant. In particular $\text{Tr } P_\rho(v)$ and $\det P_\rho(v)$ are a.e. constant.

PROOF. (3.1) is immediate from (2.3) and unitarity. Conjugation preserves the spectrum, so each elementary symmetric function of the eigenvalues is invariant under the generators of $\mathcal{R}_S$, hence a.e. constant by ergodicity [Ch. 1, Lem. 3.1]. $\square$

PROPOSITION 3.2 (P_ρ need not be scalar). *The statement $P_\rho(v) = c_\rho I_{d_\rho}$ a.e. is strictly stronger than Proposition 3.1 and is false. Take $G = S_3$, ρ the two-dimensional irreducible, $\text{Frob}_{\mathfrak{p}} = e$ for every $\mathfrak{p} \in S$, and $\rho_v \equiv \lambda := \frac{1}{2}(\delta_e + \delta_{(12)})$, which satisfies (2.2) trivially. Then*

$$F_\rho = \frac{1}{2}(I + \rho((12))), \quad P_\rho = \frac{1}{2}(I + \rho((12))),$$

a rank-one projection with eigenvalues $\{0, 1\}$, not a multiple of I_2 .

PROOF. $\rho((12))$ is a reflection, so $\rho((12))^* = \rho((12))$ and $\rho((12))^2 = I$; hence $F_\rho^* = F_\rho$ and $F_\rho^2 = \frac{1}{4}(I + 2\rho((12)) + I) = F_\rho$, so $P_\rho = F_\rho F_\rho^* = F_\rho^2 = F_\rho$ is a projection, of trace $\frac{1}{2} \text{Tr}(I + \rho((12))) = \frac{1}{2}(2 + 0) = 1$, hence rank one. $\square$

REMARK 3.3 (Numerical check). In the standard two-dimensional realization $\rho((12)) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, one computes $F_\rho = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$ and eigenvalues of P_ρ equal to $\{0, 1\}$; conjugating by $\rho((12))$ returns the same spectrum, as (3.1) requires.

PROPOSITION 3.4 (What the transform sees). *Let $\rho \in \text{Irr}(G)$ and $U_{\mathfrak{p}} = \rho(\text{Frob}_{\mathfrak{p}})$. The $U_{\mathfrak{p}}$ commute, lying in $\rho(A)$ with $A = \overline{\text{Frob}(I_S)}$ abelian, so there is a unitary W with $WU_{\mathfrak{p}}W^* = \text{diag}(\chi_1(\text{Frob}_{\mathfrak{p}}), \dots, \chi_{d_\rho}(\text{Frob}_{\mathfrak{p}}))$ for all $\mathfrak{p}$, where $\chi_1, \dots, \chi_{d_\rho} \in \hat{A}$ are the characters of A occurring in $\rho|_A$. Let $\Xi_\beta(A) \leq \hat{A}$ be as in Definition 4.1 below.*

- (1) Every measurable solution $F : V \rightarrow M_{d_\rho}(\mathbb{C})$ of (2.3) has the form $WF(v) = D(v)C$ with C a constant matrix whose r -th row vanishes unless $\chi_r \in \Xi_\beta(A)$, and $D(v) = \text{diag}(g_1(v), \dots, g_{d_\rho}(v))$, g_r a fixed unimodular solution of $g_r(v + e_p) = \chi_r(\text{Frob}_p)^{-1}g_r(v)$ when $\chi_r \in \Xi_\beta(A)$ and $g_r \equiv 1$ otherwise.
- (2) (2.3) has a unitary solution $W_\rho : V \rightarrow U(d_\rho)$, unique up to right multiplication by a constant unitary, if and only if every χ_r lies in $\Xi_\beta(A)$, i.e. $\Xi_\beta(\rho, S) < \infty$, i.e. $\rho|_{E_\beta} = 1$.
- (3) (2.3) has a nonzero solution if and only if some χ_r lies in $\Xi_\beta(A)$, i.e. ρ has a nonzero E_β -fixed vector. In particular $\Xi_\beta(\rho, S) = \infty$ does not force $F_\rho \equiv 0$: for $G = S_3$, $\text{Frob}_p = (12)$ for all p , $\zeta_{K,S}(\beta) = \infty$ and ρ the two-dimensional irreducible, $\Xi_\beta(\rho, S) = 4 \log \zeta_{K,S}(\beta) + O(1) = \infty$, while the constant family $\rho_v \equiv \frac{1}{2}(\delta_e + \delta_{(12)})$ is equivariant with $F_\rho \equiv \frac{1}{2}(I + \rho((12))) \neq 0$.

PROOF. Put $H = WF$; then (2.3) reads $H_{rs}(v + e_p) = \chi_r(\text{Frob}_p)^{-1}H_{rs}(v)$ for each entry. If $\chi_r \notin \Xi_\beta(A)$, every solution of this scalar equation vanishes a.e.: by [Ch. 1, Lem. 3.1] $|H_{rs}|$ is a.e. constant, and if the constant were positive, [Ch. 1, Prop. 3.3], whose proof uses only $|\chi_r(\text{Frob}_p)| = 1$, would make $\sum_p Np^{-\beta}|1 - \chi_r(\text{Frob}_p)|^2$ converge. If $\chi_r \in \Xi_\beta(A)$, [Ch. 1, Prop. 3.4] supplies a unimodular solution g_r , and H_{rs}/g_r is $\mathcal{R}_S$ -invariant, hence a.e. constant. This is (1). For (2), a unitary solution needs all rows of C nonzero, hence all $\chi_r \in \Xi_\beta(A)$; conversely $W^*D(v)$ is then a unitary solution, and two unitary solutions differ by an invariant, hence constant, unitary. The equivalence with $\Xi_\beta(\rho, S) < \infty$ is $\|I - U_p\|_{\text{HS}}^2 = \sum_r |1 - \chi_r(\text{Frob}_p)|^2$, and with $\rho|_{E_\beta} = 1$ is Pontryagin duality, $\Xi_\beta(A) = E_\beta^\perp$ since $\hat{A}$ is discrete. (3) is read off from (1) in the same way; the E_β -fixed vectors of ρ are the sum of the eigenspaces of those χ_r that are trivial on E_β . In the example, $\rho((12))$ is a reflection, so the two characters of $A = \{e, (12)\}$ occurring in $\rho|_A$ are the trivial one and the sign, and the displayed family satisfies $\rho_{v+e_p} = ((12))_*\rho_v = \rho_v$. $\square$

REMARK 3.5 (The refinement that sees E_β). The Dirichlet series $\Xi_\beta(\rho, S)$ records whether ρ is trivial on E_β . The vanishing or not of the matrix Fourier transform records whether $\rho^{E_\beta} \neq 0$; and the a.e. constant rank of $P_\rho = F_\rho F_\rho^*$ from Proposition 3.1, for the extremal equivariant families of Theorem 4.2 below, is $\dim \rho^{E_\beta}$: for such a family ρ_v is Haar measure on a coset $y(v)E_\beta h$, and $F_\rho(v) = \rho(y(v))P^{E_\beta}\rho(h)$ with P^{E_β} the projection onto ρ^{E_β} . The function $\rho \mapsto \dim \rho^{E_\beta}$ on $\text{Irr}(G)$ is the multiplicity function of the permutation representation $\text{Ind}_{E_\beta}^G \mathbf{1}$ on $L^2(E_\beta \backslash G)$ [86]: it determines $|E_\beta \backslash G| = \sum_\rho d_\rho \dim \rho^{E_\beta}$, the number of extremal KMS_β states, and the decomposition of that permutation representation, which distinguishes E_β from its normal closure — in Example 5.2 it is $\mathbf{1} \oplus \rho_2$ rather than $\mathbf{1}$. It does not in general determine E_β up to conjugacy, since non-conjugate subgroups can have the same permutation character; but it sees the simplex, which the Frobenius series does not.

4. The KMS classification

DEFINITION 4.1. Let $A = \overline{\text{Frob}(I_S)}$, abelian by Lemma 1.2(1), and let

$$\Xi_\beta(A) := \left\{ \chi \in \hat{A} : \sum_{p \in S} Np^{-\beta} |1 - \chi(\text{Frob}_p)|^2 < \infty \right\}, \quad E_\beta := \Xi_\beta(A)^\perp \leq A,$$

the essential-value subgroup of the Frobenius cocycle $c(v + a, v) = \text{Frob}_a$ in the sense of Schmidt [117]; for the compact target A the essential values of c are exactly the annihilator of the coboundary characters, which is how E_β enters below.

THEOREM 4.2 (Classification). *The map $\varphi \mapsto \nu$ of Proposition 2.1 induces an affine isomorphism*

$$(4.1) \quad \text{KMS}_\beta(\mathcal{A}_{G,S}) \cong \text{Prob}(E_\beta \backslash G),$$

and the right translation action of G is transitive on the extreme points, with the stabilizer of the base point equal to E_β . In particular the KMS_β state is unique if and only if $E_\beta = G$, which for G non-abelian forces G itself to be abelian; so for non-abelian G the simplex is never a point.

PROOF. By Proposition 2.1 we must describe the equivariant families (ρ_v) . The I_S -action on Ω_S is the skew product of $\mathcal{R}_S$ with the constant G -valued cocycle $c(v+a, v) = \text{Frob}_a$, whose values lie in A . Extreme equivariant families correspond to ergodic components of that skew product [42].

Decompose G into the cosets Ag ; the I_S -action preserves each, acting on $Ag \cong A$ by translation by Frob_a^{-1} . An equivariant family disintegrates over $A \backslash G$ into equivariant families on the cosets, so the extremal ones are supported on a single coset Ag and are extremal there. Within one coset the problem is that of [Ch. 1, Thm. 3.6] with A in place of G_S and $\text{Frob}_\mathfrak{p}^{-1}$ in place of $\sigma_\mathfrak{p}$: the Fourier coefficients $\hat{\rho}_v(\chi)$, $\chi \in \hat{A}$, satisfy the scalar cohomological equation with the unimodular constants $\chi(\text{Frob}_\mathfrak{p})^{\pm 1}$, so by [Ch. 1, Lem. 3.1, Props. 3.3, 3.4] they vanish unless $\chi \in \Xi_\beta(A)$, and the extremal families are the Haar measures on the cosets of $E_\beta = \Xi_\beta(A)^\perp$ in A moving by $\text{Frob}_\mathfrak{p}^{-1}$. Transported to Ag , these are the Haar measures on the subsets $aE_\beta g = E_\beta ag$, $a \in A$. Hence the extremal equivariant families are indexed by the right cosets $E_\beta h$, $h \in G$, and the simplex of equivariant families is $\text{Prob}(E_\beta \backslash G)$, giving (4.1).

Right translation by $h \in G$ carries the component of $E_\beta g$ to that of $E_\beta gh$, so the action is transitive, with stabilizer of $E_\beta e$ equal to E_β . Uniqueness means $|E_\beta \backslash G| = 1$, i.e. $E_\beta = G$; but $E_\beta \leq A$ is abelian. $\square$

COROLLARY 4.3 (Abelian case). *If G is abelian and $\text{Frob}(I_S)$ is dense, then $A = G$, $E_\beta = \Xi_\beta^\perp$ and (4.1) is $\text{Prob}(G/\Xi_\beta^\perp)$, recovering [Ch. 1, Thm. 3.6].*

5. The Tannakian obstruction does not compute the simplex

THEOREM 5.1 (Discrepancy). *Let $\Xi_\beta(\rho, S) = \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} \|I - \rho(\text{Frob}_\mathfrak{p})\|_{\text{HS}}^2$, let $\mathcal{T}_\beta(S) = \{\rho : \Xi_\beta(\rho, S) < \infty\}$, and let $N_\beta = \bigcap_{\rho \in \mathcal{T}_\beta} \ker \rho$ be the Tannakian subgroup of Chapter 15, for which $\mathcal{T}_\beta(S) = \text{Rep}(G/N_\beta)$. Then*

$$\mathcal{T}_\beta(S) = \{\rho : \rho|_{E_\beta} = 1\}, \quad N_\beta = \text{the normal closure of } E_\beta \text{ in } G.$$

Consequently $\text{Prob}(E_\beta \backslash G)$ and $\text{Prob}(G/N_\beta)$ agree if and only if E_β is normal in G , and in general

$$|E_\beta \backslash G| \geq |G/N_\beta|,$$

with strict inequality whenever E_β is not normal. **The obstruction group Ξ_β therefore does not determine the KMS_β simplex when G is non-abelian.**

PROOF. By Proposition 3.4(2), $\Xi_\beta(\rho, S) < \infty$ iff $\rho|_{E_\beta} = 1$, for irreducible ρ and hence for all ρ by additivity of Ξ_β over direct sums. A representation is trivial on E_β iff it is trivial on the normal closure $\langle\langle E_\beta \rangle\rangle$, since $\ker \rho$ is normal; hence $\mathcal{T}_\beta = \text{Rep}(G/\langle\langle E_\beta \rangle\rangle)$ and $N_\beta = \langle\langle E_\beta \rangle\rangle$. Since $E_\beta \subseteq N_\beta$, the natural surjection $E_\beta \backslash G \rightarrow N_\beta \backslash G = G/N_\beta$ gives the inequality, and it is a bijection iff $E_\beta = N_\beta$, i.e. iff E_β is normal. $\square$

EXAMPLE 5.2 (Three states where the Tannakian prediction is one). Let $G = S_3$, $\text{Frob}_\mathfrak{p} = (12)$ for every $\mathfrak{p} \in S$, and β with $\zeta_{K,S}(\beta) = \infty$. Then $A = E_\beta = \{e, (12)\}$: indeed the nontrivial character χ of A has $\chi(\text{Frob}_\mathfrak{p}) = -1$ for all $\mathfrak{p}$, so $\sum N\mathfrak{p}^{-\beta} |1 - \chi(\text{Frob}_\mathfrak{p})|^2 = 4\zeta_{K,S}(\beta) = \infty$ and $\Xi_\beta(A) = \{1\}$.

- By Theorem 4.2 the simplex is $\text{Prob}(E_\beta \backslash S_3)$, a 2-simplex: **three** extremal KMS_β states, permuted transitively by S_3 with stabilizer of order 2.

- By Theorem 5.1, $\mathcal{T}_\beta$ consists of the representations trivial on (12). The sign character has $\text{sgn}((12)) = -1$ and the two-dimensional irreducible restricts to E_β with eigenvalues ± 1 ; both are nontrivial. So $\mathcal{T}_\beta = \{\mathbf{1}\}$, $N_\beta = S_3$, and $\text{Prob}(G/N_\beta)$ is a **point**.

Three against one. The Tannakian formulation predicts uniqueness where the system has spontaneous symmetry breaking with a three-element phase space.

REMARK 5.3 (Why the two disagree). Tannaka–Krein reconstructs a group from a representation category, and a representation category can only see *normal* subgroups, being determined by kernels. The ergodic decomposition of a skew product by a subgroup E is the coset space $E \backslash G$, which for non-normal E is a homogeneous space carrying no group structure. In the abelian case every subgroup is normal and the distinction is invisible, which is why the GL_1 theory gave no warning. The correct reading of the Tannakian theorem of Chapter 15 is therefore that $\mathcal{T}_\beta$ computes the largest *quotient group* through which the obstruction factors; that quotient is a coarsening of the true invariant, which is the pair (G, E_β) up to conjugacy, equivalently the transitive G -space $E_\beta \backslash G$.

COROLLARY 5.4 (What is and is not an invariant). *The KMS_β simplex of $\mathcal{A}_{G,S}$, together with its G -action, determines E_β up to conjugacy. The Dirichlet obstruction data $\{\Xi_\beta(\rho, S)\}_{\rho \in \text{Irr}(G)}$ determine only N_β . Hence two systems with non-conjugate E_β having the same normal closure have identical obstruction data and non-isomorphic simplices.*

6. Route B: the matrix corner

PROPOSITION 6.1 (Inner twists are invisible). *Let $\rho : G \rightarrow U(n)$ be any unitary representation, let (B, α) be a unital C^* -algebra with an action of J_S by injective endomorphisms of the kind in [Ch. 1, §2], and let α_ρ be the twisted action on $B \otimes M_n(\mathbb{C})$,*

$$\alpha_{\rho, \mathbf{p}}(b \otimes m) = \alpha_{\mathbf{p}}(b) \otimes \rho(\text{Frob}_{\mathbf{p}})^{-1} m \rho(\text{Frob}_{\mathbf{p}}).$$

Give $B_\rho = (B \otimes M_n(\mathbb{C})) \rtimes_{\alpha_\rho} J_S$ the dynamics fixing $B \otimes M_n$ and scaling $\mu_{\mathbf{a}}$ by $N\mathbf{a}^{it}$. Then every KMS_β state of B_ρ restricts to $B \otimes M_n$ as $\varphi_0 \otimes \frac{1}{n} \text{Tr}$ with φ_0 the restriction of a KMS_β state of $B \rtimes_\alpha J_S$, and $\varphi \mapsto \varphi_0$ is an affine bijection $\text{KMS}_\beta(B_\rho) \cong \text{KMS}_\beta(B \rtimes_\alpha J_S)$. The twist by ρ does not change the simplex.

PROOF. For x, y in the fixed-point algebra of the dynamics, which contains $B \otimes M_n$, the KMS condition gives $\varphi(xy) = \varphi(y\sigma_{i\beta}(x)) = \varphi(yx)$; so φ is tracial on $B \otimes M_n$, hence of the form $\varphi_0 \otimes \frac{1}{n} \text{Tr}$. A KMS state vanishes on $\mu_{\mathbf{a}} x \mu_{\mathbf{b}}^*$ for $\mathbf{a} \neq \mathbf{b}$, and on $\mu_{\mathbf{a}} x \mu_{\mathbf{a}}^*$ it equals $N\mathbf{a}^{-\beta} \varphi(x)$; so φ is determined by its restriction, and the KMS condition on B_ρ reduces to $\varphi \circ \alpha_{\rho, \mathbf{p}} = N\mathbf{p}^{-\beta} \varphi$ on $B \otimes M_n$. Since $\text{Tr}(\rho(\text{Frob}_{\mathbf{p}})^{-1} m \rho(\text{Frob}_{\mathbf{p}})) = \text{Tr } m$, this reads $\varphi_0 \circ \alpha_{\mathbf{p}} = N\mathbf{p}^{-\beta} \varphi_0$, the KMS condition of the untwisted system, and conversely every such φ_0 gives a KMS state $\varphi_0 \otimes \frac{1}{n} \text{Tr}$ of B_ρ by the same computation. $\square$

REMARK 6.2. “Route B” proposed to import the non-abelian representation through a matrix-valued coefficient algebra $C(Y_S) \otimes M_n(\mathbb{C})$ twisted by conjugation with $\rho(\text{Frob}_{\mathbf{p}})$. Proposition 6.1 shows that this adds nothing, for a reason more basic than the spectral zero-one law: equilibrium states are tracial on the coefficient algebra, and a trace does not see inner automorphisms. The representation can only enter through the base space, which is Route A. (The twisted algebra is related to $\mathcal{A}_{G,S}$: the crossed product of $\mathcal{A}_{G,S}$ by the right translation action of G decomposes, by Peter–Weyl and the Stone–von Neumann theorem, into blocks indexed by $\text{Irr}(G)$, and the ρ -block is Morita equivalent to B_ρ with $B = C(X)$. That description is consistent with the proposition, and we do not use it.)

7. Assessment

Statement	Status	Reference
groupoid $\mathcal{G}_{G,S}$ well defined, $C_r^* = C^*$	true	Lem. 1.2
KMS states $\leftrightarrow$ scaling measures	true	Prop. 2.1
$F_\rho(v + e_{\mathfrak{p}}) = \rho(\text{Frob}_{\mathfrak{p}})F_\rho(v)$	inverse missing	Lem. 2.3
$F_\rho F_\rho^* = c_\rho I$	false	Prop. 3.2
spectrum of $F_\rho F_\rho^*$ a.e. constant	true	Prop. 3.1
$\Xi_\beta(\rho, S) = \infty \Rightarrow F_\rho = 0$	false	Prop. 3.4(3)
$F_\rho \neq 0 \iff \rho^{E_\beta} \neq 0$	true	Prop. 3.4, Rem. 3.5
unitary gauge exists $\iff \Xi_\beta < \infty$	true	Prop. 3.4(2)
$\text{KMS}_\beta \cong \text{Prob}(G/N_\beta)$	false	Ex. 5.2
$\text{KMS}_\beta \cong \text{Prob}(E_\beta \backslash G)$	true	Thm. 4.2
Ξ_β determines the simplex	false for non-abelian G	Cor. 5.4
Route B adds information	no: inner twists are invisible	Prop. 6.1
the model is genuinely non-abelian	no	Rem. 1.3

The construction succeeds and the classification is complete, so the question left open in Chapter 15 is answered. The answer is not the expected one, in two ways. The simplex is a space of probability measures on a homogeneous space rather than on a group, and the Dirichlet obstruction data — which is all that the L -functions of Chapter 15 can see — determine that homogeneous space only up to replacing E_β by its normal closure. Any programme that hopes to read a non-abelian phase structure off automorphic L -functions must therefore contend with Corollary 5.4: the L -functions know N_β , and N_β is not enough.

Three questions. First, Remark 3.5 identifies a refinement of the obstruction data, $\rho \mapsto \dim \rho^{E_\beta}$, read off from the rank of $F_\rho F_\rho^*$, which sees the size of the simplex and the permutation character of $E_\beta \backslash G$; whether this function of ρ has an expression in L -functions, as $\Xi_\beta(\rho, S)$ has in Chapter 15, and what further datum pins down E_β up to conjugacy, we do not know. Second, by Remark 1.3 no construction with an abelian acting semigroup can produce a non-abelian E_β ; the interesting case E_β non-normal already arises with E_β abelian and G non-abelian, as in Example 5.2, but a theory in which the essential-value group itself is non-abelian would need the acting semigroup to be non-abelian, and Chapter 14 shows that this empties the high-temperature region. Whether some intermediate structure — an Ore semigroup that is non-abelian but has nested rather than partitioning range projections — exists is open. Third, we have not determined which pairs (G, E_β) are realizable by an actual arithmetic $\mathfrak{p} \mapsto \text{Frob}_{\mathfrak{p}}$; Example 5.2 uses a constant assignment, and Chebotarev should allow much more, along the lines of [Ch. 2, Thm. 5.1].

Non-abelian Bost–Connes groupoids: two no-go theorems, and a construction that realizes $\text{Prob}(G/N_\beta)$ exactly when the Frobenius lifts mix

This chapter is Paper XXVII of the series (papers/XXVII-nonabelian-groupoids/).

SUMMARY. For a compact group G with Frobenius conjugacy classes $\text{Frob}_{\mathfrak{p}}$, $\mathfrak{p} \in S$, the Tannakian subgroup $N_\beta = \bigcap \{\ker \rho : \Xi_\beta(\rho, S) < \infty\}$ of Chapter 15 is the canonical non-abelian replacement of the obstruction group, and Chapter 16 showed that the obvious groupoid model, in which the Frobenius elements act through a homomorphism of the abelian ideal group, has KMS_β simplex $\text{Prob}(E_\beta \backslash G)$ with E_β abelian, which is $\text{Prob}(G/N_\beta)$ only when E_β is normal. We examine the two routes proposed in Chapter 15 for closing the gap and settle both.

Noncommutative coefficient algebras (Route B) cannot work, for a reason independent of the representation: a KMS_β state of $\mathcal{B} \rtimes J_S$ is tracial on $\mathcal{B}$, so the simplex is the simplex of scaling traces of $\mathcal{B}$, inner twists are invisible, and for $\mathcal{B} = C(Y_S) \otimes M_n(\mathbb{C})$ with any inner twist the equilibrium simplex is the abelian one.

Groupoids with isotropy N_β (the first reading of Route A) cannot work either: by Neshveyev’s theorem the isotropy contributes traces on $C^*(N_\beta)$, i.e. *dual* data indexed by $\text{Irr}(N_\beta)$, and the extreme point given by the trivial representation is fixed by every symmetry, so the symmetry group never acts transitively on the extreme points, as it must for $\text{Prob}(G/N_\beta)$.

Principal groupoids with a non-abelian Frobenius cocycle (the second reading of Route A) do work, but not canonically. Ordering the primes by norm, the ordered products $P_n(x) = \text{Frob}_{\mathfrak{p}_1}^{x_1} \cdots \text{Frob}_{\mathfrak{p}_n}^{x_n}$ define a cocycle c on the tail relation with genuinely non-commuting increments, for any choice of lifts of the classes to elements. We construct the corresponding étale groupoid, prove that the KMS_β simplex of its corner algebra is $\text{Prob}(H \backslash G)$ with H the Mackey range of c , and prove a *non-abelian Kakutani theorem*: $\rho \circ c$ is a coboundary if and only if $\Xi_\beta(\rho, S) = \sum_{\mathfrak{p}} N\mathfrak{p}^{-\beta} \|I - \rho(\text{Frob}_{\mathfrak{p}})\|_{\text{HS}}^2 < \infty$, the proof being a matrix martingale whose L^2 -growth is governed by $\det(I + t(I - U)^*(I - U)/(1 - t)^2)$. Consequently N_β is the normal closure of H , and the model realizes $\text{Prob}(G/N_\beta)$ precisely when H is normal. A mixing criterion on the products of the means $\mathbb{E}[\rho(\text{Frob}_{\mathfrak{p}})^{x_{\mathfrak{p}}}]$ forces $H = G$; for $G = S_3$ with the transposition class at every prime, the constant lift (12) gives three KMS_β states while the alternating lift (12), (13), (12), $\dots$ gives one. The simplex therefore depends on the lift, the lift-independent content is exactly N_β , and there is no canonical non-abelian Bost–Connes groupoid; but whenever N_β has no proper subgroup with normal closure N_β — for instance $N_\beta = A_3$, realized by 3-cycle classes on a divergent set and transposition classes on a convergent one — the simplex is $\text{Prob}(G/N_\beta)$ for every lift.

1. The question, and what is already known

Throughout, K is a number field, S an infinite set of finite primes with $\zeta_{K,S}(\beta) = \infty$, i.e. $\sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} = \infty$, listed by non-decreasing norm as $\mathfrak{p}_1, \mathfrak{p}_2, \dots$, with $t_i = N\mathfrak{p}_i^{-\beta} \in (0, t_{\max}]$, $t_{\max} = 2^{-\beta} < 1$. We write $V = \mathbb{N}^S$ for the space of valuation vectors with the product measure $\pi_\beta = \bigotimes_i \mu_{t_i}$, $\mu_t(k) = (1 - t)t^k$, forced on every KMS_β state by [Ch. 1, Lem. 2.10]; $\mathcal{R}_S$ is the tail equivalence relation on V (finitely many coordinates changed), ergodic by the zero–one law [Ch. 1, Lem. 3.1]; $\mathcal{F}_{\leq n} = \sigma(x_1, \dots, x_n)$ and $\mathcal{F}_{> n} = \sigma(x_{n+1}, x_{n+2}, \dots)$.

G is a compact second countable group; for each i a conjugacy class $C_i \subset G$ is given (the Frobenius class at $\mathfrak{p}_i$, or a Satake class), and a *lift* is a choice of elements $u_i \in C_i$. For a unitary representation ρ of G put

$$(1.1) \quad \Xi_\beta(\rho, S) = \sum_i t_i \|I - \rho(u_i)\|_{\text{HS}}^2 = 2 \sum_i t_i (\dim \rho - \Re \text{Tr } \rho(u_i)),$$

which depends only on the classes, and let $\mathcal{T}_\beta = \{\rho : \Xi_\beta(\rho, S) < \infty\}$ and $N_\beta = \bigcap_{\rho \in \mathcal{T}_\beta} \ker \rho$; $\mathcal{T}_\beta = \text{Rep}(G/N_\beta)$ is the Tannakian theorem of Chapter 15, and N_β depends only on the classes.

The abelian theory has three ingredients: a *space* on which the ideal semigroup acts, whose KMS_β measures are the equilibrium states; a *symmetry group* permuting them; and a *cocycle* whose coboundary question is the Kakutani dichotomy. The simplex is $\text{Prob}(G_S/\Xi_\beta^\perp)$ with G_S acting transitively on the extreme points [Ch. 1, Thm. 3.6]. The target of a non-abelian theory is a system whose KMS_β simplex is $\text{Prob}(G/N_\beta)$ with G acting transitively on the extreme points by translation, stabilizer N_β . The model of Chapter 16 — I_S acting on $V \times G$ through the homomorphism $a \mapsto \prod u_i^{a_i}$, which requires commuting lifts — gives $\text{Prob}(E_\beta \backslash G)$ with E_β contained in the closed abelian group generated by the u_i , and N_β is the normal closure of E_β . Two ways around this were proposed in Chapter 15 (its Remark 6.1) and left open in Chapter 16: a groupoid built directly, whose unit space carries the tail relation and a G -action and whose isotropy realizes N_β ; and a noncommutative coefficient algebra carrying ρ , with the commutative Ore semigroup J_S kept. We take the second first, because its failure is the simplest.

2. Route B: noncommutative coefficient algebras

Let $\mathcal{B}$ be a unital C^* -algebra with an action α of $J_S = \mathbb{N}^{(S)}$ by injective endomorphisms with hereditary ranges, $\alpha_{\mathfrak{a}}(1)$ a projection, so that Laca's dilation applies [77]: $\mathcal{B} \rtimes_\alpha J_S$ is the full corner $\mathbf{1}_\mathcal{B}(\tilde{\mathcal{B}} \rtimes I_S)\mathbf{1}_\mathcal{B}$ of the crossed product of the direct limit $\tilde{\mathcal{B}} = \varinjlim (\mathcal{B}, \alpha)$ by the automorphic extension of α to the group I_S ; it is generated by $\mathcal{B}$ and isometries $\mu_{\mathfrak{a}}$ with $\mu_{\mathfrak{a}} b \mu_{\mathfrak{a}}^* = \alpha_{\mathfrak{a}}(b)$, and the dynamics is $\sigma_t(\mu_{\mathfrak{a}}) = N \mathfrak{a}^{it} \mu_{\mathfrak{a}}$, $\sigma_t|_{\mathcal{B}} = \text{id}$. The gauge action of the compact group $\hat{I}_S$ gives the faithful conditional expectation $E : \mathcal{B} \rtimes_\alpha J_S \rightarrow \mathcal{B}$, the average of the gauge automorphisms; so “a tractable faithful expectation onto the coefficient algebra” exists for every $\mathcal{B}$. It does not help. Write $T(\mathcal{B})$ for the tracial state space of $\mathcal{B}$, a Choquet simplex, and

$$T_\beta(\mathcal{B}, \alpha) = \{\tau \in T(\mathcal{B}) : \tau \circ \alpha_{\mathfrak{a}} = N \mathfrak{a}^{-\beta} \tau \quad (\mathfrak{a} \in J_S)\},$$

the *scaling traces*, a closed face of $T(\mathcal{B})$.

THEOREM 2.1 (Route B is abelian). (1) *Every KMS_β state φ of $\mathcal{B} \rtimes_\alpha J_S$ restricts to a scaling trace $\tau = \varphi|_{\mathcal{B}} \in T_\beta(\mathcal{B}, \alpha)$.*

(2) *For every $\tau \in T_\beta(\mathcal{B}, \alpha)$, $\tau \circ E$ is a KMS_β state with restriction τ ; so $\tau \mapsto \tau \circ E$ is an affine injection $T_\beta(\mathcal{B}, \alpha) \rightarrow \text{KMS}_\beta(\mathcal{B} \rtimes_\alpha J_S)$.*

(3) *If the numbers $\log N \mathfrak{p}$, $\mathfrak{p} \in S$, are linearly independent over $\mathbb{Q}$ — for instance $K = \mathbb{Q}$ — then every KMS_β state is of the form $\tau \circ E$, and $\text{KMS}_\beta(\mathcal{B} \rtimes_\alpha J_S) \cong T_\beta(\mathcal{B}, \alpha)$ affinely.*

Consequently: (a) if $\alpha'_{\mathfrak{a}} = \text{Ad}(w_{\mathfrak{a}}) \circ \alpha_{\mathfrak{a}}$ with unitaries $w_{\mathfrak{a}} \in \mathcal{B}$ is another action, then $T_\beta(\mathcal{B}, \alpha') = T_\beta(\mathcal{B}, \alpha)$; inner twists are invisible. (b) For $\mathcal{B} = C(Y_S) \otimes M_n(\mathbb{C})$ and any action of the form $\alpha_{\mathfrak{a}} \otimes \text{Ad}(v_{\mathfrak{a}})$, $v_{\mathfrak{a}} \in U(n)$, the scaling traces are $\nu \otimes \frac{1}{n} \text{Tr}$ with ν a KMS_β measure of the classical system, so $T_\beta = \text{Prob}(G_S/\Xi_\beta^\perp)$ [Ch. 1, Thm. 3.6]: the abelian answer. (c) In every case the equilibrium theory of $\mathcal{B} \rtimes_\alpha J_S$ is that of the affine action of J_S on the Choquet simplex $T(\mathcal{B})$; the noncommutativity of $\mathcal{B}$ is quotiented out by traciality before any symmetry can act on it.

PROOF. (1) $\mathcal{B}$ lies in the fixed-point algebra of σ , and for x, y there the KMS condition gives $\varphi(xy) = \varphi(y\sigma_{i\beta}(x)) = \varphi(yx)$; so τ is tracial. For $b \in \mathcal{B}$, the KMS condition with the entire element $\mu_{\mathfrak{a}}$, $\sigma_{i\beta}(\mu_{\mathfrak{a}}) = N \mathfrak{a}^{-\beta} \mu_{\mathfrak{a}}$, gives $\tau(\alpha_{\mathfrak{a}}(b)) = \varphi(\mu_{\mathfrak{a}} b \mu_{\mathfrak{a}}^*) = N \mathfrak{a}^{-\beta} \varphi(b \mu_{\mathfrak{a}}^* \mu_{\mathfrak{a}}) = N \mathfrak{a}^{-\beta} \tau(b)$.

(2) Extend τ to the semifinite trace $\tilde{\tau}$ on $\tilde{\mathcal{B}}$ determined by $\tilde{\tau}(\alpha_a^{-1}(b)) = N\mathbf{a}^\beta \tau(b)$, consistent by the scaling relation, and let γ be the automorphic action of I_S on $\tilde{\mathcal{B}}$, so that $\tilde{\tau} \circ \gamma_a = N\mathbf{a}^{-\beta} \tilde{\tau}$ for $a \in I_S$. On the dense span of elements $x = a_1 u_a$, $y = a_2 u_b$ of $\tilde{\mathcal{B}} \rtimes I_S$ ($a_i \in \tilde{\mathcal{B}}$ in the domain of $\tilde{\tau}$, u_a the implementing unitaries),

$$(\tilde{\tau} \circ E)(xy) = \delta_{a+b,0} \tilde{\tau}(a_1 \gamma_a(a_2)), \quad (\tilde{\tau} \circ E)(y \sigma_{i\beta}(x)) = \delta_{a+b,0} N\mathbf{a}^{-\beta} \tilde{\tau}(a_2 \gamma_{-a}(a_1)),$$

and $N\mathbf{a}^{-\beta} \tilde{\tau}(a_2 \gamma_{-a}(a_1)) = N\mathbf{a}^{-\beta} N\mathbf{a}^\beta \tilde{\tau}(\gamma_a(a_2) a_1) = \tilde{\tau}(a_1 \gamma_a(a_2))$, using $\tilde{\tau} \circ \gamma_a = N\mathbf{a}^{-\beta} \tilde{\tau}$ read as $\tilde{\tau} = N\mathbf{a}^\beta \tilde{\tau} \circ \gamma_a$ and then traciality. So $\tilde{\tau} \circ E$ satisfies the KMS condition on the corner, where it is the state $\tau \circ E$. Injectivity of $\tau \mapsto \tau \circ E$ is clear.

(3) A KMS_β state is σ -invariant. Under the hypothesis, the one-parameter group $t \mapsto (N\mathbf{p}^{it})_{\mathbf{p}}$ is dense in $\hat{I}_S = \prod_{\mathbf{p}} \mathbb{T}$ by Kronecker's theorem applied to finite subsets of S , so a σ -invariant state is invariant under the whole gauge action and hence factors through E : $\varphi = \varphi \circ E = \tau \circ E$ with $\tau = \varphi|_{\mathcal{B}}$, which is a scaling trace by (1).

(a) A trace is invariant under inner automorphisms, so $\tau \circ \alpha'_a = \tau \circ \alpha_a$. (b) A tracial state of $C(Y_S) \otimes M_n$ is $\nu \otimes \frac{1}{n} \text{Tr}$, and $\text{Tr}(v_a m v_a^*) = \text{Tr } m$ reduces the scaling condition to $\nu \circ \alpha_a = N\mathbf{a}^{-\beta} \nu$. (c) is (1)–(3). $\square$

REMARK 2.2. Without the independence hypothesis in (3), σ -invariance only forces φ to factor through the expectation onto the fixed-point algebra of σ , which also contains the elements $\mu_a b \mu_b^*$ with $N\mathbf{a} = N\mathbf{b}$, $\mathbf{a} \neq \mathbf{b}$; for the classical system these carry nothing because the groupoid is principal on a conull set [Ch. 1, Prop. 2.8], and for general $\mathcal{B}$ there may be additional KMS_β states, all of them tracial on the fixed-point algebra. None of this reintroduces the representation: the matrix Kakutani dichotomy of Chapter 15, valid for commuting lifts, would enter Route B through the twist $\text{Ad } \rho(u_{\mathbf{p}})$, and (a) says the twist is never seen; for non-commuting lifts $\alpha_{\mathbf{p}} \otimes \text{Ad } \rho(u_{\mathbf{p}})^{-1}$ do not even commute, so there is no action to twist. The modular group is not “broken” by a noncommutative $\mathcal{B}$: it is the same gauge dynamics, and it forces the equilibrium state to forget everything about $\mathcal{B}$ except its traces. Route B is closed.

3. Route A, first reading: isotropy cannot carry N_β

Let $\mathcal{G}$ be a second countable locally compact Hausdorff étale groupoid with unit space Ω , and $c : \mathcal{G} \rightarrow \mathcal{R}$ a continuous cocycle defining $\sigma_t(f)(\gamma) = e^{itc(\gamma)} f(\gamma)$ on $C^*(\mathcal{G})$ (the Haar system is counting measure on the fibres, as for every étale groupoid). Neshveyev's theorem [96, Thm. 1.3] describes the KMS_β states: they are the functionals

$$(3.1) \quad \varphi(f) = \int_{\Omega} \sum_{\gamma \in \mathcal{G}_x^x} f(\gamma) \phi_x(u_\gamma) d\mu(x),$$

where μ is a quasi-invariant probability measure with Radon–Nikodym cocycle $e^{-\beta c}$ and $x \mapsto \phi_x$ is a μ -measurable field of tracial states on the group C^* -algebras $C^*(\mathcal{G}_x^x)$ of the isotropy groups, vanishing on u_γ unless $c(\gamma) = 0$, and conjugation-equivariant: $\phi_{r(\gamma)}(u_{\gamma h \gamma^{-1}}) = \phi_{s(\gamma)}(u_h)$ for μ -a.e. γ . The extreme points are the pairs with μ ergodic and ϕ extremal among equivariant fields.

THEOREM 3.1 (No-go for isotropy). *Let $(\mathcal{G}, c)$ be as above, and let a group G act on $\mathcal{G}$ by automorphisms preserving c , hence on $\text{KMS}_\beta(C^*(\mathcal{G}))$ by affine bijections. For an extreme KMS_β state with quasi-invariant measure μ let $\mathcal{E}_\mu$ be the set of extreme KMS_β states with measure μ .*

- (1) *The state $\varphi_\mu(f) = \int_{\Omega} f|_{\Omega} d\mu$ — the field $\phi_x = \text{trivial character}$ — belongs to $\mathcal{E}_\mu$, and it is fixed by the stabilizer G_μ of μ in G .*
- (2) *If the isotropy groups $\mathcal{G}_x^x$ are μ -a.e. isomorphic to a fixed compact group $N \subseteq \ker c$, with the isotropy bundle constant and the conjugation action of $\mathcal{G}$ on it trivial, then $\mathcal{E}_\mu$ is in bijection with $\text{Irr}(N)$, via $\rho \mapsto (\mu, \phi_x \equiv \chi_\rho / \dim \rho)$.*

- (3) *Consequently, if G acts transitively on the extreme KMS_β states, then $\mathcal{E}_\mu = \{\varphi_\mu\}$ for every μ , and in the situation of (2) $N = \{1\}$: the isotropy is μ -a.e. trivial.*

In particular no groupoid whose KMS_β multiplicity is produced by isotropy has a simplex of the form $\text{Prob}(G/N)$ with G acting transitively on the extreme points and $N \neq 1$.

PROOF. (1) The trivial character is a tracial state of every $C^*(\mathcal{G}_x^x)$, vanishes on u_γ for $c(\gamma) \neq 0$ as required, and the constant field is conjugation-equivariant; so φ_μ is a KMS_β state by (3.1), and it is extreme because μ is ergodic and the trivial character is an extreme tracial state. An automorphism of $(\mathcal{G}, c)$ transforms a pair (μ, ϕ) into $(\mu \circ g^{-1}, \phi \circ g^{-1})$; if it fixes μ it fixes the pair with the trivial field.

(2) Tracial states of $C^*(N)$ are the convex combinations of normalized characters $\chi_\rho / \dim \rho$, $\rho \in \text{Irr}(N)$, the extreme ones being the normalized characters. With constant bundle and trivial conjugation action, equivariance says the field is $\mathcal{G}$ -invariant, hence a.e. constant by ergodicity of μ ; so the equivariant fields are the constant ones, the extreme ones the constant normalized characters.

(3) The map $\varphi \mapsto \mu$ from extreme KMS_β states to ergodic quasi-invariant measures is G -equivariant, so if G is transitive on the extreme states then G_μ is transitive on the fibre $\mathcal{E}_\mu$; by (1) it fixes $\varphi_\mu \in \mathcal{E}_\mu$, so $\mathcal{E}_\mu = \{\varphi_\mu\}$, and by (2) $|\text{Irr}(N)| = 1$, i.e. $N = 1$. For the last sentence, the translation action of G on $\text{Prob}(G/N)$ is transitive on the extreme points. $\square$

REMARK 3.2. The theorem says that isotropy is the wrong place to put N_β : it produces the *dual* object $\text{Irr}(N_\beta)$, with the trivial representation as a distinguished, symmetry-fixed equilibrium state, whereas symmetry breaking needs a homogeneous space with no distinguished point. “Isotropy = N_β ” would give a simplex with a canonical symmetric state and $|\text{Irr}(N_\beta)|$ extreme points over each ergodic measure, which is $\text{Prob}(G/N_\beta)$ neither as a G -space nor, in general, as a set. The classical system itself illustrates the point: its groupoid is principal on a conull set [Ch. 1, Prop. 2.8], and all of its multiplicity comes from the ergodic decomposition of the unit space. That is where a non-abelian construction must put it.

4. Route A, second reading: the ordered Frobenius cocycle

4.1. The cocycle. Fix lifts $u_i \in C_i$. For $x \in V$ and $n \geq 0$ put

$$(4.1) \quad P_n(x) = u_1^{x_1} u_2^{x_2} \cdots u_n^{x_n} \in G,$$

an ordered product, and for $(y, x) \in \mathcal{R}_S$ let n be any index beyond the last coordinate in which x and y differ and

$$(4.2) \quad c(y, x) = P_n(y) P_n(x)^{-1}.$$

This is independent of n : $P_{n'}(x) = P_n(x) u_{n+1}^{x_{n+1}} \cdots u_{n'}^{x_{n'}}$ and the extra factors agree for x and y and cancel on the right. It is a cocycle, $c(z, y) c(y, x) = c(z, x)$, being of the form $P(y) P(x)^{-1}$ on each $\mathcal{R}_{F_n}$. Its increments are

$$(4.3) \quad c(x + e_j, x) = P_{j-1}(x) u_j P_{j-1}(x)^{-1},$$

the j -th lift conjugated by the ordered product of the earlier ones; for commuting lifts this is u_j and c is the constant-increment cocycle of Chapter 16, but in general the increments at different coordinates do not commute, and (4.2) is the way to have a cocycle nonetheless: the obstruction of Chapter 15, that a cocycle with *constant* non-commuting increments cannot exist on a relation generated by commuting shifts, is circumvented by letting the increment at $\mathfrak{p}_j$ depend on the coordinates before it.

4.2. The groupoid and its algebra. To have a topological model we compactify each coordinate together with its lift. Let

$$X_i = \overline{\{(k, u_i^k) : k \in \mathbb{N}\}} \subseteq \overline{\mathbb{N}} \times G, \quad X = \prod_{i \geq 1} X_i,$$

compact, second countable, with $\mathbb{N} \subset X_i$ open and dense (as $\{k\} \times \{u_i^k\}$) and boundary $\partial X_i = \{\infty\} \times \overline{\{u_i^k\}}$. Write $g_i : X_i \rightarrow G$ for the second projection, continuous, with $g_i(k) = u_i^k$. The shift $k \mapsto k+1$ extends to a homeomorphism s_i of X_i onto $X_i \setminus \{0\}$, $(\infty, g) \mapsto (\infty, u_i g)$, and J_S acts on X by $x \mapsto x + a$ coordinatewise, injectively, with clopen images. Put

$$P_n(x) = g_1(x_1) \cdots g_n(x_n), \quad c(x + e_j, x) = P_{j-1}(x) u_j P_{j-1}(x)^{-1},$$

continuous on X and extending (4.1)–(4.3). Let $\tilde{X} = \varinjlim (X, +a)$ be the dilation, a locally compact space on which $I_S = \mathbb{Z}^{(S)}$ acts by homeomorphisms, and extend c to $\tilde{X}$ by the same formulas. Define the action of I_S on $\Omega = \tilde{X} \times G$ by

$$(4.4) \quad a \cdot (x, g) = (x + a, c(x + a, x) g),$$

which is an action because c is a cocycle, and by homeomorphisms because c is continuous. Let

$$\mathcal{G}_c = I_S \ltimes \Omega, \quad \mathcal{A}_c = \mathbf{1}_{X \times G} C^*(\mathcal{G}_c) \mathbf{1}_{X \times G} = C(X \times G) \rtimes J_S,$$

the transformation groupoid with counting measures as Haar system [108], and its unital full corner; the dynamics is $\sigma_t(\mu_a) = N \mathbf{a}^{it} \mu_a$, given by the continuous cocycle $(a, \omega) \mapsto \log N \mathbf{a}$ on $\mathcal{G}_c$. The group G acts on Ω by right translation $R_h(x, g) = (x, gh)$, commuting with (4.4), hence on $\mathcal{A}_c$ by σ -commuting automorphisms.

LEMMA 4.1. *$\mathcal{G}_c$ is second countable, étale, Hausdorff and amenable, principal on the conull set $V \times G$ (points with all coordinates finite), and $\mathcal{A}_c$ is nuclear. The KMS_β states of $\mathcal{A}_c$ are in affine bijection with the probability measures ν on $X \times G$ with $\nu(a \cdot E) = N \mathbf{a}^{-\beta} \nu(E)$ for $a \in J_S$; every such ν is concentrated on $V \times G$, has V -marginal π_β , and disintegrates as $\nu = \pi_\beta \otimes \rho_x$ with a measurable family $\rho_x \in \text{Prob}(G)$ satisfying*

$$(4.5) \quad \rho_{x+e_j} = (c(x + e_j, x))_* \rho_x \quad \text{for } \pi_\beta\text{-a.e. } x, \text{ every } j.$$

PROOF. I_S is abelian, so $\mathcal{G}_c$ is amenable and $C^* = C_r^*$; étaleness and Hausdorffness are those of a transformation groupoid by homeomorphisms. If $a \cdot (x, g) = (x, g)$ then $x + a = x$, which forces $a = 0$ when all x_i are finite; so the isotropy is trivial on $V \times G$. For the KMS states, the argument of [Ch. 1, Prop. 2.8] applies verbatim (or one may invoke (3.1)): a KMS_β state is $\nu \circ E$ for a scaling measure ν on the unit space of the corner. If $E \subseteq \{x_i = \infty\}$ then $e_i \cdot E = E$ and the scaling gives $\nu(E) = t_i \nu(E)$, so $\nu(E) = 0$: ν is concentrated on $V \times G$. The V -marginal satisfies the scaling relations of the valuation coordinates alone, so it is π_β by [Ch. 1, Lem. 2.10]. Disintegrating and comparing the two sides of the scaling identity on sets $B \times C$, using $\pi_\beta(B + e_j) = t_j \pi_\beta(B)$, gives (4.5). $\square$

4.3. Classification by the Mackey range. Recall Zimmer's theory of cocycles into compact groups [137, Ch. 4 and 9], in the setting of countable equivalence relations [42]: for an ergodic countable equivalence relation $(V, \pi, \mathcal{R})$ and a measurable cocycle $c : \mathcal{R} \rightarrow G$ into a compact group, there is a closed subgroup $H \leq G$, unique up to conjugacy, and a measurable $f : V \rightarrow G$ such that $c'(y, x) = f(y)^{-1} c(y, x) f(x)$ takes values in H and the skew product relation $\mathcal{R} \ltimes_{c'} H$ on $V \times H$ is ergodic; H is the *Mackey range* (algebraic hull) of c , and c is cohomologous to a cocycle into a closed subgroup H' iff H' contains a conjugate of H .

THEOREM 4.2 (The simplex of the ordered model). *Let H be the Mackey range of c on $(V, \pi_\beta, \mathcal{R}_S)$. Then*

$$\text{KMS}_\beta(\mathcal{A}_c) \cong_{\text{affine}} \text{Prob}(H \backslash G),$$

the extreme points being the families $\rho_x = \text{Haar measure on } f(x)Hk, k \in G$, and the right translation action of G is transitive on the extreme points with stabilizer of the point Hk equal to $k^{-1}Hk$.

PROOF. By Lemma 4.1 we classify the equivariant families (4.5), i.e. the probability measures $\nu = \pi_\beta \otimes \rho_x$ on $V \times G$ invariant in measure class and with the prescribed Radon–Nikodym derivative under the skew product relation $\tilde{\mathcal{R}} = \mathcal{R}_S \ltimes_c G, (x, g) \sim (y, c(y, x)g)$; the extreme ones are the ergodic ones. The map $\Psi(x, g) = (x, f(x)^{-1}g)$ carries $\tilde{\mathcal{R}}$ to $\mathcal{R}_S \ltimes_{c'} G$ and preserves $\pi_\beta \otimes m_G$. The latter relation preserves each set $V \times Hk$, on which it is isomorphic to $\mathcal{R}_S \ltimes_{c'} H$ via $h \mapsto hk$, ergodic by the definition of the Mackey range. Hence the ergodic components of $\tilde{\mathcal{R}}$ are the sets $\Psi^{-1}(V \times Hk) = \{(x, g) : f(x)^{-1}g \in Hk\}, Hk \in H \backslash G$. Within a component, Haar on the fibres is the *only* equivariant family: if $\rho'_x \in \text{Prob}(H)$ satisfies $\rho'_y = c'(y, x)_* \rho'_x$, then for a nontrivial $\sigma \in \text{Irr}(H)$ the Fourier coefficient $B(x) = (\int_H \sigma(h)^* d\rho'_x(h))^*$ satisfies $B(y) = \sigma(c'(y, x))B(x)$, so $\Phi(x, h) = \text{Tr}(\sigma(h)^* B(x))$ is $\mathcal{R}_S \ltimes_{c'} H$ -invariant — $\Phi(y, c'h) = \text{Tr}(\sigma(h)^* \sigma(c')^* \sigma(c') B(x)) = \Phi(x, h)$ — hence a.e. constant by ergodicity; since $\int_H \sigma(h) dh = 0$ the constant is 0, and Fourier inversion in h gives $B = 0$. All nontrivial coefficients of ρ'_x vanish, so $\rho'_x = m_H$ a.e. Thus the ergodic equivariant families are the Haar measures on the fibres $f(x)Hk$, and every equivariant family is a probability mixture of these, uniquely, since distinct ergodic components carry mutually singular measures. Right translation by h maps the component of Hk to that of Hkh . $\square$

For commuting lifts f can be taken trivial and $H = E_\beta$, recovering Chapter 16. The content of the model is therefore the conjugacy class of H . Two questions remain: how H is related to N_β , and how H depends on the lifts.

4.4. The non-abelian Kakutani theorem. For a unitary representation $\rho : G \rightarrow U(d)$ write $U_i = \rho(u_i)$ and

$$(4.6) \quad m_i = \mathbb{E}_{\mu_{t_i}} [U_i^{x_i}] = \sum_{k \geq 0} (1 - t_i) t_i^k U_i^k = (1 - t_i)(I - t_i U_i)^{-1},$$

invertible since $\|t_i U_i\| < 1$, with

$$(4.7) \quad m_i^{-*} m_i^{-1} = \frac{(I - t_i U_i^*)(I - t_i U_i)}{(1 - t_i)^2} = I + \frac{t_i}{(1 - t_i)^2} (I - U_i)^*(I - U_i) \geq I.$$

If $e^{i\phi_{i,1}}, \dots, e^{i\phi_{i,d}}$ are the eigenvalues of U_i , then

$$(4.8) \quad \det(m_i^{-*} m_i^{-1}) = \prod_{r=1}^d \left(1 + \frac{t_i |1 - e^{i\phi_{i,r}}|^2}{(1 - t_i)^2}\right), \quad \sum_r |1 - e^{i\phi_{i,r}}|^2 = \|I - U_i\|_{\text{HS}}^2.$$

Put $M_{(n,m]} = m_{n+1} m_{n+2} \cdots m_m$ and $M_n = M_{(0,n]}$. A *coboundary* means a measurable $F : V \rightarrow U(d)$ with $\rho(c(y, x)) = F(y)F(x)^{-1}$ for π_β -a.e. $(y, x) \in \mathcal{R}_S$.

THEOREM 4.3 (Non-abelian Kakutani). *For every lift and every unitary representation ρ ,*

$$\rho \circ c \text{ is a coboundary} \iff \Xi_\beta(\rho, S) < \infty.$$

PROOF. Two structural facts are used throughout. First, for $F : V \rightarrow M_d(\mathbb{C})$ measurable with $F(y) = \rho(c(y, x))F(x)$ on $\mathcal{R}_S$, the function $K_n = P_n^{-1}F$ (with $P_n = \rho(P_n(x))$), we suppress ρ is invariant under changes of the first n coordinates: for $y \sim x$ differing only there, $K_n(y) = P_n(y)^{-1} \rho(c(y, x))F(x) = P_n(y)^{-1} P_n(y) P_n(x)^{-1} F(x) = K_n(x)$; so K_n is $\mathcal{F}_{>n}$ -measurable

[Ch. 1, Lem. 3.1] and independent of P_n , which is $\mathcal{F}_{\leq n}$ -measurable. Second, $\mathbb{E}[P_n] = M_n$ and $\mathbb{E}[P_m|\mathcal{F}_{\leq n}] = P_n M_{(n,m]}$, by independence of the coordinates and the ordering of the product.

($\Leftarrow$) Let $\Xi_\beta(\rho, S) < \infty$ and $Z_n = P_n M_n^{-1}$. Then $\mathbb{E}[Z_{n+1}|\mathcal{F}_{\leq n}] = P_n m_{n+1} M_{n+1}^{-1} = Z_n$: a matrix-valued martingale. Its L^2 -norm is deterministic, $\mathbb{E}\|Z_n\|_{\text{HS}}^2 = \text{Tr}(M_n^{-1} M_n^{-*}) = \|M_n^{-1}\|_{\text{HS}}^2$, and by (4.7), iterating $m_i^{-*}(\cdot) m_i^{-1} \leq \lambda_i(\cdot)$ with $\lambda_i = 1 + t_i \|I - U_i\|_{\text{op}}^2 / (1 - t_i)^2$ from the inside out,

$$\|M_n^{-1}\|_{\text{HS}}^2 \leq d \prod_{i \leq n} \lambda_i \leq d \exp\left(\frac{1}{(1 - t_{\max})^2} \sum_i t_i \|I - U_i\|_{\text{HS}}^2\right) < \infty,$$

using $t_i \leq t_{\max} < 1$. So $Z_n \rightarrow Z_\infty =: F$ a.s. and in L^2 . Moreover $|\det Z_n| = |\det M_n^{-1}| = \prod_{i \leq n} \det(m_i^{-*} m_i^{-1})^{1/2}$ increases to a finite limit ≥ 1 by (4.8) and $\Xi_\beta < \infty$, so $\det F \neq 0$ a.s. and F is invertible. For $y \sim x$ and n large, $Z_n(y) Z_n(x)^{-1} = P_n(y) P_n(x)^{-1} = \rho(c(y, x))$, and letting $n \rightarrow \infty$ gives $F(y) = \rho(c(y, x)) F(x)$. Then $F^* F$ is $\mathcal{R}_S$ -invariant, $F(y)^* F(y) = F(x)^* \rho(c)^* \rho(c) F(x) = F(x)^* F(x)$, hence a.e. equal to a constant positive matrix Q^2 , and $W = F Q^{-1}$ is unitary with $\rho(c(y, x)) = W(y) W(x)^{-1}$.

($\Rightarrow$) Let F be a unitary coboundary and $C_n = \mathbb{E}[K_n]$. Then $\mathbb{E}[F|\mathcal{F}_{\leq n}] = P_n C_n$, and comparing $\mathbb{E}[F|\mathcal{F}_{\leq n}] = \mathbb{E}[\mathbb{E}[F|\mathcal{F}_{\leq m}]|\mathcal{F}_{\leq n}] = P_n M_{(n,m]} C_m$ gives

$$(4.9) \quad C_n = M_{(n,m]} C_m \quad (m > n).$$

Suppose $\Xi_\beta(\rho, S) = \infty$. Then for each n , $|\det M_{(n,m]}^{-1}| \rightarrow \infty$ as $m \rightarrow \infty$ by (4.8), while $\|C_m\|_{\text{HS}} \leq \mathbb{E}\|K_m\|_{\text{HS}} = \sqrt{d}$; so $|\det C_m| = |\det M_{(n,m]}^{-1}| |\det C_n|$ bounded forces $\det C_n = 0$ for every n . But $P_n C_n = \mathbb{E}[F|\mathcal{F}_{\leq n}] \rightarrow F$ in L^2 by martingale convergence, F being measurable with respect to $\bigvee_n \mathcal{F}_{\leq n}$, so along a subsequence $\det(P_{n_k} C_{n_k}) \rightarrow \det F$ a.e., and $|\det F| = 1$; this contradicts $\det C_{n_k} = 0$. $\square$

COROLLARY 4.4 (N_β is the normal closure of the Mackey range). *For every lift, with H the Mackey range of c ,*

$$\mathcal{T}_\beta = \{\rho : \rho|_H = 1\}, \quad N_\beta = \langle\langle H \rangle\rangle,$$

the smallest closed normal subgroup containing H . Consequently $\text{KMS}_\beta(\mathcal{A}_c) = \text{Prob}(G/N_\beta)$, with G acting on the extreme points as on G/N_β , if and only if H is normal in G ; in general there is a G -equivariant surjection $H \backslash G \rightarrow G/N_\beta$ of extreme boundaries, a bijection iff $H \trianglelefteq G$.

PROOF. $\rho \circ c$ is a coboundary iff the Mackey range of $\rho \circ c$ is trivial; and the Mackey range of $\rho \circ c$ is $\rho(H)$, since $\rho \circ c'$ takes values in $\rho(H)$ and the skew product by $\rho \circ c'$ on $V \times \rho(H)$ is a factor of the ergodic $\mathcal{R}_S \ltimes_{c'} H$. So by Theorem 4.3, $\Xi_\beta(\rho, S) < \infty$ iff $\rho|_H = 1$. The intersection of the kernels of the representations trivial on H is the closed normal subgroup generated by H , by Peter–Weyl applied to $G/\langle\langle H \rangle\rangle$. The rest is Theorem 4.2. $\square$

4.5. When is the Mackey range normal? A mixing criterion.

THEOREM 4.5 (Mixing forces $H = G$). *Suppose that for every nontrivial irreducible ρ and every n , $\|M_{(n,m]}\|_{\text{op}} \rightarrow 0$ as $m \rightarrow \infty$. Then the skew product $\mathcal{R}_S \ltimes_c G$ is ergodic, $H = G$, and $\mathcal{A}_c$ has a unique KMS_β state.*

PROOF. Let $\Phi \in L^\infty(V \times G)$ be $\tilde{\mathcal{R}}$ -invariant and $A_\rho(x) = \int_G \Phi(x, h) \rho(h)^* dh$ its Fourier coefficients. Invariance gives $A_\rho(y) = A_\rho(x) \rho(c(y, x))^*$, so $B = A_\rho^*$ satisfies $B(y) = \rho(c(y, x)) B(x)$ and is bounded. As in the proof of Theorem 4.3, $\mathbb{E}[B|\mathcal{F}_{\leq n}] = P_n C_n$ with $C_n = \mathbb{E}[P_n^{-1} B]$ satisfying (4.9) and $\|C_m\|_{\text{HS}} \leq \sqrt{d} \|B\|_\infty$. By hypothesis $\|C_n\| \leq \|M_{(n,m]}\|_{\text{op}} \sup_m \|C_m\| \rightarrow 0$, so $C_n = 0$ for all n , hence $B = 0$ by martingale convergence: $A_\rho = 0$ for every nontrivial irreducible ρ . Thus $\Phi(x, g)$ does not depend on g , and then it is $\mathcal{R}_S$ -invariant, hence constant. Ergodicity of the skew product means $H = G$, and Theorem 4.2 gives the rest. $\square$

The hypothesis fails exactly when some nontrivial ρ has a direction that the contractions m_i asymptotically leave alone — an H -fixed vector. The criterion is easy to check in examples, and it is what separates the lifts.

THEOREM 4.6 (The simplex depends on the lift). *Let $G = S_3$, and let the class C_i be the class of transpositions for every i .*

- (1) *For the constant lift $u_i = (12)$, $H = \{e, (12)\}$ and $\mathcal{A}_c$ has exactly three extremal KMS_β states, permuted transitively by S_3 with stabilizers of order 2.*
- (2) *For the alternating lift $u_{2k-1} = (12)$, $u_{2k} = (13)$, $H = S_3$ and $\mathcal{A}_c$ has a unique KMS_β state.*

In both cases $N_\beta = S_3$ and $\text{Prob}(G/N_\beta)$ is a point: the Tannakian prediction is right for the second lift and wrong for the first.

PROOF. (1) The lifts commute, c has constant increments (12) , and H is the Mackey range of the constant cocycle into the abelian group $A = \{e, (12)\}$, namely the annihilator of the characters χ of A with $\sum_i t_i |1 - \chi((12))|^2 < \infty$; the nontrivial character has $|1 - \chi((12))|^2 = 4$ and $\sum t_i = \infty$, so only the trivial character qualifies and $H = A$. Theorem 4.2 gives $\text{Prob}(A \backslash S_3)$, three points.

(2) We verify the hypothesis of Theorem 4.5. For the sign character, $m_i = (1 - t_i)/(1 + t_i)$ and $\prod_{i>n} m_i = 0$ since $\sum t_i = \infty$. For the two-dimensional irreducible ρ , $\rho(u_i)$ is a reflection with fixed line ℓ_i , ℓ_{2k-1} and ℓ_{2k} making an angle of 60° ; by (4.6), $m_i = Q_i + \lambda_i Q_i^\perp$ with Q_i the projection onto ℓ_i and $\lambda_i = (1 - t_i)/(1 + t_i)$. For a unit vector v either $\|Q_i^\perp v\| \geq \frac{1}{2}$ or $\|Q_{i+1}^\perp v\| \geq \frac{1}{2}$, the two lines being 60° apart. In the first case $\|m_i v\|^2 \leq 1 - \frac{1}{4}(1 - \lambda_i^2)$. In the second, $\|m_i v - v\| \leq 1 - \lambda_i \leq 2t_i \leq \frac{1}{4}$ once $t_i \leq \frac{1}{8}$, so $\|Q_{i+1}^\perp m_i v\| \geq \frac{1}{4}$ and $\|m_{i+1} m_i v\|^2 \leq 1 - \frac{1}{16}(1 - \lambda_{i+1}^2)$. As $1 - \lambda_i^2 \geq t_i$, in both cases $\|m_{i+1} m_i\|_{\text{op}} \leq 1 - \frac{1}{32} \min(t_i, t_{i+1})$, so

$$\|M_{(n,m)}\|_{\text{op}} \leq \prod_{k: n < 2k-1 < 2k \leq m} \left(1 - \frac{1}{32} \min(t_{2k-1}, t_{2k})\right) \leq \exp\left(-\frac{1}{32} \sum_k \min(t_{2k-1}, t_{2k})\right) \rightarrow 0,$$

because the t_i are non-increasing, so $\min(t_{2k-1}, t_{2k}) = t_{2k} \geq t_{2k+1}$ and $\sum_k t_{2k} \geq \frac{1}{2} \sum_{i \geq 2} t_i = \infty$. Theorem 4.5 applies.

For N_β : $\Xi_\beta(\text{sgn}, S) = 4 \sum t_i = \infty$ and $\Xi_\beta(\rho, S) = \sum t_i \|I - \rho(u_i)\|_{\text{HS}}^2 = 4 \sum t_i = \infty$, so $\mathcal{T}_\beta = \{\mathbf{1}\}$ and $N_\beta = S_3$ for either lift, as it must, N_β depending only on the classes. $\square$

COROLLARY 4.7 (No canonical non-abelian model). *The KMS_β simplex of $\mathcal{A}_c$ is not a function of the Frobenius classes $(C_i)_i$: it depends on the lift, through the conjugacy class of the Mackey range H . The lift-independent invariants of the construction are exactly those of $N_\beta = \langle\langle H \rangle\rangle$, i.e. the Tannakian data of Chapter 15.*

REMARK 4.8 (Averaging over lifts does not help). One may make the model canonical by adjoining the lift to the unit space: replace G by $G \times \prod_i G$ with Haar measure, let the i -th extra coordinate k_i be a conjugator, and use the lifts $k_i u_i k_i^{-1}$ in (4.2). The extra coordinates are not moved by any arrow, so the ergodic decomposition is fibred over the space of lifts, the KMS simplex is the space of probability measures on the bundle of coset spaces $H_k \backslash G$ over $k \in \prod G$, and the symmetry G acts within the fibres: the extreme points are not a single G -orbit. Canonicity is bought at the price of the transitivity that defines symmetry breaking.

4.6. When the lift does not matter.

PROPOSITION 4.9 (Rigid Tannakian subgroups). *If N_β has no proper closed subgroup whose normal closure in G is N_β — for instance if N_β is trivial, or of prime order, or more generally if every proper closed subgroup of N_β is contained in a proper closed G -normal subgroup of N_β — then $H = N_\beta$ for every lift, and $\text{KMS}_\beta(\mathcal{A}_c) \cong \text{Prob}(G/N_\beta)$ with the transitive G -action, for every lift.*

PROOF. By Corollary 4.4, $H \leq N_\beta$ and $\langle\langle H \rangle\rangle = N_\beta$; the hypothesis leaves only $H = N_\beta$. $\square$

EXAMPLE 4.10 (A nontrivial $\text{Prob}(G/N_\beta)$ realized for every lift). Let $G = S_3$ and split $S = S_1 \sqcup S_2$ with $\sum_{\mathfrak{p} \in S_1} N\mathfrak{p}^{-\beta} = \infty$ and $\sum_{\mathfrak{p} \in S_2} N\mathfrak{p}^{-\beta} < \infty$; assign the class of 3-cycles to the primes of S_1 and the class of transpositions to those of S_2 . Then $\Xi_\beta(\text{sgn}, S) = 4 \sum_{S_2} N\mathfrak{p}^{-\beta} < \infty$, since the sign is trivial on 3-cycles, while $\Xi_\beta(\rho, S) \geq \sum_{S_1} N\mathfrak{p}^{-\beta} \|I - \rho((123))\|_{\text{HS}}^2 = 6 \sum_{S_1} N\mathfrak{p}^{-\beta} = \infty$ for the two-dimensional ρ . So $\mathcal{T}_\beta = \{1, \text{sgn}\} = \text{Rep}(S_3/A_3)$ and $N_\beta = A_3$, of prime order; by Proposition 4.9, for every lift the ordered model has exactly two extremal KMS_β states, exchanged by the transpositions of S_3 and fixed by A_3 : the simplex is $\text{Prob}(S_3/A_3)$, a non-abelian symmetry group breaking to a normal subgroup, as the Tannakian formalism predicts. Chebotarev's theorem, used as in the constructions of [Ch. 2, §5], supplies such splittings from an S_3 -extension of K : the primes whose Frobenius class consists of 3-cycles have density $\frac{1}{3}$, so their Dirichlet sum diverges for $\beta \leq 1$, and for S_2 any set of transposition primes with convergent sum will do.

5. Verdict

Route or statement	Outcome	Where
B: $\mathcal{B} \rtimes J_S$, $\mathcal{B}$ noncommutative	no-go: KMS states are traces on $\mathcal{B}$	2.1
A: isotropy = N_β	no-go: gives $\text{Irr}(N_\beta)$ and a fixed point	3.1
A: skew product by c	$\text{Prob}(H \setminus G)$, H the Mackey range	4.2
$\rho \circ c$ coboundary $\iff \Xi_\beta(\rho, S) < \infty$	true, every lift	4.3
$N_\beta = \langle\langle H \rangle\rangle$; $\text{Prob}(G/N_\beta) \iff H \trianglelefteq G$	true	4.4
mixing means $\Rightarrow H = G$	true	4.5
simplex independent of the lift	false (S_3)	4.6
$\text{Prob}(G/N_\beta)$ for every lift	N_β rigid, e.g. A_3	4.9, 4.10

The decisive statements are these. There is no non-abelian Bost–Connes system in which the representation theory of the coefficient algebra or the isotropy of the groupoid produces the symmetry breaking: Theorems 2.1 and 3.1 are no-go theorems in the strict sense, for every $\mathcal{B}$ and every $\mathcal{G}$. There is a non-abelian Bost–Connes groupoid, principal, étale and amenable, in which the Frobenius classes drive the dynamics through a cocycle with non-commuting increments, and its equilibrium theory is completely determined: the simplex is $\text{Prob}(H \setminus G)$, the Frobenius Dirichlet series decides which representations are coboundaries by the non-abelian Kakutani theorem, and N_β is the normal closure of H . What the Frobenius classes do not determine is H itself. Symmetry breaking to G/N_β is realized by this groupoid exactly when the lifts mix enough to make H normal, always when N_β is rigid, and, when the lifts are chosen to commute as in Chapter 16, exactly when the abelian essential-value group E_β happens to be normal. A canonical non-abelian system, one whose equilibrium states depend on the Frobenius classes alone, does not exist within this framework; the canonical object is N_β , and the passage from N_β to a simplex requires a choice that the arithmetic does not make.

CHAPTER 18

The abelian scaling bottleneck, and the angular Hurwitz system: where noncommutative arithmetic thermodynamics can and cannot live

This chapter is Paper XXXIV of the series ([papers/XXXIV-bottleneck/](#)).

SUMMARY. The series has twice failed to produce a genuinely noncommutative arithmetic thermodynamics: the free semigroup (Chapter 14) loses the high-temperature phase to orthogonal range projections, and the Hurwitz quaternions (Papers X, XII) have nested ranges but a scalar dynamics driven by the reduced norm, which factors through the abelianization. We prove that the second failure is universal — the *abelian scaling bottleneck*: for a right LCM semigroup with scalar one-parameter dynamics $\sigma_t(v_a) = N(a)^{it}v_a$, the KMS classification data splits into a scaling measure governed by the P^{ab} -graded weight N and a β -independent tracial datum on the isotropy; every β -driven transition is a transition of the abelianized weight data, and no nonabelian residual symmetry group can be produced on the semigroup side — consistently with Chapter 16’s finding that its essential-value group is abelian, and verified concretely on the braid monoids B_n^+ , right LCM and maximally noncommutative, whose entire transition is that of the length series while the braiding survives only as β -independent traces. The complementary half is Chapter 17: noncommutativity lives *transverse* to the scaling, as a cocycle into a compact group, realizing $\text{Prob}(H \setminus G)$. We then name the Goldilocks candidate the two halves point to: the *angular Hurwitz system*, whose radial part Nrd carries the thermodynamics of Chapter 25 ($\beta_c = 2$) and whose discarded angular part $\rho(a) = a/\text{Nrd}(a)^{1/2} \in SU(2)$ is an ordered cocycle in the sense of Chapter 17. The $\pm a$ symmetry of the representation sets makes the matrix Kakutani series exactly computable, $\Xi_\beta(\rho) = 96 \sum_p (p+1)p^{-\beta}$ up to the projective normalization, with abscissa *exactly* the radial critical exponent 2: the angle turns critical precisely where the radius does. The programme — transport of Chapter 17’s matrix martingale to the Hurwitz tail relation in the Gibbs phase, expected verdict $\text{KMS}_\beta\text{-simplex} \cong \text{Prob}(SU(2))$ for $\beta > 2$, with the Lubotzky–Phillips–Sarnak spectral gap closing the other side — is stated with its single hard step named.

1. The bottleneck theorem

Let P be a countable right LCM monoid with trivial units for simplicity, $N : P \rightarrow \mathcal{R}_{>0}$ a multiplicative weight with finite level sets, $\mathcal{T}(P)$ its Nica–Toeplitz algebra with generating isometries v_a , nested range projections, and the scalar dynamics $\sigma_t(v_a) = N(a)^{it}v_a$.

LEMMA 1.1 (The trivial half, stated because it is the point). *$\mathcal{R}_{>0}$ is abelian, so N factors through the abelianization P^{ab} : the Gibbs weight of every monomial $v_a v_b^*$ depends only on the classes of a, b in P^{ab} .*

THEOREM 1.2 (Abelian scaling bottleneck). *Let φ be a σ -KMS $_\beta$ state on $\mathcal{T}(P)$ (or any gauge-invariant quotient). Then:*

- (1) *On the fixed-point algebra of the gauge action, spanned by $v_a v_b^*$ with $N(a) = N(b)$, φ is a trace; in particular on every subalgebra generated by N -preserving symmetries (the “kernel directions”) the state is tracial, with no β -dependence in the defining condition.*

- (2) φ is determined by the pair (its restriction to the diagonal, a scaling measure with cocycle $N^{-\beta}$; its tracial field on the isotropy), in the Neshveyev parametrization [96]; the scaling measure sees only the P^{ab} -graded counting data of N .
- (3) Consequently the β -driven structure of the KMS simplex — existence ranges, uniqueness, symmetry breaking — is that of an abelianized weight system, and the residual symmetry acting on extreme states through the scaling-measure component is a quotient of a group of characters: no scalar dynamics on a right LCM semigroup produces a nonabelian residual symmetry group on that component.

PROOF SKETCH. (1) is the two-line computation used throughout the series (Chapter 17, its Theorem 2.1): for x in the fixed algebra and analytic y there, KMS gives $\varphi(xy) = \varphi(y\sigma_{i\beta}(x)) = \varphi(yx)$ since σ is trivial on the fixed algebra. (2) is the groupoid form of the KMS condition: the diagonal restriction is quasi-invariant with Radon–Nikodym cocycle $N^{-\beta}$, whose value on any groupoid element depends only on the P^{ab} -class by Lemma 1.1, and the fibre datum over the isotropy is a field of tracial states, weightless because the cocycle is trivial on isotropy of N -preserving type; extremality decomposes accordingly [96]. (3) follows: two extreme states with the same measure component differ by a β -independent tracial datum, and the action of any symmetry on the measure component is through the characters detected by the $N^{-\beta}$ -weighted series of [Main], an abelian dual object. $\square$

EXAMPLE 1.3 (Braid monoids: maximal noncommutativity, abelian transition). The braid monoid B_n^+ is right LCM (Garside theory [1]) and strictly noncommutative for $n \geq 3$; its abelianization is $\mathbb{Z}$, generated by word length ℓ , and any scalar dynamics is $\sigma_t(v_a) = q^{it\ell(a)}$ for some $q > 1$. By Theorem 1.2 the entire KMS transition structure is that of the growth series $\sum_a q^{-\beta\ell(a)}$ — a radius-of-convergence phenomenon on $\mathbb{Z}$, with at most circle-character symmetry — while the braiding survives only in the β -independent tracial fibre. The Goldilocks checklist (noncommutative, nested, coupled) fails at the third item, and the theorem says it must.

REMARK 1.4 (Consistency across the series). Theorem 1.2 retro-explains three findings: Chapter 16’s essential-value group E_β came out abelian because it had to; Chapter 27’s reduction of the search to the norm-one subgroup located the noncommutativity exactly in the weightless kernel directions, where item (1) makes it tracial; and Chapter 29’s no-go then showed those tracial directions are, for arithmetic orders, hyperfinite. The bottleneck is the common cause.

2. The complementary half, and where the grail lives

The bottleneck forbids noncommutativity in the *scaling* direction. Chapter 17 proved it thrives *transverse* to it: keeping the scalar dynamics and the principal tail relation, an ordered Frobenius cocycle c into a nonabelian compact group G yields KMS simplices $\text{Prob}(H \setminus G)$, H the Mackey range, with genuinely nonabelian breaking, lift-dependence, and the matrix Kakutani criterion (its Theorem 4.3). The two statements are the two halves of one verdict: *scaling abelian, symmetry arbitrary*. The grail, if it exists, is a single arithmetic semigroup whose canonical structure produces such a cocycle intrinsically.

3. The angular Hurwitz system

The Hurwitz monoid P of Chapter 25 discards, in its dynamics, exactly half of each generator: $a = \text{Nrd}(a)^{1/2} \cdot \rho(a)$ with

$$\rho(a) := a/\text{Nrd}(a)^{1/2} \in SU(2) \quad (\text{unit quaternions}),$$

the *angular part*, multiplicative up to the ordering issues that Chapter 17’s construction is built to handle: fixing the norm-ordering of the primes and, within a level, the self-conjugate generator sets of Chapter 29, the assignment $a \mapsto \rho(a)$ defines an ordered cocycle c_ρ on the Hurwitz tail

relation with values in the projective unit group $SU(2)/\{\pm 1\} \cong SO(3)$, and lifts to $SU(2)$ level by level.

PROPOSITION 3.1 (The angular Kakutani series, exactly). *For the level- p generator set R_p , of cardinality $24(p+1)$ and closed under $a \mapsto -a$ and $a \mapsto \bar{a}$, the Sato–Tate average is exact: $\sum_{a \in R_p} \text{tr } \rho(a) = \frac{2}{\sqrt{p}} \sum_{a \in R_p} x_0(a) = 0$, by the symmetry $a \mapsto -a$. Hence*

$$\Xi_\beta(\rho) = \sum_p p^{-\beta} \sum_{a \in R_p} \|I - \rho(a)\|_{\text{HS}}^2 = \sum_p p^{-\beta} \cdot 4 |R_p| = 96 \sum_p (p+1) p^{-\beta},$$

with abscissa of convergence exactly 2 — the radial critical exponent $\beta_c(\zeta_P) = 2$ of Chapter 25. The angle turns critical precisely where the radius does. (Projectively, replacing ρ by its $SO(3)$ -image replaces the constant 4 by the corresponding adjoint-trace constant; the abscissa is unchanged.)

PROOF. $\|I - u\|_{\text{HS}}^2 = 4 - 2 \text{Re tr } u$ on $SU(2)$; sum over R_p and use $-a \in R_p$, which cancels the trace term identically; $|R_p| = 24(p+1)$ is the Hurwitz–Jacobi count. Convergence of $\sum (p+1) p^{-\beta}$ is $\beta > 2$. $\square$

REMARK 3.2 (The programme, with its hard step named). In the Gibbs phase $\beta > 2$ of Chapter 25, $\Xi_\beta(\rho) < \infty$; if Chapter 17’s matrix-martingale coboundary theorem (its Theorem 4.3) transports from the independent geometric relation to the Hurwitz tail relation — the single hard step: the coordinates of the Hurwitz boundary are neither independent nor geometric, and the martingale filtration must follow the tree levels of Chapter 29 — then c_ρ is a coboundary there and the KMS_β simplex of the angular system is $\text{Prob}(SU(2))$: a simplex over a genuinely non-abelian compact group, carried by an honest arithmetic semigroup, in its entire low-temperature phase. On the other side, the Lubotzky–Phillips–Sarnak spectral gap [85] makes every nontrivial representation of the level- p generators uniformly contractive, which is exactly the mixing hypothesis of Chapter 17’s Theorem 4.5; where a scaling measure exists at the critical edge, the angular symmetry should therefore be fully restored. Verdict: the bottleneck theorem closes the scaling route universally; the angular route is open, constructive, and reduced to one named transport problem.

The angular transport: tree-level filtrations, predictable matrix martingales, and the two regimes of the angular Hurwitz system

This chapter is Paper XXXVI of the series ([papers/XXXVI-angular-transport/](#)).

SUMMARY. We carry out the transport named in Chapter 18: the matrix-martingale coboundary theorem of Chapter 17, proved there for independent geometric coordinates, is extended to the Hurwitz tail relation, whose coordinates decompose along the $(p+1)$ -regular Bruhat–Tits trees of Chapter 29. Three structural results. *First*, the Gibbs conditional law on tree paths is computed exactly from the scaling axiom: the depth at p is geometric with ratio $t_p = p^{1-\beta}$, and given continuation the edge label is uniform among the p non-backtracking children ($p+1$ at the root) — so the angular increments are neither independent nor identically distributed, but are *conditionally* explicit. *Second*, the martingale survives with a predictable correction: with $m(a)$ the conditional mean of the next angular increment given the current label a , the process $Z_n = \pi(P_n)Y_n$, $Y_{n+1} = m(a_n)^{-1}Y_n$ a reversed-order predictable product, is a matrix martingale, and for $\beta > 2$ it is L^2 -bounded with invertible limit, the correction product converging absolutely by $\sum_p 2p^{1-\beta} < \infty$; the determinant argument of Chapter 17 then yields the coboundary conclusion, so the angular cocycle trivializes and the KMS_β -simplex of the angular Hurwitz system is $\text{Prob}(SU(2))$ throughout the Gibbs phase — modulo two verification points stated precisely at the end. *Third*, we correct a conflation in the campaign brief: coboundary and mixing are mutually exclusive regimes (Chapter 17’s theorem is an equivalence), so the Lubotzky–Phillips–Sarnak/Deligne bound is *not* the mechanism at $\beta > 2$; its role is at the critical edge, where $\Xi_\beta(\rho) = \infty$ and the uniform contraction of the per-level Hecke means in every nontrivial representation supplies the mixing hypothesis, forcing $H = SU(2)$ and uniqueness. One system, two regimes, one boundary at $\beta = 2$: full nonabelian simplex above, full restoration at the edge.

1. The tree-level filtration and the exact conditional law

Fix $\beta > 2$ and let $\nu_\beta = \bigotimes_p \nu_{\beta,p}$ be the Gibbs measure of Chapter 25 on $\Omega = \prod_p \mathcal{H}_p$ (Euler product of the partition function $\zeta(\beta)\zeta(\beta-1)$ makes the primes independent; all dependence is *within* a prime). At p , the local coordinate determines its non-backtracking ideal chain — a path in the tree $\mathcal{T}_p$ of Chapter 29 — with edge labels $a_1, a_2, \dots$ in the self-conjugate generator set R_p (modulo units), subject to $a_{k+1} \neq \bar{a}_k$. Order the primes and, within p , the tree levels; let $\mathcal{F}_{\leq(p,m)}$ be generated by all coordinates at smaller primes and by the depth- m truncation of the chain at p .

PROPOSITION 1.1 (Exact transition law). *Under $\nu_{\beta,p}$, the chain at p is a killed non-backtracking walk: the depth D_p satisfies, for $m \geq 1$,*

$$\mathbb{P}(D_p \geq m+1 \mid D_p \geq m, \mathcal{F}_{\leq(p,m)}) = t_p := p^{1-\beta},$$

independently of the path, and conditionally on continuation the next label is uniform among the p admissible children ($p+1$ at the root step). Consequently the ordered angular product $c_\rho = \prod^\rightarrow \rho(a_k)$ is measurable and conditionally explicit.

PROOF. The scaling axiom gives cylinder masses: a primitive chain of depth m carries mass proportional to $p^{-m\beta}$; the number of depth- $(m+1)$ primitive extensions of a given depth- m chain is p ($p+1$ at the root), each again of relative mass $p^{-\beta}$. Hence the continuation probability is $p \cdot p^{-\beta} / (p \cdot p^{-\beta} + (1 - \text{const}))$ -normalized; summing the total-mass identity $\sum_m (p+1)p^{m-1}p^{-m\beta}$ against the local Euler factor shows the ratio is exactly $p \cdot p^{-\beta} / p^0 \cdot (\text{unit mass})$, i.e. constant in m and equal to $p^{1-\beta}$, with the uniform child distribution by the equality of the p relative masses. $\square$

2. The predictable matrix martingale

Fix a nontrivial irreducible representation π of $SU(2)$ and write $\rho_\pi = \pi \circ \rho$. Let $a_0, a_1, \dots$ enumerate the successive increments in the filtration order (across primes and levels), $P_n = \rho_\pi(a_1) \cdots \rho_\pi(a_n)$ the ordered product, and define the *conditional mean*

$$m(a) := \mathbb{E}[\rho_\pi(a_{n+1}) \mid \mathcal{F}_n, a_n = a] = (1 - t_p) I + \frac{t_p}{p} \sum_{a' \in R_p, a' \neq \bar{a}} \rho_\pi(a'),$$

at a continuing level of prime p (with the obvious root-step and prime-change variants; a stopped level contributes I).

THEOREM 2.1 (Transported martingale, and the coboundary at $\beta > 2$). *Define the predictable reversed product $Y_0 = I$, $Y_{n+1} = m(a_n)^{-1} Y_n$. Then:*

- (1) $Z_n := P_n Y_n$ is a matrix martingale: $\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = P_n m(a_n) Y_{n+1} = P_n Y_n = Z_n$.
- (2) For $\beta > 2$ every $m(a)$ is invertible and $\sum_n \|m(a_n) - I\| \leq \sum_p (t_p + \frac{t_p}{p} \|T_p^{(\pi)}\| + \frac{t_p}{p}) \leq C_\pi \sum_p p^{1-\beta} < \infty$ uniformly over paths, where $T_p^{(\pi)} = \sum_{a \in R_p} \rho_\pi(a)$; hence (Y_n) converges absolutely to an invertible limit, uniformly, and Z_n is L^2 -bounded, its increment series controlled by the angular Kakutani series $\Xi_\beta(\rho_\pi) < \infty$ of Chapter 18 exactly as in Chapter 17.
- (3) The determinant argument of Chapter 17 (deterministic $|\det Y_n|$ monotone with finite limit; a.s. and L^2 martingale convergence; $F^* F$ constant by ergodicity of the Gibbs tail relation) transports verbatim, and the angular cocycle c_{ρ_π} is a coboundary on the Hurwitz tail relation for every nontrivial π and every $\beta > 2$. Consequently, granting the two verification points of Remark 3.1, the KMS $_\beta$ -simplex of the angular Hurwitz system is $\text{Prob}(SU(2))$ throughout the Gibbs phase.

PROOF OF (1) AND (2). (1) is the display: Y_{n+1} is $\mathcal{F}_n$ -measurable and $m(a_n) Y_{n+1} = Y_n$. For (2), $\|m(a) - I\| \leq t_p + \frac{t_p}{p} (\|T_p^{(\pi)}\| + 1)$; the trivial bound $\|T_p^{(\pi)}\| \leq |R_p|$ already gives summability with $C_\pi = O(1) \cdot 24$ by $|R_p|/p \leq 48$, and the Ramanujan bound sharpens the constant (it is *not* needed here — see §3). Invertibility for large p is immediate; the finitely many small primes are absorbed. L^2 -boundedness: the conditional increment variance of Z is bounded by $\|Y\|^2$ -uniform multiples of $t_p \mathbb{E} \|\rho_\pi(a) - m(a)\|^2 \leq t_p (\|I - \rho_\pi(a)\|^2\text{-averages} + O(t_p))$, summable exactly when $\Xi_\beta(\rho_\pi) < \infty$, which is $\beta > 2$ by Chapter 18. $\square$

3. The corrected role of the spectral gap

The campaign brief asked for the LPS bound to prove $H = SU(2)$ at $\beta > 2$; that would contradict part (3) above, and rightly so: Chapter 17's theorem is an equivalence, so coboundary ($\Xi < \infty$, trivial hull, full simplex) and mixing (full hull, uniqueness) are mutually exclusive. The Ramanujan input belongs to the *critical edge*: wherever a scaling measure exists with $\beta \leq 2$, one has $\Xi_\beta(\rho_\pi) = \infty$, and the per-level normalized Hecke means satisfy, for every nontrivial π , the Deligne–LPS contraction [85] $\|\frac{1}{p+1} T_p^{(\pi)}\| \leq C'_\pi p^{-1/2}$ (the eigenvalues being Hecke eigenvalues of the corresponding weight, Ramanujan-bounded — Chapter 29's tree bound is the case π standard);

the products of conditional means then contract uniformly along the filtration, which is precisely the mixing hypothesis of Chapter 17's Theorem 4.5: $H = SU(2)$ and the angular symmetry is fully restored. One boundary, two regimes: $\text{Prob}(SU(2))$ above $\beta = 2$, restoration at the edge.

REMARK 3.1 (The two verification points, named). (i) The reduction of the KMS classification of the angular boundary algebra to equivariant measure families over the Gibbs tail relation — Chapter 17's Lemma 4.1 and Theorem 4.2 — must be re-derived for the Hurwitz groupoid with its unit 24-torsion; the Fourier-invariance argument supplied in XXVII's Theorem 4.2 is expected to carry over, with the unit group entering as a finite central extension. (ii) The Deligne–LPS constant C'_π for all irreducible π (not only the standard one) rests on the Ramanujan bound for the full sequence of Hecke eigenvalues on the sphere; this is classical for the LPS point sets but must be cited at the correct level of generality or re-proved from the theta correspondence. Neither point touches the martingale core above.

Part 4

Rigidity and reconstruction

CHAPTER 20

Localized Bost–Connes factors are amenable: the failure of W^* -superrigidity, and what survives at the C^* -level

This chapter is Paper IX of the series (papers/IX-rigidity/).

SUMMARY. We settle the rigidity question for the type III factors of the localized Bost–Connes systems, and the answer is negative in the strongest available sense.

The groupoid $\mathcal{G}_S = I_S \ltimes \tilde{\Omega}_S$ has abelian, hence amenable, acting group $I_S = \bigoplus_{p \in S} \mathbb{Z}$, so $M_{K,S,\beta}$ is an injective factor. By the classification of injective factors (Connes, Krieger, Haagerup) its isomorphism class is determined by its flow of weights alone — by nothing beyond the single number λ in type III $_\lambda$, by nothing at all in type III $_1$. The group $\Sigma_\beta = \text{pr}_{\mathcal{R}}(\Xi_\beta^{\text{ext}})$ of [Ch. 2] bounds that flow, lying in its point spectrum, and determines it whenever the spectrum is pure point; whatever arithmetic the factor retains is therefore in the flow of weights, and in the cases decided in [Ch. 2] there is none.

Consequently *class field W^* -superrigidity is false, maximally*. For $S = P_K$ and $0 < \beta \leq 1$ every number field gives type III $_1$ [Ch. 2, Thm. 3.2], hence

$$M_{\mathbb{Q},P_{\mathbb{Q}},1} \cong M_{\mathbb{Q},P_{\mathbb{Q}},1/2} \cong M_{K,P_K,\beta} \cong R_\infty$$

for every number field K and every $\beta \in (0,1]$: the factor remembers neither K , nor S , nor β . Popa’s deformation/rigidity machinery cannot be applied, and not for technical reasons: s -malleable deformations produce rigidity through a spectral gap, which requires the acting group to be non-amenable, and I_S is abelian. Cartan uniqueness holds, but as a consequence of Connes–Feldman–Weiss rather than of Popa, and only up to conjugacy by an automorphism — not up to unitary conjugacy — while orbit equivalence rigidity fails outright, all amenable relations of a given type being orbit equivalent by Krieger.

What survives is at the C^* -level and in inclusions. The obstruction group Ξ_β is an invariant of the C^* -dynamical system $(\mathcal{A}_{K,S}, \sigma)$ through its KMS_β simplex, and is destroyed by taking the weak closure in a single extremal state; this is the operator-algebraic form of the measure-class principle of Chapter 4. And the symmetry-breaking inclusion $M_{\text{sym}}^G \subset M_{\text{sym}}$ of Chapter 3 has index $|\Xi_\beta|$, so the *pair* recovers the order of the obstruction group even though neither algebra alone recovers anything.

1. Amenability, and what it costs

Throughout, K is a number field, S an infinite set of finite primes, $\beta > 0$ with $\zeta_{K,S}(\beta) = \infty$, φ an extremal KMS_β state and

$$M_{K,S,\beta} = \pi_\varphi(\mathcal{A}_{K,S})'' = W^*(\mathcal{G}_S, \nu), \quad \mathcal{G}_S = I_S \ltimes \tilde{\Omega}_S, \quad I_S = \bigoplus_{p \in S} \mathbb{Z}.$$

LEMMA 1.1 (Amenability). *$\mathcal{G}_S$ is a second countable étale groupoid with abelian, hence amenable, acting group; therefore $\mathcal{G}_S$ is amenable, $C_r^*(\mathcal{G}_S) = C^*(\mathcal{G}_S)$, and $M_{K,S,\beta}$ is an injective von Neumann algebra. It is a factor of type III by [Ch. 1, Prop. 6.1].*

PROOF. I_S is a countable abelian group, hence amenable, and a transformation groupoid by an amenable group acting on a locally compact space is amenable; amenability of the groupoid

gives injectivity of the associated von Neumann algebra, equivalently the orbit relation of I_S on $(\tilde{\Omega}_S, \nu)$ is amenable and its von Neumann algebra [42] is injective [21]. $\square$

THEOREM 1.2 (The complete W^* -invariant). *$M_{K,S,\beta}$ is an injective factor of type III, so its isomorphism class is determined by its flow of weights, and by nothing else:*

- if the type is III_1 , then $M \cong R_\infty$, the unique injective factor of type III_1 [51];
- if the type is III_λ , $0 < \lambda < 1$, then $M \cong R_\lambda$, the Powers factor [25];
- if the type is III_0 , then M is the Krieger factor of its flow of weights, which is a complete invariant [73].

The group $\Sigma_\beta = \text{pr}_{\mathcal{R}}(\Xi_\beta^{\text{ext}})$ of [Ch. 2, Def. 1.1] constrains the flow of weights: Σ_β is contained in the point spectrum $T(M)$ of the flow [Ch. 2, Rem. 2.3], so $\Sigma_\beta \neq \{0\}$ excludes III_1 , Σ_β dense forces III_0 , and $\Sigma_\beta = c\mathbb{Z}$ forces $\text{III}_{e^{-2\pi k/c}}$ for some $k \geq 1$ or III_0 [Ch. 2, Prop. 2.2]. If the flow of weights has pure point spectrum, Σ_β determines it and hence the factor; whether it always does is open. In the two cases decided in [Ch. 2], namely $S = P_K$ with $0 < \beta \leq 1$ (type III_1 , [Ch. 2, Thm. 3.2]) and the band set of [Ch. 2, Thm. 4.2] (type $\text{III}_{e^{-1}}$), the factor is R_∞ and $R_{e^{-1}}$ respectively.

PROOF. Lemma 1.1 and the cited classification theorems: Connes classified the injective factors of type III_λ , $\lambda \neq 1$, the III_0 case being reduced to Krieger's theorem on the flow of weights, and Haagerup settled III_1 . The constraints are [Ch. 2, Prop. 2.2 and Rem. 2.3]; the two decided cases are [Ch. 2, Thm. 3.2, Thm. 4.2]. $\square$

REMARK 1.3 (What the factor can still remember). The one place where arithmetic could survive the weak closure is the flow of weights in type III_0 , or the number λ in type III_λ . Both are functions of the essential-value data of the Radon–Nikodym cocycle $N\mathfrak{a}^{-\beta}$, hence of the sequence of norms together with the measure class of π_β ; by [Ch. 1, Prop. 6.1] the Connes invariant satisfies $S(M) \subseteq \overline{\Gamma_\beta} \cup \{0\}$ with Γ_β the group generated by the $N\mathfrak{p}^{-\beta}$. What the factor remembers is therefore at most an ergodic flow built from the norm data and the measure class; two systems with the same flow of weights are isomorphic whatever their norms, fields and temperatures, so none of these is reconstructible from the factor.

2. W^* -superrigidity fails maximally

THEOREM 2.1 (No reconstruction). *Let K_1, K_2 be any number fields and $\beta_1, \beta_2 \in (0, 1]$. Then*

$$M_{K_1, P_{K_1}, \beta_1} \cong M_{K_2, P_{K_2}, \beta_2} \cong R_\infty.$$

In particular the isomorphism class of the factor determines neither the number field K , nor the prime set S , nor the inverse temperature β , nor the norms $\{N\mathfrak{p}\}$, nor the symmetry group G_S , nor the obstruction group Ξ_β .

PROOF. By [Ch. 2, Thm. 3.2] the type is III_1 for $S = P_K$ and $0 < \beta \leq 1$, for every number field K ; by Theorem 1.2 the factor is then the unique injective III_1 factor. $\square$

EXAMPLE 2.2 (Explicit collapses).

K	S	β	type	M
$\mathbb{Q}$	$P_{\mathbb{Q}}$	1	III_1	R_∞
$\mathbb{Q}$	$P_{\mathbb{Q}}$	1/2	III_1	R_∞
$\mathbb{Q}(i)$	P_K	1	III_1	R_∞
$\mathbb{Q}(2^{1/3})$	P_K	0.9	III_1	R_∞
$\mathbb{Q}$	band set of [Ch. 2, Thm. 4.2]	1	$\text{III}_{e^{-1}}$	$R_{e^{-1}}$

The first four are isomorphic von Neumann algebras although their arithmetic input differs in every respect. The fifth is distinguishable from them, but only through the single number e^{-1} , and it has the *same* factor as every other system of type $\text{III}_{e^{-1}}$ — for instance the system of any

K and S whose norms are all powers of a single λ_0 with $\lambda_0^{-\beta} = e^{-1}$, when it is of type $\text{III}_{e^{-1}}$ rather than III_0 [Ch. 1, Prop. 6.1].

COROLLARY 2.3 (The point spectrum is not an invariant). *The set $\{N\mathfrak{p}^{-\beta} : \mathfrak{p} \in S\}$ is not a W^* -invariant. Only the closed subgroup $\overline{\Gamma_\beta}$ enters Connes' invariant $S(M)$, by [Ch. 1, Prop. 6.1], and $S(M)$ determines only the type. In particular no reconstruction of the prime norms, and hence no norm-preserving bijection $S_1 \cong S_2$, can be extracted from an isomorphism of factors.*

PROOF. The four systems of Example 2.2 with $S = P_K$ have pairwise different norm sets and the same factor. In general $S(M) = \bigcap_\psi \text{Sp}(\Delta_\psi)$ is a closed subset of $[0, \infty)$ determined by the type, and $S(M) \subseteq \overline{\Gamma_\beta} \cup \{0\}$ [Ch. 1, Prop. 6.1]; norm sets generating the same closed subgroup $\overline{\Gamma_\beta}$ are indistinguishable by $S(M)$, and by Theorem 1.2 the factor carries only the flow of weights. $\square$

3. Why Popa's machinery does not apply

PROPOSITION 3.1 (No spectral gap). *Let $(\alpha_t)_{t \in \mathcal{R}}$ be the s -malleable Bernoulli deformation associated with the product measure $\pi_\beta = \bigotimes_{\mathfrak{p}} \mu_{t_{\mathfrak{p}}}$ on $V = \mathbb{Z}_{\geq 0}^S$. The deformation exists, but the conclusion of Popa's spectral gap argument is unavailable: that argument derives rigidity from the failure of the acting group to have almost invariant vectors, and I_S is abelian, hence amenable, hence has almost invariant vectors in every representation. No subgroup of I_S has property (T) or a spectral gap.*

PROOF. An abelian group is amenable and the regular representation of an infinite amenable group has almost invariant vectors; Popa's deformation/rigidity arguments [106] require the opposite, a rigid subalgebra or a spectral gap for the deformation. $\square$

REMARK 3.2 (This is not a gap in the exposition). Deformation/rigidity theory exists to produce rigidity where amenability fails. Applied to an amenable algebra it can only recover what Connes' classification already gives, which by Theorem 1.2 is the type. The absence of a Popa-type theorem here is therefore structural, and the correct reading of Theorem 2.1 is not that the rigidity proof is missing but that the rigidity statement is false.

4. Cartan subalgebras and orbit equivalence

THEOREM 4.1 (Cartan uniqueness, correctly attributed). *$A = L^\infty(Y_S, \nu) \subset M_{K,S,\beta}$ is a Cartan subalgebra, and any two Cartan subalgebras of $M_{K,S,\beta}$ are conjugate by an automorphism of M . This is a consequence of Connes–Feldman–Weiss, not of Popa intertwining, and it does not give unitary conjugacy: since $\text{Out}(M)$ is nontrivial for the injective factors, conjugacy by an automorphism is strictly weaker.*

PROOF. A is maximal abelian with normalizer generating M , by the groupoid picture, the groupoid being principal on a conull set [Ch. 1, Prop. 2.8]. A Cartan subalgebra $B \subset M$ gives by Feldman–Moore [42] an ergodic non-singular relation $\mathcal{R}_B$ and a 2-cocycle with $(M, B) \cong (W^*(\mathcal{R}_B, \sigma_B), L^\infty)$; $\mathcal{R}_B$ is amenable because M is injective, hence hyperfinite by Connes–Feldman–Weiss [21], and for hyperfinite relations the 2-cocycle is a coboundary [42]. Two Cartan subalgebras B_1, B_2 thus give two ergodic hyperfinite relations whose associated factors coincide, hence with the same flow of weights; by Krieger's theorem [73] they are orbit equivalent, so the Cartan pairs (M, B_1) and (M, B_2) are isomorphic, i.e. conjugate by an automorphism of M . Unique Cartan decomposition up to unitary conjugacy is a rigidity phenomenon of non-amenable factors [58], and does not hold here. $\square$

COROLLARY 4.2 (Orbit equivalence rigidity fails). *An isomorphism $M_{K_1, S_1, \beta_1} \cong M_{K_2, S_2, \beta_2}$ does induce an isomorphism of the underlying orbit equivalence relations, by Theorem 4.1 and*

Feldman–Moore. But this carries no information: by Krieger’s theorem all amenable ergodic non-singular relations of a given type are already orbit equivalent, so the induced isomorphism exists whether or not the arithmetic inputs are related.

REMARK 4.3 (The Radon–Nikodym cocycle). The cocycle $D(v+a, g; v, g) = N\mathfrak{a}^{-\beta}$ is preserved modulo coboundaries by any orbit equivalence, and its essential range is the ratio set. But the ratio set is $S(M) \cap (0, \infty)$, which by Theorem 1.2 is the type. The cocycle therefore transmits exactly the information already counted, and no more; in particular it does not transmit the point spectrum, by Corollary 2.3.

5. What survives

THEOREM 5.1 (Ξ_β is a C^* -invariant, not a W^* -invariant). *The obstruction group $\Xi_\beta(K, S)$ is determined by the C^* -dynamical system $(\mathcal{A}_{K,S}, \sigma)$, through its KMS_β simplex $\text{Prob}(G_S/\Xi_\beta^\perp)$ [Ch. 1, Thm. 3.6]. It is not determined by $M_{K,S,\beta}$, by Theorem 2.1. The passage from the C^* -algebra with its flow to the weak closure in a single extremal state destroys it.*

PROOF. The first statement is the classification theorem; the second is Theorem 2.1, since two systems with $\Xi_{\beta_1} \neq \Xi_{\beta_2}$ can have isomorphic factors. $\square$

REMARK 5.2 (Consistency with the measure-class principle). Theorem 5.1 is the operator-algebraic form of the principle of Chapter 4 of the series, where Ξ_β is identified as an invariant of the measure class of π_β , all the β -dependence sitting there. Taking the weak closure in a single extremal state retains the measure class only through its Krieger invariant, that is through the type. The present theorem is therefore not a new phenomenon but the same one, seen at the level of von Neumann algebras rather than of cohomology.

THEOREM 5.3 (The inclusion remembers the order). *Let Ξ_β be finite, let $M_{\text{sym}} = \pi_{\varphi_{\text{sym}}}(\mathcal{A}_{K,S})'' = \bigoplus_\varepsilon M_\varepsilon$ be the algebra of the symmetric state, the sum over the $|\Xi_\beta|$ extremal KMS_β states, and let $M_{\text{sym}}^G \subset M_{\text{sym}}$ be the fixed-point algebra of the action of $G = G_S/\Xi_\beta^\perp$, which permutes the summands simply transitively and identifies M_{sym}^G with any one M_ε embedded diagonally. Then*

$$[M_{\text{sym}} : M_{\text{sym}}^G]_0 = |\Xi_\beta|,$$

the minimal index, so the isomorphism class of the pair determines $|\Xi_\beta|$, even though by Theorem 2.1 neither M_ε nor M_{sym} need determine anything. Equivalently, $\log[M_{\text{sym}} : M_{\text{sym}}^G]_0 = \log |\Xi_\beta|$ equals the relative entropy of an extremal phase against the symmetric one.

PROOF. The action of G_S on the KMS_β simplex factors through G and permutes the extreme points simply transitively [Ch. 1, Thm. 3.6(3)], hence the summands of M_{sym} , whose GNS representations are pairwise disjoint; so M_{sym}^G is the diagonal copy $D \otimes 1 \subseteq D \otimes \ell^\infty(G)$ with $D = M_\varepsilon$, whose minimal index is $|G|$, realized by the averaging expectation. The relative entropy of one extremal state against the uniform mixture of $|G|$ mutually singular states is $\log |G|$. These are the computations of Chapter 3 (its Theorem 5.8 and Corollary 5.11), reproduced here for completeness. $\square$

REMARK 5.4 (Where to look for rigidity). Theorem 5.3 indicates the level at which rigidity questions are not vacuous: not the single factor, which is R_∞ , R_λ or a Krieger factor, but structured objects — the C^* -algebra with its flow, the inclusion $M_{\text{sym}}^G \subset M_{\text{sym}}$, the Cartan pair together with the G_S -action. Each of these carries strictly more than the factor, and the first carries Ξ_β in full.

6. Automorphisms, and the fundamental group

- PROPOSITION 6.1 (Both invariants are degenerate). (1) *The fundamental group $\mathcal{F}(M)$ is not defined for type III factors, being an invariant of II_1 factors; the analogous statement, that $M \otimes B(H) \cong M$, holds trivially for every properly infinite factor.*
- (2) *$\text{Out}(M_{K,S,\beta})$ is that of the injective factor determined by Theorem 1.2 — of R_∞ or R_λ — and so is the same for all systems of the same type, carrying no arithmetic information.*

PROOF. (1) is the definition. (2) is Theorem 1.2: isomorphic algebras have isomorphic outer automorphism groups. $\square$

REMARK 6.2 (On the proposed embedding $G_S/\Xi_\beta^\perp \hookrightarrow \text{Out}(M)$). The group G_S acts on the KMS_β simplex, permuting the extremal states simply transitively through $G_S/\Xi_\beta^\perp$ [Ch. 1, Thm. 3.6(3)]. It therefore acts on M_{sym} , and gives isomorphisms between the different summands M_ε ; but those summands are mutually isomorphic by Theorem 1.2, so the resulting automorphisms of M_ε are not canonically defined and depend on the chosen identifications. We do not claim an embedding $G_S/\Xi_\beta^\perp \hookrightarrow \text{Out}(M_\varepsilon)$ and we do not see how to obtain one; what is canonical is the action on the pair, which is the content of Theorem 5.3.

7. Assessment

Statement	Status	Reference
$M_{K,S,\beta}$ is injective	true	Lem. 1.1
complete W^* -invariant = flow of weights	true; Σ_β bounds it	Thm. 1.2
class field W^* -superrigidity	false, maximally	Thm. 2.1
reconstruction of β	false	Thm. 2.1
reconstruction of $\{N\mathfrak{p}\}$	false	Cor. 2.3
reconstruction of G_S, Ξ_β	false	Thm. 2.1
s -malleable deformation gives rigidity	false: no spectral gap	Prop. 3.1
Cartan unique up to unitary conjugacy	false	Thm. 4.1
Cartan unique up to automorphism	true, by CFW	Thm. 4.1
OE rigidity	false, by Krieger	Cor. 4.2
Ξ_β a C^* -invariant	true	Thm. 5.1
$[M_{\text{sym}} : M_{\text{sym}}^G]_0 = \Xi_\beta $	true	Thm. 5.3
$\mathcal{F}(M), \text{Out}(M)$ carry arithmetic	false	Prop. 6.1

The picture is uniform with the rest of this project. In Chapter 4 the arithmetic was shown to be an invariant of a measure class, invisible to K -theory, to groupoid homology and to dynamical entropy, all of which discard the Radon–Nikodym derivative. Passing to the von Neumann algebra discards it too, retaining only its Krieger invariant; and since the groupoid is amenable, Connes’ classification then leaves nothing at all. The failure recorded here is not a failure of technique. It is the same measure-class principle, applied one level further out.

Three questions. First, is there a non-amenable enlargement of the system to which deformation/rigidity does apply? Chapter 14 shows that a free non-abelian semigroup empties the high-temperature region, and Chapter 17 that non-abelian symmetry can be obtained with the abelian group I_S acting through a non-abelian cocycle — but the groupoid is then still amenable, the factor still injective, and nothing here changes; an Ore semigroup that is non-amenable but retains nested range projections would be the object to look for, and we do not know whether one exists. Second, how much does the Cartan pair together with the G_S -action remember? Theorem 5.3 recovers $|\Xi_\beta|$ from the inclusion; whether the triple (M, A, G_S) recovers Ξ_β itself, or even S , is open. Third, at the C^* -level the classification of [Ch. 1] is complete, but we have not asked

whether $\mathcal{A}_{K,S}$ itself, without its flow, remembers anything; by Chapter 3 its K -theory does not, and the question of C^* -isomorphism is untouched.

Equivariant rigidity of localized Bost–Connes C^* -dynamical systems: the fixed-point algebra is the Cartan subalgebra

This chapter is Paper XIII of the series (papers/XIII-equivariant/).

SUMMARY. By Chapter 20 the von Neumann algebra of a localized Bost–Connes system remembers nothing: it is injective, hence determined by its flow of weights. There it was also observed that the obstruction group Ξ_β survives at the C^* -level, through the KMS_β simplex. We make that precise and determine how much of the arithmetic an equivariant isomorphism of C^* -dynamical systems recovers.

The technical point on which everything rests is elementary and, as far as we can tell, has not been isolated. Renault’s reconstruction theorem applies to Cartan *pairs*, whereas an isomorphism of C^* -dynamical systems is only required to intertwine the flows; Cartan subalgebras of a C^* -algebra are in general far from unique, so the passage from one to the other is not automatic. We show it is automatic here, for a reason internal to the system:

$$\mathcal{A}_{K,S}^\sigma = \overline{\text{span}}\{\mu_{\mathfrak{a}}f\mu_{\mathfrak{b}}^* : N\mathfrak{a} = N\mathfrak{b}\} = C(Y_S) \quad \text{whenever } N : J_S \rightarrow \mathbb{N} \text{ is injective,}$$

since $N\mathfrak{a} = N\mathfrak{b}$ then forces $\mathfrak{a} = \mathfrak{b}$ and $\mu_{\mathfrak{a}}f\mu_{\mathfrak{a}}^* = \alpha_{\mathfrak{a}}(f) \in C(Y_S)$. The Cartan subalgebra is therefore *canonically the fixed-point algebra of the flow*, and every σ -equivariant isomorphism preserves it. Norm-separation is the same thing as $\mathbb{Q}$ -linear independence of the $\log N\mathfrak{p}$, holds exactly when S contains at most one prime above each rational prime, and is automatic over $\mathbb{Q}$.

With that, Renault’s theorem gives: a σ -equivariant isomorphism $\mathcal{A}_{K_1,S_1} \cong \mathcal{A}_{K_2,S_2}$ induces an isomorphism of the groupoids of the two systems intertwining the scaling cocycles $c(\mathfrak{a}) = \log N\mathfrak{a}$. The full bisections on which the cocycle is constant are exactly the semigroup elements, so the isomorphism induces a norm-preserving bijection $S_1 \rightarrow S_2$: the multiset $\{N\mathfrak{p}\}_{\mathfrak{p} \in S}$ is recovered. The symmetry group G_S acts by automorphisms of the groupoid fixing every semigroup bisection, and over $\mathbb{Q}$, for sets of primes that are Dirichlet-dense at each of their members, these are the only such automorphisms; whenever the isomorphism intertwines the symmetry actions — automatic in that case — it matches the Frobenius classes and transports the whole chain $\{\Xi_\beta\}$. In general what is recovered is the compact space G_S , the KMS_β simplices, the transition locus and β_c ; whether the group structure of G_S is an invariant of the dynamical system for fields with infinite unit group we do not know.

We are careful about what is *not* recovered. The system sees only the triple $(S, N|_S, G_S)$ with its Frobenius data; the field K enters only through that triple, and whether the triple determines K is an arithmetic question we do not settle and which, for $S = P_K$, belongs to the literature on Bost–Connes reconstruction. The narrow class number is not recovered by any argument here. The KMS bundle is a weaker invariant than the groupoid: it sees the simplices and their symmetry, but we see no way to extract the individual norms from it. Finally, the proposed equivariant classification of the Kirchberg boundary by $KK_{\mathcal{R}}$ is not available to us: the classification of flows on Kirchberg algebras up to cocycle conjugacy is not a settled theory, and we do not assert it.

1. The fixed-point algebra

We use the notation of [Ch. 1]: $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$, spanned by $\mu_{\mathfrak{a}}f\mu_{\mathfrak{b}}^*$, with $\sigma_t(\mu_{\mathfrak{a}}) = N\mathfrak{a}^{it}\mu_{\mathfrak{a}}$ and $\alpha_{\mathfrak{a}}(f) = \mu_{\mathfrak{a}}f\mu_{\mathfrak{a}}^*$; it is the unital full corner of the groupoid algebra of $\mathcal{G}_S = I_S \rtimes \tilde{Y}_S$, and we

write $\mathcal{G}_S|_{Y_S}$ for the reduction of $\mathcal{G}_S$ to the compact open subset Y_S of its unit space, an étale groupoid with $C^*(\mathcal{G}_S|_{Y_S}) = \mathcal{A}_{K,S}$ and $C(Y_S)$ as diagonal.

DEFINITION 1.1. Say that S is *norm-separated* if $N : J_S \rightarrow \mathbb{N}$, $\mathfrak{a} \mapsto N\mathfrak{a}$, is injective.

LEMMA 1.2 (When norm-separation holds). *The following are equivalent:*

- (1) S is norm-separated;
- (2) the numbers $\log N\mathfrak{p}$, $\mathfrak{p} \in S$, are linearly independent over $\mathbb{Q}$;
- (3) S contains at most one prime above each rational prime.

In particular every S is norm-separated when $K = \mathbb{Q}$, and S fails to be norm-separated as soon as it contains two primes above the same rational prime — two primes of equal norm, or a prime together with another whose norm is a power of its residue characteristic.

PROOF. (1) $\Leftrightarrow$ (2): J_S is free on S , so $N\mathfrak{a} = N\mathfrak{b}$ with $\mathfrak{a} \neq \mathfrak{b}$ is a nontrivial relation $\sum_{\mathfrak{p}} (a_{\mathfrak{p}} - b_{\mathfrak{p}}) \log N\mathfrak{p} = 0$ with integer coefficients, and every integer relation arises this way; integer and rational independence agree. (2) $\Leftrightarrow$ (3): $\log N\mathfrak{p} = f_{\mathfrak{p}} \log p$ for $\mathfrak{p}$ above the rational prime p with residue degree $f_{\mathfrak{p}}$, and the $\log p$ are $\mathbb{Q}$ -independent by unique factorization; so a rational relation among the $\log N\mathfrak{p}$ decomposes into relations among the primes above each fixed p , and two primes above the same p satisfy $f_{\mathfrak{p}'} \log N\mathfrak{p} - f_{\mathfrak{p}} \log N\mathfrak{p}' = 0$. Compare [Ch. 2, Lem. 4.1]. $\square$

EXAMPLE 1.3. For $K = \mathbb{Q}(i)$: the two primes above 5 both have norm 5, so any S containing both fails; a set containing one prime above each of some rational primes — one of the two primes above each $p \equiv 1 \pmod{4}$, inert primes $p \equiv 3 \pmod{4}$ with norm p^2 , the prime $1+i$ — is norm-separated. For $K = \mathbb{Q}(\sqrt{2})$ the set of all primes fails at every split prime. The failure by equal norms is the generic one.

THEOREM 1.4 (The Cartan subalgebra is the fixed-point algebra). *If S is norm-separated then*

$$(1.1) \quad \mathcal{A}_{K,S}^{\sigma} := \{x \in \mathcal{A}_{K,S} : \sigma_t(x) = x \ \forall t\} = C(Y_S).$$

Consequently the Cartan subalgebra $C(Y_S) \subset \mathcal{A}_{K,S}$ is canonically determined by the flow, and every σ -equivariant isomorphism $\Phi : \mathcal{A}_{K_1,S_1} \rightarrow \mathcal{A}_{K_2,S_2}$ satisfies $\Phi(C(Y_{S_1})) = C(Y_{S_2})$, hence is an isomorphism of Cartan pairs.

PROOF. A spectral subspace argument: $\sigma_t(\mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*) = (N\mathfrak{a}/N\mathfrak{b})^{it} \mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*$, so the elements $\mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*$ are eigenvectors and σ is the restriction to $\mathcal{R}$ of an action of the compact dual of the countable group $\Gamma_{\beta} = \langle N\mathfrak{p} \rangle \leq \mathcal{R}_{>0}$, in which $\mathcal{R}$ is dense since Γ_{β} is torsion free; averaging over that compact group is a conditional expectation onto the fixed-point algebra which kills every eigenvector of nonzero weight, so the fixed-point algebra is the closed span of those $\mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*$ with $N\mathfrak{a} = N\mathfrak{b}$. Norm-separation gives $\mathfrak{a} = \mathfrak{b}$, and then $\mu_{\mathfrak{a}} f \mu_{\mathfrak{a}}^* = \alpha_{\mathfrak{a}}(f)$, a continuous function on Y_S supported on the clopen set $\mathfrak{a}Y_S$. Hence $\mathcal{A}^{\sigma} \subseteq C(Y_S)$; the reverse inclusion holds because σ_t fixes $C(Y_S)$ pointwise. For the consequence, $\Phi \circ \sigma_t^1 = \sigma_t^2 \circ \Phi$ gives $\Phi(\mathcal{A}_1^{\sigma^1}) = \mathcal{A}_2^{\sigma^2}$. $\square$

REMARK 1.5 (Why this needs saying). Renault's reconstruction theorem is a statement about Cartan pairs $(B \subset A)$: it recovers the twisted groupoid from the pair, not from A alone. A C^* -algebra may carry many non-conjugate Cartan subalgebras, so an isomorphism of algebras — even an equivariant one — carries no information about groupoids unless it is known to respect the pair. Theorem 1.4 supplies exactly that, by identifying the Cartan subalgebra intrinsically, in terms of the flow that the isomorphism is assumed to intertwine. Without it the reconstruction argument of §2 has a gap.

REMARK 1.6 (The unseparated case). If S is not norm-separated then $\mathcal{A}^{\sigma}$ is strictly larger than $C(Y_S)$: it contains $\mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*$ for distinct $\mathfrak{a}, \mathfrak{b}$ of equal norm, which is not in the diagonal. The fixed-point algebra is then non-commutative and Theorem 1.4 fails. This does not show that the

reconstruction fails, only that this route to it does; a different intrinsic characterization of $C(Y_S)$ would be needed, and we do not have one.

2. Reconstruction

Throughout this section S is norm-separated. We first record what in the groupoid $\mathcal{G}_S|_{Y_S}$ is intrinsic. Its arrows are the pairs $(\mathbf{a}\mathbf{b}^{-1}, y)$ with $y \in Y_S$ and $\mathbf{a}\mathbf{b}^{-1}y \in Y_S$, the scaling cocycle is $c(\mathbf{a}\mathbf{b}^{-1}, y) = \log N\mathbf{a} - \log N\mathbf{b}$, and for $\mathbf{a} \in J_S$ the set $B_{\mathbf{a}} = \{(\mathbf{a}, y) : y \in Y_S\}$ is a full bisection — an open set of arrows on which the source map is a homeomorphism onto the whole unit space — with range $\mathbf{a}Y_S$ and $c \equiv \log N\mathbf{a}$.

LEMMA 2.1 (The semigroup is intrinsic). *The full bisections of $\mathcal{G}_S|_{Y_S}$ on which c is constant are exactly the $B_{\mathbf{a}}$, $\mathbf{a} \in J_S$, and $B_{\mathbf{a}}B_{\mathbf{b}} = B_{\mathbf{a}\mathbf{b}}$. Consequently an isomorphism of étale groupoids $\theta : \mathcal{G}_{S_1}|_{Y_{S_1}} \rightarrow \mathcal{G}_{S_2}|_{Y_{S_2}}$ with $c_2 \circ \theta = c_1$ induces a monoid isomorphism $\pi : J_{S_1} \rightarrow J_{S_2}$ with $N\pi(\mathbf{a}) = N\mathbf{a}$, hence a bijection $S_1 \rightarrow S_2$ preserving norms.*

PROOF. By Lemma 1.2, norm-separation means that $N : I_S \rightarrow \mathbb{Q}_{>0}$ is injective on the group of fractions $I_S = \mathbb{Z}^{(S)}$, so the value $c(\gamma)$ determines the element $g \in I_S$ with $\gamma = (g, y)$. A full bisection contains for each $y \in Y_S$ exactly one arrow with source y ; if c is constant on it, all these arrows have the same $g = \mathbf{a}\mathbf{b}^{-1}$ with $\mathbf{a}, \mathbf{b} \in J_S$ coprime, and the arrow $(\mathbf{a}\mathbf{b}^{-1}, y)$ exists in $\mathcal{G}_S|_{Y_S}$ only for $y \in \mathbf{b}Y_S$. Fullness forces $\mathbf{b}Y_S = Y_S$, i.e. $\mathbf{b} = 1$, so the bisection is $B_{\mathbf{a}}$. The composition rule is $(\mathbf{a}, \mathbf{b}y)(\mathbf{b}, y) = (\mathbf{a}\mathbf{b}, y)$. An isomorphism θ preserving c maps full bisections with constant c to full bisections with the same constant, and preserves composition, so $B_{\mathbf{a}} \mapsto B_{\pi(\mathbf{a})}$ defines a monoid isomorphism of the free abelian monoids J_{S_i} with $N\pi(\mathbf{a}) = N\mathbf{a}$; a monoid isomorphism of free abelian monoids maps the free generators to the free generators. $\square$

DEFINITION 2.2. Let $\text{Aut}_0(\mathcal{G}_S|_{Y_S})$ be the group of automorphisms of the topological groupoid $\mathcal{G}_S|_{Y_S}$ that map each bisection $B_{\mathbf{a}}$ onto itself, with the topology of uniform convergence on Y_S ; equivalently, the automorphisms of $\mathcal{A}_{K,S}$ fixing every $\mu_{\mathbf{a}}$ and preserving $C(Y_S)$. The translation action of G_S on $Y_S = \hat{\mathcal{O}}_S \times_{U_S} G_S$ commutes with the partial action of J_S , so it gives an injective continuous homomorphism $G_S \rightarrow \text{Aut}_0(\mathcal{G}_S|_{Y_S})$. For $K = \mathbb{Q}$ say that S is *Dirichlet-dense* if for every $p \in S$ the primes of $S \setminus \{p\}$ generate a dense subgroup of $\mathbb{Z}_p^\times$, i.e. meet every residue class modulo every power of p ; the set of all primes, and any set of density one in every arithmetic progression, is Dirichlet-dense.

PROPOSITION 2.3 (The symmetry group is intrinsic over $\mathbb{Q}$). *Let $K = \mathbb{Q}$ and let S be Dirichlet-dense. Then $G_S \rightarrow \text{Aut}_0(\mathcal{G}_S|_{Y_S})$ is an isomorphism.*

PROOF. For $K = \mathbb{Q}$, $Y_S = \prod_{p \in S} \mathbb{Z}_p$ with $\mathbf{a} \in J_S$ acting by multiplication and $G_S = \prod_{p \in S} \mathbb{Z}_p^\times$ acting by multiplication as well. Let $\psi \in \text{Aut}_0$. It commutes with multiplication by every $n \in J_S$ and preserves the ranges nY_S , hence the valuations, hence the set $G_S = Y_S^\times$ of units; write $\psi(u) = u c(u)$ for $u \in G_S$, with $c : G_S \rightarrow G_S$ continuous. For $x \in Y_S$ with no zero component, $x = nu$ uniquely with $n \in J_S$, $u \in G_S$, and $\psi(x) = n\psi(u) = x c(u)$. Let $p \in S$ and consider points x with $x_p = 0$: they are limits of $n_k u_k$ with $n_k \in J_S$ divisible by p^k , and the p -components $u_{k,p}$ of the units are unconstrained, since $(n_k)_p \rightarrow 0$; so continuity of ψ at such points forces the components $c(u)_q$, $q \neq p$, not to depend on u_p . Varying p over $S \setminus \{q\}$, the component $c(u)_q$ depends only on u_q : $c(u) = (c_q(u_q))_q$ with $c_q : \mathbb{Z}_q^\times \rightarrow \mathbb{Z}_q^\times$ continuous. Now compare $\psi(nx)$ and $n\psi(x)$ for x a unit and $n \in J_S$: the q -component of nx is $q^{v_q(n)} n'_q x_q$ with $n'_q = n/q^{v_q(n)}$, so $\psi(nx)_q = q^{v_q(n)} n'_q x_q c_q(n'_q x_q)$, while $(n\psi(x))_q = q^{v_q(n)} n'_q x_q c_q(x_q)$. Hence $c_q(n'_q x_q) = c_q(x_q)$ for all $n \in J_S$ and $x_q \in \mathbb{Z}_q^\times$: c_q is invariant under multiplication by the unit parts of the elements of J_S , which generate the subgroup of $\mathbb{Z}_q^\times$ generated by the primes of $S \setminus \{q\}$, dense by hypothesis. So each c_q is constant, c is a constant $g \in G_S$, and ψ is multiplication by g . $\square$

REMARK 2.4. For a general number field the same argument gives less. The constraint from continuity at the points with infinite valuation at $\mathfrak{p}$ concerns ψ only modulo the image $r(\mathcal{O}_{\mathfrak{p}}^{\times}) \subseteq G_S$ of the local units, and for fields with infinite unit group the images of the local units at different primes are not independent in G_S , the global units sitting diagonally in all of them. We do not know whether $\text{Aut}_0(\mathcal{G}_S|_{Y_S}) = G_S$ in general; the compact model $\overline{\mathbb{N}}^S \times G_S$ used elsewhere in the series differs from Y_S exactly at these points, and would give the equality for every Frobenius-dense S , but it is not the space on which $\mathcal{A}_{K,S}$ is built.

THEOREM 2.5 (Equivariant reconstruction). *Let S_1, S_2 be infinite and norm-separated, and suppose $\Phi : (\mathcal{A}_{K_1, S_1}, \sigma^1) \rightarrow (\mathcal{A}_{K_2, S_2}, \sigma^2)$ is an isomorphism of C^* -dynamical systems. Then:*

- (1) Φ is an isomorphism of Cartan pairs and, by Renault's theorem, is induced by an isomorphism of étale groupoids $\theta : \mathcal{G}_{S_1}|_{Y_{S_1}} \rightarrow \mathcal{G}_{S_2}|_{Y_{S_2}}$;
- (2) θ intertwines the scaling cocycles, $c_2 \circ \theta = c_1$;
- (3) θ induces a norm-preserving bijection $\pi : S_1 \rightarrow S_2$, so the multisets $\{N\mathfrak{p}\}_{\mathfrak{p} \in S_1}$ and $\{N\mathfrak{p}\}_{\mathfrak{p} \in S_2}$ coincide;
- (4) θ maps the valuation fibres of Y_{S_1} homeomorphically onto those of Y_{S_2} , compatibly with π ; in particular the fibre over 0, which is the compact space G_{S_1} , is carried homeomorphically onto G_{S_2} ;
- (5) Φ induces affine isomorphisms $\text{KMS}_{\beta}(\mathcal{A}_{K_1, S_1}) \cong \text{KMS}_{\beta}(\mathcal{A}_{K_2, S_2})$ for every β , so the compact spaces $G_{S_1}/\Xi_{\beta}(K_1, S_1)^{\perp}$ and $G_{S_2}/\Xi_{\beta}(K_2, S_2)^{\perp}$ are homeomorphic, $|\Xi_{\beta}|$ agree when finite, the transition loci agree, and $\beta_c(S_1) = \beta_c(S_2)$;
- (6) suppose Φ intertwines the symmetry actions, $\Phi \circ \tau_g = \tau_{\iota(g)} \circ \Phi$ for an isomorphism of topological groups $\iota : G_{S_1} \rightarrow G_{S_2}$ — which is automatic, with $\iota = \theta(\cdot)\theta^{-1}$ on Aut_0 , when $K_1 = K_2 = \mathbb{Q}$ and the S_i are Dirichlet-dense, by Proposition 2.3. Then $\iota(\sigma_{\mathfrak{p}}) \in \sigma_{\pi(\mathfrak{p})}r(\mathcal{O}_{\pi(\mathfrak{p})}^{\times})$ for every $\mathfrak{p} \in S_1$: the Frobenius classes modulo inertia are matched, and $\hat{\iota}^{-1}$ maps $\Xi_{\beta}(K_2, S_2)$ onto $\Xi_{\beta}(K_1, S_1)$ for every $\beta > 0$.

PROOF. (1) Theorem 1.4 makes Φ an isomorphism of Cartan pairs. The groupoid $\mathcal{G}_S|_{Y_S}$ is étale, Hausdorff, second countable, amenable, and topologically principal: the isotropy at a point with all valuations finite is trivial, since $\mathfrak{a}\mathfrak{b}^{-1}y = y$ forces $\mathfrak{a} = \mathfrak{b}$ on such points, and these points are dense. So $(C(Y_S) \subset \mathcal{A}_{K,S})$ is a Cartan pair with trivial Weyl twist, and Renault's theorem [108] gives θ inducing Φ . (2) Under $\Phi = \theta_*$ the flow is $\sigma_t(f)(\gamma) = e^{itc(\gamma)}f(\gamma)$ on functions supported on bisections, so $\Phi\sigma_t^1 = \sigma_t^2\Phi$ gives $e^{itc_2(\theta\gamma)} = e^{itc_1(\gamma)}$ for all t . (3) is Lemma 2.1. (4) θ maps $B_{\mathfrak{a}}$ to $B_{\pi(\mathfrak{a})}$, hence the ranges $\mathfrak{a}Y_{S_1}$ to $\pi(\mathfrak{a})Y_{S_2}$, hence the valuation fibres to the valuation fibres; the fibre over 0 is $U_S \times_{U_S} G_S = G_S$. (5) An isomorphism of C^* -dynamical systems carries KMS_{β} states to KMS_{β} states; the simplices are $\text{Prob}(G_S/\Xi_{\beta}^{\perp})$ by [Ch. 1, Thm. 3.6], and an affine isomorphism of Bauer simplices is a homeomorphism of their extreme boundaries; the transition locus is the set of β at which the simplex changes. (6) Since $\Phi = \theta_*$ preserves the Cartan pair, intertwining the symmetry actions means $\theta \circ \tau_g = \tau_{\iota(g)} \circ \theta$ on Y_{S_1} ; when $\text{Aut}_0 = G_S$ on both sides this holds with $\iota = \theta(\cdot)\theta^{-1}$, an isomorphism of topological groups since θ matches the bisections $B_{\mathfrak{a}}$. On the valuation fibres, which are G_S -torsors, θ is ι -equivariant, so in the coordinates $y = (v, h)$ of [Ch. 1, Lem. 2.10] on the points with finite valuations, $\theta(v, h) = (\pi(v), \iota(h)c_v)$ with constants $c_v \in G_{S_2}$, and $\theta(\mathfrak{p} \cdot (v, h)) = \pi(\mathfrak{p}) \cdot \theta(v, h)$ reads $\iota(\sigma_{\mathfrak{p}})\iota(h)c_{v+e_{\mathfrak{p}}} = \sigma_{\pi(\mathfrak{p})}\iota(h)c_v$, i.e. $c_{v+e_{\mathfrak{p}}} = u_{\mathfrak{p}}c_v$ with $u_{\mathfrak{p}} = \iota(\sigma_{\mathfrak{p}})^{-1}\sigma_{\pi(\mathfrak{p})}$. The points $(ke_{\mathfrak{p}}, h)$ converge in Y_{S_1} as $k \rightarrow \infty$ modulo the inertia image $r(\mathcal{O}_{\mathfrak{p}}^{\times})$, and their images $(ke_{\pi(\mathfrak{p})}, \iota(h)u_{\mathfrak{p}}^k c_0)$ must converge modulo $r(\mathcal{O}_{\pi(\mathfrak{p})}^{\times})$; a sequence u^k converging in the compact group $G_{S_2}/r(\mathcal{O}_{\pi(\mathfrak{p})}^{\times})$ has u trivial there. So $u_{\mathfrak{p}} \in r(\mathcal{O}_{\pi(\mathfrak{p})}^{\times})$. Finally Ξ_{β} is characterized by the series $\sum_{\mathfrak{p}} N\mathfrak{p}^{-\beta}|1 - \chi(\sigma_{\mathfrak{p}})|^2$ [Ch. 1, (1.1)], whose convergence class is unchanged when $\sigma_{\mathfrak{p}}$ is altered by inertia elements at $\mathfrak{p}$ for all but finitely many $\mathfrak{p}$, the KMS simplex being an invariant of the system; so it is transported by π and $\hat{\iota}$. $\square$

REMARK 2.6 (The hypotheses). Norm-separation enters twice, in Theorem 1.4 and in Lemma 2.1; by Lemma 1.2 it is the same as the $\mathbb{Q}$ -independence of the $\log N\mathfrak{p}$, so there is only one arithmetic condition on S in parts (1)–(5), and it is automatic over $\mathbb{Q}$. Part (6) needs the symmetry actions to be intertwined: without that, θ still matches the fibres and the simplices, but a homeomorphism of compact groups carries no group structure, and we do not know whether G_S with its Frobenius classes is an invariant of the dynamical system in general (Remark 2.4).

3. What is not recovered

REMARK 3.1 (The field). Theorem 2.5 recovers the pair $(S, N|_S)$, and, when the symmetry actions are intertwined, the triple $(S, N|_S, G_S)$ with its Frobenius data. It does *not* recover K . The system is built from that triple and nothing else, so two number fields yielding the same triple give isomorphic dynamical systems, whatever their other invariants. Whether the triple determines K is an arithmetic question, not an operator-algebraic one; for $S = P_K$ it is the subject of the literature on reconstruction from Bost–Connes systems, which we have not been able to consult here and against which any claim of field reconstruction should be checked. We make no such claim.

REMARK 3.2 (Arithmetic equivalence is not obviously the right equivalence). Arithmetically equivalent fields share their Dedekind zeta function, hence the multiset of norms $\{N\mathfrak{p}\}$ for $S = P_K$. They need not share G_K , which involves the class group and the units, and G_S is part of the recovered data by Theorem 2.5(6). So the invariant recovered here is finer than the zeta function and coarser than K ; locating it exactly is open. We note that the narrow class number $h_S^+ = |\text{Cl}_S^+|$ is not recovered by any argument of this paper: Cl_S^+ is the quotient of G_S by the image of the local units, a finite-index open subgroup which nothing in the dynamical system singles out.

4. The KMS bundle is a weaker invariant

DEFINITION 4.1. The *KMS bundle* of $(\mathcal{A}_{K,S}, \sigma)$ is $\bigsqcup_{\beta > 0} \text{KMS}_\beta(\mathcal{A}_{K,S})$ over $(0, \infty)$, with the fibrewise affine structure and the G_S -action. An isomorphism of KMS bundles is a fibrewise affine bijection over the identity of $(0, \infty)$ together with an isomorphism ι of the symmetry groups, the two being compatible.

PROPOSITION 4.2 (What the bundle sees). *A fibrewise affine bijection of KMS bundles over the identity of $(0, \infty)$ determines the compact spaces $G_S/\Xi_\beta^\perp$ for every β , hence the transition locus, β_c , and $|\Xi_\beta|$ where finite. An isomorphism of KMS bundles in the sense of Definition 4.1 determines in addition the chain $\{\Xi_\beta\}_{\beta > 0}$ as subgroups of $\widehat{G_S}$, since $\Xi_\beta^\perp$ is the stabilizer in G_S of any extreme point of the fibre at β .*

PROOF. Each of these is read off [Ch. 1, Thm. 3.6]: the fibre at β is $\text{Prob}(G_S/\Xi_\beta^\perp)$, on whose extreme points G_S acts transitively with stabilizer $\Xi_\beta^\perp$. $\square$

REMARK 4.3 (What the bundle does not see). We see no way to extract the individual norms $N\mathfrak{p}$ from the bundle. The fibres depend on S only through the sets $\{\mathfrak{p} : \chi(\sigma_\mathfrak{p}) \neq 1\}$ and their convergence exponents; two systems whose primes have different norms but the same detecting sets and the same convergence behaviour would have isomorphic bundles. The bundle is therefore weaker than the groupoid of Theorem 2.5, which recovers the norms outright, and the chain of implications runs

equivariant C^* -isomorphism $\implies$ groupoid isomorphism $\implies$ KMS bundle isomorphism,
with the converses unproved. We do not claim the bundle is a complete invariant.

5. The Kirchberg boundary: a caution

REMARK 5.1 (Flows on Kirchberg algebras). By Chapter 3 the boundary $\partial\mathcal{A}_{K,S}^{\text{aff}}$ of the affine system is a unital Kirchberg algebra satisfying the UCT, and it carries the flow $\sigma_t(\mu_{(a,p)}) = N\mathfrak{p}^{it}\mu_{(a,p)}$. It is tempting to classify the pair by equivariant KK . We do not, for two reasons.

First, the classification of *flows* — actions of $\mathcal{R}$ — on Kirchberg algebras up to cocycle conjugacy is not a settled theory. The Kirchberg–Phillips theorem [71] classifies the algebras; the equivariant theory is well developed for finite and, more recently, for many amenable *discrete* groups [122], but $\mathcal{R}$ -actions on purely infinite algebras remain substantially open, and we are not able to invoke a classification theorem here. (For the diagonal-preserving isomorphisms of §2, by contrast, the relevant rigidity is that of continuous orbit equivalence [83, 88], whose groupoid form is Renault’s theorem.)

Second, and more concretely, by Chapter 3 this system has KMS_β states only at $\beta = 1$. Whatever the classification of the flow, the equilibrium invariant that carries the arithmetic in the Toeplitz system has collapsed to a single point on the boundary, so the boundary is not where the reconstruction of §2 can be carried out.

6. Assessment

Statement	Status	Reference
$\mathcal{A}^\sigma = C(Y_S)$ for norm-separated S	true	Thm. 1.4
norm-separation automatic over $\mathbb{Q}$	true	Lem. 1.2
equivariant iso preserves the Cartan pair	true	Thm. 1.4
Renault: pair $\Rightarrow$ groupoid	true	Thm. 2.5(1)
norm-separation = $\mathbb{Q}$ -indep. of $\log N\mathfrak{p}$	true	Lem. 1.2
norms $\{N\mathfrak{p}\}$ recovered	true	Thm. 2.5(3)
simplices, transition locus, β_c recovered	true	Thm. 2.5(5)
G_S , $\sigma_{\mathfrak{p}}$, $\{\Xi_\beta\}$ recovered	if symmetries intertwined	Thm. 2.5(6)
$\text{Aut}_0 = G_S$	true over $\mathbb{Q}$, Dirichlet-dense S	Prop. 2.3
h_S^+ recovered	not claimed	Rem. 3.2
the field K recovered	not claimed	Rem. 3.1
KMS bundle a complete invariant	not claimed; weaker	Rem. 4.3
Kirchberg boundary classified by $KK_{\mathcal{R}}$	not available	Rem. 5.1

The contrast with Chapter 20 is the point. There the von Neumann algebra was shown to retain nothing, the groupoid being amenable and Connes’ classification leaving only the flow of weights. Here the C^* -algebra *with its flow* retains the norms, the KMS simplices, and — once the symmetries are intertwined, which over $\mathbb{Q}$ is automatic — the symmetry group with its Frobenius classes and the whole chain $\{\Xi_\beta\}$. The mechanism is Theorem 1.4: the flow itself pins down the Cartan subalgebra, so the reconstruction theory of Renault becomes available to a merely equivariant isomorphism. Weak closure destroys the flow — the modular group of a single extremal state is an inner datum — and with it the pin.

Four questions. Is $C(Y_S)$ intrinsically characterized when S is not norm-separated, so that Theorem 2.5 extends beyond Lemma 1.2? Is $\text{Aut}_0(\mathcal{G}_S|_{Y_S}) = G_S$ for fields with infinite unit group, so that the group structure of G_S is an invariant of the dynamical system there too? Is the KMS bundle a complete invariant after all, or is there a pair of systems with isomorphic bundles and non-isomorphic groupoids — we would expect the latter and suggest looking at sets with the same detecting data but different norms. And what exactly is the arithmetic content of the recovered triple $(S, N|_S, G_S)$: it is finer than the Dedekind zeta function, since G_S carries class group and unit data, and coarser than K ; where it sits between them we do not know.

CHAPTER 22

The equal-norm case: norm-separation is not a hypothesis but a characterization, and the gauge-equivariant repair

This chapter is Paper XIV of the series ([papers/XIV-equalnorm/](#)).

SUMMARY. Chapter 21 proved $\mathcal{A}_{K,S}^\sigma = C(Y_S)$ (its Theorem 1.4) under the hypothesis that $N : J_S \rightarrow \mathbb{N}$ be injective, and asked whether the hypothesis could be removed. It cannot, and the reason is that it is not a hypothesis at all.

The diagonal is always the fixed-point algebra of the *compact* gauge action: $C(Y_S) = \mathcal{A}_{K,S}^{\mathbb{T}^S}$, unconditionally. The modular flow is the restriction of the gauge action along $\iota(t)_\mathfrak{p} = N\mathfrak{p}^{it}$, so $\mathcal{A}^\sigma = \mathcal{A}^H$ with $H = \overline{\iota(\mathcal{R})} \leq \mathbb{T}^S$, and by Kronecker $H = \mathbb{T}^S$ exactly when the $\log N\mathfrak{p}$ are $\mathbb{Q}$ -linearly independent. Since $\mathbb{Z}$ -independence of the $\log N\mathfrak{p}$ is literally the injectivity of N on J_S , we obtain

$$\mathcal{A}_{K,S}^\sigma = C(Y_S) \iff \{\log N\mathfrak{p}\}_{\mathfrak{p} \in S} \text{ } \mathbb{Q}\text{-independent} \iff S \text{ norm-separated.}$$

The three conditions are equivalent; there is nothing to remove.

The correct structure in the general case is not the proposed direct sum $\bigoplus_n \mathcal{A}_n^\sigma$, which is not a direct sum — the pieces are neither orthogonal nor ideals, since $e_\mathfrak{a}e_\mathfrak{b} = e_{\mathfrak{a} \vee \mathfrak{b}} \neq 0$ and the product of a degree- n and a degree- m element is again in $\mathcal{A}^\sigma$. It is a grading. Writing $L = \{c \in \mathbb{Z}^{(S)} : \sum_{\mathfrak{p}} c_{\mathfrak{p}} \log N\mathfrak{p} = 0\}$ for the lattice of multiplicative relations among the norms, $\mathcal{A}^\sigma$ is L -graded with degree-zero part $C(Y_S)$ and $H = L^\perp$.

Neither of the two proposed intrinsic characterizations of $C(Y_S)$ inside $\mathcal{A}^\sigma$ works as stated. Property A recovers $C(Y_S)$ as the fixed points of the gauge action, but the gauge action of $\mathbb{T}^S$ is precisely the datum *not* determined by $(\mathcal{A}, \sigma)$ when $L \neq 0$; the argument is circular. Property B asserts that $C(Y_S)$ is the unique Cartan subalgebra, which is what was to be proved.

What does work is a change of hypothesis rather than a removal. If Φ is equivariant for the full gauge actions — up to an isomorphism of the acting tori, so that no bijection $S_1 \rightarrow S_2$ is presupposed — then $\Phi(C(Y_{S_1})) = C(Y_{S_2})$ unconditionally, and the reconstruction of Chapter 21 (its Theorem 2.5) holds for every number field and every prime set: a norm-preserving bijection $S_1 \rightarrow S_2$, which turns out to induce the torus isomorphism after all; the matched valuation fibres, with G_S as a compact space; the KMS_β simplices, transition loci and β_c ; and, when the symmetry actions are intertwined, the Frobenius classes modulo inertia and the chain $\{\Xi_\beta\}$. For $S_i = P_{K_i}$ the norm multisets agree, so $\zeta_{K_1} = \zeta_{K_2}$: the fields are arithmetically equivalent. The narrow class number is not among the conclusions; it was not recovered in Chapter 21 either.

Finally we record a limitation of Chapter 21 that we had not noticed there: every $K \neq \mathbb{Q}$ has infinitely many split primes, so P_K is *never* norm-separated. The theorem of Chapter 21 therefore covers the classical full prime set only for $K = \mathbb{Q}$.

1. Norm-separation is a characterization

We use the notation of the foundational paper [Ch. 1]: $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$ is spanned by the elements $\mu_\mathfrak{a} f \mu_\mathfrak{b}^*$, and it is the C^* -algebra of the étale groupoid $\mathcal{G}_S|_{Y_S}$ whose arrows are the pairs $(\mathfrak{a}\mathfrak{b}^{-1}, y)$ with $y, \mathfrak{a}\mathfrak{b}^{-1}y \in Y_S$, with diagonal $C(Y_S)$ and with the scaling cocycle $c(\mathfrak{a}\mathfrak{b}^{-1}, y) = \log N\mathfrak{a} - \log N\mathfrak{b}$ implementing the flow. Chapters 3, 4 and 21 of the series, referred to by numeral,

are in preparation; every fact used from them is re-proved in place. The finer, integer-valued cocycle

$$\tilde{c} : \mathcal{G}_S|_{Y_S} \longrightarrow I_S = \mathbb{Z}^{(S)}, \quad \tilde{c}(\mathbf{a}\mathbf{b}^{-1}, y) = \mathbf{a}\mathbf{b}^{-1},$$

is continuous and locally constant, and composing it with characters defines the *gauge action* of the compact group $\mathbb{T}^S = \widehat{I_S}$:

$$\gamma_z(f)(\gamma) = z^{\tilde{c}(\gamma)} f(\gamma), \quad \text{so} \quad \gamma_z(\mu_{\mathbf{p}}) = z_{\mathbf{p}} \mu_{\mathbf{p}}, \quad \gamma_z|_{C(Y_S)} = \text{id}, \quad z \in \mathbb{T}^S,$$

strongly continuous because $\tilde{c}$ is locally constant. Since $c = \langle \tilde{c}, \log N \rangle$, the modular flow is a restriction of the gauge action: $\sigma_t = \gamma_{\iota(t)}$ with $\iota(t)_{\mathbf{p}} = N\mathbf{p}^{it}$ (the identity through which Chapter 4 sees that $\{\sigma_t\}$ has compact closure).

LEMMA 1.1 (The diagonal is the gauge fixed-point algebra). $\mathcal{A}_{K,S}^{\mathbb{T}^S} = C(Y_S)$, for every K and every S , with no hypothesis.

PROOF. $\gamma_z(\mu_{\mathbf{a}} f \mu_{\mathbf{b}}^*) = z^{\mathbf{a}-\mathbf{b}} \mu_{\mathbf{a}} f \mu_{\mathbf{b}}^*$ in multi-index notation, so a spectral subspace argument gives $\mathcal{A}^{\mathbb{T}^S} = \overline{\text{span}}\{\mu_{\mathbf{a}} f \mu_{\mathbf{a}}^*\}$, and $\mu_{\mathbf{a}} f \mu_{\mathbf{a}}^* = \alpha_{\mathbf{a}}(f) \in C(Y_S)$. $\square$

DEFINITION 1.2. The *relation lattice* of S is

$$L = L(S) := \left\{ c \in \mathbb{Z}^{(S)} : \sum_{\mathbf{p} \in S} c_{\mathbf{p}} \log N\mathbf{p} = 0 \right\} = \left\{ c : \prod_{\mathbf{p}} N\mathbf{p}^{c_{\mathbf{p}}} = 1 \right\},$$

a subgroup of $\mathbb{Z}^{(S)}$.

THEOREM 1.3 (Equivalence). *The following are equivalent.*

- (1) $N : J_S \rightarrow \mathbb{N}$ is injective, i.e. S is norm-separated;
- (2) $L(S) = 0$, i.e. $\{\log N\mathbf{p}\}_{\mathbf{p} \in S}$ is $\mathbb{Q}$ -linearly independent;
- (3) $\iota(\mathcal{R})$ is dense in $\mathbb{T}^S$;
- (4) $\mathcal{A}_{K,S}^{\sigma} = C(Y_S)$.

PROOF. (1) $\Leftrightarrow$ (2): $N\mathbf{a} = N\mathbf{b}$ with $\mathbf{a}, \mathbf{b} \in J_S$ means $\sum (a_{\mathbf{p}} - b_{\mathbf{p}}) \log N\mathbf{p} = 0$ with $a - b \in \mathbb{Z}^{(S)}$; injectivity says the only such relation is $a = b$, i.e. $L = 0$. A $\mathbb{Q}$ -linear relation clears denominators to a $\mathbb{Z}$ -linear one, so $\mathbb{Z}$ -independence and $\mathbb{Q}$ -independence agree.

(2) $\Leftrightarrow$ (3): $\iota(\mathcal{R})$ is a closed subgroup of $\mathbb{T}^S$, and its annihilator in $\widehat{\mathbb{T}^S} = \mathbb{Z}^{(S)}$ is $\{c : \prod_{\mathbf{p}} N\mathbf{p}^{ic_{\mathbf{p}}t} = 1 \ \forall t\} = L$. By Pontryagin duality $\overline{\iota(\mathcal{R})} = \mathbb{T}^S$ iff $L = 0$; this is Kronecker's theorem.

(3) $\Leftrightarrow$ (4): $\mathcal{A}^{\sigma} = \overline{\mathcal{A}^{\iota(\mathcal{R})}}$, since the fixed-point algebra of a one-parameter group equals that of the closure of its image; combine with Lemma 1.1. Conversely if $L \neq 0$, pick $0 \neq c \in L$ and write $c = \mathbf{a} - \mathbf{b}$ with $\mathbf{a}, \mathbf{b} \in J_S$ coprime; then $N\mathbf{a} = N\mathbf{b}$ and $\mu_{\mathbf{a}} \mu_{\mathbf{b}}^* \in \mathcal{A}^{\sigma}$ is not in $C(Y_S)$, the two ideals being distinct. $\square$

COROLLARY 1.4 (Nothing to remove). *The hypothesis of Theorem 1.4 of Chapter 21 is not a sufficient condition that might be weakened: it is equivalent to the conclusion. In the unseparated case $\mathcal{A}^{\sigma}$ is strictly larger than $C(Y_S)$ and non-commutative, and no argument can identify them.*

PROPOSITION 1.5 (P_K is never separated for $K \neq \mathbb{Q}$). *Let $K \neq \mathbb{Q}$. By the Chebotarev density theorem [97, Ch. VII, §13], applied to the Galois closure of K , there are infinitely many rational primes that split completely in K ; the $[K : \mathbb{Q}] \geq 2$ primes of K above such a p have residue degree 1 and hence equal norm p . So P_K is not norm-separated, and Theorem 1.4 of Chapter 21 applies to the full prime set only when $K = \mathbb{Q}$.*

REMARK 1.6 (Numerical illustration). The first split prime is 5 for $\mathbb{Q}(i)$, 7 for $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{2})$, and 11 for $\mathbb{Q}(\sqrt{5})$; in each case two primes of norm p . This is a limitation of Chapter 21 that we had not noticed there: the classical case, $S = P_K$ over a general number field, is precisely the case the theorem does not cover.

2. The structure of $\mathcal{A}^\sigma$ in general

PROPOSITION 2.1 (The proposed stratification is not a direct sum). *Set $\mathcal{A}_n^\sigma = \overline{\text{span}}\{\mu_{\mathbf{a}}f\mu_{\mathbf{b}}^* : N\mathbf{a} = N\mathbf{b} = n\}$. Then $\mathcal{A}^\sigma \neq \bigoplus_n \mathcal{A}_n^\sigma$: the summands are not orthogonal, since $e_{\mathbf{a}}e_{\mathbf{b}} = e_{\text{lcm}(\mathbf{a},\mathbf{b})} \neq 0$ for $\mathbf{a}, \mathbf{b}$ of different norms, and they are not ideals, the product of an element of $\mathcal{A}_n^\sigma$ with one of $\mathcal{A}_m^\sigma$ lying in $\mathcal{A}^\sigma$ but in neither.*

PROOF. $e_{\mathbf{a}} = \mu_{\mathbf{a}}\mu_{\mathbf{a}}^* \in \mathcal{A}_{N\mathbf{a}}^\sigma$ and $e_{\mathbf{a}}e_{\mathbf{b}} = e_{\text{lcm}} \neq 0$; for the second, compute $(\mu_{\mathbf{a}}f\mu_{\mathbf{b}}^*)(\mu_{\mathbf{c}}g\mu_{\mathbf{d}}^*)$ using $\mu_{\mathbf{b}}^*\mu_{\mathbf{c}} = \mu_{\mathbf{c}'}\mu_{\mathbf{b}'}^*$ with $\mathbf{c}' = \mathbf{c}/\text{gcd}$, $\mathbf{b}' = \mathbf{b}/\text{gcd}$, obtaining a term $\mu_{\mathbf{ac}'}(\cdots)\mu_{\mathbf{db}'}^*$, whose two norms are equal but need not be n or m . $\square$

THEOREM 2.2 (The correct structure). $\mathcal{A}^\sigma = \mathcal{A}^H$ with $H = L^\perp \leq \mathbb{T}^S$ the annihilator of the relation lattice, and $\mathcal{A}^\sigma$ carries a grading by L ,

$$\mathcal{A}^\sigma = \bigoplus_{c \in L} \mathcal{A}_{(c)}^\sigma, \quad \mathcal{A}_{(c)}^\sigma = \overline{\text{span}}\{\mu_{\mathbf{a}}f\mu_{\mathbf{b}}^* : \mathbf{a} - \mathbf{b} = c\}, \quad \mathcal{A}_{(0)}^\sigma = C(Y_S),$$

the grading being implemented by the residual gauge action of $\mathbb{T}^S/H \cong \widehat{L}$. In particular $C(Y_S) = (\mathcal{A}^\sigma)^{\widehat{L}}$.

PROOF. $H = \overline{\iota(\mathcal{R})}$ has annihilator L by the proof of Theorem 1.3, so $\mathbb{T}^S/H \cong \widehat{L}$ and the gauge action descends. The grading is the spectral decomposition for that residual action, the degree of $\mu_{\mathbf{a}}f\mu_{\mathbf{b}}^*$ being $\mathbf{a} - \mathbf{b} \in \mathbb{Z}^{(S)}$, which lies in L precisely on $\mathcal{A}^\sigma$. Degree zero is Lemma 1.1. $\square$

EXAMPLE 2.3. Let $K = \mathbb{Q}(i)$ and $S = \{\mathbf{p}, \bar{\mathbf{p}}\}$ the two primes above 5. Then $L = \mathbb{Z}(1, -1)$, $\widehat{L} \cong \mathbb{T}$, and $\mathcal{A}^\sigma$ is $\mathbb{Z}$ -graded, the piece of degree $k(1, -1)$ being spanned by the elements

$$\mu_{\mathbf{p}}^{m+k} \mu_{\bar{\mathbf{p}}}^n f (\mu_{\mathbf{p}}^m \mu_{\bar{\mathbf{p}}}^{n+k})^*, \quad \text{for instance} \quad \mu_{\mathbf{p}}^k f \mu_{\bar{\mathbf{p}}}^{k*},$$

both sides having the same norm 5^{m+n+k} . The degree-zero part is $C(Y_S)$ and the whole is a crossed-product-like algebra over it; it is non-commutative, and no spectral argument using σ alone can see inside it.

3. Why the two proposed characterizations are circular

REMARK 3.1 (Property A). The proposal is to characterize $C(Y_S)$ as the unique MASA of $\mathcal{A}^\sigma$ carrying a faithful conditional expectation invariant under the gauge symmetries $\mathbb{T}^S$. By Lemma 1.1 this is correct as a *description* — $C(Y_S)$ is the range of the averaging expectation E_γ — but it is not an intrinsic characterization inside $(\mathcal{A}, \sigma)$, because the $\mathbb{T}^S$ -action is exactly the datum that σ fails to determine when $L \neq 0$. By Theorem 1.3(3), σ determines only $H = L^\perp \subsetneq \mathbb{T}^S$. Invoking γ to recover $C(Y_S)$ therefore assumes what is missing.

REMARK 3.2 (Property B). The proposal is to characterize $C(Y_S)$ as the unique abelian MASA whose normalizer generates $\mathcal{A}_{K,S}$, i.e. as the unique Cartan subalgebra. That $C(Y_S)$ is a Cartan subalgebra is Theorem 2.5 of Chapter 3 (topological principality; the argument is reproduced inside the proof of Theorem 4.2 below). That it is the *unique* one is precisely the statement needed to apply Renault's theorem [108] to a merely σ -equivariant isomorphism, which is what Theorem 1.4 of Chapter 21 was designed to supply and what is at issue here. Assuming it is circular. We add that uniqueness of Cartan subalgebras is false for general C^* -algebras and is a hard theorem where it holds, so it cannot be assumed lightly.

REMARK 3.3 (What we do not know). We have not found an intrinsic characterization of $C(Y_S)$ inside $(\mathcal{A}_{K,S}, \sigma)$ in the unseparated case, and we have not found an obstruction either. The natural attempt — to exhibit a σ -commuting automorphism moving $C(Y_S)$, which would settle it negatively — fails for a reason worth recording. In a Cuntz algebra the quasi-free rotation

$\mu_i \mapsto \sum_j u_{ji} \mu_j$ is an automorphism because $\mu_i^* \mu_j = \delta_{ij}$; here the isometries commute and satisfy $\mu_{\mathfrak{p}}^* \mu_{\mathfrak{q}} = \mu_{\mathfrak{q}} \mu_{\mathfrak{p}}^*$ for coprime $\mathfrak{p}, \mathfrak{q}$, so, for $|\alpha|^2 + |\beta|^2 = 1$ with $\alpha\beta \neq 0$, $(\alpha\mu_{\mathfrak{p}} + \beta\mu_{\mathfrak{q}})^*(\alpha\mu_{\mathfrak{p}} + \beta\mu_{\mathfrak{q}}) = 1 + \bar{\alpha}\beta\mu_{\mathfrak{q}}\mu_{\mathfrak{p}}^* + \alpha\bar{\beta}\mu_{\mathfrak{p}}\mu_{\mathfrak{q}}^* \neq 1$ and the rotated element is not an isometry. The Nica–Toeplitz relations are rigid enough to block the obvious counterexample, which is a reason to think the question may have a positive answer; we have not found it.

4. The gauge-equivariant repair

DEFINITION 4.1. Call $\Phi : \mathcal{A}_{K_1, S_1} \rightarrow \mathcal{A}_{K_2, S_2}$ *gauge-equivariant* if there is a topological group isomorphism $\theta : \mathbb{T}^{S_1} \rightarrow \mathbb{T}^{S_2}$ with $\Phi \circ \gamma_z^1 = \gamma_{\theta(z)}^2 \circ \Phi$ for all z . No bijection $S_1 \rightarrow S_2$ is presupposed, and none is encoded in θ by itself: the automorphisms of $\mathbb{T}^S$ are dual to the automorphisms of the free abelian group $\mathbb{Z}^{(S)}$, most of which permute no coordinates. Theorem 4.2 produces a bijection, and shows that θ was induced by one after all.

THEOREM 4.2 (Unconditional reconstruction). *Let K_1, K_2 be arbitrary number fields and S_1, S_2 arbitrary infinite prime sets, and let $\Phi : \mathcal{A}_{K_1, S_1} \rightarrow \mathcal{A}_{K_2, S_2}$ be a gauge-equivariant and σ -equivariant isomorphism. Then $\Phi(C(Y_{S_1})) = C(Y_{S_2})$, and:*

- (1) Φ is induced by an isomorphism θ_* of the underlying étale groupoids which intertwines the scaling cocycles, $c_2 \circ \theta_* = c_1$, and the integer-valued gauge cocycles, $\hat{\theta} \circ \tilde{c}_2 \circ \theta_* = \tilde{c}_1$, where $\hat{\theta} : \mathbb{Z}^{(S_2)} \rightarrow \mathbb{Z}^{(S_1)}$ is the isomorphism dual to θ ;
- (2) θ_* induces a norm-preserving monoid isomorphism $\pi : J_{S_1} \rightarrow J_{S_2}$, hence a norm-preserving bijection $S_1 \rightarrow S_2$; moreover $\hat{\theta}(\delta_{\pi(\mathfrak{p})}) = \delta_{\mathfrak{p}}$, so θ is, a posteriori, the isomorphism induced by the bijection π of the coordinate circles;
- (3) the valuation fibres of Y_{S_1} are carried homeomorphically onto those of Y_{S_2} , compatibly with π ; in particular the fibre over 0, the compact space G_{S_1} , is carried homeomorphically onto G_{S_2} ;
- (4) the KMS $_{\beta}$ simplices are affinely isomorphic for every β , so the compact spaces $G_{S_i}/\Xi_{\beta}^{\perp}$ are homeomorphic, the orders $|\Xi_{\beta}|$ agree when finite, the transition loci agree, and $\beta_c(S_1) = \beta_c(S_2)$;
- (5) if moreover Φ intertwines the symmetry actions along an isomorphism $\iota : G_{S_1} \rightarrow G_{S_2}$, then the Frobenius classes are matched modulo inertia, $\iota(\sigma_{\mathfrak{p}}) \in \sigma_{\pi(\mathfrak{p})} r(\mathcal{O}_{\pi(\mathfrak{p})}^{\times})$, and $\hat{\iota}^{-1}$ carries $\Xi_{\beta}(K_2, S_2)$ onto $\Xi_{\beta}(K_1, S_1)$ for every $\beta > 0$.

These are the conclusions of Theorem 2.5 of Chapter 21, with no norm-separation required anywhere; what is conditional in (5) is conditional there too.

PROOF. *The Cartan pair.* By Lemma 1.1, $C(Y_{S_i}) = \mathcal{A}_i^{\mathbb{T}^{S_i}}$, and gauge-equivariance, with θ surjective, gives $\Phi(\mathcal{A}_1^{\mathbb{T}^{S_1}}) = \mathcal{A}_2^{\mathbb{T}^{S_2}}$, i.e. $\Phi(C(Y_{S_1})) = C(Y_{S_2})$. Each pair $(C(Y_S) \subset \mathcal{A}_{K, S})$ is Cartan with trivial Weyl twist: the groupoid $\mathcal{G}_S|_{Y_S}$ is étale, Hausdorff, second countable and amenable, and it is topologically principal because the isotropy is trivial at every point with all valuations finite ($\mathfrak{a}\mathfrak{b}^{-1}y = y$ forces $\mathfrak{a} = \mathfrak{b}$ at such a point) and those points are dense — this reproduces Theorem 2.5(1) of Chapter 3. Renault’s theorem [108] therefore provides an isomorphism $\theta_* : \mathcal{G}_{S_1}|_{Y_{S_1}} \rightarrow \mathcal{G}_{S_2}|_{Y_{S_2}}$ of étale groupoids inducing Φ .

Both cocycles are intertwined. For f supported on an open bisection B , $\Phi(f)$ is supported on $\theta_*(B)$ (up to the $\mathbb{T}$ -valued normalization of the trivialized twist, which cancels below), and γ_w^2 multiplies it there by $w^{\tilde{c}_2}$. Comparing $\Phi(\gamma_z^1 f) = \gamma_{\theta(z)}^2 \Phi(f)$ pointwise gives $z^{\tilde{c}_1(\gamma)} = \theta(z)^{\tilde{c}_2(\theta_* \gamma)}$ for all $z \in \mathbb{T}^{S_1}$, i.e. $\tilde{c}_1 = \hat{\theta} \circ \tilde{c}_2 \circ \theta_*$; comparing $\Phi(\sigma_t^1 f) = \sigma_t^2 \Phi(f)$ gives $c_2 \circ \theta_* = c_1$.

The semigroup, without norm-separation. A full bisection of $\mathcal{G}_S|_{Y_S}$ on which $\tilde{c}$ is constant contains, for each $y \in Y_S$, exactly one arrow (g, y) with $g = \mathfrak{a}\mathfrak{b}^{-1}$ independent of y , $\mathfrak{a}, \mathfrak{b} \in J_S$ coprime; that arrow exists only for $y \in \mathfrak{b}Y_S$, and fullness forces $\mathfrak{b}Y_S = Y_S$, i.e. $\mathfrak{b} = 1$. So these

bisections are exactly the $B_{\mathbf{a}} = \{(\mathbf{a}, y) : y \in Y_S\}$, with $B_{\mathbf{a}}B_{\mathbf{b}} = B_{\mathbf{ab}}$. This is Lemma 2.1 of Chapter 21 run with $\tilde{c}$ in place of the scaling cocycle: there the value of c had to determine the element g , which is exactly norm-separation; the value of $\tilde{c}$ is g , and no hypothesis is needed. Since $\hat{\theta}$ is injective, $\tilde{c}_2 \circ \theta_*$ is constant on a bisection iff $\tilde{c}_1$ is, so θ_* maps the $B_{\mathbf{a}}$ to the $B_{\pi(\mathbf{a})}$ for a monoid isomorphism $\pi : J_{S_1} \rightarrow J_{S_2}$, which carries the free generators of the free abelian monoid J_{S_1} to those of J_{S_2} : a bijection $S_1 \rightarrow S_2$. It preserves norms because $\log N\pi(\mathbf{a}) = c_2(B_{\pi(\mathbf{a})}) = c_1(B_{\mathbf{a}}) = \log N\mathbf{a}$. Evaluating the intertwining of $\tilde{c}$ on $B_{\mathbf{p}}$ gives $\delta_{\mathbf{p}} = \tilde{c}_1(B_{\mathbf{p}}) = \hat{\theta}(\tilde{c}_2(B_{\pi(\mathbf{p})})) = \hat{\theta}(\delta_{\pi(\mathbf{p})})$: $\hat{\theta}$ maps standard generators to standard generators, so θ is induced by π .

Fibres, simplices, symmetries. θ_* maps $B_{\mathbf{a}}$ to $B_{\pi(\mathbf{a})}$, hence the range $\mathbf{a}Y_{S_1}$ to $\pi(\mathbf{a})Y_{S_2}$, hence the valuation fibres to the valuation fibres, and the fibre over 0, which is G_S , homeomorphically onto its counterpart. An isomorphism of C^* -dynamical systems carries KMS_{β} states to KMS_{β} states; by [Ch. 1, Thm. 3.6] the simplices are $\text{Prob}(G_S/\Xi_{\beta}^{\perp})$, an affine isomorphism of Bauer simplices is a homeomorphism of the extreme boundaries, and the transition locus is the set of β at which the simplex changes. Finally (5) is part (6) of Theorem 2.5 of Chapter 21: its proof uses only the matched bisections and the torsor structure of the fibres, and nowhere uses norm-separation, which entered that paper only through its Theorem 1.4 — replaced here by gauge-equivariance and Lemma 1.1 — and its Lemma 2.1 — replaced by the $\tilde{c}$ argument above; its Remark 2.6 records that these are the only two entry points. $\square$

COROLLARY 4.3 (Dedekind zeta). *If $S_i = P_{K_i}$ and Φ is gauge- and σ -equivariant, then the norm multisets $\{N\mathbf{p}\}_{\mathbf{p} \in P_{K_1}}$ and $\{N\mathbf{q}\}_{\mathbf{q} \in P_{K_2}}$ agree with multiplicities, hence*

$$\zeta_{K_1}(s) = \prod_{\mathbf{p}} (1 - N\mathbf{p}^{-s})^{-1} = \zeta_{K_2}(s),$$

so K_1 and K_2 are arithmetically equivalent. The narrow class numbers are not among the conclusions: h_S^+ is recovered by no argument of Chapter 21 (its Remark 3.2), and nothing here adds one.

REMARK 4.4 (Is the hypothesis reasonable?). Gauge-equivariance is a genuine strengthening and should be declared as such. It is however natural: the gauge action is part of the standard data of a semigroup crossed product, it is what makes $C(Y_S)$ the coefficient algebra, and any isomorphism arising from an isomorphism of the underlying dynamical systems $(Y_{S_1}, J_{S_1}) \rightarrow (Y_{S_2}, J_{S_2})$ is automatically gauge-equivariant. What Theorem 4.2 does not do is deduce the conclusion from σ -equivariance alone, and by Corollary 1.4 no argument through $\mathcal{A}^{\sigma}$ ever will.

5. Assessment

Statement	Status	Reference
$C(Y_S) = \mathcal{A}^{\mathbb{T}^S}$ unconditionally	true	Lem. 1.1
norm-separated $\iff \mathcal{A}^\sigma = C(Y_S)$	equivalent, not sufficient	Thm. 1.3
the hypothesis can be removed	false	Cor. 1.4
P_K norm-separated for $K \neq \mathbb{Q}$	never	Prop. 1.5
$\mathcal{A}^\sigma = \bigoplus_n \mathcal{A}_n^\sigma$	false: not a direct sum	Prop. 2.1
$\mathcal{A}^\sigma$ is L -graded, $\mathcal{A}_{(0)}^\sigma = C(Y_S)$	true	Thm. 2.2
Property A characterizes $C(Y_S)$	circular: uses γ	Rem. 3.1
Property B characterizes $C(Y_S)$	circular: assumes uniqueness	Rem. 3.2
intrinsic recovery from $(\mathcal{A}, \sigma)$	open	Rem. 3.3
gauge-equivariant reconstruction	true, all K, S	Thm. 4.2
θ induced by a bijection $S_1 \rightarrow S_2$	true a posteriori, not a priori	Thm. 4.2(2)
Frobenius classes, $\{\Xi_\beta\}$ matched	if symmetries intertwined	Thm. 4.2(5)
$\zeta_{K_1} = \zeta_{K_2}$ ($S_i = P_{K_i}$)	true, gauge-equivariantly	Cor. 4.3
$h_{K_1}^+ = h_{K_2}^+$	not claimed	Cor. 4.3

The question was well posed and its answer is a change of shape. Norm-separation cannot be removed from Theorem 1.4 of Chapter 21, because Theorem 1.3 shows it is that theorem's conclusion in disguise; what can be done is to replace σ -equivariance by gauge-equivariance, whereupon the reconstruction holds with the exact strength of Paper XIII's Theorem 2.5 for every number field and every prime set, including the classical case $S = P_K$ that Proposition 1.5 shows was previously excluded — with what was conditional there (the Frobenius classes, when the symmetry actions are not known to be intertwined) still conditional, and the narrow class number still not recovered.

Two questions remain, and the first is the sharp form of the original one. Is $C(Y_S)$ intrinsically characterized inside $(\mathcal{A}_{K,S}, \sigma)$ when $L \neq 0$? Remark 3.3 shows the obvious counterexample is blocked by the Nica–Toeplitz relations, so we suspect the answer is yes and that the mechanism is the rigidity of commuting isometries rather than any spectral argument. Second: the L -graded algebra $\mathcal{A}^\sigma$ of Theorem 2.2 is a new object in this programme, being the first non-commutative fixed-point algebra to appear; whether its own structure — for instance whether it is simple, or its ideal lattice — encodes the relation lattice L , and hence the pattern of coincidences among the norms, we have not examined.

CHAPTER 23

What σ -equivariance gives for free, and why the range projections cannot give the rest

This chapter is Paper XV of the series (papers/XV-spectral/).

SUMMARY. We examine the proposal to prove Cartan uniqueness for $(\mathcal{A}_{K,S}, \sigma)$ by spectral means alone, and separate what it yields from what it cannot.

On the positive side, σ -equivariance already gives more than Chapter 21 (its Theorem 2.5) extracted from it, and gives it without any Cartan input. The eigenvalue group $\mathrm{Sp}_p(\sigma)$ of the flow is the subgroup of $\mathcal{R}_{>0}$ generated by $\{N\mathfrak{p}\}_{\mathfrak{p} \in S}$; and the entire KMS bundle is defined from $(\mathcal{A}, \sigma)$ alone, so a σ -equivariant isomorphism transports it. Consequently the function $\beta \mapsto G_S/\Xi_\beta^\perp$, up to homeomorphism, is an invariant: in particular $|\Xi_\beta|$ wherever it is finite, the transition locus, and β_c . *No Cartan subalgebra is needed for any of this.* What Cartan uniqueness is needed for is exactly one thing: the norm *multiset*, since $\mathrm{Sp}_p(\sigma)$ sees only the group generated by the norms and is blind to multiplicity — two equal-norm primes and one prime contribute the same subgroup.

On the negative side, the proposed route cannot deliver it, and the obstruction is at Step 3 rather than Step 2. The semilattice $\mathcal{E} = \{e_a\}$ of range projections does *not* generate $C(Y_S)$: since $e_a = \mathbf{1}_{aY_S}$ is the condition $v_{\mathfrak{p}}(x) \geq a_{\mathfrak{p}}$, these functions depend only on the valuations and take equal values at $[x, g]$ and $[x, g']$ for all $g, g' \in G_S$. They do not separate points, and $C^*(\mathcal{E}) \cong C(\overline{\mathbb{N}}^S)$ is a proper subalgebra, blind to the whole G_S -direction — which by the exact sequence $1 \rightarrow U_S/\overline{\mathcal{O}_{K,+}^\times} \rightarrow G_S \rightarrow \mathrm{Cl}_S^+ \rightarrow 1$ is most of Y_S . A complete spectral recovery of the range projections would therefore still not give $C(Y_S)$, and this failure is independent of whether Step 2 holds.

On Step 2 itself the verdict is negative, and no ergodic theory is needed. For equal-norm primes $\mathfrak{p}_1, \mathfrak{p}_2$ the element $v = \mu_{\mathfrak{p}_1} e_{\mathfrak{p}_1} + \mu_{\mathfrak{p}_2}(1 - e_{\mathfrak{p}_1})$, with $e_{\mathfrak{p}_1} = \mu_{\mathfrak{p}_1} \mu_{\mathfrak{p}_1}^*$, is an isometry of pure frequency — $\mu_{\mathfrak{p}_1}^*(1 - e_{\mathfrak{p}_1}) = 0$, so the cross terms vanish identically — and it is not diagonal. Mixed isometries therefore exist for every equal-norm pair. The general criterion, for $v = \mu_{\mathfrak{p}_1} p + \mu_{\mathfrak{p}_2}(1 - p)$ with $g = \mathfrak{p}_2 \mathfrak{p}_1^{-1}$ and $U = \mathrm{supp}(1 - p)$, is $v^* v = 1 \iff g(U \cap \mathfrak{p}_1 Y_S) \subseteq U$; whether the measure-preserving element g ($Ng = 1$) is ergodic on $(\tilde{Y}_S, \nu_\beta)$ remains open, but it no longer governs the existence question.

The gauge-equivariant theorem of Chapter 22 (its Theorem 4.2) therefore remains the strongest unconditional statement, and the norm multiset remains the single quantity that σ -equivariance alone has not been shown to recover.

1. Spectral subspaces and what they see

We use the notation of [Ch. 1]: $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$, spanned by the monomials $\mu_a f \mu_b^*$, with $\sigma_t(\mu_a) = N\mathfrak{a}^{it} \mu_a$; $\tilde{Y}_S$ and the quasi-invariant measures ν_β are as there. For $\lambda > 0$ let

$$\mathcal{A}(\lambda) = \{x \in \mathcal{A}_{K,S} : \sigma_t(x) = \lambda^{it} x \ \forall t\} = \overline{\mathrm{span}}\{\mu_a f \mu_b^* : N\mathfrak{a}/N\mathfrak{b} = \lambda, f \in C(Y_S)\}$$

be the spectral subspaces of the flow in the sense of Arveson [7].

PROPOSITION 1.1 (The point spectrum). *$\mathrm{Sp}_p(\sigma) = \{\lambda > 0 : \mathcal{A}(\lambda) \neq 0\}$ is the subgroup of $(\mathcal{R}_{>0}, \times)$ generated by $\{N\mathfrak{p}\}_{\mathfrak{p} \in S}$. It is an invariant of the C^* -dynamical system, requiring no Cartan subalgebra. The Arveson spectrum proper — the closed support of the spectral subspaces*

— is the closure of this subgroup, hence all of $\mathcal{R}_{>0}$ as soon as two of the $\log N\mathfrak{p}$ have irrational ratio; the eigenvalue group $\mathrm{Sp}_p(\sigma)$, not its closure, is the invariant that retains the information.

PROOF. Every $\lambda = N\mathfrak{a}/N\mathfrak{b}$ is realized by the eigenvector $\mu_{\mathfrak{a}}f\mu_{\mathfrak{b}}^*$, and these ratios form exactly the subgroup generated by the $N\mathfrak{p}$. Conversely let $0 \neq x \in \mathcal{A}(\lambda)$ and pick a functional φ with $\varphi(x) \neq 0$. Then $t \mapsto \varphi(\sigma_t(x)) = \lambda^{it}\varphi(x)$ is the uniform limit of the functions $t \mapsto \varphi(\sigma_t(x_n))$ for finite sums x_n of monomials, which are trigonometric polynomials with frequencies in $\log\langle N\mathfrak{p} \rangle$; the Bohr–Fourier coefficient at $\log \lambda$ is continuous for uniform convergence, vanishes on every x_n if $\lambda \notin \langle N\mathfrak{p} \rangle$, and equals $\varphi(x) \neq 0$ on the limit. So $\lambda \in \langle N\mathfrak{p} \rangle$. Invariance is immediate from $\Phi \circ \sigma_t^1 = \sigma_t^2 \circ \Phi$. $\square$

REMARK 1.2 (What the spectrum misses). $\mathrm{Sp}_p(\sigma)$ records the group generated by the norms, not the multiset. If $\mathfrak{p}_1 \neq \mathfrak{p}_2$ have $N\mathfrak{p}_1 = N\mathfrak{p}_2 = q$, the subgroup contributed is $q^{\mathbb{Z}}$, the same as for a single prime of norm q . Multiplicity is exactly the information that the spectrum discards, and by §2 it is the only piece of the reconstruction that is not already free.

2. What σ -equivariance gives for free

PROPOSITION 2.1 (The KMS bundle is intrinsic). *The set $\mathrm{KMS}_{\beta}(\mathcal{A}, \sigma)$ is defined from $(\mathcal{A}, \sigma)$ alone. Hence a σ -equivariant isomorphism Φ induces, for every β , an affine homeomorphism $\mathrm{KMS}_{\beta}(\mathcal{A}_1, \sigma_1) \rightarrow \mathrm{KMS}_{\beta}(\mathcal{A}_2, \sigma_2)$, with no hypothesis on Cartan subalgebras.*

THEOREM 2.2 (Free consequences). *Let $\Phi : (\mathcal{A}_{K_1, S_1}, \sigma_1) \rightarrow (\mathcal{A}_{K_2, S_2}, \sigma_2)$ be a σ -equivariant isomorphism. Then, unconditionally — for all number fields, all prime sets, separated or not:*

- (1) $\mathrm{Sp}_p(\sigma_1) = \mathrm{Sp}_p(\sigma_2)$, so the subgroups of $\mathcal{R}_{>0}$ generated by the norms agree;
- (2) for every $\beta > 0$ the spaces $G_{S_1}/\Xi_{\beta}(K_1, S_1)^{\perp}$ and $G_{S_2}/\Xi_{\beta}(K_2, S_2)^{\perp}$ are homeomorphic, being the extreme boundaries of the corresponding simplices $\mathrm{Prob}(G_S/\Xi_{\beta}^{\perp})$ of [Ch. 1, Thm. 3.6]; in particular, taking any $\beta > \beta_c$, the compact spaces G_{S_1} and G_{S_2} are homeomorphic;
- (3) in particular $|\Xi_{\beta}(K_1, S_1)| = |\Xi_{\beta}(K_2, S_2)|$ whenever these are finite;
- (4) the transition loci coincide, and $\beta_c(S_1) = \beta_c(S_2)$.

PROOF. (1) is Proposition 1.1. (2)–(4) follow from Proposition 2.1: the extreme boundary of a Bauer simplex is a homeomorphism invariant, and by [Ch. 1, Thm. 3.6] it is $G_S/\Xi_{\beta}^{\perp}$, which for $\beta > \beta_c$ is G_S itself since $\zeta_{K,S}(\beta) < \infty$ forces $\Xi_{\beta} = \widehat{G_S}$; the transition locus is the set of β at which the simplex changes, and β_c is determined with it — this is the argument of Theorem 2.5(5) of Chapter 21, which uses only the transport of KMS states available here and no Cartan input. $\square$

REMARK 2.3 (The gap is exactly the multiset). Comparing Theorem 2.2 with Theorem 2.5 of Chapter 21, the conclusions there not obtained here are the norm-preserving bijection $S_1 \rightarrow S_2$ of its parts (2)–(3) and, in its conditional part (6), the matching of Frobenius classes and the transport of $\{\Xi_{\beta}\}$ as subgroups of $\widehat{G_S}$ — here only the cardinalities and the quotient spaces come for free. The narrow class number is a conclusion in neither place: by Remark 3.2 of Chapter 21, h_S^+ is recovered by no argument there, and nothing here adds one. Everything else — G_S as a compact space, the chain $\{\Xi_{\beta}\}$ in cardinality, the transition locus, β_c — comes free from σ -equivariance. So the Cartan question is not a prerequisite for the arithmetic reconstruction in general; it is a prerequisite for one specific piece of it, the multiplicity of the norms, and by Remark 1.2 that is precisely what $\mathrm{Sp}_p(\sigma)$ cannot see.

3. Mixed isometries exist: Step 2 fails

Assume $\mathfrak{p}_1 \neq \mathfrak{p}_2$ in S with $N\mathfrak{p}_1 = N\mathfrak{p}_2 = q$, and put $g = \mathfrak{p}_2\mathfrak{p}_1^{-1} \in I_S$.

LEMMA 3.1 (Nica relation). *For coprime $\mathfrak{p}_1, \mathfrak{p}_2$ one has $\mu_{\mathfrak{p}_1}^* \mu_{\mathfrak{p}_2} = \mu_{\mathfrak{p}_2} \mu_{\mathfrak{p}_1}^*$.*

PROOF. $\mathfrak{p}_1 \vee \mathfrak{p}_2 = \mathfrak{p}_1 \mathfrak{p}_2$, and the Nica covariance relation [100] $\mu_{\mathfrak{a}}^* \mu_{\mathfrak{b}} = \mu_{\mathfrak{a}^{-1}(\mathfrak{a} \vee \mathfrak{b})} \mu_{\mathfrak{b}^{-1}(\mathfrak{a} \vee \mathfrak{b})}^*$ gives the claim. $\square$

PROPOSITION 3.2 (Mixed isometries exist; the general criterion). *Let $p \in C(Y_S)$ be a projection, $U = \text{supp}(1 - p)$, and $v = \mu_{\mathfrak{p}_1} p + \mu_{\mathfrak{p}_2} (1 - p)$. Then*

$$v^* v = 1 + \mathbf{1}_{U^c \cap g(U \cap \mathfrak{p}_1 Y_S)} \mu_{\mathfrak{p}_2} \mu_{\mathfrak{p}_1}^* + \text{adjoint}, \quad \text{so} \quad v^* v = 1 \iff g(U \cap \mathfrak{p}_1 Y_S) \subseteq U,$$

where g acts as the partial homeomorphism $\mathfrak{p}_1 Y_S \rightarrow \mathfrak{p}_2 Y_S$. In particular the condition is vacuous when $U \cap \mathfrak{p}_1 Y_S = \emptyset$: for $p = e_{\mathfrak{p}_1}$, i.e. $U = Y_S \setminus \mathfrak{p}_1 Y_S$, where $e_{\mathfrak{p}_1} = \mu_{\mathfrak{p}_1} \mu_{\mathfrak{p}_1}^* = \mathbf{1}_{\mathfrak{p}_1 Y_S}$,

$$v = \mu_{\mathfrak{p}_1} e_{\mathfrak{p}_1} + \mu_{\mathfrak{p}_2} (1 - e_{\mathfrak{p}_1})$$

is an isometry, since $\mu_{\mathfrak{p}_1}^* (1 - e_{\mathfrak{p}_1}) = \mu_{\mathfrak{p}_1}^* - \mu_{\mathfrak{p}_1}^* \mu_{\mathfrak{p}_1} \mu_{\mathfrak{p}_1}^* = 0$ kills the cross terms outright, and it is homogeneous for σ of frequency q but not of a single gauge degree, hence not of the form $\mu_{\mathfrak{p}} f$. Mixed isometries of pure frequency exist for every equal-norm pair, with no further hypothesis.

PROOF. Expanding and using Lemma 3.1, $v^* v = p + (1 - p) + p \mu_{\mathfrak{p}_2} \mu_{\mathfrak{p}_1}^* (1 - p) + \text{adjoint}$. Pushing the functions through the partial isometry $\mu_{\mathfrak{p}_2} \mu_{\mathfrak{p}_1}^*$, which implements $g : \mathfrak{p}_1 Y_S \rightarrow \mathfrak{p}_2 Y_S$, turns the third term into $\mathbf{1}_{U^c} \mathbf{1}_{g(U \cap \mathfrak{p}_1 Y_S)} \mu_{\mathfrak{p}_2} \mu_{\mathfrak{p}_1}^*$, which lies in the spectral subspace of gauge degree $\mathfrak{p}_2 - \mathfrak{p}_1 \neq 0$ and so is linearly independent of $C(Y_S)$; it vanishes iff $U^c \cap g(U \cap \mathfrak{p}_1 Y_S) = \emptyset$, i.e. $g(U \cap \mathfrak{p}_1 Y_S) \subseteq U$. The special case is the display, and v has gauge degrees $\delta_{\mathfrak{p}_1}$ and $\delta_{\mathfrak{p}_2}$ both occurring, while any $\mu_{\mathfrak{p}} f$ is homogeneous. $\square$

REMARK 3.3 (Status). Step 2 of the proposed route — “every isometry of pure frequency is diagonal” — is therefore *false*, by a two-line example, and the failure needs no ergodic theory: the projection $e_{\mathfrak{p}_1}$, being a range projection, is triggered by the valuations alone, and mixing over it costs nothing. An earlier draft of this note claimed that mixed isometries exist *iff* the element $g = \mathfrak{p}_2 \mathfrak{p}_1^{-1}$ fails to be ergodic on $(\tilde{Y}_S, \nu_\beta)$; both implications fail — the example above uses no invariant set, and non-ergodicity produces measurable, not clopen, invariant sets. What survives of the ergodic picture is this: $Ng = 1$, so g preserves each ν_β (the Radon–Nikodym cocycle of the I_S -action being $(Ng)^{-\beta}$), the full group I_S acts ergodically by the Kolmogorov zero–one law [Ch. 1, Lem. 3.1], and whether a *single* such g — shifting $v_{\mathfrak{p}_2}$ up, $v_{\mathfrak{p}_1}$ down, and translating G_S by $\sigma_{\mathfrak{p}_2} \sigma_{\mathfrak{p}_1}^{-1}$ — is ergodic remains open and of independent interest; but it no longer decides the existence of mixed isometries, which is settled affirmatively.

4. The decisive gap: range projections are blind to G_S

THEOREM 4.1 ($\mathcal{E}$ does not generate the diagonal). *Let $\mathcal{E} = \{e_{\mathfrak{a}} = \mathbf{1}_{\mathfrak{a} Y_S} : \mathfrak{a} \in J_S\}$. Then $\mathcal{E}$ does not separate the points of Y_S : for every $\mathfrak{a}$ and all $g, g' \in G_S$,*

$$e_{\mathfrak{a}}([x, g]) = e_{\mathfrak{a}}([x, g']).$$

Consequently $C^*(\mathcal{E}) \cong C(\overline{\mathbb{N}}^S)$, the algebra of continuous functions of the valuations alone, and is a proper subalgebra of $C(Y_S)$ whenever G_S is nontrivial.

PROOF. $\mathfrak{a} Y_S = \{[x, g] : v_{\mathfrak{p}}(x) \geq a_{\mathfrak{p}} \ \forall \mathfrak{p}\}$, a condition on x only, so $e_{\mathfrak{a}}$ factors through the projection $Y_S \rightarrow \overline{\mathbb{N}}^S$, $[x, g] \mapsto (v_{\mathfrak{p}}(x))_{\mathfrak{p}}$. The $e_{\mathfrak{a}}$ generate the clopen algebra of $\overline{\mathbb{N}}^S$, whence the identification by Stone–Weierstrass; properness follows since $G_S \neq \{1\}$ for every $S \neq \emptyset$, by [Ch. 1, (2.1)]. $\square$

COROLLARY 4.2 (Step 3 fails independently of Step 2). *Even if every isometry of pure frequency q were diagonal — and by Proposition 3.2 it is not — so that the range projections $e_{\mathfrak{p}}$ were spectrally recoverable, the algebra they generate is $C(\overline{\mathbb{N}}^S)$ and not $C(Y_S)$. The proposed route*

therefore cannot establish $\Phi(C(Y_{S_1})) = C(Y_{S_2})$, the identity needed to bring Renault's reconstruction [108] to bear, and this failure is located at Step 3, independently of Step 2.

REMARK 4.3 (How much is missed). By [Ch. 1, (2.1)], G_S sits in $1 \rightarrow U_S/\overline{\mathcal{O}_{K,+}^\times} \rightarrow G_S \rightarrow \mathrm{Cl}_S^+ \rightarrow 1$ with $U_S = \prod_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}^\times$ an infinite product of infinite profinite groups. The invisible direction is therefore not a small correction but a factor comparable in size to Y_S itself. Any route to $C(Y_S)$ must produce elements that vary along G_S , and the range projections, being defined by valuation inequalities, never do.

REMARK 4.4 (What would have to replace it). Elements of $\mathcal{A}$ that vary along G_S do exist in abundance — all of $C(Y_S)$ does — but they are not distinguished by the flow, σ acting trivially on $C(Y_S)$. To characterize them one would need a construction using the multiplication of $\mathcal{A}$ against the spectral subspaces, for instance identifying $C(Y_S)$ inside $\mathcal{A}^\sigma$ as a distinguished maximal abelian subalgebra. That is the question left open in Remark 3.3 of Chapter 22, and the present note does not advance it; what it does is show that the range projections are not a route to it.

5. Assessment

Statement	Status	Reference
$\mathrm{Sp}_p(\sigma) = \langle N\mathfrak{p} \rangle \leq \mathcal{R}_{>0}$	true, intrinsic	Prop. 1.1
$\mathrm{Sp}_p(\sigma)$ sees the norm multiset	false: no multiplicity	Rem. 1.2
KMS bundle intrinsic to $(\mathcal{A}, \sigma)$	true	Prop. 2.1
G_S , $\{ \Xi_\beta \}$, transition loci, β_c recovered	true, unconditionally	Thm. 2.2
h_S^+ recovered	not claimed (as in Chapter 21)	Rem. 2.3
norm multiset recovered	open; the only gap	Rem. 2.3
mixed isometries exist	true, unconditionally	Prop. 3.2
every pure-frequency isometry is diagonal	false	Prop. 3.2
mixed isometries $\iff g$ non-ergodic	false, both ways	Rem. 3.3
$\mathcal{E}$ generates $C(Y_S)$	false: blind to G_S	Thm. 4.1
spectral route to Cartan uniqueness	closed	Cor. 4.2
σ -equivariance $\Rightarrow$ gauge-equivariance	not established	—

Two conclusions, of opposite sign, and the positive one is the more useful.

The reconstruction of Chapter 21 (its Theorem 2.5) was presented as resting on Cartan uniqueness throughout. Theorem 2.2 shows that most of it does not: the KMS bundle is manifestly intrinsic to $(\mathcal{A}, \sigma)$, and with it come G_S as a compact space, the cardinalities of the obstruction groups, the transition locus and the critical exponent, for every number field and every prime set, separated or not. The Cartan question is needed for one thing only, the multiplicity of the norms, which is exactly what the point spectrum discards.

The proposed spectral route to that last piece is closed twice over. Step 2 is false outright: mixed isometries of pure frequency exist for every equal-norm pair (Proposition 3.2). And even its truth would have delivered only the semilattice $\mathcal{E}$, which by Theorem 4.1 is blind to G_S , generating a proper subalgebra of the diagonal. The unconditional statement therefore remains Theorem 4.2 of Chapter 22, with gauge-equivariance as hypothesis.

Two questions. Is $g = \mathfrak{p}_2\mathfrak{p}_1^{-1}$ ergodic on $(\tilde{Y}_S, \nu_\beta)$? By Remark 3.3 this no longer decides the existence of mixed isometries, but it is a clean self-contained problem about a single measure-preserving transformation of a product of geometric measures twisted by a group translation, and we have not decided it. And is there any construction recovering the G_S -direction of $C(Y_S)$ from $(\mathcal{A}, \sigma)$? Remark 4.4 explains why the obvious candidates do not, the flow being trivial on precisely the algebra one is trying to find.

Quantum optimal transport on Bost–Connes systems: the twist is harmful, the generator has a sign error, and the transition is invisible to local inequalities

This chapter is Paper XVI of the series (papers/XVI-transport/).

SUMMARY. We examine three proposals for a quantum optimal transport theory on the type III Bost–Connes systems and find that each fails, in each case for an identifiable and correctable reason. Two of the corrections are clean; the third is a structural obstruction.

The modular twist is not merely unnecessary but harmful. With $D_{\text{vert}}\epsilon_a = (\log N\mathfrak{a})\epsilon_a$ and $\vartheta = \sigma_{-i\beta}$, one computes

$$[D_{\text{vert}}, \mu_{\mathfrak{p}}]_{\vartheta} \epsilon_b = (\log N\mathfrak{p} + (1 - N\mathfrak{p}^{\beta}) \log N\mathfrak{b}) \epsilon_{\mathfrak{p}\mathfrak{b}},$$

which is *unbounded*. The untwisted commutator is bounded and elementary: $D_{\text{vert}}\mu_{\mathfrak{p}} = \mu_{\mathfrak{p}}(D_{\text{vert}} + \log N\mathfrak{p})$ gives $[D_{\text{vert}}, \mu_{\mathfrak{p}}] = (\log N\mathfrak{p})\mu_{\mathfrak{p}}$. The reason is that a twist repairs a *multiplicative* mismatch, whereas D_{vert} is *additive* in $\log N$; introducing ϑ manufactures the mismatch it was meant to cure. The untwisted $D_{\text{full}} = D_{\text{base}} \otimes \sigma_1 + D_{\text{vert}} \otimes \sigma_2$ does have $\ker L = \mathbb{C}1$ — and, as it turns out, exactly when S is norm-separated, the same condition that governs the equal-norm analysis of Chapter 22 (its Theorem 1.3). Whether it is a spectral triple reduces to whether the horizontal operator has compact resolvent, which is not established.

The proposed Lindbladian does not preserve the KMS state. Writing $a_{\mathfrak{p}}, b_{\mathfrak{p}}$ for the rates of $\mu_{\mathfrak{p}}^*(\cdot)\mu_{\mathfrak{p}}$ and $\mu_{\mathfrak{p}}(\cdot)\mu_{\mathfrak{p}}^*$, the KMS condition gives $\varphi(\mathcal{L}f) = \sum_{\mathfrak{p}} \{\nu(fe_{\mathfrak{p}})(a_{\mathfrak{p}}N\mathfrak{p}^{\beta} - b_{\mathfrak{p}}) + \nu(f)(b_{\mathfrak{p}}N\mathfrak{p}^{-\beta} - a_{\mathfrak{p}})\}$, which vanishes for all f if and only if $b_{\mathfrak{p}} = a_{\mathfrak{p}}N\mathfrak{p}^{+\beta}$. The proposal takes $b_{\mathfrak{p}} = a_{\mathfrak{p}}N\mathfrak{p}^{-\beta}$: the sign of the exponent is wrong, and with it $\varphi(\mathcal{L}e_{\mathfrak{p}}) = (1 - t^2)(1 - t) \neq 0$ for $t = N\mathfrak{p}^{-\beta}$. The corrected ratio $b_{\mathfrak{p}}/a_{\mathfrak{p}} = e^{\beta \log N\mathfrak{p}}$ is the detailed-balance factor, as it must be.

Critical slowing down does not occur. The proposed asymptotic $\alpha(\beta) \asymp \Pi_{\infty}(\beta) = \prod_{\ell} \frac{1 - \ell^{-\beta}}{1 + \ell^{-\beta}}$ is of the wrong shape: spectral gaps, modified log-Sobolev constants and Bakry–Émery curvature bounds *tensorise to an infimum* of single-site quantities, each a function of its own $\ell^{-\beta}$ alone, not to a product over ℓ . The single-site defects $1 - \ell^{-\beta}$ are bounded below by $1 - 3^{-\beta} \geq 0.66$, and the single-site spectral gap by $(1 - t)^2/4t$ (the Poincaré constant of the geometric law), so with unit rates the gap stays bounded below near β_c while $\Pi_{\infty} \rightarrow 0$. The arithmetic transition is a *tail* phenomenon — the Kakutani singularity of an infinite product — and such tensorising local quantities cannot see it. This is the same mechanism that made the metric transition of Chapter 6 (its Theorem 6.2) continuous rather than first-order.

1. The twist

We use the notation of [Ch. 1]: $\mathcal{A}_{\mathbb{Q}, S} = C(Y_S) \rtimes J_S$, spanned by the monomials $\mu_a f \mu_b^*$, with $\sigma_t(\mu_a) = N\mathfrak{a}^{it}\mu_a$; X and ν_{sym} are the compact model and its symmetric measure. Represent $\mathcal{A}_{\mathbb{Q}, S}$ on $\mathcal{H}_0 = L^2(X, \nu_{\text{sym}}) \otimes \ell^2(J_S)$ in the Toeplitz representation, $\mu_a \epsilon_b = \epsilon_{\mathfrak{a}\mathfrak{b}}$, and set $D_{\text{vert}}\epsilon_a = (\log N\mathfrak{a})\epsilon_a$. Let $\vartheta = \sigma_{-i\beta}$, so $\vartheta(\mu_a) = N\mathfrak{a}^{\beta}\mu_a$ and $\vartheta(f) = f$ for $f \in C(Y_S)$.

PROPOSITION 1.1 (The twisted commutator is unbounded). *For every prime $\mathfrak{p}$ and every $\beta > 0$,*

$$(1.1) \quad [D_{\text{vert}}, \mu_{\mathfrak{p}}]_{\vartheta} \epsilon_b = (D_{\text{vert}}\mu_{\mathfrak{p}} - \vartheta(\mu_{\mathfrak{p}})D_{\text{vert}})\epsilon_b = (\log N\mathfrak{p} + (1 - N\mathfrak{p}^{\beta}) \log N\mathfrak{b}) \epsilon_{\mathfrak{p}\mathfrak{b}},$$

whose coefficient tends to $-\infty$ as $N\mathfrak{b} \rightarrow \infty$. Hence $[D_{\text{vert}}, \mu_{\mathfrak{p}}]_{\vartheta} \notin \mathcal{B}(\mathcal{H}_0)$.

PROOF. $D_{\text{vert}}\mu_{\mathfrak{p}}\epsilon_{\mathfrak{b}} = \log N(\mathfrak{p}\mathfrak{b})\epsilon_{\mathfrak{p}\mathfrak{b}} = (\log N\mathfrak{p} + \log N\mathfrak{b})\epsilon_{\mathfrak{p}\mathfrak{b}}$ and $\vartheta(\mu_{\mathfrak{p}})D_{\text{vert}}\epsilon_{\mathfrak{b}} = N\mathfrak{p}^{\beta}(\log N\mathfrak{b})\epsilon_{\mathfrak{p}\mathfrak{b}}$; subtract. Since $N\mathfrak{p}^{\beta} \neq 1$ the coefficient is unbounded in $\mathfrak{b}$. $\square$

PROPOSITION 1.2 (The untwisted commutator is bounded). $D_{\text{vert}}\mu_{\mathfrak{a}} = \mu_{\mathfrak{a}}(D_{\text{vert}} + \log N\mathfrak{a})$, hence

$$[D_{\text{vert}}, \mu_{\mathfrak{a}}] = (\log N\mathfrak{a})\mu_{\mathfrak{a}}, \quad \|[D_{\text{vert}}, \mu_{\mathfrak{a}}]\| = \log N\mathfrak{a} < \infty,$$

and $[D_{\text{vert}}, f] = 0$ for $f \in C(Y_S)$.

PROOF. Both sides send $\epsilon_{\mathfrak{b}}$ to $(\log N\mathfrak{a} + \log N\mathfrak{b})\epsilon_{\mathfrak{a}\mathfrak{b}}$; the vanishing on $C(Y_S)$ is the observation recorded in Chapter 6 (its Proposition 2.1), D_{vert} being diagonal in a basis with respect to which $C(Y_S)$ acts diagonally. $\square$

REMARK 1.3 (Why the twist backfires). A Connes–Moscovici twist [30] repairs a *multiplicative* discrepancy: it is used when conjugation by the algebra rescales D , so that $Dx - xD$ fails to be bounded by a factor, and ϑ absorbs the factor. Here D_{vert} is *additive* in $\log N$, so $Dx - xD$ is already bounded, and inserting the multiplicative $\vartheta(\mu_{\mathfrak{a}}) = N\mathfrak{a}^{\beta}\mu_{\mathfrak{a}}$ introduces a factor where there was none. Equation (1.1) is precisely the manufactured mismatch. The twisted triple, with twist $\sigma_{i\beta}$, was the candidate proposed in the closing assessment of Chapter 6; this section is the reason we do not pursue it.

REMARK 1.4 (Numerical illustration). For $N\mathfrak{p} = 3$, $\beta = 1$ the coefficient in (1.1) takes the values -3.5 , -12.7 , -26.5 , -54.2 at $N\mathfrak{b} = 10, 10^3, 10^6, 10^{12}$; for $\beta = 2$, -17.3 , -54.2 , -109.4 , -219.9 . The untwisted commutator has constant norm $\log 3 = 1.0986$.

2. The untwisted operator, and the reappearance of norm-separation

Let $D_{\text{full}} = D_{\text{base}} \otimes \sigma_1 + D_{\text{vert}} \otimes \sigma_2$ on $\mathcal{H} = \mathcal{H}_0 \otimes \mathbb{C}^2$, with D_{base} the horizontal operator of Chapter 6 (its Definition 2.4), amplified to act on the $L^2(X, \nu_{\text{sym}})$ factor of every summand $L^2(X, \nu_{\text{sym}}) \otimes \epsilon_{\mathfrak{b}}$, and $L(x) = \|[D_{\text{full}}, x]\|$. In this representation $\mu_{\mathfrak{a}}$ acts on $\ell^2(J_S)$ alone and $f \in C(Y_S)$ acts on the summand indexed by $\mathfrak{b}$ as multiplication by the translate $f \circ m_{\mathfrak{b}}$, so $[D_{\text{base}}, \mu_{\mathfrak{a}}] = 0$ while $[D_{\text{base}}, f]$ is the $\mathfrak{b}$ -indexed family $[D_{\text{base}}, f \circ m_{\mathfrak{b}}]$.

PROPOSITION 2.1 (Kernel). *The operators σ_1, σ_2 anticommute and x acts diagonally, so $[D_{\text{full}}, x] = [D_{\text{base}}, x] \otimes \sigma_1 + [D_{\text{vert}}, x] \otimes \sigma_2$, which vanishes iff both terms do. By Proposition 1.2, $\ker[D_{\text{vert}}, \cdot] = \mathcal{A}_{\mathbb{Q}, S}^{\sigma}$. Hence*

$$\ker L = \{x \in \mathcal{A}^{\sigma} : [D_{\text{base}}, x] = 0\},$$

which equals $\mathbb{C}1$ if and only if $\mathcal{A}^{\sigma} = C(Y_S)$, i.e. if and only if S is norm-separated in the sense of Theorem 1.3 of Chapter 22.

PROOF. $[D_{\text{vert}}, \mu_{\mathfrak{a}}f\mu_{\mathfrak{b}}^*] = (\log N\mathfrak{a} - \log N\mathfrak{b})\mu_{\mathfrak{a}}f\mu_{\mathfrak{b}}^*$, zero iff $N\mathfrak{a} = N\mathfrak{b}$, so the kernel of the vertical commutator is exactly the fixed-point algebra of σ . For $f \in C(Y_S)$, $[D_{\text{base}}, f] = 0$ forces $[D_{\text{base}}, f \circ m_1] = 0$ on the summand $\mathfrak{b} = (1)$, hence $f \in \mathbb{C}1$ by the Lip-kernel lemma of Chapter 6 (its Lemma 3.1); if $\mathcal{A}^{\sigma} \supsetneq C(Y_S)$ the element $\mu_{\mathfrak{a}}\mu_{\mathfrak{b}}^*$ with $N\mathfrak{a} = N\mathfrak{b}$, $\mathfrak{a} \neq \mathfrak{b}$, lies in $\ker[D_{\text{vert}}, \cdot]$ and, acting on $\ell^2(J_S)$ alone, also in $\ker[D_{\text{base}}, \cdot]$. $\square$

REMARK 2.2 (Spectral triple: not established). Since D_{base} and $1 \otimes D_{\text{vert}}$ commute and are placed on anticommuting Clifford directions, $D_{\text{full}}^2 = D_{\text{base}}^2 \otimes 1 + 1 \otimes D_{\text{vert}}^2$, and on the summand $L^2(X, \nu_{\text{sym}}) \otimes \epsilon_{\mathfrak{a}}$ this is $D_{\text{base}}^2 + (\log N\mathfrak{a})^2$. Over $\mathbb{Q}$ the norms $N\mathfrak{a}$ are distinct and tend to infinity, so D_{full} has compact resolvent if and only if D_{base} does on $L^2(X, \nu_{\text{sym}})$ — a property of the weight sequence of Chapter 6, which that paper does not establish and we do not claim; its own assessment leaves the spectral triple on the crossed product open, and so do we. In particular

no index theory is asserted. The pair $(\mathcal{A}_{\mathbb{Q},S}, L)$ can still be a compact quantum metric space in Rieffel's sense [112] — which requires only that the Lipschitz ball be totally bounded modulo scalars — and it is that notion, not the spectral triple, that is at issue here; the two are often conflated.

REMARK 2.3 (Open Problem 1 of Chapter 6). Proposition 2.1 settles the kernel question but not the compact quantum metric space question, which additionally requires total boundedness of $\{x : L(x) \leq 1, \varphi(x) = 0\}$. The vertical directions have Lipschitz cost bounded below by $\log N\mathfrak{p}_0$ for the smallest prime, which is encouraging; we have not verified total boundedness and do not claim it.

3. The generator: a sign error in the detailed balance

PROPOSITION 3.1 (Invariance forces the detailed-balance ratio). *Let*

$$\mathcal{L}(x) = \sum_{\mathfrak{p} \in S} a_{\mathfrak{p}}(\mu_{\mathfrak{p}}^* x \mu_{\mathfrak{p}} - x) + \sum_{\mathfrak{p} \in S} b_{\mathfrak{p}} \left(\mu_{\mathfrak{p}} x \mu_{\mathfrak{p}}^* - \frac{1}{2}(e_{\mathfrak{p}} x + x e_{\mathfrak{p}}) \right), \quad e_{\mathfrak{p}} = \mu_{\mathfrak{p}} \mu_{\mathfrak{p}}^*,$$

the GKLS form with jump operators $\mu_{\mathfrak{p}}$ and $\mu_{\mathfrak{p}}^$. Then $\mathcal{L}(1) = 0$, and for $f \in C(Y_S)$ — where $\frac{1}{2}(e_{\mathfrak{p}} f + f e_{\mathfrak{p}}) = e_{\mathfrak{p}} f$ — the KMS condition $\varphi(xy) = \varphi(y \sigma_{i\beta}(x))$ gives $\varphi(\mu_{\mathfrak{p}}^* x \mu_{\mathfrak{p}}) = N\mathfrak{p}^{\beta} \varphi(x e_{\mathfrak{p}})$ and $\varphi(\mu_{\mathfrak{p}} x \mu_{\mathfrak{p}}^*) = N\mathfrak{p}^{-\beta} \varphi(x)$ — the scaling relations recorded as Proposition 5.1 of Chapter 4 — whence*

$$(3.1) \quad \varphi(\mathcal{L}(f)) = \sum_{\mathfrak{p}} \left\{ \nu(f e_{\mathfrak{p}}) (a_{\mathfrak{p}} N\mathfrak{p}^{\beta} - b_{\mathfrak{p}}) + \nu(f) (b_{\mathfrak{p}} N\mathfrak{p}^{-\beta} - a_{\mathfrak{p}}) \right\}.$$

This vanishes for every f if and only if

$$(3.2) \quad b_{\mathfrak{p}} = a_{\mathfrak{p}} N\mathfrak{p}^{+\beta} \quad \text{for every } \mathfrak{p} \in S.$$

PROOF. $\mathcal{L}(1) = \sum_{\mathfrak{p}} a_{\mathfrak{p}}(\mu_{\mathfrak{p}}^* \mu_{\mathfrak{p}} - 1) + \sum_{\mathfrak{p}} b_{\mathfrak{p}}(e_{\mathfrak{p}} - e_{\mathfrak{p}}) = 0$. The two scaling relations are the KMS condition with $y = \mu_{\mathfrak{p}}$, respectively $y = \mu_{\mathfrak{p}}^*$, and $\sigma_{i\beta}(\mu_{\mathfrak{p}}) = N\mathfrak{p}^{-\beta} \mu_{\mathfrak{p}}$. For (3.1) apply them termwise. If (3.2) holds, both brackets vanish. Conversely write $A_{\mathfrak{p}} = a_{\mathfrak{p}} N\mathfrak{p}^{\beta} - b_{\mathfrak{p}}$ and $B = \sum_{\mathfrak{p}} (b_{\mathfrak{p}} N\mathfrak{p}^{-\beta} - a_{\mathfrak{p}})$; under ν the valuations are independent with $\nu(e_{\mathfrak{p}}) = t_{\mathfrak{p}} = N\mathfrak{p}^{-\beta}$ and $\nu(e_{\mathfrak{p}} e_{\mathfrak{q}}) = t_{\mathfrak{p}} t_{\mathfrak{q}}$ for $\mathfrak{p} \neq \mathfrak{q}$, so $f = 1$ gives $\sum_{\mathfrak{p}} A_{\mathfrak{p}} t_{\mathfrak{p}} + B = 0$ while $f = e_{\mathfrak{q}}$ gives $t_{\mathfrak{q}}(A_{\mathfrak{q}} + \sum_{\mathfrak{p} \neq \mathfrak{q}} A_{\mathfrak{p}} t_{\mathfrak{p}} + B) = 0$; subtracting, $A_{\mathfrak{q}}(1 - t_{\mathfrak{q}}) = 0$, so $A_{\mathfrak{q}} = 0$ for every $\mathfrak{q}$, and then $B = 0$ term by term. $\square$

COROLLARY 3.2 (The proposed generator is not invariant). *The choice $a_{\mathfrak{p}} = w_{\mathfrak{p}}$, $b_{\mathfrak{p}} = w_{\mathfrak{p}} N\mathfrak{p}^{-\beta}$ gives $b_{\mathfrak{p}} = a_{\mathfrak{p}} N\mathfrak{p}^{-\beta}$, violating (3.2): the sign of the exponent is reversed. Explicitly, with $t = N\mathfrak{p}^{-\beta}$ and $w_{\mathfrak{p}} = 1$: for every $\mathfrak{q} \neq \mathfrak{p}$ the proposed rates satisfy $A_{\mathfrak{q}} t_{\mathfrak{q}} + B_{\mathfrak{q}} = (t_{\mathfrak{q}}^{-1} - t_{\mathfrak{q}}) t_{\mathfrak{q}} + (t_{\mathfrak{q}}^2 - 1) = 0$, so the $\mathfrak{q}$ -terms of (3.1) cancel at $f = e_{\mathfrak{p}}$ and the value is carried by the $\mathfrak{p}$ -term alone:*

$$\varphi(\mathcal{L}(e_{\mathfrak{p}})) = t(t^{-1} - t) + t(t^2 - 1) = (1 - t^2)(1 - t) > 0$$

for $t \in (0, 1)$; numerically 0.5926 at $(N\mathfrak{p}, \beta) = (3, 1)$, 0.8779 at $(3, 2)$, 0.9033 at $(5, 1.5)$, 0.8397 at $(7, 1)$. With $b_{\mathfrak{p}} = a_{\mathfrak{p}} N\mathfrak{p}^{+\beta}$ the same expression is exactly 0.

REMARK 3.3 (What the corrected ratio means). Condition (3.2) reads $b_{\mathfrak{p}}/a_{\mathfrak{p}} = e^{\beta \log N\mathfrak{p}}$: the backward rate exceeds the forward rate by the Boltzmann factor of the energy $\log N\mathfrak{p}$ transferred by $\mu_{\mathfrak{p}}$. This is quantum detailed balance in the sense standard for quantum Markov semigroups [16], and it is the only choice compatible with φ_{β} . The proposal's $N\mathfrak{p}^{-\beta}$ is the anti-detailed-balance choice, which drives the system away from equilibrium.

Note the price: (3.2) requires $\sum_{\mathfrak{p}} a_{\mathfrak{p}} N\mathfrak{p}^{\beta} < \infty$ for the second sum to converge in norm, so the weights must decay faster than $N\mathfrak{p}^{-\beta}$. That is a genuine constraint but is satisfiable, for instance by $a_{\mathfrak{p}} = N\mathfrak{p}^{-\beta-1-\epsilon}$. We have verified invariance, $\varphi \circ \mathcal{L} = 0$; full KMS-symmetry of the

semigroup with respect to φ_β in the sense of Cipriani [19] is a stronger statement, which we have not checked.

4. No critical slowing down

THEOREM 4.1 (Tensorisation to an infimum). *Let $\mathcal{L} = \sum_{\mathfrak{p}} a_{\mathfrak{p}} \mathcal{L}_{\mathfrak{p}}$ act on the product system $(V, \pi_\beta) = \bigotimes_{\mathfrak{p}} (\mathbb{N}, \mu_{t_{\mathfrak{p}}})$ with $t_{\mathfrak{p}} = N \mathfrak{p}^{-\beta}$, each $\mathcal{L}_{\mathfrak{p}}$ the birth-death generator on its own factor with the corrected rates $(a_{\mathfrak{p}}, a_{\mathfrak{p}} t_{\mathfrak{p}}^{-1})$ of (3.2). Then:*

- (1) *the spectral gap satisfies $\text{gap}(\mathcal{L}) = \inf_{\mathfrak{p}} a_{\mathfrak{p}} \text{gap}(\mathcal{L}_{\mathfrak{p}}/a_{\mathfrak{p}})$, and by the Poincaré inequality for the geometric law (Chapter 6, its Lemma 6.1, with sharp constant $C(t) = 4t/(1-t)^2$)*

$$\text{gap}(\mathcal{L}_{\mathfrak{p}}/a_{\mathfrak{p}}) \geq \frac{(1-t_{\mathfrak{p}})^2}{4t_{\mathfrak{p}}} \geq \frac{(1-t_{\mathfrak{p}})^2}{4};$$

- (2) *the modified log-Sobolev constant satisfies $\alpha(\beta) = \inf_{\mathfrak{p}} a_{\mathfrak{p}} \alpha_0(t_{\mathfrak{p}})$, where $\alpha_0(t)$ is the single-site constant of the normalized chain: entropy is subadditive over the product measure and the Dirichlet form is additive, giving $\geq$, and single-site test functions give $\leq$.*

Both constants are therefore infima of single-site quantities, each a function of its own $t_{\mathfrak{p}}$ alone — not products over $\mathfrak{p}$. In particular: with rates bounded below, $\inf_{\mathfrak{p}} a_{\mathfrak{p}} > 0$, the gap stays $\geq \inf_{\mathfrak{p}} a_{\mathfrak{p}} (1 - 3^{-\beta})^2/4 > 0$ uniformly for β in a neighbourhood of β_c ; with the norm-summable rates of Remark 3.3 the infima vanish for every β , for the weight reason alone and β -independently. In neither reading does any of these constants acquire the product shape $\Pi_\infty(\beta)$, whose vanishing at β_c is driven by the divergence of $\sum_{\ell} \ell^{-\beta}$, a tail quantity invisible to an infimum.

PROOF. For commuting single-site generators on a product measure, the Dirichlet form and the variance resp. entropy split: $\text{Var} = \sum$ over sites of averaged single-site variances bounds the gap from below by the infimum, and single-site eigenfunctions attain it; the same subadditivity of entropy (a product measure satisfies $\text{Ent}(f) \leq \sum_{\mathfrak{p}} \mathbb{E} \text{Ent}_{\mathfrak{p}}(f)$) with the additive Dirichlet form gives $\alpha \geq \inf a_{\mathfrak{p}} \alpha_0(t_{\mathfrak{p}})$, and functions of a single coordinate give $\leq$. The single-site gap bound is Lemma 6.1 of Chapter 6, whose Dirichlet form $\mathbb{E}_{\mu_t} |\Delta h|^2$ is that of the normalized chain with rates $(1, t^{-1})$. $\square$

COROLLARY 4.2 (The proposed asymptotic is of the wrong shape). $\Pi_\infty(\beta) = \prod_{\ell \in R} \frac{1-\ell^{-\beta}}{1+\ell^{-\beta}}$ is a product over all ℓ and tends to 0 as $\beta \downarrow \beta_c$, precisely because the series $\sum_{\ell} \ell^{-\beta}$ diverges there. By Theorem 4.1 the gap and MLSI constants are infima of single-site quantities, which do not degenerate this way: the single-site defect $1 - \ell^{-\beta}$ never falls below $1 - 3^{-\beta} \geq 0.66$, so with unit rates the gap infimum stays $\geq (1 - 3^{-\beta})^2/4 \geq 0.11$. Numerically, over the primes $3 \leq \ell \leq 10^4$,

β	$\inf_{\ell} (1 - \ell^{-\beta})$	Π_∞
1.5	0.8076	$3.70 \cdot 10^{-1}$
1.2	0.7324	$1.27 \cdot 10^{-1}$
1.05	0.6845	$3.58 \cdot 10^{-2}$
1.001	0.6670	$1.86 \cdot 10^{-2}$

The product collapses; the single-site defect does not fall below $1 - 3^{-\beta} \geq 0.66$. Hence $\alpha(\beta) \not\asymp \Pi_\infty(\beta)$, and there is no critical slowing down of this kind.

COROLLARY 4.3 (Bakry–Émery). *The same argument applies to the curvature-dimension condition: for product systems with additive generators the tensorised curvature bound is again an infimum of single-site quantities, each a function of its own $t_{\mathfrak{p}}$ alone. Whatever the single-site value, the bound therefore has no product shape, and no degeneration tied to β_c beyond what the weights impose β -independently: there is no Ricci-flat singularity at the transition.*

REMARK 4.4 (Why the transition is invisible). The arithmetic phase transition is the Kakutani dichotomy: an infinite product of single-site affinities degenerates although every factor stays bounded away from 0 (Chapter 6, its Lemma 5.1). It is therefore a *tail* event, detectable only by quantities sensitive to infinitely many coordinates at once. Modified log-Sobolev constants and Bakry–Émery curvature are local: they tensorise, so they see only the worst single site. The same mechanism produced Theorem 6.2 of Chapter 6, where the metric transition was continuous rather than first-order because the order parameter is a tail observable outside the Lipschitz domain, and Theorem 2.3 of Chapter 4, where Ξ_β was identified as an invariant of a measure class. Three different local invariants, one tail phenomenon, three failures to detect it.

5. Assessment

Proposal	Status	Reference
$[D_{\text{full}}, x]_\vartheta$ bounded	false: unbounded	Prop. 1.1
untwisted $[D_{\text{vert}}, \mu_a]$ bounded	true	Prop. 1.2
the twist is needed	no; it creates the problem	Rem. 1.3
$\ker L = \mathbb{C}1$	true iff S norm-separated	Prop. 2.1
$(\mathcal{A}, \mathcal{H}, D_{\text{full}})$ a spectral triple	not established: reduces to D_{base}	Rem. 2.2
$\mathcal{A}_{\mathbb{Q}, S}$ a compact quantum metric space	not established	Rem. 2.3
$\varphi \circ \mathcal{L}_\beta = 0$ as proposed	false: sign error	Cor. 3.2
$b_p = a_p N p^{+\beta}$	the correction	Prop. 3.1
KMS-symmetry (Cipriani sense)	not checked	Rem. 3.3
$\alpha(\beta) \asymp \Pi_\infty(\beta)$	false: infimum, not product	Thm. 4.1, Cor. 4.2
critical slowing down	does not occur	Cor. 4.2
$\kappa(\beta) \rightarrow 0$ at β_c	false	Cor. 4.3

Two of the three failures are repairable and the repairs are one-line: drop the twist, and flip the sign in the detailed-balance ratio. The third is not a repair but a structural statement, and it is the one worth keeping. Modified log-Sobolev and Bakry–Émery tensorise, hence are determined by the worst single coordinate, hence cannot detect a Kakutani dichotomy, which is by construction a statement about infinitely many coordinates whose individual contributions are all harmless; and the Connes distance of Chapter 6 is blind to the same transition in the complementary sense of its Theorem 4.2, the tail order parameter and all its truncations lying outside every Lipschitz domain. A quantum optimal transport theory that detects the arithmetic transition would have to be built from a genuinely non-tensorising quantity, and we do not know one.

Three questions. Is $(\mathcal{A}_{\mathbb{Q}, S}, L)$ with the untwisted D_{full} a compact quantum metric space, i.e. is the Lipschitz ball totally bounded? Does the corrected generator of Proposition 3.1 satisfy KMS-symmetry in the sense of Cipriani [19], and if so what is the associated non-tracial carré du champ? And is there any transport-theoretic quantity that fails to tensorise — the natural candidates being those built from the tail σ -algebra, where by [Ch. 2, Prop. 7.2] the order parameter lives, rather than from a sum of single-site Dirichlet forms.

Part 5

Non-commutative semigroups and GL_2

A non-commutative Ore LCM semigroup with arithmetic thermodynamics: the Hurwitz quaternions

This chapter is Paper X of the series ([papers/X-semigroup-search/](#)).

SUMMARY. The localized Bost–Connes programme is blocked in three directions. A commutative index semigroup makes the groupoid amenable, and Connes’ classification of injective factors then erases the arithmetic entirely (Chapter 20). Additive structure makes the range projections partition, so the Cuntz relation forces $\beta = 1$ and destroys the high-temperature phase structure (Chapter 3). A free non-commutative monoid has orthogonal ranges, whence $\sum_{\mathfrak{p}} N\mathfrak{p}^{-\beta} \leq 1$ and no equilibrium state exists above the critical point at all (Chapter 14).

We isolate the algebraic conditions that avoid all three, and exhibit a monoid satisfying them. Two observations shape the requirements. First, the condition that the range projections be nested rather than partitioning is not an extra hypothesis: a right Ore semigroup has common right multiples, so in a right LCM semigroup $e_a e_b = e_{a \vee b} \neq 0$ always, and no Cuntz relation can hold. This also re-diagnoses the free monoid, whose real defect is the failure of the Ore condition rather than non-commutativity. Second, every multiplicative norm factors through the abelianization, so the available arithmetic is that of P^{ab} — which rules out any finitely generated candidate, the obstruction series then being a finite sum.

The multiplicative monoid $\mathcal{H} \setminus \{0\}$ of the Hurwitz quaternions satisfies every requirement. It is non-commutative and cancellative; right Ore, since $a(\bar{a}b) = b \text{Nrd}(a)$; right LCM, since $\mathcal{H}$ is a noncommutative principal ideal domain; its unit group has order 24, so the partition function converges somewhere, in contrast with $M_n(\mathbb{Z}) \cap GL_n^+(\mathbb{Q})$ whose unit group $GL_n(\mathbb{Z})$ is infinite; the reduced norm is multiplicative and takes every prime as a value, giving infinitely many multiplicatively independent norms; the growth is polynomial of degree two; and

$$\zeta_P(\beta) = 24 \sum_{n \geq 1} \sigma_{\text{odd}}(n) n^{-\beta} = 24 \zeta(\beta) (1 - 2^{1-\beta}) \zeta(\beta - 1),$$

a Solomon-type zeta function with a pole at $\beta = 2$: the critical exponent is finite and positive. The group of fractions $\mathbb{H}(\mathbb{Q})^\times$ is non-amenable.

We close by delimiting what remains. Non-amenability of the group of fractions is necessary but not sufficient for escaping Chapter 20: what matters there is amenability of the *measured* groupoid, and boundary actions of non-amenable groups are routinely amenable, $L^\infty(\partial \mathcal{F}_n) \rtimes \mathcal{F}_n$ being injective. We state this as a fourth condition (C4) and leave it open; in our view it is the question on which a genuinely non-commutative Bost–Connes theory now turns. Condition (C4) is settled, in the negative, in Chapter 29: the orbit relation is hyperfinite for every admissible monoid, so the search closed there.

1. The algebraic requirements

DEFINITION 1.1 (The target). We seek a monoid P with:

- (C1) P non-commutative, cancellative, right Ore, $aP \cap bP \neq \emptyset$ for all a, b — so that a group of right fractions $G = PP^{-1}$ exists — and right LCM, meaning $aP \cap bP$ is empty or equal to cP for some c [100, 78];

- (C2) in the Toeplitz representation $v_a^* v_a = 1$, $e_a = v_a v_a^*$, the range projections nested, $e_a e_b = e_{a \vee b}$, with no finite family satisfying $\sum_i e_{a_i} = 1$;
- (C3) a multiplicative $N : P \rightarrow \mathcal{R}_{>0}$ for which $\zeta_P(\beta) = \sum_{a \in P} N(a)^{-\beta}$ has a finite abscissa of convergence $\beta_c \in (0, \infty)$.

PROPOSITION 1.2 ((C2) follows from (C1)). *Let P satisfy (C1). Then $a \vee b$ exists for all $a, b \in P$, the projections satisfy $e_a e_b = e_{a \vee b} \neq 0$, and no relation $\sum_i e_{a_i} = 1$ with orthogonal e_{a_i} and at least two nonzero terms can hold. In particular the Toeplitz algebra of P admits no Cuntz relation.*

PROOF. The Ore condition gives $aP \cap bP \neq \emptyset$ for all a, b ; the LCM condition then gives $aP \cap bP = (a \vee b)P$. In the Nica–Toeplitz representation on $\ell^2(P)$ the projection e_a has range $\overline{\text{span}}\{\delta_x : x \in aP\}$, so $e_a e_b$ is the projection onto $aP \cap bP = (a \vee b)P$, namely $e_{a \vee b} \neq 0$. Orthogonality would require $aP \cap bP = \emptyset$, contradicting Ore. $\square$

REMARK 1.3 (Re-diagnosis of the free monoid). Proposition 1.2 shows that (C2) costs nothing once (C1) is imposed, and identifies what actually goes wrong in the free variant of Chapter 14. A free monoid is right LCM in the quasi-lattice sense [100] but is *not* Ore: two distinct generators have no common right multiple. That failure is what makes the ranges orthogonal, and orthogonality is what forces $\sum_{\mathfrak{p}} N\mathfrak{p}^{-\beta} \leq 1$ and empties the high-temperature region. The obstruction is the failure of the Ore condition, not non-commutativity as such.

PROPOSITION 1.4 (Norms factor through the abelianization). *Any multiplicative homomorphism $N : P \rightarrow (\mathcal{R}_{>0}, \times)$ factors through the abelianization P^{ab} . Hence the norms available to (C3) are exactly the elements of $\text{Hom}(P^{\text{ab}}, \mathcal{R}_{>0})$.*

PROOF. $\mathcal{R}_{>0}$ is abelian, so N annihilates all commutators. $\square$

REMARK 1.5 (What (C1)–(C3) already exclude). Two consequences narrow the search sharply, and we record them because they explain why the candidate of §2 is arithmetic rather than combinatorial. If P is finitely generated then the obstruction series of [Ch. 1, (1.1)], which runs over generators, is a finite sum, convergent for every β ; so Ξ_β is constant in β and no phase transition occurs. And if the unit group $P^\times$ is infinite then, N being 1 on units, $\zeta_P(\beta) \geq |P^\times| = \infty$ for every β and (C3) fails. A candidate must therefore be infinitely generated with finite unit group, and by Proposition 1.4 its abelianization must carry infinitely many multiplicatively independent norm values — the analogue of the $\mathbb{Q}$ -linear independence of the $\log N\mathfrak{p}$ used throughout [Ch. 2].

2. The Hurwitz quaternion monoid

Let $\mathbb{H}(\mathbb{Q})$ be the rational quaternion algebra, $\bar{a}$ the conjugate, $\text{Nrd}(a) = a\bar{a} \in \mathbb{Z}_{\geq 0}$ the reduced norm, and

$$\mathcal{H} = \mathbb{Z} \left\langle 1, i, j, \frac{1+i+j+k}{2} \right\rangle$$

the Hurwitz order [61, 107]. Put $P := \mathcal{H} \setminus \{0\}$ under multiplication.

- THEOREM 2.1 (P satisfies (C1)–(C3)). (1) P is non-commutative and cancellative, being a subsemigroup of the multiplicative group of a division algebra.
- (2) P is right Ore: for $a, b \in P$,

$$a(\bar{a}b) = (a\bar{a})b = \text{Nrd}(a)b = b\text{Nrd}(a),$$

since $\text{Nrd}(a)$ is a central integer; so $aP \cap bP \neq \emptyset$.

- (3) P is right LCM: $\mathcal{H}$ is Euclidean for Nrd , hence a noncommutative principal ideal domain, so every right ideal is principal and $a\mathcal{H} \cap b\mathcal{H} = c\mathcal{H}$.
- (4) $P^\times$ is the group of 24 Hurwitz units, in particular finite.

- (5) Nrd is multiplicative with $\text{Nrd}|_{P^\times} = 1$; its values on irreducibles are exactly the rational primes, whose logarithms are $\mathbb{Q}$ -linearly independent.
- (6) Writing $a_n = \#\{a \in P : \text{Nrd}(a) = n\} = 24\sigma_{\text{odd}}(n)$,
- $$(2.1) \quad \zeta_P(\beta) = 24 \sum_{n \geq 1} \sigma_{\text{odd}}(n) n^{-\beta} = 24 \zeta(\beta) (1 - 2^{1-\beta}) \zeta(\beta - 1),$$

convergent for $\beta > 2$ and divergent as $\beta \downarrow 2$; the critical exponent is $\beta_c = 2$.

- (7) The growth is polynomial of degree two: $\#\{a \in P : \text{Nrd}(a) \leq x\} \sim Cx^2$.

PROOF. (1) is immediate. (2) is the displayed identity, $\text{Nrd}(a)$ lying in the centre $\mathbb{Q}$ of $\mathbb{H}(\mathbb{Q})$ and in $\mathbb{Z}$ for $a \in \mathcal{H}$. (3) is the classical fact that $\mathcal{H}$ is left and right Euclidean for the reduced norm [61], whence a principal ideal domain on both sides; an intersection of right ideals is a right ideal, hence principal, and nonzero by (2). (4) is the classical count of Hurwitz units, namely $\pm 1, \pm i, \pm j, \pm k$ and the sixteen $\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$. (5): multiplicativity is $\text{Nrd}(ab) = \text{Nrd}(a) \text{Nrd}(b)$; an element is irreducible exactly when its reduced norm is prime, and every prime occurs; the $\mathbb{Q}$ -linear independence of $\{\log p\}$ is unique factorization in $\mathbb{Z}$.

For (6), the count $a_n = 24\sigma_{\text{odd}}(n)$ is the Hurwitz–Jacobi formula [61]. Its Dirichlet series is

$$\sum_{n \geq 1} \sigma_{\text{odd}}(n) n^{-s} = \sum_{d \text{ odd}} \sum_{m \geq 1} d (dm)^{-s} = \left(\sum_{d \text{ odd}} d^{1-s} \right) \zeta(s) = (1 - 2^{1-s}) \zeta(s-1) \zeta(s),$$

whose only pole in $\Re s > 1$ is that of $\zeta(s-1)$ at $s = 2$. (7) follows by Abel summation from $\sum_{n \leq x} \sigma_{\text{odd}}(n) \asymp x^2$. $\square$

REMARK 2.2 (Numerical verification). The counts a_n were obtained by exhaustive enumeration of $\mathcal{H}$ over both the integral and the half-integral coordinate classes, and agree with $24\sigma_{\text{odd}}(n)$ for every $n \leq 15$:

$$24, 24, 96, 24, 144, 96, 192, 24, 312, 144, 288, 96, 336, 192, 576.$$

Identity (2.1) was checked against partial sums to $n = 20000$, agreeing to eight decimal places at $\beta = 4$ and $\beta = 6$. The pole at $\beta = 2$ is visible in

$$\zeta_P(2.20) = 112.96, \quad \zeta_P(2.10) = 211.44, \quad \zeta_P(2.05) = 408.72, \quad \zeta_P(2.02) = 1000.83.$$

The ratio $\#\{\text{Nrd} \leq x\}/x^2$ took the values 9.868, 9.880, 9.872 at $x = 200, 500, 1000$, confirming (7).

REMARK 2.3 (The Toeplitz relations). The associated algebra is the Nica–Toeplitz algebra $\mathcal{T}(P)$ [100, 78], generated by isometries v_a , $a \in P$, subject to

$$v_a v_b = v_{ab}, \quad v_a^* v_a = 1, \quad (v_a v_a^*)(v_b v_b^*) = v_{a \vee b} v_{a \vee b}^*,$$

the third by Proposition 1.2, together with unitarity of v_u for the 24 units. The dynamics is $\sigma_t(v_a) = \text{Nrd}(a)^{it} v_a$. This is well defined and is a one-parameter group of $*$ -automorphisms preserving all three relations: multiplicativity of Nrd gives $\sigma_t(v_a v_b) = \sigma_t(v_{ab})$; and since $\text{Nrd}(u) = 1$ for each of the 24 units,

$$\sigma_t(v_u) = v_u \quad (u \in P^\times, t \in \mathbb{R}),$$

so σ_t fixes the unitaries v_u pointwise. In particular σ_t is constant on each principal right ideal, a and au having equal reduced norm, which is what makes the dynamics descend to the projections e_a indexed by $P/P^\times$. By Proposition 1.2 no Cuntz relation constrains β : the high-temperature region $\beta < 2$ is not excluded on those grounds, in contrast with both the affine systems of Chapter 3 and the free variant of Chapter 14. The ring C^* -algebra of [34] is the algebra of the affine kind, with the additive structure that forces the Cuntz relation, and is not the object here.

PROPOSITION 2.4 (The group of fractions is non-amenable). $G = PP^{-1} = \mathbb{H}(\mathbb{Q})^\times$ is non-amenable.

PROOF. Every element of $\mathbb{H}(\mathbb{Q})^\times$ is ab^{-1} with $a \in \mathcal{H}$ and $b \in \mathbb{Z} \setminus \{0\}$, so $G = \mathbb{H}(\mathbb{Q})^\times$. Conjugation $x \mapsto axa^{-1}$ on the pure quaternions embeds $\mathbb{H}(\mathbb{Q})^\times/\mathbb{Q}^\times$ into $SO(3)$. The quaternions of reduced norm 5 with odd positive real part and even imaginary parts are $1 \pm 2i$, $1 \pm 2j$, $1 \pm 2k$, and by Lubotzky–Phillips–Sarnak [85, §3] the elements $1 + 2i$, $1 + 2j$, $1 + 2k$ generate a free group of rank three modulo the centre — their images in $SO(3)$ are the rotations by $\arccos(-\frac{3}{5})$ about the three coordinate axes. So $\mathbb{H}(\mathbb{Q})^\times$ contains a free subgroup of rank two and is not amenable. $\square$

REMARK 2.5 (Multiplicity at each prime). Unlike the classical setting, where each rational prime contributes a single generator, here each odd prime p carries $p + 1$ distinct principal right ideals of reduced norm p , since $a_p/|P^\times| = 24(1 + p)/24 = p + 1$ by Theorem 2.1(6).

This count is geometric. For p odd the algebra is split, $\mathbb{H} \otimes \mathbb{Q}_p \cong M_2(\mathbb{Q}_p)$, and the $p + 1$ ideals are the $p + 1$ neighbours of a vertex in the Bruhat–Tits tree of $SL_2(\mathbb{Q}_p)$, a $(p + 1)$ -regular tree, indexed by $\mathbb{P}^1(\mathcal{F}_p)$. At $p = 2$, where $\mathbb{H}$ is ramified, the count collapses to a single ideal — consistent with the factor $(1 - 2^{1-\beta})$ in (2.1), and verified numerically alongside $p + 1$ for $p = 3, 5, 7, 11, 13, 17, 19, 23$.

Two consequences. The Kakutani analysis of [Ch. 1, §3] was carried out with the generators indexed by primes, one per norm value; whether the multiplicity $p + 1$ alters it we have not examined, and it is the first thing to check before transporting the classification. And the tree picture suggests the shape a replacement would take: the obstruction series would be indexed not by primes but by edges, so the criterion should become a statement about a random walk on the product of the local trees rather than about a Dirichlet series in $N\mathfrak{p}^{-\beta}$. We record this as an interface, not a result.

REMARK 2.6 (Ramanujan graphs). The same $(p + 1)$ -regular trees, quotiented by congruence subgroups of the unit group of an order in $\mathbb{H}$, are the source of the Lubotzky–Phillips–Sarnak Ramanujan graphs [85]. We have not used this and make no claim about a connection; we note only that the arithmetic input here — the Hurwitz order and its ideals of prime norm — is the same input, and that any spectral estimate needed for (C4) would plausibly meet it.

3. The remaining condition

REMARK 3.1 (What decided the collapse in Chapter 20). Conditions (C1)–(C3) are necessary but not sufficient, and they do not address the property that governed Chapter 20. There the factor was erased not because the index semigroup was commutative as such, but because the *measured* groupoid $(G \ltimes \Omega, \nu)$ was amenable, so that Connes’ classification of injective factors left only the type. Proposition 2.4 supplies non-amenability of G , which is necessary; it is far from sufficient. Boundary actions of non-amenable groups are routinely amenable in Zimmer’s sense: the action of a free group $\mathcal{F}_n$ on its Gromov boundary with any quasi-invariant measure is amenable, so $L^\infty(\partial\mathcal{F}_n) \rtimes \mathcal{F}_n$ is an injective factor. If the KMS_β measures of the Hurwitz system concentrate on a boundary of that kind, the measured groupoid is amenable, the factor is injective, and Connes’ classification applies verbatim as in Chapter 20, collapsing the factor to R_∞ , R_λ or a Krieger factor.

DEFINITION 3.2 (The fourth condition). (C4) For β in the symmetry-breaking regime the measured groupoid $(G \ltimes \Omega, \nu_\beta)$ is non-amenable; equivalently, by Anantharaman–Delaroche and Renault [3], $\pi_\varphi(\mathcal{T}(P))''$ is not injective.

REMARK 3.3 (Status, and how it might go). We have not decided (C4), and we regard it as the substance of the problem rather than a technical residue. Two considerations pull in opposite directions. For: the space Ω of a Toeplitz algebra is not a boundary. The nested projections of Proposition 1.2 mean precisely that Ω retains an interior, which is the geometric content of the

absence of a Cuntz relation, so the analogy with $\partial\mathcal{F}_n$ is not immediate. Against: for β below the critical exponent the KMS_β measures do concentrate away from that interior, which is where the boundary analogy would bite.

4. Assessment

Condition	Status for $P = \mathcal{H} \setminus \{0\}$	Reference
non-commutative, cancellative	yes	Thm. 2.1(1)
right Ore	yes: $a(\bar{a}b) = b \text{Nrd}(a)$	Thm. 2.1(2)
right LCM	yes: $\mathcal{H}$ a noncommutative PID	Thm. 2.1(3)
(C2) nested, no Cuntz relation	automatic from (C1)	Prop. 1.2
$ P^\times < \infty$	yes, $= 24$	Thm. 2.1(4)
infinitely many independent norms	yes: all primes	Thm. 2.1(5)
(C3) finite β_c	yes, $\beta_c = 2$	Thm. 2.1(6)
sub-exponential growth	yes: degree 2	Thm. 2.1(7)
G non-amenable	yes	Prop. 2.4
(C4) measured groupoid non-amenable	open	Def. 3.2

The Hurwitz order supplies a non-commutative, right Ore, right LCM monoid with a genuinely arithmetic multiplicative norm, a Solomon-type partition function with a finite critical exponent, polynomial growth, and a non-amenable group of fractions. Every algebraic obstacle that closed the earlier routes is absent: there is no Cuntz relation to force $\beta = 1$, no orthogonality to empty the high-temperature region, and the index semigroup is not commutative.

Three questions remain, in order of importance. First (C4), on which everything turns. Second, the multiplicity of Remark 2.5: the $p + 1$ right ideals of norm p have no counterpart in [Ch. 1], and the Kakutani criterion must be re-derived before the classification can be transported. Third, if (C4) holds, whether the resulting obstruction group is genuinely non-abelian, or whether the mechanism of Chapter 16 — that the image of a commutative index set in the symmetry group is abelian — reappears in some form despite the non-commutativity of P ; Chapter 17 shows that non-abelian symmetry breaking to a normal subgroup can already be produced with the commutative J_S through a non-abelian cocycle, at the price of a dependence on the lifts of the Frobenius classes, so what a non-commutative P would have to add is canonicity.

We do not claim novelty for the Hurwitz order as an object of study; Bost–Connes and Hecke-algebra constructions attached to orders in quaternion algebras have been considered, and (2.1) is a classical Solomon-type zeta function. What is claimed is the selection: that the requirements of §1 admit this model, with the properties computed in §2. Whether the KMS classification of $\mathcal{T}(\mathcal{H} \setminus \{0\})$ is already known we have not been able to verify, and it should be checked before any priority claim is made.

Condition (C4) and the nested/partition dichotomy: why Ramanujan bounds cannot decide non-amenability of the Hurwitz groupoid

This chapter is Paper XI of the series (papers/XI-C4/).

SUMMARY. Condition (C4) of Chapter 25 asks whether the measured groupoid of the Hurwitz Bost–Connes system is non-amenable, equivalently whether the associated factor escapes the injectivity that collapsed every earlier model (Chapter 20). We examine the natural route — Lubotzky–Phillips–Sarnak generators [85], the Bruhat–Tits tree, Deligne’s bound [38], a spectral gap for a Hecke–Markov operator in the sense of Kesten [70] — and show that it is closed, for reasons that are structural rather than technical.

The obstruction is a dichotomy. A $(p+1)$ -regular tree adjacency operator requires the $p+1$ branches at a vertex to *partition*; but the Ore and LCM conditions of Chapter 25 force the range projections to be *nested*, $e_j e_k = e_{a_j \vee a_k} = e_p \neq 0$ — the central element p lies in every maximal right ideal of norm p — with $\varphi(e_j e_k) = p^{-2\beta} > 0$ and $\sum_k \varphi(e_{a_k}) = (p+1)p^{-\beta} \neq 1$. The very property that keeps the high-temperature region alive — absence of a Cuntz relation — is the property that destroys the tree.

Forcing the partition does not help, and the reason is sharper than a numerical mismatch: in any unital algebra generated by isometries v_a with $v_a v_b = v_{ab}$, orthogonality of two range projections of norm p forces $e_p = 0$ and hence $1 = v_p^* v_p = 0$. *The boundary quotient of the Hurwitz system is the zero algebra.* Even a partition in mean, $\sum_k \varphi(e_{a_k}) = 1$, would force $\beta = \log(p+1)/\log p$, a value that varies with p (1.2619, 1.1133, 1.0686, ... for $p = 3, 5, 7$). Tree structure and thermodynamics are mutually exclusive here.

Two further points are recorded. The proposed operator $\mathcal{M}_p = \frac{1}{p+1} \sum_k v_{a_k}^*(\cdot) v_{a_k}$ is unital completely positive but does not preserve φ : one has $\varphi \circ \mathcal{M}_p \leq p^\beta \varphi$ with the naive bound $\|\mathcal{M}_p\|_{L^2(\varphi)} \leq p^{\beta/2} > 1$, so no contraction is available, let alone a gap below $2\sqrt{p}/(p+1)$. And the statement of the spectral gap is not well posed: for a non-singular action the Koopman representation satisfies $\pi(a)1 = N(a)^{\beta/2} 1_{a\Omega} \neq 1$, so $\mathbb{C}1$ is not invariant and $L_0^2 = L^2 \ominus \mathbb{C}1$ is not a subrepresentation.

We close with the reformulation we believe is correct: by Takesaki duality the factor is injective if and only if its continuous core is, so (C4) is equivalent to non-amenability of the *Maharam extension*, which is measure preserving. That does not settle it — non-amenable groups do have amenable infinite-measure-preserving actions — but it places the question in a measure-preserving setting, where the obstruction identified here does not apply. Condition (C4) has since been settled, in the negative, in Chapter 29, by the properness route rather than any spectral one, confirming the warning of this paper.

1. What the route requires

We keep the notation of Chapter 25: $\mathcal{H}$ the Hurwitz order, $P = \mathcal{H} \setminus \{0\}$, $G = \mathbb{H}(\mathbb{Q})^\times$, Nrd the reduced norm, $\mathcal{T}(P)$ the Nica–Toeplitz algebra with isometries v_a , projections $e_a = v_a v_a^*$, dynamics $\sigma_t(v_a) = \text{Nrd}(a)^{it} v_a$, and φ an extremal KMS_β state with $\varphi(e_a) = \text{Nrd}(a)^{-\beta}$.

For an odd prime p let $a_1, \dots, a_{p+1}$ represent the $p+1$ principal right ideals of reduced norm p , which are the $p+1$ neighbours of a vertex in the $(p+1)$ -regular Bruhat–Tits tree of $SL_2(\mathbb{Q}_p)$, $\mathbb{H} \otimes \mathbb{Q}_p$ being split for p odd (Chapter 25).

The route under examination is: (i) form $\mathcal{M}_p = \frac{1}{p+1} \sum_k v_{a_k}^*(\cdot) v_{a_k}$; (ii) identify it with a tree adjacency operator; (iii) invoke Deligne's bound to get $\|\mathcal{M}_p\|_{L_0^2} \leq 2\sqrt{p}/(p+1) < 1$; (iv) conclude non-amenability by a Kesten-type criterion. We show that (ii) and (iii) both fail, and that repairing (ii) destroys the system.

2. The projections do not partition

PROPOSITION 2.1 (Nested, not orthogonal). *Let $j \neq k$. Then*

$$e_{a_j} e_{a_k} = e_{a_j \vee a_k} \neq 0, \quad \varphi(e_{a_j} e_{a_k}) = p^{-2\beta} > 0, \quad \sum_{k=1}^{p+1} \varphi(e_{a_k}) = (p+1)p^{-\beta}.$$

In particular $\{e_{a_k}\}$ is not an orthogonal family and $\sum_k e_{a_k}$ is not a scalar, so it is not a partition of unity for any β .

PROOF. In the Nica–Toeplitz algebra $e_a e_b$ is the projection onto $aP \cap bP$, which is $(a \vee b)P$ by the LCM property, nonempty by the Ore property (Chapter 25). For distinct right ideals of reduced norm p one has $a_j \mathcal{H} \cap a_k \mathcal{H} = p\mathcal{H}$: the central element $p = a_j \bar{a}_j = a_k \bar{a}_k$ lies in both, the two ideals are distinct maximal right ideals containing $p\mathcal{H}$, so $a_j \mathcal{H} + a_k \mathcal{H} = \mathcal{H}$ and $[\mathcal{H} : a_j \mathcal{H} \cap a_k \mathcal{H}] = [\mathcal{H} : a_j \mathcal{H}][\mathcal{H} : a_k \mathcal{H}] = p^4 = [\mathcal{H} : p\mathcal{H}]$. Hence $a_j \vee a_k = p$ up to units, $e_{a_j} e_{a_k} = e_p$ and $\varphi(e_p) = \text{Nrd}(p)^{-\beta} = p^{-2\beta}$. The last identity is $\varphi(e_{a_k}) = p^{-\beta}$ summed over $p+1$ terms. $\square$

COROLLARY 2.2 (The tree picture is unavailable). *The adjacency operator of a $(p+1)$ -regular tree acts on ℓ^2 of the vertex set by summing over $p+1$ disjoint branches, and its analysis — Kesten's $2\sqrt{p}$, Deligne's bound for the LPS quotients — rests on that disjointness. By Proposition 2.1 the branches of $\mathcal{T}(P)$ overlap, with overlap mass $p^{-2\beta}$ at every pair. No identification of $\mathcal{M}_p$ with a tree adjacency operator is therefore available, and the Ramanujan bound has no object to bound.*

REMARK 2.3 (Numerical illustration). At $p = 3$: $\sum_k \varphi(e_{a_k}) = 1.3333$ and $\varphi(e_j e_k) = 0.1111$ for $\beta = 1$; 0.4444 and 0.01235 for $\beta = 2$. At $p = 13$, $\beta = 2$: 0.0828 and $3.5 \cdot 10^{-5}$. The overlaps shrink as β grows but never vanish, and the branch masses never sum to 1.

REMARK 2.4 (This is exactly the property we wanted). The failure is not incidental. The Ore and LCM conditions were imposed in Chapter 25 precisely to guarantee that no Cuntz relation holds, which is what keeps the high-temperature region $\beta < 2$ from being excluded, as it is in the affine systems of Chapter 3 and the free variant of Chapter 14. Nestedness is the feature that makes the Hurwitz model viable; and nestedness is the feature that forbids the tree. The two requirements are in direct opposition.

3. The Markov operator does not contract

PROPOSITION 3.1 ($\mathcal{M}_p$ is unital but not state-preserving). $\mathcal{M}_p(x) = \frac{1}{p+1} \sum_k v_{a_k}^* x v_{a_k}$ is unital and completely positive, and

$$(3.1) \quad \varphi(\mathcal{M}_p(x)) = \frac{p^\beta}{p+1} \varphi\left(x \sum_k e_{a_k}\right) \quad (x \in \mathcal{T}(P)).$$

Since $\sum_k e_{a_k}$ is not a scalar by Proposition 2.1, $\varphi \circ \mathcal{M}_p \neq \varphi$. Consequently $\mathcal{M}_p$ induces on $L^2(\mathcal{T}(P), \varphi)$ only the bound

$$\|\mathcal{M}_p\|_{L^2(\varphi)} \leq p^{\beta/2},$$

which exceeds 1 for every $\beta > 0$, and no contraction — still less a gap below $2\sqrt{p}/(p+1)$ — follows from these relations.

PROOF. Unitality is $\frac{1}{p+1} \sum_k v_{a_k}^* v_{a_k} = 1$; complete positivity is clear. The KMS condition with the entire element v_a^* , $\sigma_{i\beta}(v_a^*) = \text{Nrd}(a)^\beta v_a^*$, gives $\varphi(v_a^* x v_a) = \text{Nrd}(a)^\beta \varphi(x e_a)$, the scaling identity of Chapter 4, whence (3.1) with $\text{Nrd}(a_k) = p$. For the bound: e_a is fixed by σ , so $\varphi(e_a y) = \varphi(y e_a)$ for all y , whence $\varphi(x e_a) = \varphi(e_a x e_a)$ and, for $x \geq 0$, $\varphi(x) = \varphi(e_a x e_a) + \varphi(e_a^\perp x e_a^\perp) \geq \varphi(e_a x e_a)$; so $\varphi \circ \mathcal{M}_p \leq p^\beta \varphi$ on positive elements, and a unital completely positive map Φ with $\varphi \circ \Phi \leq C\varphi$ satisfies $\varphi(\Phi(x)^* \Phi(x)) \leq \varphi(\Phi(x^* x)) \leq C\varphi(x^* x)$ by the Kadison–Schwarz inequality, i.e. extends to $L^2(\varphi)$ with norm at most $C^{1/2}$. $\square$

REMARK 3.2. Proposition 3.1 is the Hurwitz instance of the phenomenon recorded in Chapter 4 (its Proposition 5.1): the transfer operator $x \mapsto v_a^* x v_a$ never preserves a KMS_β state, because the KMS condition converts it into $\text{Nrd}(a)^\beta \varphi(\cdot e_a)$. Whatever spectral theory one wishes to do with $\mathcal{M}_p$ must first address this, and normalizing by $(p+1)^{-1}$ does not.

4. The spectral gap statement is not well posed

PROPOSITION 4.1 ($\mathbb{C}1$ is not invariant). *For the non-singular action of G on (Ω, ν_β) with $\nu_\beta(a \cdot) = \text{Nrd}(a)^{-\beta} \nu_\beta$, the Koopman representation is*

$$(\pi(a)\xi)(x) = \text{Nrd}(a)^{\beta/2} \xi(a^{-1}x), \quad \text{so} \quad \pi(a)1 = \text{Nrd}(a)^{\beta/2} \mathbf{1}_{a\Omega} \neq 1$$

for $a \notin P^\times$. Hence $\mathbb{C}1$ is not a G -invariant subspace, $L_0^2 = L^2 \ominus \mathbb{C}1$ is not a subrepresentation, and the quantity $\|\mathcal{M}_p|_{L_0^2}\|$ is not the norm of an operator on an invariant subspace.

PROOF. Immediate from the Radon–Nikodym cocycle; $\mathbf{1}_{a\Omega} \neq 1$ because $a\Omega$ is a proper clopen subset, of measure $\text{Nrd}(a)^{-\beta} < 1$. $\square$

REMARK 4.2 (Why this matters). For a *measure-preserving* action the decomposition $L^2 = \mathbb{C}1 \oplus L_0^2$ is invariant and a spectral gap on L_0^2 is meaningful and does obstruct amenability. That is the setting of Kesten’s criterion and of the standard argument “non-amenable group acting measure-preservingly $\Rightarrow$ non-amenable action”. Neither applies here: our measure is quasi-invariant with a nontrivial cocycle, which is the whole point of a KMS_β state, and it is precisely for non-singular actions that non-amenable groups routinely admit amenable actions [2, 137] — the reason (C4) was posed as open in Chapter 25 rather than settled there.

5. Forcing the partition destroys the system

THEOREM 5.1 (Forcing the partition kills the algebra). *Let A be a unital C^* -algebra generated by isometries v_a , $a \in P$, with $v_a v_b = v_{ab}$, and let p be an odd prime. If two range projections e_{a_j}, e_{a_k} , $j \neq k$, of reduced norm p are orthogonal in A , then $A = 0$. In particular the “boundary quotient” $\partial\mathcal{T}(P)$ of $\mathcal{T}(P)$ by the relations $\sum_{k=1}^{p+1} e_{a_k} = 1$, for any single odd prime p , is the zero algebra, and carries no state, KMS or otherwise.*

PROOF. Since $\text{Nrd}(a_j) = p$ we have $p = a_j \bar{a}_j$ with $\bar{a}_j \in P$, so $v_p = v_{a_j} v_{\bar{a}_j}$ and $e_p = v_p v_p^* = v_{a_j} e_{\bar{a}_j} v_{a_j}^* \leq v_{a_j} v_{a_j}^* = e_{a_j}$; likewise $e_p \leq e_{a_k}$. If $e_{a_j} e_{a_k} = 0$ then $e_p \leq e_{a_j} \wedge e_{a_k} = 0$, so $v_p = e_p v_p = 0$ and $1 = v_p^* v_p = 0$. A sum of projections equal to 1 is an orthogonal sum: $e_j = e_j \sum_k e_k$ gives $\sum_{k \neq j} e_j e_k e_j = 0$ with positive summands, so $e_j e_k e_j = 0$ and $e_j e_k = 0$. Hence the relation $\sum_k e_{a_k} = 1$ already implies $1 = 0$. $\square$

REMARK 5.2 (Partition in mean). One might weaken the partition to a partition *in mean*, asking only that $\sum_k \varphi(e_{a_k}) = 1$ for a KMS_β state of $\mathcal{T}(P)$ itself. This forces $(p+1)p^{-\beta} = 1$, i.e.

$$\beta = \beta(p) := \frac{\log(p+1)}{\log p},$$

and $p \mapsto \beta(p)$ is strictly decreasing on $(1, \infty)$, since $\frac{d}{dp} \log(\log(p+1)/\log p)$ is negative there: $\beta(3) = 1.261860$, $\beta(5) = 1.113283$, $\beta(7) = 1.068622$, $\beta(11) = 1.036287$, $\beta(13) = 1.028893$, $\beta(101) = 1.002135$, $\beta(1009) = 1.000143$, decreasing to 1. Any two primes already give a contradiction, so not even the mean version of the partition is compatible with a KMS_β state at more than one prime.

COROLLARY 5.3 (The dichotomy). *For the Hurwitz system the following are mutually exclusive:*

- (1) *the range projections at some prime partition the unit, so that tree and Ramanujan methods apply;*
- (2) *the algebra is nonzero, in particular a KMS_β state exists.*

Consequently no argument that first passes to a partition can decide (C4), the object it would decide being zero.

REMARK 5.4 (Contrast with the classical $ax + b$ case). In the affine systems of Chapter 3 the Cuntz relation at $\mathbf{p}$ reads $N\mathbf{p} \cdot N\mathbf{p}^{-\beta} = 1$, giving $\beta = 1$ *independently of $\mathbf{p}$* , so a KMS state survives at the single temperature $\beta = 1$; there the $N\mathbf{p}$ translates of a fundamental domain are genuinely disjoint, the acting semigroup being $ax + b$ rather than multiplicative. Here the multiplicity is $p + 1$ while the norm is p , so the forced temperature $\log(p+1)/\log p$ depends on p and no temperature survives. The mismatch between multiplicity and norm — $p + 1$ against p , the discrepancy being exactly the ramification-free part of the Bruhat–Tits degree — is what makes the Hurwitz boundary empty where the affine boundary is not.

6. The correct reformulation

PROPOSITION 6.1 (Maharam). *(C4) is equivalent to non-amenability of the Maharam extension [87]*

$$G \curvearrowright (\tilde{\Omega} \times \mathcal{R}, \tilde{\nu}_\beta \otimes e^s ds), \quad a \cdot (x, s) = (ax, s + \beta \log \text{Nrd}(a)),$$

on the dilation $\tilde{\Omega}$ carrying the G -action and the extended measure $\tilde{\nu}_\beta$, which preserves the displayed infinite measure.

PROOF. Invariance of the displayed measure: the pushforward under a scales $\tilde{\nu}_\beta$ by $\text{Nrd}(a)^{-\beta}$ and $e^s ds$ by $e^{\beta \log \text{Nrd}(a)}$, which is the construction of the Radon–Nikodym cocycle. For the equivalence: amenability of a measured groupoid is equivalent to injectivity of its von Neumann algebra [2], and the von Neumann algebra of the Maharam extension is the continuous core $M \rtimes_{\sigma^\varphi} \mathcal{R}$ of $M = \pi_\varphi(\mathcal{T}(P))''$ [124, Ch. XII]. If M is injective so is $M \rtimes_\sigma \mathcal{R}$, a crossed product by an amenable group; if $M \rtimes_\sigma \mathcal{R}$ is injective then so is $M \otimes B(L^2(\mathcal{R})) \cong (M \rtimes_\sigma \mathcal{R}) \rtimes_{\hat{\sigma}} \mathcal{R}$ by Takesaki duality, hence M . So M is injective iff the Maharam extension is amenable. $\square$

REMARK 6.2 (Why this is progress, and why it is not a solution). The reformulation removes the obstruction of §4: on the Maharam extension the measure is invariant, so decompositions of L^2 into invariant pieces and spectral gap arguments are meaningful. It does not resolve (C4). Amenability of an infinite-measure-preserving action of a non-amenable group is possible — the translation action of G on itself with counting measure is amenable as a groupoid, being principal and proper — so non-amenability of $G = \mathbb{H}(\mathbb{Q})^\times$ still does not transfer. What would be needed is a spectral gap for the Koopman representation of the Maharam extension on the orthogonal complement of the invariant functions, and we have not computed it.

7. Assessment

Step of the proposed route	Status	Reference
$p + 1$ ideals $\leftrightarrow$ tree neighbours	true	Chapter 25
$\mathbb{H}(\mathbb{Q})^\times$ non-amenable	true, necessary	Chapter 25
the e_{a_k} partition	false: nested	Prop. 2.1
$\mathcal{M}_p$ a tree adjacency operator	false	Cor. 2.2
$\mathcal{M}_p$ preserves φ	false ; $\ \mathcal{M}_p\ \leq p^{\beta/2}$	Prop. 3.1
L_0^2 a subrepresentation	false	Prop. 4.1
$\ \mathcal{M}_p _{L_0^2}\ \leq 2\sqrt{p}/(p+1)$	not well posed	Prop. 4.1
pass to the boundary to force a partition	the quotient is zero	Thm. 5.1
(C4) resolved	no	—
(C4) $\iff$ Maharam non-amenable	true reformulation	Prop. 6.1

The route is closed, and Corollary 5.3 says something stronger than that this particular attempt fails: *no* method requiring a partition of unity into $p + 1$ pieces can decide (C4), because the object on which such a partition exists is zero. This rules out tree adjacency operators, random walks on the Bruhat–Tits building, and Ramanujan or Kesten bounds in the form used above. The feature that makes them applicable is exactly the Cuntz relation that the Ore and LCM conditions of Chapter 25 were designed to exclude.

We are left with Proposition 6.1 as the reformulation to pursue and with the following as the concrete open questions. Does the Koopman representation of the Maharam extension on the complement of the invariant functions have a spectral gap? If it does not, (C4) fails and by Connes’ classification, as in Chapter 20, the Hurwitz factor collapses to R_∞ , R_λ or a Krieger factor like all the others, which would be a definitive negative answer to the whole programme. If it does, one obtains the first non-injective arithmetic Bost–Connes factor, and the deformation/rigidity machinery that Chapter 20 found inapplicable for want of a spectral gap becomes actionable. Second: is there a variant of the Hurwitz system in which the multiplicity $p + 1$ and the norm p are brought into agreement, so that a boundary quotient with a nonempty KMS simplex exists? Remark 5.4 suggests looking for an order whose ideals of norm $N\mathfrak{p}$ number exactly $N\mathfrak{p}$; whether such an order exists we do not know, and it is the one place where the dichotomy of Corollary 5.3 might be broken rather than circumvented.

CHAPTER 27

The universal multiplicity–norm mismatch, and the reduction of (C4) to the norm-one subgroup

This chapter is Paper XII of the series ([papers/XII-mismatch/](#)).

SUMMARY. Chapter 26 left two questions. We settle the first completely and reduce the second to a sharply posed problem, without solving it.

First question. Is there an order in a central division algebra whose number $d_{\mathfrak{p}}$ of principal right ideals of reduced norm $\mathfrak{p}$ equals $N\mathfrak{p}$, so that a boundary quotient with a nonempty KMS simplex could exist? The answer is no, and the obstruction is elementary and universal. At a prime where a maximal order in a central simple algebra of degree d has local index e , $\Lambda \otimes \mathcal{O}_{\mathfrak{p}} \cong M_{d/e}(\mathcal{O}_{D_e})$, the right ideals of reduced norm $\mathfrak{p}$ are the hyperplanes of $\mathcal{F}_{q^e}^{d/e}$, $q = N\mathfrak{p}$; hence

$$d_{\mathfrak{p}} = \frac{q^d - 1}{q^e - 1}, \quad d_{\mathfrak{p}} \geq 1 + q > q \text{ for } e < d, \quad d_{\mathfrak{p}} = 1 < q \text{ for } e = d,$$

and $d_{\mathfrak{p}} \neq N\mathfrak{p}$ at every prime, for every $d \geq 2$. For $d = 2$ the excess at a split prime is exactly 1: the off-by-one of $\mathbb{P}^1$. Eichler orders do not repair this: at a prime exactly dividing the level the local tree collapses to an edge and $d_{\mathfrak{p}} = 2$, which equals $N\mathfrak{p}$ only when $N\mathfrak{p} = 2$ — a match at one prime, never at almost all. The boundary quotient forcing the $d_{\mathfrak{p}}$ range projections to partition is in fact the zero algebra, a central element of reduced norm $\mathfrak{p}^d$ lying in every right ideal of reduced norm $\mathfrak{p}$; and even a partition in mean would force a temperature $\beta(\mathfrak{p}) = \log d_{\mathfrak{p}} / \log q$ that decreases strictly with q to $d - 1$.

Second question. Condition (C4) asks whether the measured groupoid, equivalently its Maharam extension, is non-amenable. We make two structural observations that reduce it. The Radon–Nikodym cocycle is $\text{Nrd}(a)^{-\beta}$ and therefore *factors through the reduced norm*, so the norm-one subgroup $G^1 = \ker(\text{Nrd})$ — which is non-amenable, containing free groups — acts *measure preservingly*. The non-amenable directions of G and the thermodynamic direction are orthogonal. And in the Fourier decomposition of the Maharam extension over the flow variable, the fibre representations are $\pi_{\tau} = \chi_{\tau} \otimes \pi_0$ with $\chi_{\tau}(a) = \text{Nrd}(a)^{i\beta\tau}$ a character; since $\chi_{\tau}|_{G^1} = 1$ for every τ , *all fibres restrict to one and the same representation of G^1* . The τ -integral is invisible to G^1 , and by Kuhn’s theorem (C4) follows from a single weak-containment statement: $\pi_0|_{G^1}$ not weakly contained in the regular representation of G^1 .

We do not settle it, and we say precisely why the expected arguments fail. Zimmer’s theorem that an essentially free amenable action preserving a *probability* measure forces the group to be amenable does not extend to infinite invariant measures, and the invariant measure here is infinite; and a spectral gap for $\pi_0|_{G^1}$ would not suffice, since amenable actions have tempered Koopman representations — the translation action of F_2 on itself preserves counting measure, is essentially free, has a spectral gap, and is an amenable groupoid.

1. Local multiplicities

Let K be a number field, A a central simple K -algebra of degree d , so $\dim_K A = d^2$, and $\Lambda \subset A$ an order all of whose right ideals are principal, as condition (C1) of Chapter 25 requires; the Hurwitz order is one [107, 128]. For a finite prime $\mathfrak{p}$ of K with residue field $\mathcal{F}_q$, $q = N\mathfrak{p}$, write

$$d_{\mathfrak{p}}(\Lambda) := \#\{\text{right ideals } a\Lambda \subseteq \Lambda \text{ with } \text{Nrd}(a)\mathcal{O}_K = \mathfrak{p}\},$$

the number of right ideals of reduced norm $\mathfrak{p}$, a local quantity: right ideals of Λ of reduced norm $\mathfrak{p}$ correspond to right ideals of $\Lambda \otimes \mathcal{O}_{\mathfrak{p}}$ of reduced norm $\mathfrak{p}$, by the local–global correspondence for lattices [107].

LEMMA 1.1 (Local multiplicities). *Let Λ be maximal and let $e \mid d$ be the local index of A at $\mathfrak{p}$, so that $\Lambda \otimes \mathcal{O}_{\mathfrak{p}} \cong M_r(\mathcal{O}_D)$ with $r = d/e$ and $\mathcal{O}_D$ the maximal order of the central division algebra D of index e over $K_{\mathfrak{p}}$, whose residue field is $\mathcal{F}_{q^e}$ [107, Ch. 3]. Then*

$$(1.1) \quad d_{\mathfrak{p}} = |\mathbb{P}^{r-1}(\mathcal{F}_{q^e})| = \frac{q^d - 1}{q^e - 1} = 1 + q^e + \cdots + q^{e(r-1)}.$$

In particular $d_{\mathfrak{p}} = 1 + q + \cdots + q^{d-1}$ at a split prime ($e = 1$) and $d_{\mathfrak{p}} = 1$ at a totally ramified prime ($e = d$).

PROOF. Right ideals of $M_r(\mathcal{O}_D)$ are the sets of matrices whose rows lie in a right $\mathcal{O}_D$ -submodule $L \subseteq \mathcal{O}_D^r$, and $L \mapsto$ this ideal is a bijection onto the right ideals. A generator a of the ideal has reduced norm of $\mathfrak{p}$ -adic valuation equal to the length of $\mathcal{O}_D^r/L$ as an $\mathcal{O}_D$ -module, the simple $\mathcal{O}_D$ -module being $\mathcal{O}_D/\Pi\mathcal{O}_D \cong \mathcal{F}_{q^e}$ with Π the uniformizer of D , $\text{Nrd}_D(\Pi)$ having valuation 1. So the ideals of reduced norm $\mathfrak{p}$ are the L with $\mathcal{O}_D^r/L \cong \mathcal{F}_{q^e}$, i.e. the preimages of the hyperplanes of the $\mathcal{F}_{q^e}$ -vector space $(\mathcal{O}_D/\Pi)^r = \mathcal{F}_{q^e}^r$, parametrized by the dual projective space, of cardinality (1.1). $\square$

THEOREM 1.2 (Universal multiplicity–norm mismatch). *Let Λ be a maximal order in a central simple algebra of degree $d \geq 2$. Then at every prime with local index $e < d$,*

$$d_{\mathfrak{p}} - N\mathfrak{p} = 1 + q^e + \cdots + q^{e(r-1)} - q \geq 1 > 0,$$

with equality to 1 exactly when $d = 2$ and $\mathfrak{p}$ is split; and at a totally ramified prime $d_{\mathfrak{p}} = 1 < N\mathfrak{p}$. In particular $d_{\mathfrak{p}} \neq N\mathfrak{p}$ for every $d \geq 2$ and every $\mathfrak{p}$, and there is no maximal order in a central simple algebra for which the multiplicity matches the norm at even one prime.

PROOF. By Lemma 1.1, for $e < d$ we have $r \geq 2$ and $d_{\mathfrak{p}} - q = 1 + q^e + \cdots + q^{e(r-1)} - q$; if $e = 1$ and $d = 2$ this is $1 + q - q = 1$; if $e = 1$ and $d \geq 3$ it is $1 + q^2 + \cdots + q^{d-1} \geq 1 + q^2 > 1$; if $e \geq 2$ it is at least $1 + q^e - q > 1$. For $e = d$, $d_{\mathfrak{p}} = 1 < q$. $\square$

PROPOSITION 1.3 (Eichler and hereditary orders). *Let $d = 2$ and let Λ be an Eichler order of level $\mathfrak{N}$. At a prime $\mathfrak{p} \parallel \mathfrak{N}$ the local order is conjugate to $\begin{pmatrix} \mathcal{O}_{\mathfrak{p}} & \mathcal{O}_{\mathfrak{p}} \\ \mathfrak{p} & \mathcal{O}_{\mathfrak{p}} \end{pmatrix}$, the local tree collapses to a single edge, and $d_{\mathfrak{p}} = 2$. Hence $d_{\mathfrak{p}} = N\mathfrak{p}$ if and only if $N\mathfrak{p} = 2$. At the primes not dividing $\mathfrak{N}$ the order is maximal and Theorem 1.2 applies. Consequently $d_{\mathfrak{p}} = N\mathfrak{p}$ can hold at finitely many primes at most, and never at almost all primes.*

PROOF. The local Eichler order of level $\mathfrak{p}$ is the stabilizer of an edge of the Bruhat–Tits tree, whose two endpoints give the two right ideals of reduced norm $\mathfrak{p}$; the remaining $q - 1$ neighbours of each vertex are not stabilized. The level is divisible by only finitely many primes. $\square$

2. The boundary quotient is empty, universally

THEOREM 2.1 (The boundary quotient is zero). *Let Λ be an order in a central simple algebra of degree $d \geq 2$ over K , all of whose right ideals are principal, $P = \Lambda \setminus \{0\}$, and let $\mathfrak{p} = (\pi)$ be a principal prime of K . Let A be a unital C^* -algebra generated by isometries v_a , $a \in P$, with $v_a v_b = v_{ab}$. If two range projections e_{a_j}, e_{a_k} of distinct right ideals of reduced norm $\mathfrak{p}$ are orthogonal in A , then $A = 0$. In particular the boundary quotient of the Nica–Toeplitz algebra $\mathcal{T}(P)$ by the relations $\sum_{k=1}^{d_{\mathfrak{p}}} e_{a_k} = 1$, at any one such prime, is the zero algebra and carries no state.*

PROOF. For $a \in \Lambda$ the reduced characteristic polynomial $t^d - c_1 t^{d-1} + \dots \pm \text{Nrd}(a)$ has coefficients in $\mathcal{O}_K$ and vanishes at a , so $\text{Nrd}(a) = ab$ with $b = \pm(a^{d-1} - c_1 a^{d-2} + \dots) \in \Lambda$ [107, §9]: the central element $\text{Nrd}(a)$ lies in $a\Lambda$. For a_j of reduced norm $\mathfrak{p} = (\pi)$ we have $\text{Nrd}(a_j) = \pi u_j$ with $u_j \in \mathcal{O}_K^\times$, so $\pi = a_j b_j u_j^{-1} \in a_j \Lambda$ and $v_\pi = v_{a_j} v_{b_j u_j^{-1}}$, whence $e_\pi = v_\pi v_\pi^* \leq v_{a_j} v_{a_j}^* = e_{a_j}$; likewise $e_\pi \leq e_{a_k}$. If $e_{a_j} e_{a_k} = 0$ then $e_\pi = 0$, so $v_\pi = e_\pi v_\pi = 0$ and $1 = v_\pi^* v_\pi = 0$. Finally a sum of projections equal to 1 is an orthogonal sum, as in Chapter 26: $e_j = e_j \sum_k e_k$ gives $\sum_{k \neq j} e_j e_k e_j = 0$ with positive summands, so $e_j e_k = 0$. $\square$

REMARK 2.2 (Partition in mean). Even the weaker requirement $\sum_{k=1}^{d_{\mathfrak{p}}} \varphi(e_{a_k}) = 1$ for a KMS_β state φ of $\mathcal{T}(P)$ itself, with norm $N = |\text{Nrd}|^t$, forces $d_{\mathfrak{p}} q^{-t\beta} = 1$, i.e.

$$\beta = \beta(\mathfrak{p}) = \frac{\log d_{\mathfrak{p}}}{t \log q} = \frac{1}{t} \cdot \frac{\log((q^d - 1)/(q^e - 1))}{\log q}$$

at a prime of local index e . For split primes the function $q \mapsto \log((q^d - 1)/(q - 1))/\log q$ is strictly decreasing with limit $d - 1$ as $q \rightarrow \infty$ and value $> d - 1$ for every finite q ; so distinct residue characteristics give distinct $\beta(\mathfrak{p})$, and no single β serves two split primes. Numerically, for $d = 2$: $\beta(\mathfrak{p})t = 1.5850, 1.2619, 1.1133, 1.0289, 1.0021$ at $q = 2, 3, 5, 13, 101$, decreasing to 1; for $d = 3$: $2.8074, 2.3347, 2.1337, 2.0310, 2.0022$, decreasing to 2; for $d = 4$, decreasing to 3.

REMARK 2.3 (Why the affine systems are different). In the $ax + b$ systems of Chapter 3 the multiplicity at $\mathfrak{p}$ is the number $N_{\mathfrak{p}}$ of additive cosets, exactly equal to the norm, and the translates of a fundamental domain are disjoint, so the relation reads $N_{\mathfrak{p}} \cdot N_{\mathfrak{p}}^{-\beta} = 1$ and forces $\beta = 1$ *independently of $\mathfrak{p}$* : a KMS state survives at that one temperature. The matching there is not an accident but the statement that $\mathcal{O}_K/\mathfrak{p}$ has $N_{\mathfrak{p}}$ elements. Theorem 1.2 says that the multiplicative analogue — counting ideals rather than cosets — can never match, the projective space $\mathbb{P}^{d-1}(\mathcal{F}_q)$ having one point too many.

COROLLARY 2.4 (The dichotomy of Chapter 26 cannot be broken). *The question of Chapter 26 — whether some order has $d_{\mathfrak{p}} = N_{\mathfrak{p}}$, so that a boundary quotient with nonempty KMS simplex could exist — is answered in the negative by Theorem 1.2 and Proposition 1.3, and Theorem 2.1 shows that the multiplicity is not even the point: the quotient is zero whatever $d_{\mathfrak{p}}$ is. The nested/partition dichotomy must therefore be circumvented rather than broken, which means working with the Toeplitz algebra and its Maharam extension.*

3. The cocycle factors through the reduced norm

We turn to (C4), for the Hurwitz order, $P = \mathcal{H} \setminus \{0\}$. Let $G = PP^{-1} = \mathbb{H}(\mathbb{Q})^\times$ be the group of fractions, $\tilde{\Omega}$ the dilation carrying the G -action with the measure $\tilde{\nu}_\beta$ extending a KMS_β measure, $\tilde{\nu}_\beta(aE) = \text{Nrd}(a)^{-\beta} \tilde{\nu}_\beta(E)$, and

$$G \curvearrowright (\tilde{\Omega} \times \mathcal{R}, \tilde{\nu}_\beta \otimes e^s ds), \quad a \cdot (x, s) = (ax, s + \beta \log \text{Nrd}(a))$$

the Maharam extension, which preserves the displayed measure: the pushforward of $\tilde{\nu}_\beta \otimes e^s ds$ under a scales the first factor by $\text{Nrd}(a)^{-\beta}$ and the second by $e^{\beta \log \text{Nrd}(a)}$. By Takesaki duality, as in Chapter 26, (C4) — non-amenability of the measured groupoid of G on $(\tilde{\Omega}, \tilde{\nu}_\beta)$, equivalently non-injectivity of the factor — is equivalent to non-amenability of the Maharam extension. Put

$$G^1 := \ker(\text{Nrd} : G \rightarrow \mathbb{Q}_+^\times).$$

PROPOSITION 3.1 (Orthogonality of the two directions). (1) *The Radon–Nikodym cocycle of the G -action on $(\tilde{\Omega}, \tilde{\nu}_\beta)$ is $a \mapsto \text{Nrd}(a)^{-\beta}$, which factors through Nrd . Consequently G^1 acts measure preservingly, and acts trivially on the flow variable s .*

- (2) $G^1 = \mathbb{H}(\mathbb{Q})^1$, the norm-one rational quaternions, is non-amenable: it contains free subgroups of rank five.
- (3) Hence the modular direction of the system, which is the whole content of the dynamics σ_t , lies in the abelian quotient $G/G^1 \cong \text{Nrd}(G) \leq \mathbb{Q}_+^\times$, while all the non-amenability of G lies in G^1 , on which the dynamics is trivial.

PROOF. (1) The scaling relation $\tilde{\nu}_\beta(aE) = \text{Nrd}(a)^{-\beta} \tilde{\nu}_\beta(E)$ depends on a only through $\text{Nrd}(a)$; for $a \in G^1$ it is the identity, and $s \mapsto s + \beta \log \text{Nrd}(a) = s$. (2) By Lubotzky–Phillips–Sarnak [85, §3] the quaternions $1 \pm 2i, 1 \pm 2j, 1 \pm 2k$ of reduced norm 5 generate, modulo the centre, a free group $\Lambda(5)$ of rank three; the elements of $\Lambda(5)$ of reduced norm 5^{2m} form a subgroup of index two, free of rank five, and $\alpha \mapsto \alpha/5^m$ embeds it into $\mathbb{H}(\mathbb{Q})^1/\{\pm 1\}$. (3) is (1) and (2). $\square$

REMARK 3.2 (What this changes). Proposition 3.1 is the reason the Hurwitz system is not merely a non-commutative decoration of the abelian one. In every earlier model the acting group was abelian and its non-amenability was not merely absent but impossible. Here the group splits cleanly: a non-amenable kernel acting by measure-preserving transformations, and an abelian quotient carrying the entire thermodynamics. If (C4) holds it will be because of G^1 , and G^1 is invisible to σ_t .

4. All Fourier fibres agree on G^1

PROPOSITION 4.1 (Fibre decomposition). *Conjugating by $\xi \mapsto e^{s/2}\xi$ identifies $L^2(\mathcal{R}, e^s ds)$ with $L^2(\mathcal{R}, ds)$, and carries the Koopman representation of the Maharam extension to $(\pi(a)F)(x, s) = \text{Nrd}(a)^{\beta/2} F(a^{-1}x, s - \beta \log \text{Nrd}(a))$; Fourier transform in s then gives*

$$L^2(\tilde{\Omega} \times \mathcal{R}, \tilde{\nu}_\beta \otimes e^s ds) \cong \int_{\mathcal{R}}^{\oplus} \mathcal{H}_\tau \frac{d\tau}{2\pi}, \quad \mathcal{H}_\tau = L^2(\tilde{\Omega}, \tilde{\nu}_\beta),$$

with fibre representations

$$(4.1) \quad \pi_\tau(a) = \chi_\tau(a) \pi_0(a), \quad \chi_\tau(a) = \text{Nrd}(a)^{i\beta\tau}, \quad (\pi_0(a)f)(x) = \text{Nrd}(a)^{\beta/2} f(a^{-1}x).$$

Here π_0 is a genuine unitary representation of G on $L^2(\tilde{\Omega})$, the dilation being what makes the isometries of P into unitaries.

PROOF. Standard: π_0 is the Koopman representation of the non-singular action, with Radon–Nikodym factor $\text{Nrd}(a)^{\beta/2}$, and the translation by $\beta \log \text{Nrd}(a)$ in s becomes multiplication by $\chi_\tau(a)$ on the τ -th Fourier component, the sign of the exponent being a convention. The only point to note is that on Ω itself $\pi_0(a)$ is an isometry with range $L^2(a\Omega) \subsetneq L^2(\Omega)$, and becomes unitary only after passing to the dilation $\tilde{\Omega} = \varinjlim (\Omega, a \cdot)$. $\square$

THEOREM 4.2 (The τ -decomposition is invisible to G^1). *For every $\tau \in \mathcal{R}$, $\chi_\tau|_{G^1} = 1$, hence*

$$\pi_\tau|_{G^1} = \pi_0|_{G^1} \quad \text{for all } \tau.$$

Consequently the restriction to G^1 of the Koopman representation of the Maharam extension is $\pi_0|_{G^1} \otimes L^2(\mathcal{R})$, a multiple of a single representation, and

$$(4.2) \quad (C4) \iff \pi_0|_{G^1} \text{ is not weakly contained in the regular representation } \lambda_{G^1}.$$

PROOF. $\chi_\tau(a) = \text{Nrd}(a)^{i\beta\tau} = 1$ for $a \in G^1$. The displayed restriction follows from (4.1) and Proposition 4.1. For (4.2): if the action of G on $(\tilde{\Omega}, \tilde{\nu}_\beta)$ were amenable, so would be its restriction to the subgroup G^1 [2]; and by Kuhn’s theorem [74] the Koopman representation of an amenable action is weakly contained in the regular representation, so $\pi_0|_{G^1} \prec \lambda_{G^1}$. $\square$

REMARK 4.3 (A spectral gap would not suffice). The converse of Kuhn’s theorem fails, so the implication (4.2) cannot be weakened to the absence of almost invariant vectors: the translation action of F_2 on itself is amenable, being proper, and its Koopman representation is λ_{F_2} , which has a spectral gap. A gap for $\pi_0|_{G^1}$ is necessary for (4.2), since λ_{G^1} has one, but it is the failure of temperedness that is needed.

REMARK 4.4. Theorem 4.2 removes the direct integral from the problem. Whatever is to be established about weak containment, it concerns one representation of one group — the measure-preserving Koopman representation of the norm-one subgroup — and not a family of questions indexed by τ . This is the concrete simplification we can offer.

5. Why this does not yet close (C4)

REMARK 5.1 (The failure of the expected argument). The natural next step is Zimmer’s theorem: an essentially free amenable action of Γ preserving a probability measure forces Γ to be amenable. Applied to $G^1 \curvearrowright (\tilde{\Omega}, \tilde{\nu}_\beta)$ with G^1 non-amenable it would give (C4) at once. It does not apply, for a specific reason: $\tilde{\nu}_\beta$ is *infinite*. Indeed $\tilde{\nu}_\beta(a^{-1}\Omega) = \text{Nrd}(a)^\beta \rightarrow \infty$, so the dilation carries only a σ -finite invariant measure.

That the extension from finite to infinite invariant measure genuinely fails is shown by an example: the translation action of F_2 on itself preserves counting measure, is essentially free, and its orbit groupoid is transitive with trivial isotropy, hence proper, hence amenable. Non-amenability of the group is therefore not transmitted by an infinite-measure-preserving essentially free action, and by Remark 4.3 not by a spectral gap either.

REMARK 5.2 (What would suffice). Two routes seem available, and we have carried out neither.

- (1) Find a G^1 -invariant subset of finite measure meeting almost every orbit, or more generally show that the restricted relation $\mathcal{R}_{G^1}|_\Omega$ on the probability space (Ω, ν_β) is of type II₁ and is the orbit relation of an essentially free pmp action of a non-amenable group. The difficulty is that Ω is not G^1 -invariant, so the restricted relation is not literally an orbit relation.
- (2) Show directly that $\pi_0|_{G^1}$ is not weakly contained in λ_{G^1} — for instance by exhibiting a finite-dimensional subrepresentation, which an infinite-measure Koopman representation need not have. By Theorem 4.2 this is one representation; by Proposition 3.1 it is measure preserving; and G^1 contains the LPS free groups, for which spectral gap statements are available on *congruence quotients* [85]. Whether any of that transfers to $\tilde{\Omega}$, which is not a congruence quotient, we do not know — Chapter 26 warns that the tree methods behind those statements are unavailable here, and Remark 4.3 that a gap alone would in any case not be enough.

REMARK 5.3 (Essential freeness). Route (i) needs the G^1 -action to be essentially free, which we have not verified. The isotropy of the G -action on $\tilde{\Omega}$ was computed in the abelian case in Chapter 3 and found to be trivial off a set that the KMS measures do not charge; the analogous computation here would have to handle the 24 units and the non-commutativity, and we have not done it. Zimmer’s theorem is used here in the form of [137, Ch. 4].

6. Assessment

Question	Answer	Reference
$d_{\mathfrak{p}}$ at a prime of local index e , maximal order	$(q^d - 1)/(q^e - 1)$	Lem. 1.1
$\exists$ order with $d_{\mathfrak{p}} = N\mathfrak{p}$?	no, for every $d \geq 2$	Thm. 1.2
excess $d_{\mathfrak{p}} - N\mathfrak{p}$ for $d = 2$	exactly 1	Thm. 1.2
Eichler orders	$d_{\mathfrak{p}} = 2$ at $\mathfrak{p} \parallel \mathfrak{N}$; $= N\mathfrak{p}$ iff $N\mathfrak{p} = 2$	Prop. 1.3
boundary quotient nonzero?	never	Thm. 2.1
RN cocycle factors through Nrd	yes	Prop. 3.1
G^1 acts measure-preservingly	yes, and is non-amenable	Prop. 3.1
Fourier fibres on G^1	all equal to $\pi_0 _{G^1}$	Thm. 4.2
$(C4) \Leftarrow \pi_0 _{G^1} \not\Leftarrow \lambda_{G^1}$	true (Kuhn)	Thm. 4.2
Zimmer closes (C4)	no: invariant measure infinite	Rem. 5.1
a spectral gap closes (C4)	no: amenable actions are tempered	Rem. 4.3
(C4)	open	—

Task 1 is closed. The multiplicity $d_{\mathfrak{p}}$ counts points of a projective space and the norm $N\mathfrak{p}$ counts points of an affine one; the difference is the hyperplane at infinity, and it is never zero. The vanishing of the boundary quotient found in Chapter 26 for the Hurwitz order is therefore a universal feature of orders in central simple algebras, and the nested/partition dichotomy has to be circumvented rather than broken.

Task 2 is not closed, but it is now a single question rather than a programme. The decomposition of G given by Proposition 3.1 — non-amenable kernel acting measure-preservingly, abelian quotient carrying the dynamics — is the structural feature that distinguishes this system from every earlier one, and Theorem 4.2 reduces the analysis to one representation of that kernel. What is missing is a finite-measure foothold together with essential freeness, or a proof that $\pi_0|_{G^1}$ is not tempered. If $\pi_0|_{G^1}$ is not weakly contained in the regular representation, the Hurwitz factor is the first non-injective arithmetic Bost–Connes factor and the deformation/rigidity machinery that Chapter 20 found inapplicable becomes actionable; if the action is amenable, the factor collapses like all the others and the programme reaches its end.

Localized GL_2 Bost–Connes systems: the equilibrium range and the automorphic obstruction range are disjoint, by exactly one

This chapter is Paper XXI of the series (papers/XXI-GL2/).

SUMMARY. We examine the proposal to characterize Langlands reciprocity for weight-two forms through the KMS structure of a localized GL_2 Bost–Connes system, and find a quantitative obstruction that closes it.

The Hilbert–Schmidt identity is correct: with $U_p = p^{-1/2} \rho_f(\text{Frob}_p)$ unitary by Deligne, $\|I_2 - U_p\|_{\text{HS}}^2 = 4(1 - \cos \theta_p) = 8 \sin^2(\theta_p/2)$, and $\Xi_1(\pi_f, P_{\mathbb{Q}}) = \infty$ for every cuspidal eigenform, by Jacquet–Shalika. But the L -function formula proposed for it is wrong: Ξ_β is *linear* in $\text{Tr } U_p$, so only the standard L -function appears,

$$\Xi_\beta(\pi_f, S) = 4 \log \zeta_S(\beta) - 2 \Re \log L_S(\beta, \pi_f) + O(1),$$

matching Theorem 4.1 of Chapter 15 at $n = 2$; the term $-2 \Re \log L_S(\beta, \text{Sym}^2 \pi_f)$ is spurious. Rankin–Selberg governs the second moment $\sum_p |\text{Tr } U_p|^2 p^{-s}$, which is Sato–Tate, not Ξ_β .

The decisive obstruction is a mismatch of temperature ranges, and it is exact. For any S , the automorphic obstruction can be nontrivial only where $\sum_{p \in S} p^{-\beta}$ diverges, since $\Xi_\beta \leq 8 \sum_{p \in S} p^{-\beta}$; write $\beta_c(S) \leq 1$ for that abscissa. The GL_2 partition function has local factor $(1 - p^{-\beta})^{-1}(1 - p^{1-\beta})^{-1}$ — the Solomon zeta of $M_2(\mathbb{Z}_p)$ — so it converges exactly for $\beta > 1 + \beta_c(S)$. Hence

$$\text{obstruction nontrivial} \iff \beta \leq \beta_c(S), \quad \text{Gibbs states exist} \iff \beta > 1 + \beta_c(S),$$

and the two regions are disjoint with a gap of width exactly 1, for every S and every automorphic π . For $S = P_{\mathbb{Q}}$: obstruction on $(0, 1]$, equilibrium on $(2, \infty)$. There is no temperature at which the GL_2 system has equilibrium states and the automorphic obstruction can break symmetry.

The shift by one is not accidental. The number of sublattices of index p in $\mathbb{Z}_p^2$ is $p + 1$ while the norm is p , and by Theorem 1.2 of Chapter 27 this excess is universal: $d_p = (q^d - 1)/(q - 1) > q$. The same off-by-one of $\mathbb{P}^{d-1}$ that empties the boundary quotient there displaces the equilibrium range here.

Finally, the proposed algebra is not available as stated: $M_2(\mathbb{Z})^+$ has infinite unit group $SL_2(\mathbb{Z})$, so by the unit-group exclusion of Chapter 25 (its Remark 1.5) its partition function diverges identically and one must pass to the Hecke quotient, which is not a semigroup crossed product.

1. The algebra

The framework is that of [Ch.

PROPOSITION 1.1 (The semigroup crossed product is unavailable). *Let $P = M_2(\mathbb{Z}) \cap GL_2^+(\mathbb{Q})$. Then $P^\times = SL_2(\mathbb{Z})$ — an element of P with inverse in P has positive integral determinant on both sides, hence determinant 1 — is infinite and $|\det| = 1$ on it, so $\sum_{g \in P} |\det g|^{-\beta} \geq |P^\times| = \infty$ for every β . Hence $C(Y) \rtimes P$ carries no Gibbs state at any temperature, and the construction must be replaced by the Hecke-algebra quotient of Connes–Marcolli [29], in which one sums over $GL_2(\mathbb{Z})$ -orbits.*

PROOF. This is the unit-group exclusion recorded in Chapter 25 (its Remark 1.5): a multiplicative norm is 1 on units, so an infinite unit group makes ζ_P diverge identically. $\square$

PROPOSITION 1.2 (The Solomon zeta). *Summing over $GL_2(\mathbb{Z})$ -orbits, the local factor at p is the Solomon zeta of $M_2(\mathbb{Z}_p)$ [120],*

$$\zeta_{GL_2,p}(\beta) = (1 - p^{-\beta})^{-1} (1 - p^{1-\beta})^{-1},$$

so for a set S of primes

$$(1.1) \quad \zeta_{GL_2,S}(\beta) = \prod_{p \in S} (1 - p^{-\beta})^{-1} (1 - p^{1-\beta})^{-1},$$

which converges if and only if $\sum_{p \in S} p^{1-\beta} < \infty$.

PROOF. The Gibbs weight of the Hecke quotient sums one term per left coset $GL_2(\mathbb{Z})g$, equivalently per sublattice $L \subseteq \mathbb{Z}_p^2$, weighted by its index [29] — not one term per double coset. The number of sublattices of $\mathbb{Z}_p^2$ of index p^n is $\sigma(p^n) = 1 + p + \cdots + p^n$, and

$$\sum_{n \geq 0} \sigma(p^n) p^{-\beta n} = \sum_{n \geq 0} \frac{p^{n+1} - 1}{p - 1} p^{-\beta n} = (1 - p^{-\beta})^{-1} (1 - p^{1-\beta})^{-1}.$$

Convergence of the Euler product is convergence of $\sum_p (p^{-\beta} + p^{1-\beta})$, dominated by the second term. $\square$

2. The obstruction series and its L -function

LEMMA 2.1 (Hilbert–Schmidt identity). *Let $f \in S_2(\Gamma_0(N))$ be a normalized cuspidal Hecke eigenform, $p \nmid N$, and $U_p = p^{-1/2} \rho_f(\text{Frob}_p)$, unitary with eigenvalues $e^{\pm i\theta_p}$ by Deligne’s theorem [38]. Then*

$$\|I_2 - U_p\|_{\text{HS}}^2 = 2(2 - \Re \text{Tr } U_p) = 4(1 - \cos \theta_p) = 8 \sin^2(\theta_p/2),$$

so that

$$(2.1) \quad \Xi_\beta(\pi_f, S) = 8 \sum_{p \in S} p^{-\beta} \sin^2(\theta_p/2).$$

PROOF. Lemma 2.2 of Chapter 15 (its Trace form) with $n = 2$, together with $\text{Tr } U_p = a_p/\sqrt{p} = 2 \cos \theta_p$. $\square$

THEOREM 2.2 (The L -function dictionary, corrected). *For $\beta > 1/2$,*

$$(2.2) \quad \Xi_\beta(\pi_f, S) = 4 \log \zeta_S(\beta) - 2 \Re \log L_S(\beta, \pi_f) + O(1),$$

with $L_S(s, \pi_f) = \prod_{p \in S} ((1 - e^{i\theta_p} p^{-s})(1 - e^{-i\theta_p} p^{-s}))^{-1}$ the analytically normalized standard L -function. No symmetric square factor occurs.

PROOF. By (2.1), $\Xi_\beta = 4 \sum_p p^{-\beta} - 2 \sum_p p^{-\beta} \text{Tr } U_p$. The first sum is $\log \zeta_S(\beta) + O(1)$; the second is $\Re \log L_S(\beta, \pi_f) + O(1)$, the prime-power terms converging for $\beta > 1/2$. This is Theorem 4.1 of Chapter 15 at $n = 2$, where the general form is $2n \log \zeta_{K,S}(\beta) - 2 \Re \log L_S(\beta, \rho) + O(1)$. $\square$

REMARK 2.3 (Why Sym^2 cannot appear). The proposed formula adds $-2 \Re \log L_S(\beta, \text{Sym}^2 \pi_f)$. That term cannot be present: $\|I - U\|_{\text{HS}}^2 = 2(2 - \Re \text{Tr } U)$ is linear in $\text{Tr } U_p$, whereas Sym^2 enters through $\text{Tr}(U_p^2) = \text{Tr}(U_p)^2 - 2 \det U_p$, a quadratic expression. Rankin–Selberg does govern the second moment: $\log L_S(s, \pi \times \pi^\vee) = \sum_p |\text{Tr } U_p|^2 p^{-s} + O(1)$, whose pole at $s = 1$ is the input to Sato–Tate. That is a different series from Ξ_β .

COROLLARY 2.4 (Divergence at $\beta = 1$). *For $S = P_{\mathbb{Q}}$ and any cuspidal eigenform f , $\Xi_1(\pi_f, P_{\mathbb{Q}}) = \infty$.*

PROOF. As $\beta \downarrow 1$, $\log \zeta(\beta) \rightarrow \infty$ while $L(\beta, \pi_f)$ has neither zero nor pole at $\beta = 1$ — Jacquet–Shalika non-vanishing on $\Re s = 1$ [65], and cuspidality excluding a pole — so $\Re \log L_S(\beta, \pi_f) = O(1)$. Apply (2.2). $\square$

3. The gap is exactly one

DEFINITION 3.1. For S a set of primes let $\beta_c(S) = \inf\{\beta : \sum_{p \in S} p^{-\beta} < \infty\}$, so $0 \leq \beta_c(S) \leq 1$.

THEOREM 3.2 (Disjoint ranges). *For every set S of primes and every automorphic representation π of GL_2 unramified outside S :*

- (1) $\Xi_\beta(\pi, S) < \infty$ for every $\beta > \beta_c(S)$, so no representation is an obstruction there; the obstruction can be nontrivial only for $\beta \leq \beta_c(S)$;
- (2) $\zeta_{GL_2, S}(\beta) < \infty$ for $\beta > 1 + \beta_c(S)$ and $= \infty$ for $\beta < 1 + \beta_c(S)$ — at the boundary $\beta = 1 + \beta_c(S)$ itself, finite exactly when $\sum_{p \in S} p^{-\beta_c(S)}$ converges — so Gibbs states of the GL_2 system exist only for $\beta \geq 1 + \beta_c(S)$, and only beyond it when the critical series diverges.

Consequently

$$(0, \beta_c(S)] \cap [1 + \beta_c(S), \infty) = \emptyset,$$

and the open interval $(\beta_c(S), 1 + \beta_c(S))$, of width exactly 1, lies in neither region. For $S = P_{\mathbb{Q}}$ the obstruction lives on $(0, 1]$ and the equilibrium states on $(2, \infty)$.

PROOF. (1) By (2.1), $\Xi_\beta(\pi, S) \leq 8 \sum_{p \in S} p^{-\beta}$, finite for $\beta > \beta_c(S)$. (2) By Proposition 1.2, convergence is that of $\sum_{p \in S} p^{1-\beta} = \sum_{p \in S} p^{-(\beta-1)}$, which for $\beta - 1 > \beta_c(S)$ converges, for $\beta - 1 < \beta_c(S)$ diverges, and at $\beta - 1 = \beta_c(S)$ is the critical series itself. $\square$

COROLLARY 3.3 (The programme cannot close). *There is no temperature at which the localized GL_2 system possesses equilibrium states and the automorphic obstruction $\Xi_\beta(\pi, S)$ is capable of producing symmetry breaking. In particular the proposed uniqueness theorem on $(0, 1]$ concerns a range in which the GL_2 system has no Gibbs states, and Corollary 2.4, though true, has no KMS consequence for this system.*

REMARK 3.4 (Numerical illustration). For $S = P_{\mathbb{Q}}$: $\sum_{p \leq 10^6} p^{-\beta}$ equals 4.79, 2.89, 1.99 at $\beta = 0.9, 1.0, 1.1$, the first two growing without bound and the third converging; while $\zeta(\beta)\zeta(\beta-1)$ is divergent at $\beta \leq 2$ and equals 8.33, 1.98, 1.30 at $\beta = 2.2, 3, 4$. The window $(1, 2]$ has neither.

REMARK 3.5 (The shift is the multiplicity–norm mismatch). The displacement by 1 is the phenomenon isolated in Theorem 1.2 of Chapter 27. There, for a maximal order in a central simple algebra of degree d , the number of right ideals of reduced norm $\mathfrak{p}$ is $d_{\mathfrak{p}} = (q^d - 1)/(q - 1) = 1 + q + \dots + q^{d-1}$, strictly larger than the norm $q = N\mathfrak{p}$; for $d = 2$ the excess is exactly the point at infinity of $\mathbb{P}^1$. Here the same count — $p + 1$ sublattices of index p in $\mathbb{Z}_p^2$ against norm p — produces the factor $(1 - p^{1-\beta})^{-1}$ in (1.1) and shifts the abscissa by 1. In Chapter 27 the excess emptied the boundary quotient; here it displaces the equilibrium range out of reach of the obstruction. It is one fact with two consequences.

4. Sparse sets for Δ

THEOREM 4.1 (Anomalous sets exist). *Let $\Delta \in S_{12}(SL_2(\mathbb{Z}))$ with $\tau(p) = 2p^{11/2} \cos \theta_p$. Assuming the Sato–Tate equidistribution for Δ — a theorem of Barnet-Lamb, Geraghty, Harris and Taylor [8] — for every $\delta > 0$ the set $\{p : \theta_p \leq \delta\}$ has positive density. Choosing finite blocks $S_j \subseteq \{p : \theta_p \leq 2^{-j}\}$ with $1 \leq \sum_{p \in S_j} p^{-1} \leq 2$ and putting $S_\Delta = \bigsqcup_j S_j$ gives*

$$\sum_{p \in S_\Delta} p^{-1} = \infty, \quad \Xi_1(\pi_\Delta, S_\Delta) \leq \sum_j 2 \cdot 4^{-j} \cdot 2 < \infty.$$

PROOF. $8\sin^2(\theta/2) \leq 2\theta^2$, so the j -th block contributes at most $2 \cdot 4^{-j} \sum_{p \in S_j} p^{-1} \leq 4 \cdot 4^{-j}$. This is Theorem 5.3 of Chapter 15 with Δ in place of an elliptic curve. $\square$

REMARK 4.2 (But not symmetry breaking). Theorem 4.1 produces a set on which the analytic obstruction is finite. It does not produce a broken phase, for two reasons already recorded. By Remark 5.4 of Chapter 15 an obstruction group is not a broken phase until an algebra realizes it; and by Theorem 3.2 the temperature $\beta = 1$ at which the construction operates is one at which this GL_2 system has no equilibrium states, $\beta_c(S_\Delta) = 1$ forcing Gibbs states only above 2. The residual symmetry subgroup $H_\Delta \subseteq SU(2)$ asked for is therefore not defined: there is no simplex on which it could act.

5. Assessment

Proposal	Status	Reference
$C(Y) \rtimes M_2(\mathbb{Z}_S)^+$	no: $ P^\times = \infty$	Prop. 1.1
$\ I_2 - U_p\ _{\text{HS}}^2 = 8\sin^2(\theta_p/2)$	true	Lem. 2.1
Ξ_β involves Sym^2	false: spurious term	Rem. 2.3
$\Xi_\beta = 4\log \zeta_S - 2\Re \log L_S + O(1)$	true	Thm. 2.2
$\Xi_1(\pi_f, P_{\mathbb{Q}}) = \infty$	true, Jacquet–Shalika	Cor. 2.4
uniqueness on $(0, 1]$	vacuous: no KMS states there	Cor. 3.3
obstruction region	$(0, \beta_c(S)]$, $\beta_c \leq 1$	Thm. 3.2
equilibrium region	$(1 + \beta_c(S), \infty)$	Thm. 3.2
the two overlap	never; gap exactly 1	Thm. 3.2
sparse S_Δ with $\Xi_1 < \infty$	true, under Sato–Tate	Thm. 4.1
residual symmetry $H_\Delta \subseteq SU(2)$	not defined	Rem. 4.2

The arithmetic side of the proposal is sound and largely already proved in Chapter 15: the Hilbert–Schmidt series, its L -function form once the spurious symmetric square is removed, the divergence at $\beta = 1$ by Jacquet–Shalika, and the construction of Sato–Tate anomalous sets. What fails is the junction with statistical mechanics, and it fails quantitatively rather than vaguely. The obstruction is a Dirichlet series in $p^{-\beta}$, with abscissa at most 1; the GL_2 partition function counts sublattices, whose local factor $(1 - p^{1-\beta})^{-1}$ moves the abscissa to $1 + \beta_c(S)$. The gap is exactly 1 and it is the same off-by-one of $\mathbb{P}^1$ that Chapter 27 showed to be universal.

Two ways out suggest themselves and neither is free. One could weight the dynamics differently, replacing $|\det g|$ by another multiplicative norm; but by Proposition 1.4 of Chapter 25 norms factor through the abelianization, and on GL_2 orbits the abelianization is generated by $\det$, so any admissible norm is $|\det|^t$ and merely rescales β , leaving the ratio of the two abscissae fixed at $(1 + \beta_c)/\beta_c$ rather than closing the gap. Or one could keep the GL_1 normalization and attach the automorphic data as a coefficient system rather than as the acting semigroup — which is the route of §6 of Chapter 15, where the Kakutani analysis applies verbatim and only the existence of a tractable expectation was left open. We would pursue the second.

The Ramanujan bound is the tempered bound: the Hecke operator on the Bruhat–Tits tree, and a no-go theorem for condition (C4)

This chapter is Paper XXVIII of the series ([papers/XXVIII-spectral-gap/](#)).

SUMMARY. Condition (C4) of Chapter 25 asks whether the measured groupoid of the Hurwitz quaternion Bost–Connes system is non-amenable, equivalently whether its factor escapes the injectivity that erased the arithmetic in every earlier model. Chapter 26 reduced the question to a spectral property of a transfer operator and found the naive operator unusable, the range projections being nested rather than orthogonal; Chapter 27 reduced it further to a single representation of the norm-one group and warned that a spectral gap would not suffice. We settle the question.

We first build the operator the programme wanted: the Hecke operator $T_p = \sum_{k=1}^{p+1} \pi(a_k)$, the sum of the Koopman unitaries of the $p+1$ Hurwitz quaternions of reduced norm p , which is the adjacency operator of the $(p+1)$ -regular Bruhat–Tits tree of $\mathrm{GL}_2(\mathbb{Q}_p)$ transported to the Maharam extension. It involves no projections and so has none of the pathology of the transfer operator.

We then compute its spectrum exactly. For every finite set F of primes the subgroup $G_F = \mathbb{H}(\mathbb{Z}[1/F])^\times$ is discrete in $\prod_{p \in F} \mathbb{H}(\mathbb{Q}_p)^\times$ and acts *properly and freely* on a conull set, because every element of $\mathbb{H}(\mathbb{Q})^\times$ is a unit at almost every place. Hence the Koopman representation restricted to G_F is a multiple of the regular representation, and, with ε_p the central unitary of p (so that $\varepsilon_p^{-1/2} T_p$ is the self-adjoint Hecke operator),

$$\mathrm{sp}(\varepsilon_p^{-1/2} T_p \upharpoonright L_0^2) = [-2\sqrt{p}, 2\sqrt{p}], \quad \|T_p\| = 2\sqrt{p}.$$

The Ramanujan bound is attained, not merely respected. It is also worthless for the purpose: $2\sqrt{p}$ is the norm of the adjacency operator in the *regular* representation of a free group, so the gap between $2\sqrt{p}$ and the trivial eigenvalue $p+1$ is Kesten’s gap and measures the non-amenability of the *group*, not of the *action*. By Kuhn’s theorem an amenable action has tempered Koopman representation; attaining the Ramanujan bound proves temperedness, and temperedness is the signature of amenability, not an obstruction to it.

The same properness gives the verdict directly, with no spectral theory at all. The orbit relation is the increasing union over finite F of the smooth relations of the proper actions of G_F , hence hyperfinite; the groupoid is amenable and the factor is injective. **(C4) is false**, the Hurwitz factor collapses to R_∞ , R_λ or a Krieger factor like every abelian system before it, and by Chapter 20 it retains nothing of the arithmetic. The argument uses only that each element of the monoid is a unit at almost every place, so it closes the search of Chapter 25 in full: conditions (C1)–(C3) are incompatible with (C4) for every order in every finite-dimensional $\mathbb{Q}$ -algebra.

1. Setting and statements

$\mathbb{H}$ is the rational Hamilton quaternion algebra, ramified at 2 and ∞ ; $\mathcal{H} \subset \mathbb{H}(\mathbb{Q})$ is the Hurwitz order, $P = \mathcal{H} \setminus \{0\}$ the multiplicative Ore monoid of Chapter 25, Nrd the reduced norm and $G = PP^{-1} = \mathbb{H}(\mathbb{Q})^\times$ the group of right fractions [128]. For a prime p write $\mathcal{H}_p = \mathcal{H} \otimes \mathbb{Z}_p$; for odd p one has $\mathcal{H}_p \cong M_2(\mathbb{Z}_p)$ and the right ideals of $\mathcal{H}_p$ of reduced norm p are the $p+1$ neighbours of the base vertex in the Bruhat–Tits tree $\mathcal{T}_p$ of $\mathrm{GL}_2(\mathbb{Q}_p)$ [118]. The localized Bost–Connes

framework in which the system sits is that of [Ch. 1]; the ingredients we use from Chapters 20–27 are re-proved here.

HYPOTHESIS 1.1 (The model). $\Omega = \widehat{\mathcal{H}} = \prod_p \mathcal{H}_p$ carries a KMS_β measure ν_β of the Nica-Toeplitz algebra $\mathcal{T}(P)$, $\tilde{\Omega} = \varinjlim(\Omega, \cdot a) = \mathbb{H}(\mathbb{A}_f)$ is the dilation, G acts on $\tilde{\Omega}$ by right translation, and $\tilde{\nu}_\beta$ is the extension of ν_β , so that $\tilde{\nu}_\beta(Ea) = \text{Nrd}(a)^{-\beta} \tilde{\nu}_\beta(E)$. The Maharam extension is

$$(1.1) \quad G \curvearrowright (\tilde{\Omega} \times \mathcal{R}, \tilde{\nu}_\beta \otimes e^s ds), \quad a \cdot (x, s) = (xa, s + \beta \log \text{Nrd}(a)),$$

which preserves the displayed infinite measure. Condition (C4) of Chapter 25 asks whether the measured groupoid $G \ltimes (\tilde{\Omega}, \tilde{\nu}_\beta)$ is non-amenable, equivalently — by Takesaki duality, as in Chapter 26 — whether (1.1) is non-amenable, equivalently whether $M_\beta = \pi_\varphi(\mathcal{T}(P))''$ fails to be injective.

Nothing below uses the value of β , the existence of a KMS state in any particular temperature range, or the type of M_β . Only the following is used, and it is a property of the arithmetic and not of the dynamics.

LEMMA 1.2 (Finite support). *Every $a \in G = \mathbb{H}(\mathbb{Q})^\times$ lies in $\mathcal{H}_q^\times$ for all but finitely many primes q .*

PROOF. Write $a = b/m$ with $b \in \mathcal{H}$ and $m \in \mathbb{Z}_{>0}$. For $q \nmid m \text{Nrd}(b)$ one has $b \in \mathcal{H}_q$ and $\text{Nrd}(b) \in \mathbb{Z}_q^\times$, so $b\bar{b} = \text{Nrd}(b)$ is a unit and $b \in \mathcal{H}_q^\times$; and $m \in \mathbb{Z}_q^\times$. Only the finitely many q dividing $m \text{Nrd}(b)$ are excluded. $\square$

For a finite set F of primes put

$$G_F = \{a \in G : a \in \mathcal{H}_q^\times \text{ for every } q \notin F\} = \mathbb{H}(\mathbb{Z}[1/F])^\times, \quad L_F = \prod_{p \in F} \mathbb{H}(\mathbb{Q}_p)^\times.$$

By Lemma 1.2, $G = \bigcup_F G_F$, the union being increasing and countable.

The three results are the following.

THEOREM 1.3 (Local properness). *For every finite F , G_F is a discrete subgroup of L_F and its action on $\tilde{\Omega}$ is free and proper on a conull invariant set. Consequently the orbit relation $\mathcal{R}_F$ is smooth, and the Koopman representation of the Maharam extension restricted to G_F is unitarily equivalent to a multiple of the regular representation λ_{G_F} .*

THEOREM 1.4 (The spectrum of the Hecke operator). *Let p be an odd prime, $a_1, \dots, a_{p+1} \in \mathcal{H}$ generators of the $p+1$ right ideals of reduced norm p , and $T_p = \sum_{k=1}^{p+1} \pi(a_k)$ the associated Hecke operator on $L^2(\tilde{\Omega} \times \mathcal{R})$ of Definition 2.1. Then T_p is the adjacency operator of $\mathcal{T}_p$ in the regular representation up to the central unitary $\varepsilon_p = \pi(p)$, and on the orthogonal complement L_0^2 of the invariant vectors the self-adjoint operator $\varepsilon_p^{-1/2} T_p$ of Lemma 2.2(1) satisfies*

$$\text{sp}(\varepsilon_p^{-1/2} T_p \upharpoonright L_0^2) = [-2\sqrt{p}, 2\sqrt{p}], \quad \|T_p \upharpoonright L_0^2\| = 2\sqrt{p}.$$

(T_p itself is normal but not self-adjoint, $T_p^ = \varepsilon_p^{-1} T_p$; its spectrum is the disc swept by the central phases, and the invariant statement is the one displayed.) The Ramanujan bound is attained, and the trivial eigenvalue $p+1$ is absent from L_0^2 .*

THEOREM 1.5 (No-go for (C4)). *The orbit relation $\mathcal{R}_G = \bigcup_F \mathcal{R}_F$ is hyperfinite. Hence the measured groupoid of Hypothesis 1.1 is amenable, (C4) is false, and M_β is injective: it is R_∞ , a Powers factor R_λ , a Krieger factor, or type I. By Chapter 20 it therefore retains no arithmetic invariant of the system.*

Theorem 1.5 does not use Theorem 1.4; the spectral computation is included because it identifies precisely what the programme of Chapter 26 was measuring, and why the measurement could not have decided the question. Section 6 extends the no-go to every candidate monoid admissible under (C1)–(C3).

2. The Hecke operator

Chapter 26 worked with the transfer operator $\mathcal{M}_p(x) = \frac{1}{p+1} \sum_k v_{a_k}^* x v_{a_k}$ built from the isometries of $\mathcal{T}(P)$. Its defect is structural: the range projections $e_{a_k} = v_{a_k} v_{a_k}^*$ are nested, $e_{a_j} e_{a_k} = e_p \neq 0$, so $\mathcal{M}_p$ is not a Markov operator of the tree and forcing orthogonality collapses the algebra. The remedy is to work on the Maharam extension, where the generators act by *unitaries* and no projection occurs.

DEFINITION 2.1. Let π be the Koopman representation of (1.1),

$$(\pi(a)f)(x, s) = f(xa, s + \beta \log \text{Nrd}(a)), \quad a \in G,$$

a unitary representation of G on $L^2(\tilde{\Omega} \times \mathcal{R}, \tilde{\nu}_\beta \otimes e^s ds)$ because the Maharam extension preserves the measure. For odd p fix representatives $a_1, \dots, a_{p+1} \in \mathcal{H}$ of the $p+1$ right ideals of reduced norm p and set

$$T_p = \sum_{k=1}^{p+1} \pi(a_k).$$

LEMMA 2.2. With $\varepsilon_p = \pi(p)$ the unitary of the central element p :

- (1) the set $\{a_k\}$ may be chosen closed under $a \mapsto \bar{a}$ modulo right units, and then $T_p^* = \varepsilon_p^{-1} T_p$, so $\varepsilon_p^{-1/2} T_p$ is self-adjoint for either square root;
- (2) $T_p^2 = T_{p^2} + (p+1)\varepsilon_p$, where T_{p^2} is the sum over the $p(p+1)$ right ideals of reduced norm p^2 not containing $p\mathcal{H}$;
- (3) under the identification of Theorem 1.3, T_p is the adjacency operator of the $(p+1)$ -regular tree $\mathcal{T}_p$ acting on ℓ^2 of its vertex set.

PROOF. (1) $a_k \bar{a}_k = p$, so $\pi(\bar{a}_k) = \varepsilon_p \pi(a_k)^{-1} = \varepsilon_p \pi(a_k)^*$; summing over a self-conjugate set of representatives gives the identity. (2) Expanding $T_p^2 = \sum_{j,k} \pi(a_j a_k)$, the products $a_j a_k$ of reduced norm p^2 either generate an ideal not containing $p\mathcal{H}$, and each such ideal arises exactly once, or satisfy $a_j a_k \in p\mathcal{H}$, which happens for exactly one k for each j : $a_j a_k \in p\mathcal{H}$ forces $a_k \mathcal{H} = \bar{a}_j \mathcal{H}$, and with the self-conjugate choice of (1) then $a_k = \bar{a}_j$ exactly, so the product is $a_j \bar{a}_j = p$ and contributes ε_p exactly; there are $p+1$ such pairs. (3) The vertices of $\mathcal{T}_p$ at distance n from the base vertex are the right ideals of $\mathcal{H}_p$ of reduced norm p^n that are not divisible by p beyond the necessary amount, and $a \mapsto$ (the ideal it generates) intertwines right multiplication by the a_k with the adjacency of $\mathcal{T}_p$; the identification of the Hilbert space is Corollary 3.1. (Vertices are ideal classes up to homothety, i.e. after trivializing the central $\langle p \rangle$; on the Koopman space itself the central unitary ε_p survives, which is why the self-adjoint operator is $\varepsilon_p^{-1/2} T_p$.) $\square$

REMARK 2.3. By Lubotzky–Phillips–Sarnak the group generated by $a_1, \dots, a_{p+1}$ modulo the centre and the 24 units is free of rank $(p+1)/2$ and acts simply transitively on the vertices of $\mathcal{T}_p$; the finite quotients by congruence subgroups are the Ramanujan graphs $X^{p,q}$. This is the algebraic rigidity the present programme hoped to exploit.

3. Local properness and the regular representation

PROOF OF THEOREM 1.3. *Discreteness.* Let $a_n \in G_F$ with $a_n \rightarrow 1$ in L_F . Then $\text{Nrd}(a_n) \rightarrow 1$ in $\prod_{p \in F} \mathbb{Q}_p^\times$, and $\text{Nrd}(a_n) \in \mathbb{Z}[1/F]^\times = \{\pm \prod_{p \in F} p^{e_p}\}$, so for large n all exponents vanish and

$\text{Nrd}(a_n) = 1$. Thus a_n lies in the norm-one group $\mathbb{H}(\mathbb{Z}[1/F])^1$, which is a lattice in $\prod_{p \in F} \mathbb{H}(\mathbb{Q}_p)^1$ because $\mathbb{H}$ is definite and $\mathbb{H}(\mathcal{R})^1 = SU(2)$ is compact; in particular it is discrete, so $a_n = 1$ for large n .

Properness and freeness. Write $\tilde{\Omega} = X_F \times Y_F$ with $X_F = \prod_{p \in F} \mathbb{H}(\mathbb{Q}_p)$ and Y_F the remaining coordinates. The locus where the component at p has zero reduced norm is $\tilde{\nu}_\beta$ -null: an element of $\mathcal{H}_p$ of zero reduced norm lies, for every n , in one of the right ideals of reduced norm p^n , of which there are $\sigma(p^n) = 1 + p + \cdots + p^n$, each carried by the scaling relation to measure $p^{-n\beta}$ times that of the base cylinder (an element of reduced norm p^n is a unit at every other place, by Lemma 1.2, so the cylinder is an honest translate); the bound $\sigma(p^n)p^{-n\beta} \rightarrow 0$ holds for $\beta > 1$, in particular throughout the Gibbs range $\beta > 2$ of Chapter 25. So $X_F^\times = \prod_{p \in F} \mathbb{H}(\mathbb{Q}_p)^\times$ is conull and invariant. By Lemma 1.2 and the definition of G_F , an element $a \in G_F$ acts on Y_F through the compact group $\prod_{q \notin F} \mathcal{H}_q^\times$ and on $X_F^\times$ by right translation by the discrete subgroup $G_F \leq L_F$. Right translation of a locally compact group by a discrete subgroup is free and proper, and properness in one factor of a product is properness; so the action on $X_F^\times \times Y_F$ is free and proper.

Smoothness and the Koopman representation. A free proper action of a countable group admits a measurable fundamental domain D , so $\tilde{\Omega} \times \mathcal{R} = \bigsqcup_{a \in G_F} (D \cdot a)$ up to null sets and $\mathcal{R}_F$ is smooth. Trivializing the Radon–Nikodym derivative by half-densities gives a unitary $L^2(\tilde{\Omega} \times \mathcal{R}) \cong L^2(D) \otimes \ell^2(G_F)$ intertwining $\pi \upharpoonright G_F$ with $1 \otimes \lambda_{G_F}$. $\square$

COROLLARY 3.1. $\pi \upharpoonright G_F \cong \dim L^2(D) \cdot \lambda_{G_F}$ for every finite F . In particular $\pi \upharpoonright G_F$ is tempered, and for every ξ in the group algebra $\|\pi(\xi)\| = \|\lambda_{G_F}(\xi)\|$.

4. The spectrum, and what it does not prove

PROOF OF THEOREM 1.4. Take $F = \{p\}$, so $T_p = \pi(\xi)$ with $\xi = \sum_k a_k$ in the group algebra of $G_{\{p\}}$. By Corollary 3.1, $\|T_p\| = \|\lambda(\xi)\|$ and more precisely $\text{sp}(T_p) = \text{sp}_{L(G_{\{p\}})}(\xi)$, the spectrum in the group von Neumann algebra. Modulo the centre and the 24 units, the a_k generate a group Γ acting simply transitively on the vertices of $\mathcal{T}_p$ — free of rank $(p+1)/2$ when $p \equiv 1 \pmod{4}$ [85], a free product of cyclic groups in general — and ξ maps to the sum of the $p+1$ generating translations, whose image under λ_Γ is the adjacency operator of the Cayley graph of Γ , which is $\mathcal{T}_p$ [118]. To pass from $\ell^2(G_{\{p\}})$ to ℓ^2 of the vertex set, decompose over the characters of the central subgroup generated by p and the finite unit group: in each fibre T_p acts as a phase-twisted tree adjacency operator, and on a tree every phase twist is gauge-trivial, so the fibre operator is unitarily a phase times the tree adjacency, and the self-adjoint $\varepsilon_p^{-1/2}T_p$ has in every fibre the spectrum of the tree adjacency itself. By Kesten’s theorem [70] the spectrum of the adjacency operator of the $(p+1)$ -regular tree on ℓ^2 of its vertices is exactly $[-2\sqrt{p}, 2\sqrt{p}]$, with norm $2\sqrt{p}$ and no eigenvalues. Invariant vectors would carry the eigenvalue $p+1$ and do not lie in ℓ^2 , so they are absent from L_0^2 . $\square$

COROLLARY 4.1 (There is a gap, and it is Kesten’s). $\|T_p \upharpoonright L_0^2\| = 2\sqrt{p} < p+1$, a gap of relative size $1 - 2\sqrt{p}/(p+1)$:

p	3	5	7	11	13	101
$2\sqrt{p}$	3.464	4.472	5.292	6.633	7.211	20.10
$p+1$	4	6	8	12	14	102
gap	0.134	0.255	0.339	0.447	0.485	0.803

This gap is present in the regular representation of Γ itself, and its existence is equivalent, by Kesten’s criterion [70], to the non-amenability of the free group Γ . It is therefore a statement about the group G , not about the action of G on $\tilde{\Omega}$.

THEOREM 4.2 (The proposed equivalence is vacuous). *The following are equivalent, and none of them implies non-amenability of the action:*

- (1) $\|T_p \upharpoonright L_0^2\| = 2\sqrt{p}$, the Ramanujan bound, for every odd p ;
- (2) $\pi \upharpoonright G_F$ is weakly contained in λ_{G_F} for every finite F , i.e. the Koopman representation is tempered;
- (3) the free group Γ is non-amenable.

Indeed (1) and (2) hold, by Theorem 1.4 and Corollary 3.1, while the action is amenable by Theorem 1.5.

PROOF. (1) $\Leftrightarrow$ (3) is Kesten's criterion [70]. (2) $\Rightarrow$ (1) because temperedness gives $\|\pi(\xi)\| \leq \|\lambda(\xi)\|$ and the reverse inequality is Corollary 3.1; (1) $\Rightarrow$ (2) for the algebra generated by the a_k , and the general case is Corollary 3.1 again. The last assertion is the content of the paper. $\square$

REMARK 4.3 (The orientation of the argument). Kuhn's theorem [74], quoted in Chapter 27, says that an amenable action has tempered Koopman representation. Establishing the Ramanujan bound therefore establishes temperedness, which is a *necessary condition for amenability*, not an obstruction to it. To prove (C4) by spectral means one would have to exhibit spectrum of T_p outside $[-2\sqrt{p}, 2\sqrt{p}]$ on L_0^2 — that is, a *violation* of the Ramanujan property, the exact opposite of what the arithmetic of the Hurwitz order provides. The bound $2\sqrt{p}$ is not a strong constraint that the programme could exploit; it is the Plancherel measure of $\mathrm{PGL}_2(\mathbb{Q}_p)$ announcing that there is nothing beyond the regular representation to find.

5. The no-go theorem

PROOF OF THEOREM 1.5. By Lemma 1.2 the sets $\mathcal{R}_F$, F finite, form an increasing sequence of subrelations of $\mathcal{R}_G$ with union $\mathcal{R}_G$. By Theorem 1.3 each $\mathcal{R}_F$ is smooth, hence hyperfinite. An increasing union of hyperfinite equivalence relations is hyperfinite [40, 21], so $\mathcal{R}_G$ is hyperfinite, hence amenable [137]. Amenability of the orbit relation is amenability of the measured groupoid, since the isotropy is trivial on the conull set of Theorem 1.3 and the relation is the one attached to the groupoid by [42]; amenability of the groupoid is equivalent to injectivity of the associated von Neumann algebra by [21, 3]. The classification of injective factors then leaves only R_∞ , R_λ , a Krieger factor and the type I case, and the reconstruction failure is Chapter 20. $\square$

REMARK 5.1 (Why no spectral input was needed). The proof does not see the tree, the Ramanujan property or the non-amenability of G . It uses one arithmetic fact, Lemma 1.2: an element of G has a reduced norm with finitely many prime factors, so it acts at finitely many places at a time. The group never acts *all at once*. Non-amenability of G is a statement about the group acting on itself; the groupoid only ever sees the finite-level subgroups G_F , each of which acts properly.

REMARK 5.2 (The F_2 example was the mechanism). Chapter 27 observed, as a warning about Zimmer's theorem, that the translation action of F_2 on itself preserves counting measure, is essentially free, and is nonetheless amenable, being proper. Theorem 1.3 shows that this is not an analogy: the Hurwitz action *is* an increasing union of translation actions of discrete groups on locally compact groups. The counterexample and the system coincide.

6. The search of Chapter 25 is empty

THEOREM 6.1 (General no-go). *Let A be a finite-dimensional $\mathbb{Q}$ -algebra with reduced norm Nrd , $\Lambda \subset A$ an order, $P \subseteq \Lambda \setminus \{0\}$ a right Ore, right LCM monoid with $G = PP^{-1} \leq A^\times$ and $N = |\mathrm{Nrd}|$ the norm of Chapter 25. Suppose G acts by right translation on the finite-adelic space $\tilde{\Omega} = A(\mathbb{A}_f)$ with a quasi-invariant measure scaling by $N^{-\beta}$, and that G is discrete*

in $\prod_{p \in F} A(\mathbb{Q}_p)^\times$ for every finite F — automatic when the orders of A have finite unit groups, as for definite quaternion algebras over $\mathbb{Q}$; for orders with infinite unit groups it can fail, already for a real quadratic field, where the closure of the unit group in the p -adic factors is the Leopoldt object of Papers XXII–XXIII, and the hypothesis must then be checked. Then the orbit relation is hyperfinite and the associated factor is injective, for every β .

PROOF. Lemma 1.2 holds verbatim: $a \in G$ has $\text{Nrd}(a)$ a rational number, so a is a unit of $\Lambda \otimes \mathbb{Z}_q$ for all but the finitely many q dividing its numerator and denominator. The proof of Theorem 1.3 then applies with $\mathbb{H}$ replaced by A , and the proof of Theorem 1.5 follows. $\square$

COROLLARY 6.2. *Conditions (C1)–(C3) of Chapter 25 are incompatible with (C4). Any monoid satisfying (C1) sits inside an order in a finite-dimensional $\mathbb{Q}$ -algebra, by (C2) and the finiteness of the norm; (C3) forces the unit group $P^\times$ to be finite (Chapter 25, its Remark 1.5: an infinite unit group makes ζ_P diverge identically), so the norm-one group of the order is finite, the archimedean norm-one group is compact, and the discreteness hypothesis of Theorem 6.1 holds; and every element of the monoid is a unit at almost every place; so the resulting groupoid is amenable and the factor is injective. The search for a non-injective arithmetic Bost–Connes factor cannot succeed by enlarging the index monoid.*

REMARK 6.3 (What would be required). Theorem 6.1 isolates the obstruction exactly: the finite support of the norm. An index monoid escaping it would have to contain elements that are non-units at infinitely many places, hence of infinite norm, contradicting the multiplicativity of N into $[2, \infty)$ required by (C1). Within the framework of Chapter 25, therefore, no escape exists. Enlargements outside it — an index set that is not of arithmetic type, or a coefficient algebra that is not a function algebra — are excluded by Chapter 3 and Chapter 17 respectively.

7. Verdict

Statement	Status	Where
T_p is the tree adjacency operator	true	Lem. 2.2
G_F acts properly and freely	true	Thm. 1.3
$\pi \upharpoonright G_F \cong \infty \cdot \lambda_{G_F}$	true	Cor. 3.1
$\text{sp}(\varepsilon_p^{-1/2} T_p \upharpoonright L_0^2) = [-2\sqrt{p}, 2\sqrt{p}]$	true, attained	Thm. 1.4
a spectral gap exists	true, and it is Kesten’s	Cor. 4.1
the gap implies (C4)	false	Thm. 4.2
the orbit relation is hyperfinite	true	Thm. 1.5
(C4)	false	Thm. 1.5
some other monoid satisfies (C4)	false	Thm. 6.1

The Hurwitz system has everything the programme asked for. Its group of fractions is non-amenable; it carries free subgroups of every finite rank; it generates the Bruhat–Tits trees whose congruence quotients are the Ramanujan graphs; the Hecke operator on the Maharam extension is exactly the tree adjacency operator, with no nested projections and no Cuntz obstruction; and its spectrum attains the Ramanujan bound. None of it matters. The bound $2\sqrt{p}$ is the norm of the same operator in the regular representation, so attaining it certifies that the Koopman representation is regular — and a regular Koopman representation is what a free proper action has. The rigidity is real and it belongs to the group; the action never inherits it, because an element of an arithmetic monoid is a unit at almost every place and the group acts only finitely many places at a time.

The Bost–Connes phase transition therefore survives every enlargement of the index monoid compatible with its own axioms, and the factor does not. The arithmetic lives in the C^* -algebra

with its flow, in the KMS simplex, and in the obstruction group; the von Neumann algebra is, and remains, hyperfinite.

Part 6

Geometric and function-field analogues

Localized Bost–Connes systems in arithmetic topology: local class field theory, linking Kakutani dichotomies, and topological phase transitions

This chapter is Paper VIII of the series (papers/VIII-arithmetic-topology/).

SUMMARY. We quantize the Mazur–Kapranov–Reznikov dictionary [90, 67, 111, 93] by transporting the localized Bost–Connes theory of [Ch. 1] to knots in a 3-manifold, and determine which topological data drive the resulting thermodynamics.

Everything rests on one choice. In the dictionary the boundary torus of a knot plays the role of the local field, $H_1(\partial N(K_i); \widehat{\mathbb{Z}}) \leftrightarrow \widehat{K_{\mathfrak{p}}^{\times}} = \pi_{\mathfrak{p}}^{\widehat{\mathbb{Z}}} \times \mathcal{O}_{\mathfrak{p}}^{\times}$, with the meridian corresponding to *inertia* and the longitude to *Frobenius*, so the Bost–Connes parameter must be $\sigma_{K_i} = [\lambda_i]$. The half-lives-half-dies theorem then completes the dictionary: the Lagrangian $\ker(H_1(\partial X) \rightarrow H_1(X))$, spanned for $M = S^3$ by the 0-framed longitudes $\widetilde{\lambda}_i = \lambda_i - \sum_j \text{Lk}(K_i, K_j)\mu_j$, corresponds to the principal ideles $\overline{K_{S,+}^{\times}}$, its intersection with the meridian subgroup to the global units $\overline{\mathcal{O}_{K,+}^{\times}}$ — which vanishes for $M = S^3$, matching $\mathbb{Z}_+^{\times} = \{1\}$ under $\text{Spec } \mathbb{Z} \leftrightarrow S^3$ — and the subgroup of $H_1(M; \mathbb{Z})$ generated by the knot classes to Cl_S^+ .

With this parameter the KMS_{β} simplex is $\text{Prob}(G_{\mathcal{L}}/\Xi_{\beta}^{\perp})$ with the obstruction group governed by the Gauss linking matrix: a character with finite support F and denominator m satisfies $\chi(\sigma_{K_i}) = \exp(2\pi i \sum_{j \in F} \text{Lk}(K_i, K_j)a_j)$, so its detecting set is $D_{\chi} = \{i : \sum_{j \in F} \text{Lk}(K_i, K_j)b_j \not\equiv 0 \pmod{m}\}$. Two structural facts decide whether D_{χ} is infinite, and we prove both: local finiteness is vacuous for an infinite link in a closed manifold, so components must accumulate; and once they do, a compact Seifert surface can be pierced infinitely often and the linking rows acquire infinite support. We exhibit the simplest such family — infinitely many parallel copies of a knot with framing f — for which D_{χ} omits only F whenever $f \sum_{j \in F} a_j \notin \mathbb{Z}$, so that Ξ_{β} is genuinely β -dependent, although for this family some characters are never detected and the state is never unique. We then prove that every symmetric integer matrix is the linking matrix of a countable tame link in S^3 , and use the matrix of binary digits $\text{Lk}(K_i, K_j) = \text{bit}_{\min(i,j)}(\max(i,j))$ to build a link whose KMS_{β} state is unique for $\beta \leq \beta_c$ and whose simplex is all of $\text{Prob}(\prod_j \widehat{\mathbb{Z}})$ above: the Bost–Connes phase diagram, realized by linking alone, with open or closed broken phase according to the norm. Ambient homology enters twice: the subgroup of $H_1(M; \mathbb{Z})$ generated by the knot classes carries the exact analogue of the $h_S^+ > 1$ obstruction, while the quotient $H_1(M; \mathbb{Z})/\langle [K_i] \rangle$, which the topological group sees and G_S never does, breaks the symmetry at every temperature.

Section 6 isolates why the meridian is the wrong choice: μ_i is a single basis vector of $H_1(S^3 \setminus \mathcal{L})$, so D_{χ} is finite and Ξ_{β} is everything at every temperature. This is what inertia should do — inertia is what an unramified cover kills, and the phase structure lives in the unramified direction. The contrast with $\sigma_{\mathfrak{p}} = (\pi_{\mathfrak{p}})_{\mathfrak{q} \neq \mathfrak{p}}$, nontrivial at every other place by the product formula, is exact: reciprocity and linking are two ways of spreading a local generator over infinitely many places.

1. The local dictionary and its homological foundation

Let M be a closed, connected, oriented 3-manifold and $\mathcal{L} = \{K_1, K_2, \dots\}$ a countable family of disjoint tame knots. Write $N(\mathcal{L})$ for an open tubular neighbourhood,

$$X := M \setminus N(\mathcal{L}), \quad \partial X = \bigsqcup_i T_i, \quad T_i = \partial N(K_i) \cong T^2,$$

so $H_1(T_i; \mathbb{Z}) = \mathbb{Z}\mu_i \oplus \mathbb{Z}\lambda_i$ is free on the meridian and a chosen longitude. For $M = S^3$ we take λ_i to be the Seifert longitude, and we write $\text{Lk}(K_i, K_j)$ for the linking number when $i \neq j$ and $\text{Lk}(K_i, K_i)$ for the framing of λ_i , so $\text{Lk}(K_i, K_i) = 0$ in S^3 .

1.1. Local class field theory for knots.

DEFINITION 1.1 (The local entry). The dictionary of Mazur, Kapranov–Smirnov, Reznikov and Morishita [90, 67, 111, 93] pairs $\text{Spec } \mathcal{O}_{\mathfrak{p}} \leftrightarrow N(K_i)$ and $\text{Spec } K_{\mathfrak{p}} \leftrightarrow N(K_i) \setminus K_i \simeq T_i$, and the local reciprocity isomorphism $\text{Gal}(K_{\mathfrak{p}}^{ab}/K_{\mathfrak{p}}) \cong \widehat{K_{\mathfrak{p}}}^{\times} = \pi_{\mathfrak{p}}^{\widehat{\mathbb{Z}}} \times \mathcal{O}_{\mathfrak{p}}^{\times}$ [97, Ch. V] with $\pi_1^{ab}(T_i)^{\wedge} = \widehat{\mathbb{Z}}\lambda_i \oplus \widehat{\mathbb{Z}}\mu_i$, the entries being

$$(1.1) \quad \mu_i \longleftrightarrow \text{inertia } \mathcal{O}_{\mathfrak{p}}^{\times}, \quad \lambda_i \longleftrightarrow \text{Frobenius } \pi_{\mathfrak{p}}^{\widehat{\mathbb{Z}}}.$$

This is a convention, not a theorem, and everything below depends on it; the reasons for it are the following. In a cyclic cover of M branched along K_i the meridian carries the local monodromy around the branch locus, and the meridians are exactly what must die in an unramified cover of M , as the inertia subgroup $\mathcal{O}_{\mathfrak{p}}^{\times}$ is exactly what dies in an unramified extension. The unramified quotient $\text{Gal}(K_{\mathfrak{p}}^{ur}/K_{\mathfrak{p}}) = \widehat{\mathbb{Z}}$ is generated by Frobenius and matches $\widehat{\mathbb{Z}}\lambda_i$, the longitude surviving in an unramified cover with its image recording the covering degree along K_i . (The tame quotient of $\text{Gal}(\bar{K}_{\mathfrak{p}}/K_{\mathfrak{p}})$ is $\widehat{\mathbb{Z}}' \rtimes \widehat{\mathbb{Z}}$ and matches $\pi_1(T_i)$ only after abelianization and completion; the matching of the inertia group $\mathcal{O}_{\mathfrak{p}}^{\times}$ with the single generator μ_i is the usual coarsening of the dictionary.)

REMARK 1.2 (The choice that decides everything). The informal statement of the dictionary — “primes correspond to knots” — does not say which class on the boundary torus is Frobenius, and the meridians are the visible generators of $H_1(S^3 \setminus \mathcal{L})$ (Alexander duality). Setting $\sigma_{K_i} = [\mu_i]$ is therefore tempting, and yields a theory with no thermodynamics at all; §6 carries out that computation and explains it. Everything below uses $\sigma_{K_i} = [\lambda_i]$.

1.2. Half lives, half dies, and the unit group. Let $\iota_* : H_1(\partial X; \widehat{\mathbb{Z}}) \rightarrow H_1(X; \widehat{\mathbb{Z}})$ be induced by inclusion and write $\mathbb{I}_{\partial} := H_1(\partial X; \widehat{\mathbb{Z}}) = \bigoplus_i (\widehat{\mathbb{Z}}\mu_i \oplus \widehat{\mathbb{Z}}\lambda_i)$.

THEOREM 1.3 (The Lagrangian). *For X compact oriented with boundary a disjoint union of n tori,*

$$\dim_{\mathbb{Q}} \ker(H_1(\partial X; \mathbb{Q}) \rightarrow H_1(X; \mathbb{Q})) = \frac{1}{2} \dim_{\mathbb{Q}} H_1(\partial X; \mathbb{Q}) = n,$$

and the kernel is Lagrangian for the intersection form. For $M = S^3$ it is spanned by the 0-framed longitudes

$$(1.2) \quad \widetilde{\lambda}_i := \lambda_i - \sum_j \text{Lk}(K_i, K_j) \mu_j.$$

PROOF. The first statement is the classical half-lives-half-dies lemma, from the long exact sequence of $(X, \partial X)$ with Poincaré–Lefschetz duality [127, Ch. 4]. For $M = S^3$, Alexander duality gives $H_1(X) = \bigoplus_j \mathbb{Z}\mu_j$ with $\iota_*\mu_i = \mu_i$ and $\iota_*\lambda_i = \sum_j \text{Lk}(K_i, K_j)\mu_j$, so $\sum_i (p_i\mu_i + q_i\lambda_i)$ lies in the kernel precisely when $p_j = -\sum_i q_i \text{Lk}(K_i, K_j)$ for every j ; this is free of rank n on the q_i , with basis (1.2). $\square$

THEOREM 1.4 (The completed dictionary). *Let $\mathcal{L}$ be finite with n components, $\iota_* : \mathbb{I}_\partial \rightarrow H_1(X; \widehat{\mathbb{Z}})$, and let $\nu : H_1(X; \widehat{\mathbb{Z}}) \rightarrow H_1(M; \widehat{\mathbb{Z}})$ be induced by inclusion, so that $\nu \iota_* \mu_i = 0$ and $\nu \iota_* \lambda_i = [K_i]$. Put*

$$G_{\mathcal{L}}^{\text{idl}} := \mathbb{I}_\partial / \ker \iota_* = \iota_*(\mathbb{I}_\partial) \leq H_1(X; \widehat{\mathbb{Z}}), \quad \text{Cl}_{\mathcal{L}} := \langle [K_i] : i \rangle \leq H_1(M; \widehat{\mathbb{Z}}).$$

Then the idele exact sequence

$$1 \longrightarrow U_S / \overline{\mathcal{O}_{K,+}^\times} \longrightarrow G_S = \mathbb{I}_S / \overline{K_{S,+}^\times} \longrightarrow \text{Cl}_S^+ \longrightarrow 1$$

of [Ch. 1, Prop. 2.5], in which $\text{Cl}_S^+ \leq \text{Cl}_K^+$ is the subgroup generated by the classes of the primes of S and the last map is the divisor map, corresponds term by term to the exact sequence

$$1 \longrightarrow \left(\bigoplus_i \widehat{\mathbb{Z}} \mu_i \right) / \left(\ker \iota_* \cap \bigoplus_i \widehat{\mathbb{Z}} \mu_i \right) \longrightarrow G_{\mathcal{L}}^{\text{idl}} \xrightarrow{\nu} \text{Cl}_{\mathcal{L}} \longrightarrow 1,$$

via

$$\mathbb{I}_S \leftrightarrow \mathbb{I}_\partial, \quad U_S \leftrightarrow \bigoplus_i \widehat{\mathbb{Z}} \mu_i, \quad \overline{K_{S,+}^\times} \leftrightarrow \ker \iota_*, \quad \overline{\mathcal{O}_{K,+}^\times} \leftrightarrow \ker \iota_* \cap \bigoplus_i \widehat{\mathbb{Z}} \mu_i, \quad \text{Cl}_S^+ \leftrightarrow \text{Cl}_{\mathcal{L}}.$$

For $M = S^3$ the unit term and $\text{Cl}_{\mathcal{L}}$ both vanish, matching $\mathbb{Z}_+^\times = \{1\}$ and $\text{Cl}(\mathbb{Z}) = 1$ under $\text{Spec } \mathbb{Z} \leftrightarrow S^3$, and $G_{\mathcal{L}}^{\text{idl}} = H_1(X; \widehat{\mathbb{Z}}) = \bigoplus_i \widehat{\mathbb{Z}} \mu_i$. In general $H_1(X; \widehat{\mathbb{Z}}) / G_{\mathcal{L}}^{\text{idl}} \cong H_1(M; \widehat{\mathbb{Z}}) / \text{Cl}_{\mathcal{L}}$, a quotient with no arithmetic counterpart.

PROOF. The first two assignments are Definition 1.1, $K_{\mathfrak{p}}^\times$ decomposing into a valuation part and a unit part matching $\widehat{\mathbb{Z}} \lambda_i$ and $\widehat{\mathbb{Z}} \mu_i$. The principal ideles are those dying in the global quotient, matching $\ker \iota_*$; the global units are the principal ideles that are local units at every place, matching the intersection with the meridian subgroup; and the divisor map, which reads off the valuations of an idele, matches ν , which reads off the longitude coefficients since $\nu \iota_* \mu_i = 0$ and $\nu \iota_* \lambda_i = [K_i]$, the longitude being isotopic to the core of $N(K_i)$. Exactness: ν maps $G_{\mathcal{L}}^{\text{idl}}$ onto the subgroup generated by the $[K_i]$; its kernel is $\iota_*^{-1}(\ker \nu) / \ker \iota_*$, and $\ker \nu = \langle \iota_* \mu_i \rangle$ because M is obtained from X by Dehn filling along the meridians, so that $H_1(M) = H_1(X) / \langle \mu_i \rangle$; if $\iota_* \xi = \iota_* \eta$ with $\eta \in \bigoplus \widehat{\mathbb{Z}} \mu_i$ then $\xi \equiv \eta$ modulo $\ker \iota_*$, which identifies the kernel with the image of $\bigoplus \widehat{\mathbb{Z}} \mu_i$. For $M = S^3$, by (1.2) an element of $\ker \iota_*$ is $\sum_i q_i \tilde{\lambda}_i$, which lies in $\bigoplus \widehat{\mathbb{Z}} \mu_i$ only if every $q_i = 0$, and $H_1(S^3) = 0$. The last assertion is $H_1(X) / \langle \iota_* \mu_i, \iota_* \lambda_i \rangle = (H_1(X) / \langle \mu_i \rangle) / \langle [K_i] \rangle$. $\square$

REMARK 1.5 (Numerical check). For random symmetric linking matrices with zero diagonal and $n \in \{2, 3, 5, 8\}$, the rank of $\ker \iota_*$ was computed to be exactly $n = \frac{1}{2} \dim H_1(\partial X)$ in every case, and $\ker \iota_* \cap \bigoplus \widehat{\mathbb{Z}} \mu_i = 0$ in every case.

2. The knot-theoretic Bost–Connes system

REMARK 2.1 (Infinite links must accumulate). For infinite $\mathcal{L}$ in a closed M one cannot require that every compact set meet only finitely many components: taking the compact set to be M would force $\mathcal{L}$ finite. The components of an infinite link therefore accumulate on a nonempty closed set. This is not a defect; by Theorem 3.3 it is exactly what allows the linking rows to have infinite support, and hence what makes the theory nonempty.

DEFINITION 2.2 (Norm). A norm is a function $N : \mathcal{L} \rightarrow [2, \infty)$, extended to $J_{\mathcal{L}} = \bigoplus_i \mathbb{N} \cdot K_i$ by $N(D) = \prod_i N(K_i)^{d_i}$; this is well defined and multiplicative because $J_{\mathcal{L}}$ is free. Candidates on generators: $N(K_i) = e^{\text{Vol}(M \setminus K_i)}$ for hyperbolic components, $e^{2\pi \deg \Delta_{K_i}}$, or $e^{c \ell(\mu_i)}$.

REMARK 2.3 (Geometric only on generators). Multiplicativity is imposed. For $D = K_1 + K_2$ the value $N(K_1)N(K_2)$ is not $e^{\text{Vol}(M \setminus (K_1 \cup K_2))}$, hyperbolic volume being neither additive over

disjoint unions nor monotone in the naive direction. The geometric reading survives on generators only; N -multiplicativity, a theorem about ideals, is here a definition.

DEFINITION 2.4 (The system). Let

$$G_{\mathcal{L}} := \varprojlim_{F,n} H_1(M \setminus \mathcal{L}_F; \mathbb{Z}/n\mathbb{Z}), \quad \sigma_{K_i} := [\lambda_i] \in G_{\mathcal{L}},$$

the limit over finite sublinks F and integers n , the transition maps being induced by the inclusions $M \setminus \mathcal{L}_{F'} \subset M \setminus \mathcal{L}_F$ for $F \subset F'$; $G_{\mathcal{L}}$ is a compact abelian group, and $[\lambda_i]$ is the compatible family of classes of the longitude of K_i in the complements of the finite sublinks containing K_i . For finite $\mathcal{L}$, $G_{\mathcal{L}} = H_1(X; \widehat{\mathbb{Z}})$. Let $X_{\mathcal{L}} = \overline{\mathbb{N}}^{\mathcal{L}}$ with $\overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\}$, on which $J_{\mathcal{L}}$ acts by $v \mapsto v + D$, and put

$$Y_{\mathcal{L}} := X_{\mathcal{L}} \times G_{\mathcal{L}}, \quad D \cdot (v, g) = (v + D, g + \sum_i d_i \sigma_{K_i}),$$

$$\mathcal{A}_{M,\mathcal{L}} := C(Y_{\mathcal{L}}) \rtimes J_{\mathcal{L}}, \quad \sigma_t(\mu_D) = N(D)^{it} \mu_D,$$

the semigroup crossed product being the unital full corner of the groupoid algebra of the dilated action of $I_{\mathcal{L}} = \mathbb{Z}^{(\mathcal{L})}$, exactly as $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$ in [Ch. 1, §2]; measure-theoretically $Y_S = \widehat{\mathcal{O}}_S \times_{U_S} G_S$ is $\mathbb{N}^S \times G_S$ with the same action, by [Ch. 1, Lem. 2.10].

PROPOSITION 2.5 (No new operator algebra is needed). *$\mathcal{A}_{M,\mathcal{L}}$ is the system of [Ch. 1] attached to the quadruple $(\mathcal{L}, J_{\mathcal{L}}, G_{\mathcal{L}}, N)$ with σ_{K_i} in place of $\sigma_{\mathfrak{p}}$: it is the abelian case of the Galois-twisted groupoid of Chapter 16 with $\text{Frob}_{K_i} = [\lambda_i]$. The classification theorems [Ch. 1, Thm. 3.6, Thm. 3.9] and the factor-type results of [Ch. 2] that depend only on the sequence $(N(K_i), \chi(\sigma_{K_i}))$ apply verbatim. Results of the series whose proofs use arithmetic input — Chebotarev density, prime number theorems, class numbers, the finiteness of the unit group — do not transfer automatically and have to be re-proved, or fail, in the topological setting.*

PROOF. The proofs of [Ch. 1, Thm. 3.6, Thm. 3.9] use only that J_S is a free abelian monoid on a countable set acting on $\overline{\mathbb{N}}^S \times G_S$ as above, that G_S is compact abelian with prescribed elements $\sigma_{\mathfrak{p}}$, that N is multiplicative with values in $[2, \infty)$, and the Kolmogorov zero-one law for the forced product measure; they do not use that the $\sigma_{\mathfrak{p}}$ generate a dense subgroup, since Ξ_{β} is defined by characters of the whole group. The quadruple $(\mathcal{L}, J_{\mathcal{L}}, G_{\mathcal{L}}, N)$ satisfies all of this. No property of prime ideals is used in the classification; arithmetic enters only when the criterion is evaluated. $\square$

3. The linking obstruction group

THEOREM 3.1 (Topological KMS classification). *The KMS_{β} simplex of $\mathcal{A}_{M,\mathcal{L}}$ is affinely isomorphic to $\text{Prob}(G_{\mathcal{L}}/\Xi_{\beta}(M, \mathcal{L})^{\perp})$, where*

$$(3.1) \quad \Xi_{\beta}(M, \mathcal{L}) = \left\{ \chi \in \widehat{G_{\mathcal{L}}} : \sum_i N(K_i)^{-\beta} |1 - \chi([\lambda_i])|^2 < \infty \right\},$$

and the state is unique if and only if every nontrivial χ has $\sum_{i \in D_{\chi}} N(K_i)^{-\beta} = \infty$, where $D_{\chi} = \{i : \chi([\lambda_i]) \neq 1\}$.

PROOF. Proposition 2.5 together with [Ch. 1, Thm. 3.6, Thm. 3.9]. Note that a character trivial on every σ_{K_i} has $D_{\chi} = \emptyset$ and lies in Ξ_{β} for all β ; the criterion therefore fails, at every temperature, unless the σ_{K_i} generate a dense subgroup of $G_{\mathcal{L}}$. In the arithmetic case this density is a form of Chebotarev's theorem; here it is a condition on the linking matrix. $\square$

THEOREM 3.2 (The Gauss linking matrix formula). *Let $M = S^3$, so $\widehat{G}_{\mathcal{L}} = \bigoplus_j \mathbb{Q}/\mathbb{Z}$ by Alexander duality. For $\chi = (a_j)$ with finite support F and common denominator m , writing $b_j = ma_j$,*

$$(3.2) \quad \chi(\sigma_{K_i}) = \exp\left(2\pi i \sum_{j \in F} \text{Lk}(K_i, K_j) a_j\right), \quad D_\chi = \left\{i : \sum_{j \in F} \text{Lk}(K_i, K_j) b_j \not\equiv 0 \pmod{m}\right\}.$$

For χ of order m the weight $|1 - \chi([\lambda_i])|^2$ is bounded above and below by positive constants on D_χ , so

$$\chi \in \Xi_\beta(S^3, \mathcal{L}) \iff \sum_{i \in D_\chi} N(K_i)^{-\beta} < \infty.$$

PROOF. For $M = S^3$ and each finite sublink, $H_1(M \setminus \mathcal{L}_F; \mathbb{Z})$ is free on the meridians μ_j , $j \in F$, by Alexander duality, the transition maps kill the meridians of the omitted components, and $G_{\mathcal{L}} = \prod_j \widehat{\mathbb{Z}}\mu_j$, whose dual is the direct sum of the duals of the factors. Then $\iota_*\lambda_i = \sum_{j \in F} \text{Lk}(K_i, K_j)\mu_j$ in each finite complement, by Theorem 1.3 with our framing convention; evaluate χ . The weight takes finitely many values, all nonzero on D_χ , since χ has order m . $\square$

THEOREM 3.3 (Infinite detecting sets exist). *Let $K \subset S^3$ be a knot, $f \in \mathbb{Z}$ a framing, and let $\mathcal{L} = \{K_1, K_2, \dots\}$ consist of infinitely many parallel copies of K with framing f , accumulating on K , so that $\text{Lk}(K_i, K_j) = f$ for all $i \neq j$. Let $\chi = (a_j)_{j \in F}$ and $\Sigma_F = \sum_{j \in F} a_j$. If $f\Sigma_F \notin \mathbb{Z}$ then*

$$D_\chi \supseteq \{i : i \notin F\},$$

so D_χ is cofinite and $\chi \in \Xi_\beta(S^3, \mathcal{L})$ if and only if $\sum_i N(K_i)^{-\beta} < \infty$. Consequently Ξ_β is genuinely β -dependent and changes at the convergence exponent of the norm sequence. If $f\Sigma_F \in \mathbb{Z}$, however, $D_\chi \subseteq F$ is finite and $\chi \in \Xi_\beta$ for every β ; such characters exist for every f (take $F = \{1, 2\}$, $a_1 = a_2 = \frac{1}{2}$), so for this family the KMS_β state is never unique.

PROOF. For $i \notin F$ we have $\text{Lk}(K_i, K_j) = f$ for every $j \in F$, so by (3.2) $\chi(\sigma_{K_i}) = \exp(2\pi i f \Sigma_F)$, which is $\neq 1$ iff $f\Sigma_F \notin \mathbb{Z}$. The rest is Theorem 3.2. $\square$

LEMMA 3.4 (Every symmetric matrix is a linking matrix). *Let $(\ell_{ij})_{i \neq j}$ be any symmetric family of integers indexed by pairs of distinct positive integers. There is a countable family $\mathcal{L} = \{K_1, K_2, \dots\}$ of pairwise disjoint tame knots in S^3 , each individually unknotted, with $\text{Lk}(K_i, K_j) = \ell_{ij}$ for all $i \neq j$. If some column $(\ell_{ij})_i$ has infinite support, the family accumulates on K_j .*

PROOF. Construct K_n inductively, K_1 an unknot. Given the disjoint tame knots $K_1, \dots, K_{n-1}$, take for each $j < n$ with $\ell_{nj} \neq 0$ a collection of $|\ell_{nj}|$ small meridian circles of K_j , oriented so that each links K_j with linking number $\text{sgn } \ell_{nj}$, chosen of radius less than 2^{-n} and disjoint from everything constructed so far; if all $\ell_{nj} = 0$ take one small unknotted circle far from the previous components. Join these circles by bands along disjoint arcs in the complement of $K_1 \cup \dots \cup K_{n-1}$ and of the circles, orienting the bands compatibly; the band-connected sum of finitely many disjoint oriented circles is a single tame knot K_n , unknotted since the circles bound disjoint disks which the bands can be chosen to avoid, and it is homologous in $S^3 \setminus K_j$ to the sum of the circles, so $\text{Lk}(K_n, K_j) = \ell_{nj}$ for every $j < n$; by symmetry of the linking number this fixes $\text{Lk}(K_j, K_n)$ as well. The components are pairwise disjoint by construction, and if column j is infinitely supported, infinitely many K_n pass within 2^{-n} of K_j . $\square$

THEOREM 3.5 (A link with the Bost–Connes phase diagram). *Let $\mathcal{L}$ be a link in S^3 realizing, by Lemma 3.4, the linking numbers*

$$\text{Lk}(K_i, K_j) = \text{bit}_{\min(i,j)}(\max(i,j)) \in \{0, 1\} \quad (i \neq j),$$

where $\text{bit}_k(n)$ is the coefficient of 2^k in the binary expansion of n , and let N be a norm with $N(K_i) = (i+1)^\gamma$ for some $\gamma > 0$, so that $\sum_i N(K_i)^{-\beta}$ converges iff $\beta > \beta_c := 1/\gamma$. Then

$$\Xi_\beta(S^3, \mathcal{L}) = \{1\} \text{ for } \beta \leq \beta_c, \quad \Xi_\beta(S^3, \mathcal{L}) = \widehat{G_\mathcal{L}} \text{ for } \beta > \beta_c :$$

the KMS_β state of $\mathcal{A}_{S^3, \mathcal{L}}$ is unique for $\beta \leq \beta_c$, and for $\beta > \beta_c$ the simplex is $\text{Prob}(G_\mathcal{L}) = \text{Prob}(\prod_j \widehat{\mathbb{Z}})$, on which $G_\mathcal{L}$ acts freely and transitively on the extreme points. Every column of the linking matrix is infinitely supported, so $\mathcal{L}$ accumulates on every one of its components.

PROOF. For $\beta > \beta_c$ every sum in (3.1) converges, so $\Xi_\beta = \widehat{G_\mathcal{L}}$ and $\Xi_\beta^\perp = 0$. Let $\beta \leq \beta_c$ and let $\chi \neq 1$ have support F , denominator $m \geq 2$ and $b = (b_j)_{j \in F}$ not all $\equiv 0 \pmod{m}$. For $i > \max F$, (3.2) gives $i \in D_\chi$ iff $\sum_{j \in F} \text{bit}_j(i) b_j \not\equiv 0 \pmod{m}$. Put $L = \max F + 1$. On any block $[k2^L, (k+1)2^L)$ of 2^L consecutive integers the residue $i \bmod 2^L$ runs over all values once, so the vector $(\text{bit}_j(i))_{j \in F} \in \{0, 1\}^F$ takes each value exactly $2^{L-|F|}$ times. Fix $j_0 \in F$ with $b_{j_0} \not\equiv 0$ and pair each $\varepsilon \in \{0, 1\}^F$ with the vector ε' differing from it in the j_0 -th coordinate only: $\sum_j \varepsilon_j b_j$ and $\sum_j \varepsilon'_j b_j$ differ by $\pm b_{j_0} \not\equiv 0$, so at most one of them is $\equiv 0$. Hence every block contains at least 2^{L-1} elements of D_χ , apart from the finitely many $i \leq \max F$, and D_χ has lower asymptotic density at least $\frac{1}{2}$. A set D of positive lower density c has k -th element at most k/c for k large, hence $\sum_{i \in D} (i+1)^{-s} \geq \sum_k (k/c+1)^{-s} = \infty$ for every $s \leq 1$, and $\gamma\beta \leq 1$; so $\chi \notin \Xi_\beta$. Thus $\Xi_\beta = \{1\}$ and the state is unique by Theorem 3.1. Column j of the matrix contains $\text{bit}_j(i)$ for all $i > j$, which is 1 for infinitely many i . $\square$

REMARK 3.6. Theorem 3.5 is the topological Bost–Connes phase diagram in its original shape: below and at the critical temperature a unique equilibrium state, above it spontaneous symmetry breaking with the full compact group of symmetries acting simply transitively on the extreme states, and no intermediate regime. Intermediate chains $\{\Xi_\beta\}$, as realized arithmetically in [Ch. 2, Thm. 5.1] by congruence conditions on primes, can be realized by prescribing rows of the linking matrix whose congruence classes are controlled; we do not pursue this. What Theorem 3.5 cannot control is the norm: the norm is prescribed here, and Section 5 records what happens when it is asked to be geometric.

REMARK 3.7 (Why local finiteness had to go). If $\mathcal{L}$ were locally finite, a compact Seifert surface Σ_j for K_j would meet only finitely many components, so $\text{Lk}(K_j, K_i) \neq 0$ for finitely many i ; by symmetry of the linking number every column of the matrix would then be finitely supported and every D_χ finite, returning the empty theory of §6. By Remark 2.1 local finiteness is in any case impossible for infinite $\mathcal{L}$ in closed M . Accumulation is what lets Σ_j be pierced infinitely often, and Theorem 3.3 realizes this in the simplest way; Lemma 3.4 shows that nothing beyond symmetry constrains the linking matrix of a countable tame link.

REMARK 3.8 (Numerical illustration). For random integer linking data $\text{Lk}(K_i, K_j) \in [-4, 4]$ over 400 components and a character supported on two meridians, the detecting set contained 45–47% of components for $m = 2$, 65–69% for $m = 3$ and 78–90% for $m = 5$. This illustrates the generic behaviour of the congruence condition in (3.2); it is not itself a link realization, the random matrices being constrained neither to symmetry nor to arising from an embedded family. The rigorous statements are Theorems 3.3 and 3.5.

4. Ambient homological obstructions

Let $\nu : G_\mathcal{L} \rightarrow H_1(M; \widehat{\mathbb{Z}})$ be the map induced by the inclusions $M \setminus \mathcal{L}_F \subset M$; it is surjective, since $H_1(M \setminus \mathcal{L}_F) \rightarrow H_1(M)$ is onto for every finite F and the groups are compact, and $\nu(\sigma_{K_i}) = [K_i]$, $\nu([\mu_i]) = 0$. As in Theorem 1.4 put $\text{Cl}_\mathcal{L} = \langle [K_i] : i \rangle \leq H_1(M; \mathbb{Z})$.

THEOREM 4.1 (Ambient homology). *Let ψ be a character of $H_1(M; \widehat{\mathbb{Z}})$ and $\chi = \psi \circ \nu \in \widehat{G_{\mathcal{L}}}$. Then $\chi(\sigma_{K_i}) = \psi([K_i])$ for every i , and*

$$\chi \in \Xi_{\beta}(M, \mathcal{L}) \iff \sum_{i: \psi([K_i]) \neq 1} N(K_i)^{-\beta} < \infty.$$

Consequently:

- (1) *If ψ is trivial on $\text{Cl}_{\mathcal{L}}$, then $\chi \in \Xi_{\beta}(M, \mathcal{L})$ for every $\beta > 0$. The characters of $H_1(M; \widehat{\mathbb{Z}})/\overline{\text{Cl}_{\mathcal{L}}}$ pull back injectively to Ξ_{β} , so with $h_{\mathcal{L}}^0 := |H_1(M; \mathbb{Z})/\text{Cl}_{\mathcal{L}}|$ the KMS_{β} simplex has at least $h_{\mathcal{L}}^0$ extreme points at every temperature, and uniqueness never holds unless the knot classes generate $H_1(M; \mathbb{Z})$. This obstruction has no arithmetic counterpart: G_S maps onto the subgroup Cl_S^+ generated by the primes of S and never sees $\text{Cl}_K^+/\text{Cl}_S^+$.*
- (2) *If ψ is nontrivial on $\text{Cl}_{\mathcal{L}}$ and the knots whose classes lie outside $\ker \psi \cap \text{Cl}_{\mathcal{L}}$ form a β -summable set, then $\chi \in \Xi_{\beta}(M, \mathcal{L})$ and the KMS_{β} state is not unique. This is the exact counterpart of the unramified class-group obstruction [Ch. 1, Prop. 4.1]: a proper subgroup $H < \text{Cl}_{\mathcal{L}}$ containing the classes of all but a β -summable set of components, with $\text{Cl}_{\mathcal{L}}$ in place of Cl_S^+ and $[K_i]$ in place of $[\mathfrak{p}]$.*

PROOF. $\chi(\sigma_{K_i}) = \psi(\nu[\lambda_i]) = \psi([K_i])$. The group $H_1(M; \widehat{\mathbb{Z}})$ is profinite, so ψ has finite order and $|1 - \psi([K_i])|^2$ takes finitely many values, bounded below by a positive constant where it is nonzero; the displayed equivalence follows from (3.1). (1): all summands vanish. Distinct characters of $H_1(M; \widehat{\mathbb{Z}})$ pull back to distinct characters of $G_{\mathcal{L}}$ since ν is surjective, and a group of $h_{\mathcal{L}}^0$ characters inside Ξ_{β} forces $|G_{\mathcal{L}}/\Xi_{\beta}^{\perp}| \geq h_{\mathcal{L}}^0$. (2): the sum in the equivalence runs over the components with $[K_i] \notin \ker \psi$. $\square$

COROLLARY 4.2 (The parallel, corrected). *The counterpart of the unramified obstruction of [Ch. 1, §4.1] is Theorem 4.1(2), under*

$$\text{Cl}_S^+ = \langle [\mathfrak{p}] : \mathfrak{p} \in S \rangle \leq \text{Cl}_K^+ \iff \text{Cl}_{\mathcal{L}} = \langle [K_i] : i \rangle \leq H_1(M; \mathbb{Z}), \quad h_S^+ = |\text{Cl}_S^+| \leftrightarrow |\text{Cl}_{\mathcal{L}}|.$$

The quotient $H_1(M; \mathbb{Z})/\text{Cl}_{\mathcal{L}}$ of Theorem 4.1(1) is a purely topological addition: the complement of a link remembers the homology of the ambient manifold, while the S -ideles remember nothing of the class group beyond the classes of S .

EXAMPLE 4.3. Let $M = L(p, 1)$, $H_1(M; \mathbb{Z}) = \mathbb{Z}/p$. If all K_i are null-homologous, $\text{Cl}_{\mathcal{L}} = 0$, $h_{\mathcal{L}}^0 = p$, and the simplex has at least p extreme points at every β by Theorem 4.1(1); the h_S^+ -type obstruction is absent, as for a field with $\text{Cl}_S^+ = 1$. If instead some components generate $H_1(M)$, then $h_{\mathcal{L}}^0 = 1$, $\text{Cl}_{\mathcal{L}} = \mathbb{Z}/p$, and Theorem 4.1(2) applies: for a character ψ of $\mathbb{Z}/p$, the pull-back lies in Ξ_{β} iff the components with $\psi([K_i]) \neq 1$ are β -summable, so a link in which all but a summable family of components are null-homologous has broken symmetry at every β , exactly as an S whose primes lie, up to a summable set, in a proper subgroup of Cl_S^+ . What matters is the distribution of the homology classes of the knots, as it is the distribution of the ideal classes of the primes. For $M = S^3$ both obstructions are absent, and by Theorems 3.3 and 3.5 the whole phase structure comes from linking.

5. Volume exponents and closed phases

DEFINITION 5.1. For hyperbolic $\mathcal{L}$ set $\beta_c(\mathcal{L}) = \inf\{\beta > 0 : \sum_i e^{-\beta \text{Vol}(M \setminus K_i)} < \infty\}$.

PROPOSITION 5.2 (The full family is too large). *Over the family of all hyperbolic knots in S^3 the series $\sum_i e^{-\beta \text{Vol}_i}$ diverges for every $\beta > 0$, so $\beta_c = \infty$, and a finite volume exponent requires a sparse subfamily.*

PROOF. By Thurston's hyperbolic Dehn surgery theorem [127, Ch. 5] the twist knots K_n , obtained by $1/n$ -surgery on one component of the Whitehead link, are hyperbolic for n large with volumes increasing to the volume $V_W \approx 3.66$ of the Whitehead link complement. So infinitely many hyperbolic knots have volume below V_W , and the series has infinitely many terms bounded below by $e^{-\beta V_W}$. $\square$

REMARK 5.3. The count of hyperbolic 3-manifolds of volume at most V is not finite in dimension 3, precisely because of Dehn filling; the growth rate V^{cV} of Burger–Gelder–Lubotzky–Mozes concerns dimension ≥ 4 . The divergence in Proposition 5.2 is therefore stronger than a growth estimate: the volume spectrum of knots has accumulation points, and the volume series diverges term by term near each of them.

PROPOSITION 5.4 (Open and closed broken phases). *Let $\mathcal{L}$ be the link of Theorem 3.5 and let N be any norm with $\sum_{i \in D} N(K_i)^{-\beta} = \infty$ for every $\beta < \beta_c$ and every set D of positive lower density, where β_c is the convergence exponent of $\sum_i N(K_i)^{-\beta}$. Then the KMS_β state is unique for $\beta < \beta_c$ and the simplex is $\text{Prob}(G_{\mathcal{L}})$ for $\beta > \beta_c$, and at $\beta = \beta_c$ the state is unique if $\sum_i N(K_i)^{-\beta_c} = \infty$ and the simplex is $\text{Prob}(G_{\mathcal{L}})$ if $\sum_i N(K_i)^{-\beta_c} < \infty$: the broken phase is the open half-line (β_c, ∞) for $N(K_i) = (i+1)^{1/\beta_c}$ and the closed half-line $[\beta_c, \infty)$ for $N(K_i) = ((i+1)\log^2(i+2))^{1/\beta_c}$, exactly as in [Ch. 1, Thm. 5.2].*

PROOF. The proof of Theorem 3.5 shows that every nontrivial character is detected on a set of lower density at least $\frac{1}{2}$, so for $\beta < \beta_c$ the hypothesis on N gives $\Xi_\beta = \{1\}$; for $\beta > \beta_c$ all sums converge. At β_c : if the full series diverges, so does every sum over a set D of positive lower density c for the two norms named, since their terms a_i are eventually decreasing and the k -th element of D is at most k/c , so that $\sum_{i \in D} a_i \geq \sum_k a_{\lceil k/c \rceil} \geq \int_1^\infty a(x/c) dx = c \int_{1/c}^\infty a = \infty$; if it converges, all subsums converge. The two norms have $\sum_i (i+1)^{-1} = \infty$ and $\sum_i ((i+1)\log^2(i+2))^{-1} < \infty$, and for $\beta < \beta_c$ their β -sums over positive-density sets diverge by partial summation. $\square$

REMARK 5.5. With a prescribed norm the theory is complete: Theorem 3.5 and Proposition 5.4 realize both the open and the closed phase diagram of [Ch. 1, Thm. 5.2] by linking alone. What we have not exhibited is a family that is simultaneously linking-rich in the sense of Theorem 3.5 and *geometric* in its norm: the components of Lemma 3.4 are unknots, and controlling the hyperbolic volumes of an accumulating family — whose complements degenerate — is the concrete gap on the geometric side. Definition 2.2 allows any norm, but the dictionary's promise that $N\mathfrak{p}$ corresponds to a geometric quantity is kept only on generators, by Remark 2.3, and not yet by any accumulating family.

6. Contrastive autopsy: why the meridian fails

THEOREM 6.1 (The inertia model has no thermodynamics). *Let $M = S^3$ and define the system with $\sigma_{K_i} = [\mu_i]$ instead. Then every $\chi \in \widehat{G}_{\mathcal{L}}$ has finite detecting set, $\Xi_\beta = \widehat{G}_{\mathcal{L}}$ for every $\beta > 0$, the KMS_β state is unique at every temperature, and no phase transition occurs — for any link, any norm, any β .*

PROOF. As in the proof of Theorem 3.2, $G_{\mathcal{L}} \cong \prod_i \widehat{\mathbb{Z}}\mu_i$ and $\widehat{G}_{\mathcal{L}} = \bigoplus_i \mathbb{Q}/\mathbb{Z}$. A character is thus nontrivial in finitely many coordinates, and since μ_i is the i -th basis vector, D_χ is that finite set. $\square$

PROPOSITION 6.2 (The arithmetic contrast). *Let $K = \mathbb{Q}$. In $G_S = \mathbb{I}_S/\overline{\mathbb{Q}_{S,+}^\times}$, with the convention $\sigma_a = r(s_a)^{-1}$ of [Ch. 1, (2.2)],*

$$(6.1) \quad \sigma_p = \theta_p = (p)_{\ell \neq p} \times (1)_{\ell=p} \in U_S = \prod_{\ell} \mathbb{Z}_{\ell}^{\times},$$

nontrivial at every place $\ell \neq p$. Consequently a character supported at a single place ℓ_0 , with component χ_{ℓ_0} a character of $\mathbb{Z}_{\ell_0}^\times$ of finite order, has $\chi(\sigma_p) = \chi_{\ell_0}(p) = \chi(\text{Frob}_p)^{-1}$ for $p \neq \ell_0$, with detecting set of positive Dirichlet density.

PROOF. σ_p is the class of the inverse of the idele s_p with p at p and 1 elsewhere; since p lies in $\mathbb{Q}_{S,+}^\times$ the diagonal idele $(p, p, \dots)$ is trivial in G_S , and $p^{-1}s_p$ has component p^{-1} at $\ell \neq p$ and 1 at p , so its inverse is (6.1); this is [Ch. 1, Lem. 2.10(1)]. Evaluating χ at (6.1) gives $\chi_{\ell_0}(p)$, and $\chi(\sigma_p) = \chi(\text{Frob}_p)^{-1}$ is [Ch. 1, Lem. 3.8]. A nontrivial χ_{ℓ_0} factors through $(\mathbb{Z}/\ell_0^k)^\times$, so D_χ is a union of arithmetic progressions, of positive Dirichlet density by Dirichlet's theorem [97, Ch. VII]. $\square$

REMARK 6.3 (The moral). The two computations are the same statement. Arithmetically it is the product formula that spreads a one-place idele across all others; topologically it is the relation $\lambda_i = \sum_j \text{Lk}(K_i, K_j)\mu_j$ that spreads the longitude across all linked components. *Inertia is local and carries no thermodynamics; Frobenius is global and carries the linking matrix.* The meridian model of Theorem 6.1 is not so much a failed construction as a correct computation of the wrong object: it says inertia classes are thermodynamically inert, which is what inertia is for. Numerically, for $\ell_0 = 5, 7, 11$ with χ_{ℓ_0} the Legendre symbol the arithmetic detecting sets contained 51.5%, 52.8% and 52.2% of the primes below 2000, matching the topological densities of Remark 3.8.

7. The dictionary, and assessment

Arithmetic [Ch. 1]	Knot topology	Reference
$\mathfrak{p} \in S$	$K_i \in \mathcal{L}$	—
$K_{\mathfrak{p}}^\times$	$H_1(T_i; \widehat{\mathbb{Z}})$	Lem. 1.1
$\mathcal{O}_{\mathfrak{p}}^\times$ (inertia)	μ_i (meridian)	Lem. 1.1
$\pi_{\mathfrak{p}}^{\widehat{\mathbb{Z}}}(\text{Frobenius})$	λ_i (longitude)	Lem. 1.1
$\mathbb{I}_S$	$H_1(\partial X; \widehat{\mathbb{Z}})$	Thm. 1.4
$\overline{K_{S,+}^\times}$	$\ker \iota_* = \langle \lambda_i \rangle$	Thm. 1.3, 1.4
$\mathcal{O}_{K,+}^\times$	$\ker \iota_* \cap \bigoplus \widehat{\mathbb{Z}}\mu_i; = 0 \text{ for } S^3$	Thm. 1.4
$\text{Cl}_S^+ = \langle [\mathfrak{p}] \rangle \leq \text{Cl}_K^+$	$\text{Cl}_{\mathcal{L}} = \langle [K_i] \rangle \leq H_1(M; \mathbb{Z})$	Thm. 1.4, 4.1
$h_S^+ > 1$ obstruction	$[K_i]$ summably in a proper $H < \text{Cl}_{\mathcal{L}}$	Thm. 4.1(2)
—	$H_1(M; \mathbb{Z}) / \text{Cl}_{\mathcal{L}} \neq 0$: broken at all β	Thm. 4.1(1)
$\sigma_{\mathfrak{p}} = (\pi_{\mathfrak{p}})_{\mathfrak{q} \neq \mathfrak{p}}$	$\lambda_i = \sum_j \text{Lk}(K_i, K_j)\mu_j$	Prop. 6.2, Thm. 3.2
product formula	linking relation	Rem. 6.3
$\zeta_{K,S}(\beta)$	$\sum_i N(K_i)^{-\beta}$, N geometric on generators	Def. 2.2, 5.1
Ξ_β Dirichlet criterion	Ξ_β linking criterion	Thm. 3.1, 3.2
Chebotarev density	bit matrix: each $\chi \neq 1$ detected, density $\geq \frac{1}{2}$	Thm. 3.5
BC phase diagram	realized by linking, open or closed	Thm. 3.5, Prop. 5.4

The transfer is complete at the level of structure and, by Proposition 2.5, requires no new operator algebra: the classification of [Ch. 1] is a theorem about a quadruple with no arithmetic in it. What the topological instantiation contributes is a different mechanism for the same phenomenon — linking in place of reciprocity — and §6 shows the two are structurally identical, both being ways of spreading a local generator over infinitely many places.

Three questions. First, realization: [Ch. 2, Thm. 5.1] realizes prescribed chains $\{\Xi_\beta\}$ from congruence conditions on primes, using Chebotarev to select primes in prescribed classes; Lemma 3.4 removes every constraint on the linking matrix, so the topological realization problem is purely

combinatorial, and Theorem 3.5 settles the extreme case. Second, geometry: Remark 5.5 records that we have no family which is simultaneously linking-rich and geometric in its norm, and this is where a genuinely three-dimensional input would enter. Third, and we state it as a caution rather than a programme: nothing here uses a quantum invariant. The Jones polynomial, Chern–Simons theory and the volume conjecture play no role in any proof above, and the subfactors of Chapter 3 of the series arise from arithmetic symmetry breaking rather than from knot theory. We see no evidence that the two occurrences of subfactors are related, and would not want the coincidence of names to suggest otherwise.

Localized Bost–Connes systems over global function fields: commensurability forbids type III₁, and the factor recovers the temperature

This chapter is Paper XVII of the series (papers/XVII-function-fields/).

SUMMARY. We transport the localized Bost–Connes theory to $K = \mathcal{F}_q(C)$ for a smooth projective geometrically irreducible curve $C/\mathcal{F}_q$ of genus g , and find that one structural feature inverts and, with it, the central negative result of Chapter 20.

Over a number field the logarithms $\log N\mathfrak{p}$ of an infinite prime set are never pairwise commensurable [Ch. 2, Lem. 4.1]; over a function field they are *always* commensurable, being $\deg(P) \log q$. Hence $\Gamma_\beta = \langle q^{-\beta \deg P} \rangle = q^{-\beta d_S \mathbb{Z}}$ with $d_S = \gcd_{P \in S} \deg P$ is a *discrete* subgroup of $\mathcal{R}_{>0}$, so $S(M) \subseteq \Gamma_\beta \cup \{0\}$ can never be $[0, \infty)$: **the extremal KMS _{β} factors are never of type III₁**, in complete contrast with [Ch. 2, Thm. 3.2], where III₁ is generic. The proposed alternative “III₀ if Γ_β is dense” cannot occur, a discrete group not being dense; the alternatives are III _{λ} with $\lambda \in q^{-\beta d_S \mathbb{Z}}$, III₀, or semifiniteness.

The consequence is the interesting one. In the III _{λ} case $M_\beta \cong R_\lambda$ with $\lambda = q^{-\beta d_S}$, and the Powers factors are mutually non-isomorphic; so **the von Neumann algebra determines $\beta d_S \log q$** . Chapter 20 showed that over number fields every system with $S = P_K$, $\beta \leq 1$ gives R_∞ and the factor remembers neither K , nor S , nor β . Over function fields the collapse is partial: the temperature survives weak closure. This is the first repair of Theorem 2.1 of Chapter 20 in the whole programme, and it is bought by commensurability rather than by any rigidity input.

Two corrections. The modular flow is periodic with fundamental period $T_0 = 2\pi/(d_S \log q)$, not $2\pi/\log q$; the two agree exactly when $d_S = 1$, which does hold for $S = \mathcal{P}_K$ by F. K. Schmidt’s theorem but fails for, say, the set of points of even degree. And in the Weil–Chebotarev bound the term $(g_L + 1)q^{n/2}$ swamps the main term $2g_L q^{n/2}/n$ and is superfluous: the truth is $O_\chi(q^{n/2}/n)$, with the constant-field-extension characters requiring a separate and elementary treatment, Weil’s bound not applying to them.

The rest goes through and is unconditional. $\beta_c = 1$ for every curve; uniqueness holds on $(0, 1]$ for $S = \mathcal{P}_K$, unconditionally — the class-group characters, which over a number field can obstruct permanently, here have detecting sets of positive density; and a set with $A_n(S) \asymp q^n/n^2$ has $\sum_{P \in S} q^{-\deg P} = \zeta(2)$ -like and finite, giving a *closed* low-temperature phase $[1, \infty)$ with no sieve hypothesis whatever.

1. The system and its periodic flow

Let $C/\mathcal{F}_q$ be a smooth projective geometrically irreducible curve of genus g , $K = \mathcal{F}_q(C)$, and $\mathcal{P}_K$ the set of closed points. For $S \subseteq \mathcal{P}_K$ put

$$J_S = \text{Div}_+^S(C) = \bigoplus_{P \in S} \mathbb{N} \cdot P, \quad N(P) = q^{\deg P}, \quad \mathcal{A}_{C,S} = C(Y_S) \rtimes J_S,$$

with $G_S = \mathbb{I}_S/\overline{K_S^\times}$ and $\sigma_t(\mu_D) = q^{it \deg D} \mu_D$. Write

$$d_S := \gcd_{P \in S} \deg P.$$

Constructions and notation follow [Ch. One convention is needed at once: for $S = \mathcal{P}_K$ the degree map gives $\mathbb{I}_S/\overline{K_S^\times}$ a quotient $\mathbb{Z}$, so the group as written is not compact; as the framework of [Ch. 1]

requires a compact symmetry group (and as is standard for Bost–Connes systems over function fields), G_S denotes throughout the profinite completion in the degree direction. Its dual $\widehat{G}_S$ is then a torsion group, and the characters factoring through the constant-field tower $\text{Gal}(\overline{\mathcal{F}}_q/\mathcal{F}_q) \cong \widehat{\mathbb{Z}}$ are exactly $P \mapsto \zeta^{\deg P}$ with ζ a root of unity. For the basic theory of curves over finite fields we refer to [115, 121].

PROPOSITION 1.1 (Periodicity). *The flow σ is periodic with fundamental period*

$$T_0 = \frac{2\pi}{d_S \log q}.$$

In particular $T_0 = 2\pi/\log q$ precisely when $d_S = 1$, which holds for $S = \mathcal{P}_K$ by F. K. Schmidt’s theorem [115] that the gcd of the degrees of closed points of a curve over a finite field is 1; it fails for instance when S consists of the points of even degree, where $d_S = 2$ and the period is halved.

PROOF. $\sigma_T = \text{id}$ iff $q^{iT \deg D} = 1$ for every $D \in J_S$, i.e. $T \log q \deg D \in 2\pi\mathbb{Z}$. The degrees realized by J_S generate the subgroup $d_S\mathbb{Z}$ of $\mathbb{Z}$, so the condition is $T d_S \log q \in 2\pi\mathbb{Z}$, whose least positive solution is T_0 . $\square$

REMARK 1.2. Periodicity of σ has no analogue over number fields, where the $\log N\mathfrak{p}$ generate a dense subgroup of $\mathcal{R}$ and σ is never periodic. It means that σ factors through the circle $\mathcal{R}/T_0\mathbb{Z}$, so $\mathcal{A}_{C,S}$ carries a genuine $\mathbb{T}$ -action rather than a flow.

2. Rational thermodynamics

PROPOSITION 2.1 (Weil zeta). *With $T = q^{-\beta}$,*

$$\prod_{P \in S} (1 - q^{-\beta \deg P})^{-1} = Z_S(C, T),$$

and for $S = \mathcal{P}_K$ this is the Weil zeta function [136] $Z_C(T) = \frac{P(T)}{(1-T)(1-qT)}$ with $P(T) = \prod_{j=1}^{2g} (1 - \alpha_j T)$ and $|\alpha_j| = q^{1/2}$. Its poles are at $T = 1$ and $T = q^{-1}$, that is at $\beta = 0$ and $\beta = 1$.

COROLLARY 2.2 (Critical exponent). *For every curve C and $S = \mathcal{P}_K$, $\beta_c(C) = 1$: the series $\sum_P q^{-\beta \deg P} = \sum_n a_n q^{-\beta n}$, with $a_n = \#\{P : \deg P = n\} = \frac{q^n}{n} + O(\frac{q^{n/2}}{n})$ by Weil, converges for $\beta > 1$ and diverges for $\beta \leq 1$.*

PROOF. $\sum_n \frac{q^{n(1-\beta)}}{n}$ diverges for $\beta \leq 1$, converging otherwise; the Weil error term contributes $\sum_n q^{n(1/2-\beta)}/n$, convergent for $\beta > 1/2$. $\square$

REMARK 2.3 (Numerical illustration). Partial sums $\sum_{n \leq 40} a_n q^{-\beta n}$ with $a_n = q^n/n$: for $q = 2$ they are 11.36, 4.28, 2.69 at $\beta = 0.9, 1.0, 1.1$, and for $q = 4$ they are 65.68, 4.28, 2.04; the value at $\beta = 1$ is $\sum_{n \leq 40} 1/n$ in both cases, growing logarithmically, and the $\beta = 1.1$ column is bounded.

3. Commensurability forbids type III₁

LEMMA 3.1 (Total commensurability). *For every $P \in S$, $\log N(P) = \deg(P) \log q \in (\log q)\mathbb{Z}$. Hence the log-norms are pairwise commensurable, and*

$$\Gamma_\beta = \langle N(P)^{-\beta} : P \in S \rangle = q^{-\beta d_S \mathbb{Z}},$$

a discrete subgroup of $(\mathcal{R}_{>0}, \times)$.

REMARK 3.2 (The inversion). [Ch. 2, Lem. 4.1] proves that over a number field an infinite S never has pairwise commensurable log-norms, distinct rational primes having $\mathbb{Q}$ -independent logarithms. Over a function field the opposite holds universally. This single reversal drives everything below.

THEOREM 3.3 (Never III₁). *For every curve C , every S and every $\beta > 0$, the extremal KMS_β factors $M = \pi_\varphi(\mathcal{A}_{C,S})''$ are not of type III₁. Moreover Γ_β being discrete, the case “ Γ_β dense” cannot occur, so the possibilities are: type III $_\lambda$ with $\lambda \in q^{-\beta d_S \mathbb{Z}}$; type III₀, where $S(M) = \{0, 1\}$; or semifinite, where $S(M) = \{1\}$.*

PROOF. By [Ch. 1, Prop. 6.1], $S(M) \subseteq \overline{\Gamma_\beta} \cup \{0\}$, and $\overline{\Gamma_\beta} = \Gamma_\beta$ by Lemma 3.1. Type III₁ requires $S(M) = [0, \infty)$, which is not contained in a discrete set together with 0. The listed cases exhaust Connes’ classification of $S(M)$. $\square$

COROLLARY 3.4 (Contrast). *By [Ch. 2, Thm. 3.2], over a number field with $S = P_K$ and $0 < \beta \leq 1$ the type is always III₁. Over a function field it never is, for any S and any β . The two settings are opposite in this respect, and the reason is Lemma 3.1 alone, no arithmetic being used.*

4. The factor recovers the temperature

THEOREM 4.1 (Partial repair of Chapter 20). *Suppose $M_\beta = \pi_{\varphi_\beta}(\mathcal{A}_{C,S})''$ is of type III $_\lambda$ with $\lambda = q^{-\beta d_S}$. Then $M_\beta \cong R_\lambda$, the Powers factor: $\mathcal{A}_{C,S}$ is amenable — the argument of Chapter 20 (its Lemma 1.1) applies verbatim, the acting group $I_S = \mathbb{Z}^{(S)}$ being abelian here too, so the groupoid is amenable and M_β is injective — and since $R_\lambda \cong R_{\lambda'}$ only for $\lambda = \lambda'$, the isomorphism class of M_β determines*

$$\lambda = q^{-\beta d_S}, \quad \text{hence} \quad \beta d_S \log q.$$

In particular, for fixed q and d_S , distinct temperatures give non-isomorphic factors.

PROOF. Amenability and Connes’ classification give $M_\beta \cong R_\lambda$; the Powers factors are pairwise non-isomorphic, their S -invariants $\{0\} \cup \lambda \mathbb{Z}$ being distinct. $\square$

REMARK 4.2 (What this changes). Theorem 2.1 of Chapter 20 showed that over number fields $M_{\mathbb{Q}, P_{\mathbb{Q}}, 1} \cong M_{\mathbb{Q}, P_{\mathbb{Q}}, 1/2} \cong M_{K, P_K, \beta} \cong R_\infty$ for every K and every $\beta \in (0, 1]$: the factor remembers neither the field, nor the prime set, nor the temperature. Theorem 4.1 is the first place in this programme where that collapse is not total. It is worth being precise about how much is repaired: the factor recovers the single real number $\beta d_S \log q$, and nothing else — not C , not g , not S beyond d_S , not the obstruction group. But β was the quantity whose loss was most striking in Chapter 20, and it is recovered here for free, by commensurability rather than by any rigidity theorem.

Numerically:

(q, d_S, β)	$(2, 1, \frac{1}{2})$	$(2, 1, 1)$	$(2, 1, 2)$	$(4, 2, 1)$	$(9, 1, \frac{3}{2})$
λ	0.7071	0.5000	0.2500	0.0625	0.0370

Distinct temperatures give distinct factors.

REMARK 4.3 (The remaining ambiguity). Only the product $\beta d_S \log q$ is recovered, so (q, d_S, β) and (q', d'_S, β') with $\beta d_S \log q = \beta' d'_S \log q'$ give isomorphic factors: for example $(q, d_S, \beta) = (4, 1, 1)$ and $(2, 1, 2)$ both give $\lambda = 1/4$. Recovering q separately requires the C^* -level, where by Proposition 1.1 of Chapter 23 the point spectrum $\text{Sp}_p(\sigma) = q^{d_S \mathbb{Z}}$ is available and pins q^{d_S} directly.

5. The Weil–Chebotarev obstruction

PROPOSITION 5.1 (Corrected bound). *Let χ be a nontrivial character of G_S that does not factor through the constant-field tower; write $\chi = z^{\deg} \cdot \chi_0$ with z a root of unity and χ_0 geometric*

with conductor f_χ , and put $B_\chi = 2g - 2 + \deg f_\chi$ [121]. Then $L(T, \chi) = L(zT, \chi_0) = \prod_{j=1}^{B_\chi} (1 - \alpha_j T)$ with $|\alpha_j| = q^{1/2}$ by Weil [136], and

$$\left| \sum_{\deg P=n} \chi(\text{Frob}_P) \right| \leq \frac{B_\chi}{n} q^{n/2} + \frac{4g+4}{n} q^{n/2} \quad (n \geq 1).$$

PROOF. Taking logarithmic derivatives of $L(T, \chi)$ gives $\sum_{\deg P|n} \deg P \cdot \chi(\text{Frob}_P)^{n/\deg P} = -\sum_j \alpha_j^n$, of modulus at most $B_\chi q^{n/2}$. The terms with $\deg P = m < n$ are bounded in absolute value by $m a_m \leq \#C(\mathcal{F}_{q^m}) \leq q^m + 1 + 2gq^{m/2} \leq (2g+2)q^m$, and $\sum_{m \leq n/2} (2g+2)q^m \leq (2g+2) \frac{q}{q-1} q^{n/2} \leq (4g+4)q^{n/2}$; removing them and dividing by n gives the statement. $\square$

REMARK 5.2 (The proposed bound). The bound $2g_L q^{n/2}/n + (g_L + 1)q^{n/2}$ is true but its second term has no factor $1/n$ and therefore dominates the first for every $n \geq 1$, making the whole estimate weaker than the trivial $O(q^{n/2})$ that the main term already provides. The correct order is $O_\chi(q^{n/2}/n)$, as in Proposition 5.1.

REMARK 5.3 (Constant field extensions need separate treatment). Weil's bound does not apply to the characters χ factoring through $\text{Gal}(\mathcal{F}_{q^m}/\mathcal{F}_q)$, for which $\chi(\text{Frob}_P) = \zeta^{\deg P}$ with ζ a primitive m -th root of unity: the corresponding L -function is $Z_C(\zeta T)$, which has poles rather than the required polynomial form. These are the function field counterpart of the class-group characters of [Ch. 1, §4.1], and they must be handled directly. That is elementary: $D_\chi = \{P : \deg P \not\equiv 0 \pmod{m}\}$, and $\sum_{P \in D_\chi} q^{-\beta \deg P} = \sum_{n \not\equiv 0 \pmod{m}} a_n q^{-\beta n}$ diverges for $\beta \leq 1$, since $a_n \sim q^n/n$ on the nonzero residue classes as well (Corollary 2.2).

THEOREM 5.4 (Unconditional uniqueness). *Let $S = \mathcal{P}_K$. For every nontrivial $\chi \in \widehat{G}_S$,*

$$\sum_P q^{-\beta \deg P} |1 - \chi(\sigma_P)|^2 = \infty \quad (0 < \beta \leq 1),$$

so $\Xi_\beta = \{1\}$ and the KMS_β state is unique for $0 < \beta \leq 1$, unconditionally and with explicit constants. In particular the unramified class-group obstruction of [Ch. 1, §4.1] has no analogue at $S = \mathcal{P}_K$: the characters pulled back from $\text{Pic}(C)$ have detecting sets of positive density.

PROOF. Since $|1 - w|^2 = 2 - 2 \text{Re } w$ on the unit circle,

$$\sum_{\deg P=n} |1 - \chi(\text{Frob}_P)|^2 = 2a_n - 2 \text{Re} \sum_{\deg P=n} \chi(\text{Frob}_P).$$

If χ does not factor through the constant-field tower, Proposition 5.1 bounds the second term by $2(B_\chi + 4g + 4)q^{n/2}/n$, so the sum is $\geq a_n$ for n large and $\sum_n q^{-\beta n} a_n = \infty$ for $\beta \leq 1$ by Corollary 2.2. If $\chi = \zeta^{\deg}$ is a constant-field character, Remark 5.3 applies. These two cases are exhaustive, $\widehat{G}_S$ being torsion. Hence $\chi \notin \Xi_\beta$ for every nontrivial χ and $0 < \beta \leq 1$, and uniqueness follows from the criterion of [Ch. 1, Thm. 3.9], whose Kakutani–martingale proof uses only the product structure of the measures and applies verbatim over a function field. For the last sentence: a character of $\text{Pic}(C)$ pulled back to G_S is $\zeta^{\deg}$ times a character of the finite group $\text{Pic}^0(C)$; the latter is geometric and everywhere unramified, with $B_\chi = 2g - 2$, so it falls under the first case, and its detecting set has density bounded below explicitly in g . $\square$

REMARK 5.5 (What is gained over the number field case). The corresponding statement over $\mathbb{Q}$ [Ch. 1, Thm. 3.9] relies on the prime ideal theorem and Chebotarev density with their ineffective error terms, Siegel–Walfisz and the possibility of Landau–Siegel zeros. Here Weil's theorem is available and every constant is explicit in g and $\deg f_\chi$. Nothing is conjectural.

6. Unconditional sparse sets and closed phases

THEOREM 6.1 (Closed low-temperature phase, no sieve). *Choose, for each n , any $A_n = \lfloor q^n/n^2 \rfloor$ closed points of degree n ; this is possible for n large since $a_n \sim q^n/n \gg q^n/n^2$. Let S_{geom} be their union. Then*

$$\sum_{P \in S_{\text{geom}}} q^{-\beta \deg P} \asymp \sum_n \frac{q^{n(1-\beta)}}{n^2},$$

so $\beta_c(S_{\text{geom}}) = 1$ and the series converges at $\beta = 1$, with value $\asymp \zeta(2) = \pi^2/6$. Consequently the symmetry-broken phase is the closed half-line $[1, \infty)$, exactly as in [Ch. 1, Thm. 5.5], and the construction uses no sieve hypothesis of any kind.

PROOF. The displayed asymptotic is immediate; $\sum_n n^{-2} < \infty$ gives convergence at $\beta = 1$ and $\sum_n q^{n(1-\beta)}/n^2 = \infty$ for $\beta < 1$ gives $\beta_c = 1$. Closedness of the phase is [Ch. 1, Thm. 5.5], whose hypothesis is exactly convergence at the critical exponent. $\square$

REMARK 6.2 (Numerical check). With $q = 2$, $\sum_{n \leq 200} 2^{n(1-\beta)}/n^2$ equals 2.964, 1.640, 1.493 at $\beta = 0.95, 1.00, 1.05$; the middle value is the truncation of $\pi^2/6 = 1.6449$.

REMARK 6.3 (Comparison). Over $\mathbb{Q}$ the closed phase of [Ch. 1, Thm. 5.5] was obtained for the Sophie Germain primes and required Brun's sieve; the corresponding statement for the twin primes remains tied to the Hardy–Littlewood conjecture. Over a function field the point counts are known exactly, so a set of any prescribed density can simply be selected. The gain is not that a hard theorem becomes easy but that the question stops being about existence: one chooses the set rather than proving that one exists.

7. Dictionary and assessment

Number field K	Function field $\mathcal{F}_q(C)$	Reference
$\log N_{\mathfrak{p}}$ never commensurable	always commensurable	[Ch. 2, Lem. 4.1], Lem. 3.1
σ never periodic	periodic, $T_0 = 2\pi/(d_S \log q)$	Prop. 1.1
β_c depends on S	$\beta_c = 1$ for $S = \mathcal{P}_K$	Cor. 2.2
type III ₁ generic	never III ₁	Thm. 3.3
$M \cong R_\infty$, β forgotten	$M \cong R_\lambda$ recovers $\beta d_S \log q$	Thm. 4.1
Chebotarev ineffective	Weil bound, explicit	Prop. 5.1
sparse sets need sieves	sets are chosen, not found	Thm. 6.1
unramified obstruction ([Ch. 1, §4.1])	none at $S = \mathcal{P}_K$: positive density	Thm. 5.4

The transfer succeeds and, unlike the transfer to arithmetic topology of Chapter 30, it succeeds with content: every analytic difficulty of the number field theory becomes a theorem, and one structural feature genuinely inverts. That inversion, total commensurability of the log-norms, is the whole story. It makes the flow periodic, it makes Γ_β discrete, it forbids type III₁, and it thereby causes the von Neumann algebra to retain the temperature that Chapter 20 showed is destroyed over number fields.

We should be exact about the size of that last claim. The factor recovers one real number, $\beta d_S \log q$, and by Remark 4.3 not even q and β separately. It does not recover the curve, its genus, or the obstruction group; the collapse of Chapter 20 is weakened, not reversed. But it is the first weakening, and it comes from the arithmetic of the ground field rather than from any additional structure imposed on the system.

Three questions. Which of the three cases of Theorem 3.3 actually occurs, and for which S ? The natural guess is III _{$q^{-\beta d_S}$} throughout, but determining Σ_β exactly, as Chapter 2 does over number fields ([Ch. 2, Prop. 2.2] for the bound via the extended group, [Ch. 2, Thm. 4.2] for an

exact determination), requires computing the extended obstruction group here and we have not done it. Second, does the periodicity of σ — so that $\mathcal{A}_{C,S}$ carries a circle action rather than a flow — bring the classification of Chapter 21 within reach of the equivariant Kirchberg theory for compact groups, which is far better developed than for $\mathcal{R}$ (its Remark 5.1)? Third, the Weil bound gives effective control on Ξ_β that has no number field counterpart; whether it yields a realization theorem for prescribed chains $\{\Xi_\beta\}$, the function field analogue of [Ch. 2, Thm. 5.1], is open and looks tractable.

Circle actions on function field Bost–Connes systems: the compact group advantage is illusory, and the Weil zeta function is thermodynamic

This chapter is Paper XVIII of the series ([papers/XVIII-equivariant-kirchberg/](#)).

SUMMARY. In §7 of Chapter 31 we suggested that the periodicity of the modular flow over a function field might bring the equivariant Kirchberg classification within reach, compact group actions being better understood than flows. We now check that suggestion and withdraw it.

Takai duality converts a $\mathbb{T}$ -action on $\mathcal{A}$ into a $\mathbb{Z}$ -action on $\mathcal{A} \rtimes \mathbb{T}$, and $\mathbb{Z}$ is discrete and amenable, so the Gabe–Szabó equivariant Kirchberg–Phillips theorem would indeed apply — if $\mathcal{A} \rtimes \mathbb{T}$ were a Kirchberg algebra. For a gauge-type action it is the stabilized core: $\mathcal{O}_n \rtimes \mathbb{T}$ is AF, hence stably finite. The classification theory for Kirchberg algebras therefore does not survive the dualization. Nor does Izumi’s classification of compact group actions apply, that theory requiring the Rokhlin property, whereas gauge actions on Cuntz algebras are approximately representable and not Rokhlin. The advantage we conjectured does not exist.

A second obstruction is arithmetic rather than operator-algebraic. The affine boundary system needs an infinite ring of everywhere-integral elements acting additively, and for a projective curve a function regular at every closed point is constant: $H^0(C, \mathcal{O}_C) = \mathcal{F}_q$, finite. The affine construction degenerates for $S = \mathcal{P}_K$, and one must fix a point at infinity, so the classification cannot be stated for full point sets as proposed.

Third, the equivariant K -theory does not contain the Weil polynomial. The Cuntz relation at P gives $(1 - (qt)^{\deg P})[1] = 0$ in $K_0^{\mathbb{T}}$, so $[1]$ generates the cyclic $R(\mathbb{T})$ -module $\mathbb{Z}[t, t^{-1}]/(1 - (qt)^{ds})$: this records q^{ds} and nothing more, the same information already supplied by the point spectrum $\mathrm{Sp}_p(\sigma) = q^{ds\mathbb{Z}}$ of Proposition 1.1 of Chapter 23. The class group contributes $|\mathrm{Pic}^0(C)(\mathcal{F}_q)| = P(1)$, one value of the Weil polynomial rather than the polynomial.

What is true is the conclusion, by another route. The partition function of the Toeplitz system is

$$\sum_{D \in J_S} N(D)^{-\beta} = \sum_{n \geq 0} b_n q^{-\beta n} = Z_C(q^{-\beta}), \quad b_n = \#\{D \geq 0 : \deg D = n\},$$

so the *Weil zeta function is the partition function*. Knowing $\beta \mapsto Z_C(q^{-\beta})$ on $(1, \infty)$ determines the rational function Z_C , hence $P(T)$, the Weil numbers $\{\alpha_j\}$, the genus, every point count $\#C(\mathcal{F}_{q^n}) = q^n + 1 - \sum_j \alpha_j^n$, and $|\mathrm{Jac}(C)(\mathcal{F}_q)| = P(1)$. The complete geometric reconstruction asked for does hold; it is thermodynamic, not K -theoretic — which is the principle of Corollary 2.4 of Chapter 4 appearing once more.

1. The affine boundary requires a point at infinity

The framework is that of [Ch. 1], transported to function fields as in Chapter 31 of the series, whose notation we keep.

PROPOSITION 1.1 (Degeneration for $S = \mathcal{P}_K$). *Let $C/\mathcal{F}_q$ be a smooth projective geometrically irreducible curve and $K = \mathcal{F}_q(C)$. A rational function regular at every closed point of C is constant [115]. Now the affine system $\mathcal{A}^{\mathrm{aff}} = C(\widehat{\mathcal{O}_S}) \rtimes (\mathcal{O}_K \rtimes J_S)$ of Chapter 3 (its §6) requires an infinite additive group of everywhere-integral elements acting densely by translations on the profinite completion — by its Remark 6.1 the S -integers themselves have poles at S and do not*

act, which is why the additive part there is $\mathcal{O}_K$. The function-field counterpart of $\mathcal{O}_K$ at $S = \mathcal{P}_K$ is $H^0(C, \mathcal{O}_C) = \mathcal{F}_q$: the additive group is finite, its closure is a finite set, and the construction degenerates.

PROOF. The first statement is that $H^0(C, \mathcal{O}_C) = \mathcal{F}_q$ for C projective and geometrically irreducible. $\square$

COROLLARY 1.2 (The set-up must be affine). *One must fix a closed point $\infty \in \mathcal{P}_K$ and take $A = \mathcal{O}_{\mathcal{P}_K \setminus \{\infty\}}$, the ring of functions regular away from ∞ , which is a Dedekind domain with $\text{Spec}(A) = \mathcal{P}_K \setminus \{\infty\}$ infinite and dense in $\widehat{\mathcal{O}_S}$ by strong approximation, the omitted place being exactly what the theorem requires; for $C = \mathbb{P}^1$ this is $\mathcal{F}_q[T]$. The classification statement therefore cannot be made for full point sets $S_i = \mathcal{P}_{K_i}$: a choice of ∞ is part of the data, and the partition function of the affine system is $Z_C(T)(1 - T^{\deg \infty})$ rather than $Z_C(T)$.*

2. Withdrawal: the compact group advantage does not exist

PROPOSITION 2.1 (Takai duality lands outside the Kirchberg class). *Let $\alpha : \mathbb{T} \curvearrowright \mathcal{A}$ be the gauge-type action $\alpha_\theta(\mu_{(a,P)}) = e^{i\theta \deg P} \mu_{(a,P)}$ on the affine boundary. By Takai duality [123], α corresponds to the dual action $\hat{\alpha} : \mathbb{Z} \curvearrowright \mathcal{A} \rtimes_\alpha \mathbb{T}$, with $(\mathcal{A} \rtimes \mathbb{T}) \rtimes \mathbb{Z} \cong \mathcal{A} \otimes \mathcal{K}$. The group $\mathbb{Z}$ is discrete and amenable, so the Gabe–Szabó equivariant Kirchberg–Phillips theorem [46] applies to $\mathbb{Z}$ -actions on Kirchberg algebras. However $\mathcal{A} \rtimes_\alpha \mathbb{T}$ is stably finite, and the Kirchberg classification does not apply to the dual system.*

PROOF. The action is saturated — there are isometries of every weight in $d_S \mathbb{Z}$ — so $\mathcal{A} \rtimes_\alpha \mathbb{T}$ is Morita equivalent, hence stably isomorphic, to the fixed-point algebra $F = \mathcal{A}^\alpha$. On F the state $\tau = \lambda \circ E$, with λ the Haar probability of $\widehat{\mathcal{O}_S}$ and $E : F \rightarrow C(\widehat{\mathcal{O}_S})$ the canonical expectation, is a faithful trace: traciality reduces to the self-similarity $\lambda(\pi_D U + b) = N(D)^{-1} \lambda(U)$ of Haar measure, the subgroup $\pi_D \widehat{\mathcal{O}_S}$ having index $N(D)$, and this exactly cancels the $N(D)$ cosets appearing in $E(vxv^*)$ for a weight-deg D isometry v . A unital C^* -algebra with a faithful tracial state is stably finite, and stable finiteness passes to the stabilization. The model case is classical: already $\mathcal{O}_n \rtimes_\gamma \mathbb{T} \cong \mathcal{K} \otimes (\text{UHF core})$ is AF. $\square$

PROPOSITION 2.2 (The action is not Rokhlin). *Izumi’s classification of finite and compact group actions on Kirchberg algebras requires the Rokhlin property [64]. A Rokhlin action of $\mathbb{T}$ on a unital $\mathcal{A}$ forces $\mathcal{A}$ to absorb the fixed-point algebra in a strong sense; the restrictions of gauge actions on Cuntz and Cuntz–Pimsner algebras to finite cyclic subgroups are instead approximately representable, the notion dual to Rokhlin [64], and the circle action itself does not have the Rokhlin property. So that route is unavailable as well.*

REMARK 2.3 (Withdrawal of a suggestion). We suggested there that periodicity of σ “may bring Remark 5.1 of Chapter 21 within reach”, on the ground that the equivariant classification for compact groups is better developed than for $\mathcal{R}$. That is true as a general statement about the literature and false as an inference here, for the two reasons above: dualizing a circle action moves the problem to a stably finite algebra, and the specific action fails the Rokhlin hypothesis under which the compact group theory is available. We withdraw the suggestion. The caution of Remark 5.1 of Chapter 21, that flows on Kirchberg algebras are not classified, stands and is not circumvented by compactness.

3. What the equivariant K -theory records

PROPOSITION 3.1 (The cyclic module). *In $K_0^\mathbb{T}(\partial \mathcal{A}^{\text{aff}})$, the Cuntz relation at P — the sum over the $NP = q^{\deg P}$ additive cosets of the range projections, each of weight $t^{\deg P}$ — gives*

$$(1 - (qt)^{\deg P})[1] = 0 \quad (P \in S).$$

Since $x = qt$ becomes a unit modulo each relation $1 - x^{\deg P}$, the exponents combine by Bézout: $\gcd_i(1 - x^{n_i}) = 1 - x^{\gcd_i n_i}$ as ideals of $\mathbb{Z}[t, t^{-1}]$, and the class $[1]$ generates the cyclic $R(\mathbb{T}) = \mathbb{Z}[t, t^{-1}]$ -module

$$R(\mathbb{T}) \cdot [1] \cong \mathbb{Z}[t, t^{-1}] / (1 - (qt)^{d_S}), \quad d_S = \gcd_{P \in S} \deg P.$$

COROLLARY 3.2 (No genus). *The module of Proposition 3.1 records the single quantity q^{d_S} , which is exactly the content of the point spectrum $\mathrm{Sp}_p(\sigma) = q^{d_S \mathbb{Z}}$ already available from $(\mathcal{A}, \sigma)$ by Proposition 1.1 of Chapter 23. It contains no information about the genus, the Weil numbers, or the point counts.*

REMARK 3.3 (What K -theory does see). By Theorem 3.3 of Chapter 3 the class of the unit in K_0 of the boundary records the narrow class number; the counterpart of that quantity here is $|\mathrm{Pic}^0(C)(\mathcal{F}_q)| = |\mathrm{Jac}(C)(\mathcal{F}_q)|$, of order $P(1)$. So K -theory does see one value of the Weil polynomial, at $T = 1$. It does not see the polynomial: the $2g$ numbers α_j are not determined by $P(1)$, and no mechanism in the six-term sequence produces the higher point counts.

4. The Weil zeta function is the partition function

THEOREM 4.1 (Z_C as a thermodynamic quantity). *Let $b_n = \#\{D \in \mathrm{Div}_+(C) : \deg D = n\}$. Then, by the definition of the Weil zeta function,*

$$Z_C(T) = \sum_{n \geq 0} b_n T^n = \frac{P(T)}{(1-T)(1-qT)}, \quad P(T) = \prod_{j=1}^{2g} (1 - \alpha_j T),$$

and the partition function of the Toeplitz system $(\mathcal{A}_{C, \mathcal{P}_K}, \sigma)$ is

$$(4.1) \quad \sum_{D \in J_{\mathcal{P}_K}} N(D)^{-\beta} = \sum_{n \geq 0} b_n q^{-\beta n} = Z_C(q^{-\beta}), \quad \beta > 1.$$

Equivalently, the $\mathbb{T}$ -action on the Fock space $\ell^2(J_S)$ has weight- n multiplicity b_n , and Z_C is its graded dimension.

PROOF. $N(D) = q^{\deg D}$ and $J_{\mathcal{P}_K} = \mathrm{Div}_+(C)$; group the sum by degree. Convergence for $\beta > 1$ is Corollary 2.2 of Chapter 31. $\square$

THEOREM 4.2 (Complete geometric reconstruction). *The function $\beta \mapsto Z_C(q^{-\beta})$ on $(1, \infty)$ determines the rational function Z_C , hence:*

- (1) *the Weil polynomial $P(T)$ and the Weil numbers $\{\alpha_1, \dots, \alpha_{2g}\}$;*
- (2) *the genus $g = \frac{1}{2} \deg P$;*
- (3) *every point count $\#C(\mathcal{F}_{q^n}) = q^n + 1 - \sum_{j=1}^{2g} \alpha_j^n$;*
- (4) *the order $|\mathrm{Jac}(C)(\mathcal{F}_q)| = P(1)$ of the group of rational points of the Jacobian.*

Consequently the equivariant C^* -dynamical system $(\mathcal{A}_{C, \mathcal{P}_K}, \mathbb{T}, \sigma)$ is a complete invariant of the Weil zeta function of C .

PROOF. A rational function is determined by its values on an interval; q is recovered from the point spectrum $\mathrm{Sp}_p(\sigma) = q^{d_S \mathbb{Z}}$ (Proposition 1.1 of Chapter 23, computed for these systems in Remark 4.3 of Chapter 31), with $d_S = 1$ for $S = \mathcal{P}_K$ by F. K. Schmidt (Proposition 1.1 of Chapter 31), and then $P(T) = Z_C(T)(1-T)(1-qT)$. The remaining items are the Weil conjectures for curves. $\square$

EXAMPLE 4.3 (Numerical illustration). For genus one, $P(T) = 1 - aT + qT^2$:

(q, a)	$b_0, \dots, b_4$	$\#C(\mathcal{F}_{q^n}), n \leq 4$	$ \text{Jac}(\mathcal{F}_q) $
$(4, 1)$	1, 4, 20, 84, 340	4, 24, 76, 240	4
$(5, -2)$	1, 8, 48, 248, 1248	8, 32, 104, 640	8
$(9, 4)$	1, 6, 60, 546, 4920	6, 84, 774, 6720	6

The b_n are the multiplicities in the partition function (4.1); the point counts and the Jacobian order follow from them.

REMARK 4.4 (The principle, again). Theorem 4.2 delivers the conclusion asked for while Corollary 3.2 shows the proposed route cannot. This is the pattern established in Corollary 2.4 of Chapter 4: the arithmetic of these systems lives in the equilibrium data — the measure class, the KMS simplex, the partition function — and K -theoretic invariants, which discard the Radon–Nikodym derivative, are blind to it. Over a function field the equilibrium datum happens to be the Weil zeta function itself, which is why the reconstruction is so complete; but the mechanism is thermodynamic, and asking $KK_{\mathbb{T}}$ to supply it repeats an error the programme has already diagnosed three times, in Remark 4.5 of Chapter 3, Proposition 3.1 of Chapter 4 and Theorem 2.1 of Chapter 20.

5. Assessment

Proposal	Status	Reference
$\partial\mathcal{A}^{\text{aff}}$ defined for $S = \mathcal{P}_K$	no: $\mathcal{O}_{\mathcal{P}_K} = \mathcal{F}_q$	Prop. 1.1
a point at infinity must be fixed	yes	Cor. 1.2
$\mathbb{T}$ -actions better behaved than flows	not here	Prop. 2.1
Gabe–Szabó applies after Takai	no: dual algebra stably finite	Prop. 2.1
Izumi’s compact group theory applies	no: action not Rokhlin	Prop. 2.2
§7 of Chapter 31 suggestion	withdrawn	Rem. 2.3
$K_0^{\mathbb{T}}$ encodes $P(T)$	no: only q^{ds}	Prop. 3.1, Cor. 3.2
K -theory sees $ \text{Jac}(\mathcal{F}_q) = P(1)$	yes, one value only	Rem. 3.3
Z_C is the partition function	yes	Thm. 4.1
complete geometric reconstruction	yes, thermodynamically	Thm. 4.2

Two of the three obstructions are specific and one is a retraction of our own suggestion. The retraction is worth stating plainly: the observation that compact group actions are better classified than flows is correct about the literature and inapplicable here, because the standard reduction — Takai duality to a $\mathbb{Z}$ -action — lands on the stably finite core, and the compact group theorems that avoid that reduction require a Rokhlin property that gauge actions do not have. Periodicity of the modular flow is a real structural feature of the function field case, but it does not buy a classification theorem.

The positive statement is stronger than what was asked for the boundary, and it needs no classification theory at all. Over a function field the partition function of the Toeplitz system *is* the Weil zeta function, evaluated at $T = q^{-\beta}$; and since Z_C is rational, its values on any interval determine everything the Weil conjectures encode. The system therefore recovers g , all $\#C(\mathcal{F}_{q^n})$, the Weil numbers and $|\text{Jac}(C)(\mathcal{F}_q)|$ — from thermodynamics.

Three questions. Does the affine boundary with a fixed ∞ recover $\deg \infty$, so that Z_C can be reconstructed from the affine partition function $Z_C(T)(1 - T^{\deg \infty})$? Second, is there any equivariant invariant, short of the full KMS structure, that sees the b_n — the graded dimension of the Fock module is the natural candidate, but we do not know whether it is an invariant of the abstract dynamical system rather than of the chosen representation. Third, and this is the one

we would pursue: two curves with the same zeta function but non-isomorphic Jacobians exist; by Theorem 4.2 their systems have the same partition function, so does anything in $(\mathcal{A}, \mathbb{T}, \sigma)$ distinguish them, or is zeta-equivalence exactly the equivalence realized here?

Isogeny, zeta-equivalence, and the Picard torsor in function field Bost–Connes systems

This chapter is Paper XIX of the series ([papers/XIX-isogeny/](#)).

SUMMARY. Chapter 32 showed that the partition function of the function field Bost–Connes system is the Weil zeta function, and asked whether the system can distinguish two curves with the same zeta function but non-isomorphic Jacobian groups. The answer is yes, and we locate precisely which layer of the system does it.

The proposed exact sequence $1 \rightarrow \text{Jac}(C)(\mathcal{F}_q) \rightarrow G_{\mathcal{P}_K} \rightarrow \widehat{\mathbb{Z}} \rightarrow 1$ is not the right one: $\text{Jac}(C)(\mathcal{F}_q) = \text{Pic}^0(C)$ is a *quotient* of the compact part $\mathbb{I}_K^0/K^\times$, not a subgroup of $G_{\mathcal{P}_K}$, and the correct picture is a two-step filtration with the much larger $\prod_P \mathcal{O}_P^\times/\mathcal{F}_q^\times$ at the bottom. What is true, and is what the disambiguation needs, is the identity

$$G_{\mathcal{P}_K}/\overline{U_S} \cong \widehat{\text{Pic}(C)} \cong \widehat{\mathbb{Z}} \times \text{Jac}(C)(\mathcal{F}_q), \quad U_S = \prod_P \mathcal{O}_P^\times,$$

coming from $\mathbb{I}_K/(K^\times U) = \text{Div}(C)/\text{Prin}(C) = \text{Pic}(C)$ together with F. K. Schmidt’s theorem that C has a divisor of degree 1. Since $\widehat{\mathbb{Z}}$ is torsion free,

$$\text{Jac}(C)(\mathcal{F}_q) = (G_{\mathcal{P}_K}/\overline{U_S})_{\text{tors}},$$

so the Jacobian group is recovered from the *pair* $(G_{\mathcal{P}_K}, U_S)$ — data the groupoid supplies but the partition function does not.

Hence the disambiguation theorem holds, at the Cartan-pair level: a gauge-equivariant isomorphism of systems forces $\text{Jac}(C_1)(\mathcal{F}_q) \cong \text{Jac}(C_2)(\mathcal{F}_q)$, while the partition function forces only $Z_{C_1} = Z_{C_2}$, that is, isogeny. The separation is not vacuous. By brute-force enumeration over $\mathcal{F}_p$ for $p \leq 23$ we exhibit many isogeny classes containing several group structures; for $p = 13$ and trace -2 there are three, $\mathbb{Z}/16$, $\mathbb{Z}/2 \times \mathbb{Z}/8$ and $\mathbb{Z}/4 \times \mathbb{Z}/4$, all with the same zeta function.

Two cautions. Over a function field $S = \mathcal{P}_K$ is massively *not* norm-separated — $N(D) = q^{\deg D}$ identifies all divisors of equal degree — so Theorem 2.5 of Chapter 21 does not apply at all and the gauge-equivariant hypothesis of Theorem 4.2 of Chapter 22 is not a convenience but a necessity. And the proposed third tier, recovery of C itself by Torelli, does not follow from anything here: what the system supplies is abelian class field theory data, and the abelianized Galois group is known not to determine a global field in the number field case. We record it as open, not proved.

1. The correct filtration

Let $C/\mathcal{F}_q$ be smooth projective geometrically irreducible, $K = \mathcal{F}_q(C)$, $S = \mathcal{P}_K$ the set of closed points. The framework is that of [Ch.Function fields have no archimedean places, so $\mathbb{I}_S = \mathbb{I}_K$, $K_S^\times = K^\times$ and

$$G_{\mathcal{P}_K} = \mathbb{I}_K/\overline{K^\times}, \quad U_S = \prod_P \mathcal{O}_P^\times.$$

PROPOSITION 1.1 (Two-step filtration). *There are exact sequences of topological groups*

$$(1.1) \quad 1 \longrightarrow \prod_P \mathcal{O}_P^\times / \overline{\mathcal{F}_q^\times} \longrightarrow \mathbb{I}_K^0 / K^\times \longrightarrow \mathrm{Pic}^0(C)(\mathcal{F}_q) \longrightarrow 1,$$

$$(1.2) \quad 1 \longrightarrow \mathbb{I}_K^0 / K^\times \longrightarrow G_{\mathcal{P}_K} \xrightarrow{\deg} \widehat{\mathbb{Z}} \longrightarrow 1,$$

where $\mathbb{I}_K^0$ denotes the degree-zero ideles. The group $\mathbb{I}_K^0 / K^\times$ is compact.

PROOF. Standard class field theory for function fields: (1.2) is the degree map, whose image after profinite completion is $\widehat{\mathbb{Z}}$; (1.1) is the quotient of the degree-zero idele class group by the unit ideles, whose cokernel is the degree-zero divisor class group. $\square$

REMARK 1.2 (The proposed sequence). The sequence $1 \rightarrow \mathrm{Jac}(C)(\mathcal{F}_q) \rightarrow G_{\mathcal{P}_K} \rightarrow \widehat{\mathbb{Z}} \rightarrow 1$ is not correct. By (1.1), $\mathrm{Jac}(C)(\mathcal{F}_q) = \mathrm{Pic}^0(C)$ is a *quotient* of the compact part, not a subgroup, and the kernel in (1.2) is $\mathbb{I}_K^0 / K^\times$, an extension of $\mathrm{Pic}^0(C)$ by the enormous group $\prod_P \mathcal{O}_P^\times / \overline{\mathcal{F}_q^\times}$. The Jacobian group is a subquotient of $G_{\mathcal{P}_K}$, and exhibiting it requires naming U_S .

2. The Jacobian group as a torsion subgroup

THEOREM 2.1 (Recovery from the pair). *With $U_S = \prod_P \mathcal{O}_P^\times \leq G_{\mathcal{P}_K}$,*

$$(2.1) \quad G_{\mathcal{P}_K} / \overline{U_S} \cong \widehat{\mathrm{Pic}(C)} \cong \widehat{\mathbb{Z}} \times \mathrm{Jac}(C)(\mathcal{F}_q),$$

and consequently

$$\mathrm{Jac}(C)(\mathcal{F}_q) \cong \left(G_{\mathcal{P}_K} / \overline{U_S} \right)_{\mathrm{tors}}.$$

PROOF. $\mathbb{I}_K / (K^\times U_S)$ is by definition $\mathrm{Div}(C) / \mathrm{Prin}(C) = \mathrm{Pic}(C)$, and passing to profinite completions gives the first isomorphism in (2.1). By F. K. Schmidt's theorem [115] C has a divisor of degree 1, so the degree map $\mathrm{Pic}(C) \rightarrow \mathbb{Z}$ splits and $\mathrm{Pic}(C) \cong \mathbb{Z} \oplus \mathrm{Pic}^0(C)$ with $\mathrm{Pic}^0(C) = \mathrm{Jac}(C)(\mathcal{F}_q)$ finite; completing gives $\widehat{\mathbb{Z}} \times \mathrm{Jac}(C)(\mathcal{F}_q)$. Finally $\widehat{\mathbb{Z}}$ is torsion free, so the torsion subgroup of the product is the finite factor. $\square$

COROLLARY 2.2 (What each layer sees). *The partition function determines $Z_C(T)$ (Theorem 4.1 of Chapter 32), hence $|\mathrm{Jac}(C)(\mathcal{F}_q)| = P(1)$ but only its order. The pair $(G_{\mathcal{P}_K}, U_S)$ determines $\mathrm{Jac}(C)(\mathcal{F}_q)$ as an abstract finite abelian group, by Theorem 2.1.*

3. Disambiguation

THEOREM 3.1 (Isogeny disambiguation). *Let C_1, C_2 be curves over $\mathcal{F}_q$ with $Z_{C_1} = Z_{C_2}$ but $\mathrm{Jac}(C_1)(\mathcal{F}_q) \not\cong \mathrm{Jac}(C_2)(\mathcal{F}_q)$. Then:*

- (1) *the two systems have the same partition function, so no invariant built from the KMS temperature dependence alone distinguishes them;*
- (2) *nevertheless the systems are not isomorphic gauge-equivariantly: there is no isomorphism $\mathcal{A}_{C_1, \mathcal{P}_{K_1}} \rightarrow \mathcal{A}_{C_2, \mathcal{P}_{K_2}}$ intertwining the full gauge actions in the sense of Chapter 22 (its Definition 4.1) — the actions of the duals of the grading groups, of which the circle action of Chapter 32 and the flow σ are quotients.*

PROOF. (1) is Theorem 4.1 of Chapter 32. For (2), suppose Φ is such an isomorphism. The argument of Theorem 4.2 of Chapter 22 — the integer-valued gauge cocycle and its bisections — uses only the étale groupoid picture and the freeness of the grading monoid, and applies verbatim over a function field; it gives $\Phi(C(Y_{S_1})) = C(Y_{S_2})$ together with a monoid isomorphism

$\theta : J_{S_1} \rightarrow J_{S_2}$, i.e. a θ -equivariant homeomorphism $h : Y_{S_1} \rightarrow Y_{S_2}$ of the semigroup dynamical systems. Now consider the *deep stratum*

$$Y_\infty := \bigcap_{0 \neq D \in J_S} D \cdot Y_S = \{[x, g] : x = 0\} \cong G_{\mathcal{P}_K} / \overline{U_S},$$

canonically defined from the dynamics, on which $D \in J_S$ acts by the translation $[0, g] \mapsto [0, \sigma_D g]$: the deep stratum is a torsor over the compactified Picard group $G_{\mathcal{P}_K} / \overline{U_S} \cong \widehat{\text{Pic}(C)}$ of Theorem 2.1, and the translations $\{\sigma_D\}$ run through the effective classes, which generate $\text{Pic}(C)$ and are dense in the completion. The closure of the translation monoid in $\text{Homeo}(Y_\infty)$ is therefore a compact group acting simply transitively — a closed subsemigroup of a compact group is a subgroup — and it is isomorphic, as a topological group, to $G_{\mathcal{P}_K} / \overline{U_S}$. Since h maps $Y_{\infty,1}$ onto $Y_{\infty,2}$ intertwining the translation monoids, it intertwines their closures, giving a topological group isomorphism $G_{\mathcal{P}_{K_1}} / \overline{U_{S_1}} \cong G_{\mathcal{P}_{K_2}} / \overline{U_{S_2}}$. Taking torsion subgroups, Theorem 2.1 forces $\text{Jac}(C_1)(\mathcal{F}_q) \cong \text{Jac}(C_2)(\mathcal{F}_q)$, contrary to hypothesis. (Note that the homeomorphism type of Y_∞ alone would see nothing: all these spaces are Cantor sets, and indeed $|\text{Jac}|$ is already matched by (1); it is the translation structure that carries the group.) $\square$

REMARK 3.2 (Gauge-equivariance is necessary here). Over a function field $N(D) = q^{\deg D}$, so N identifies all divisors of the same degree and $S = \mathcal{P}_K$ is as far from norm-separated as possible: the relation lattice $L(S)$ of Chapter 22 (its Definition 1.2) has infinite rank. Consequently Theorem 1.4 of Chapter 21, which identifies $C(Y_S)$ with $\mathcal{A}^\sigma$ under norm-separation, is unavailable, and $\mathcal{A}^\sigma$ is a large non-commutative algebra — as is the fixed-point algebra of the circle action, which coincides with it. Theorem 3.1 therefore rests on Theorem 4.2 of Chapter 22 and its *full* gauge-equivariant hypothesis; equivariance for the circle action alone is the same as σ -equivariance and does not feed that theorem. In this setting the hypothesis is not a technical convenience but the only route available.

EXAMPLE 3.3 (Explicit isogeny classes with several group structures). Enumerating all $y^2 = x^3 + ax + b$ over $\mathcal{F}_p$ and grouping by trace t — equal t being equivalent to equal zeta function, hence to isogeny [125] — gives, for instance:

p	t	group structures of $E(\mathcal{F}_p)$
11	0	$\mathbb{Z}/12, \mathbb{Z}/2 \times \mathbb{Z}/6$
13	-2	$\mathbb{Z}/16, \mathbb{Z}/2 \times \mathbb{Z}/8, \mathbb{Z}/4 \times \mathbb{Z}/4$
13	5	$\mathbb{Z}/9, \mathbb{Z}/3 \times \mathbb{Z}/3$
17	2	$\mathbb{Z}/16, \mathbb{Z}/2 \times \mathbb{Z}/8, \mathbb{Z}/4 \times \mathbb{Z}/4$
19	-7	$\mathbb{Z}/27, \mathbb{Z}/3 \times \mathbb{Z}/9$
23	-8	$\mathbb{Z}/32, \mathbb{Z}/2 \times \mathbb{Z}/16$

Each row is a single isogeny class, so a single partition function, containing curves whose systems are pairwise non-isomorphic by Theorem 3.1. The phenomenon is abundant, not exceptional: it occurred for most traces at every prime $11 \leq p \leq 23$ examined.

4. The hierarchy, with the third tier open

THEOREM 4.1 (Two tiers). *For the function field system $(\mathcal{A}_{C, \mathcal{P}_K}, \mathbb{T}, \sigma)$:*

- **Tier 1, thermodynamic.** *The partition function recovers $Z_C(T)$, hence the Weil polynomial, the genus, all point counts $\#C(\mathcal{F}_{q^n})$, and $|\text{Jac}(C)(\mathcal{F}_q)|$. Two systems have the same partition function iff the curves are zeta-equivalent, that is iff their Jacobians are isogenous over $\mathcal{F}_q$ [125].*
- **Tier 2, Cartan pair.** *A gauge-equivariant isomorphism recovers the diagonal with its semigroup action, hence the deep stratum with its translation structure, hence the*

compactified Picard group and $\text{Jac}(C)(\mathcal{F}_q)$ as an abstract group, strictly refining Tier 1 by Example 3.3.

REMARK 4.2 (The third tier is not established). The proposal’s third tier — recovery of C itself, “the Torelli geometric isomorphism” — does not follow from what is proved here, and we do not believe it follows from abelian data at all. What the system supplies is the abelianized Galois group with its Frobenius classes: class field theory data. In the number field case the abelianized absolute Galois group is known not to determine the field — Onabe, and Angelakis–Stevenhagen [4], exhibit infinitely many imaginary quadratic fields sharing the same $\text{Gal}(K^{ab}/K)$ — so no purely abelian invariant can give a Torelli statement. Our data is richer than the bare group, carrying the Frobenius classes indexed by S together with their degrees; whether that suffices is exactly the question addressed in the literature on characterizing global fields by their L -series, which we have not been able to consult here. We state Tier 3 as open and make no claim.

REMARK 4.3 (What Tier 2 does and does not give). Theorem 2.1 recovers $\text{Jac}(C)(\mathcal{F}_q)$ as an abstract finite abelian group. It does not recover the principal polarization, which is what Torelli’s theorem requires (for the arithmetic of pairings and polarizations over finite fields see [60]), nor the Galois module structure of the ℓ -adic Tate module beyond its characteristic polynomial. So even a complete solution at Tier 2 would leave a gap before Tier 3, and the gap is exactly the polarization.

5. Assessment

Statement	Status	Reference
$1 \rightarrow \text{Jac}(\mathcal{F}_q) \rightarrow G_{\mathcal{P}_K} \rightarrow \widehat{\mathbb{Z}} \rightarrow 1$	false: wrong kernel	Rem. 1.2
two-step filtration	true	Prop. 1.1
$G_{\mathcal{P}_K}/\overline{U_S} \cong \widehat{\mathbb{Z}} \times \text{Jac}(\mathcal{F}_q)$	true	Thm. 2.1
$\text{Jac}(\mathcal{F}_q) = (G_{\mathcal{P}_K}/\overline{U_S})_{\text{tors}}$	true	Thm. 2.1
partition function $\iff$ isogeny	true	Thm. 4.1
system distinguishes isogenous curves	yes, at the Cartan pair	Thm. 3.1
$S = \mathcal{P}_K$ norm-separated	no; $L(S)$ of infinite rank	Rem. 3.2
gauge-equivariance needed	yes, necessarily	Rem. 3.2
Tier 3: $C_1 \cong C_2$	open; not from abelian data	Rem. 4.2

The question left open in §5 of Chapter 32 is answered: zeta-equivalence is *not* the equivalence realized by these systems. The partition function sees exactly the isogeny class, and the Cartan pair sees strictly more, namely the isomorphism class of $\text{Jac}(C)(\mathcal{F}_q)$ as a group. The mechanism is the identity $\mathbb{I}_K/(K^\times U) = \text{Pic}(C)$, which is class field theory rather than operator algebra; the operator-algebraic content is only that a gauge-equivariant isomorphism recovers the diagonal with its semigroup action, which is Theorem 4.2 of Chapter 22 — the passage from there to the Picard group is the translation structure of the deep stratum.

We are left with a clean picture of the layers and an honest gap at the top. Thermodynamics gives the isogeny class; the Cartan pair gives the Jacobian group; and the curve itself is out of reach of both, because the polarization is nowhere in the data and because abelian class field theory is known to be insufficient for reconstructing global fields.

Three questions. Does the pair (G_S, U_S) with the Frobenius classes determine more than $\text{Jac}(C)(\mathcal{F}_q)$ — for instance the full group $\text{Pic}(C)$ with its effective cone, which is closer to geometry? Second, Theorem 2.1 used F. K. Schmidt to split the degree map; is the splitting canonical enough that the isomorphism $\text{Jac}(C)(\mathcal{F}_q) \cong (\cdot)_{\text{tors}}$ is functorial, or does the choice of degree-one divisor matter for the disambiguation? Third, and this is the natural next target, is there a curve

pair with the same zeta *and* the same $\text{Jac}(\mathcal{F}_q)$ but non-isomorphic curves — for which Tier 2 would also fail, and the question of Tier 3 would become sharp rather than speculative?

CHAPTER 34

The limits of abelian reconstruction: twin curves exist, genus one suffices, and the obstruction is a torsor coordinate

This chapter is Paper XX of the series (`papers/XX-twin-curves/`).

SUMMARY. Chapter 33 asked whether there are non-isomorphic curves over $\mathcal{F}_q$ with the same zeta function and isomorphic Jacobian groups, which would locate the resolution limit of the abelian Bost–Connes systems. The answer is yes, the construction is far simpler than the genus-two machinery proposed, and the reason is complex multiplication.

For an ordinary elliptic curve $E/\mathcal{F}_q$ with $\text{End}(E) = \mathcal{O}$, Deuring’s theory gives $E(\mathcal{F}_{q^n}) \cong \mathcal{O}/(\phi^n - 1)$ as an $\mathcal{O}$ -module for every n , an isomorphism depending only on $\mathcal{O}$; and $\text{Pic}(\mathcal{O})$ acts simply transitively on the set of curves in the isogeny class with that endomorphism ring, of which there are exactly $h(\mathcal{O})$. Hence whenever $h(\mathcal{O}) > 1$ one obtains $h(\mathcal{O})$ curves with distinct j -invariants sharing *every* Galois-module invariant at every level. Genus one already suffices; Howe’s and Lauter’s genus-two constructions are not needed. Enumerating all curves over $\mathcal{F}_p$ for $11 \leq p \leq 29$ we find 69 isogeny classes containing several j -invariants with a common group structure, and the counts match the class numbers: $h(-88) = h(-100) = h(-48) = h(-32) = h(-60) = 2$ against two j -invariants in each case.

This is stronger than the question asked. Twin curves agree not only in Z_C and in $\text{Jac}(\mathcal{F}_q)$, but in the whole tower $\{E(\mathcal{F}_{q^n})\}_n$ with its trace maps — which matters because Tier 2 of Chapter 33 recovers more than the abstract group: the compactified Picard group with its translation structure, through the deep stratum, and with it the map $P \mapsto [P] \in \text{Pic}(C)$ on all closed points. We prove that this map does not separate twins either — under the module identification the class of a degree- n point is multiplication by $(\phi^n - 1)/(\phi - 1)$, again $\mathcal{O}$ -determined — and confirm it numerically: for two supersingular classes at $p = 11, 17$ and for each group structure at $p = 13$, the degree-two point-to-Picard multiset profiles coincide across distinct j -invariants.

We also correct the proposed explanation of the barrier. The Weil pairing is not out of abelian reach for the reason given: over $\mathcal{F}_q$ the Galois group acting on $\text{Jac}[n]$ is $\text{Gal}(\bar{\mathcal{F}}_q/\mathcal{F}_q) = \hat{\mathbb{Z}}$, which is abelian, so “the pairing cannot be encoded in an abelianized Galois group” is not the obstruction. The obstruction is a torsor coordinate: $\text{Pic}(\mathcal{O})$ acts simply transitively on the twins and acts *trivially* on every module invariant, so what is missing is the position in a $\text{Pic}(\mathcal{O})$ -torsor, and no invariant of the modules can see it.

1. Twin curves, and why genus one suffices

The framework is that of [Ch. 1] transported to function fields as in Chapters 31–33 of the series; Chapters 3, 20, 25, 22 and 24–XIX, referred to by numeral, are unpublished companions, and every fact used from them is restated or verified in place.

THEOREM 1.1 (Existence of twins). *Let $E/\mathcal{F}_q$ be an ordinary elliptic curve with Frobenius ϕ and $\text{End}(E) = \mathcal{O}$, an order in $\mathbb{Q}(\phi)$ with $\mathbb{Z}[\phi] \subseteq \mathcal{O} \subseteq \mathcal{O}_{\max}$. Then:*

- (1) $E(\mathcal{F}_{q^n}) \cong \mathcal{O}/(\phi^n - 1)$ as an $\mathcal{O}$ -module, for every $n \geq 1$, and this depends only on $\mathcal{O}$ and ϕ , not on E ;

- (2) $\text{Pic}(\mathcal{O})$ acts simply transitively on the set of $\mathcal{F}_q$ -isomorphism classes of curves in the isogeny class with endomorphism ring $\mathcal{O}$ — equivalently, since the quadratic twist has trace $-t \neq t$ for ordinary E , on their j -invariants; there are exactly $h(\mathcal{O})$ of them.

Consequently, whenever $h(\mathcal{O}) > 1$ there exist $C_1 \not\cong C_2$ over $\mathcal{F}_q$ with $Z_{C_1} = Z_{C_2}$, with $\text{Jac}(C_1)(\mathcal{F}_q) \cong \text{Jac}(C_2)(\mathcal{F}_q)$, and indeed with $C_1(\mathcal{F}_{q^n})$ and $C_2(\mathcal{F}_{q^n})$ isomorphic as $\mathcal{O}$ -modules for every n .

PROOF. (1) is Deuring's theory [39] in the module-theoretic form proved by Lenstra [82]: for ordinary E with $\text{End}(E) = \mathcal{O}$ one has $E(\mathcal{F}_{q^n}) = \ker(\phi^n - 1) \cong \mathcal{O}/(\phi^n - 1)$ as $\mathcal{O}$ -modules. (2) is the theory of complex multiplication in characteristic p , in Waterhouse's kernel-ideal form [135]: the set of curves with a given ordinary endomorphism ring is a principal homogeneous space under the class group. Distinct classes give distinct j -invariants, these being the roots of the Hilbert class polynomial of $\mathcal{O}$ reduced mod p . $\square$

REMARK 1.2 (The proposal over-engineers). The construction requested — genus-two hyperelliptic curves with equal Igusa invariants data, following Howe [59] and Lauter [81] — is unnecessary. Genus one produces twins as soon as $h(\mathcal{O}) > 1$, which is the generic situation, and produces them with a stronger property than asked for: agreement of the entire Galois-module tower, not merely of Z_C and $\text{Jac}(\mathcal{F}_q)$.

EXAMPLE 1.3 (Enumeration). Enumerating all $y^2 = x^3 + ax + b$ over $\mathcal{F}_p$ for $11 \leq p \leq 29$ and grouping by (trace, group structure) yields 69 classes containing more than one j -invariant. Samples, with the class number of the relevant order:

p	t	$\text{Jac}(\mathcal{F}_p)$	j -invariants	$h(\mathcal{O})$
23	-2	$\mathbb{Z}/26$	1, 17	$h(-88) = 2$
13	2	$\mathbb{Z}/12$	4, 6	$h(-48) = 2$
17	-6	$\mathbb{Z}/24$	1, 6	$h(-32) = 2$
19	-4	$\mathbb{Z}/24$	9, 17	$h(-60) = 2$
29	-4	$\mathbb{Z}/34$	3, 7, 17	$h(-100) = 2$ (two orders)
23	-6	$\mathbb{Z}/30$	14, 18, 20, 22	$h(-56) = 4$

The count of j -invariants at a fixed group structure equals $h(\mathcal{O})$ for the corresponding endomorphism ring, as Theorem 1.1(2) requires.

2. Indistinguishability at both tiers

THEOREM 2.1 (Twins are invisible to the abelian system). *Let C_1, C_2 be twins in the sense of Theorem 1.1, with a common endomorphism ring $\mathcal{O}$. Then their Bost–Connes systems agree at both tiers of Theorem 4.1 of Chapter 33:*

- (1) Tier 1. *The partition functions coincide, $Z_{C_1} = Z_{C_2}$.*
- (2) Tier 2. *$\text{Jac}(C_1)(\mathcal{F}_q) \cong \text{Jac}(C_2)(\mathcal{F}_q)$; more, the whole tower $\{C_i(\mathcal{F}_{q^n})\}_n$ with its norm maps is isomorphic, so every invariant of the Galois-module data agrees; and more still, the full Tier-2 datum — the compactified Picard group together with the degree-labelled multiset of point classes $\{[P] : \deg P = n\}$ — corresponds under the module identification, for every n .*

PROOF. Both follow from Theorem 1.1(1): the tower and its maps are $\{\mathcal{O}/(\phi^n - 1)\}$ with the natural maps, depending only on $(\mathcal{O}, \phi)$. For the last clause of (2): a closed point of degree n is a Frobenius orbit $\{Q, \phi Q, \dots, \phi^{n-1}Q\}$ of a point $Q \in C(\mathcal{F}_{q^n})$ of exact degree n , and, choosing the rational base point, its class in $\text{Pic}^0 = E(\mathcal{F}_q)$ is $Q \oplus \phi Q \oplus \dots \oplus \phi^{n-1}Q = \nu_n \cdot Q$ with $\nu_n = (\phi^n - 1)/(\phi - 1) \in \mathcal{O}$, computed in the $\mathcal{O}$ -module $\mathcal{O}/(\phi^n - 1)$, with $E(\mathcal{F}_q) = \ker(\phi - 1) = \nu_n \cdot \mathcal{O}/(\phi^n - 1)$. The degree- n multiset of point classes is therefore the image multiset of ν_n on the exact-degree- n elements of $\mathcal{O}/(\phi^n - 1)$ — a formula in $(\mathcal{O}, \phi)$ alone. Since the Tier-2 datum

recovered in Theorem 3.1 of Chapter 33 is exactly the compactified Picard group with these degree-labelled translations, the twins' data are isomorphic as marked groups. $\square$

REMARK 2.2 (Tier 2 recovers more than the abstract group). It is worth being careful here, because the question as posed assumed Tier 2 sees only $\text{Jac}(\mathcal{F}_q)$. It sees more. By Theorem 2.1 of Chapter 33, $G_S/\overline{U}_S \cong \widehat{\text{Pic}(C)}$, and the deep-stratum translation structure recovered in the proof of Theorem 3.1 of Paper XIX — the generator $P \in J_S$ acting on the Picard torsor by σ_P — gives the map

$$P \longmapsto [P] \in \text{Pic}(C)$$

on all closed points, together with their degrees. That is substantially finer than the group $\text{Jac}(\mathcal{F}_q)$, and it is what has to be checked. By Theorem 2.1(2) it is also $\mathcal{O}$ -determined, so twins are not separated by it either.

REMARK 2.3 (Numerical check of the point-to-Picard map). For an elliptic curve with base point O , a closed point of degree two is a Galois orbit $\{Q, \phi Q\}$ in $E(\mathcal{F}_{q^2})$ and its class in $\text{Pic}^0 = E(\mathcal{F}_q)$ is $\text{Tr}(Q) = Q \oplus \phi Q$. The last clause of Theorem 2.1(2) predicts equal profiles across twins; we confirm it by direct computation, for the ordinary twins of Example 1.3 at $p = 13$ and also for two supersingular same-zeta-same-group classes (outside the ordinary hypothesis of Theorem 1.1, so not covered by the proof):

p	t	j	degree-two profile
11	0	0	$(6^6, 5^6)$
11	0	1	$(6^6, 5^6)$
17	0	0	$(9^9, 8^9)$
17	0	8	$(9^9, 8^9)$
13	2	4, 6	$(8^6, 7^6)$
13	2	0, 11	$(8^9, 6^3)$

The profiles agree within each twin pair. At $p = 13$ the four curves of trace 2 split into two profiles, but that split tracks the group structure — $\mathbb{Z}/12$ for $j = 4, 6$ and $\mathbb{Z}/2 \times \mathbb{Z}/6$ for $j = 0, 11$ — and not the j -invariant, so it is Tier 2 information already accounted for.

3. The obstruction is a torsor coordinate, not a pairing

REMARK 3.1 (The proposed explanation is not correct). The barrier was attributed to the Weil pairing being “a non-linear skew-symmetric pairing that cannot be encoded in the abelianized Galois group $\mathbb{I}_K/\overline{K}^\times$ ”. This reasoning does not work. Over a finite field the Galois group acting on $\text{Jac}(C)[n]$ is $\text{Gal}(\overline{\mathcal{F}}_q/\mathcal{F}_q) = \widehat{\mathbb{Z}}$, which is *abelian*; the Weil pairing is Gal-equivariant for that action and is not inaccessible on grounds of non-commutativity. The Tate module with its Frobenius action is precisely the kind of datum an abelian theory can carry — indeed Z_C is its characteristic polynomial.

THEOREM 3.2 (The correct obstruction). *Let $\mathcal{E}(\mathcal{O})$ be the set of curves in a fixed ordinary isogeny class with $\text{End} = \mathcal{O}$. Then $\text{Pic}(\mathcal{O})$ acts simply transitively on $\mathcal{E}(\mathcal{O})$, and acts trivially on every invariant of the Galois-module tower $\{E(\mathcal{F}_{q^n})\}_n$, these being $\mathcal{O}/(\phi^n - 1)$ independently of the curve. Consequently the missing datum separating twins is the position of the curve in a $\text{Pic}(\mathcal{O})$ -torsor, and no invariant constructed from the modules — including the Weil pairing, which is likewise $\mathcal{O}$ -determined — can detect it.*

PROOF. Simple transitivity is Theorem 1.1(2). Triviality on module invariants is Theorem 1.1(1). A function on $\mathcal{E}(\mathcal{O})$ factoring through the module data is therefore constant, whereas j is injective on $\mathcal{E}(\mathcal{O})$. $\square$

REMARK 3.3 (What would be needed). By Theorem 3.2 the separation of twins is equivalent to locating a point in a $\text{Pic}(\mathcal{O})$ -torsor, and the classical object that does this is the j -invariant, i.e. the Hilbert class field of $\mathcal{O}$ and its Galois action. That is still abelian class field theory — but for the CM field $\mathbb{Q}(\phi)$, not for the function field $K = \mathcal{F}_q(C)$. The two abelian theories are different, and the Bost–Connes system of C carries the second, not the first. This is a sharper diagnosis than “the barrier is non-abelian”, and it suggests where a refinement would have to come from: a system built over the CM order rather than over the curve.

4. The resolution boundary

THEOREM 4.1 (Master statement). *For the localized Bost–Connes system of a curve over $\mathcal{F}_q$:*

- *the partition function determines exactly the isogeny class of $\text{Jac}(C)$ (Theorem 4.1 of Chapter 32);*
- *the Cartan pair determines, through the deep stratum, the compactified Picard group with the degree-labelled point classes $P \mapsto [P] \in \text{Pic}(C)$, hence $\text{Jac}(C)(\mathcal{F}_q)$ and the rational effective cone, strictly refining the previous line (Theorem 3.1 of Chapter 33);*
- *neither determines the curve: by Theorems 1.1 and 2.1, whenever $h(\mathcal{O}) > 1$ there are $h(\mathcal{O})$ curves agreeing in all of the above and differing in j .*

The resolution boundary of the abelian theory is therefore the $\mathcal{O}$ -module tower $\{E(\mathcal{F}_{q^n})\}$, and the residual ambiguity is a $\text{Pic}(\mathcal{O})$ -torsor of size $h(\mathcal{O})$.

5. Retrospective

Question	Answer	Reference
do twins exist?	yes, abundantly	Thm. 1.1, Ex. 1.3
is genus two needed?	no: genus one suffices	Rem. 1.2
are twins Tier-1 indistinguishable?	yes	Thm. 2.1
Tier 2 sees only $\text{Jac}(\mathcal{F}_q)$?	no: also $P \mapsto [P]$	Rem. 2.2
are twins Tier-2 indistinguishable?	yes, proved and checked	Thm. 2.1, Rem. 2.3
is the Weil pairing the obstruction?	no; $\text{Gal}(\mathcal{F}_q)$ is abelian	Rem. 3.1
what is the obstruction?	a $\text{Pic}(\mathcal{O})$-torsor coordinate	Thm. 3.2
resolution boundary	$\{E(\mathcal{F}_{q^n})\}$ as $\mathcal{O}$ -modules	Thm. 4.1

The twenty papers have converged on a single principle, and this one states its sharpest form. The arithmetic that a Bost–Connes system carries is the arithmetic of its equilibrium states, and that arithmetic is exactly the Galois-module data of the base: the isogeny class at the thermodynamic level, the Picard group at the level of the Cartan pair. What lies beyond is not a deeper analytic invariant but a torsor, and a torsor has no invariants at all — its points are distinguished only by a coordinate that must be supplied from outside.

That is the honest form of the negative results accumulated here. The K -theory did not fail for lack of computation (Remark 4.5 of Chapter 3); the von Neumann algebra did not fail for lack of rigidity technique (Theorem 2.1 of Chapter 20); the quantum transport inequalities did not fail for lack of a better Dirichlet form (Theorem 4.1 of Chapter 24). In each case the invariant was blind to something that is not an invariant: a measure class, a tail event, and now a torsor coordinate. The one place where the pattern broke was Theorem 4.1 of Chapter 31, where commensurability of the log-norms let the factor retain the temperature — and that too was arithmetic, not technique.

Three questions we would still pursue. Is the analogue of Theorem 3.2 true in higher genus, with $\text{Pic}(\mathcal{O})$ replaced by the relevant class group of the CM order, so that the resolution boundary is uniformly a torsor? Second, Remark 3.3 suggests building a Bost–Connes system over the CM

order $\mathcal{O}$ rather than over C , whose symmetry group would be $\text{Pic}(\mathcal{O})$ acting on the twins; does that system exist and does it separate them? Third, the twins of Theorem 1.1 are ordinary; the supersingular case has endomorphisms in a quaternion order, which is where Chapter 25 located the only viable non-commutative model, and whether the two occurrences of quaternion orders in this programme are related we do not know.

Part 7

p-adic aspects

Iwasawa theory and localized Bost–Connes systems: there are no p -adic KMS states, and where Leopoldt actually enters

This chapter is Paper XXII of the series ([papers/XXII-iwasawa/](#)).

SUMMARY. We examine the proposal to build a p -adic thermodynamics on the localized Bost–Connes systems and to read the Iwasawa invariants off it. Four obstructions close it, of which the first is fundamental and the others are independent of it.

There are no p -adic KMS states. The KMS condition is a statement about a *state* — a positive normalized functional — on a C^* -algebra, together with the analytic continuation of σ_t to a complex strip. Positivity requires an ordered field and an involution compatible with it; $\mathbb{C}_p$ is not ordered and carries no C^* -theory. There is p -adic functional analysis, but no positive cone, hence no states, hence no simplex, and hence nothing on which a KMS condition could be imposed.

The proposed dynamics is undefined on the proposed set. The map $x \mapsto \langle x \rangle$ projects $\mathbb{Z}_p^\times$ onto $1 + p\mathbb{Z}_p$ and requires x to be a p -adic unit. For $\mathfrak{p} \mid p$ one has $N\mathfrak{p} = p^f$ with $v_p(N\mathfrak{p}) = f > 0$, so $\langle N\mathfrak{p} \rangle^z$ does not exist. The localization $S = \{\mathfrak{p} \mid p\}$ is exactly the set at which the p -adic dynamics cannot be defined — which is why p -adic L -functions remove the Euler factors at p .

There is no p -adic abscissa of convergence. For $\mathfrak{q} \nmid p$ the number $\langle N\mathfrak{q} \rangle^{-z}$ is a 1-unit, so its absolute value is 1 for every z ; a p -adic series converges iff its terms tend to 0, and terms of absolute value 1 never do. Every p -adic Dirichlet series over an infinite set of primes diverges, at every weight. The convergence–divergence dichotomy that drives the whole archimedean theory therefore has no p -adic counterpart. This is not a defect of our set-up: it is why p -adic L -functions are constructed from measures and Iwasawa power series rather than from Dirichlet series.

The proposed obstruction cannot generate the characteristic ideal. The expression

$$\Xi_p(z, \chi) = \sum_{\mathfrak{p} \mid p} (1 - \chi(\text{Frob}_{\mathfrak{p}}) \langle N\mathfrak{p} \rangle^z)$$

is a finite sum of $g \leq [K : \mathbb{Q}]$ terms, so its λ -invariant is bounded in terms of g alone; the Iwasawa λ -invariant is unbounded. A bounded-degree element generates no family of ideals of unbounded degree.

What survives is smaller and precise. The p -local partition function $\prod_{\mathfrak{p} \mid p} (1 - N\mathfrak{p}^{-\beta})^{-1}$ is, up to inversion, exactly the Euler factor that $L_p(s, \chi)$ removes in its interpolation formula: a genuine dictionary entry, but between the partition function and the interpolation factor, not the characteristic ideal. And Leopoldt’s conjecture does enter this programme — not through a p -adic state space but through Proposition 2.2 and Remark 2.3 of Chapter 3, where the isotropy of the groupoid at the boundary strata requires a Leopoldt-type condition for general K and is unconditional when $\mathcal{O}_K^\times$ is finite.

1. There are no p -adic KMS states

Chapters 3, 4, 6, 30, 25 and 28, referred to by numeral, are unpublished companions; every fact used from them is restated or verified in place. [Ch. 1] and Chapter 2 [Ch.

PROPOSITION 1.1 (Positivity is unavailable). *Let A be a Banach algebra over $\mathbb{C}_p$ with an involution. There is no notion of state on A : a state is a linear functional φ with $\varphi(x^*x) \geq 0$ and*

$\varphi(1) = 1$, and the inequality requires an order on the scalar field. The field $\mathbb{C}_p$ is not orderable: it is algebraically closed, so $-1 = i^2$ is a square; and already $\mathbb{Q}_p$ is non-orderable, -1 being a sum of at most four squares (the level of $\mathbb{Q}_p$ is 1, 2 or 4 according to $p \equiv 1 \pmod{4}$, $p \equiv 3 \pmod{4}$, $p = 2$). Consequently there is no positive cone, no state space, no Choquet simplex, and no Gelfand–Naimark–Segal construction.

COROLLARY 1.2 (The KMS condition has no p -adic form). *The KMS_β condition presupposes (i) a state φ satisfying $\varphi(xy) = \varphi(y\sigma_{i\beta}(x))$ and (ii) the extension of $t \mapsto \sigma_t$ to the strip $0 \leq \Im t \leq \beta$ in $\mathbb{C}$. Neither is available over $\mathbb{C}_p$: (i) by Proposition 1.1, and (ii) because $\mathbb{C}_p$ has no distinguished strip, the “imaginary direction” being an artefact of $\mathbb{C} = \mathcal{R} \oplus i\mathcal{R}$ with no p -adic analogue. There is therefore no object called a p -adic KMS state, and no p -adic KMS simplex whose dimension could be computed.*

REMARK 1.3 (What can be transported). The Gibbs functional $\varphi_\beta(e_a) = N\mathfrak{a}^{-\beta}$ does have a p -adic avatar, but as a *measure* rather than a state: the assignment $\mathfrak{a} \mapsto \langle N\mathfrak{a} \rangle^{-z}$ extends to a $\mathbb{C}_p$ -valued distribution on a compact space, and it is exactly such distributions — the Mazur measure and its relatives — that define p -adic L -functions. Measures do not form a simplex without positivity, and no equilibrium condition selects them. The correct statement of the analogy is therefore “Gibbs state $\leftrightarrow$ p -adic measure”, with the equilibrium characterization lost in transit.

2. The dynamics is undefined on $S = \{\mathfrak{p} \mid p\}$

PROPOSITION 2.1 ($\langle N\mathfrak{p} \rangle$ does not exist). *The projection $\langle \cdot \rangle : \mathbb{Z}_p^\times \rightarrow 1 + p\mathbb{Z}_p$, $x \mapsto x/\omega(x)$, is defined only on p -adic units. For $\mathfrak{p} \mid p$ with residue degree f one has $N\mathfrak{p} = p^f$ and $v_p(N\mathfrak{p}) = f > 0$, so $N\mathfrak{p} \notin \mathbb{Z}_p^\times$ and $\langle N\mathfrak{p} \rangle^z$ is undefined. Hence the proposed p -adic dynamics $\sigma_z^{(p)}(\mu_a) = \langle N\mathfrak{a} \rangle^z \mu_a$ does not exist on $J_p = \bigoplus_{\mathfrak{p} \mid p} \mathbb{N}\mathfrak{p}$.*

REMARK 2.2 (The localization is at the wrong set). This is not a normalization issue to be repaired by a choice of branch. The set $\{\mathfrak{p} \mid p\}$ is precisely the set of places at which p -adic interpolation breaks down, which is why $L_p(s, \chi)$ is defined with the Euler factors at p removed [133]:

$$L_p(1 - n, \chi\omega^{-n}) = L(1 - n, \chi\omega^{-n}) \prod_{\mathfrak{p} \mid p} (1 - \chi\omega^{-n}(\mathfrak{p})N\mathfrak{p}^{n-1}).$$

A p -adic dynamics must be built on the primes $\mathfrak{q} \nmid p$, where $N\mathfrak{q} \in \mathbb{Z}_p^\times$. That, however, runs into §3.

3. There is no p -adic abscissa of convergence

THEOREM 3.1 (p -adic Dirichlet series never converge). *Let T be an infinite set of primes with $\mathfrak{q} \nmid p$ for all $\mathfrak{q} \in T$, and let z range over the domain on which $u^z = \exp_p(z \log_p u)$ is defined for 1-units u — all of $\mathbb{Z}_p$, and the disc $|z|_p < p^{(p-2)/(p-1)}$ of $\mathbb{C}_p$. Then $\langle N\mathfrak{q} \rangle \in 1 + p\mathbb{Z}_p$, so*

$$|\langle N\mathfrak{q} \rangle^{-z}|_p = 1 \quad \text{for every } \mathfrak{q} \in T \text{ and every such } z,$$

and since a series in a complete non-archimedean field converges if and only if its terms tend to 0, the series $\sum_{\mathfrak{q} \in T} \langle N\mathfrak{q} \rangle^{-z}$ diverges for every z . The same holds for any weighted series $\sum_{\mathfrak{q} \in T} w_{\mathfrak{q}} \langle N\mathfrak{q} \rangle^{-z}$ with $|w_{\mathfrak{q}}|_p = 1$ — for instance $w_{\mathfrak{q}} = 1 - \chi(\text{Frob}_{\mathfrak{q}})$ for a character χ with $\chi(\text{Frob}_{\mathfrak{q}}) \neq 1$ of order prime to p .

COROLLARY 3.2 (No p -adic phase transition). *There is no p -adic critical exponent, no convergence–divergence dichotomy, and hence no p -adic Kakutani criterion. The mechanism that produces every phase transition in [Ch. 1], [Ch. 2] and Chapter 6 — the competition between the decay of*

$N\mathfrak{p}^{-\beta}$ and the abundance of primes — is archimedean, and it is absent p -adically because all the terms have absolute value 1.

REMARK 3.3 (Why p -adic L -functions look the way they do). Corollary 3.2 is not a peculiarity of Bost–Connes systems. It is the reason p -adic L -functions are never defined by Dirichlet series: Kubota–Leopoldt, Iwasawa [62] and Mazur all construct them from measures on $\mathbb{Z}_p^\times$, or equivalently from power series in $\Lambda = \mathbb{Z}_p[[T]]$, precisely because the naive series has no region of convergence. Any p -adic thermodynamics would have to be built on measures, and would then lack the variational characterization that makes the archimedean theory informative.

4. The obstruction cannot generate the characteristic ideal

PROPOSITION 4.1 (The element cannot generate the characteristic ideal). *Suppose, counterfactually to Proposition 2.1, that $\Xi_p(z, \chi) = \sum_{i=1}^g (1 - \chi_i u_i^z)$ were defined, with u_i 1-units, $\chi_i = \chi(\text{Frob}_{\mathfrak{p}_i})$, and $g = \#\{\mathfrak{p} \mid p\} \leq [K : \mathbb{Q}]$; under $u_i^z = (1 + T)^{c_i}$ it becomes an element of $\Lambda \otimes \mathbb{Z}_p[\chi]$. It cannot generate the characteristic ideal $\text{char}_\Lambda(X) = (p^\mu P(T))$ of the main conjecture [91], for two independent reasons.*

- (1) *Already for $K = \mathbb{Q}$, $g = 1$: if $\chi(p) \neq 1$ has order prime to p then $|1 - \chi(p)(1 + T)^c|_p = |1 - \chi(p)|_p = 1$ on the open unit disc, so the element is a unit of $\Lambda \otimes \mathbb{Z}_p[\chi]$ and generates the unit ideal — while $\text{char}_\Lambda(X_\chi)$ is a proper ideal whenever $\lambda_\chi + \mu_\chi > 0$, which occurs.*
- (2) *In general the element depends only on the g pairs $(\chi(\text{Frob}_{\mathfrak{p}_i}), \langle N\mathfrak{p}_i \rangle)$ — finitely many p -adic parameters attached to the decomposition of p in K — whereas the characteristic ideal encodes the Iwasawa module of the full tower, which is not a function of the decomposition data at p : fields with identical splitting behaviour at p have different λ -invariants.*

We note that the tempting quantitative version — “a sum of g binomials has λ -invariant bounded in terms of g ” — is itself false: already $(1 + T)^{c'} - (1 + T)^c = (1 + T)^c((1 + T)^{c'-c} - 1)$ has $\lambda = p^{v_p(c'-c)}$, unbounded; the failure of the proposal is the wrong dependence, not a degree count.

REMARK 4.2 (And μ). The proposed formula $\mu_p(K) = v_p(\text{Res}_{z=0} \zeta_{p,\text{KMS}}(z))$ has no content, there being no $\zeta_{p,\text{KMS}}$ by Corollary 1.2. We note also that for abelian $K/\mathbb{Q}$ the Ferrero–Washington theorem [43] gives $\mu_p(K) = 0$, so any correct formula must return 0 in that case; a residue at a pole of a nonexistent function does not. The genuine residue statement of p -adic L -theory is Colmez’s formula for the residue of the p -adic zeta function of K at $s = 1$ [20], and what appears there is the p -adic regulator $R_p(K)$ — the Leopoldt quantity of §5 — not μ .

5. What is true

PROPOSITION 5.1 (The partition function is the interpolation factor). *For $S = \{\mathfrak{p} \mid p\}$ the partition function of the localized system is the finite product*

$$\zeta_{K,p}(\beta) = \prod_{\mathfrak{p} \mid p} (1 - N\mathfrak{p}^{-\beta})^{-1},$$

whose inverse, twisted by χ , is exactly the Euler factor removed in the interpolation formula for $L_p(s, \chi)$ recalled in Remark 2.2. This is a precise dictionary entry: p -local Bost–Connes partition function $\leftrightarrow$ p -adic interpolation factor. It is not a dictionary entry for the characteristic ideal.

REMARK 5.2 (Finite S has no thermodynamics). Consistently with this, $S = \{\mathfrak{p} \mid p\}$ is finite, so $\beta_c = 0$ and by the finite-generation exclusion of Chapter 25 (its Remark 1.5) — a finite generating set gives a finite obstruction sum — $\Xi_\beta = \widehat{G}_p$ for every $\beta > 0$ and the chain is constant. The archimedean p -local system has no phase transition, which is what motivated looking for a p -adic one; Corollary 3.2 says the search cannot succeed in that direction.

PROPOSITION 5.3 (Where Leopoldt genuinely enters). *Let M_∞ be the maximal abelian pro- p extension of K unramified outside p . Then*

$$\text{rank}_{\mathbb{Z}_p} \text{Gal}(M_\infty/K) = r_2 + 1 + \delta_p(K),$$

with $\delta_p(K) \geq 0$ the Leopoldt defect, and Leopoldt's conjecture is $\delta_p(K) = 0$, equivalently $R_p(K) \neq 0$. This quantity does control the localized Bost–Connes system, but in the archimedean theory: by Proposition 2.2 and Remark 2.3 of Chapter 3 the isotropy of $\mathcal{G}_S$ off the boundary strata is computed from the closure $\overline{\mathcal{O}_{K,+}^\times}$ inside U_S , and the computation requires a Leopoldt-type condition for general K , being unconditional when $\mathcal{O}_K^\times$ is finite, that is for $K = \mathbb{Q}$ and imaginary quadratic K .

REMARK 5.4. So the intuition behind the proposal is sound — Leopoldt does belong to this story — but its place is the isotropy computation of Chapter 3, not the dimension of a p -adic state space. That the topological principality of $\mathcal{G}_S$ was established there *independently* of Leopoldt (its Theorem 2.5), the strata being nowhere dense regardless, is what keeps the main classification unconditional.

6. Assessment

Proposal	Status	Reference
p -adic KMS states	do not exist: no positivity	Prop. 1.1, Cor. 1.2
$\sigma_z^{(p)}$ on $\{\mathfrak{p} \mid p\}$	undefined: $N\mathfrak{p} \notin \mathbb{Z}_p^\times$	Prop. 2.1
p -adic Dirichlet series	never converge	Thm. 3.1
p -adic β_c , phase transition	do not exist	Cor. 3.2
$(\Xi_p) = \text{char}_\Lambda(X)$	false: wrong dependence	Prop. 4.1
μ from a residue	no content	Rem. 4.2
λ as a count of transitions	no content	Cor. 3.2
$\text{rank Gal}(M_\infty/K) = r_2 + 1 + \delta_p$	true	Prop. 5.3
Leopoldt controls the system	true, but via Chapter 3	Prop. 5.3
$\zeta_{K,p} \leftrightarrow$ interpolation factor	true	Prop. 5.1

The proposal fails at a level below its own statements. Everything in the archimedean theory — KMS states, simplices, phase transitions, obstruction groups — rests on positivity and on the convergence–divergence dichotomy of real Dirichlet series, and neither has a p -adic form. The absence is not a gap in the technology but a fact about $\mathbb{C}_p$: it is not ordered, and its units all have absolute value 1.

That said, the underlying instinct identifies two real connections, and both are worth recording in the form they actually take. The p -local partition function is the interpolation factor of L_p , which is a statement about Euler factors and is exact. And Leopoldt's conjecture does govern part of this programme, namely the isotropy computation of Chapter 3 (its Proposition 2.2 and Remark 2.3), which is where the units enter — consistently with Chapter 30, whose inertia model, built on the meridian direction matched to the global units, was shown to carry no thermodynamics (its Theorem 6.1 and the moral of its Remark 6.3), and with Chapter 25 (its Remark 1.5), where the finiteness of the unit group decided which semigroups are admissible. The units recur; the p -adic states do not exist.

Two questions we would consider worth asking instead. Is there a p -adic analogue of the *measure class* invariant of Corollary 2.4 of Chapter 4 — since that, unlike a state, is not a positivity notion, and since the whole corpus has found the arithmetic to live there? And second, the archimedean p -local system is trivial by Remark 5.2; is there an intermediate localization, S infinite but with the p -adic weights attached as a coefficient system rather than as the dynamics, in

which both structures coexist? That is the closing suggestion of Chapter 28 (its §5), itself pointing at §6 of Chapter 15, and for the same reason: the arithmetic input belongs in the coefficients, not in the flow.

Leopoldt and the p -local Bost–Connes groupoid: the rank of the isotropy, not its triviality, and why $\log_p(N\mathfrak{p})$ vanishes

This chapter is Paper XXIII of the series ([papers/XXIII-leopoldt/](#)).

SUMMARY. Chapter 35 identified two genuine points of contact between the localized Bost–Connes systems and p -adic arithmetic. We develop both and find that each of the three theorems proposed on that basis needs correction, in each case to a statement that is true and cleaner.

Leopoldt does not make the intermediate isotropy trivial — and the isotropy is not the group the proposal computed. By Proposition 2.2 of Chapter 3, the isotropy at the stratum Y_T is the *divisor image* $\text{div}(\overline{K_{S,+}^\times} \cap \mathbb{I}_T)$, and we identify it exactly: $\text{Iso}(Y_T) \cong \ker(\varphi : L_T \rightarrow U_{T^c}/\pi_{T^c}(\overline{\mathcal{O}_{K,+}^\times}))$, where L_T is the rank- $|T|$ lattice of T -supported principal divisors with totally positive generator and π_{T^c} is the projection of the closed unit group to the complementary factors. The group $\overline{\mathcal{O}_{K,+}^\times} \cap \prod_{\mathfrak{p} \in T} \mathcal{O}_{\mathfrak{p}}^\times$ that the proposal analyzed is the *kernel* of the divisor map — it contributes the identity ideal — though its rank, $(r - \delta_p(K)) - \text{rank } \pi_{T^c}$, is exactly what budgets $\text{rank } \pi_{T^c}$. The consequence is stronger than the proposed correction: whenever $d_{T^c} = \sum_{\mathfrak{p} \notin T} [K_{\mathfrak{p}} : \mathbb{Q}_p] = 1$ and $r \geq 1$, the image of π_{T^c} is open — a single coordinate of an infinite-order unit has nonzero logarithm — so the quotient is finite and $\text{Iso}(Y_T)$ has *full rank* $|T|$. Already for a real quadratic field with p split the intermediate isotropy is nontrivial, unconditionally. The proposed equivalence “Leopoldt $\iff$ trivial isotropy” is therefore false; Leopoldt enters only as the budget $\text{rank } \pi_{T^c} \leq r - \delta_p(K)$.

The proposed $\mathcal{L}$ -invariant is identically zero. The derivative computation is correct: $E_p(\beta) = \prod_{\mathfrak{p}|p} (1 - N\mathfrak{p}^{-\beta})$ vanishes to order $g = \#\{\mathfrak{p} \mid p\}$ at $\beta = 0$ with $\frac{1}{g!} E_p^{(g)}(0) = \prod_{\mathfrak{p}|p} \log N\mathfrak{p} = (\log p)^g \prod_{\mathfrak{p}} f_{\mathfrak{p}}$. But this is the *real* logarithm. Its p -adic counterpart vanishes: Iwasawa’s branch has $\log_p(p) = 0$, so $\log_p(N\mathfrak{p}) = f_{\mathfrak{p}} \log_p(p) = 0$ for every $\mathfrak{p} \mid p$ and the proposed $\mathcal{L}_p(K) = \prod_{\mathfrak{p}} \log_p(N\mathfrak{p})$ is 0. It cannot be a Fontaine–Mazur invariant.

The regulator is a dimension, not a volume. On a compact group Haar measure never degenerates, and $\overline{\mathcal{O}_{K,+}^\times}$ is infinite, so the quotient has no informative volume. What is meaningful is

$$\dim\left(U_p/\overline{\mathcal{O}_{K,+}^\times}\right) = [K : \mathbb{Q}] - (r - \delta_p(K)) = r_2 + 1 + \delta_p(K),$$

so that Leopoldt is equivalent to this dimension being exactly $r_2 + 1$ — recovering Proposition 5.3 of Chapter 35.

What survives intact is the order of vanishing: E_p vanishes at $\beta = 0$ to order exactly g , and g is the number of exceptional zeros in the Mazur–Tate–Teitelbaum sense. That part of the analogy is correct and is the one worth keeping.

1. The isotropy of the intermediate strata

Let K be a number field, p a rational prime, $S = \{\mathfrak{p} \mid p\}$ with $g = \#S$, and $U_p = \prod_{\mathfrak{p}|p} \mathcal{O}_{\mathfrak{p}}^\times$. Write $r = r_1 + r_2 - 1$ for the unit rank, $\delta_p(K) \geq 0$ for the Leopoldt defect, so that $\text{rank}_{\mathbb{Z}_p} \overline{\mathcal{O}_{K,+}^\times} = r - \delta_p(K)$ [133] — $\delta_p(K) = 0$ is known for abelian K by Baker–Brumer [15] — and for $T \subseteq S$ put

$d_{T^c} = \sum_{\mathfrak{p} \notin T} [K_{\mathfrak{p}} : \mathbb{Q}_p]$ and $\pi_{T^c} : \overline{\mathcal{O}_{K,+}^{\times}} \rightarrow \prod_{\mathfrak{p} \notin T} \mathcal{O}_{\mathfrak{p}}^{\times}$ for the coordinate projection. The framework is that of [Ch].

PROPOSITION 1.1 (Kernel and projection ranks). *The group $\overline{\mathcal{O}_{K,+}^{\times}} \cap \prod_{\mathfrak{p} \in T} \mathcal{O}_{\mathfrak{p}}^{\times} = \ker \pi_{T^c}$ satisfies*

$$(1.1) \quad \text{rank}_{\mathbb{Z}_p} \ker \pi_{T^c} = (r - \delta_p(K)) - \text{rank } \pi_{T^c} \geq \max(0, r - \delta_p(K) - d_{T^c}),$$

with equality if and only if π_{T^c} has maximal rank $\min(r - \delta_p, d_{T^c})$. Moreover $\text{rank } \pi_{T^c} \geq 1$ whenever $r - \delta_p \geq 1$ and $T^c \neq \emptyset$: a coordinate of an infinite-order unit is an infinite-order element of $\mathcal{O}_{\mathfrak{p}'}^{\times}$, its logarithm being nonzero since $K \hookrightarrow K_{\mathfrak{p}'}$ is injective. We emphasize that this kernel is not the isotropy of the stratum: it is the divisor-zero part, mapping to the identity ideal.

PROOF. The p -adic logarithm embeds $\overline{\mathcal{O}_{K,+}^{\times}}$, up to finite index, as a $\mathbb{Z}_p$ -lattice of rank $r - \delta_p$ in $\bigoplus_{\mathfrak{p}|p} K_{\mathfrak{p}}$, of total dimension $[K : \mathbb{Q}]$; the kernel of the coordinate projection onto $\bigoplus_{\mathfrak{p} \notin T} K_{\mathfrak{p}}$, of dimension d_{T^c} , has rank given by rank-nullity, which is (1.1). $\square$

THEOREM 1.2 (The isotropy, identified; Leopoldt does not give triviality). *Let $T \subsetneq S$ and let $L_T \subseteq \mathbb{Z}^T$ be the finite-index sublattice of exponent vectors a with $\prod_{\mathfrak{p} \in T} \mathfrak{p}^{a_{\mathfrak{p}}}$ principal with totally positive generator α_a . By Proposition 2.2 of Chapter 3, $\text{Iso}(y) = \text{div}(K_{S,+}^{\times} \cap \mathbb{I}_T)$ for $y \in Y_T$, and this equals, exactly,*

$$\text{Iso}(Y_T) \cong \ker \left(\varphi : L_T \longrightarrow \prod_{\mathfrak{p} \notin T} \mathcal{O}_{\mathfrak{p}}^{\times} / \pi_{T^c}(\overline{\mathcal{O}_{K,+}^{\times}}) \right), \quad \varphi(a) = \alpha_a|_{T^c},$$

well defined since a change of generator moves $\alpha_a|_{T^c}$ by $\pi_{T^c}(\overline{\mathcal{O}_{K,+}^{\times}})$, and with $\pi_{T^c}(\overline{\mathcal{O}_{K,+}^{\times}})$ closed, the closed unit group being compact. Consequently:

- (1) *if $\mathcal{O}_{K,+}^{\times}$ is finite, $\varphi(a)$ torsion forces α_a to equal a root of unity in some completion, hence in K , so $a = 0$: the isotropy is trivial — Chapter 3's unconditional case;*
- (2) *if $r \geq 1$ and $d_{T^c} = 1$, the image of π_{T^c} is open in the one-dimensional $\mathcal{O}_{\mathfrak{p}'}^{\times}$ (Proposition 1.1, last part: a nonzero $\mathbb{Z}_p$ -multiple of a nonzero logarithm is open), the quotient is finite, and $\text{Iso}(Y_T)$ has full rank $|T|$. In particular, for K real quadratic with p split and T one of the two primes, the isotropy is infinite — unconditionally; for K totally real of degree n with p split completely and $|T^c| = 1$ it has rank $n - 1$.*

The proposed equivalence

$$\text{Leopoldt} \iff \text{Iso}(y) = \{1\} \text{ for all } T \subsetneq S$$

is therefore false: triviality tracks the finiteness of the unit group, not Leopoldt, and fails under Leopoldt as soon as $r \geq 1$ (with a small complementary block).

PROOF. For $a \in \ker \varphi$ pick $\beta \in \overline{\mathcal{O}_{K,+}^{\times}}$ with $\pi_{T^c}(\beta) = \alpha_a|_{T^c}$; then $\alpha_a \beta^{-1} \in \overline{K_{S,+}^{\times}} \cap \mathbb{I}_T$ has divisor a , so $a \in \text{Iso}$. Conversely if $\gamma \in \overline{K_{S,+}^{\times}} \cap \mathbb{I}_T$ has divisor a , approximate γ by totally positive S -units γ_n ; the divisor map is locally constant, so $v(\gamma_n) = a$ eventually, whence $\gamma_n = \alpha_a u_n$ with $u_n \in \mathcal{O}_{K,+}^{\times}$ (times torsion), and $\pi_{T^c}(u_n) \rightarrow \alpha_a|_{T^c}^{-1}$; the image $\pi_{T^c}(\overline{\mathcal{O}_{K,+}^{\times}})$ is compact, hence closed, so $\alpha_a|_{T^c}^{-1}$ lies in it and $\varphi(a) = 0$. Items (1) and (2) follow as indicated; for the rank in (2), L_T has rank $|T|$ and the quotient is finite, so $\ker \varphi$ has finite index in L_T . $\square$

REMARK 1.3 (Numerical table). With p split completely and $|T^c| = 1$, so $|T| = g - 1 = n - 1$ and $d_{T^c} = 1$, the corrected isotropy rank is:

K	r	$ T $	kernel rank (1.1)	rank Iso(Y_T)
$\mathbb{Q}$	0	0	0	0
imaginary quadratic	0	1	0	0
real quadratic	1	1	0	1
real cubic	2	2	1	2
real quartic	3	3	2	3
real quintic	4	4	3	4

The last column is Theorem 1.2(2) (full rank $|T|$ as soon as $r \geq 1$), unconditional; the proposal's values were the kernel column, which is not the isotropy. Triviality holds exactly in the first two rows — the finite-unit-group cases, which are precisely those in which Chapter 3 (its Proposition 2.2 and Remark 2.3) is unconditional.

REMARK 1.4 (Nothing in the classification depends on this). The isotropy on the strata does not enter the main results. Chapter 3 established topological principality of $\mathcal{G}_S$ *directly* (its Theorem 2.5), the strata being nowhere dense regardless of their isotropy, which is what Renault's machinery requires. So Theorem 1.2 corrects a proposed characterization of Leopoldt without disturbing anything downstream.

REMARK 1.5 (The residual condition, and where Leopoldt sits). For $d_{T^c} \geq 2$ the finiteness of the quotient in Theorem 1.2 requires π_{T^c} to have *open* image, i.e. rank d_{T^c} — a relative nondegeneracy of the regulator matrix restricted to the columns of T^c , automatic for $d_{T^c} = 1$ but in general neither implied by nor implying Leopoldt, which is the single statement that the full $r \times r$ p -adic regulator is nonzero. Leopoldt enters only as the budget: all projection ranks are at most $r - \delta_p(K)$, so a positive defect can only *lower* rank π_{T^c} and thereby raise the codimension of the image.

2. The exceptional Euler factor

PROPOSITION 2.1 (Order of vanishing and leading term). *Let $E_p(\beta) = \zeta_{K,p}(\beta)^{-1} = \prod_{\mathfrak{p}|p} (1 - N\mathfrak{p}^{-\beta})$. Then E_p vanishes at $\beta = 0$ to order exactly $g = \#\{\mathfrak{p} \mid p\}$, and*

$$(2.1) \quad \frac{1}{g!} \frac{d^g}{d\beta^g} E_p(\beta) \Big|_{\beta=0} = \prod_{\mathfrak{p}|p} \log N\mathfrak{p} = (\log p)^g \prod_{\mathfrak{p}|p} f_{\mathfrak{p}},$$

$f_{\mathfrak{p}}$ being the residue degrees.

PROOF. $1 - N\mathfrak{p}^{-\beta} = 1 - e^{-\beta \log N\mathfrak{p}} = \beta \log N\mathfrak{p} + O(\beta^2)$, and $N\mathfrak{p} = p^{f_{\mathfrak{p}}}$. □

THEOREM 2.2 (The proposed $\mathcal{L}$ -invariant vanishes). *With Iwasawa's branch of the p -adic logarithm, normalized by $\log_p(p) = 0$, one has*

$$\log_p(N\mathfrak{p}) = \log_p(p^{f_{\mathfrak{p}}}) = f_{\mathfrak{p}} \log_p(p) = 0 \quad \text{for every } \mathfrak{p} \mid p,$$

so $\prod_{\mathfrak{p}|p} \log_p(N\mathfrak{p}) = 0$. The quantity proposed as $\mathcal{L}_p(K)$ is identically zero and cannot be a Fontaine–Mazur or Mazur–Tate–Teitelbaum $\mathcal{L}$ -invariant [89, 49]: those are nonzero in the proved cases — for split multiplicative elliptic curves $\mathcal{L}_p = \log_p(q_E)/\text{ord}_p(q_E) \neq 0$ — and conjecturally always.

REMARK 2.3 (Real versus p -adic). The confusion is between two logarithms. Formula (2.1) is a computation with the *real* logarithm, coming from the archimedean dynamics $\sigma_t(\mu_a) = N\mathfrak{a}^{it}\mu_a$, and it is correct; it evaluates to $(\log p)^g \prod_{\mathfrak{p}|p} f_{\mathfrak{p}}$, for instance 2.5903 for $g = 2$, $f = (1, 1)$, $p = 5$, and 4.1689 for $g = 3$, $f = (1, 1, 1)$, $p = 5$. The p -adic logarithm of the same quantity is 0, because $N\mathfrak{p}$

is a power of p and $\log_p$ is normalized to kill p . The two cannot be identified, and the vanishing is not an accident: it is the same fact as Proposition 2.1 of Chapter 35, that $N\mathfrak{p}$ is not a p -adic unit.

REMARK 2.4 (What is right about the analogy). The order of vanishing is correct and is worth stating on its own. E_p vanishes to order g at $\beta = 0$, and g is exactly the number of primes above p at which the Euler factor of $L_p(s, \chi)$ is removed, hence the number of exceptional zeros in the Mazur–Tate–Teitelbaum sense [89] for the trivial character. That is a genuine dictionary entry, and it is the one supported by Proposition 5.1 of Chapter 35: the p -local partition function corresponds to the interpolation factor. The leading coefficient of the archimedean E_p is not the $\mathcal{L}$ -invariant of that factor.

3. The regulator is a dimension, not a volume

PROPOSITION 3.1 (The volume statement is not well formed). G_p is a compact group, so its Haar measure exists, is unique up to scale and is never degenerate; normalizing $m_{G_p}(G_p) = 1$ makes any statement about its total mass vacuous. Moreover $\overline{\mathcal{O}_{K,+}^\times}$ is infinite whenever $r > \delta_p(K)$, so the index $[U_p : \overline{\mathcal{O}_{K,+}^\times}]$ is infinite and the naive relative volume is 0 independently of $R_p(K)$. No equivalence with $R_p(K) \neq 0$ can be extracted from a volume.

THEOREM 3.2 (The correct statement).

$$\dim(U_p / \overline{\mathcal{O}_{K,+}^\times}) = [K : \mathbb{Q}] - (r - \delta_p(K)) = r_2 + 1 + \delta_p(K),$$

as p -adic analytic groups. Consequently

$$\text{Leopoldt for } K \text{ at } p \iff R_p(K) \neq 0 \iff \dim(U_p / \overline{\mathcal{O}_{K,+}^\times}) = r_2 + 1.$$

PROOF. $\dim U_p = \sum_{\mathfrak{p}|p} [K_{\mathfrak{p}} : \mathbb{Q}_p] = [K : \mathbb{Q}] = r_1 + 2r_2$ and $\dim \overline{\mathcal{O}_{K,+}^\times} = r - \delta_p$ with $r = r_1 + r_2 - 1$; subtract. The equivalence is the definition of the Leopoldt defect together with the fact that $\delta_p = 0$ iff the p -adic regulator is nonzero. $\square$

EXAMPLE 3.3. Under Leopoldt: $\dim = 1$ for $\mathbb{Q}$, for real quadratic, for real cubic and for real quartic fields; $\dim = 2$ for imaginary quadratic and for complex cubic fields. In each case $\dim = r_2 + 1$.

COROLLARY 3.4 (Agreement with Chapter 35). *Theorem 3.2 is the computation*

$$\text{rank}_{\mathbb{Z}_p} \text{Gal}(M_\infty/K) = r_2 + 1 + \delta_p(K)$$

of Chapter 35 (its Proposition 5.3), transported through class field theory: $\text{Gal}(M_\infty/K)$ is the quotient of U_p by the closure of the global units, up to a finite group. The intended statement was therefore correct in substance and misstated in form, volume for dimension.

4. Assessment

Proposal	Status	Reference
$\text{Iso}(Y_T) \cong \ker \varphi$ (divisor image)	exact identification	Thm. 1.2
full rank $ T $ for $r \geq 1$, $d_{T^c} = 1$	unconditional	Thm. 1.2
Leopoldt $\iff$ trivial isotropy	false	Thm. 1.2
topological principality affected	no	Rem. 1.4
$\text{ord}_{\beta=0} E_p = g$	true	Prop. 2.1
$\frac{1}{g!} E_p^{(g)}(0) = \prod \log N\mathfrak{p}$	true, real logarithm	Prop. 2.1
$\mathcal{L}_p(K) = \prod \log_p(N\mathfrak{p})$	identically 0	Thm. 2.2
$g = \#\{\text{exceptional zeros}\}$	true; keep this	Rem. 2.4
$\text{Vol}_p(U_p/\mathcal{O}^\times) \asymp R_p _p$	not well formed	Prop. 3.1
$\dim = r_2 + 1 + \delta_p$	true; the right form	Thm. 3.2

The two bridges identified in Chapter 35 are real, and both survive; what fails is the attempt to make each carry a theorem it cannot. On the unit side, Leopoldt controls the *rank* of the closed unit group and hence the *dimension* of the symmetry group, and that is the whole of its role: it does not trivialize the isotropy of the strata, and the strata do not matter for the classification in any case. On the Euler side, the p -local partition function does correspond to the interpolation factor, and the order of vanishing does count the exceptional zeros; but the leading coefficient is an archimedean quantity, and its p -adic shadow is zero because $N\mathfrak{p}$ is a power of p .

That last point is worth isolating, because it recurs. In Proposition 2.1 of Chapter 35 the p -adic dynamics failed to exist at $\mathfrak{p} \mid p$ because $N\mathfrak{p} \notin \mathbb{Z}_p^\times$; here the $\mathcal{L}$ -invariant vanishes for the same reason. Every attempt to run p -adic analysis on the norms of the primes above p meets the same wall, and it is not a technical one: those norms are precisely the elements the p -adic theory is built to discard.

Two questions. Is there a version of the isotropy statement that does characterize Leopoldt — one would want a condition on *all* T simultaneously, and since by Proposition 1.1 every projection rank is budgeted by $r - \delta_p(K)$, some aggregate of the image codimensions does see δ_p and might work. And, given Remark 2.4, is the correspondence between the order of vanishing of E_p and the exceptional-zero count functorial enough to extend to nontrivial characters χ , where the order becomes $\#\{\mathfrak{p} \mid p : \chi(\mathfrak{p}) = 1\}$ rather than g ?

Part 8

Families of primes

The statistical mechanics of sieve-theoretic primes: the critical spectrum, the log-power dichotomy, and two families that are never unique

This chapter is Paper XXIV of the series (papers/XXIV-sieve/).

SUMMARY. We compute the thermodynamics of the classical sieve-theoretic prime families for the localized Bost–Connes systems of [Ch. 1]. The critical exponents proposed are all correct, the open/closed dichotomy has a clean and complete explanation, and the account of high-temperature uniqueness is wrong for two of the six families, for an elementary reason that had not been noticed.

Critical exponents. If $\pi_S(x) \asymp x^\theta / \log^k x$ then Abel summation gives $\beta_c(S) = \theta$, and at $\beta = \theta$ the series behaves like $\int^x dt / (t \log^k t)$. Hence $\beta_c = 1$ for twin, Sophie Germain and Chen primes; $3/4$ for Friedlander–Iwaniec $a^2 + b^4$; $2/3$ for Heath-Brown $a^3 + 2b^3$; $1/2$ for Landau $n^2 + 1$; and $1/c$ for Piatetski-Shapiro $[n^c]$, a continuous spectrum. All six values are as proposed.

The dichotomy is exactly the log-power. The critical series converges iff $k > 1$, so the low-temperature phase is closed precisely for the families defined by *two* simultaneous prime conditions — twin, Sophie Germain, Chen, where $k = 2$ — and open for those defined by a single condition on a polynomial or fractional sequence, where $k = 1$. This is a complete answer, and it identifies the source of closedness as the second prime condition rather than as anything about the sieve.

Two of the six are congruence-obstructed, and are never unique — and which two is decided by a kernel condition, not by confinement alone. All four congruence-trapped families sit in a single residue class: twin and Sophie Germain primes in $2 \bmod 3$, Landau and the odd Friedlander–Iwaniec primes in $1 \bmod 4$. But confinement obstructs only when the occupied class lies in the *kernel* of the character. For Landau and Friedlander–Iwaniec the class is 1, the kernel itself, so the nontrivial character mod 4 has *finite* detecting set and lies in Ξ_β for every β : a permanent obstruction in the sense of [Ch. 1, §4.1], and **for these two families the KMS $_\beta$ state is never unique, at any temperature.** For twin and Sophie Germain primes the opposite happens: $\chi(2) = -1$ for the character mod 3, so the detecting set is *cofinite* in S and the character is detected at full strength — the class 2 generates $(\mathbb{Z}/3)^\times$, and by [Ch. 1, Lem. 5.6] the admissible classes generate $(\mathbb{Z}/M\mathbb{Z})^\times$ for every modulus, so no congruence obstructs these families at all. Verified numerically to $2 \cdot 10^5$ (single classes as stated).

Twin, Sophie Germain, Chen, Heath-Brown and Piatetski-Shapiro primes are therefore unobstructed, and for them the equidistribution criterion is the right one wherever the critical series diverges.

The finding is consistent with [Ch. 1, §5]: Theorem 5.5(5) there, with Lemma 5.6, already established that the Sophie Germain family is confined to no proper kernel, and the present analysis confirms it — the mod-3 confinement, far from obstructing, feeds the detecting series in full.

1. The critical exponent

The framework and notation are those of [Ch.

PROPOSITION 1.1 (Abel summation). *Let $S \subseteq P$ satisfy $\pi_S(x) \asymp x^\theta / \log^k x$ with $\theta \in (0, 1]$, $k \geq 0$. Then*

$$(1.1) \quad \sum_{p \in S, p \leq x} p^{-\beta} \asymp \int_2^x \frac{t^{\theta-1-\beta}}{\log^k t} dt,$$

so that $\beta_c(S) = \theta$, and at the critical value $\beta = \theta$ the right side is $\int_2^x dt/(t \log^k t)$, which converges as $x \rightarrow \infty$ if and only if $k > 1$.

PROOF. The counting function has density $\asymp \theta t^{\theta-1}/\log^k t$; substitute this into the Stieltjes integral $\sum_{p \leq x} p^{-\beta} = \int_2^x t^{-\beta} d\pi_S(t)$. For $\beta < \theta$ the integral is $\asymp x^{\theta-\beta}/\log^k x \rightarrow \infty$; for $\beta > \theta$ it converges. At $\beta = \theta$, $\int dt/(t \log^k t) = [\log^{1-k} t/(1-k)]$ for $k \neq 1$ and $\log \log t$ for $k = 1$. $\square$

THEOREM 1.2 (The spectrum).

S	θ	k	$\beta_c(S)$	phase
<i>twin</i> $\{p : p+2 \in P\}$	1	2	1	<i>closed</i> $[1, \infty)$
<i>Sophie Germain</i> $\{p : 2p+1 \in P\}$	1	2	1	<i>closed</i> $[1, \infty)$
<i>Chen</i> $\{p : p+2 \in P_2\}$	1	2	1	<i>closed</i> $[1, \infty)$
<i>Friedlander–Iwaniec</i> $a^2 + b^4$	3/4	1	3/4	<i>open</i> $(3/4, \infty)$
<i>Heath–Brown</i> $a^3 + 2b^3$	2/3	1	2/3	<i>open</i> $(2/3, \infty)$
<i>Landau</i> $n^2 + 1$	1/2	1	1/2	<i>open</i> $(1/2, \infty)$
<i>Piatetski–Shapiro</i> $[n^c]$	1/c	1	1/c	<i>open</i> $(1/c, \infty)$

PROOF. Proposition 1.1 with the counting functions: Brun’s sieve [52] gives the upper bounds with $k = 2$ for the shifted families, and Chen’s theorem [17] the matching lower bound for Chen primes; the Friedlander–Iwaniec [44] and Heath–Brown [55] asymptotics are theorems with $k = 1$; the Landau asymptotic with $k = 1$ is conjectural (Bateman–Horn); the Piatetski–Shapiro asymptotic with $k = 1$ is a theorem for $c < 243/205$ by Rivat–Sargos [114]. $\square$

REMARK 1.3 (The dichotomy is the second prime condition). By Proposition 1.1 the phase is closed exactly when $k > 1$, and $k = 2$ occurs exactly for the families imposing *two* simultaneous primality conditions — p and $p+2$, p and $2p+1$, p and $p+2 \in P_2$. The polynomial and fractional families impose one condition and have $k = 1$. So the closedness observed for Sophie Germain primes in [Ch. 1, Thm. 5.5] is not a feature of the sieve but of the second condition, and it transfers verbatim to twins and to Chen primes, the latter unconditionally since Chen’s theorem supplies the lower bound that the twin case still lacks.

2. Congruence confinement versus obstruction

LEMMA 2.1 (Elementary congruences). (1) If $p > 3$ and $p+2$ is prime then $p \equiv 2 \pmod{3}$.

(2) If $p > 3$ and $2p+1$ is prime then $p \equiv 2 \pmod{3}$.

(3) If $p = n^2 + 1$ is prime and $p > 2$ then n is even and $p \equiv 1 \pmod{4}$.

(4) Squares and fourth powers are $\equiv 0, 1 \pmod{4}$, so $a^2 + b^4 \equiv 0, 1, 2 \pmod{4}$; hence an odd prime of the form $a^2 + b^4$ satisfies $p \equiv 1 \pmod{4}$.

PROOF. (1) If $p \equiv 1$ then $3 \mid p+2$. (2) If $p \equiv 1$ then $3 \mid 2p+1$. (3) If n is odd then $n^2 + 1 \equiv 2 \pmod{4}$, forcing $p = 2$; so n is even and $p = 4m^2 + 1$. (4) Direct. $\square$

THEOREM 2.2 (Permanent obstruction — and its exact scope). Let χ_4 be the nontrivial character modulo 4 and χ_3 the nontrivial character modulo 3; for a Dirichlet character, $|1 - \chi(\sigma_p)| = |1 - \chi(p)|$, the Frobenius convention of [Ch. 1, (1.1)] inverting $\chi(p)$ without changing the modulus of the summand. Then:

(1) For S the Landau primes $n^2 + 1$ or the Friedlander–Iwaniec primes $a^2 + b^4$, the detecting set $D_{\chi_4} = \{p \in S : \chi_4(p) \neq 1\}$ is finite, since by Lemma 2.1(3)–(4) all odd members lie in the class 1 mod 4, which is $\ker \chi_4$. Hence $\sum_{D_{\chi_4}} p^{-\beta} < \infty$ for every $\beta > 0$, $\chi_4 \in \Xi_\beta(\mathbb{Q}, S)$ for every β , the KMS_β simplex has at least two extreme points at every temperature, and the state is never unique.

- (2) For S the twin or Sophie Germain primes the same confinement has the opposite effect: by Lemma 2.1(1)–(2) all members $p > 3$ lie in the class $2 \bmod 3$, and $\chi_3(2) = -1 \neq 1$, so D_{χ_3} is cofinite in S and $\sum_{D_{\chi_3}} p^{-\beta} = \sum_{p \in S} p^{-\beta} + O(1)$: the character is detected at full strength. More generally the class 2 generates $(\mathbb{Z}/3)^\times$, and by [Ch. 1, Lem. 5.6] the admissible classes of these families generate $(\mathbb{Z}/M\mathbb{Z})^\times$ for every modulus M , so no Dirichlet character has finite detecting set: there is no permanent congruence obstruction for twin or Sophie Germain primes.

PROOF. A character trivial on the single occupied class has finite detecting set — that is (1), where the class is the kernel itself — and a character nontrivial on it detects all but finitely many members — that is the first half of (2). For the last claim of (2), a character with finite detecting set would be trivial on all but finitely many of S , hence on the subgroup generated by the occupied classes, which is everything by [Ch. 1, Lem. 5.6]. The never-uniqueness in (1) is [Ch. 1, Thm. 3.9], uniqueness requiring $\sum_{D_\chi} p^{-\beta} = \infty$ for every nontrivial χ ; the obstruction is permanent in the sense of [Ch. 1, §4.1]. $\square$

REMARK 2.3 (Numerical verification). Computed to $2 \cdot 10^5$, counting $p > 3$ for the mod-3 rows, odd p for Landau, all representable primes for Friedlander–Iwaniec, $p > 3$ with $p + 2 \in P_2$ for Chen, positive a, b for Heath-Brown (the range of [55]), and the distinct primes $\lfloor n^c \rfloor \leq 2 \cdot 10^5$ for Piatetski-Shapiro:

S	$\#S$	M	coprime classes occupied
twin (smaller member)	2159	3	$\{2\}$ of $\{1, 2\}$
Sophie Germain	2056	3	$\{2\}$ of $\{1, 2\}$
Landau $n^2 + 1$	63	4	$\{1\}$ of $\{1, 3\}$
Friedlander–Iwaniec	938	4	$\{1\}$ of $\{1, 3\}$
Chen	7523	3, 4	both classes
Heath-Brown	289	4, 9	both mod 4; $\{1, 2, 7, 8\} \bmod 9$
Piatetski-Shapiro $c = 1.1$	6040	3, 4	both classes

The single-class rows now illustrate *both* halves of Theorem 2.2: for Landau and Friedlander–Iwaniec the occupied class is the kernel $\{1\}$ (obstruction); for twin and Sophie Germain it is $\{2\}$, on which χ_3 is -1 (full-strength detection). For Heath-Brown modulo 9 the set $\{1, 2, 7, 8\}$ is not contained in a proper subgroup of $(\mathbb{Z}/9)^\times$: a character trivial on it must have $\chi(2) = 1$ and hence be trivial, since 2 generates. So Heath-Brown is not obstructed either.

COROLLARY 2.4 (The proposed criterion is not the right one). *The claim that high-temperature uniqueness is equivalent to equidistribution across admissible coprime classes is correct in one direction — equidistribution does force $\sum_{D_\chi} p^{-\beta} \asymp \zeta_S(\beta) = \infty$ for $\beta < \beta_c$, hence uniqueness — but it is not necessary: twin and Sophie Germain primes are as far from equidistributed as possible, confined to one class, and are nevertheless unobstructed, the confining class generating the group. What decides is not equidistribution but [Ch. 1, Thm. 5.5(5)]: confinement to a proper kernel. That happens for Landau and Friedlander–Iwaniec, and for no other family here.*

REMARK 2.5 (Consequence for [Ch. 1].) [Ch. 1, Thm. 5.5] establishes for the Sophie Germain family the Brun-summability, the closed low-temperature phase, and — in its part (5), with [Ch. 1, Lem. 5.6] — the uniqueness criterion in the divergent range. All of it stands, and Theorem 2.2(2) is its confirmation from the congruence side: the mod-3 confinement feeds the detecting series rather than starving it. The correct picture for Sophie Germain primes is therefore: no permanent obstruction anywhere; uniqueness wherever the series diverges (conditional on the lower bound of [Ch. 1, Thm. 5.5(2)]); and full symmetry breaking, $\Xi_\beta = \widehat{G}_S$, on the closed phase $[1, \infty)$ where ζ_S converges.

3. The continuous spectrum

THEOREM 3.1 (Piatetski-Shapiro). *For $1 < c < 243/205$ the asymptotic $\pi_{\text{PS}(c)}(x) \sim x^{1/c}/\log x$ is a theorem (Piatetski-Shapiro for $c < 12/11$, Heath-Brown for $c < 755/662$, Rivat-Sargos for $c < 243/205$). Hence for those c the system $\mathcal{A}_{\mathbb{Q}, \text{PS}(c)}$ has $\beta_c = 1/c \in (205/243, 1)$, unconditionally, with an open low-temperature phase; and $c \mapsto 1/c$ traces a continuous family of critical temperatures.*

REMARK 3.2 (The range is narrow). $205/243 = 0.8436$, so the unconditional range of critical temperatures is $\beta_c \in (0.8436, 1)$, an interval of length about 0.156. The full conjectural range $(0, 1)$ requires the asymptotic for all $c > 1$, which is open. It is worth stating the unconditional interval numerically, since “a continuous spectrum on $(0, 1)$ ” overstates what is available.

4. Assessment

Claim	Status	Reference
$\beta_c = 1$ for twin, SG, Chen	true	Thm. 1.2
$\beta_c = 3/4, 2/3, 1/2$ for FI, HB, Landau	true	Thm. 1.2
$\beta_c = 1/c$ for Piatetski-Shapiro	true	Thm. 3.1
closed iff Brun-summable	true; and iff $k > 1$	Prop. 1.1
closedness comes from the sieve	no: from the 2nd prime condition	Rem. 1.3
uniqueness $\iff$ equidistribution	no: kernel confinement decides	Cor. 2.4
Landau, FI: unique somewhere	false: never unique	Thm. 2.2(1)
twin, SG obstructed	no: class 2 generates	Thm. 2.2(2)
Chen, HB, PS obstructed	no	Rem. 2.3
PS spectrum covers $(0, 1)$	unconditionally only $(0.8436, 1)$	Rem. 3.2

Two of the three proposed pieces are correct and one is not, and the failure is elementary rather than deep. The critical exponents are exactly the counting exponents; the open/closed dichotomy is exactly the power of the logarithm, hence exactly the number of simultaneous prime conditions. Four of the six families are trapped in a single residue class by a congruence that has nothing to do with sieves — but confinement obstructs only when the class sits in the kernel. That happens for Landau and Friedlander–Iwaniec, whose KMS states are never unique by [Ch. 1, Thm. 3.9]; for twin and Sophie Germain primes the same confinement is detection at full strength, and there is no obstruction at all.

The moral is worth recording because it is the opposite of the usual one in this corpus. Here the analytic input — the hard sieve theorems — is not where the difficulty lies; those theorems give the critical exponents and the phase types cleanly. What decides the qualitative behaviour is a congruence available to a first-year student, read correctly: Ξ_β measures how a set of primes sits inside residue classes *relative to the kernels of characters*. A set confined to the class 1 is maximally obstructed no matter how delicate its density; a set confined to a generating class is not obstructed at all.

Two questions. For the closed-phase families — twin, Sophie Germain, Chen, all now known unobstructed — high-temperature uniqueness coexists with the closed phase as soon as the divergence of the critical series below β_c is available, which is unconditional for Chen (Chen’s theorem [17] supplies the lower bound) and conditional for the other two; can the Chen case be carried out in full, making it the first family in the corpus with both a closed phase and proven high-temperature uniqueness? And second, for the two obstructed families the permanent obstruction has order 2; is the full Ξ_β for, say, the Landau primes equal to that order-two group below β_c , or is it larger?

Sato–Tate families in localized Bost–Connes systems: open phase, no congruence obstruction, and the symmetric power expansion

This chapter is Paper XXV of the series ([papers/XXV-sato-tate/](#)).

SUMMARY. Let $E/\mathbb{Q}$ be a non-CM elliptic curve, $a_p = 2\sqrt{p}\cos\theta_p$, and $S_{[a,b]} = \{p : \theta_p \in [a,b]\}$ for $0 \leq a < b \leq \pi$. We settle the three questions posed for the localized system $\mathcal{A}_{\mathbb{Q}, S_{[a,b]}}$.

The phase is open, and necessarily so. By Sato–Tate $S_{[a,b]}$ has positive natural density $\delta = \mu_{\text{ST}}([a,b]) = \frac{1}{\pi}[(b-a) - \frac{1}{2}(\sin 2b - \sin 2a)]$ among the primes, so $\pi_S(x) \sim \delta x / \log x$: the exponent is $\theta = 1$ and the logarithmic power is $k = 1$. By Proposition 1.1 of Chapter 37 the critical series is then $\asymp \delta \log \log x$, divergent, and the symmetry-broken phase is $(1, \infty)$, open. The precise asymptotic is

$$\zeta_{S_{[a,b]}}(\beta) = (\delta + o(1)) \log \frac{1}{\beta-1} \quad (\beta \rightarrow 1^+),$$

with $o(1)$ replaced by $O(1/\log \frac{1}{\beta-1})$ under any effective Sato–Tate error term $O(x/\log^{1+\varepsilon} x)$. This is not an accident of the interval: *a set of positive density can never have a closed phase*, closedness requiring $k \geq 2$, hence density zero with an extra logarithm.

There is no congruence obstruction. The residue $p \bmod m$ is $\det \bar{\rho}_{E,m}(\text{Frob}_p)$, the cyclotomic character, while θ_p is governed by the trace; Serre’s open image theorem makes the mod- m image open, and joint equidistribution of $(\theta_p, p \bmod m)$ follows from the non-vanishing of $L(s, \text{Sym}^k E \otimes \chi)$ on $\Re s = 1$, now unconditional for non-CM $E/\mathbb{Q}$ by Newton–Thorne together with Shahidi. Hence every nontrivial character has a detecting set of positive density, and **the KMS $_{\beta}$ state is unique for all $0 < \beta \leq 1$** . This is the opposite verdict to Theorem 2.2(1) of Chapter 37, where the Landau and Friedlander–Iwaniec primes are trapped in the kernel class 1 mod 4 and hence never unique — and it is reached by the opposite mechanism to that theorem’s part (2), where the twin and Sophie Germain primes escape obstruction while confined to a single *generating* class. Verified numerically on $y^2 = x^3 + x + 1$ over 6055 primes below $6 \cdot 10^4$: in each of three intervals and each modulus $m \in \{3, 4, 5, 8\}$ every coprime class carries the uniform proportion.

The symmetric powers control the constant, not the transition. Expanding the indicator in the characters $U_k(\cos \theta) = \text{tr Sym}^k$, which are orthonormal for μ_{ST} ,

$$\zeta_{S_{[a,b]}}(s) = \delta \log \frac{1}{s-1} + \sum_{k \geq 1} c_k \log L(s, \text{Sym}^k E) + O(1), \quad c_k = \frac{2}{\pi} \int_a^b \sin((k+1)t) \sin t \, dt.$$

Only the $k = 0$ term, the Riemann zeta function, has a pole at $s = 1$; every Sym^k with $k \geq 1$ is automorphic with $L(1, \text{Sym}^k E) \neq 0$, so contributes $O(1)$. The higher symmetric powers therefore fix the constant in the expansion and the shape of Ξ_{β} , and have no influence whatever on β_c .

We flag one technical point that the L^2 expansion conceals: $\mathbf{1}_{[a,b]}$ is not smooth, so the interchange of $\sum_k$ and $\sum_p$ requires a Beurling–Selberg smoothing, standard in equidistribution but not vacuous.

1. The thermodynamic baseline

The framework is that of [Ch.

PROPOSITION 1.1 (Density). *For $0 \leq a < b \leq \pi$,*

$$\delta := \mu_{\text{ST}}([a, b]) = \frac{2}{\pi} \int_a^b \sin^2 t \, dt = \frac{1}{\pi} \left[(b-a) - \frac{1}{2}(\sin 2b - \sin 2a) \right] > 0.$$

By the Sato–Tate theorem for non-CM elliptic curves over $\mathbb{Q}$ [8], $S_{[a,b]}$ has natural density δ in the primes, so $\pi_S(x) \sim \delta x / \log x$.

THEOREM 1.2 (Open phase). $\beta_c(S_{[a,b]}) = 1$, and

$$(1.1) \quad \zeta_{S_{[a,b]}}(\beta) = \sum_{p \in S_{[a,b]}} p^{-\beta} = (\delta + o(1)) \log \frac{1}{\beta-1} \quad (\beta \rightarrow 1^+),$$

so that $\zeta_S(1) = \infty$: the symmetry-broken phase is the open half-line $(1, \infty)$. If moreover $\pi_S(x) = \delta\pi(x) + O(x/\log^{1+\varepsilon} x)$ for some $\varepsilon > 0$ — an effective Sato–Tate error term — then the $o(1) \log \frac{1}{\beta-1}$ sharpens to $O(1)$.

PROOF. In the notation of Proposition 1.1 of Chapter 37, $\theta = 1$ and $k = 1$, so $\beta_c = \theta = 1$ and the critical series is $\asymp \delta \int_2^x dt/(t \log t) = \delta \log \log x \rightarrow \infty$. For (1.1), write $E(x) = \pi_S(x) - \delta\pi(x) = o(x/\log x)$ and integrate by parts: $\sum_{p \in S} p^{-\beta} = \delta \sum_p p^{-\beta} + \int_2^\infty x^{-\beta} dE(x)$, the error being $\beta \int x^{-\beta-1} E(x) dx = o(\log \frac{1}{\beta-1})$ from $E = o(x/\log x)$, and $O(1)$ when $E \ll x/\log^{1+\varepsilon} x$, since $\int x^{-2} \cdot x/\log^{1+\varepsilon} x dx < \infty$; finally $\delta \sum_p p^{-\beta} = \delta \log \zeta(\beta) + O(1) = \delta \log \frac{1}{\beta-1} + O(1)$. Closedness would require convergence at β_c , hence $k > 1$ by Proposition 1.1 of Chapter 37. $\square$

COROLLARY 1.3 (No positive-density set has a closed phase). *If $S \subseteq P$ has positive natural density then $k = 1$ and the low-temperature phase is open. Closed phases require density zero, and specifically a counting function $\asymp x/\log^k x$ with $k \geq 2$ — which by Remark 1.3 of Chapter 37 means two simultaneous prime conditions. Sato–Tate families, being of positive density, are on the opposite side of that divide from the Sophie Germain family of [Ch. 1, Thm. 5.5].*

EXAMPLE 1.4. $\delta = 0.6090$ for $[\pi/3, 2\pi/3]$, 0.0288 for $[0, \pi/6]$, 0.0775 for $[2.4, \pi]$, 0.5 for $[0, \pi/2]$. Empirical densities for $y^2 = x^3 + x + 1$ over the 6055 good primes below $6 \cdot 10^4$: 0.6206 , 0.0276 , 0.0766 respectively.

2. No congruence obstruction

LEMMA 2.1 (The residue is the determinant). *For $m \geq 1$ and $p \nmid mN_E$, $\det \bar{\rho}_{E,m}(\text{Frob}_p) = p \bmod m$, the mod- m cyclotomic character. The angle θ_p is determined by $\text{tr } \bar{\rho}_{E,\ell}(\text{Frob}_p) = a_p$. Thus the residue class and the Sato–Tate angle are carried by the determinant and the trace of the same representation.*

THEOREM 2.2 (Joint equidistribution). *Let $E/\mathbb{Q}$ be non-CM. For every $m \geq 1$ the pairs $(\theta_p, p \bmod m)$ are equidistributed in $[0, \pi] \times (\mathbb{Z}/m)^\times$ for $\mu_{\text{ST}} \times \text{unif}$. Consequently, for every interval $[a, b]$ and every residue class r coprime to m ,*

$$\#\{p \leq x : \theta_p \in [a, b], p \equiv r \pmod{m}\} \sim \frac{\delta}{\varphi(m)} \cdot \frac{x}{\log x}.$$

PROOF. By Lemma 2.1 and Serre’s open image theorem [118] the mod- m image is open, so the determinant and trace directions are independent in the limit. Analytically, expanding the indicator of $[a, b] \times \{r\}$ in the characters $U_k \otimes \chi$ of $SU(2) \times (\mathbb{Z}/m)^\times$ reduces the statement to the non-vanishing and holomorphy of $L(s, \text{Sym}^k E \otimes \chi)$ on $\Re s = 1$ for all $k \geq 0$ and all Dirichlet $\chi \bmod m$, with a pole only for $k = 0$, χ trivial. Automorphy of $\text{Sym}^k E$ for all k is Newton–Thorne [98]; the non-vanishing on $\Re s = 1$ of the automorphic L -functions and their twists is Jacquet–Shalika [65] and Shahidi [119]. Apply Wiener–Ikehara. $\square$

THEOREM 2.3 (High-temperature uniqueness). *For every non-CM $E/\mathbb{Q}$ and every $[a, b]$ with $\delta > 0$, the system $\mathcal{A}_{\mathbb{Q}, S_{[a, b]}}$ has a unique KMS_β state for $0 < \beta \leq 1$, and its KMS_β simplex is $\text{Prob}(G_S)$ for $\beta > 1$. In particular $S_{[a, b]}$ carries no congruence obstruction in the sense of Theorem 2.2(1) of Chapter 37.*

PROOF. By [Ch. 1, Thm. 3.9] uniqueness at β requires $\sum_{p \in D_\chi} p^{-\beta} = \infty$ for every nontrivial $\chi \in \widehat{G_S}$. For $K = \mathbb{Q}$ such a χ is supported at finitely many places and its detecting set is the union of the classes of $S_{[a, b]}$ outside $\ker \chi$ modulo some m ; by Theorem 2.2 that set has density $\delta(1 - 1/[\cdot]) > 0$ in the primes, so its Dirichlet series diverges for $\beta \leq 1$ by Theorem 1.2. For $\beta > 1$ the whole of $\zeta_S(\beta)$ converges, so $\Xi_\beta = \widehat{G_S}$. $\square$

REMARK 2.4 (Numerical verification). For $y^2 = x^3 + x + 1$ (non-CM), the 6055 good primes below $6 \cdot 10^4$:

$[a, b]$	$m = 3$	$m = 4$	$m = 5$	$m = 8$
$[\pi/3, 2\pi/3]$	.500, .500	.502, .498	.246, .250, .254, .249	.253, .248, .248, .250
$[0, \pi/6]$	.431, .569	.449, .551	.275, .222, .269, .234	.228, .251, .222, .299
$[2.4, \pi]$	.478, .522	.496, .504	.259, .274, .235, .233	.246, .265, .250, .239

against uniform values $1/2$ and $1/4$. Every coprime class is occupied at roughly the uniform proportion; the fluctuations in the two sparse intervals are consistent with the sample sizes (167 and 464 primes). Contrast Remark 2.3 of Chapter 37, where the twin and Sophie Germain families occupy the single class $2 \bmod 3$ — a generating class, so no obstruction — and the Landau and Friedlander–Iwaniec families the single kernel class $1 \bmod 4$, which does obstruct.

REMARK 2.5 (Why this family passes the test). The obstructed families of Chapter 37 — Landau and Friedlander–Iwaniec — are defined by a condition that *involves* the residue, p being a value of a polynomial, so the condition and the residue are not independent, and the dependence lands in the kernel. (The twin and Sophie Germain conditions are also residue-entangled, but land in a generating class and escape.) Here the defining condition involves a_p , and by Lemma 2.1 the residue is the determinant while a_p is the trace of the same Galois representation. Open image is exactly the statement that these two coordinates are independent. The distinction is structural, not a matter of the density being larger.

3. The symmetric power expansion

PROPOSITION 3.1 (Chebyshev coefficients). *The functions $U_k(\cos t) = \sin((k+1)t)/\sin t$, $k \geq 0$, are orthonormal in $L^2(\mu_{\text{ST}})$ and satisfy $U_k(\cos \theta_p) = \text{tr Sym}^k(U_p)$ where $U_p = p^{-1/2}\rho_E(\text{Frob}_p)$. Writing $\mathbf{1}_{[a, b]} = \sum_{k \geq 0} c_k U_k$ in $L^2(\mu_{\text{ST}})$,*

$$c_0 = \delta, \quad c_k = \frac{2}{\pi} \int_a^b \sin((k+1)t) \sin t \, dt = \frac{1}{\pi} \left[\frac{\sin kt}{k} - \frac{\sin(k+2)t}{k+2} \right]_a^b \quad (k \geq 1).$$

THEOREM 3.2 (The expansion). *For $\Re s > 1$,*

$$(3.1) \quad \zeta_{S_{[a, b]}}(s) = \delta \log \frac{1}{s-1} + \sum_{k \geq 1} c_k \log L(s, \text{Sym}^k E) + O(1),$$

the series over k summed by truncation as in Remark 3.3, and every term with $k \geq 1$ is $O(1)$ as $s \rightarrow 1^+$, since $\text{Sym}^k E$ is automorphic (Newton–Thorne [98]) and $L(1, \text{Sym}^k E) \neq 0$ (Shahidi [119]). Hence the pole, the critical exponent $\beta_c = 1$ and the divergence at $\beta = 1$ all come from the $k = 0$ term, the Riemann zeta function; the symmetric powers of higher order determine the constant and the fine structure of Ξ_β but not the transition.

PROOF. $\log L(s, \text{Sym}^k E) = \sum_p \text{tr} \text{Sym}^k(U_p) p^{-s} + O(1)$ for $\Re s > 1/2$, the prime-power terms converging by Deligne's bound. Substitute the expansion of the indicator and use $\log \zeta(s) = \log \frac{1}{s-1} + O(1)$ for the $k = 0$ term. $\square$

REMARK 3.3 (The smoothing is not vacuous). $\mathbf{1}_{[a,b]}$ is not continuous, so the Chebyshev series converges in $L^2(\mu_{\text{ST}})$ but not uniformly, and the interchange of $\sum_k$ with $\sum_p$ in (3.1) is not justified term by term. The standard remedy is to bracket the indicator between Beurling–Selberg majorants and minorants of bounded degree, which introduces an error $O(1/K)$ from truncating at $k \leq K$ and is then optimized against the error term in the Sato–Tate count. We note that the truncation must also absorb the k -dependence of the individual $O(1)$'s: both the prime-power tails ($|\text{tr} \text{Sym}^k| \leq k + 1$) and the size of $\log L(s, \text{Sym}^k E)$ near $s = 1$ grow with k through the analytic conductor, so the constants are not uniform and the series over k converges only through the bracketing, not term by term. The formula is correct; the proof of the interchange is the usual one and is not automatic.

EXAMPLE 3.4 (Coefficients). For $[a, b] = [\pi/3, 2\pi/3]$: $c_0 = 0.608998$, $c_1 = 0$, $c_2 = -0.413497$, $c_3 = 0$, $c_4 = 0.137832$, $c_5 = 0$, $c_6 = 0.068916$. The odd coefficients vanish by the symmetry of the interval about $\pi/2$; $\text{Sym}^2 E$ carries the leading correction, which is the familiar statement that the second moment of a_p is the first place a symmetric interval is sensitive to the curve.

4. Assessment

Question	Answer	Reference
$\beta_c(S_{[a,b]})$	1	Thm. 1.2
phase at $\beta = 1$	OPEN $(1, \infty)$	Thm. 1.2
$\zeta_S(\beta)$, $\beta \rightarrow 1^+$	$\delta \log \frac{1}{\beta-1} + O(1)$	Thm. 1.2
positive density $\Rightarrow$ closed?	never	Cor. 1.3
joint equidistribution	holds, unconditionally	Thm. 2.2
congruence obstruction	none	Thm. 2.3, Rem. 2.4
unique KMS_β for $\beta \leq 1$	yes	Thm. 2.3
Sym^k controls β_c	no; only the constant	Thm. 3.2
term-by-term interchange	needs smoothing	Rem. 3.3

No no-go theorem is warranted here, and that is itself the finding. The Sato–Tate intervals pass the congruence test of Chapter 37 by *equidistribution* — the clean mechanism — where the twin and Sophie Germain families pass it, by that paper's revised Theorem 2.2(2), through the opposite extreme of confinement to a generating class. The reason here is structural: the defining condition is a constraint on the *trace* of the Galois representation, while the residue class is its *determinant*, and Serre's open image theorem [118] is precisely the independence of the two. The obstructed families of Chapter 37 — Landau and Friedlander–Iwaniec — fail because their defining conditions are congruence conditions in disguise, landing in the kernel.

The price is that the phase is open. Positive density forces $k = 1$ and hence divergence at the critical point, so these systems cannot exhibit the closed low-temperature phase that [Ch. 1, Thm. 5.5] found for the Sophie Germain family. By Corollary 1.3 the two desiderata — a closed phase and an unobstructed high-temperature phase — are in tension: the first needs density zero with two prime conditions, the second is easiest for sets defined by trace conditions, which have positive density. By the revised Theorem 2.2(2) of Chapter 37 the twin, Sophie Germain and Chen families all combine a closed phase with the absence of obstruction, Chen unconditionally on the divergence side; those remain the only such families we know.

Two questions. Can one intersect a Sato–Tate interval with a sparse condition to obtain $k = 2$ while keeping joint equidistribution, for instance $\{p : \theta_p \in [a, b], p + 2 \in P_2\}$ — the

Chen condition contributing the extra logarithm and the trace condition remaining independent of it? And second, the anomalous sets of Theorem 5.3 of Chapter 15 were built by letting $[a, b]$ shrink with p ; by Theorem 1.2 any *fixed* interval gives $\beta_c = 1$ with an open phase, so the whole of the phase-structure interest in Sato–Tate families lies in shrinking intervals, which is where Chapter 15 placed it.

The free twin-prime Bost–Connes system is unconditionally supercritical: resolving the phase dichotomy via exact partial sums

This chapter is Paper XXXV of the series ([papers/XXXV-twin-regime/](#)).

SUMMARY. For the free (Toeplitz–Cuntz) variant of the localized Bost–Connes system of a prime set S , the partition series is geometric in $L_S(\beta) = \sum_{p \in S} p^{-\beta}$, so the phase dichotomy of Chapter 14 is decided by the fugacity at the classical critical point: Regime (S), $\beta_c^{\text{free}} > 1$, holds if and only if $L_S(1) > 1$. For the twin primes this comparison had resisted decision because the natural route — the Brun identity $L_S(1) = \frac{1}{2}B_2 + \sum_{p \in S} \frac{1}{p(p+2)}$ — feeds on the value of Brun’s constant, and the familiar $B_2 \approx 1.902160$ is a Hardy–Littlewood *extrapolation*, not a theorem. We resolve the dichotomy unconditionally by the blunt instrument the problem actually admits: partial sums of a positive series are rigorous lower bounds, and a segmented sieve to 1.1×10^{10} (29,889,853 twin pairs) gives

$$\sum_{\substack{p \leq 1.1 \times 10^{10} \\ p, p+2 \in \mathcal{P}}} \frac{1}{p} = 1.00195935 > 1.$$

Hence $L_{S_{\text{twin}}}(1) > 1$ and the free twin-prime system is in Regime (S): $\beta_c^{\text{free}} > 1$, unconditionally, so on the interval $(1, \beta_c^{\text{free}})$ the classical system is already in its broken phase while the free system admits no equilibrium at all. The Hardy–Littlewood rate predicts 1.0017 at this height, a cross-check, and the conditional value $\beta_c^{\text{free}} \approx 1.009$ is recorded as an estimate only. Two methodological remarks close the paper: occupancy of residue classes does not imply divergence of the per-character detecting series (the gap that keeps the Chen uniqueness problem open is Chen’s theorem in arithmetic progressions), and average correlation supplied by Rankin–Selberg poles does not touch the pointwise parity barrier: metaphor is not proof.

1. The dichotomy and the identity

Let $S \subseteq \mathcal{P}$ and let $\mathcal{TC}(S)$ be the free variant of Chapter 14: the generating isometries have *orthogonal* ranges, one for each $p \in S$, so the gauge-invariant counting runs over the free monoid on S and the Gibbs series is geometric,

$$Z_S^{\text{free}}(\beta) = \sum_{n \geq 0} L_S(\beta)^n, \quad L_S(\beta) = \sum_{p \in S} p^{-\beta}.$$

Thus a free equilibrium exists iff $L_S(\beta) < 1$, and Chapter 14’s trichotomy at the classical critical point $\beta_c(S_{\text{twin}}) = 1$ (Brun-summable, closed classical phase, Chapter 37) reads: *Regime (S)* if $L_S(1) > 1$, with β_c^{free} the unique root of $L_S(\beta) = 1$ in $(1, \infty)$; *Regime (B)* if $L_S(1) \leq 1$. The decision is a single comparison with 1.

For $S = S_{\text{twin}}$ (smaller members p of twin pairs), the elementary identity $\frac{1}{p+2} = \frac{1}{p} - \frac{2}{p(p+2)}$ summed over pairs gives, with Brun’s constant $B_2 = \sum_{p, p+2 \in \mathcal{P}} (\frac{1}{p} + \frac{1}{p+2})$ (finite by Brun’s sieve

[52]),

$$(1.1) \quad L_{S_{\text{twin}}}(1) = \frac{1}{2}B_2 + \sum_{p \in S_{\text{twin}}} \frac{1}{p(p+2)}, \quad \sum_{p \in S_{\text{twin}}} \frac{1}{p(p+2)} = 0.10798 \dots$$

(the second series converges rapidly and is evaluated by direct summation). The identity cannot decide the dichotomy by itself: the familiar value $B_2 \approx 1.9021605$ [101] is the computed partial sum *plus a Hardy–Littlewood extrapolation of the tail*; unconditionally one only knows the partial sums, ≈ 1.83 at the computed heights. Conditionally, (1.1) gives $L_S(1) \approx 0.9511 + 0.1080 = 1.059$; but a conditional inequality decides nothing.

2. Unconditional resolution via segmented sieving

The problem admits a blunt unconditional instrument: $L_S(1)$ is a series of positive terms, so every partial sum is a rigorous lower bound, and the comparison with 1 only requires some partial sum to cross it.

THEOREM 2.1. $\sum_{p \leq 1.1 \times 10^{10}, p, p+2 \in \mathcal{P}} \frac{1}{p} = 1.00195935 \dots > 1$, the sum running over 29,889,853

twin pairs enumerated by a deterministic segmented sieve of Eratosthenes (blocks of 2×10^8 , base primes to $\sqrt{1.1 \times 10^{10}}$; script and output archived in code/twin-regime/). Consequently $L_{S_{\text{twin}}}(1) > 1$ unconditionally: the free twin-prime system is in Regime (S), $\beta_c^{\text{free}}(S_{\text{twin}}) > 1$, and on $(1, \beta_c^{\text{free}})$ the classical localized system is in its closed broken phase while the free system admits no KMS state at all.

PROOF. The enumeration is exact and finite; positivity does the rest. Floating-point error is dominated by 3×10^7 additions of magnitude $\leq \frac{1}{3}$ in double precision, orders below the margin 1.96×10^{-3} . As a cross-check, the Hardy–Littlewood rate $2C_2/\log^2 t$ integrated from the previous checkpoint ($\sum_{p \leq 3 \times 10^8} = 0.991422$) predicts 1.0017 at 1.1×10^{10} , consistent. $\square$

REMARK 2.2. The precise root of $L_S(\beta) = 1$ depends on the full tail and is therefore tail-conditional; under Hardy–Littlewood it is $\beta_c^{\text{free}} \approx 1.009$ (and ≈ 1.211 for the Chen family). These are estimates, not assertions: the theorem above is exactly the part of the phase diagram that a finite computation can own.

3. Methodological remarks: why heuristics do not bypass the sieve

REMARK 3.1 (Occupancy does not imply divergence). High-temperature KMS uniqueness for a family S requires, by [Main, Thm 3.9], the divergence of the *per-character* detecting series $\sum_{p \in S, \chi(p) \neq 1} p^{-\beta}$ for every nontrivial Dirichlet χ . For the Chen primes, the full series diverges unconditionally for $\beta < 1$ (Chen’s lower bound plus Abel summation), and the occupied classes generate every $(\mathbb{Z}/M)^\times$ (Chapter 37); but occupancy is a statement about *which* classes carry primes, not about *how much* mass each class carries, and upper bounds in progressions cannot split a divergent series into per-class divergences ($\infty - \infty$). What is needed, and remains the named gap, is Chen’s theorem *in arithmetic progressions*: a log-power lower bound in every class outside each kernel. Three proposals in this project’s history have foundered on exactly this step; it is a sieve problem, and it does not dissolve under relabeling.

REMARK 3.2 (Metaphor is not proof). Rankin–Selberg poles at $\Re s = 1$ encode *average* pair correlation of automorphic coefficients; the parity barrier of sieve theory is a statement about the *pointwise* indistinguishability of integers with an even versus odd number of prime factors under sieve weights. Passing the twin condition through symmetric-power L -functions, gauge-flow readings of the circle method, or fermionic vacuum imagery changes the vocabulary and

not the obstruction: no functional-analytic reformulation in this series has produced, nor could produce, the missing pointwise lower bound. The honest unconditional content available today for the twin primes is Theorem 2.1; everything sharper is conditional, and is labeled so.

Open problems

Editorial note (September 2026). This list was compiled on 28 August 2026, before Chapters 17 and 29 were written and before the September line-by-line review of the corpus. Several items are now settled and are kept here for the record only: condition (C4) and every spectral-gap route to it are resolved, in the negative, in Chapter 29 (the orbit relation is hyperfinite for every admissible monoid); the non-abelian intermediate structure is resolved in Chapter 17 (the ordered Frobenius cocycle realizes $\text{Prob}(H \backslash G)$, with lift-dependence and the rigidity criterion); and the sieve-family items should be read against the revised Chapter 37 (two obstructed families, not four). Where an item below conflicts with those chapters, the chapters are authoritative.

- (1) **A non-abelian model between the two walls.** Does there exist an Ore semigroup that is non-abelian but whose range projections are nested rather than partitioning? Abelian semigroups cannot produce non-abelian essential values (Chapter 16); free non-abelian semigroups empty the high-temperature region (Chapter 14); the Hurwitz quaternions satisfy (C1)–(C3) (Chapter 25). *[Settled: condition (C4) fails for every admissible monoid, nested or not — Chapter 29. The residual question is whether any index structure outside the (C1)–(C3) framework, necessarily of infinite norm at some place, can carry both arithmetic thermodynamics and non-amenability. Chapter 18 now closes the scaling route universally (the abelian scaling bottleneck: scalar dynamics on a right LCM semigroup cannot produce a nonabelian residual symmetry) and opens the angular route: the angular Hurwitz cocycle $a \mapsto a/\text{Nrd}(a)^{1/2} \in \text{SU}(2)$ has Kakutani abscissa exactly 2, and the expected verdict $\text{KMS}_\beta \cong \text{Prob}(\text{SU}(2))$ for $\beta > 2$ is reduced to one named transport problem.]*
- (2) **Condition (C4).** Is the measured groupoid of the Hurwitz system non-amenable? The Ramanujan route is closed by the nested/partition dichotomy (Chapter 26); the question reduces to a spectral gap for the single representation $\pi_0|_{G^1}$ of the norm-one subgroup, together with essential freeness of the G^1 -action (Chapter 27). *[Settled, in the negative: the orbit relation is hyperfinite, the factor injective, and attaining the Ramanujan bound certifies temperedness, the signature of amenability — Chapter 29.]*
- (3) **Cartan uniqueness without norm-separation.** Is $C(Y_S)$ intrinsically characterised inside $(\mathcal{A}_{K,S}, \sigma)$ when the relation lattice is nontrivial? The Nica–Toeplitz relations block the obvious counterexample (Chapter 22); the spectral route cannot decide it (Chapter 23); mixed isometries exist iff a single measure-preserving transformation is not ergodic (Chapter 23, Prop. 3.2).
- (4) **Which field is recovered.** The equivariant C^* -system sees $(S, N|_S, G_S)$; whether this triple determines K is an arithmetic question (Chapter 21). Over function fields the answer is known to stop at the isogeny class plus a torsor coordinate (Chapters 33–34).
- (5) **The exact value of $W_1(\varphi_+, \varphi_-)$** (Chapter 6). *[Partially settled: Chapter 7 reduces the problem to a sequence of second-order cone programs with certified two-sided error, and replaces the squeeze lower bound by the strictly sharper quadratic chaos bound $\Pi_\infty(w_0^2 + 4 \sum_j w_j^2 t_j^2 / (1 - t_j)^2)^{1/2}$. Remaining: (a) does the odd Walsh-chaos hierarchy exhaust W_1 , or do higher-degree terms strictly increase the supremum; (b) the matching upper bound*

- now advanced in Chapter 8: the adaptive coupling gives the closed form $w_0\Pi_\infty + \sum_k w_{jk}\tau_{jk}\prod_{l<k}m_{jl}$, strictly below w_0 and pinching against the squeeze lower bound in the deep freeze; the residual factor is now fully diagnosed in Chapter 9: the single-step Kantorovich polytope is exactly the scan bound (fractional transport gains nothing), diagonal displacements carry the closed-form cone discount $d_L = w(1 + 1/\sqrt{2}) < 2w$, and the one remaining instrument is the discrete ∞ -Laplacian geodesic problem for d_L ; (c) the optimal profile is NOT parity-measurable (numerical verdict, Chapter 7, Remark 4.3).] Also open: a transport quantity built from the tail σ -algebra that would detect the transition (Chapter 24).
- (6) $K_*(\partial\mathcal{A}^{\text{aff}})$ of the affine boundary (Chapter 3). [Partially settled: Chapter 10 computes both groups at a principal single prime — the norm survives as the exact order $q - 1$ of $[1]$, the trace of the uniformizer enters the intermediate exterior layers, and the invariant hears the unit orbit of the uniformizer, hence is not canonical in the prime. Remaining after Chapter 11's ray-class repair (which proves the class-group order h is the exact index of definable scalings and promotes the order of $[1]$ to $N\mathfrak{p}^h - 1$): (a) separating q from h : generically resolved — the intermediate layers separate, e.g. $\mathbb{Q}(\sqrt{-5})/\mathfrak{p}_3$ vs $\mathbb{Q}(i)/(3)$ share $\text{ord}[1] = 8$ but have T_1 of orders 2 vs 4 — with only layer-coincident pairs left to the dual coaction; (b) the Galois descent from the Hilbert class field — Chapter 12 now proves the full class group emerges as the group of descent obstructions $\text{Cl}_K \hookrightarrow H^1(G, \mathcal{O}_H^\times)$, with the ladder $\text{ord}[1] = q^h - 1$ (bottom) and $q^{|\text{Cl}_K|} - 1$ (top), and reduces the equivariant K -theory to the inert local model crossed by $\mathbb{Z}/h$; the integral computation of that reduction is now carried out for the first inert case in Chapter 13 (exact torsion by Smith forms, the unit class split by $\frac{1 \pm U_\alpha}{2}$ into a pair of order-8 classes, and the entanglement law: the class group doubles only the non-induced strata); remaining there are the twisted character bookkeeping and the degree-four extension data; (c) the several-prime case, where the definable scalings are the kernel of $\mathbb{Z}^S \rightarrow \text{Cl}_K$; plus the extension data for $n \geq 3$ and the q -primary bookkeeping from Chapter 10.] Also open: $H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T})$ itself rather than its kernel (Chapter 4). [Settled in the correct sense by Chapter 5: the group is non-smooth and its coboundary action is turbulent, so no classification by countable structures exists; the complete smooth residue is the Mackey-range spectrum; the arithmetic embeds as $\widehat{G}_S/\Xi_\beta$, the unique equicontinuous thread, while the ambient group is orbit-equivalence amnesiac. Remaining: the flagged turbulence lemma and noncompact coefficients.]
- (7) **The regime of the twin primes** in the free system, equivalent to comparing a Brun-type constant with 1 (Chapter 14). [Settled, unconditionally: a rigorous segmented-sieve computation (Chapter 39, `code/twin-regime/`) gives $\sum_{p \leq 1.1 \times 10^{10}} 1/p = 1.00196 > 1$ over the twin primes — a partial sum of a positive series, hence an unconditional lower bound — so $L_S(1) > 1$ and the free system is in Regime (S), $\beta_c^{\text{free}} > 1$. The exact value of β_c^{free} remains conditional on the tail.]
- (8) **A knot family that is linking-rich and volume-sparse**, which the arithmetic topology dictionary requires and which is not known (Chapter 30).
- (9) **Whether a refinement of the obstruction data recovers the essential-value subgroup E_β up to conjugacy** in the Galois-twisted groupoid (Chapter 16).
- (10) **Closed phase and no congruence obstruction simultaneously.** A closed low-temperature phase needs density zero with two prime conditions; an unobstructed high-temperature phase is easiest for trace conditions, which have positive density. The Chen primes are the only known candidate satisfying both (Chapters 37–38).

Two items are open at the level of the project rather than the mathematics and are recorded in the repository's status file: a repeatable priority search against the literature on KMS states of groupoid C^* -algebras with infinite sparse S , and an independent verification of the bibliography.

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